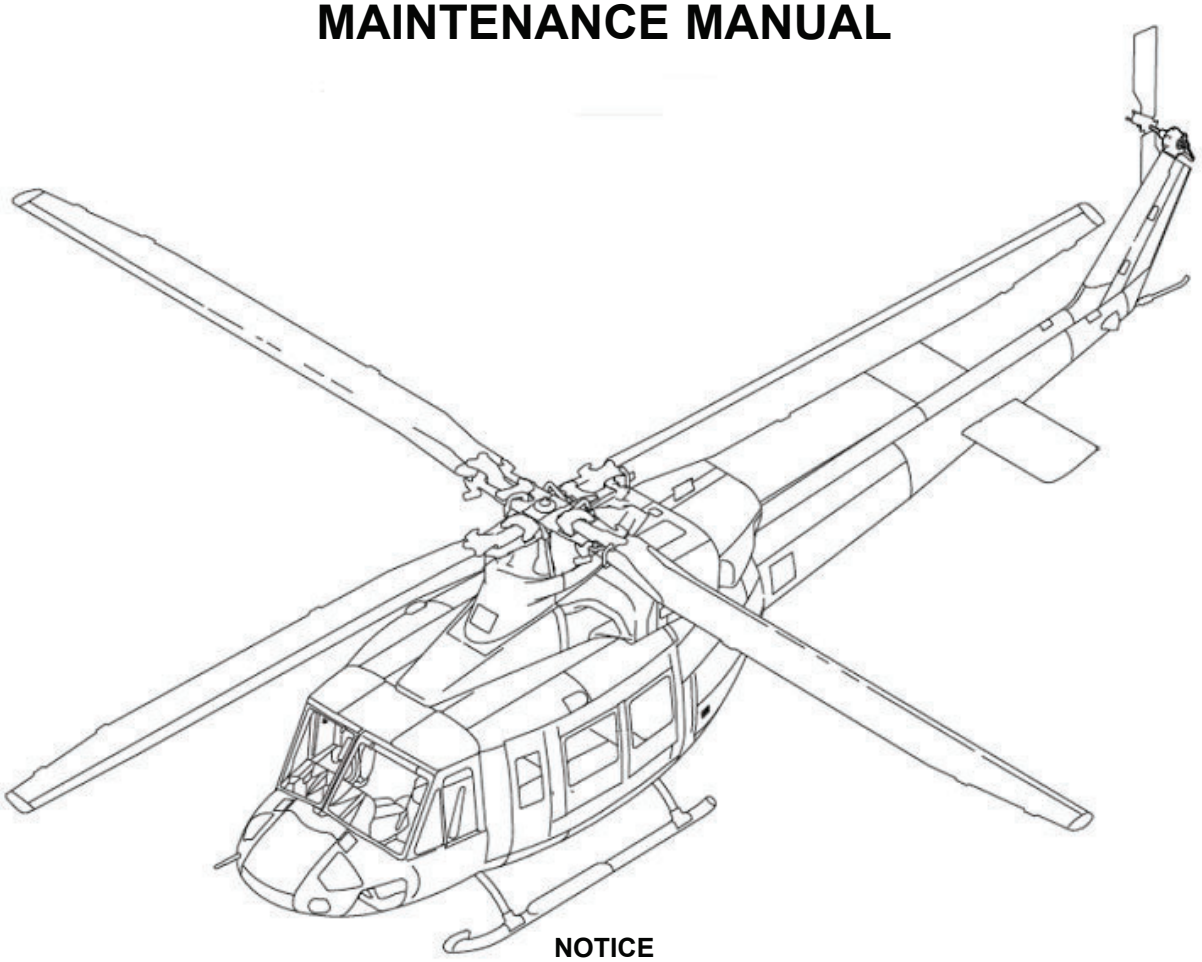
Bell MODEL 412/412EP

MAINTENANCE MANUAL



NOTICE

The instructions set forth in this manual, as supplemented or modified by Alert Service Bulletins (ASB) or other directions issued by Bell and Airworthiness Directives (AD) issued by the applicable regulatory agencies, shall be strictly followed.

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Highlights

The listed changes are introduced in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

N/C	Data module	Reason for update
Chapter 05 - Inspections:		
C	412-A-05-40-00-00A-281A-A / 00022, Scheduled Inspections - Daily Inspection - Part A	Added 412-SI-97 applicability to the applicable steps.
C	412-A-05-40-00-01A-281A-A / 00023, Scheduled Inspections - 100 Hour/12 Month Inspection - Part A	Added new step 1 to Fuselage section to add: "Inspect the aft transmission mount friction dampers for condition and security". Added 412-SI-97 applicability to steps 4 and 5 of section TAIL ROTOR DRIVESHAFT.
C	412-A-05-40-00-04A-281A-A / 00026, Scheduled Inspections - 25 Hour/30 Day Inspection - Part B	Added 412-SI-97 applicability to the applicable steps.
C	412-A-05-40-00-05A-281A-A / 00027, Scheduled Inspections - 300 Hour/12 Month Inspection - Part B	Modified step 7.6 of section FUSELAGE - CABIN from 'Move pylon fore and aft, using mast as lever. Check friction dampers for freedom of movement and smooth operation'. Added 412-SI-97 applicability to steps 7 and 8 of section TAILBOOM notes.
C	412-A-05-47-00-11A-283A-A / 00044, Special Inspections - Each 600 Hours of Tail Rotor Driveshaft Operation or 12 Months	Added 412-SI-97 applicability to the applicable notes.
C	412-A-05-47-00-12A-283A-A / 00046, Special Inspections - Each 600 Hours of Main Driveshaft Operation or 12 Months	Added 412-SI-97 applicability to the step 3 note.
C	412-A-05-47-00-19A-283A-A / 00054, Special Inspections - Each 2500 or 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-109 and Subsequent) Operation	Modified Note to change "412-040-366-129" to "412-040-366-127/-129". Modified step 11 to add "and mast assembly 412-040-366-127/-129 and subsequent" after mast configuration. Added 412-SI-97 applicability to common notes.
C	412-A-05-47-00-20A-283A-A / 00055, Special Inspections - Each 2500 Hours of Tail Rotor Drive System Operation	Added 412-SI-97 applicability to the common note.
C	412-A-05-47-00-21A-283A-A / 00056, Special Inspections - Each 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-103 and -105) Operation	Added 412-SI-97 applicability to the common note.
C	412-A-05-47-00-22A-283A-A / 00057, Special Inspections - Each 3000 Hours of Transmission (412-040-002) Operation	Added 412-SI-97 applicability to the common note.
C	412-A-05-47-00-23A-283A-A / 00058, Special Inspections - Each 3000 Hours of Transmission (412-040-004/412-040-007/412-040-802) Operation	Added 412-SI-97 applicability to the common note.
C	412-A-05-47-00-28A-283A-A / 00060, Special Inspections - Each 3000 Hours or 5 Years Of Transmission Mount Assembly (204-031-927) And Friction Damper Assembly (204-031-920) Operation	Added the transmission mounts detailed inspection link.
C	412-A-05-50-00-03A-284A-A / 00066, Conditional Inspection - After Overtorque	Added 412-SI-97 applicability to steps 4 and 5.
C	412-A-05-11-00-00A-042A-A / 00070, Component Overhaul Schedule - Description	Modified Mast Assembly Part number from "412-040-366-129" to "412-040-366-127/-129". Modified Note 4 to add "or 5 years" after 2500 hours. Modified Notes 19, 22 and 23 to add "or 5 years" after 3000 hours. Added 412-SI-97 applicability to the applicable notes.

N/C	Data module	Reason for update
Chapter 12 - Servicing:		
C	412-A-12-00-00-00A-040A-A / 00126, Servicing - General	Added 412-570-001-103, BHT-412-SI-74 and BHT-412-SI-97 applicability to the note 5. Added BHT-412-SI-97 applicability to the note 6.
C	412-A-12-01-00-00A-040A-A / 00141, Lubrication - Description	Revised Table 1 Note to add the applicability BHT-412-SI-97. Revised Note 7 and 8 of Figure 1, Sheet 2 to add the applicability BHT-412-SI-97.
C	412-A-12-01-00-00A-240B-A / 00143, Lubrication - Flex Coupling Lubrication Log	Revised the contents of Table 1. Deleted Figure 1.
Chapter 18 - Vibration and Noise Analysis:		
C	412-A-18-11-00-00A-042B-A / 00169, Balancing Main Driveshaft - Control of Main Driveshaft Vibration	Added 412-SI-97 applicability to the step 3 and Table 1 title. Added 412-570-001-103, BHT-412-SI-74 and 412-SI-97 applicability to the step 2.
C	412-A-18-11-00-09A-271A-A / 00173, Balancing Main Driveshaft - Making Adjustments	Added 412-SI-97 applicability to the steps 1.3, 1.4 and Table 1 note.
C	412-A-18-11-00-14A-271A-A / 00188, Main Driveshaft - Vibration Control	Added 412-SI-97 applicability to the Support equipment remarks, step 2, Note in step 2.19, Table 1 title and Table 2 note. Revised step 2.5 to add 'Refer to applicable Flight Manual' instead of '(BHT-412-FM)'.
Chapter 52 - Doors and Windows:		
C	412-A-52-01-00-09A-520A-A / 00892, Heated Windshield Kit (412-706-010/-138) - Removal	Added new step 16. Modified step 11 to add solid shim (24). Revised the Figure 1 to add solid shim (24).
C	412-A-52-01-00-09A-280A-A / 00894, Heated Windshield Kit (412-706-010/-138) - Inspection	Updated the figure to add shim (24).
C	412-A-52-01-00-09A-720A-A / 00896, Heated Windshield Kit (412-706-010/-138) - Installation	Added new steps 3.3, 3.4 and 4. Deleted step 3.25. Modified steps 1, 2, 5, 6 and 7 to add solid shim (24). Revised the figure 1 to add solid shim (24). Added consumable C-317 and C-406.
Chapter 63 - Main Rotor Drive System:		
C	412-A-63-10-01-01A-520A-A / 00994, Engine-to-Transmission (Main) Driveshaft - Removal	Added 412-SI-97 applicability to the steps 4 and 5.
C	412-A-63-10-01-01A-720A-A / 00999, Engine-to-Transmission (Main) Driveshaft - Installation	Added 412-SI-97 applicability to steps 1, 2 and notes of step 8. Revised step 5 to add 'Refer to applicable Flight Manual' instead of '(BHT-412-FM)'.
C	412-A-63-10-01-01A-042A-A / 01000, Engine-to-Transmission (Main) Driveshaft - Balancing	Added 412-SI-97 applicability to the step 1 Note.
C	412-A-63-10-01-02A-520A-A / 01001, Engine-to-Driveshaft Coupling Adapter - Removal	Revised Support equipment to add 412-SI-97 applicability. Revised note 3 of Figure 2 to add 412-SI-97 applicability.
C	412-A-63-10-01-02A-280A-A / 01002, Engine-to-Driveshaft Coupling Adapter - Inspection	Revised Note 6 of Figure 3, Sheet 3 add 412-SI-97 applicability. Added 412-SI-97 applicability to steps 1 and 2.
C	412-A-63-10-01-02A-720A-A / 01004, Engine-to-Driveshaft Coupling Adapter - Installation	Revised Support equipment to add 412-SI-97 applicability. Revised note 3 of Figure 2 to add 412-SI-97 applicability.
C	412-A-63-21-01-00A-028A-A / 01005, Main Rotor Mast - Description	Revised Note 1 of Figure 1, Sheet 3 to add 412-SI-97 applicability.

N/C	Data module	Reason for update
C	412-A-63-21-01-00A-520A-A / 01006, Main Rotor Mast - Removal	Revised Note 1 of Figure 1, Sheet 3 to add 412-SI-97 applicability. Revised Procedural Step 3 Caution and 6 to add 412-SI-97 applicability.
C	412-A-63-21-01-00A-251A-A / 01007, Main Rotor Mast - Cleaning	Revised Note 1 of Figure 1, Sheet 3 to add 412-SI-97 applicability.
C	412-A-63-21-01-00A-720A-A / 01009, Main Rotor Mast - Installation	Revised Note 1 of Figure 2, Sheet 3 add 412-SI-97 applicability. Revised steps 14.1, 15, 20 and step 4 Caution to add 412-SI-97 applicability.
C	412-A-63-21-01-00A-272A-A / 01010, Torque Tube Assembly - Alignment	Revised Note 1 of Figure 1, Sheet 2 to add 412-SI-97 applicability. Revised Note 1 to add 412-SI-97 applicability.
C	412-A-63-21-02-00A-520A-A / 01013, Transmission - Removal	Added 412-SI-97 applicability to step 6 Caution and step 7.
C	412-A-63-21-02-00A-720A-A / 01015, Transmission - Build-up	Added Classic to HP applicability to the step 13.9.
C	412-A-63-21-02-00A-720B-A / 01016, Transmission - Installation	Added 412-SI-97 applicability to the steps 8 and 9.
C	412-A-63-22-01-18A-042A-A / 01107, Full Flow Debris Monitor - Description	Updated DMC applicability to add BHT-412-SI-97. Updated para 1 to add BHT-412-SI-97 applicability.
C	412-A-63-22-01-18A-520A-A / 01108, Full Flow Debris Monitor - Removal	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-63-22-01-18A-280A-A / 01109, Full Flow Debris Monitor - Inspection	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-63-22-01-18A-251A-A / 01110, Full Flow Debris Monitor - Cleaning	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-63-22-01-18A-720A-A / 01111, Full Flow Debris Monitor - Installation	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-63-21-02-02A-520A-A / 01140, Main Input Quill Magnetic Seal - Removal	Updated DMC applicability to add BHT-412-SI-97. Updated Figure 1 and 2 to add 412-SI-97 applicability.
C	412-A-63-21-02-02A-520B-A / 01141, Main Input Quill Magnetic Seal - Removal	Updated DMC applicability to add BHT-412-SI-97. Updated Figure 1 and 2 to add 412-SI-97 applicability.
C	412-A-63-21-02-02A-720A-A / 01145, Main Input Quill Magnetic Seal - Installation	Updated DMC applicability to add BHT-412-SI-97. Updated Figure 1 and 3 to add 412-SI-97 applicability.
C	412-A-63-21-02-02A-720B-A / 01146, Main Input Quill Magnetic Seal - Installation	Updated DMC applicability to add BHT-412-SI-97. Updated Figure 1 and 3 to add 412-SI-97 applicability.
C	412-A-63-21-02-07A-042A-A / 01170, Tail Rotor Drive Quill - Description	Added 412-SI-97 applicability to the steps 1 and 2.
C	412-A-63-21-02-07A-280A-A / 01172, Tail Rotor Drive Quill - Inspection	Added 412-SI-97 applicability to steps 1 and 2.
C	412-A-63-21-02-08A-720A-A / 01181, Rotor Brake Quill - Installation	Added 412-SI-97 applicability to step 6 Caution and step 9.
Chapter 64 - Tail Rotor:		
C	412-A-64-00-00-00A-280B-A / 01230, Tail Rotor Hub and Blade Assembly - Inspection	Modified step 9 to change the DMC reference from ‘DMC-412-A-64-00-00-00A-280A-B’ to ‘DMC-412-A-64-00-00-05A-280A-B’..
Chapter 65 - Tail Rotor Drive System:		
C	412-A-65-10-00-00A-422A-A / 01258, Tail Rotor Driveshaft - Troubleshooting	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-00-00A-520A-A / 01259, Tail Rotor Driveshaft - Removal	Updated DMC applicability to add BHT-412-SI-97.

N/C	Data module	Reason for update
C	412-A-65-10-00-00A-251A-A / 01260, Tail Rotor Driveshaft - Cleaning	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-00-00A-280A-A / 01261, Tail Rotor Driveshaft - Inspection	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-00-00A-685A-A / 01262, Tail Rotor Driveshaft - Repair	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-00-00A-720A-A / 01263, Tail Rotor Driveshaft - Installation	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-01-00A-520A-A / 01264, Driveshaft Hanger Assemblies - Removal	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-01-00A-251A-A / 01265, Driveshaft Hanger Assemblies - Cleaning	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-01-00A-280A-A / 01266, Driveshaft Hanger Assemblies - Inspection	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-01-00A-240A-A / 01267, Driveshaft Hanger Assemblies - Lubrication	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-01-00A-720A-A / 01268, Driveshaft Hanger Assemblies - Installation	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-02-00A-520A-A / 01270, Tail Rotor Driveshaft and Hangers - Removal	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-02-00A-251A-A / 01271, Tail Rotor Driveshaft and Hangers - Cleaning	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-02-00A-280A-A / 01272, Tail Rotor Driveshaft and Hangers - Inspection	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-02-00A-685A-A / 01273, Tail Rotor Driveshaft and Hangers - Repair	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-02-00A-240A-A / 01274, Tail Rotor Driveshaft and Hangers - Lubrication	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-10-02-00A-720A-A / 01275, Tail Rotor Driveshaft and Hangers - Installation	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-65-20-01-11A-280A-A / 01297, Flexible Couplings - Inspection and Lubrication	Added 412-SI-97 applicability to Note of steps 1 and 12.
C	412-A-65-20-01-00A-520A-A / 01302, Tail Rotor Gearbox - Removal	Revised Support Equipment to add 412-SI-97 applicability. Revised Procedural step 7 to add 412-SI-97 applicability.
C	412-A-65-20-01-00A-720A-A / 01306, Tail Rotor Gearbox - Installation	Revised Support Equipment to add 412-SI-97 applicability. Revised Caution of steps 1 and 8 to add 412-SI-97 applicability.
Chapter 71 - Power Plant:		
C	412-A-71-00-00-01A-272A-A / 01591, Engine to Transmission - Alignment	Updated DMC applicability to add BHT-412-SI-97. Updated Figure 2 to add 412-SI-97 applicability.
C	412-A-71-00-00-01A-272B-A / 01592, Engine to Transmission (412Classic/SP to 412HP Upgrade) - Alignment	Updated DMC applicability to add BHT-412-SI-97. Updated the common note to add 412Classic. Removed applicability from Figure 3 title.
Chapter 76 - Engine Controls:		
C	412-A-76-00-00-00A-028A-A / 01723, Engine Controls - Description	Revised Procedural step 7 and 8 to add 412-SI-97 applicability.
C	412-A-76-10-00-00A-271A-A / 01727, Engine Torque Transmitter - Adjustment and Functional Check	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-76-10-00-00A-271B-A / 01728, Engine Torque Transmitter - Adjustment and Functional Check	Updated DMC applicability to add BHT-412-SI-97. Removed applicability from Figure 1 title.

N/C	Data module	Reason for update
C	412-A-76-10-00-00A-271C-A / 01730, Torque Limit - Adjustment	Added 412-SI-97 applicability to the 3rd row of step 4.1, title of Table 3 and Figure 5.
C	412-A-76-11-00-00A-028A-A / 01740, Droop Compensator - Description	Added 412-SI-97 applicability to the Figure 4 title and step 2.
C	412-A-76-11-00-06A-276B-A / 01756, N2 Control - Rigging	Updated DMC applicability to add BHT-412-SI-97. Added 412-SI-97 applicability to note of steps 4 and 11.
C	412-A-76-11-00-07A-720A-A / 01759, Droop Compensator - Installation	Revised steps 7 and 8 to add 412-SI-97 applicability.
C	412-A-76-11-00-08A-720A-A / 01762, Droop Compensator Controls - Installation	Revised steps 20 and 21 to add 412-SI-97 applicability
C	412-A-76-11-00-09A-276B-A / 01764, N2 Control - Rigging	Updated DMC applicability to add BHT-412-SI-97.
Chapter 79 - Engine Oil System:		
C	412-A-79-00-00-00A-422A-A / 01767, Engine Oil System - Troubleshooting	Revised the entire corrective action of the Probable causes: Normal wear and Excessive wear from table 1. Removed 'Chips in oil cooler' from the probable cause section of table 1.
Chapter 95 - Instrument System:		
C	412-A-95-61-02-00A-042A-A / 01845, Triple Torque Pressure Indicator - Description	Updated DMC applicability to add BHT-412-SI-97.
Chapter 96 - Electrical System:		
C	412-A-96-00-02-04A-042A-A / 01904, Electrical and Avionics Equipment - Reference Designator, Nomenclature, and Location - Description	Added 412-SI-97 applicability to the applicable rows.
C	412-A-96-12-00-01A-042A-A / 01980, Mast Torque Measurement System (412 Classic/SP to 412 HP Upgrade) - Description	Updated DMC applicability to add BHT-412-SI-97. Updated DMC title to add ‘Classic’.
C	412-A-96-12-00-01A-280A-A / 01981, Mast Torque - Indication Check	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-96-12-00-01A-320A-A / 01982, Mast Torque System - Operational Check	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-96-12-00-01A-273A-A / 01983, Mast Torque System - Calibration Check	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-96-12-00-01A-273B-A / 01984, Mast Torque System - Zero Torque Calibration	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-96-12-00-01A-280B-A / 01985, Mast Torque System - Flight Check	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-96-12-00-02A-042A-A / 01986, Engine Torque Pressure Indicating System - Description	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-96-12-00-02A-320A-A / 01987, Engine Torque Pressure Indicating System - Operational Check	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-96-04-00-08A-042A-A / 02120, Governor RPM Actuator System - Description	Updated DMC applicability to add BHT-412-SI-74 and BHT-412-SI-97.
C	412-A-96-04-00-08A-320A-A / 02121, Governor RPM Actuator System - Operational Check	Updated DMC applicability to add BHT-412-SI-74 and BHT-412-SI-97.
C	412-A-96-04-00-08A-042B-A / 02122, Governor RPM Actuator System - Description	Updated DMC applicability to add BHT-412-SI-97.
C	412-A-96-04-00-08A-320B-A / 02123, Governor RPM Actuator System - Operational Check	Updated DMC applicability to add BHT-412-SI-97.
Chapter 99 - Kits:		

N/C	Data module	Reason for update
C	412-A-97-00-02-00A-042B-A / 02655, Recorder Download Unit (RDU) - Description	Modified the contact information from "Smiths Industries Aerospace" to "Extant Aerospace".
C	412-A-97-00-03-02A-042A-A / 02681, Recorder Download Unit (RDU) - Description	Modified the contact information from "GE aviation" to "Extant Aerospace".

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LIST OF CHAPTERS

Chapter	Data module code
Chapter 01 - Introduction	412-A-01-00-00-00A-009A-A / 00007
Chapter 04 - Airworthiness Limitations Schedule	412-A-04-00-00-00A-009A-A / 00011
Chapter 05 - Inspections	412-A-05-00-00-00A-009A-A / 00014
Chapter 06 - Dimensions and Charts	412-A-06-00-00-00A-009A-A / 00071
Chapter 07 - Lifting and Jacking	412-A-07-00-00-00A-009A-A / 00074
Chapter 08 - Weight and Balance	412-A-08-00-00-00A-009A-A / 00081
Chapter 09 - Towing	412-A-09-00-00-00A-009A-A / 00102
Chapter 10 - Parking and Mooring	412-A-10-00-00-00A-009A-A / 00108
Chapter 11 - Placards and Markings	412-A-11-00-00-00A-009A-A / 00122
Chapter 12 - Servicing	412-A-12-00-00-00A-009A-A / 00125
Chapter 18 - Vibration and Noise Analysis	412-A-18-00-00-00A-009A-A / 00144
Chapter 21 - Air Distribution (Ventilation)	412-A-21-00-00-00A-009A-A / 00189
Chapter 25 - Equipment and Furnishings	412-A-25-00-00-00A-009A-A / 00244
Chapter 26 - Fire Protection	412-A-26-00-00-00A-009A-A / 00291
Chapter 28 - Fuel System	412-A-28-00-00-00A-009A-A / 00323
Chapter 29 - Hydraulic System	412-A-29-00-00-00A-009A-A / 00610
Chapter 30 - Ice and Rain Protection	412-A-30-00-00-00A-009A-A / 00677
Chapter 32 - Landing Gear	412-A-32-00-00-00A-009A-A / 00694
Chapter 52 - Doors and Windows	412-A-52-00-00-00A-009A-A / 00774
Chapter 53 - Fuselage	412-A-53-00-00-00A-009A-A / 00897
Chapter 62 - Main Rotor	412-A-62-00-00-00A-009A-A / 00932
Chapter 63 - Main Rotor Drive System	412-A-63-00-00-00A-009A-A / 00987
Chapter 64 - Tail Rotor	412-A-64-00-00-00A-009A-A / 01225
Chapter 65 - Tail Rotor Drive System	412-A-65-00-00-00A-009A-A / 01257
Chapter 67 - Flight Controls	412-A-67-00-00-00A-009A-A / 01319
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Chapter 79 - Engine Oil System	412-A-79-00-00-00A-009A-A / 01765
Chapter 95 - Instrument System	412-A-95-00-00-00A-009A-A / 01787
Chapter 96 - Electrical System	412-A-96-00-00-00A-009A-A / 01870
Chapter 97 - Avionics	412-A-97-00-00-00A-009A-A / 02235
Chapter 98 - Wiring Diagrams	412-A-98-00-00-00A-009A-A / 02454
Chapter 99 - Kits	412-A-99-00-00-00A-009A-A / 02622
Chapter 01 - Introduction	

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Bulletin Record - Technical standard records	DMC-412-A-00-71-00-00A-008B-A / 00002	2025-05-13	ALL
Temporary revision record - Technical standard records	DMC-412-A-00-72-00-00A-008A-A / 00003	2023-05-31	ALL
Customer support and services - Function	DMC-BHT-A-00-00-00-00A-011A-A / 00004	2023-05-31	ALL
Warranty and Customer Feedback - Administrative forms and data	DMC-BHT-A-00-00-00-00A-023A-A / 00005	2023-05-31	ALL
Use of a S1000D publication - Introduction	DMC-BHT-A-00-00-00-00A-018A-A / 00006	2023-05-31	ALL

Bulletin Record - Technical standard records

This Bulletin Record provides a current listing of applicable bulletins that have been incorporated in this manual. Subsequent applicable bulletins will be incorporated in future revisions/reissues.

Table 1. ALERT SERVICE BULLETINS

ASB NUMBER	SUBJECT	DATE
(1) 412-22-188	Air Conditioning STC SR00066DE Aft Evaporator Return Screen Installation, Verification of	30 March 2022
1 This and all previously issued applicable bulletins have been incorporated, except 412-10-138, 412-10-143, 412-12-151 and 412-15-163.		

Table 2. *TECHNICAL BULLETINS*

TB NUMBER	*SUBJECT*	*DATE*
(1) *412-20-244*	*Collective Lever Assembly, 412-010-464-101, Introduction of*	*9 December 2020*
1 *This, and all previously issued applicable bulletins, have been incorporated except 412-01-178, 412-04-199, 412-07-211, 412-08-216, 412-09-220, 412-13-230 and 412-16-237.*		

Temporary revision record - Technical standard records

This temporary revision record provides a current listing of active temporary revisions against the manual. Temporary revisions which have been canceled/incorporated will only be maintained on the record until the next revision is issued. If there are no temporary revisions shown on the record, this is confirmation that there are no temporary revisions issued against the manual.

Temporary Revision No.	Title	Date issued	Date canceled

Customer support and services - Function

1. Customer support and services

Flying smart means that no matter where you are, or what time it is, you can make a call and get additional information, clarification, or advice on a technical or operational issue concerning your helicopter or information contained in our Technical Publications. Product Support Engineering (PSE) is just a phone call away and may be contacted as follows:

Phone:	450-437-2862 or 800-363-8023 (US/Canada)
Fax:	450-433-0272
E-mail:	productsupport@bellflight.com

Warranty and Customer Feedback - Administrative forms and data



THIS MANUAL APPLIES ONLY TO HELICOPTERS AND COMPONENTS MAINTAINED IN ACCORDANCE WITH BELL APPROVED PROCEDURES USING BELL APPROVED PARTS.

ALL INSPECTION, REPAIR AND OVERHAUL PROCEDURES PUBLISHED BY BELL, INCLUDING PART RETIREMENT LIFE, ARE BASED SOLELY ON THE USE OF BELL PARTS THAT HAVE BEEN MAINTAINED USING BELL APPROVED DATA. THE DATA PUBLISHED HEREIN OR OTHERWISE SUPPLIED BY BELL IS NOT APPLICABLE TO NON-BELL PARTS OR PARTS THAT HAVE BEEN REPAIRED USING DATA AND/OR PROCESSES NOT APPROVED BY BELL.

BELL IS NOT RESPONSIBLE FOR ANY PART OTHER THAN THOSE THAT IT HAS APPROVED.

BEFORE PERFORMING ANY PROCEDURE CONTAINED IN THIS MANUAL YOU MUST INSPECT THE AFFECTED PARTS AND RECORDS FOR EVIDENCE OF ANY MANUFACTURE, REPAIR, REWORK, OR USE OF A PROCESS NOT APPROVED BY BELL.

IF YOU IDENTIFY OR SUSPECT THE USE OF PARTS NOT AUTHORIZED BY BELL, EITHER REMOVE THE AFFECTED ITEM FROM THE AIRCRAFT OR OBTAIN INSTRUCTIONS FOR CONTINUED AIRWORTHINESS FROM THE MANUFACTURER OR THE ORGANIZATION THAT APPROVED THE REPAIR.

For Spare Parts Warranty, refer to Table 1

For New Helicopter Warranty, refer to Table 2

Table 1. *Spare Parts Warranty*

SPARE PARTS WARRANTY

WARRANTY: *Seller warrants each new helicopter part or helicopter part reconditioned by Seller to be free from defect in material and workmanship under normal use and service and if installed on Bell model helicopters for up to 1000 hours of operation, one (1) year from date of installation, or two (2) years from date of shipment by Seller, whichever occurs first. Seller assigns each manufacturer's warranty to Buyer to the extent such manufacturer's warranty exists and is assignable.*

Seller's sole obligation under this warranty is limited to the repair or replacement of parts which are determined to Seller's reasonable satisfaction to have been defective within the applicable warranty period as described above. Replacement of parts may be either new or reconditioned at Seller's election and at the lowest allowable maintenance level contained in Seller's manuals, service bulletins or applicable supplier manuals.

Seller shall also reimburse reasonable freight charges, excluding insurance, customs fees, duties, handling fees, and taxes. Seller shall not reimburse Buyer for any parts repaired or replaced outside of the Seller's Warranty Claims Process unless express prior written authorization is granted by Seller's Warranty Department to Buyer for such repair or replacement.

Parts, components and assemblies of all helicopter parts may have been restored or reworked due to mars, blemishes, dents or other irregularities during the manufacturing process. Such restoration and/or rework are permitted under Seller's approved manufacturing and engineering processes and guidelines. The restoration and/or rework so completed do not render such items defective in material or workmanship

NO FAULT FOUND:*In the event Seller determines, after evaluation of a returned part, that a defect does not exist, then Buyer shall pay all expenses incurred by Seller related to the return including, but not limited to, costs incurred in shipping and evaluating the part and cost for any replacement part and restocking of the part, which may include a fifteen percent (15%) restocking fee, per item, No Fault Found fee charged to Seller or a minimum of $300.00 USD. In addition, Seller shall not reimburse Buyer for any costs related to the removal or reinstallation of such a part.*

SELLER'S WARRANTY CLAIM PROCESS: *Defective parts must be reported in writing to the Seller's Warranty Administration within thirty (30) days of being found defective. Parts may be repaired or replaced with new or reconditioned parts, at Seller's election. Warranty adjustment is contingent upon the Buyer complying with the Seller's Warranty Process and with the Seller's Warranty Administration disposition instructions for defective parts. Failure to properly comply with Seller's Warranty Process may, at Seller's sole option, void Seller's warranty as to the allegedly defective part.*

RETURN SHIPMENT: *Parts returned to Seller will be eligible for remedy under this warranty only if the part is carefully packed by the Buyer for the return shipment. Damage occurring to a part due to improper packaging may result in the denial of a warranty claim. In the event that Seller determines a returned part to be damaged or unsalvageable due to improper packaging, the Buyer will be billed repair or replacement cost incurred by Seller. The party initiating shipment bears the risk of loss or damage to parts in transit.*

Table 1. ***Spare Parts Warranty*** **(continued)**

CORE RETURNS: *Per Seller's determination that a core return is required for which Seller has furnished a replacement part through the Warranty Process shall be shipped by Buyer, with all historical service records, to a facility designated by Seller, within thirty (30) days of shipment by Seller of the replacement part. Buyer shall provide Seller with proof of shipment within thirty (30) days following shipment of the replacement part. In the event that Buyer fails to provide Seller with such proof of shipment within the thirty (30) days, fails to provide the applicable historical service records or provide the correct RMA, Buyer shall be charged the new list price of the replacement part. If the Buyer returns the core, and provides the historical records and the RMA after the 31st day, the Buyer will be credited the invoiced amount upon receipt of the core minus a restocking fee of fifteen percent (15%), or minimum of $300.00 USD,per item, of the then said standard list price of the material provided If Historical Records are not provided and can be recreated by Seller, a debit of $500.00 USD per item will occur to Buyers account.*

WARRANTY AND LIABILITY DISCLAIMERS AND EXCLUSIONS: THIS WARRANTY IS GIVEN AND ACCEPTED IN PLACE OF (i) ALL OTHER WARRANTIES OR CONDITIONS, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTIES OR CONDITIONS OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE AND (ii) ANY OBLIGATION, LIABILITY, RIGHT, CLAIM OR REMEDY IN CONTRACT OR IN TORT, INCLUDING PRODUCT LIABILITIES BASED UPON STRICT LIABILITY, NEGLIGENCE, OR IMPLIED WARRANTY IN LAW.

This warranty is the only warranty made by Seller. The Buyer's sole remedy for a breach of this warranty or any defect in a part is the repair or replacement of the helicopter part and reimbursement of reasonable freight charges. Seller excludes liability, whether as a result of a breach of contract or warranty, negligence or strict product liability, for incidental, special, consequential and/or punitive damages, including without limitation, damage to the helicopter or other property, costs and expenses resulting from required changes or modifications to helicopter components and assemblies, changes in retirement lives and overhaul periods, local customs fees and taxes, and costs or expenses for commercial losses or lost profits due to loss of use or grounding of helicopters or otherwise.

Seller makes no warranty and disclaims all liability in contract or in tort, including, without limitation, negligence and strict tort liability, with respect to work performed by third parties at Buyer's request and with respect to engines, engine accessories, batteries, paint, radios, any and all customizing equipment, and Buyer furnished equipment or equipment manufactured by others (e.g. STC kits) and installed at Buyer's request.

Seller makes no warranty and disclaims all liability with respect to components or parts damaged by, or worn due to, normal wear and tear, erosion or corrosion. Seller makes no warranty and disclaims all liability for consumables which are defined as items required for normal and routine maintenance or replaced at scheduled intervals shorter than the warranty period. "Consumables" include but are not limited to engine and hydraulic oil, oil filters, packings and o-rings, anti-corrosion and/or sealing compounds, brush plating material, nuts, bolts, washers, screws, fluids, compounds, and standard aircraft hardware that is readily available to aircraft operators from sources other than Seller.

This warranty shall not apply to any helicopter part which has been repaired or altered outside Seller's factory in any way so as, in Seller's sole judgment, to affect its stability, safety or reliability. This warranty shall not apply to any helicopter part which has been subject to misuse, negligence or accident, or which has been installed in any aircraft which has been destroyed. Repairs and alterations which use or incorporate parts and components other than genuine Bell parts or parts approved by Bell for direct acquisition from sources other than Bell itself are not warranted by Bell, and this warranty shall be void to the extent that such repairs and alterations, in Seller's sole judgment, affect the stability, safety or reliability of the helicopter or any part thereof, or damage genuine Bell or Bell-approved parts. No person, corporation or organization, including Bell Authorized Customer Service Facilities, is authorized by Seller to assume for it any other liability in connection with the sale of its helicopters and parts.

NO STATEMENT, WHETHER WRITTEN OR ORAL, MADE BY ANY PERSON, CORPORATION OR ORGANIZATION, INCLUDING BELL AUTHORIZED CUSTOMER SERVICE FACILITIES, MAY BE TAKEN AS A WARRANTY NOR WILL IT BIND SELLER.

CHOICE OF LAW AND JURISDICTION: *This warranty shall be interpreted under and governed by the laws of the State of Texas. All legal actions based upon claims or disputes pertaining to or involving this warranty including, but not limited to, Seller's denial of any claim or portion thereof under this warranty, must be filed in the courts of general jurisdiction of Tarrant County, Texas or in the United States District Court for the Northern District of Texas, Ft. Worth Division located in Ft. Worth, Tarrant County, Texas. In the event that Buyer files such an action in either of the court systems identified above, and a final judgment in Seller's favor is rendered by such court, then Buyer shall indemnify Seller for all costs, expenses and attorneys' fees incurred by Seller in defense of such claims. In the event Buyer files such a legal action in a court other than those specified, and Seller successfully obtains dismissal of that action or transfer thereof to the above described court systems, then Buyer shall indemnify Seller for all costs, expenses and attorneys' fees incurred by Seller in obtaining such dismissal or transfer.*

Table 2. ***New Helicopter Warranty***

NEW HELICOPTER WARRANTY

Table 2. *New Helicopter Warranty* (continued)

WARRANTY:*Seller warrants each new helicopter to be free from defect in material or workmanship under normal use and service for 1,000 hours of operation or three (3) years from acceptance, whichever occurs first. Spare parts installed as warranty replacement on helicopters which are covered by this New Helicopter Warranty will only be warranted for the balance of the original aircraft warranty. Seller assigns each manufacturer's warranty to Buyer to the extent such manufacturer's warranty exists and is assignable.*

In addition, during the first year or within 1,000 hours of total aircraft time, whichever occurs first, the Seller shall reimburse Buyer at a regional labor rate, as determined solely by Seller, for reasonable actual labor costs incurred directly related to removal and reinstallation of parts determined to be defective. Seller shall not reimburse labor costs incurred for any other activity, such as troubleshooting, inspection, testing or similar activities; any reimbursement amount issued in accordance with this warranty shall be in Seller's sole discretion. Seller shall not reimburse any labor costs (1) which it does not deem to be reasonable, (2) which it does not deem directly related to removal and reinstallation, or (3) for buyer furnished equipment or equipment manufactured by other (e.g. STC).

Seller shall also reimburse reasonable freight charges, excluding insurance, customs fees, duties, handling fees, and taxes. Seller shall not reimburse Buyer for any parts repaired or replaced outside of the Seller's Warranty Claims Process unless express prior written authorization is granted by Seller's Warranty Department to Buyer for such repair or replacement.

Parts, components and assemblies of all helicopter parts may have been restored or reworked due to mars, blemishes, dents or other irregularities during the manufacturing process. Such restoration and/or rework are permitted under Seller's approved manufacturing and engineering processes and guidelines. The restoration and/or rework so completed do not render such items defective in material or workmanship.

Seller's sole obligation under this warranty is limited to the repair or replacement of parts which are determined to Seller's reasonable satisfaction to have been defective within the applicable warranty period as described above. Replacement of parts may be either new or reconditioned at Seller's election and at the lowest allowable maintenance level contained in Seller's manuals, service bulletins or applicable Supplier manuals.

NO FAULT FOUND: *In the event Seller determines, after evaluation of a returned part, that a defect does not exist, then Buyer shall pay all expenses incurred by Seller related to the return including, but not limited to, costs incurred in shipping, evaluating the part, cost for any replacement part and restocking of the part which is a minimum of $300.00USD or up to fifteen percent (15%) list price of the part, as well as any No Fault Found fees incurred by Seller. In addition, Seller shall not reimburse Buyer for any costs related to the removal or reinstallation of such a part.*

SELLER'S WARRANTY CLAIM PROCESS: *Defective parts must be reported in writing to the Seller's Warranty Administration within thirty (30) days of being found defective. Parts may be repaired or replaced with new or reconditioned parts, at Seller's election. Warranty adjustment is contingent upon the Buyer complying with the Seller's Warranty Process and with the Seller's Warranty Administration disposition instructions for defective parts. Failure to properly comply with Seller's Warranty Process may, at Seller's sole option, void Seller's warranty as to the allegedly defective part.*

RETURN SHIPMENT: *Defective parts returned to Seller will be eligible for remedy under this warranty only if the part is carefully packed by the Buyer for the return shipment. Damage occurring to a part due to improper packaging may result in the denial of a warranty claim. In the event that Seller determines a returned part to be damaged or unsalvageable due to improper packaging, the Buyer will be billed repair or replacement cost incurred by Seller. The party initiating shipment bears the risk of loss or damage to parts in transit.*

CORE RETURNS: *Per Seller's determination that a core return is required for which Seller has furnished a replacement part through the Warranty Process, the core shall be shipped by Buyer, with all historical service records, to a facility designated by Seller, within thirty (30) days of shipment by Seller of the replacement part. Buyer shall provide Seller with proof of shipment within thirty (30) days following shipment of the replacement part. In the event that Buyer fails to provide Seller with such proof of shipment within the thirty (30) days, fails to provide the applicable historical service records or provide the correct RMA, Buyer shall be charged the new list price of the replacement part. If the Buyer returns the core, and provides the historical records and the RMA after the 31st day, the Buyer will be credited the invoiced amount upon receipt of the core minus a restocking fee minimum of $300.00 USD or up to fifteen percent (15%), ,per item, of the then standard list price of the material provided If Historical Records are not provided and can be recreated by Seller, a debit of $500.00 USD per item will be made to Buyers account.*

WARRANTY AND LIABILITY DISCLAIMERS AND EXCLUSIONS: *THIS WARRANTY IS GIVEN AND ACCEPTED IN PLACE OF (i) ALL OTHER WARRANTIES OR CONDITIONS, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTIES OR CONDITIONS OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE AND (ii) ANY OBLIGATION, LIABILITY, RIGHT, CLAIM OR REMEDY IN CONTRACT OR IN TORT, INCLUDING PRODUCT LIABILITIES BASED UPON STRICT LIABILITY, NEGLIGENCE, OR IMPLIED WARRANTY IN LAW.*

Table 2. ***New Helicopter Warranty*** **(continued)**

This warranty is the only warranty made by Seller. The Buyer's sole remedy for a breach of this warranty or any defect in a part is the repair or replacement of the helicopter part, reimbursement of reasonable freight charges, and reimbursement of reasonable labor costs directly related to removal and reinstallation as provided herein. Seller excludes liability, whether as a result of a breach of contract or warranty, negligence or strict product liability, for incidental or consequential damages, including without limitation, damage to the helicopter or other property, costs and expenses resulting from required changes or modifications to helicopter components and assemblies, changes in retirement lives and overhaul periods, local customs fees and taxes, and costs or expenses for commercial losses or lost profits due to loss of use or grounding of helicopters or otherwise.

Seller makes no warranty and disclaims all liability in contract or in tort, including, without limitation, negligence and strict tort liability, with respect to work performed by third parties at Buyer's request and with respect to engines, engine accessories, batteries, paint, radios, any and all customizing equipment, and Buyer furnished equipment or equipment manufactured by others (e.g. STC kits) and installed at Buyer's request.

Seller makes no warranty and disclaims all liability with respect to components or parts damaged by, or worn due to, normal wear and tear, erosion or corrosion. Seller makes no warranty and disclaims all liability for consumables which are defined as items required for normal and routine maintenance or replaced at scheduled intervals shorter than the warranty period. "Consumables" include but are not limited to engine and hydraulic oil, oil filters, packings and o-rings, anti-corrosion and/or sealing compounds, brush plating material, nuts, bolts, washers, screws, fluids, compounds, and standard aircraft hardware that is readily available to aircraft operators from sources other than Seller.

This warranty shall not apply to any helicopter part which has been repaired or altered outside Seller's factory in any way so as, in Seller's sole judgment, to affect its stability, safety or reliability. This warranty shall not apply to any helicopter part which has been subject to misuse, negligence or accident, or which has been installed in any aircraft which has been destroyed. Repairs and alterations which use or incorporate parts and components other than genuine Bell parts or parts approved by Bell for direct acquisition from sources other than Bell itself are not warranted by Bell, and this warranty shall be void to the extent that such repairs and alterations, in Seller's sole judgment, affect the stability, safety or reliability of the helicopter or any part thereof, or damage genuine Bell or Bell-approved parts. No person, corporation or organization, including Bell Authorized Customer Service Facilities, is authorized by Seller to assume for it any other liability in connection with the sale of its helicopters and parts.

NO STATEMENT, WHETHER WRITTEN OR ORAL, MADE BY ANY PERSON, CORPORATION OR ORGANIZATION, INCLUDING BELL AUTHORIZED CUSTOMER SERVICE FACILITIES, MAY BE TAKEN AS A WARRANTY NOR WILL IT BIND SELLER.

CHOICE OF LAW AND JURISDICTION: *This warranty shall be interpreted under and governed by the laws of the State of Texas. All legal actions based upon claims or disputes pertaining to or involving this warranty including, but not limited to, Seller's denial of any claim or portion thereof under this warranty, must be filed in the courts of general jurisdiction of Tarrant County, Texas or in the United States District Court for the Northern District of Texas, Ft. Worth Division located in Ft. Worth, Tarrant County, Texas. In the event that Buyer files such an action in either of the court systems identified above, and a final judgment in Seller's favor is rendered by such court, then Buyer shall indemnify Seller for all costs, expenses and attorneys' fees incurred by Seller in defense of such claims. In the event Buyer files such a legal action in a court other than those specified, and Seller successfully obtains dismissal of that action or transfer thereof to the above described court systems, then Buyer shall indemnify Seller for all costs, expenses and attorneys' fees incurred by Seller in obtaining such dismissal or transfer.*

H ELP

E VALUATE

L OGISTICS

P UBLICATIONS

Have you found something wrong with this manual — an error, an inconsistency, unclear instructions, etc.? Although we strive for accuracy and clarity, we may make errors on occasion. If we do and you discover it, we would appreciate your telling us about it so that we can change whatever is incorrect or unclear. Please be as specific as possible.

Your complaint or suggestion will be acknowledged and we will tell you what we intend to do.

Please email publications@bellflight.com, to inform us where we have erred.

Your assistance is sincerely appreciated.

ICN-BHT-A-000000-A-97499-00007-F-002-01

Figure 1. HELP page (Sheet 1 of 1)

Use of a S1000D publication - Introduction

1. Use of a S1000D publication - General information

1.1. Document structure

The breakdown of the helicopter into systems within the manuals is done under the S1000D specification (S1000D-I9005-01000-00, Issue No. 4.0,4, may 12, 2009).

This publication is written in Simplified English and obeys the instructions given in document (ASD-STE100, Issue 6, January 2013). The European Association of Aerospace Industries (ASD) prepared these documents.

All the data are included in data modules. A code identifies each data module.

1.1.1. Data Module Code (DMC)

The Data Module Code (DMC) structure is shown in Table 1.

Table 1. DMC structure

BHT	A	29-10-01	00A	520A	A
Paragraph 1.1.1.1	Paragraph 1.1.1.2	Paragraph 1.1.1.3	Paragraph 1.1.1.4	Paragraph 1.1.1.5	Paragraph 1.1.1.6

1.1.1.1. Model identification code

The model identification code is BHT.

1.1.1.2. System difference code

The system difference code is a letter (letters I and 0 are not used).

This letter identifies two or more sub-systems that can be installed as alternative items because they do the same function.

1.1.1.3. Standard Numbering System (SNS) code

The Standard Numbering System (SNS) code includes three pairs of digits that show:

- The system (e.g., 29, hydraulic power)
- The subsystem and the sub-subsystem (e.g., 10, main hydraulic)

NOTE

When the sub-subsystem code is zero, the two digits show all of the subsystem.

- The component (e.g., 01, hydraulic reservoir filter module)

1.1.1.4. Disassembly code and disassembly code variant

1.1.1.4.1. Disassembly code

The disassembly code is a two-digit code.

The disassembly code shows the breakdown of a component, as follows:

- 00 - Data module for all of the helicopter, system, subsystem, or component.
- 01 - Data module for the first assembly that you remove from the component.
- 02 - Data module for the second assembly that you remove from the component.

Code 03 and the subsequent codes refer to the third, fourth, etc assembly that you remove from the component.

The disassembly code also identifies the data modules in sequence.

1.1.1.4.2. Disassembly code variant

The disassembly code variant is a letter (letters I and 0 are not used) that identifies alternative items.

These items are different, but the difference is too small to cause a change in the system difference code.

1.1.1.5. Information code and information code variant

1.1.1.5.1. Information code

The information code is a three-digit code.

The information code identifies the types of information you can find in the data module.

1.1.1.5.2. Information code variant

The information code variant is a letter (letters I and 0 are not used).

The information code variant is used for different data modules that are applicable to the same component and type of operation.

1.1.1.6. Item location code

You can find these item location codes:

- A on the helicopter.
- B on the removed assembly/component.
- D applicable to all the locations.
- Z is used as a generic item location code.

1.1.2. List of Effective Data Modules (LOEDM)

Each chapter includes a List of Effective Data Modules (LOEDM) . The list shows the data module codes of all the data modules in the chapter.

The LOEDM gives the information that follows for each data module:

- Technical name and information name columns. These two columns together give the title of the data module.
- Document identifier column. This column gives the data module code.
- The status of the data module. This column gives the information about the status of the data module (N for new and C for changed).
- Issue date. This column gives the date of the data module changes.
- Applicability column. This column gives information about the applicability of the data module to different versions, configurations, and modifications of the helicopter.

CHAPTER 01
INTRODUCTION

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Scope of the Manual - Description	DMC-412-A-00-00-00-00A-028A-A / 00008	2023-05-31	ALL
Use of this Manual - Description	DMC-412-A-00-00-00-00A-042A-A / 00009	2023-05-31	ALL
Helicopter - Description	DMC-412-A-00-00-00-01A-042A-A / 00010	2023-05-31	ALL

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Scope of the Manual - Description

This Maintenance Manual (MM) contains the necessary data to do maintenance on the Model 412 helicopters S/N 33001 through 33213, 36001 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999. For the systems, subsystems, or units that are applicable only to the 412EPI (S/N 37002 through 37999), maintenance must be done in accordance with the instructions in the applicable chapter of the BHT-412-MMS-EPI. For the systems, subsystems, or units that are applicable only to the 412EPX (S/N 38001 through 38999 and 39101 through 39999), maintenance must be done in accordance with the instructions in the applicable chapter of the BHT-412-MMS-EPX.

The following technical manuals contain the Instructions for Continued Airworthiness (ICA) for the Model 412 and are necessary to maintain the continued airworthiness of the helicopter.

- Maintenance Manual (BHT-412-MM)
- Maintenance Manual Supplement (BHT-412-MMS-EPI) -Applicable to 412EPI
- Maintenance Manual Supplement (BHT-412-MMS-EPX) -Applicable to 412EPX
- Illustrated Parts Breakdown Manual (BHT-412-IPB)
- Illustrated Parts Breakdown Supplement (BHT-412-IPBS-EPI) - Applicable to 412EPI
- Illustrated Parts Breakdown Supplement (BHT-412-IPBS-EPX) - Applicable to 412EPX
- Component Repair and Overhaul Manual (BHT-412-CR&O)
- Special Tools Illustrated Parts Breakdown (BHT-SPECTOOL-IPB)
- Standard Practices Manual (BHT-ALL-SPM, Standard Practices Manual)
- Electrical Standard Practices Manual (BHT-ELEC-SPM, Electrical Standard Practices Manual)
- Structural Repair Manual (BHT-MED-SRM)
- Corrosion Control Guide (CSSD-PSE-87-001, Corrosion Control Guide).

NOTE

For helicopters S/N 34001 through 34024, use information contained within this manual applicable to S/N 33108 through 33213 and S/N 36001 through 36019.

Generally, information applicable to S/N 36020 through 36086 applies to S/N 34025 and Subsequent. For specific differences in S/N 34025 and Subsequent, for instruments, electrical, and avionics, refer to NB-412-MM basic maintenance manual. Copies of this manual may be obtained from:

PT Dirgantara Indonesia

(www.indonesian-aerospace.com)

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Use of this Manual - Description

The instructions in this manual and those that have been changed by Alert Service Bulletins, issued by Bell Helicopter Textron, and by Airworthiness Directives, issued by the local Aviation Authority, shall be strictly followed.

1.1. Customer Comments

Bell Helicopter Textron strives to provide accurate and straightforward manuals. Sometimes, we may make mistakes. If you find any mistakes, we would appreciate it if you told us. Your observation, suggestion, or complaint will be acknowledged.

For your convenience, we include a Customer Feedback Form at the beginning of this manual. You can send it by fax, e-mail, or mail it to us. When you tell us about a mistake that is found in the Maintenance Manual, please be as specific as possible. Your help to make sure that the manual is correct is very much appreciated.

2. Content and Arrangement

The breakdown of the helicopter by systems in this manual is done under the Airline Transportation Association (ATA) chapter system. The manual is separated into 33 chapters to make it easy to find data.

2.1. Revision Status and List of Effective Pages

The revision status of the manual is given on the log of revisions (page A) and on the title page of the manual. The log of pages gives the revision status of every page in the manual.

2.2. Change Marks

The change marks show the changed material (text, tables, and figures). The change mark is a light blue background.

2.3. Chapters

		GENERAL INFORMATION
412-A-00-00-00-00A-009A-A / 00001	-	Introduction
		HANDLING AND SERVICING
412-A-04-00-00-00A-009A-A / 00011	-	Airworthiness Limitations Schedule
412-A-05-00-00-00A-009A-A / 00014	-	Inspections
412-A-06-00-00-00A-009A-A / 00071	-	Dimensions and Charts
412-A-07-00-00-00A-009A-A / 00074	-	Lifting and Jacking
412-A-08-00-00-00A-009A-A / 00081	-	Weight and Balance
412-A-09-00-00-00A-009A-A / 00102	-	Towing
412-A-10-00-00-00A-009A-A / 00108	-	Parking and Mooring
412-A-11-00-00-00A-009A-A / 00122	-	Placards and Markings
412-A-12-00-00-00A-009A-A / 00125	-	Servicing
412-A-18-00-00-00A-009A-A / 00144	-	Vibration and Noise
		EQUIPMENT
412-A-21-00-00-00A-009A-A / 00189	-	Air Distribution (Ventilation)
412-A-25-00-00-00A-009A-A / 00244	-	Equipment and Furnishings
412-A-26-00-00-00A-009A-A / 00291	-	Fire Protection
		FUEL/HYDRAULICS
412-A-28-00-00-00A-009A-A / 00323	-	Fuel System
412-A-29-00-00-00A-009A-A / 00610	-	Hydraulic System
		AIRFRAME
412-A-30-00-00-00A-009A-A / 00677	-	Ice and Rain Protection
412-A-32-00-00-00A-009A-A / 00694	-	Landing Gear
412-A-52-00-00-00A-009A-A / 00774	-	Doors and Windows
412-A-53-00-00-00A-009A-A / 00897	-	Fuselage

		MAIN ROTOR/MAIN ROTOR DRIVE SYSTEM
412-A-62-00-00-00A-009A-A / 00932	-	Main Rotor
412-A-63-00-00-00A-009A-A / 00987	-	Main Rotor Drive System
		TAIL ROTOR/TAIL ROTOR DRIVE SYSTEM
412-A-64-00-00-00A-009A-A / 01225	-	Tail Rotor
412-A-65-00-00-00A-009A-A / 01257	-	Tail Rotor Drive System
		FLIGHT CONTROLS
412-A-67-00-00-00A-009A-A / 01319	-	Flight Controls
		POWER PLANT
412-A-71-00-00-00A-009A-A / 01584	-	Power Plant
412-A-76-00-00-00A-009A-A / 01722	-	Engine Controls
412-A-79-00-00-00A-009A-A / 01765	-	Engine Oil System
		INSTRUMENTS/ELECTRICAL
412-A-95-00-00-00A-009A-A / 01787	-	Instrument System
412-A-96-00-00-00A-009A-A / 01870	-	Electrical System
		AVIONICS
412-A-97-00-00-00A-009A-A / 02235	-	Avionics
		WIRING DIAGRAMS
412-A-98-00-00-00A-009A-A / 02454	-	Wiring Diagrams
		KITS
412-A-99-00-00-00A-009A-A / 02622	-	Kits

2.4. Table of Contents

To locate the required information, every chapter has a table of contents.

2.5. Paragraph, Page, Figure, and Table Numbering

Paragraphs, pages, figures, and tables start at the number 1 within a chapter.

3. Bulletins

Technical Bulletins (TBs) and Alert Service Bulletins (ASBs) are published when they are necessary. These documents supplement the Maintenance Manual and give instructions and data to change components, systems, and maintenance practices on the helicopter. When a bulletin affects the way the helicopter is maintained, it is incorporated in the manual at the next available opportunity. Refer to the Bulletin Record (page BR) for the list of bulletins that have been incorporated in this manual.

4. Temporary Revisions

Temporary revisions are published when necessary. Refer to the Temporary Revision Record (page TR) for a listing of active temporary revisions against this manual.

5. Consumable Materials

NOTE

Handling, storage, and safety precautions concerning consumable materials shall be in accordance with manufacturers instructions unless otherwise specified in this manual.

The consumable materials that are necessary for you to do maintenance tasks are given by name and by item number in the text (i.e., drycleaning solvent (C-304)). The number refers to the item 304. The data on this item 304 is found in Table 13-2 in BHT-ALL-SPM, Chapter 13, Standard Practices Manual.

On occasion, the materials that are used in maintenance tasks will change properties, change suppliers, or be discontinued. Also, new improved materials become available. If there is ever a difference between the Maintenance Manual and Standard Practices Manual, the manual that has the latest date of issue shows the preferred material. However, either material may be used for the maintenance task unless it is stated otherwise in the procedure.

6. Special Tools

Some maintenance procedures require the use of special tools. These tools are given in a table at the beginning of the applicable maintenance paragraph. A description and illustration of these tools are found in the Special Tools Illustrated Parts Breakdown Manual (BHT-SPECTOOL-IPB).

7. Torques

Where applicable, special torques are specified within the text (or on illustrations) within this manual.

When a torque value is not shown, then the standard torque value for that combination of fasteners shall be used. Standard torque values for various combinations of fasteners are found in the Standard Practices Manual (BHT-ALL-SPM, Standard Practices Manual).

8. Commonly Used Symbols

Some illustrations have symbols to highlight common processes such as the application of sealant, primer, corrosion preventive compound. Other processes are the use of heat, assembly fluid, or unreduced wet primer. Figure 1 shows some of the symbols used. Always use the consumable material (C-XXX) that is called out (Paragraph 5).

9. Terminology

9.1. Warnings, Cautions and Notes



IF YOU DO NOT FOLLOW THE INSTRUCTIONS THAT ARE GIVEN IN A WARNING, PERSONAL INJURY CAN OCCUR.



IF YOU DO NOT FOLLOW THE INSTRUCTIONS THAT ARE GIVEN IN A CAUTION, YOU CAN CAUSE DAMAGE TO THE HELICOPTER OR TO THE COMPONENTS.

NOTE

A note includes supplemental data about the procedure, the practice, the condition, etc. for the maintenance task that you do.

Warnings, cautions and notes are used throughout this manual to emphasize important and critical instructions.

10. Standard Practices

Standard maintenance practices and procedures that are not given in this manual are found in the Standard Practices Manual (BHT-ALL-SPM, Standard Practices Manual) and the Electrical Standard Practices Manual (BHT-ELEC-SPM, Electrical Standard Practices Manual).

11. Replacement Parts and Assemblies

Replacement parts and assemblies that are necessary for the correct and proper maintenance of the helicopter and its components are given in the Illustrated Parts Breakdown Manual (BHT-412-IPB).

12. Wear Limits



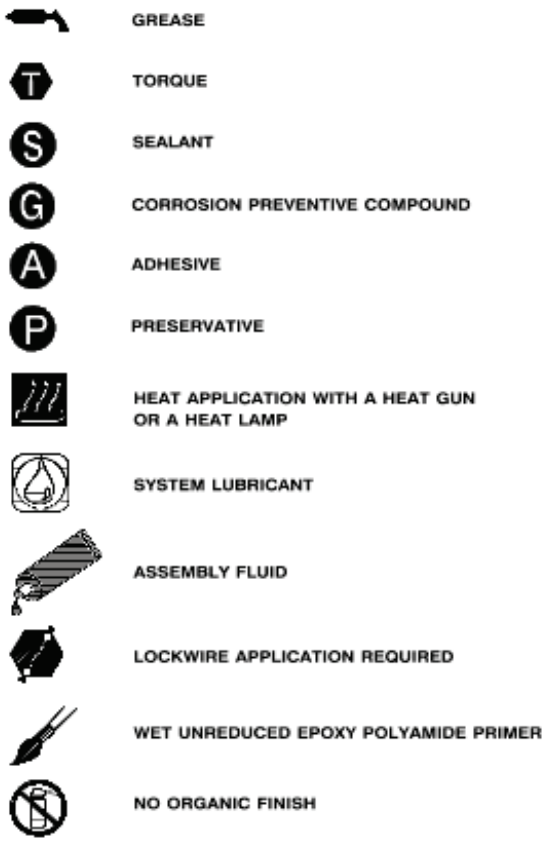
THE METRIC EQUIVALENTS TO U.S. STANDARD WEIGHTS AND MEASURES ARE GIVEN IN THIS MANUAL. HOWEVER, USE ONLY THE U.S. STANDARD VALUES WHEN YOU TAKE A MEASUREMENT TO ESTABLISH A SPECIFIC DIMENSION OR TO FIND THE SERVICEABILITY OF A COMPONENT.

Throughout this manual, wear limits are given to show the required fit between the mating parts. It is not intended that all dimensions be examined as a prescribed maintenance procedure. However, you must examine the dimensions of parts that show signs of wear or physical damage.

Wear limits, fits, and tolerances are integrated into the inspection, repair, and assembly procedures.

Unless otherwise specified, dimensions shall carry the following tolerances on decimals:

DECIMAL	TOLERANCE
.XXX	±0.010 inch
.XX	±0.03 inch
.X	±0.1 inch



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Figure 1. Commonly Used Symbols (Sheet 1 of 1)

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Helicopter - Description

The Model 412 Series Helicopter (Figure 1) consists of two major assemblies: The forward fuselage and the tailboom. The forward fuselage is of semimonocoque and reinforced shell construction with transverse bulkheads and metal and fiberglass covering. Two longitudinal main beams provide the primary structural support.

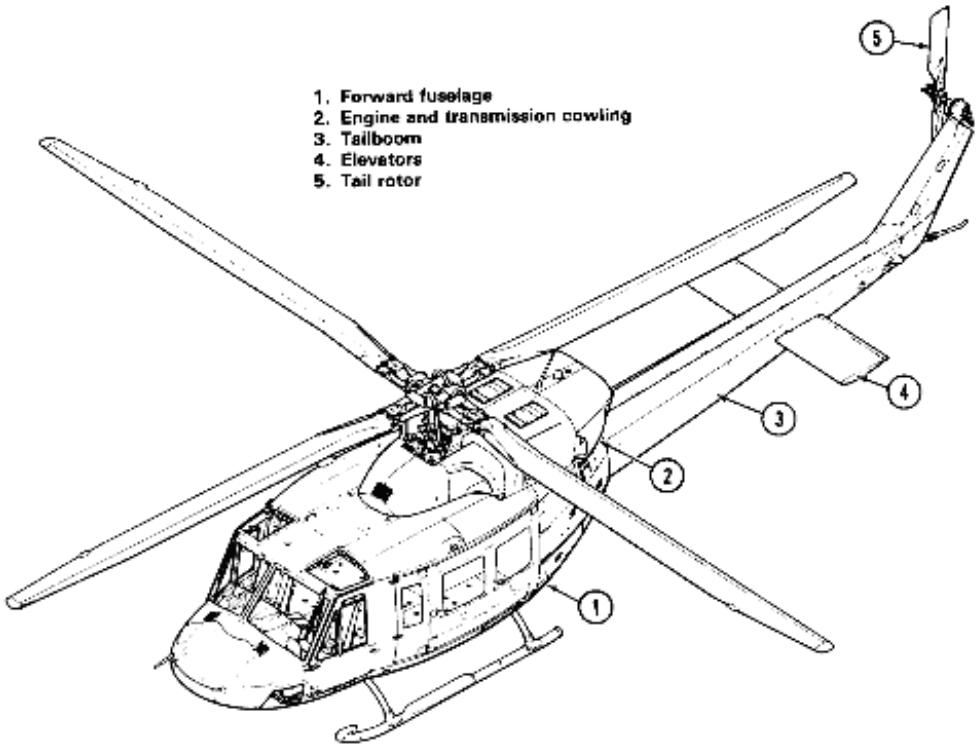
A hinged door on either side of the forward area permits direct access to the crew area and a large sliding door permits access to the cargo/passenger area. Additionally, a hinged cargo door is located immediately ahead of the sliding door. This door increases the width of access to the cargo/passenger area. Seating is provided for the pilot and forward passenger/copilot in the crew area (cockpit) and up to 13 passengers in the cargo/passenger (cabin) area.

The engine deck, located above and aft of the passenger/cargo area, is designed to accommodate the engines, reduction (combining) gearbox, firewalls, and air management system.

The tailboom is of semimonocoque construction which provides support for a vertical fin, aerodynamically actuated elevator, tail rotor and tail rotor drive system, tail skid, and baggage compartment.

The powerplant is a Pratt & Whitney Canada. PT6T-3 series twin turboshaft (Twin-Pac) engine consisting of two identical free turbine power sections. A common reduction (combining) gearbox mounted across the aft end of the power sections has an engine-to-transmission output shaft (main driveshaft) extending forward along the centerline of the engine to provide power to the transmission.

The main rotor system is a soft-in-plane, flex-beam type with four interchangeable blades. For storage and handling, two of the blades may be folded back along the fuselage. When folded, the blades do not extend beyond the width of the fuselage.



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Figure 1. Model 412 Series Helicopter (Typical) (Sheet 1 of 1)

CHAPTER 04
AIRWORTHINESS LIMITATIONS SCHEDULE

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Airworthiness Limitations Schedule - Log of FAA/ODA* approved revisions	DMC-412-A-04-00-00-00A-023A-A / 00012	2023-05-31	ALL
Airworthiness Limitations Schedule - Airworthiness Limitations Schedule	DMC-412-A-04-00-00-00A-288A-A / 00013	2023-05-31	ALL

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Airworthiness Limitations Schedule - Log of FAA/ODA* approved revisions

ISSUE NUMBER	DATE OF SIGNATURE	FAA/ODA* SIGNATURE
ISSUE 001	21 JAN 2021	SARAH F COX Digitally signed by SARAH F COX Date: 2021.01.21 07:31:33 -06'00'

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Airworthiness Limitations Schedule - Airworthiness Limitations Schedule

The Airworthiness Limitations section is FAA approved and specifies maintenance required under Secs. 43.16 and 91.403 of the Federal Aviation Regulations unless an alternative program has been FAA approved.



ALL REPAIR AND OVERHAUL PROCEDURES LIVES PUBLISHED BY BELL HELICOPTER TEXTRON, INCLUDING COMPONENT RETIREMENT LIFE, ARE BASED SOLELY ON THE USE OF BELL APPROVED PARTS AND PROCESSES. IF PARTS OR PROCESSES DEVELOPED OR APPROVED BY PARTIES OTHER THAN BELL HELICOPTER ARE USED, THEN THE DATA PUBLISHED OR OTHERWISE SUPPLIED BY BELL HELICOPTER ARE NOT APPLICABLE. THE USER IS WARNED TO NOT RELY ON BELL HELICOPTER DATA FOR PARTS AND PROCESSES NOT APPROVED BY BELL HELICOPTER. ALL APPLICABLE INSPECTIONS AND REPAIR METHODS MUST BE OBTAINED FROM THE SUPPLIER OF THE PARTS OR PROCESSES NOT APPROVED BY BELL HELICOPTER. BELL HELICOPTER IS NOT RESPONSIBLE FOR PARTS OR PROCESSES OTHER THAN THOSE WHICH IT HAS ITSELF DEVELOPED OR APPROVED.



CONTINUED USE OF ANY TIME/CALENDAR/RIN LIFE LIMITED COMPONENT PAST ITS STATED LIMITS IS NOT PERMITTED AS IT COULD SERIOUSLY AFFECT THE AIRWORTHINESS OF THE HELICOPTER.

THE COMPONENT MUST BE REMOVED FROM THE HELICOPTER NOT LATER THAN THE END OF THE LIFE LIMIT. IT MUST EITHER BE MADE UNSERVICEABLE OR MARKED IN A CONSPICUOUS MANNER TO PREVENT CONTINUED USE, EVEN INADVERTENT USE.

WHEN A REPLACEMENT PART IS INSTALLED, THE PART NAME, PART NUMBER, SERIAL NUMBER, AND CURRENT OPERATING HOURS, IF APPLICABLE, MUST BE RECORDED IN THE HISTORICAL SERVICE RECORD FOR THE ASSEMBLY OR HELICOPTER.

The mandatory airworthiness limitations schedule (Table 1) summarizes the mandatory maximum life, in hours, years or by Retirement Index Number (RIN) of components with a limited airworthiness life. Parts that are not on the schedule have an unlimited airworthiness life.

Refer to the engine manufacturer's publications for the airworthiness limitations schedule of the engine and components. For the PT6T-3 Series Engine, refer to the Pratt and whitney Service Bulletin No. 5002. For the PT6T-9 Engine, refer to the Pratt and Whitney Maintenance Manual airworthiness limitation section.



SOME PARTS ARE INSTALLED AS ORIGINAL EQUIPMENT ON BOTH MILITARY AND COMMERCIAL HELICOPTERS AND MAY HAVE A LOWER AIRWORTHINESS LIFE AND/OR OVERHAUL SCHEDULE WHEN USED ON A MILITARY HELICOPTER. IN ADDITION, CIRCUMSTANCES SURROUNDING THEIR USE MAY CALL FOR OPERATION OF THE MILITARY HELICOPTER OUTSIDE OF THE APPROVED COMMERCIAL FLIGHT ENVELOPE. CONSEQUENTLY, PARTS THAT HAVE BEEN USED ON MILITARY HELICOPTERS SHOULD NOT BE USED ON COMMERCIAL HELICOPTERS.



AIRWORTHINESS LIFE OF SOME KIT COMPONENTS MAY NOT BE COVERED IN THIS SCHEDULE. REFER TO APPLICABLE SERVICE INSTRUCTION (SI) OR MAINTENANCE MANUAL SUPPLEMENT (MMS) FOR KIT COMPONENTS SCHEDULE.

NOTE

The airworthiness life or inspection interval for any part number contained in this schedule applies to all the successive dash numbers for that component unless it is otherwise specified.

NOTE

The airworthiness life given or the failure to give an airworthiness life to a component does not constitute a warranty of any kind. The only warranty applicable to the helicopter or any component is the warranty included in the Purchase Agreement for the helicopter or the component.

The airworthiness lives given to the components and assemblies are determined by experience, tests and the judgement of Bell Helicopter engineers. The airworthiness lives and inspection intervals cannot be changed without the approval of the FAA.

Prior to disposing of unsalvageable helicopter parts and materials, caution should be exercised to make sure that the parts and materials are disposed of in a manner that does not allow them to be returned for service. Refer to FAA Advisory Circular 21-38 for guidance on the disposal of unsalvageable aircraft parts and materials.

Table 1. Airworthiness Limitations Schedule

NOMENCLATURE	PART NUMBER (1)	AIRWORTHINESS LIFE
MAIN ROTOR HUB		
Yoke Assembly	412-010-101-109	700 hours
Yoke Assembly	412-010-101-123	(5) (7) (38) 5000 hours
Yoke Assembly	412-010-101-137	(5) (33) 4500 hours
Yoke Assembly	412-010-101-139	(5) (7) 3000 hours
Spindle Assembly	412-010-156-105	(22) 5000 hours
Spindle and Damper Bearing Assembly	412-010-190-101	5000 hours
Spindle and Damper Bearing Assembly	412-010-190-103	10,000 hours
Spindle and Shear Bearing Assembly	412-310-102-101	10,000 hours
Pitch Horn Assembly	412-010-149-105	5000 hours
Pitch Horn Assembly	412-010-149-111	15,000 hours
Retention Bolt	412-010-124-105	(27) 5000 hours
Retention Bolt (Expandable)	412-010-137-103	5000 hours
Damper Bridge	412-010-104-101	5000 hours
Damper Bridge	412-010-183-101	10,000 hours
Damper Bridge	412-010-183-109	15,000 hours
Damper Bridge	412-010-185-101	10,000 hours
Damper Bridge	412-010-185-109	15,000 hours
Damper Bridge	412-018-068-101	(2) 10,000 hours or 180 months
Damper Bridge	412-010-170-101	(2) 10,000 hours or 180 months
Pivot Bearing	412-010-106-101	(4) On condition
Fitting	412-010-111-101	5000 hours
Damper Yoke Set	412-010-145-101	(4) On condition
Damper Yoke Set	412-310-146-103	(4) On condition
MAIN ROTOR DROOP RESTRAINT		
Bolt (8)	MS21250H05006	(2) 120 months
MAIN ROTOR SIMPLE PENDULUM ABSORBERS		
Bolt, Bracket (16)	EWB0420D-7-36	15,000 hours
MAIN ROTOR CONTROLS		
Rephasing Lever Assembly	412-010-403-109	1250 hours
Rephasing Lever Assembly	412-010-403-113	5000 hours
Drive Link Assembly	412-010-405-101	5000 hours
Swashplate Outer Ring	412-010-407-105	(32) 2500 hours
Swashplate Outer Ring	412-010-407-117	10,000 hours
Pitch Link Rod End Bearing	412-010-412-101	5000 hours
Pitch Link Rod End Bearing	412-010-438-101	5000 hours
Swashplate Link Rod End Bearing	412-010-412-101	5000 hours

Table 1. Airworthiness Limitations Schedule (continued)

NOMENCLATURE	PART NUMBER (1)	AIRWORTHINESS LIFE
MAIN ROTOR HUB		
Swashplate Link Rod End Bearing	412-010-448-101	5000 hours
Swashplate Link Rod End Bearing	412-310-400-103	(24) 5000 hours
Swashplate Support Assembly	412-010-409-105	(36) 5000 hours
Swashplate Support Assembly	412-010-443-101	(35) 5000 hours
Swashplate Support Assembly	412-010-453-101	(37) 5000 hours
Swashplate Support Assembly	412-010-453-105	15,000 hours
Gimbal Ring Assembly	204-010-404-001	9000 hours
Gimbal Ring Assembly	212-010-416-101	9000 hours
Collective Sleeve	204-011-408-003	9000 hours
Collective Sleeve	212-011-412-101	9000 hours
Collective Lever Assembly	412-010-408-101	(39) 10,000 hours
Collective Lever Assembly	412-010-408-105	(40) 2500 hours
Collective Lever Assembly	412-010-464-101	20,000 hours
Collective Lever Pin	412-010-465-101	20,000 hours
MAIN ROTOR CONTROL SYSTEM BOLTS		
Pitch Horn to Pitch Link (4)	20-057-5-25D	(19) (28) (29) 2500 hours
Pitch Horn to Pitch Link (4)	20-057-5-28D	(30) 2500 hours
Pitch Link-to-Rephasing Lever (4)	20-057-6-36D	(28) (29) (30) 2500 hours
Pitch Link-to-Rephasing Lever (4)	20-057-6-52D	(29) (30) 2500 hours
Drive Link-to-Rephasing Lever (4)	20-057-6-36D	(28) (29) (30) 2500 hours
Swashplate Link-to-Rephasing Lever (3)	20-057-6-38D	(28) (29) (30) 2500 hours
Swashplate Link-to-Rephasing Lever (2)	20-057-6-52D	(29) (30) 2500 hours
Swashplate Link-to-Rotating Ring (2)	20-057-6-36D	(28) (29) (30) 2500 hours
Gimbal Ring-to-Swashplate Support (2)	204-011-463-105	(28) (29) (30) 2500 hours
Gimbal Ring-to-Swashplate Inner Ring (2)	204-011-463-001	(25) (28) (29) 1000 hours
Gimbal Ring-to-Swashplate Inner Ring (2)	204-011-463-109	(30) 2500 hours
Collective Lever-to-Swashplate Support (1)	20-057-8-92D	(28) (29) (30) 2500 hours
Boost Tube-to-Universal (3)	20-057-5-24D	(28) (29) (30) 2500 hours
Universal-to-Boost Cylinder (3)	20-057-5-24D	(28) (29) (30) 2500 hours
Boost Cylinder-to-Lower Support (3)	212-001-323-001	(28) (29) (30) 2500 hours
Rephasing Lever-to-Hub (4)	20-057-6-82D	(29) (30) 1250 hours
Rephasing Lever-to-Hub (4)	20-057-8-84D	(28) (29) (30) 2500 hours
MAIN ROTOR SUPPLEMENTAL CONTROL SYSTEM BOLTS (31)		
Swashplate Inner Ring-to-R.H. Cyclic Boost (1)	20-057-5-24D	2500 hours
Swashplate Inner Ring-to-L.H. Cyclic Boost (1)	20-057-5-24D	2500 hours
Collective Lever-to-Collective Boost (1)	20-057-5-24D	2500 hours
PROPULSION AND DRIVE SYSTEM		
Planetary Spider	204-040-785-003	(6) On condition

Table 1. Airworthiness Limitations Schedule (continued)

NOMENCLATURE	PART NUMBER (1)	AIRWORTHINESS LIFE
	MAIN ROTOR HUB	
Planetary Spider	412-040-785-101	2500 hours
Planetary Spider	412-040-785-103	(45) On condition
Main Rotor Mast Assembly	412-040-101-105	(7) (8) (9) 10,000 hours or 80,000 RIN
Main Rotor Mast Assembly (used on helicopters with Mast Torque Systems)	412-040-101-121	(7) (8) (10) 10,000 hours or 60,000 RIN
Main Rotor Mast Assembly	412-040-101-127	(7) (8) (9) 10,000 hours or 80,000 RIN
Main Rotor Mast Assembly (used on helicopters with Mast Torque Systems)	412-040-101-129	(7) (8) (10) 10,000 hours or 60,000 RIN
Main Rotor Mast Assembly (used on helicopters with Mast Torque Systems and with BHT-412-SI-62, Slope Landing Kit)	412-040-101-135	(7) (8) (14) 5000 hours or 60,000 RIN
Main Rotor Mast Assembly	412-040-114-101	(7) (8) (46) 2,500 hours or 25,000 RIN
Main Rotor Mast Assembly (used on helicopters with Mast Torque Systems and with BHT-412-SI-62, Slope Landing Kit)	412-704-012-105	(7) (8) (14) 5000 hours or 60,000 RIN
Cap Retention	412-010-161-101	10,000 hours
Cap Retention	412-010-171-101	10,000 hours
Cap Retention	412-010-171-103	(47) 10,000 hours
Mast Cap	412-010-171-105	5,000 hours
Cone	412-010-165-101	(47) 10,000 hours
Cone	412-010-165-103	5,000 hours
Drive Pin	412-010-166-101	(47) 10,000 hours
Drive Pin	412-010-166-103	5,000 hours
Upper Cone Seat	412-010-164-101	10,000 hours
Upper Cone Seat	412-010-174-101	10,000 hours
Upper Cone Seat	412-010-186-101	10,000 hours
Upper Cone Seat	412-010-186-107	5,000 hours
Upper Cone Seat	412-010-286-101	2,500 hours
Splined Plate Assembly	412-010-177-101	(8) (11) (12) Life limited (see notes)
Splined Plate Assembly	412-010-177-105	(8) (10) 10,000 hours or 60,000 RIN
Splined Plate Assembly	412-010-177-109	(8) (9) 10,000 hours or 80,000 RIN
Splined Plate Assembly	412–010-177-121	(8) (14) 5,000 hours or 60,000 RIN
Splined Plate Assembly	412-010-177-113	(8) (10) 10,000 hours or 60,000 RIN
Splined Plate Assembly	412-010-277-101	(8) (46) 2,500 hours or 25,000 RIN
Splined Plate Assembly	412-010-167-105	(8) (11) (12) Life limited (see notes)
Cone	412-010-169-103	10,000 hours
Cone	412-010-179-101	10,000 hours
Cone	412-010-179-105	(47) 10,000 hours
Cone	412-010-179-107	5,000 hours
Lower Cone Seat	412-010-168-105	(7) 10,000 hours
Lower Cone Seat	412-010-178-101	(7) 10,000 hours
Lower Cone Seat	412-010-056-101	(7) 2,500 hours

Table 1. Airworthiness Limitations Schedule (continued)

NOMENCLATURE	PART NUMBER (1)	AIRWORTHINESS LIFE
MAIN ROTOR HUB		
Lower Cone Seat	412-018-056-103	(7) 10,000 hours
Lower Cone Seat	412-018-056-109	(7) 5,000 hours
ELEVATOR AND CONTROLS		
Horn Assembly	205-001-914-103	(3) On condition
AIRFRAME		
Upper Aft Beam Cap	212-030-191-001	(41) On condition
TAIL ROTOR INSTALLATION		
Blade Assembly	212-010-750-009	5000 hours
Blade Assembly	412-016-100-111	5000 hours
Yoke Assembly	212-011-702-001	(17) 5000 hours
LANDING GEAR		
High Aft Crosstube Assembly	412-050-011-101	(15) (34) (42) 10,000 events
High Aft Crosstube Assembly	412-050-045-107	(15) (34) (42) 20,000 events
High Aft Crosstube Assembly	412-321-104	(18) (34) (42) 20,000 events
High Aft Crosstube Assembly	412-321-901	(34) (42) (43) (44) 12,000 events
FIRE EXTINGUISHER CARTRIDGES		
HTL Industries P/N 13083-5	209-062-908-015	(2) (20) 6 years service life/9 years total life
HTL Industries P/N 30900400	209-062-908-019	(2) (20) 6 years service life/9 years total life
W. Kidde and Co. P/N 804943	209-062-908-013	(20) 6 years total life
W. Kidde and Co. P/N 804944	209-062-908-017	(20) 6 years total life
W. Kidde and Co. P/N 895408-1	209-062-908-113	(20) 10 years total life
W. Kidde and Co. P/N 895409-1	209-062-908-115	(20) 10 years total life
FLOATS		
Float Hose	70-072M000C144	(21) 6 years calendar life
Float Hose	70-072M000C222	(21) 6 years calendar life
Float Hose	412-073-800-115A (uses hose P/N 70-072L000B204)	(21) 6 years calendar life
Squib Cartridges, Floats	30908	(16) (20) 15 years total life
Squib Cartridges, Floats	29022968	(20) (26) 15 years total life
Cylinder Assembly, Floats	212-073-920-107	(20) 10 years total life
INTERNAL HOIST KIT		
Cable Cutter Assembly (Squib)	1810-033-01	(2) (20) 5 years service life/10 years total life

Table 1. Airworthiness Limitations Schedule (continued)

NOMENCLATURE	PART NUMBER (1)	AIRWORTHINESS LIFE
	MAIN ROTOR HUB	

NOTES:

1 The airworthiness life or inspection interval for any part number contained in this schedule applies to all the successive dash numbers for that component unless it is otherwise specified.

2 In-service life begins the day the component is entered into service. In-service life only remains in effect while the component is installed on the helicopter.

3 Fail safe, refer to 412-A-05-40-00-01A-281A-A / 00023 for detailed 100-hour inspection.

4 On-condition subject to daily or 25-hour visual inspection interval (as applicable) per scheduled inspections (412-A-05-40-00-00A-281A-A / 00022 or 412-A-05-40-00-04A-281A-A / 00026).

5 The bushing 412-010-222 is installed on the main rotor yoke assemblies 412-010-101-127/-129/-133/-135/-137/-139. The bushing is assigned an unlimited airworthiness life. If removed, it is not reusable.

6 An on-condition retirement is assigned to spider 204-040-785-003 subject to passing a magnetic particle inspection every 3100 hours per ASB 412-91-52A.

7 An additional 4 hours must be logged for each hoist operation performed within penalty CG region (Figure 1). Refer to Chapter 5 and BHT-412-FM.

8 Retirement Index Number (RIN) is the retirement life based on fatigue damage from normal helicopter lifts and takeoffs. New components will begin with an accumulated RIN of zero that will be increased as lifts and takeoffs are performed. Operators must record the number of lifts and takeoffs and increase the accumulated RIN accordingly. When the maximum RIN or retirement flight hours is/are reached, whichever occurs first, the component will be removed from service. Refer to Information Letter GEN-94-54/Information Letter GEN-03-94.

9 Retire when the part has 10,000 flight hours or when the accumulated RIN = 80,000, whichever occurs first. For manual tracking, increase the RIN count by 1 for each takeoff/lift recorded. If logging, increase RIN count by 2 for each lift recorded.

10 Retire when the part has 10,000 flight hours or when the accumulated RIN = 60,000, whichever occurs first. For manual tracking, increase the RIN count by 1 for each takeoff/lift recorded. If logging, increase RIN count by 2 for each lift recorded.

11 This splined plate assembly can be used on all 412 models. When used on 412 HP/EP, it will be vibroetched “412 HP" and will be retired when part has 10,000 flight hours or when the accumulated RIN = 60,000, whichever occurs first. For manual tracking, increase the RIN count by 1 for each takeoff/lift recorded. If logging, increase RIN count by 2 for each lift recorded (see (12)).

12 This splined plate assembly can be used on all 412 models. When used on 412/412SP, it will not be vibroetched "412 HP". Retire when part has 10,000 flight hours or when the accumulated RIN = 80,000, whichever occurs first. For manual tracking, increase the RIN count by 1 for each takeoff/lift recorded. If logging, increase RIN count by 2 for each lift recorded (see (11)).

13 Deleted.

14 Retire when the part has 5000 flight hours or when the accumulated RIN = 60,000, whichever occurs first. For manual tracking, increase the RIN count by 1 for each takeoff/lift recorded. If logging, increase RIN count by 2 for each lift recorded.

15 Per ASB 412-99-97.

16 Contained in inflation valve 30905. 30908 includes squib 94455 and O-ring 568-905.

17 Per ASB 412-96-89, the tail rotor flapping stop/yield indicator 212-011-713-103 must be used with the tail rotor yoke. The stop/yield indicator 212-011-713-103 must be inspected every 25 hours and at each incident involving a hard landing, sudden stoppage, or any other incident involving excessive tail rotor flapping. Refer to 412-A-05-40-00-04A-281A-A / 00026, 412-A-05-50-00-00A-284A-A / 00063 or 412-A-05-50-00-01A-284A-A / 00064.

18 Manufactured by Aeronautical Accessories, Inc. Refer to AAI Report No. ICA AA-01136 Rev. C, or Subsequent, Inspection and Maintenance Instructions. Recurring visual inspection per AAI Bulletin AA-07109 and ASB 412-08-129.

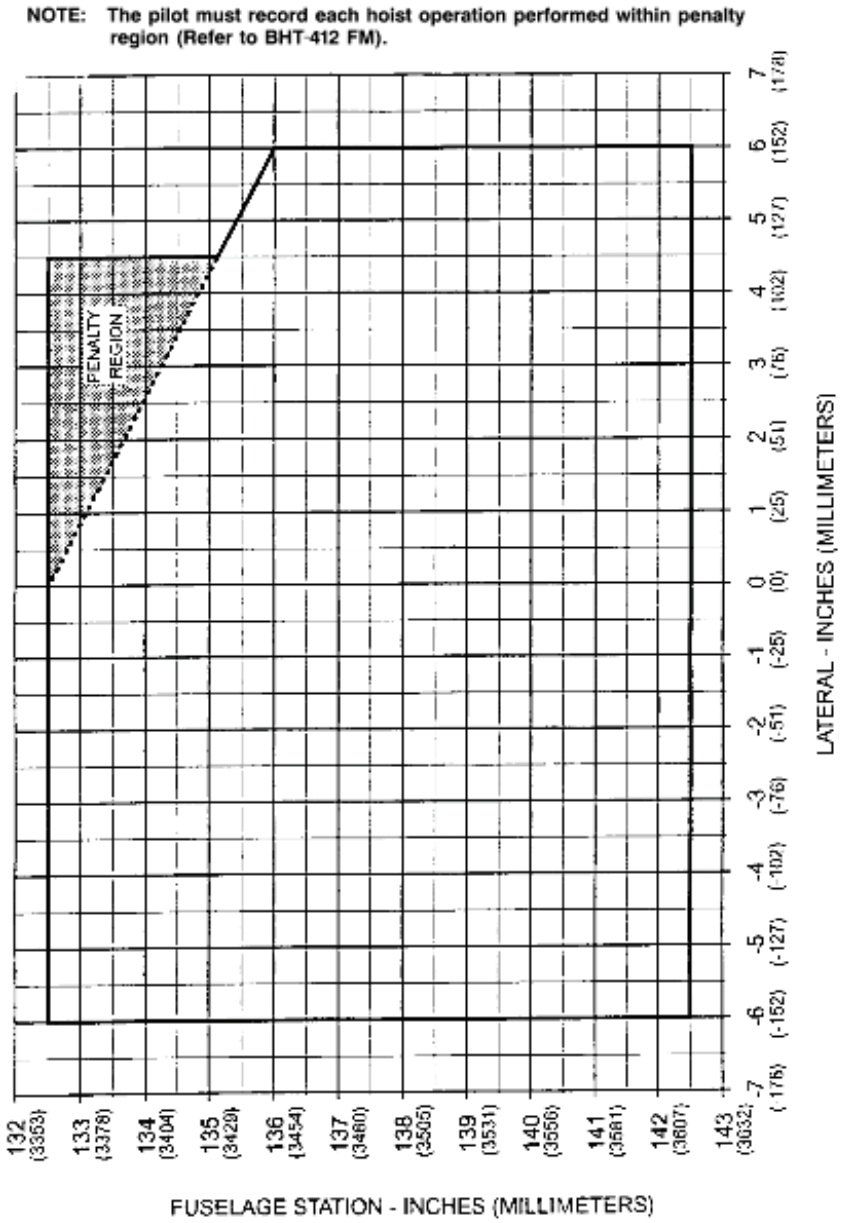
19 As per ASB 412-98-92, bolt 20-057-5-25D is replaced by 20-057-5-28D.

20 Total life begins from the date of component manufacture and cannot be interrupted.

21 Calendar life begins the day the component is entered into service and cannot be interrupted.

Table 1. Airworthiness Limitations Schedule (continued)

NOMENCLATURE	PART NUMBER (1)	AIRWORTHINESS LIFE
	MAIN ROTOR HUB	
22	Not applicable to subsequent dash numbers.	
23	Deleted.	
24	Replacement bearing cartridges 412-310-400-107 are on-condition. Refer to IL 412-IL-03-49.	
25	As per ASB 412-02-110, bolt 204-011-463-001 is replaced by 204-011-463-109.	
26	29022968 contained in valve assembly 20022988 Rev. E and refurbishment kit 29022971.	
27	As per ASB 412-93-73, bolt 412-010-124-105 is replaced by 412-010-124-109.	
28	Included in bolt kit 412-704-112-101. Bolt kit 412-704-112-101 is replaced by bolt kit 412-704-112-105.	
29	Included in bolt kit 412-704-112-103. Bolt kit 412-704-112-103 is replaced by bolt kit 412-704-112-105.	
30	Included in bolt kit 412-704-112-105.	
31	Not included in bolt kits 412-704-112-101/-103/-105.	
32	Life reduction to 2500 hours per ASB 412-08-131; remove from service by 1 June 2009.	
33	Refer to ASB 412-08-128.	
34	Use aft crosstube retention strap kit per BHT-412-SI-58 when towing helicopters with a gross weight of 8900 pounds or greater.	
35	Swashplate support assembly 412-010-443-101 was removed from service by ASB 412-93-71. All subsequent dash numbers remain in service.	
36	Swashplate support assembly 412-010-409-105 was removed from service by ASB 412-93-71.	
37	Swashplate support assembly 412-010-453-101 was removed from service by ASB 412-92-61. All subsequent dash numbers remain in service.	
38	500-hour penalty to be applied in accordance with AD 99-23-23 and ASB 412-98-93.	
39	AD 2012-22-11 imposes on collective lever 412-010-408-101 a visual inspection at intervals not to exceed 100 hours Time-In-Service (TIS) (ASB 412-11-148 Rev. A).	
40	Visual inspection at intervals not to exceed 100 hours Time-In-Service (TIS) . Inspection requirements same as listed in ASB 412-11-148 Rev. A.	
41	AD 2000-18-09 imposes on the upper aft beam cap 212-030-191-001 a recurring visual inspection at intervals not to exceed 100 hours Time-In-Service (TIS) (ASB 412-00-100). All aircraft with serial numbers 38001–38999 and 39101–39999 require a visual inspection in intervals not to exceed 50 hours Time-In-Service (TIS) . Refer to AD 2000-18-09 for inspection procedures.	
42	For manual tracking, increase the event count by 1 for each takeoff recorded.	
43	Part manufactured and part number assigned by AAI of Bristol, Tennessee. Retire part when event = 12,000 (event life).	
44	1000-event penalty to be applied against accumulated events of high aft crosstube assembly if used with increased gross weight kit 412-706-140 (BHT-412-FMS-74.5). 1000 event penalty to be applied on high aft crosstube assembly 412-321-901 on all aircraft serial numbers 38001–38999 and 39101–39999.	
45	On Condition indicates that there is no retirement life (i.e., unlimited). Any necessary inspections will be addressed by additional specific notes.	
46	Retire when the part has 2,500 flight hours or when the accumulated RIN is 25,000, whichever occurs first. For manual tracking, increase the RIN count by 1 for each takeoff/lift recorded. If logging, increase RIN count by 2 for each lift recorded.	
47	Apply a one time 7,500 hours penalty for all aircraft with serial numbers 38001–38999 and 39101–39999 on retention cap, cones, and drive pins.	



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Figure 1. Longitudinal/Lateral CG Envelope for Hoist Operations (Sheet 1 of 1)

CHAPTER 05
INSPECTIONS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number		Issue date	Applicable to
Inspections - General Description	DMC-412-A-05-00-00-00A-028A-A / 00015		2023-05-31	ALL
Inspection Requirements - Description	DMC-412-A-05-00-00-00A-042A-A / 00016		2023-05-31	ALL
Inspections - Crash Damage	DMC-412-A-05-00-00-01A-042A-A / 00017		2023-05-31	ALL
Inspections - Types of Inspections	DMC-412-A-05-00-00-02A-042A-A / 00018		2023-05-31	ALL
Inspections - Definitions	DMC-412-A-05-00-00-03A-042A-A / 00019		2023-05-31	ALL
Inspections - Inspection and Overhaul Tolerance	DMC-412-A-05-00-00-04A-042A-A / 00020		2023-08-28	ALL
Scheduled Inspections - Description	DMC-412-A-05-40-00-00A-042A-A / 00021		2023-05-31	ALL
Scheduled Inspections - Daily Inspection - Part A	DMC-412-A-05-40-00-00A-281A-A / 00022	C	2025-11-18	ALL
Scheduled Inspections - 100 Hour/12 Month Inspection - Part A	DMC-412-A-05-40-00-01A-281A-A / 00023	C	2025-11-18	ALL
Scheduled Inspections - 1000 Hour Inspection - Part A	DMC-412-A-05-40-00-02A-281A-A / 00024		2023-05-31	ALL
Scheduled Inspections - 5000 Hour/5 Year Inspection - Part A	DMC-412-A-05-40-00-03A-281A-A / 00025		2025-05-13	ALL
Scheduled Inspections - 25 Hour/30 Day Inspection - Part B	DMC-412-A-05-40-00-04A-281A-A / 00026	C	2025-11-18	ALL
Scheduled Inspections - 300 Hour/12 Month Inspection - Part B	DMC-412-A-05-40-00-05A-281A-A / 00027	C	2025-11-18	ALL
Scheduled Inspections - 600 Hour/12 Month Inspection - Part B	DMC-412-A-05-40-00-06A-281A-A / 00028		2024-12-19	ALL
Scheduled Inspections - 5000 Hour/5 Year Inspection - Part B	DMC-412-A-05-40-00-07A-281A-A / 00029		2025-05-13	ALL
Special Inspections - Description	DMC-412-A-05-47-00-00A-042A-A / 00030		2025-05-13	ALL
Special Inspections - Daily/10 Hour Inspection, Whichever Occurs First Until 250 Hours	DMC-412-A-05-47-00-00A-283A-A / 00031		2023-05-31	ALL
Special Inspections - Between 1 and 5 Flight Hours After Main Rotor Hub Installation	DMC-412-A-05-47-00-01A-283A-A / 00032		2023-05-31	ALL
Special Inspections - Between 1 and 5 Flight Hours After Tailboom Installation or Attachment Bolt Replacement	DMC-412-A-05-47-00-00A-283B-A / 00033		2023-05-31	ALL
Special inspections - Between 1 and 5 Flight Hours After Tail Rotor Gearbox or Intermediate Gearbox Installation	DMC-412-A-05-47-00-32A-283A-A / 00033.1		2024-12-19	ALL
Special Inspections - Between 1 and 25 Flight Hours After Expandable Blade Bolt Installation	DMC-412-A-05-47-00-02A-283A-A / 00034		2023-05-31	ALL
Special Inspections - Between 5 and 10 Hours of Flight After Tail Rotor Hub and Blade Assembly Installation	DMC-412-A-05-47-00-03A-283A-A / 00035		2024-12-19	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Special Inspections - Each 25 Hours For the Next Four Inspections	DMC-412-A-05-47-00-04A-283A-A / 00036		2023-05-31	ALL
Special Inspections - Each 25 Hours of Tail Rotor Operation	DMC-412-A-05-47-00-05A-283A-A / 00037		2025-05-13	ALL
Special Inspections - Each 25 Hours Tailboom Attachment Inspection	DMC-412-A-05-47-00-30A-283A-A / 00039		2025-05-13	ALL
Special Inspections - 50 Hours After Installation of Components	DMC-412-A-05-47-00-06A-283A-A / 00038		2023-05-31	ALL
Special Inspections - Each 50 Hours Tailboom Attachment Inspection	DMC-412-A-05-47-00-51A-283A-A / 00038.1		2025-05-13	ALL
Special Inspections - Each 100 Hours Tailboom Vertical Fin Spar Cap Inspection	DMC-412-A-05-47-00-31A-283A-A / 00040		2023-05-31	ALL
Special Inspections - Each 100 Hours of Collective Lever Operation	DMC-412-A-05-47-00-08A-283A-A / 00041		2023-05-31	ALL
Special Inspections - Each 150 Hours of Starter Generator (200SG119Q) Operation	DMC-412-A-05-47-00-09A-283A-A / 00042		2023-05-31	ALL
Special Inspections - Each 300 Hours or 6 Months of Expandable Blade Bolt Operation	DMC-412-A-05-47-00-10A-283A-A / 00043		2023-05-31	ALL
Special Inspections - Each 600 Hours of Tail Rotor Driveshaft Operation or 12 Months	DMC-412-A-05-47-00-11A-283A-A / 00044	C	2025-11-18	ALL
Special Inspections - Each 600 Hours or 12 Months Inspection of Tailboom Attachment Bolts	DMC-412-A-05-47-00-00B-283B-A / 00045		2023-08-28	ALL
Special Inspections - Each 600 Hours of Main Driveshaft Operation or 12 Months	DMC-412-A-05-47-00-12A-283A-A / 00046	C	2025-11-18	ALL
Special Inspections - Each 600 Hours or 12 Months Inspection Of Magnetic Brake Assembly	DMC-412-A-05-47-00-50A-283A-A / 00047		2024-12-19	ALL
Special Inspections - Each 1000 Hours of Component Operation	DMC-412-A-05-47-00-13A-283A-A / 00048		2023-05-31	ALL
Special Inspections - Each 12 Months or 2500 Landings of Aft High Crosstube Operation	DMC-412-A-05-47-00-14A-283A-A / 00049		2023-05-31	ALL
Special Inspections - Each 24 Months of Control Bolt Operation	DMC-412-A-05-47-00-15A-283A-A / 00050		2023-05-31	ALL
Special Inspections - Each 24 Months of Main Rotor Mast Operation	DMC-412-A-05-47-00-16A-283A-A / 00051		2023-05-31	ALL
Special Inspections - Each 2500 Hours of Main Rotor Hub Assembly Operation	DMC-412-A-05-47-00-17A-283A-A / 00052		2023-05-31	ALL
Special Inspections - Each 2500 Hours of Main Rotor Blade Operation	DMC-412-A-05-47-00-18A-283A-A / 00053		2023-05-31	ALL
Special Inspections - Each 2500 or 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-109 and Subsequent) Operation	DMC-412-A-05-47-00-19A-283A-A / 00054	C	2025-11-18	ALL
Special Inspections - Each 2500 Hours of Tail Rotor Drive System Operation	DMC-412-A-05-47-00-20A-283A-A / 00055	C	2025-11-18	ALL
Special Inspections - Each 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-103 and -105) Operation	DMC-412-A-05-47-00-21A-283A-A / 00056	C	2025-11-18	ALL
Special Inspections - Each 3000 Hours of Transmission (412-040-002) Operation	DMC-412-A-05-47-00-22A-283A-A / 00057	C	2025-11-18	ALL
Special Inspections - Each 3000 Hours of Transmission (412-040-004/412-040-007/412-040-802) Operation	DMC-412-A-05-47-00-23A-283A-A / 00058	C	2025-11-18	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Special Inspections - 3600 Hours Total Airframe Time and Each 300 Hours/12 Months Inspection	DMC-412-A-05-47-00-24A-283A-A / 00059		2023-05-31	ALL
Special Inspections - Each 3000 Hours or 5 Years Of Transmission Mount Assembly (204-031-927) And Friction Damper Assembly (204-031-920) Operation	DMC-412-A-05-47-00-28A-283A-A / 00060	C	2025-11-18	ALL
Special Inspections - Each 2500 Hours or 10 Years Inspection Of Collective Stick Tube	DMC-412-A-05-47-00-29A-283A-A / 00061		2024-05-09	ALL
Conditional Inspection - Description	DMC-412-A-05-50-00-00A-042A-A / 00062		2023-05-31	ALL
Conditional Inspection - After Hard Landing	DMC-412-A-05-50-00-00A-284A-A / 00063		2023-05-31	ALL
Conditional Inspection - Sudden Stoppage - Power ON or OFF	DMC-412-A-05-50-00-01A-284A-A / 00064		2025-05-13	ALL
Conditional Inspection - After Overspeed	DMC-412-A-05-50-00-02A-284A-A / 00065		2023-05-31	ALL
Conditional Inspection - After Overtorque	DMC-412-A-05-50-00-03A-284A-A / 00066	C	2025-11-18	ALL
Conditional Inspection - After Compressor Stall or Surge	DMC-412-A-05-50-00-04A-284A-A / 00067		2023-05-31	ALL
Conditional Inspection - After Lightning Strikes	DMC-412-A-05-50-00-05A-284A-A / 00068		2023-05-31	ALL
Conditional Inspection - After Engine Combining Gearbox Clutch Non-Engagement, Misengagement, or In-flight Slippage	DMC-412-A-05-50-00-06A-284A-A / 00069		2023-05-31	ALL
Component Overhaul Schedule - Description	DMC-412-A-05-11-00-00A-042A-A / 00070	C	2025-11-18	ALL

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Inspections - General Description

This chapter contains the requirements for Scheduled, Special, and Conditional Inspections, and a Component Overhaul Schedule.



FAILURE TO CORRECT CONDITIONS SUCH AS, BUT NOT LIMITED TO, CORROSION, EROSION, MECHANICAL DAMAGE, OR OBVIOUS WEAR FOUND DURING A SCHEDULED INSPECTION COULD SERIOUSLY AFFECT THE AIRWORTHINESS OF THE HELICOPTER.

These inspection requirements constitute an approved inspection program designed and recommended by Bell Helicopter Textron for the Model 412 Series. They can be utilized by registered owners or operators who desire to develop a progressive inspection system to be used as part of the complete maintenance program. A written request for approval to use the progressive inspection system shall be submitted to the governing civil aviation authority having jurisdiction over the area in which the owner or operator is located. Relief from the requirements in this chapter can be requested through the local aviation authority. For the convenience of the operator, two separate scheduled inspections are provided as follows:

- Part A Scheduled inspections consist of a daily inspection, 100 hour/12 calendar month inspection, 1000 hour inspection, and 5000 hour/5 year inspection.
- Part B Scheduled inspections consist of a 25 hour/30 day inspection, 300 hour/12 month, 600 hour/12 month inspection, and 5000 hour/5 year inspection.

Either series of inspections may be utilized. However, once a helicopter has been started on either inspection program, it shall be maintained on that program except as follows:

- If a helicopter is being inspected on the Part A inspection program and it is desired to change to the Part B inspection program, a complete Part A 1000 hour inspection shall be accomplished. The helicopter may then be changed to the Part B inspection program beginning with a 25 hour/30 day inspection.
- If a helicopter is being inspected on the Part B inspection program and it is desired to change to the Part A inspection program, a complete 600 hour inspection shall be accomplished. The helicopter may then be changed to the Part A inspection program beginning with a daily inspection.

The inspection intervals given in this chapter are the maximum permitted. Do not exceed these intervals. The owner/operator is responsible for increasing the scope and the frequency of the inspections as necessary. Make sure the helicopter is maintained safely during all unusual local changes, such as environmental conditions, helicopter use, etc. You can request changes to the requirements in this chapter through the local aviation authority.

The inspection intervals and the component overhaul schedule provided in this chapter are applicable only to Bell Helicopter Textron approved parts.

NOTE

The time period given for the overhaul of a component (or the failure to give a time period for the overhaul of a component) does not constitute a warranty of any kind. The only warranty applicable to the helicopter or any component is the warranty included in the Purchase Agreement for the helicopter or the component.

The Time Between Overhaul (TBO) and the inspection periods are determined through experience, tests, Lead The Fleet (LTF) , or any other special programs and the judgement of Bell Helicopter Textron engineers. They are subject to change only by Bell Helicopter Textron or an approved Airworthiness Authority.

Changes to the TBO will be introduced by either revisions to this chapter or a Technical Bulletin.

Every calendar and hourly inspection is a thorough visual inspection to determine the airworthiness of the helicopter and the components. Qualified persons must do the inspections in accordance with quality standard aircraft practices and the applicable maintenance manuals. Bell Helicopter Textron considers that it is mandatory to obey all the applicable Alert Service Bulletins (ASB) and the Airworthiness Directives (AD) .

Component operating time records are necessary for components that have scheduled maintenance procedures, which are different from those of the airframe. It is the owner/operator’s responsibility to keep the Historical Service Records (HSR) for the applicable component and to do the necessary maintenance procedures.

Before each inspection, remove or open the necessary cowlings, fairing, inspection doors, and panels.

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Inspection Requirements - Description

This manual does not include specific inspections required by the FAA or other government regulatory authorities. These specific inspections are given by your government regulatory authority. Refer to their requirements for these specific inspections.

The owner/operator of the helicopter is responsible for the maintenance done on the helicopter. It is the owner/operator's responsibility to:

Establish, maintain, and review the log books for discrepancies.

Make sure the Alert Service Bulletins (ASB) , the Airworthiness Directives (AD) , and the special inspections are done when they are required to be done.

Make sure the scheduled inspections, the special inspections, and the required tests for all of the installed kits are complied with.

Make sure all parts and components for which Historical Service Records are required have documented traceability to their original installation in the helicopter.



ALL PARTS REMOVED, DUE TO REACHING THEIR LIMITS OR AS A RESULT OF AN ACCIDENT/INCIDENT INSPECTION AND DEEMED UNAIRWORTHY, SHALL BE PERMANENTLY MARKED AS SCRAP OR PHYSICALLY DESTROYED TO THE EXTENT THAT THERE IS NO CHANCE OF REPAIR OR INSTALLATION ON ANY HELICOPTER OR COMPONENT.

Make sure all limited life parts that have completed their published operating limits are replaced.

Make sure all of the components that have completed their published overhaul periods are overhauled.

Make sure all of the maintenance that is done on the helicopter is done by an approved maintenance organization.

The maintenance organization/person doing the maintenance is responsible for the quality of the maintenance done.

The owner/operator may choose to ask the maintenance organization/person doing the maintenance to perform the tasks listed by prior arrangement through a separate formal agreement.

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Inspections - Crash Damage

Because of the many possible combinations that can result from crash damage, it is not possible to include the specific repair tasks in this category. The helicopter mechanic must make an analysis of the crash damage for each situation. Do the repair in accordance with the degree of damage to the specific part and the applicable repair procedures in this manual. Call Product Support Engineering with your analysis of the crash damage.

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Inspections - Types of Inspections

The maintenance procedures may include scheduled inspections, special inspections, conditional inspections, component interim inspections, and component overhaul inspections.

Scheduled inspections must occur at specified operating intervals. The intervals may be in operating time (hours), cycles, torque events (RIN), calendar (days, months, years) or other assigned units. This ensures the helicopter is airworthy.

- Scheduled inspections - Part A consists of:
  - Daily inspection - Accomplish daily before flight operation.
  - 100 hours/12 calendar months - Accomplish each 100 hours of flight operation or 12 calendar months, whichever comes first.
  - 1000 hours - Accomplish each 1000 hours of flight operation.
  - 5000 hours/5 years - Accomplish each 5000 hours of flight operation or each 60 calendar months, whichever comes first.
- Scheduled inspections - Part B consists of the following:
  - 25 hours/30 days - Accomplish each 25 hours of flight operation or each 30 days, whichever comes first.
  - 300 hours/12 month inspection - Accomplish each 300 hours of flight operation or each 12 calendar months, whichever comes first.
  - 600 hours/12 months - Accomplish each 600 hours of flight operation or each 12 calendar months, whichever comes first.
  - 5000 hours/5 years - Accomplish each 5000 hours of flight operation or each 60 calendar months, whichever comes first.

Special inspections are of a temporary nature or of a special interval that is not consistent with the scheduled inspections.

Conditional inspections do not occur at a specified time. A conditional inspection is the result of known or suspected unusual event, known or suspected malfunctions, or defects.

An interim inspection occurs between overhauls.

The component overhaul schedule gives the elapsed operating time at which a component must be removed, disassembled, examined for condition, and overhauled, in accordance with data approved by Bell Helicopter Textron.

Airworthiness limitations - Replace components in accordance with Airworthiness Limitations Schedule (412-A-04-00-00-00A-288A-A / 00013).

Lubrication and servicing requirements are in addition to those stated in this chapter (412-A-12-01-00-00A-040A-A / 00141).

For corrosion control, refer to the Corrosion Control Guide, CSSD-PSE-87-001, Corrosion Control Guide and the BHT-ALL-SPM, Standard Practices Manual.

For the PT6T Series engines, refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for the scheduled inspections, special inspections, conditional inspections, and component overhaul schedule.

For the common Bell Helicopter Textron approved optional equipment that is integrated into this maintenance manual, refer to this chapter for the scheduled inspection, conditional inspection, component interim inspection, and component overhaul inspection.

For all other Bell Helicopter Textron approved equipment, refer to the applicable Service Instruction for the scheduled inspection, special inspection, conditional inspection, component interim inspection, and component overhaul inspection.

For the inspection requirements for optional equipment approved under Supplement Type Certificate (STC) , refer to the applicable STC documentation. Maintenance and inspection of these items are the responsibility of the owner/operator.

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Inspections - Definitions

- Inspect, check, examine - Determine condition relative to an established standard.
- Condition - The state of being of an item as related to serviceable or unserviceable standard.
- Security - The presence of attaching parts that are properly tightened or appear to be, and the presence of properly installed (as required) locking devices such as lockwire, cotter pins, or other.
- Standard - An established rule or measure to determine condition.
- Damage - Physical deterioration whereby the standard renders the condition or an item acceptable or not acceptable for continuous use.
- Discard - Reject a component that has damage that cannot be repaired. To permanently remove from service.
- Preventive maintenance - Simple or minor preservation and the replacement of small standard parts not involving complex assembly operations.
- Maintenance - Inspection, overhaul, repair, preservation, and the replacement of parts, but excludes preventive maintenance.
- Operating time - Time required to be recorded in historical record sheets or helicopter logs. Operating time to be recorded may be identified as follows:
  - Time in service - Time from the moment a helicopter leaves the surface of the earth until it touches down at the next point of landing. Time during which the engine and rotor are turning with the helicopter on the ground is not taken into account.
  - Calendar time - Elapse time starts the day the inspection is accomplished, the component is installed, or the rotor is turned for the first time and ends on the last day of the month that the time limit expires. Calendar time shall be recorded without interruption. Removal of the component or storage of the helicopter etc. does not stop calendar time.
- Planned event - Occurrence of interval in which a specific action is to be taken as in the case of preventive maintenance, scheduled overhaul, or replacement in accordance with maximum airworthiness life guidelines.
- Lead The Fleet (LTF) Program - This is a program to validate the performance of an approved product improvement or a change to a maintenance interval. The engineering aspects of this change are approved. The program is closely monitored by Bell Helicopter Textron in an operational environment with selected operators.
- Special Programs - These are approved programs that may be initiated under certain special conditions to meet specific requirements. These programs will be clearly defined through a plan and the engineering and maintenance aspects will be approved by the regulatory authorities.

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Inspections - Inspection and Overhaul Tolerance

WARNING

DO NOT APPLY THESE TOLERANCES TO PARTS WITH A LIMITED AIRWORTHINESS LIFE OR PARTS SUBJECT TO AN INSPECTION LISTED IN THE AIRWORTHINESS LIMITATION SCHEDULE (412-A-04-00-00-00A-288A-A / 00013).

The Bell Helicopter Textron approved tolerance for scheduled inspections, special inspections, interim inspections, and overhaul intervals, unless otherwise stated, is 10% or up to a maximum of 300 hours operating time/6 months calendar time, whichever is less. The tolerances are established for maintenance scheduling convenience only.

Scheduled inspections, special inspections, interim inspections, or overhaul intervals required beyond the stated tolerances must be approved by Product Support Engineering.

NOTE

The following is only applicable for those operators whose governing aviation authority requires to specifically approve the inspection and overhaul tolerance.

If approval of the inspection and overhaul tolerance is necessary by the applicable governing aviation authority, this is the responsibility of the owner/operator.

Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for inspection and overhaul tolerances.

The following provides examples of when hourly, calendar, or hourly/calendar inspection tolerances have been applied:

Table 1. ***Hourly Example (10% or up to a Maximum of 300 Hours, Whichever is Less):***

300-HOUR INSPECTION DUE AT:	*MAXIMUM ALLOWED TOLERANCE*	*INSPECTION CARRIED OUT AT:*	*NEXT 300-HOUR INSPECTION DUE AT:*
3400 Hours	*10% of 300 hours = 30 hours*	*3430 Hours (10% tolerance applied)*	*3730 Hours*
3730 Hours	*10% of 300 hours = 30 hours*	*3750 Hours (within 10% tolerance)*	*4050 Hours*
4050 Hours	*10% of 300 hours = 30 hours*	*4050 Hours (tolerance not applied)*	*4350 Hours*

Table 2. ***Calendar Example (10% or up to a Maximum of 6 Months Calendar Time, Whichever is Less):***

12-MONTH INSPECTION DUE ON:	*MAXIMUM ALLOWED TOLERANCE*	*INSPECTION CARRIED OUT ON:*	*NEXT 12-MONTH INSPECTION DUE ON:*
June 10, 2010	*10% of 12 months = 1.2 months*	*July 8, 2010 (within 10% tolerance)*	*July 31, 2011* (1)
July 31, 2011	*10% of 12 months = 1.2 months*	*June 15, 2011 (completed early)*	*June 30, 2012* (1)
June 30, 2012	*10% of 12 months = 1.2 months*	*June 30, 2012 (tolerance not applied)*	*June 30, 2013* (1)

NOTE:

1 *The last day of the month applies for the next inspection (412-A-05-00-00-03A-042A-A / 00019, calendar time). For inspection intervals that are measured in days rather than months, the calendar time ends on the day that the time limit expires on, not the end of the month.*

Table 3. ***Hourly/Calendar Example (10% or up to a Maximum of 300 Hours Operating Time/6 Months Calendar Time, Whichever is Less):***

5000-HOUR/5 YEAR INSPECTION DUE AT:	*MAXIMUM ALLOWED TOLERANCE*	*INSPECTION CARRIED OUT AT:*	*NEXT 5000-HOUR/5 YEAR INSPECTION DUE AT:*
6000 Hours/December 31, 2010	*10% of 5000 hours = ~~500 hours~~ (maximum allowed is 300 hours) or* *10% of 60 months = 6 months*	*6000 Hours/January 15, 2011 (within 6 months calendar tolerance)*	*11,000 Hours/January 31, 2016* (1)
11,000 Hours/January 31, 2016	*10% of 5000 hours = ~~500 hours~~ (maximum allowed is 300 hours) or* *10% of 60 months = 6 months*	*11,100 Hours/January 2, 2016 (300 hour tolerance applied)*	*16,100 Hours/January 31, 2021* (1)
16,100 Hours/January 31, 2021	*10% of 5000 hours = ~~500 hours~~ (maximum allowed is 300 hours) or* *10% of 60 months = 6 months*	*16,250/April 25, 2021 (within 300 hour and 6 months calendar tolerance)*	*21,250/April 30, 2026* (1)

NOTE:

1 *The last day of the month applies for the next inspection (412-A-05-00-00-03A-042A-A / 00019, calendar time). For inspection intervals that are measured in days rather than months, the calendar time ends on the day that the time limit expires on, not the end of the month.*

Scheduled Inspections - Description

Part A - Inspect helicopter daily, each 100 hours/12 months, each 1000 hours, each 5000 hours/5 years.

Part B - Inspect helicopter each 25 hours/30 days, each 300 hours/12 months, each 600 hours/12 months, each 5000 hours/5 years.

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Scheduled Inspections - Daily Inspection - Part A

Procedure

Table 1. Daily Inspection - Part A

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME:_____		
	SIGNATURE:_____		
	DATE: _____W.O. _____		
	FACILITY: _____		
	NOTE For helicopters on the Part A inspection program, accomplish the following checks daily before flight operation.		
	GENERAL		
	1. Each listed inspection item or maintenance function is to be performed in accordance with the BHT-412-MM chapter specified or the BHT-412-CR&O.		
	2. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for engine inspection requirements.		
	PRELIMINARY REQUIREMENTS		
Corrosion Control Guide, CSSD-PSE-87-001	1. Use medium helicopter corrosion control guide to establish helicopter corrosion control program.		
412-A-04-00-00-00A-288A-A / 00013	2. Replace all finite life components that have completed published operating limitations.		
412-A-05-11-00-00A-042A-A / 00070	3. Overhaul all components that have completed published overhaul periods.		
412-A-12-01-00-00A-040A-A / 00141	4. Lubricate and service helicopter as required.		
412-A-05-47-00-00A-283A-A / 00031	5. Review Special Inspections and carry out applicable inspections.		
	6. Examine all inspection windows and sight glasses for cracking, crazing, and discoloration. If any of these conditions are present, the part must be removed and replaced prior to returning helicopter for service.		
Service Instruction (SI)	7. Ensure all required inspections of installed BHT kits not covered in this inspection have been performed, as applicable.		
412-A-21-00-00-00A-009A-A / 00189	**BLEED AIR HEATING SYSTEM COMPONENTS**		
	1. Visually inspect heater compartment for cleanliness, condition, and security of heating system components, wiring, ducts, supports, and structure for damage and corrosion.		
	2. Visually inspect overhead ventilating system components for condition and security.		
	3. Visually inspect heat/vent air ducts for condition and security.		
	4. Visually inspect ventilation/defog components for condition and security.		
412-A-25-00-00-00A-009A-A / 00244	**CREW/PASSENGERS SEATS**		
	1. Visually inspect crew seats for condition, security, and operation.		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2. Visually inspect crew seats restraints for condition, security, and operation.		
	3. Crew seat attenuator for compression - inspect witness line. If line is not visible, repair attenuator assembly.		
	4. Visually inspect passenger seats for condition and security.		
	5. Visually inspect passenger seats restraints for condition and security.		
412-A-25-00-00-00A-009A-A / 00244	**MISCELLANEOUS FURNISHINGS**		
	Visually inspect miscellaneous furnishing (map and data case, first aid kit, and emergency equipment) for condition and security.		
412-A-26-00-00-00A-009A-A / 00291	**FIRE EXTINGUISHERS**		
	Visually inspect cockpit and cabin portable fire extinguishers and engine compartment fire extinguishers containers for security and condition.		
412-A-28-00-00-00A-009A-A / 00323	**FUEL SYSTEM**		
	Visually inspect fuel samples for contamination.		
412-A-29-00-00-00A-009A-A / 00610	**HYDRAULIC SYSTEMS**		
	1. Visually inspect the following:		
	1.1. Hydraulic system 1 and 2 filter bypass indicator buttons - not extended.		
	1.2. Collective and cyclic servo actuators and boost tubes for leaks, damage, and security.		
	1.3. Hydraulic system 1 and 2 pumps for leaks, damage, and security.		
	1.4. Hydraulic system 1 and 2 valve and filter modules for leaks, damage, and security.		
	1.5. Hydraulic system 1 and 2 lines, hoses, and fittings for leaks, damage, chafing, kinking, and security.		
	1.6. Hydraulic system 1 and 2 reservoirs for proper fluid levels, damage, corrosion, and security.		
	1.7. Tail rotor hydraulic actuator and hoses for leakage, corrosion, and security.		
	2. Remote hydraulic filter bypass indicator (located in right nose window) - confirm not tripped.		
412-A-30-00-00-00A-009A-A / 00677	**WINDSHIELD WIPER**		
	Visually inspect windshield wiper blades for serviceability and security.		
412-A-32-00-00-00A-009A-A / 00694	**LANDING GEAR SYSTEM**		
	1. Visually inspect landing gear as follows:		
	1.1. Forward crosstube and cap fittings for condition and security of attachment.		
	1.2. Aft crosstube, cap fittings, and wear strips for condition and security of attachment.		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	1.3. Aft crosstube support beam for wear, cracks, damage, corrosion, and security of attachment.		
	1.4. Skid tubes and skid shoes for condition and security of attachment. Replace skid shoes that are worn into shoe surface. Repair weld beads as required.		
	1.5. Fuselage supports for wear, damage, and security of attachment.		
	1.6. If installed, landing gear forward crosstube crew step fittings for corrosion, damage, and security in area of attachment. If edge sealant is damaged or missing, remove crew step and inspect crosstube for corrosion or damage.		
	1.7. Emergency float reservoir pressure indicator for proper charge indication (if installed).		
	1.8. Floats for proper stowage and condition (if installed).		
	2. Visually inspect tail skid for deformation and security of attachment.		
412-A-52-00-00-00A-009A-A / 00774	**DOORS, WINDOWS, AND EMERGENCY EXIT**		
	1. Visually inspect nose door for obvious damage, security of attachment, proper latching, and seal for condition.		
	2. Visually inspect all windows for damage. Crew door windows, cargo hinged door window, passenger sliding door windows, cabin roof windows, cabin lower nose windows, and windshields.		
	3. Visually inspect crew door emergency release pins for security.		
	4. Visually inspect cabin doors for condition, security, and freedom of operation.		
	5. Avionics/electrical and heater compartment doors for condition and security of attachment. Latches for proper operation.		
412-A-53-00-00-00A-009A-A / 00897	**FUSELAGE**		
	1. Visually inspect fuselage exterior for condition and damage to protective finish.		
	2. Visually inspect fuselage underside for evidence of fuel and hydraulic fluid leakage		
	3. Visually inspect all cowlings and fairings for condition and security, missing fasteners, cracks, and proper operation of latches.		
	4. Inspect fuselage tailboom attachment points for security.		
	5. Visually inspect pitot tubes and static ports for obstruction and damage.		
	6. Visually inspect fuselage interior for evidence of water entrapment in following areas:		
	6.1. Nose compartment		
	6.2. Pilot and passenger cabin		
	6.3. Electrical compartment		
	6.4. Heater compartment		
	6.5. Baggage compartment		
412-A-53-00-00-00A-009A-A / 00897	**TAILBOOM**		
	1. Inspect tailboom exterior structure for general condition.		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2. Inspect baggage compartment interior for condition and cleanliness. 3. Check baggage compartment door for damage, proper operation, and security. 4. Inspect driveshaft and intermediate gearbox covers for damage and security.		
412-A-62-00-00-00A-009A-A / 00932	**MAIN ROTOR BLADES** Visually inspect main rotor blades for condition, damage, security, and cleanliness.		
412-A-62-00-00-00A-009A-A / 00932	**MAIN ROTOR HUB** 1. Visually inspect main rotor hub assembly for condition and security, paying particular attention to the condition of elastomeric components, and the integrity of sealing. 2. Visually inspect pitch horns for condition, security, and cracks in areas surrounding torque transfer pin holes. No cracks permitted. 3. Inspect pivot bearings and damper yoke set for elastomeric bearing delamination, cracking, and elastomer degradation.		
412-A-62-00-00-00A-009A-A / 00932	**MAIN ROTOR CONTROLS** 1. Visually inspect swashplate and support assembly and collective lever for condition and security. 2. Visually inspect hub and sleeve assembly for condition and security. **NOTE** Use swashplate link and drive link bearings - Work aid (412-A-62-00-00-00A-009A-A / 00932) to assist with determining condition of main rotor control bearings. 3. Visually inspect drive hub and rephasing levers for bearing looseness and security of attachment. 4. Visually inspect pitch links for condition, security, and bearing looseness. 5. Visually inspect mast assembly for corrosion and mechanical damage. Mast seal for leakage.		
412-A-63-00-00-00A-009A-A / 00987	**ENGINE-TO-TRANSMISSION (MAIN) DRIVESHAFT** 1. Visually inspect main driveshaft for condition and security.  **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.**		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	2. Visually inspect main driveshaft forward and aft couplings, boots, seals (and surrounding area), for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of any yellow bordered TEMP-PLATE, from white/off-white to black, will indicate a possible overheat condition and/or component degradation (Paragraph). If one of more of the red bordered TEMP-PLATE dots change from white/off-white to black, an overheat condition is present and may require replacement of both outer and inner couplings. 3. Visually inspect overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of overtemperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.		
412-A-63-00-00-00A-009A-A / 00987	**TRANSMISSION** 1. Check for proper oil level. 2. Check external oil filter for bypass indication. 3. Visually inspect transmission cases for damage, condition, and evidence of leaking. 4. Visually inspect accessories for condition, damage, and security of attachment. 5. Visually inspect external oil lines and hoses for condition, damage, chafing and leaks, paying particular attention to lines running aft of the thermostatic relief valve. 6. Visually inspect mast assembly for corrosion and mechanical damage, and mast seal for leakage. 7. Transmission mount dust boots for condition.  **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.** 8. Visually inspect transmission tail rotor output quill coupling (and surrounding area) for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black, indicates an overheat condition, and may require replacement of both outer and inner couplings (Paragraph). 9. Visually inspect overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of overtemperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.		
412-A-63-00-00-00A-009A-A / 00987	**ROTOR BRAKE QUILL AND BRAKE ASSEMBLY** 1. Visually inspect rotor brake assembly for condition, damage, and security. 2. Visually inspect rotor brake quill for condition, damage, and leakage. 3. Visually inspect rotor brake disc for warpage.		
412-A-63-00-00-00A-009A-A / 00987	**TRANSMISSION OIL COOLING AND COMBINING GEARBOX**		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	1. Visually inspect oil coolers for leaking, damage, and obstruction. 2. Visually inspect oil cooler hoses and tubes for leaking, chafing, fraying or other damage. 3. Visually inspect oil cooler blowers for damage and obstruction.		
412-A-64-00-00-00A-009A-A / 01225	**TAIL ROTOR HUB AND BLADE ASSEMBLY**  **NO CRACKS ARE PERMITTED ON ANY SURFACE OF THE TAIL ROTOR BLADES.** 1. Tail rotor blades for condition of bond lines, cracks, corrosion leading edge erosion, damage, security, and cleanliness. Clean blades are required to maintain enhanced visibility for safety. 2. Tail rotor hub for security, corrosion and condition. 3. Flapping bearings and pitch change bearings for excessive looseness and freedom of movement through full range of travel with anti-torque pedals positioned full right and then full left. 4. Tail rotor pitch change links, crosshead, pitch horns, counterweight arms, and links for security, corrosion, and condition. 5. Pitch link bearing and counterweight link bearing looseness shall not exceed 0.015 inch (0.38 mm) axial. Check freedom of movement through full range of travel with antitorque pedals positioned full right and then full left. 6. Pitch change links for binding with tail rotor blade moved to both full flapping positions. Binding is not acceptable.		
412-A-65-00-00-00A-009A-A / 01257	**TAIL ROTOR DRIVESHAFT**  **(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):** ***CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.*** (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *1. Hanger assemblies - bearings and surrounding areas for evidence of grease leakage, condition, damage, security, corrosion, and overheating. Discoloration of bearing (color change to blue to blue/black in color) or multi-color appearance of hanger that darkens adjacent to bearing is evidence of overheating. Brown color of bearing shield is normal and is not evidence of overheating.*		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	NOTE **(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):** *Do not mistake empty imprints in bonding material next to balance strip, as an indication of a missing balance strip. This spot results from removal of a test coupon to inspect for bonding voids. If a tail rotor driveshaft has more than a single empty bonding imprint, remove and send to Bell for repair and balancing.* (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *2. Driveshaft sections for cracks, rivet failure, distortion, dents, corrosion, damage to anodized finish, and loose or missing weights.* (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *3. Driveshaft for missing balance strips.* (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *4. Clamp sets for condition, security, and proper installation.*  **(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):** ***CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.*** (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *5. Flexible couplings and surrounding areas for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 6).* (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *6. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.*		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	 **(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed):** ***CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.*** (SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *7. Hanger assemblies - bearings and surrounding areas for evidence of grease leakage, condition, damage, security, corrosion, and overheating. Discoloration of bearing (blue or blue/black in color) or multi-color appearance of hanger that darkens adjacent to bearing is evidence of overheating. Brown color of bearing shield is normal and is not evidence of overheating.* (SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *8. Driveshaft sections for cracks, rivet failure, distortion, dents, corrosion, and damage to anodized finish.* (SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *9. Disc pack couplings for separation, deflection, distortion, cracks, or missing sections, and correct installation of bolts and washers.*  **(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed):** ***CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.*** (SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *10. Coupling assembly at first hanger and adjacent to transmission and surrounding area for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 11).* (SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *11. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.*		
412-A-65-00-00-00A-009A-A / 01257	**INTERMEDIATE GEARBOX**		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *1. Gearbox assembly for security, condition, corrosion, damage, and oil leaking.*  **(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):** ***CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.*** (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *2. Input and output quill flexible couplings (and surrounding area) for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 3).* (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *3. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.* (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *4. Gearbox for proper oil level and oil for evidence of contamination.* (SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *5. Gearbox assembly for security, condition, corrosion, damage, and oil leaking.*  **(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed):** ***CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.*** (SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *6. Output quill outer couplings (and surrounding area) for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 7).*		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *7. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.*		
	(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *8. Gearbox for proper oil level and oil for evidence of contamination.*		
412-A-65-00-00-00A-009A-A / 01257	**TAIL ROTOR GEARBOX**		
	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *1. Gearbox assembly for security, condition, corrosion, damage, and oil leaking.*		
	 **(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):** ***CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.***		
	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *2. Input quill flexible coupling (and surrounding area) for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 3).*		
	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *3. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of overtemperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.*		
	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *4. Gearbox for proper oil level and oil for evidence of contamination.*		
	(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *5. Gearbox assembly for security, condition, corrosion, damage, and oil leaking.*		
	(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *6. Gearbox for proper oil level and oil for evidence of contamination.*		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-67-00-00-00A-009A-A / 01319	**FLIGHT CONTROLS** 1. Inspect flight control tubes for condition, corrosion, and security. 2. Visually inspect elevators for damage and security. Check elevator control tube for positive spring action and freedom of movement.		
412-A-71-00-00-00A-009A-A / 01584	**LEFT POWER SECTION** 1. Inspect gas generator case for cracks, buckled areas, and hot spots. 2. Inspect oil and fuel hoses and tubes for chafing, leaking, and security. 3. Inspect electrical wiring for fraying, chafing, and security. 4. Check for proper oil level.		
412-A-71-00-00-00A-009A-A / 01584	**RIGHT POWER SECTION** 1. Inspect gas generator case for cracks, buckled areas, and hot spots. 2. Inspect oil and fuel hoses and tubes for chafing, leaking, and security. 3. Inspect electrical wiring for fraying, chafing, and security. 4. Check for proper oil level.		
412-A-71-00-00-00A-009A-A / 01584	**REDUCTION GEARBOX** 1. Inspect hoses and lines for chafing, leaking, and security. 2. Check oil filter impending bypass indicator button - not extended. 3. Check for proper oil level.		
412-A-71-00-00-00A-009A-A / 01584	**ENGINE AND REDUCTION GEARBOX MOUNTS** Inspect mounts for loose bearings and security.		
412-A-71-00-00-00A-009A-A / 01584	**ENGINE FIREWALLS** Visually inspect firewalls for cracks, distortion, missing rivets, broken spot welds, and deteriorating seals or gaskets.		
412-A-71-00-00-00A-009A-A / 01584	**ENGINE AIR MANAGEMENT SYSTEM** 1. Visually inspect exhaust ejectors and ducts for damage and security. 2. Visually inspect air ducts and plenum for condition, obstruction, and security.		
412-A-79-00-00-00A-009A-A / 01765	**ENGINE OIL SYSTEM** 1. Visually inspect oil coolers for leaks, damage and obstructions. 2. Visually inspect engine accessory and reduction gearboxes for oil leaks.		
412-A-95-00-00-00A-009A-A / 01787	**INSTRUMENTS** 1. Visually inspect instrument panel for cleanliness.		

Table 1. Daily Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2. Visually inspect all instruments, placards, decals, and markings for appearance and legibility.		
	3. Check magnetic compass for condition and security.		
	4. Visually inspect compass cards for validity.		
412-A-96-00-00-00A-009A-A / 01870	**ELECTRICAL SYSTEM**		
	1. Inspect electrical equipment in nose compartment for condition and security.		
	2. Inspect pedestal mounted avionics/electrical equipment for condition, cleanliness, and security.		
	3. Inspect overhead console for condition, cleanliness, and security.		
	4. Inspect all instrument lights, integrally lit panels, secondary lights, and map lights serviceability.		
	5. Inspect all caution and warning lights for proper operation by using master caution, fire test, and baggage compartment smoke detector press-to-test functions.		
	6. Check landing and search lights for condition and security.		
	7. Navigation and anticollission lights for condition and security.		
412-A-96-00-00-00A-009A-A / 01870	**BATTERY SYSTEM**		
	1. Visually inspect battery and external connections for condition and security.		
	2. Visually inspect battery vent and drain tubes for obstruction and security.		
412-A-97-00-00-00A-009A-A / 02235	**ANTENNAS**		
	Visually inspect all antennas located on fuselage and tailboom for condition and security.		
412-A-97-00-00-00A-009A-A / 02235	**AVIONICS EQUIPMENT**		
	Visually inspect all avionics located in fuselage and tailboom for condition and security.		

Scheduled Inspections - 100 Hour/12 Month Inspection - Part A

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
MEK	C-309	AR	
Sealant	C-308	AR	
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	
Hydraulic fluid	C-002	AR	
Hydraulic fluid	C-072	AR	

Procedure

Table 1. 100 Hour/12 Month Inspection - Part A

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME:_____		
	SIGNATURE:_____		
	NOTE For helicopters on the Part A inspection program, accomplish each 100 hours of flight operation or after 12 calendar months, whichever comes first.		
	GENERAL		
	1. Each listed inspection item or maintenance function is to be performed in accordance with the BHT-412-MM chapter specified or the BHT-412-CR&O.		
	2. Refer to the necessary Pratt & Whitney Canada Engine Maintenance Manual for engine inspection requirements.		
	3. Record all work accomplished during inspection in the helicopter maintenance record.		
	4. Check the helicopter records for recorded discrepancies.		
412-A-05-40-00-00A-281A-A / 00022	5. Accomplish complete Daily Inspection - Part A.		
412-A-05-47-00-00A-042A-A / 00030	6. Review the special inspections and perform any special inspection as necessary.		
412-A-04-00-00-00A-288A-A / 00013	7. Replace all finite life components that have completed published operating limitations.		
412-A-05-11-00-00A-042A-A / 00070	8. Overhaul all components that have completed published overhaul periods.		
412-A-12-01-00-00A-040A-A / 00141	9. Perform the lubrication requirements.		
	10. Remove all panels, fairings, and cowlings as necessary to satisfactorily carry out the inspection.		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	NOTE The following step is to be performed every 12 calendar months.		
	11. Inspect all fuel system, oil system, and hydraulic system filler caps for proper functioning and sealing. Make sure that the sealing O-rings within the filler caps are in good condition. Repair or replace the filler caps or replace sealing O-rings as necessary.		
	AFTER COMPLETION OF INSPECTION		
	1. Reinstall all panels, fairings, and cowlings removed during inspection.		
	2. Carry out a ground run and leak check prior to releasing the helicopter back to service.		
412-A-21-00-00-00A-009A-A / 00189	**HEAT/VENT AIR DUCTS**		
	1. Visually inspect all heating, ventilation, cooling ducts, and controls for cracks, security, and for proper operation.		
	Every sixth 100 hour inspection (600 hours) or 12 months:		
	1. Bleed air heating and ventilation/defog system components:		
412-A-96-06-00-01A-320A-A / 02141 or 412-A-96-06-00-01A-320B-A / 02142	1.1. Perform the operational check of bleed air heating system and components.		
412-A-96-06-00-02A-320A-A / 02146	1.2. Perform the operational check of defog blower.		
412-A-25-00-00-00A-009A-A / 00244	**MISCELLANEOUS FURNISHINGS**		
	1. Check all safety equipment for inspection due dates and operation.		
412-A-26-00-00-00A-009A-A / 00291	**FIRE PROTECTION**		
	1. Make sure that the fire extinguishers are properly charged.		
	2. Baggage compartment smoke detector for condition and security.		
	Every sixth 100 hour inspection (600 hours) or 12 months:		
412-A-26-20-00-03A-280A-A / 00320	1. Weight check crew and passenger cabin portable fire extinguishers.		
ASB 412-12-152	2. Visually inspect the loop clamp installed on modified discharge tubes 212-060-921-001FM and -002FM for condition and security until Part III of referenced ASB is accomplished.		
412-A-96-07-00-02A-320A-A / 02151	3. Perform the operational check of engine fire extinguishing system.		
412-A-26-10-00-03A-320A-A / 00307	4. Perform the operational check of baggage compartment smoke detector.		
412-A-04-00-00-00A-288A-A / 00013	5. Replace the engine fire extinguisher container firing cartridges in accordance with specified service life.		
412-A-28-00-00-00A-009A-A / 00323	**FUEL SYSTEM**		
	1. Inspect all exposed fuel lines and connections for leakage, damage, and security.		
	Every sixth 100 hour inspection (600 hours) or 12 months:		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	2. Remove the four access doors, located just forward of the aft crosstube, from the underside of the fuselage. Inspect the fuel lines for evidence of leakage, damage, and security.		
412-A-29-00-00-00A-009A-A / 00610	**HYDRAULIC SYSTEMS**		
	1. Inspect all lines and hoses for security and general condition.		
	Every sixth 100 hour inspection (600 hours) or 12 months:		
	NOTE Hydraulic filter elements 205-076-034-003 are non-cleanable. It is recommended they be replaced every 600 hours or 12 months. Filter elements 205-076-034-007 are cleanable. The 212-076-006-007 assembly must have -007 element only. The -003 and -007 elements are interchangeable on the assembly 212-076-006-105; intermixing of different element types is not permitted.		
	1. Remove and inspect the hydraulic filter elements.		
	2. Discard or clean the filter elements as necessary.		
	3. Install the hydraulic filter elements.		
412-A-63-00-00-00A-009A-A / 00987	4. Install 0 to 500 PSI (0 to 3447 kPa) gauge in quick disconnect at rotor brake caliper. Slowly pull rotor brake handle through travel. Cylinder should generate 240 to 260 PSI (1655 to 1793 kPa) at any handle velocity and shall maintain 150 PSI (1034 kPa) minimum in over center position for 3 minutes. Adjust as necessary.		
	5. Inspect the brake linings for wear (minimum thickness 0.150 inch (3.81mm)).		
412-A-32-00-00-00A-009A-A / 00694	**LANDING GEAR**		
	1. Inspect landing gear crosstube assemblies, skid tubes, and skid shoes, for condition and security of attachment. Inspect the crosstube retention cap (rubber bumper pads) for condition and security of attachment.		
	2. Aft crosstube assemblies support beam for cracks, damage, corrosion, and security of attachment.		
	3. Crosstube to skid tube saddle bolts for tightness.		
	4. Torque check the forward and aft cross tube support fittings U-bolts to 80 to 100 inch-pounds (9.04 to 11.29 Nm).		
	5. Torque check the ground handling wheel attachment eyebolts on the skid tubes 40 to 58 foot-pounds (54 to 79 Nm). Eyebolts that rotate prior to reaching 40 foot-pounds (54 Nm) shall be removed and inspected for evidence of bending. No bending is permitted.		
	Every third 100 hour inspection (300 hours) or 12 months:		
BHT-412-SI-58	6. If gross weight towing kit provisions are installed, inspect the fitting assembly for condition and security. Torque check the attachment bolts 160 to 190 inch-pounds (18.1 to 21.4 Nm). If the bolts are found loose, inspect for condition.		
412-A-52-00-00-00A-009A-A / 00774	**DOORS AND WINDOWS**		
	1. Crew and passenger doors structure for corrosion, damage, distortion, and positive locking mechanisms. Seals for adherence, tears, separations, and deterioration.		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	Every third 100 hour inspection (300 hours) or 12 months:		
	2. Crew door emergency jettison mechanism for condition and security. Perform the operational check.		
	3. Passenger door window retainers and fillers for damage.		
	4. Nose, electrical, and equipment compartment access doors for corrosion, damage, distortion, and positive locking mechanisms.		
	5. Baggage compartment door for corrosion, damage, and positive locking.		
	6. Inspect heated windshield, if installed, for condition and proper operation.		
412-A-53-00-00-00A-009A-A / 00897	**FUSELAGE**		
	1. Inspect fuselage tailboom attachment points for evidence of fretting.		
	2. With all panels removed, inspect the pylon side walls in the area of each opening for mechanical or corrosion damage and cracks. Pay attention around fastener hole where cracks are susceptible to develop.		
	3. Examine the top and bottom areas of the left and right pylon support channels (205-030-919) (1, Figure 1) with a bright light and mirror for loose fasteners and cracks.		
	4. Inspect transmission mounts, mounting brackets, and structure for cracked or broken parts.		
	Every third 100 hour inspection (300 hours) or every 12 months:		
412-A-63-21-04-01A-280A-A / 01189	*1. Inspect the aft transmission mount friction dampers for condition and security.*		
	2. Inspect lift link and fitting for cracks and security.		
	3. Cabin interior and exterior for corrosion and damage.		
	4. All compartments for evidence of water entrapment and corrosion.		
	5. Ensure all drain holes are open.		
	6. Inspect underside of fuselage:		
	6.1. Fuselage structure for damage, corrosion, and working rivets.		
	6.2. Exterior finish for condition and cleanliness.		
	6.3. Evidence of excessive fluid leakage.		
	6.4. Structure around landing gear for condition.		
	6.5. Forward crosstube support structure for working fasteners, cracks, corrosion, and excessive wear.		
	6.6. Aft crosstube support structure for working fasteners, cracks, corrosion, and excessive wear.		
	7. Fuselage bonded panels for damage and delamination.		
	8. Inspect cabin roof structure:		
	8.1. Cabin roof structure, cowlings, and fairing for damage, delamination, and general condition.		
	8.2. Cabin roof and cowling/fairing mounted antennas for condition and security.		
	Every sixth 100 hour inspection (600 hours) or 12 months:		
	1. Inspect fuselage cabin structure:		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	1.1. Fuselage structure for corrosion and damage to finish and sealant. Pay particular attention to edges of joints and seams and around fastener heads.		
	1.2. Engine compartment floor and service deck, FS 155.06 to 241.22, for damage, delamination, and corrosion.		
	NOTE To adequately inspect the following, it will be necessary to remove the auxiliary fuel tanks if installed.		
	2. Pylon wall for damage, working fasteners, cracks, delamination and corrosion. With all panels removed, inspect the pylon side walls in the area of each opening for mechanical or corrosion damage and cracks. Pay attention around fastener holes where cracks are susceptible to develop.		
412-A-53-00-00-00A-009A-A / 00897	**TAILBOOM**		
	1. Inspect tailboom attachment points for cracks and security.		
	2. Inspect tailboom exterior and interior structure for corrosion and damage.		
	3. Ensure that all the drain holes are open.		
412-A-65-00-00-00A-009A-A / 01257	4. Torque check the tail rotor gearbox mounting nuts by applying torque in tightening direction, do not exceed 200 inch-pounds (22.6 Nm). If any nut rotates under torque, inspect attachment hardware and tail rotor gearbox for serviceability.		
	5. Torque check the intermediate gearbox mounting bolts by applying torque in tightening direction, do not exceed 50 inch-pounds (5.65 Nm). If any bolt turns under torque, inspect the attachment hardware and intermediate gearbox for serviceability.		
412-A-67-00-00-00A-009A-A / 01319	6. Inspect elevator attachment lugs at each end of elevator horn for cracks and security of attachment. Check the elevator horn to elevator spar attaching bolts and surrounding joint for cracks and security of the attachment.		
412-A-53-00-00-00A-009A-A / 00897	7. Inspect vertical fin spar caps for cracks and corrosion.		
	NOTE If necessary, clean the area with cloth dampened with MEK (C-309) or equivalent. Ventilate the area to prevent breathing fumes.		
	8. Inspect vertical fin spar caps forward side and web from upper tailboom skin to approximately 4.0 inches (101.6 mm) below upper tailboom skin as follows: Remove the aft tailboom access door. Face aft and use bright light and small mirror to inspect area for cracks, especially near rivets holes.		
	NOTE Step 9 and step 10 pertain to helicopters modified with the BLR FastFin™ system kit 412-705-040-101. For helicopters equipped with the Dual Tailboom Strakes and FastFin™ system Supplemental Type Certificate (STC) , refer to the necessary BLR Aerospace documents for inspection criteria.		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	9. Inspect upper and lower tail fairings for visible damage to the exterior skin and cracks at the attachment points. If paint is missing locally or has scratches, inspect the fairing for delamination and voids with a tap hammer. Inspect fairings for loose or missing attaching hardware.		
	10. Inspect upper and lower strakes for visible damage, deformation, cracks at the attachment points, and loose or missing rivets.		
	Every third 100 hour inspection (300 hours) or 12 months:		
	1. Inspect tailboom joints, splices, longerons, attach fittings, and attaching hardware for corrosion, damage, and cracks.		
ASB 412-90-49	2. Remove plug button at BS 99.00.		
	NOTE Paint fissures are common in the splice area, cracks will be evident by black powder emanating from cracked area, and corrosion will leave evidence of white powder.		
	3. Using a borescope or other suitable means, inspect inside of longeron paying particular attention to cracking and/or corrosion in area of splice.		
	4. Report any cracks to the Product Support Engineering.		
	5. Install the plug button with sealant (C-308).		
412-A-67-00-00-00A-009A-A / 01319	6. Disconnect both ends of control tube 212-001-055-101 from bellcranks 212-001-705-001 and 212-001-759-101.		
ASB 412-15-166	a. Without removing the control tube 212-001-055-101, visually inspect the control tube for cracks. No cracks are permitted.		
	b.On bellcranks 212-001-705-001 and 212-001-759-101, inspect bearings DAS-14A1-512 for freedom of movement, bearings shall move freely in all directions and rotate freely. If a defective bearing is found, replace defective bearing in accordance with BHT-ALL-SPM, Standard Practices Manual.		
412-A-62-00-00-00A-009A-A / 00932	**MAIN ROTOR BLADES**		
	1. Wash the main rotor blades.		
	2. Inspect the main rotor blades' upper and lower surfaces for condition of bond lines and doublers. Leading edge for erosion. Blade surfaces for corrosion, cracks, damage, and voids.		
	3. Inspect the blade retention bolts for condition and security.		
	NOTE Waxing is not necessary for blades 412-015-300-109 and Subsequent or for any blade finished with polyurethane paint.		
	4. Apply a coat of wax to main rotor blades.		
412-A-62-00-00-00A-009A-A / 00932	**MAIN ROTOR HUB**		
	1. Inspect the hub assembly for condition, corrosion, and security, paying particular attention to the integrity of sealing.		
	2. Inspect the pitch horns for corrosion and cracks in areas surrounding torque transfer pin holes. No cracks permitted.		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	3. Inspect the pivot bearings, spindle bearings, and damper set for elastomeric bearing delamination, elastomer degradation, shim cracks, crazing, cracking, or sheeting. 4. Inspect the damper bearing bumper pads for deterioration or deformation due to contact with yoke. 5. Torque check the eight yoke assembly through bolt nuts to 70 to 80 foot-pounds (95 to 108 Nm). 6. Make sure bolts EWB0420 are installed (16 places) for simple pendulum absorber bracket. Torque check bolts. Apply 29 to 33 foot-pounds (39 to 45 Nm). 7. Functionally check the arm and bearing (pendulum) assemblies. Apply a minimum of 5 pounds (22.24 N) of force to each arm and bearing assembly (force toward center of yoke). Cycle arm and bearing through full travel. 8. Apply force away from the center of yoke. Arm and bearing assembly should cycle free and smooth. Roughness and binding are not acceptable. 9. Functionally check the droop restraint system. Check for engagement of restraint arms in lower detent of cam window. Relieve load on each blade and verify lower clevis moves freely through cam window envelope. Binding and roughness in inboard clevis is not acceptable.		
412-A-62-00-00-00A-009A-A / 00932	**MAIN ROTOR CONTROLS** **NOTE** Disconnect the pitch link upper rod ends from the pitch horns to determine main rotor control bearing wear. 1. Swashplate and support assembly: 1.1. Inspect for condition and security with special attention to the gimbal ring attachment lugs. **NOTE** The presence of black oxide powder will require investigation to determine the cause. 1.2. Inspect the gimbal ring bearings, liners, and attaching hardware for excessive looseness and wear. Gimble ring bearing and attaching bolt axial or radial looseness shall not exceed 0.005 inch (0.13 mm) with no binding allowed. 1.3. Inspect the trunnion bearings and attaching hardware for excessive looseness and wear. Trunnion bearings shall not exceed 0.020 inch (0.51 mm) axial looseness. 2. Hub and sleeve assembly: 2.1. Inspect for condition and security. 2.2. Inspect the drive hub and rephasing levers for wear and bearing looseness. 2.3. Drive the links spherical bearings shall not exceed 0.007 inch (0.18 mm) radial or 0.015 inch (0.38 mm) axial looseness. 2.4. Swashplate link non-elastomeric bearing looseness shall not exceed 0.015 inch (0.38 mm) axial.		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	3. Inspect the collective lever for condition and security. Trunnion bearing shall not exceed 0.020 inch (0.51 mm) axial looseness.		
	4. Inspect the pitch links for condition and security. Non-elastomeric bearing looseness shall not exceed 0.007 inch (0.18 mm) radial or 0.015 inch (0.38 mm) axial.		
	Every third 100 hour inspection (300 hours) or 12 months:		
	1. Torque check the swashplate link elastomeric rod end bearings (as necessary) attachments bolts to 165 to 200 inch-pounds (18.65 to 22.59 Nm).		
	2. Torque check the swashplate outer ring through bolts to 165 to 175 inch-pounds (18.65 to 19.77 Nm).		
	3. Torque check the four hub and sleeve to rephasing lever attachment bolts to 24 to 34 foot-pounds (33 to 46 Nm) for 1/2 inch (13 mm) bolts or 160 to 200 inch-pounds (18.08 to 22.59 Nm) for 3/8 inch bolts (if installed).		
	4. Torque check the drive link to rephasing lever attachment bolts to 165 to 200 inch-bolts (18.65 to 22.59 Nm).		
	5. Torque check the pitch link elastomeric rod end bearings (as necessary) attachment bolts to 165 to 200 inch-pounds (18.65 to 22.59 Nm).		
	6. Torque check the swashplate support to mast plate bolts (AN5H-6A) to 130 to 140 inch-pounds (14.69 to 15.81 Nm).		
412-A-63-00-00-00A-009A-A / 00987	**MAIN ROTOR MAST**		
	1. Inspect the mast assembly for security, corrosion, and mechanical damage.		
	2. Inspect the mast torque sensor (if installed) installation for security and condition		
	3. Inspect for evidence of oil leaks at the mast bearing cap.		
412-A-63-00-00-00A-009A-A / 00987	**TRANSMISSION**		
	1. Inspect the mounts for damage and security. Check the boots for condition and make sure that the boots are in place.		
	2. Inspect the lift link and attachments for corrosion, damage, and security. Check the bearings for looseness.		
	3. Inspect for evidence of oil leakage.		
	4. Check all transmission chip detectors for debris and clean.		
	5. Test all transmission chip detectors electrical circuits.		
	6. Check the torque of the brake disc attachment bolts 50 to 70 inch-pounds (5.65 to 7.90 Nm).		
	Every third 100 hour inspection (300 hours) or 12 months:		
	1. Check the internal oil filter or full flow oil monitor for debris and clean.		
	2. Torque check the top case to ring gear case nuts, ring gear case to main case nuts, and main case to support case nuts to 230 to 250 inch-pounds (25.99 to 28.25 Nm). Torque as necessary.		
412-A-64-00-00-00A-009A-A / 01225	**TAIL ROTOR HUB AND BLADE ASSEMBLY**		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	 **COUNTERWEIGHT BELLCRANK RETENTION NUT MS14145L6 OR ITS EQUIVALENT MS17826-6 IS DESIGNATED AS ONE TIME USE ONLY. IF MAINTENANCE ACTION REQUIRES THAT IT BE REMOVED, THE COUNTERWEIGHT BELLCRANK NUT IS TO BE DISCARDED AND REPLACED WITH A NEW NUT. OVERTORQUING OR INCORRECT APPLICATION OF THE NUT COULD CAUSE MECHANICAL FAILURE RESULTING IN EXCESSIVE VIBRATION OR DAMAGE TO THE TAIL ROTOR BLADES.** 1. Inspect the tail rotor counterweight bellcrank 212-011-705-001 retention nuts for cracks, corrosion, and security. Reapply corrosion preventive compound (C-101) if nuts are not completely covered.		
	Every third 100 hour inspection (300 hours) or 12 months:		
412-A-18-00-00-00A-009A-A / 00144	1. Dynamically balance tail rotor.		
	2. Torque check the nuts on tail rotor blade retention bolts.		
	3. Torque check the tail rotor hub retaining-nut.		
412-A-65-00-00-00A-009A-A / 01257	**TAIL ROTOR GEARBOX**		
	1. Inspect the chip detector for debris. If metallic particles are found, determine and correct cause.		
	2. Clean the chip detector.		
	3. Test the chip detector electrical circuit.		
412-A-65-00-00-00A-009A-A / 01257	**TAIL ROTOR DRIVESHAFT**		
	1. Hanger assemblies for cracks at mounting ears, corrosion, and security of attachment.		
	2. Hanger supports for condition and security of attachment.		
	3. Driveshaft sections and attaching hardware for condition and security.		
	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *4. Do the following check:*		
	4.1. Condition of the clamps for tail rotor driveshaft coupling to gearbox couplings (90° apart).		
	4.2. Inspect the clamps for cracks in or near the bolt hole lugs.		
	(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *5. Check disc pack couplings and attaching hardware for condition and security.*		
412-A-65-00-00-00A-009A-A / 01257	**INTERMEDIATE GEARBOX**		
	1. Inspect the chip detector for debris. If metallic particles are found, determine and correct cause.		
	2. Clean the chip detector.		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	3. Test the chip detector electrical circuit.		
412-A-67-00-00-00A-009A-A / 01319	**FLIGHT CONTROLS** 1. Inspect all control tube assemblies for clearance, security, and general condition. 2. Inspect the tube bellcranks, supports, and attaching hardware for corrosion, security, wear, and damage. 3. Collective and cyclic flight control actuators, check: 3.1. Universal bearings for looseness. 3.2. Input lever bearings and bolts for wear and looseness. 3.3. Fasteners attaching cylinder lower supports to structure for looseness. 3.4. Cylinder lower bearings for looseness. 3.5. Actuator assemblies for condition, leakage, and security. 3.6. Cylinder extension tubes for condition and security. 3.7. Upper cylinder housing mounting bracket for condition and security. Check mounting bracket cavity for corrosion. Cavity drain hole for obstruction. Dust boot for condition. Reapply corrosion preventive compound (C-104) as necessary. 3.8. Actuator linkage for wear and security. 3.9. Clean the exposed area of actuator piston with hydraulic fluid (C-002) or hydraulic fluid (C-072) and lint-free cloth. 3.10. Check the cyclic and collective hydraulic cylinders' piston rods for evidence of excessive wear and scoring. Wear of the piston rods indicates cylinder assembly is incorrectly aligned to lower supports and requires adjustment. 4. Antitorque controls check: 4.1. Inspect the tail rotor control tubes, bellcranks, supports, and attaching hardware between tail rotor pedals and tail rotor gearbox for corrosion, wear, and damage. 4.2. Inspect the tail rotor hydraulic actuator for leaks, security of attachment, damage, corrosion, wear, bearings, and linkages for looseness. **Every third 100 hour inspection (300 hours) or 12 months:** 1. Collective and cyclic flight control actuators: 1.1. Torque check the nuts attaching actuator cylinder to upper supports. 1.2. Remove the bolts attaching actuator to lower supports. Check the cylinder alignment to lower support. Rod end bearing side loads are not permitted. Reinstall the bolts. 2. Aerodynamically actuated elevator check: 2.1. Inspect the elevator control tube between forward support bracket and elevator horn for corrosion, wear, and damage. 2.2. Inspect the elevator control tube support and attaching hardware for security, corrosion, wear, and damage. 2.3. Inspect the elevators and support structure for loose or missing rivets, cracks in skin, and corrosion. 2.4. Inspect the trailing edge tabs and tip caps for condition and security.		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.5. Functionally check the operation of control tube as follows:  **DO NOT RELEASE ELEVATOR AND ALLOW THE SPRING TO DRIVE ELEVATOR BACK TO ORIGINAL POSITION.** 2.5.1. Move the elevator to extreme nose down position and while holding allow to slowly return to original position. Failure to return to original position, binding and/or lock-up are not acceptable. 2.5.2. If any of these conditions exist, replace the tube. 2.6. Remove the elevator control and measure spring breakout force for 90 to 100 pounds (400.34 to 444.80 N). 2.7. Functionally check the elevator rigging for proper travel. Measure the trailing edge for 16° ± 1/4° angle down from upper surface of elevator. **Every sixth 100 hour inspection (600 hours) or 12 months:** 1. Inspect the collective flight controls: 1.1. Flight control systems bellcrank and control tubes for damage, chafing, corrosion, and security of attachment. 1.2. Friction shoes and liners for condition. 1.3. Check the collective lever friction adjuster for operation. 1.4. Check for proper collective minimum friction. 1.5. Check the collective flight controls for smooth movement throughout full range of travel. 2. Inspect the flight controls: 2.1. Flight control systems bellcrank and control tubes for damage, chafing, corrosion, and security of attachment. 2.2. Check the cyclic stick friction adjuster for proper operation. 2.3. Check for the proper cyclic minimum friction. 2.4. Check the cyclic flight controls for smooth movement throughout full range of travel. 3. Inspect the anti-torque flight controls: 3.1. Flight control systems bellcrank and control tubes for damage, chafing, corrosion, and security of attachment. 3.2. Check the anti-torque friction adjuster for proper friction. 3.3. Check the anti-torque flight controls for smooth movement throughout full range of travel. 4. Aerodynamically actuated elevator: 4.1. Remove the left and right elevators. 4.2. Clean the elevator spars and inspect for corrosion. 4.3. Clean the internal bore of elevator horn and inspect for corrosion. 4.4. With elevator control removed, attach a spring scale to hole in arm of horn where control tube is normally installed. Pull the scale at right angle to arm, verify the friction is 26 to 32 pounds (115.65 to 142.34 N). If friction is not within limits, adjust the shim thickness. 4.5. Install the left and right elevators.		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

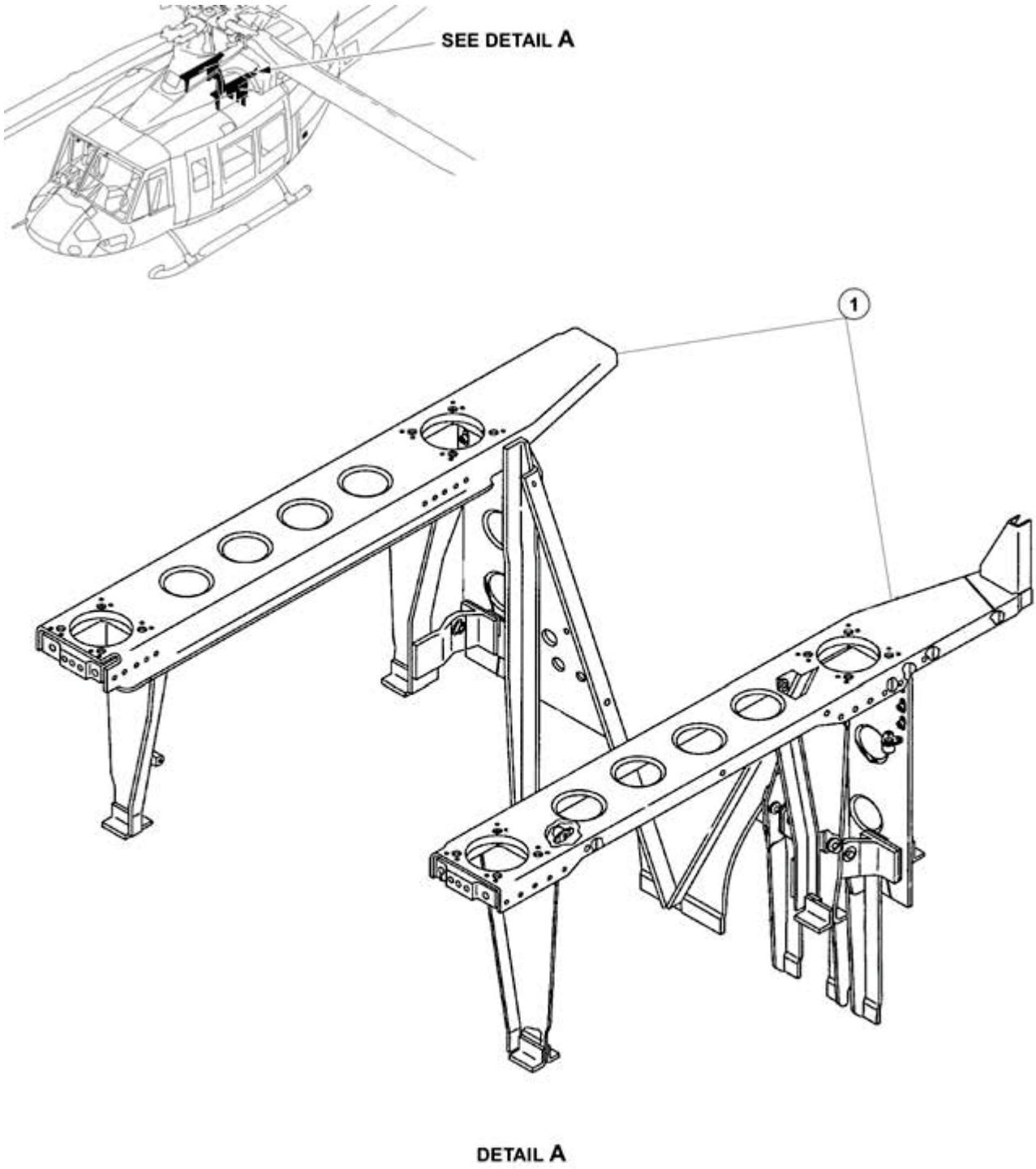
DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-71-00-00-00A-009A-A / 01584	**POWER PLANT** **Every third 100 hour inspection (300 hours) or 12 months:** **NOTE** Refer to the necessary Pratt & Whitney Canada Engine Maintenance Manual for engine inspection requirements. 1. No. 1 (left) engine power section: 1.1. Chip detectors for debris. 1.2. Clean the chip detectors. 1.3. Test the chip detector electrical circuits. 2. No. 2 (right) engine power section: 2.1. Chip detectors for debris. 2.2. Clean the chip detectors. 2.3. Test the chip detector electrical circuits. 3. Reduction (combining) gearbox: 3.1. Chip detectors for debris. 3.2. Clean the chip detectors. 3.3. Test the chip detector electrical circuits. 4. Engine electrical connections: 4.1. Power section 1 ignition leads for corrosion, chafing, and security. 4.2. Power section 2 ignition leads for corrosion, chafing, and security. 5. Starter generator: 5.1. Power section 1 starter generator brushes allowable wear. 5.2. Power section 2 starter generator brushes allowable wear. 5.3. Power section 1 starter generator cooling ducts for obstruction, kinking, and security. 5.4. Power section 2 starter generator cooling ducts for obstruction, kinking, and security. 6. Engine fire detection system elements for condition and security. **NOTE** If necessary, refer to BHT-412-SI-87 for installation of adjustable seals or refer to 412-A-71-10-00-02A-271A-A / 01613 for seal adjustment.		
412-A-71-00-00-00A-009A-A / 01584 ASB 412-10-140	7. Inspect the left-hand and right-hand engine air inlet cowlings for gaps between the lower surface of the cowling firewall and the mating horizontal engine firewall.		
412-A-76-00-00-00A-009A-A / 01722	**ENGINE CONTROLS** 1. Check the controls for smooth movement through full range of travel. **Every third 100 hour inspection (300 hours) or 12 months:** 1. Inspect the engine fuel and power controls:		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	1.1. Engine control linkages for looseness, lost motion, chafing, damage, and security of attachment.		
	1.2. Bellcranks, mounts, and jackshafts for looseness, damage, and security of attachment.		
BHT-412-FM	1.3. Perform the engine fuel control check (not applicable to PT6T-9 engine).		
	Every sixth 100 hour inspection (600 hours) or 12 months:		
	NOTE Maximum N_G topping check is not necessary to Pratt & Whitney PT6T-9 engine.		
Pratt & Whitney Canada PT6T-3 Series Maintenance Manual	1. Perform the maximum N_G topping check.		
412-A-96-00-00-00A-009A-A / 01870	**ELECTRICAL SYSTEM**		
	1. Check the inverters for security of mounting and connections.		
	2. Check the generator control units for damage and security of mounting and terminals.		
	3. Inspect all exposed wire bundles, bundle supports, and connectors for damage, corrosion, and security. Pay particular attention to the engine deck connectors and harnesses.		
	4. Inspect the relays and main bus area for security of mounting and connections.		
	5. Inspect the bus insulation for damage and condition.		
412-A-95-00-00-00A-009A-A / 01787	**INSTRUMENT SYSTEMS**		
	NOTE Following checks are necessary every 12 months only.		
	1. Calibrate the pilot, copilot compass systems, and standby magnetic compass.		
	2. Drain the pitot-static system of any accumulated moisture.		
	3. Test the pitot-static system for leaks.		
412-A-96-00-00-00A-009A-A / 01870	**BATTERY SYSTEM**		
	1. Service the battery in accordance with battery manufacturer's recommendation.		
	2. Inspect the battery compartment for general condition.		
	3. Check the battery mount for security and corrosion.		
	4. Perform the functional test of battery temperature sensor caution light.		
412-A-96-00-00-00A-009A-A / 01870	**EMERGENCY BUS AND POWER DIODES**		
	Every third 100 hour inspection (300 hours) or every 12 months:		
	1. Perform the operational check of emergency bus system as follows:		

Table 1. 100 Hour/12 Month Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	NOTE Helicopters S/N 33001 through 33107, except those modified for CAA, remove screw, spacer, and washer installed in the emergency load switch guard. 1.1. Inspect EMERG BUS INTC circuit breaker for wear. Operate the circuit breaker manually and check by feel. Worn or loose circuit breaker is not acceptable. Close circuit breaker. 1.2. Place all switches in OFF or NORM position and close all circuit breakers. 1.3. Connect the battery. 1.4. Place BATTERY BUS 1 switch and BATTERY BUS 2 switch in the ON position. Position INV 1 and INV 2 switches in the ON position. Caution panel INVERTER 1 and INVERTER 2 segments should not be illuminated. 1.5. Check the emergency bus power by pressing the FIRE EXT PRESS TO TEST button momentarily. Observe both FIRE PULL handle warning lights illuminate. 1.6. Position emergency bus switch to EMERG LOAD position. Check emergency bus power by pressing the FIRE EXT PRESS TO TEST button momentarily. Observe both FIRE PULL handles illuminate. Verify INVERTER 1 and INVERTER 2 segments are illuminated on the caution panel. **NOTE** S/N 33108 through 33213 and 36001 through 36999 — Check INVERTER 1 light segment on the caution panel is not illuminated and INVERTER 2 segment is illuminated. 1.7. Position BATTERY BUS 1 and BATTERY BUS 2 switches off. Check emergency bus power by pressing FIRE EXT PRESS TO TEST button momentarily. Observe FIRE PULL handles illuminate. 1.8. Position INVERTER 1 and INVERTER 2 switches OFF. Position emergency bus switch to NORMAL. Disconnect battery. 2. Install the screw, spacer, and washer in the emergency load switch guard if removed. 3. Perform the functional check of power diodes.		



1. Pylon Support Channel (205-030-919)

ICN-412-A-054000-B-97499-02877-A-001-01

Figure 1. Pylon Support Channel - Scheduled Inspection (Sheet 1 of 1)

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Scheduled Inspections - 1000 Hour Inspection - Part A

Procedure

Table 1. 1000 Hour Inspection - Part A

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ ___ HELICOPTER S/N: _____ ___ REGISTRY NO.: _____ ___ TOTAL TIME:_____ ___ SIGNATURE:_____		
	NOTE For helicopters on the Part A inspection program, accomplish each 1000 hours of flight operation.		
	GENERAL		
	1. Each listed inspection item or maintenance function is to be performed in accordance with the BHT-412-MM chapter specified or the BHT-412-CR&O.		
	2. Record all work accomplished during inspection in the helicopter maintenance record.		
412-A-05-40-00-01A-281A-A / 00023	3. Accomplish complete 100 Hour/12 Month Inspection - Part A.		
412-A-62-00-00-00A-009A-A / 00932	**SWASHPLATE AND SUPPORT ASSEMBLY - HUB AND SLEEVE ASSEMBLY**		
	1. Disconnect drive and swashplate links from swashplate outer ring, and drive plate set from hub assembly.		
	2. Disconnect pitch links from main rotor hub pitch horns.		
	3. Check swashplate duplex bearing and collective sleeve hub bearing set for roughness and ease of rotation.		
	4. Purge lubricate bearings.		
	5. Reconnect drive and swashplate links and tubes, and drive plate set to hub assembly.		

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Scheduled Inspections - 5000 Hour/5 Year Inspection - Part A

Procedure

Table 1. 5000 Hour/5 Year Inspection - Part A

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ ___ HELICOPTER S/N: _____ ___ REGISTRY NO.: _____ ___ TOTAL TIME:_____ ___ SIGNATURE:_____ ___		
	NOTE For helicopters on the Part A inspection program, accomplish each 5000 hours flight time or 60 months, whichever comes first. This inspection is meant to be a full visual inspection of the airframe primarily for environmental damage and corrosion.		
	GENERAL		
	1. Each listed inspection item or maintenance function is to be performed in accordance with the BHT-412-MM chapter specified or the BHT-412-CR&O.		
	PRELIMINARY REQUIREMENTS		
	1. All work accomplished during this inspection shall be recorded in the helicopter maintenance records.		
	2. Remove all panels, interior coverings, fairings, and cowlings required to satisfactorily carry out the inspection.		
	3. Remove the following items:		
	3.1. Main battery.		
	3.2. Electrical and avionics equipment from the nose area, as required, to allow a full visual inspection of the nose area to be done.		
	3.3. Main rotor transmission.		
	3.4. Intermediate gearbox.		
	3.5. Tail rotor gearbox.		
	3.6. Tailboom (to allow a full inspection of the aft fuselage bulkhead and forward tailboom bulkhead).		
	3.7. Landing gear assembly (to allow a complete inspection of the landing gear attachment points and support structure).		
412-A-71-61-00-01A-520A-A / 01701	*3.8. Air management system ejectors.*		
	4. In addition to the 5000 Hour/5 Year Inspection - Part A items, carry out the following:		

Table 1. 5000 Hour/5 Year Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
SN: 33001-33213 or 36001-36999: 412-A-05-40-00-00A-281A-A / 00022 SN: 37002-37999: 412-MMS-EPI SN: 38001-38999 or 39101-39999: 412-MMS-EPX	4.1. Complete Daily Inspection - Part A.		
SN: 33001-33213 or 36001-36999: 412-A-05-40-00-01A-281A-A / 00023 SN: 37002-37999: 412-MMS-EPI SN: 38001-38999 or 39101-39999: 412-MMS-EPX	4.2. Complete 100 Hour/12 Month Inspection - Part A.		
SN: 33001-33213 or 36001-36999: 412-A-05-40-00-02A-281A-A / 00024 SN: 37002-37999: 412-MMS-EPI SN: 38001-38999 or 39101-39999: 412-MMS-EPX	4.3. Complete 1000 Hour Inspection - Part A.		
	AFTER COMPLETION OF INSPECTION		
	1. Install the following items:		
	1.1. Main battery.		
	1.2. Electrical and avionics equipment on the nose area.		
	1.3. Main rotor transmission.		
	1.4. Intermediate gearbox.		
	1.5. Tail rotor gearbox.		
	1.6. Tailboom.		
	1.7. Landing gear assembly.		
412-A-71-61-00-01A-720A-A / 01704	*1.8. Air management system ejectors.*		
	2. Install all panels, interior coverings, fairings, and cowlings removed during inspection.		
	3. Do a ground run and leak check before you return the helicopter to service.		
412-A-53-00-00-00A-009A-A / 00897	**FORWARD FUSELAGE**		
	1. Nose panels in area of pitot tube attachment, areas inside and outside for disbonding and cracking.		
	2. Pitot tubes attachment for looseness.		

Table 1. 5000 Hour/5 Year Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	3. Nose shelves for damage, corrosion, and disbonding.		
	4. Nose shelf attachment locations for damage, corrosion, and cracking.		
	5. Right and left BL 14.00 main beams, forward of STA 23.00 for damage, corrosion, and cracking.		
	6. Instrument panel attachment structure for damage, corrosion, and cracking, primarily in the door frame attachment area.		
	7. Instrument panel glareshield for chafing and cracking.		
	8. Crew door hinge attachment structure for damage and cracks.		
	9. Outboard cabin post (1, Figure 1), cabin roof longeron (2), and windshield frame (5) upper corner area for damage and cracks.		
	10. Striker plates (3 and 4) attachment areas for damage and cracks.		
	11. Striker plates (3 and 4) for excessive wear, damage, and corrosion.		
	12. Cabin floor area under crew doors for chafing and damage.		
	13. Crew door hinges for wear, damage, corrosion, and security of attachment.		
	14. Crew doors for proper fit and operation.		
	15. Crew seat rails for excessive wear and damage.		
	16. Center and outboard bulkhead, below WL 22.00 at FS 23.00, for cracks, deformation, and corrosion along structure joints.		
	17. Collective and cyclic control jackshafts support intercostals for cracks and corrosion.		
	18. Cyclic support for corrosion and cracks.		
	19. Collective, cyclic, and antitorque control system bellcranks, levers, and support brackets for corrosion, cracks, and indication of wear at attaching points.		
	20. Crew seat support beams, at BL 14.00 and 30.00 between FS 23.00 and 74.30, for cracks and corrosion.		
	21. Forward crosstube attachment points and support structure at FS 63.33 and 74.30 for damage, corrosion, and cracks.		
	22. Aft crosstube attachment points and support structure at FS 155.06 and 166.00 for damage, corrosion, and cracks.		
	23. Crew and passenger cabin floor, FS 23.00 to 155.06, for corrosion and damage. Floor bonded panels for voids. Seat fasteners and cargo tie-down rings for condition. Area around pilot and copilot ICS foot switches for cracks.		
	NOTE In the following step, removal of the fuel cells is not required unless there is evidence of fuel leakage or damage to the fuel cells or structure is suspected.		

Table 1. 5000 Hour/5 Year Inspection - Part A (continued)

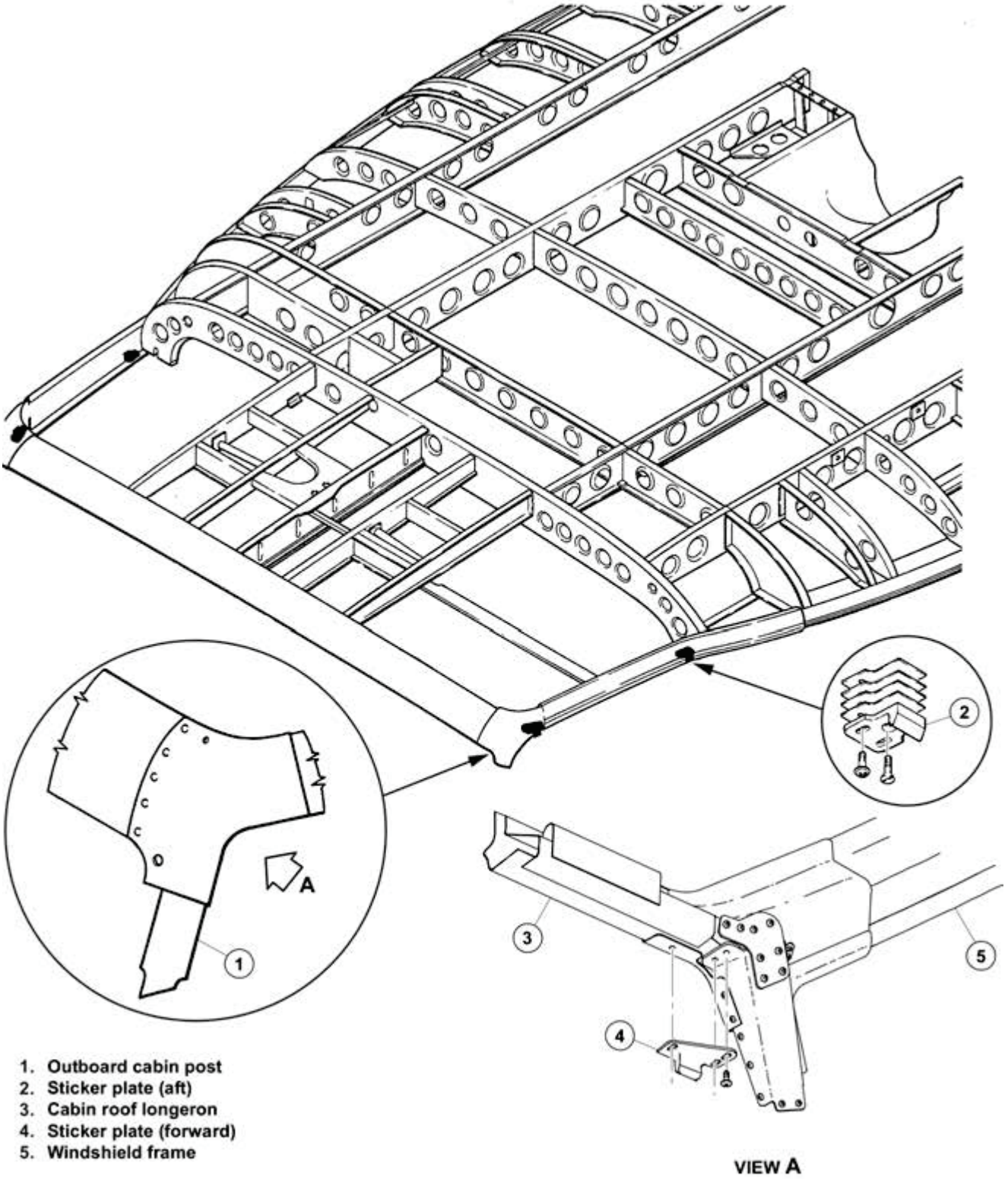
DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	24. All structure under floor panels and in forward fuel cell cavities including caps, angles, transverse bulkheads, and bonded panels between FS 74.30 and 155.06 for damage, corrosion, cracks, and voids. **NOTE** In the following step, removal of the fuel cells is not required unless there is evidence of fuel leakage or damage to the fuel cells or structure is suspected. Structure that is hidden by the fuel cells only needs to be inspected if the fuel cells are removed for some reason or structural damage is suspected. 25. All structure under floor panels and in aft fuel cell cavities including caps, angles, and panels between FS 155.06 and 200.00 for damage, corrosion, cracks, and disbonding. 26. Bottom skin, FS 23.00 to 243.937, for damage. Joints for corrosion. Access panels and covers for condition and attachment. 27. Main transmission compartment interior, FS 129.00 to 166.00 and WL 7.44 to 76.00, for corrosion. Bulkheads and main beam panels for damage and corrosion. Attaching supports and brackets for condition and security. 28. Fluid lines in transmission compartment for evidence of leaks, condition, wear, and chafing. Clamps for condition (fluid contamination and swelling) and security of attachment. 29. Lift beam fitting for condition and working rivets, lift link attachment bolt hole bores for excessive wear, and bushings for looseness. 30. Pylon (transmission) support structure for wear and cracks. Pay particular attention to right and left horizontal supports, forward vertical supports, and aft vertical webs. 31. Aft cabin bulkheads, FS 166.00, for damage, upper aft flange for cracks, and bulkhead panel for damage and disbonding. 32. Electrical/avionics compartment, FS 166.00 to 243.937, doors for condition and security. Door frames for cracks and corrosion. Compartment interior for condition and joints for corrosion. Skin for damage and cracks. 33. Oil cooler compartment, FS 166.00 to 243.937, doors for condition and security. Main beam panels and bulkheads for condition and corrosion. 34. Engine deck below the reduction gearbox, FS 211.00 to 238.00 for damage, cracks, and disbonding. *35. Inspect the ejectors shim AAL-238-011 (if installed) for damage, cracks, delamination and proper bonding to the firewall. If any damage is found, replace with new shim (412-A-71-61-00-01A-720A-A / 01704).* 36. Firewalls, FS 155.06 to 241.22, for damage, cracks, security of attachment, seals for condition, and seal attachment angles for cracks.		

Table 1. 5000 Hour/5 Year Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	37. Cabin roof, FS 35.00 to 166.00 above WL 68.00, BL 50.00 left and right side, for damaged skin and joints for corrosion, bonded panels for disbonding and corrosion. Structure above overhead console (circuit breaker panels) for cracks and corrosion.		
	38. Cargo door tracks for wear, cracks, and corrosion.		
	39. Engine and transmission cowlings and fairings for evidence of chafing, cracks, corrosion, and seals for condition.		
	40. Bulkhead, FS 241.22, for damage, corrosion, and cracks. Tailboom attachment fittings for damage, cracks, and loose fasteners. Tailboom attachment bolt holes for damage and corrosion.		
ASB 412-21-187	41. Discard the upper left-hand attachment bolts.		
	42. Do a detailed visual inspection of the remaining three attachment bolts for mechanical or corrosion damage. Any corrosion including surface corrosion is cause for rejection. Damaged threads and detectable wear or mechanical damage on the bolt shank or bolt head radii are cause for rejection.		
	43. Apply a light coat of grease (C-561) or grease (C-172) on the shanks of the bolts only. Both threads are to remain dry.		
412-A-53-00-00-00A-009A-A / 00897	44. Install tailboom after completion of inspection with a new upper left-hand bolt and serviceable bolts at the three other locations.		
412-A-53-00-00-00A-009A-A / 00897	**TAILBOOM**		
	1. Bulkhead, BS 17.42, for corrosion and cracks. Fuselage attachment bolt holes for damage and corrosion. Barrel nuts for damage and corrosion.		
	2. Tailboom and fin driveshaft covers for condition and attachment.		
	3. Tailboom and fin external skin for damage, corrosion, and cracks.		
	4. Tailboom access panels and covers for condition and attachment. Tail rotor driveshaft cover hinges for wear.		
	5. Tailboom baggage compartment door for condition, attachment, and door handle operation.		
	6. Tailboom baggage compartment interior for damage, panels for disbonding, and corrosion. Longerons above roof panel for corrosion and cracks. Longerons below floor panel for corrosion and cracks.		
	7. Tailboom interior bulkheads and stringers for cracks and corrosion.		
	8. Intermediate gearbox mount pad for damage, cracks, fretting, and corrosion.		
	9. Tail rotor gearbox support fitting assembly for damage, cracks, fretting, and corrosion. Pay particular attention to mounting holes and lugs.		
	10. Tailboom exterior bonded panels adjacent to baggage compartment for damage, delamination, and corrosion.		
	11. With a borescope or equivalent, inspect interior structure of vertical fin for cracks, corrosion, fretting, and damage.		

Table 1. 5000 Hour/5 Year Inspection - Part A (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	12. Control tube fairleads (grommets and guides) for wear and security of attachment.		



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Figure 1. Cabin Frame Inspection - Part A (Sheet 1 of 1)

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Scheduled Inspections - 25 Hour/30 Day Inspection - Part B

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

Table 1. 25 Hour/30 Day Inspection - Part B

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME:_____		
	SIGNATURE:_____		
	NOTE For helicopters on the Part B inspection program, accomplish the following checks each 25 hours of flight operation or 30 days, whichever occurs first.		
	GENERAL		
	1. Each listed inspection item or maintenance function is to be performed in accordance with the BHT-412-MM chapter specified or the BHT-412-CR&O.		
	2. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for engine inspection requirements.		
	PRELIMINARY REQUIREMENTS		
Corrosion Control Guide, CSSD-PSE-87-001.	1. Use medium helicopter corrosion control guide to establish helicopter corrosion control program.		
412-A-04-00-00-00A-288A-A / 00013	2. Replace all finite life components that have completed published operating limitations.		
412-A-05-11-00-00A-042A-A / 00070	3. Overhaul all components that have completed published overhaul periods.		
412-A-12-01-00-00A-040A-A / 00141	4. Lubricate and service helicopter as necessary.		
412-A-05-00-00-00A-009A-A / 00014	5. Review Special Inspections and carry out applicable inspections.		
	6. Examine all inspection windows and sight glasses for cracking, crazing, and discoloration. If any of these conditions are present, the part must be removed and replaced prior to returning helicopter for service.		
Service Instruction (SI)	7. Make sure that all the necessary inspections of installed BHT kits not covered in this inspection have been performed, as necessary.		
	FUSELAGE - NOSE SECTION		
412-A-95-00-00-00A-009A-A / 01787	1. Pitot tubes and static ports for visible obstruction and damage.		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-52-00-00-00A-009A-A / 00774	2. Nose doors for damage, corrosion, security of attachment, and for missing or damaged twist fasteners, seal for condition.		
412-A-53-00-00-00A-009A-A / 00897	3. Fuselage:		
	3.1. Forward fuselage area structure and skin for damage, corrosion, cleanliness, and damage to protective finish.		
	3.2. Avionics and electrical compartment for water entrapment.		
412-A-96-00-00-00A-009A-A / 01870	4. Battery installation:		
	4.1. Battery and external connections for security, corrosion, and condition.		
	4.2. Battery vent and drain tubes - unobstructed.		
	4.3. Every fourth 25 hour/30 Day Inspection (100 hours), check the battery mounts for corrosion and service battery in accordance with battery manufacturers recommendation.		
412-A-96-00-00-00A-009A-A / 01870	5. Electrical equipment for condition, corrosion, and security.		
412-A-97-00-00-00A-009A-A / 02235	6. Avionics equipment for condition and security.		
412-A-52-00-00-00A-009A-A / 00774	7. All windows for damage and cleanliness.		
412-A-30-00-00-00A-009A-A / 00677	8. Windshield wiper arms and blades for serviceability and security.		
412-A-29-00-00-00A-009A-A / 00610	SN: 33001-33213 or 36001-36999: 9. Remote hydraulic filter bypass indicator-check for bypass indication.		
412-A-97-00-00-00A-009A-A / 02235	10. Antennas for condition and security.		
412-A-52-00-00-00A-009A-A / 00774	11. Crew doors (and surrounding structure) for damage, corrosion, and proper operation, emergency release pins for security.		
	Fuselage - Cabin Section		
412-A-53-00-00-00A-009A-A / 00897	1. Fuselage structure and compartments for condition, corrosion, water entrapment, and damage to protective finish.		
412-A-53-00-00-00A-009A-A / 00897	2. Fuselage underside for evidence of fuel and hydraulic fluid leakage.		
412-A-28-00-00-00A-009A-A / 00323	3. Visually inspect the fuel samples for contamination.		
412-A-32-00-00-00A-009A-A / 00694	4. Landing gear:		
	4.1. Forward crosstube and cap fittings for condition, corrosion, and security of attachment.		
	4.2. Aft crosstube, cap fittings, and wear strips for condition, corrosion, and security of attachment.		
	4.3. Aft crosstube support beam for wear, cracks, damage, corrosion, and security of attachment.		
	4.4. Skid tubes and skid shoes for condition, corrosion, and security of attachment. Replace skid shoes that are worn into shoe surface. Repair weld beads as necessary.		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	4.5. Fuselage supports for wear, damage, and security of attachment.		
	4.6. If installed, landing gear forward crosstube crew step fittings for corrosion, damage, and security in area of attachment. If edge sealant is damaged or missing, remove crew step and inspect crosstube for corrosion or damage.		
	4.7. Tail skid for deformation and security of attachment.		
	4.8. Emergency float reservoir pressure indicator for proper charge indication (if installed).		
	4.9. Floats for proper stowage and condition (if installed).		
412-A-97-00-00-00A-009A-A / 02235	5. Antennas for damage, cleanliness, and security.		
412-A-52-00-00-00A-009A-A / 00774	6. Passenger/cargo doors (and surrounding structure) for damage, corrosion, and proper operation. Window seals for condition.		
412-A-25-00-00-00A-009A-A / 00244	7. Crew seats:		
	7.1. Seats for condition, security, and proper operation.		
	7.2. Cushions and backs for cleanliness, excessive deterioration, and tears.		
	7.3. Crew seat restraint assemblies for condition, security, and proper operation.		
	7.4. Crew seat attenuator for compression; inspect witness line. If line is not visible, repair attenuator assembly.		
412-A-25-00-00-00A-009A-A / 00244	8. Passenger seats:		
	8.1. Seats for condition and security.		
	8.2. Cushions and backs for cleanliness, excessive deterioration, and tears.		
	8.3. Passenger seat restraint assemblies for condition, security, and proper operation.		
412-A-95-00-00-00A-009A-A / 01787	9. Instruments:		
	9.1. Instrument panel for cleanliness.		
	9.2. All instrument, placards, decals, and markings for appearance and legibility.		
	9.3. Check magnetic compass for condition and security.		
	9.4. All compass cards for validity.		
412-A-96-00-00-00A-009A-A / 01870	10. Avionics and electrical equipment:		
412-A-97-00-00-00A-009A-A / 02235			
	10.1. Pedestal mounted avionics/electrical equipment for condition, cleanliness, and security.		
	10.2. Overhead console for condition, cleanliness, and security.		
	10.3. All instrument lights, integrally lit panels, secondary lights, and map lights for serviceability.		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	10.4. All caution and warning lights for proper operation by using master caution, fire test, and baggage compartment smoke detector press to test functions.		
	10.5. Landing and search lights for condition and security.		
	10.6. Navigation and anti-collision lights for condition and security.		
	10.7. Every fourth 25 Hour/30 Day Inspection, operationally check cabin heater system, bleed air shutoff, and line check valve.		
412-A-26-00-00-00A-009A-A / 00291	11. Two portable fire extinguishers for condition, mounting, and valid inspection certificate.		
412-A-21-00-00-00A-009A-A / 00189	12. Cabin heating and ventilating system:		
	12.1. Ventilating system components for condition and security.		
	12.2. Heat/vent air ducts for condition and proper operation.		
	12.3. Ventilation/defog components for condition and security.		
412-A-25-00-00-00A-009A-A / 00244	13. Miscellaneous furnishings (map and data case, first aid kit, and emergency equipment) for condition and security.		
412-A-63-00-00-00A-009A-A / 00987	14. Main rotor transmission:		
	14.1. Proper oil level.		
	14.2. External oil filter for bypass indication.		
	14.3. Cases for damage, corrosion, condition, and evidence of leakage.		
	14.4. Accessories for condition, corrosion, damage, and security of attachment.		
	14.5. External oil lines and hoses for condition, damage, chafing, and leaks, paying particular attention to lines running aft of the thermostatic relief valve.		
	14.6. Transmission mount dust boots for condition.		
	14.7. Every fourth 25 Hour/30 Day Inspection (100 hours), with all panels removed, inspect the pylon side walls in the area of each opening for mechanical or corrosion damage and cracks. Pay attention around fastener hole where cracks are susceptible to develop.		
	14.8. Examine the top and bottom areas of the left and right pylon support channels (205-030-919) (1, Figure 1) with a bright light and mirror for loose fasteners and cracks.  **WARNING** **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.**		
	14.9. Transmission tail rotor output quill coupling (and surrounding area) for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black, indicates an overheat condition, and may require replacement of both outer and inner couplings (see Paragraph).		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	14.10. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of overtemperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.		
412-A-29-00-00-00A-009A-A / 00610	15. Hydraulic systems:		
	15.1. Hydraulic system 1 and 2 filter bypass indicator buttons - not extended.		
	15.2. Collective and cyclic servo actuators for leaks, corrosion, damage, and security.		
	15.3. Hydraulic system 1 and 2 valve and filter modules for leaks, damage, and security.		
	15.4. Hydraulic system 1 and 2 lines, hoses, and fittings for leaks, damage, chafing, kinking, and security.		
	16. Flight control tubes for condition, corrosion, and security.		
	FUSELAGE AFT OF CABIN LEFT AND RIGHT SIDE		
412-A-52-00-00-00A-009A-A / 00774	1. Avionics/electrical and heater compartment doors for condition and security of attachment. Latches for proper operation.		
412-A-53-00-00-00A-009A-A / 00897	2. Fuselage structure:		
	2.1. Avionics/electrical and heater compartments for evidence of water entrapment.		
	2.2. Engine decks for condition and evidence of delamination.		
412-A-96-00-00-00A-009A-A / 01870	3. Avionics and electrical equipment for security and condition.		
412-A-97-00-00-00A-009A-A / 02235			
412-A-21-00-00-00A-009A-A / 00189	4. Heater compartment for cleanliness, condition, and security of heating system components, wiring, ducts and supports. Structure for damage and corrosion (including connections and fasteners).		
412-A-29-00-00-00A-009A-A / 00610	5. Tail rotor hydraulic actuator and hoses for leakage, corrosion, and security.		
412-A-67-00-00-00A-009A-A / 01319	6. Tail rotor control tubes for condition, corrosion, and security.		
412-A-63-00-00-00A-009A-A / 00987	7. Transmission oil cooling system:		
	7.1. Oil coolers for leakage, damage, and obstruction.		
	7.2. Oil cooler hoses and tubes for leakage, damage, chafing, and fraying.		
	7.3. Oil cooler blowers for damage, corrosion, and obstruction.		
412-A-79-00-00-00A-009A-A / 01765	8. Engine oil system:		
	8.1. Oil coolers for leakage, corrosion, damage, and obstruction.		
	8.2. Oil cooler hoses and tubes for leakage, damage, chafing, and fraying.		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	8.3. Reduction gearbox oil filter impending bypass indicator button - not extended.		
	8.4. Power section 1 for proper oil level.		
	8.5. Power section 2 for proper oil level.		
	8.6. Reduction gearbox for proper oil level.		
412-A-26-00-00-00A-009A-A / 00291	9. Engine compartment fire extinguisher containers for proper charge, condition, and mounting.		
412-A-71-00-00-00A-009A-A / 01584	10. Engine and reduction gearbox mounts for loose bearings and security.		
412-A-71-00-00-00A-009A-A / 01584	11. No.1 (left) engine power section:		
	11.1. Exhaust ejector and duct for damage and security.		
	11.2. Gas generator case for cracks, buckled areas, and hot spots.		
	11.3. Oil and fuel hoses and tubes for chafing, leaking, and security.		
	11.4. Electrical wiring for fraying, chafing, and security.		
412-A-71-00-00-00A-009A-A / 01584	12. No.2 (right) engine power section:		
	12.1. Exhaust ejector and duct for damage and security.		
	12.2. Gas generator case for cracks, buckled areas, and hot spots.		
	12.3. Oil and fuel hoses and tubes for chafing, leaking, and security.		
	12.4. Electrical wiring for fraying, chafing, and security.		
412-A-71-00-00-00A-009A-A / 01584	13. Engine firewalls, air intake ducts, and plenum for cracks, distortion, missing rivets, broken spot welds, and deteriorating seals or gaskets.		
412-A-71-00-00-00A-009A-A / 01584	14. Engine cowling for missing fasteners and cracks. Latches for proper operation.		
412-A-53-00-00-00A-009A-A / 00897	**FUSELAGE AFT/TAILBOOM ATTACHMENT**		
	1. Inspect fuselage tailboom attachment points for security and evidence of fretting.		
	TAILBOOM		
412-A-53-00-00-00A-009A-A / 00897	1. Tailboom structure:		
	1.1. Exterior structure for condition, damage, and corrosion.		
	1.2. Baggage compartment interior for condition and cleanliness.		
412-A-52-00-00-00A-009A-A / 00774	2. Baggage compartment door for damage, proper operation, and security.		
412-A-52-00-00-00A-009A-A / 00774	3. Driveshaft and intermediate gearbox covers for damage and security.		
412-A-67-00-00-00A-009A-A / 01319	4. Elevators for damage and security. Elevator control tube for positive spring action and freedom of movement.		
412-A-97-00-00-00A-009A-A / 02235	5. Tailboom mounted avionics equipment for condition and security.		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-53-00-00-00A-009A-A / 00897	6. Every fourth 25 Hour/30 Day Inspection: 6.1. Inspect vertical fin caps for cracks and/or corrosion. 6.2. Inspect interior structure (including joints, splices, longerons, attach fittings, and hardware) for condition, corrosion, damage, and cracks. **NOTE** **BLR FastFin Kit:** For helicopters equipped with the Dual Tailboom Strakes and FastFin™ system Supplemental Type Certificate (STC) , refer to the applicable BLR Aerospace documents for inspection criteria. BLR FastFin Kit: 6.3. Inspect upper and lower tail fairings for visible damage to the exterior skin and cracks at the attachment points. If paint is missing locally or has scratches, inspect the fairing for delamination and voids with a tap hammer. Inspect fairings for loose or missing attaching hardware. BLR FastFin Kit: 6.4. Inspect upper and lower strakes for visible damage, deformation, cracks at the attachment points, and loose or missing rivets.		
412-A-67-00-00-00A-009A-A / 01319	6.5. Check elevator attachment lugs at each end of elevator horn for cracks, corrosion, and security. 6.6. Check elevator horn to elevator spar attaching bolts and surrounding joint for cracks, corrosion, and security. 6.7. Check elevator control tube, support, attaching hardware, and surrounding structure, for corrosion, damage, and security.		
412-A-65-00-00-00A-009A-A / 01257	6.8. Torque check the tail rotor gearbox mounting nuts by applying torque in tightening direction, do not exceed 200 inch-pounds (22.6 Nm). If any nut rotates under torque, inspect attachment hardware and tail rotor gearbox for serviceability. 6.9. Torque check the intermediate gearbox mounting bolts by applying torque in tightening direction, do not exceed 50 inch-pounds (5.65 Nm). If any bolt turns under torque, inspect attachment hardware and intermediate gearbox for serviceability.		
412-A-65-00-00-00A-009A-A / 01257	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *7. Tail rotor driveshaft:*  **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.**		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	7.1. Hanger assemblies - bearings and surrounding areas for evidence of grease leakage, condition, damage, security, corrosion, and overheating. Discoloration of bearing (color change to blue or blue/black) or multi-color appearance of hanger that darkens adjacent to bearing is evidence of overheating. Brown color of bearing shield is normal and is not evidence of overheating. 7.2. Driveshaft sections for cracks, rivet failure, distortion, dents, corrosion, and damage to anodized finish. **NOTE** Do not mistake empty imprints in bonding material next to balance strip, as an indication of a missing balance strip. This spot results from removal of a test coupon to inspect for bonding voids. If a tail rotor driveshaft has more than a single empty bonding imprint, remove and send to Bell for repair and balancing. 7.3. Driveshaft for missing balance strips. 7.4. Clamp sets for condition, security, and proper installation. **WARNING** **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.** 7.5. Flexible couplings and surrounding areas for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 7.6). 7.6. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.		
412-A-65-00-00-00A-009A-A / 01257	(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *8. Tail rotor driveshaft:* **WARNING** **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.** 8.1. Hanger assemblies - bearings and surrounding areas for evidence of grease leakage, condition, damage, security, corrosion, and overheating. Discoloration of bearing (blue to blue/black in color) or multi-color appearance of hanger that darkens adjacent to bearing is evidence of overheating. Brown color of bearing shield is normal and is not evidence of overheating.		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	8.2. Driveshaft sections for cracks, rivet failure, distortion, dents, corrosion, and damage to anodized finish. 8.3. Disc pack couplings for separation, deflection, distortion, cracks, or missing sections, and correct installation of bolts and washers.  **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.** 8.4. Coupling assembly at first hanger and adjacent to transmission and surrounding area for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 8.5). 8.5. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.		
412-A-65-00-00-00A-009A-A / 01257	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *9. Intermediate gearbox:* 9.1. Gearbox assembly for security, condition, corrosion, damage, and oil leaking.  **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.** 9.2. Input and output quill flexible couplings (and surrounding area) for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 9.3). 9.3. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating. 9.4. Gearbox for proper oil level and oil for evidence of contamination.		
412-A-65-00-00-00A-009A-A / 01257	(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *10. Intermediate gearbox:* 10.1. Gearbox assembly for security, condition, corrosion, damage, and oil leaking.		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	 **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.**		
	10.2. Output quill outer couplings (and surrounding area) for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 10.3).		
	10.3. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.		
	10.4. Gearbox for proper oil level and oil for evidence of contamination.		
412-A-65-00-00-00A-009A-A / 01257	(SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)): *11. Tail rotor gearbox:*		
	11.1. Gearbox assembly for security, condition, corrosion, damage, and oil leaking.		
	 **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.**		
	11.2. Input quill flexible coupling (and surrounding area) for grease leakage, damage, corrosion, and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of one or more of the red bordered TEMP-PLATE dots from white/off-white to black indicates an overheat condition and may require replacement of both outer and inner couplings (step 11.3).		
	11.3. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of over temperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.		
	11.4. Gearbox for proper oil level and oil for evidence of contamination.		
412-A-65-00-00-00A-009A-A / 01257	(SN: 36020-36999 or SN: 37002-37999 or SN: 39101-39999) (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed): *12. Tail rotor gearbox:*		
	12.1. Gearbox assembly for security, condition, corrosion, damage, and oil leaking.		
	12.2. Gearbox for proper oil level and oil for evidence of contamination.		
412-A-64-00-00-00A-009A-A / 01225	13. Tail rotor hub and blade assembly:		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	 **NO CRACKS ARE PERMITTED ON ANY SURFACE OF THE TAIL ROTOR BLADES.** 13.1. Tail rotor blades for condition of bond lines, cracks, corrosion, leading edge erosion, damage, security, and cleanliness. Clean blades are necessary to maintain enhanced visibility for safety. 13.2. Tail rotor hub for security, corrosion, and condition. 13.3. Flapping bearings and pitch change bearings for excessive looseness and freedom of movement through full range of travel with antitorque pedals positioned full right and then full left. 13.4. Tail rotor pitch change links, crosshead, pitch horns, counterweight arms, and links for security, corrosion, and condition. 13.5. Pitch link bearing and counterweight link bearing looseness shall not exceed 0.015 inch (0.38 mm) axial. Check freedom of movement through full range of travel with antitorque pedals positioned full right and then full left. 13.6. Pitch change links for binding with tail rotor blade moved to both full flapping positions. Binding is not acceptable. 14. Every fourth 25 Hour/30 Day inspection:  **COUNTERWEIGHT BELLCRANK RETENTION NUT MS14145L6, OR ITS EQUIVALENT MS17826-6, IS DESIGNATED AS ONE TIME USE ONLY. IF MAINTENANCE ACTION REQUIRES THAT IT BE REMOVED, THE COUNTERWEIGHT BELLCRANK NUT IS TO BE DISCARDED AND REPLACED WITH A NEW NUT. OVERTORQUING OR INCORRECT APPLICATION OF THE NUT COULD CAUSE MECHANICAL FAILURE RESULTING IN EXCESSIVE VIBRATION OR DAMAGE TO THE TAIL ROTOR BLADES.** 14.1. Inspect the tail rotor counterweight bellcrank 212-011-705-001 retention nuts for cracks, corrosion, and security. Reapply corrosion preventive compound (C-101) if nuts are not completely covered.		
	CABIN ROOF		
412-A-52-00-00-00A-009A-A / 00774 412-A-53-00-00-00A-009A-A / 00897	1. Cabin structure and cowlings/fairings for condition.		
412-A-63-00-00-00A-009A-A / 00987	2. Transmission and transmission oil lines for condition, corrosion, and leaks.		
412-A-63-00-00-00A-009A-A / 00987 BHT-412-SI-12	3. Rotor brake quill, disc, and brake assembly (if installed):		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

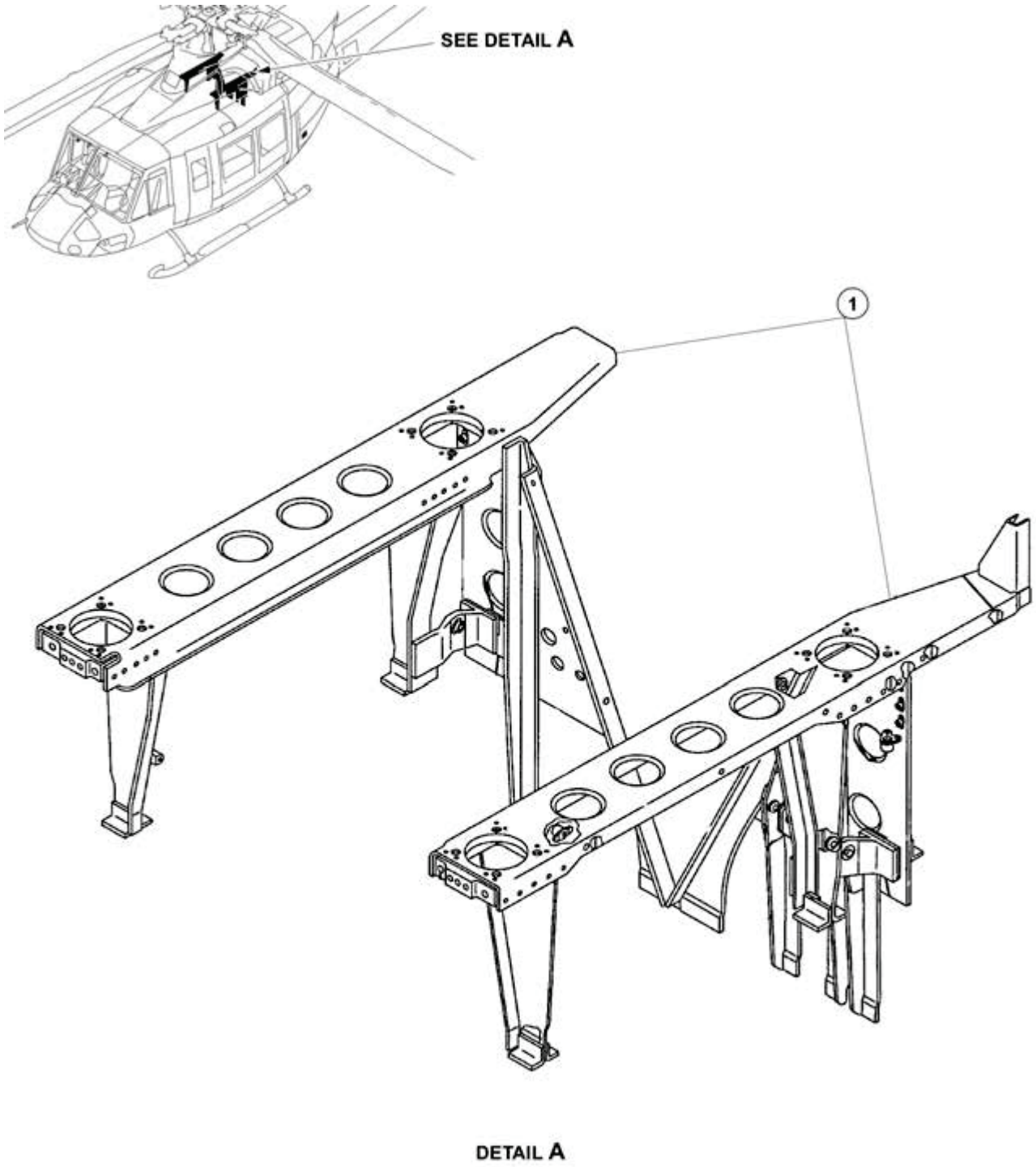
DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	3.1. Brake quill for condition, damage, and leakage.		
	3.2. Brake disc for condition, damage, security, warpage, and evidence of overheat.		
	3.3. Caliper switches and wiring for condition and security of attachment.		
	3.4. Every fourth 25 hour/30 Day Inspection (100 hours), check the torque of the brake disc attachment bolts 50 to 70 inch-pounds (5.65 to 7.90 Nm).		
412-A-29-00-00-00A-009A-A / 00610	4. Hydraulic systems:		
	4.1. Hydraulic system 1 and 2 pumps for leakage, damage, and security.		
	4.2. Hydraulic lines for leaks, chafing, and kinking.		
	4.3. Hydraulic system 1 and 2 reservoirs for proper fluid levels, damage, corrosion, and security.		
412-A-62-00-00-00A-009A-A / 00932	5. Main rotor blades:		
	5.1. Wash the main rotor blades.		
	5.2. Main rotor blades for condition, damage, and security.		
	NOTE Waxing is not necessary for blades 412-015-300-109 and subsequent or for any blade finished with polyurethane paint.		
	5.3. Apply a coat of wax to main rotor blades.		
412-A-62-00-00-00A-009A-A / 00932	6. Main rotor hub:		
	6.1. Inspect the hub assembly for condition, corrosion, and security, paying particular attention to the integrity of sealing.		
	6.2. Inspect the pitch horns for condition, security, and corrosion and cracks in areas surrounding torque transfer pin holes. No cracks permitted.		
	6.3. Inspect the pivot bearings, spindle bearings, and damper set for elastomeric bearing delamination, elastomer degradation, shim cracks, crazing, cracking, or sheeting.		
	6.4. Inspect the damper bearing bumper pads for deterioration or deformation due to contact with yoke.		
	6.5. Functionally check the arm and bearing (pendulum) assemblies. Apply a minimum of 5 pounds (22.24 N) of axial force to each arm and bearing assembly. Cycle arm and bearing through full travel.		
	6.6. Apply force away from the center of yoke. Arm and bearing assembly should cycle free and smooth. Roughness and binding are not acceptable.		
	6.7. Functionally check the droop restraint system. Check for engagement of the restraint arms in lower detent of cam window. Relieve the load on each blade and verify lower clevis moves freely through cam window envelope. Binding and roughness in inboard clevis is not acceptable.		
412-A-62-00-00-00A-009A-A / 00932	7. Main rotor controls:		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	7.1. Visually inspect the swashplate and support assembly, and collective lever for condition and security. 7.2. Visually inspect the hub and sleeve assembly for condition and security. **NOTE** Use swashplate link and drive link bearings - workaid (412-A-62-00-00-00A-009A-A / 00932) to assist with determining condition of main rotor control bearings. 7.3. Visually inspect the drive hub and rephasing levers for bearing looseness and security of attachment. 7.4. Visually inspect the pitch links for condition, security, and bearing looseness. 8. Every fourth 25 Hour/30 Day Inspection: **NOTE** Disconnect the pitch link upper rod ends from the pitch horns to determine main rotor control bearing wear. 8.1. Inspect the swashplate and support assembly for condition and security with special attention to gimbal ring attachment lugs. **NOTE** The presence of black oxide powder will require investigation to determine the cause. 8.2. Inspect the gimbal ring bearings, liners, and attaching hardware for excessive looseness and wear. Gimble ring bearing and attaching bolt axial or radial looseness shall not exceed 0.005 inch (0.13 mm) with no binding allowed. 8.3. Inspect the trunnion bearings and attaching hardware for excessive looseness and wear. Trunnion bearings shall not exceed 0.020 inch (0.51 mm) axial looseness. 8.4. Inspect the drive hub and rephasing levers for wear and bearing looseness. 8.5. Drive links' spherical bearings shall not exceed 0.007 inch (0.18 mm) radial or 0.015 inch (0.38 mm) axial looseness. 8.6. Swashplate link non-elastomeric bearing looseness shall not exceed 0.015 inch (0.38 mm) axial. 8.7. Collective lever trunnion bearing shall not exceed 0.020 inch (0.51 mm) axial looseness. 8.8. Pitch link's non-elastomeric bearing looseness shall not exceed 0.007 inch (0.18 mm) radial or 0.015 inch (0.38 mm) axial.		
412-A-63-00-00-00A-009A-A / 00987	9. Main rotor mast: 9.1. Inspect the mast assembly for security, corrosion, and mechanical damage. 9.2. Inspect for evidence of oil leaks at mast bearing cap.		
412-A-63-00-00-00A-009A-A / 00987	10. Engine-to-transmission (main) driveshaft:		

Table 1. 25 Hour/30 Day Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	10.1. Main driveshaft for corrosion, condition, and security.  **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.** 10.2. Main driveshaft forward and aft couplings, boots, seals (and surrounding area), for grease leakage, damage corrosion and evidence of overheat as indicated by TEMP-PLATES or coupling discoloration. A change in color of any yellow bordered TEMP-PLATE, from white/off-white to black, will indicate a possible overheat condition and /or component degradation (Paragraph). If one or more of the red bordered TEMP-PLATE dots change from white/off-white to black, an overheat condition is present and may require replacement of both outer and inner couplings. 10.3. Overheat TEMP-PLATES for condition and security. TEMP-PLATES must not show evidence of overtemperature indication, deterioration, debonding, or excessive discoloration of the epoxy overcoating.		



1. Pylon Support Channel (205-030-919)

ICN-412-A-054000-B-97499-02877-A-001-01

Figure 1. Pylon Support Channel - Scheduled Inspection (Sheet 1 of 1)

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Scheduled Inspections - 300 Hour/12 Month Inspection - Part B

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	
Sealant	C-308	AR	
MEK	C-309	AR	

Procedure

Table 1. 300 Hour/12 Month Inspection - Part B

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME:_____		
	SIGNATURE:_____		
	DATE: _____W.O. _____		
	NOTE For helicopters on the Part B inspection program, accomplish the following checks each 300 hours of flight operation or 12 calendar months, whichever occurs first.		
	GENERAL		
	Each listed inspection item or maintenance function is to be performed in accordance with the chapter specified.		
	PRELIMINARY REQUIREMENTS		
	1. All work accomplished during inspection shall be recorded in helicopter maintenance records.		
	2. Remove all panels, fairings, and cowlings required to satisfactorily carry out the inspection.		
	AFTER COMPLETION OF INSPECTION		
	1. Reinstall all panels, fairings, and cowlings removed during inspection.		
	2. Carry out a ground run and leak check prior to releasing the helicopter back to service.		
	FUSELAGE - CABIN		
412-A-53-00-00-00A-009A-A / 00897	1. Forward fuselage:		
	1.1. Cabin exterior structure for corrosion and damage.		
	1.2. Cabin interior structure for corrosion and damage.		
	1.3. Ensure all drain holes are open.		
	1.4. All compartments for evidence of water entrapment and corrosion.		
412-A-52-00-00-00A-009A-A / 00774	2. Doors and windows:		
	2.1. Crew doors structure for corrosion, damage, distortion, and positive locking mechanisms. Seals for tears and separations.		

Table 1. 300 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.2. Crew door emergency jettison mechanism for condition and security. Perform operational check.		
	2.3. Passenger door structure for corrosion, damage, distortion, and positive locking mechanisms.		
	2.4. Nose, electrical, and equipment compartment access doors for corrosion, damage, distortion, and positive locking mechanisms.		
	2.5. Passenger door window retainers and fillers for damage. Seals for adherence, tears, separations, and deterioration.		
	2.6. If installed, inspect heated windshield for condition and proper installation.		
412-A-67-00-00-00A-009A-A / 01319	3. Collective flight control actuator:		
	3.1. Universal bearings for looseness.		
	3.2. Input lever bearings and bolts for wear and looseness.		
	3.3. Fasteners attaching cylinder lower supports to structure for looseness.		
	3.4. Cylinder lower bearings for looseness.		
	3.5. Actuator assembly for damage and leakage.		
	3.6. Cylinder extension tube for condition and security.		
	3.7. Actuator linkage, balance spring, and bracket for wear and security.		
	3.8. Clean exposed area of actuator piston with hydraulic fluid and a clean lint-free cloth.		
	3.9. Check hydraulic cylinder piston rod for evidence of excessive wear and scoring. Wear of the piston rod indicates cylinder assembly is incorrectly aligned to lower support and requires adjustment.		
	3.10. Remove bolts attaching actuator to lower supports. Check cylinder alignment to lower support. Rod end bearing side loads are not permitted. Install bolts.		
	3.11. Torque check nuts attaching actuator cylinder to upper supports.		
BHT-ALL-SPM, Standard Practices Manual	3.12. Upper cylinder housing mounting bracket cavity for corrosion. Cavity drain hole for obstruction. Reapply corrosion preventive compound (C-104) as required. Dust boot for condition.		
412-A-67-00-00-00A-009A-A / 01319	4. Collective control tubes:		
	4.1. Control tubes between collective jackshaft and collective control actuator pilot valve input lever for corrosion, wear, and damage.		
	4.2. Control tube bellcrank, supports, and attaching hardware for corrosion, wear, and damage.		
412-A-67-00-00-00A-009A-A / 01319	5. Cyclic flight control actuators:		
	5.1. Universal bearings for looseness.		
	5.2. Input lever bearings and bolts for wear and looseness.		
	5.3. Fasteners attaching cylinder lower supports to structure for looseness.		

Table 1. 300 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	5.4. Cylinder lower bearings for looseness.		
	5.5. Actuator assembly for damage and leakage.		
	5.6. Cylinder extension tube for condition and security.		
	5.7. Actuator linkage for wear and security.		
	5.8. Clean exposed area of actuator piston with hydraulic fluid and a clean lint-free cloth.		
	5.9. Check hydraulic cylinder piston rod for evidence of excessive wear and scoring. Wear of the piston rod indicates cylinder assembly is incorrectly aligned to lower support and requires adjustment.		
	5.10. Remove bolts attaching actuator to lower supports. Check cylinder alignment to lower support. Rod end bearing side loads are not permitted. Install bolts.		
	5.11. Torque check nuts attaching actuator cylinder to upper supports.		
BHT-ALL-SPM, Standard Practices Manual	5.12. Upper cylinder housing mounting bracket cavity for corrosion. Cavity drain hole for obstruction. Re-apply corrosion preventive compound (C-104) as required. Dust boot for condition.		
412-A-67-00-00-00A-009A-A / 01319	6. Cyclic control tubes:		
	6.1. Control tubes between cyclic jackshaft and cyclic control actuator pilot valve input lever for corrosion, wear, and damage.		
	6.2. Control tube bellcranks, mixing levers, supports, and attaching hardware for corrosion, wear, and damage.		
412-A-63-00-00-00A-009A-A / 00987	7. Transmission:		
	7.1. All transmission chip detectors for debris.		
	7.2. Clean all transmission chip detectors.		
	7.3. Test all transmission chip detector electrical circuits.		
	7.4. Transmission internal oil filter or full flow oil debris monitor for debris.		
	7.5. Clean transmission internal oil filter or full flow oil debris monitor.		
412-A-63-21-04-01A-280A-A / 01189	*7.6. Inspect the transmission aft mount friction dampers for condition and security.*		
	7.7. Mounts for damage and security.		
	7.8. Lift link and attachments for corrosion, damage, and security. Bearings for looseness.		
	7.9. Mast torque sensor (if installed) installation for condition and security.		
BHT-412-CR&O	7.10. Torque check: top case to ring gear case nuts, ring gear case to main case nuts and main case to support case nuts 230 to 250 inch-pounds (25.99 to 28.25 Nm). Re-torque as required.		
412-A-32-00-00-00A-009A-A / 00694	8. Landing gear assembly:		
	8.1. Landing gear skid shoes for excessive wear, cracks, damage, corrosion, and security of attachment.		
	8.2. Aft cross tube support beam for cracks, damage, corrosion, and security of attachment.		

Table 1. 300 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	8.3. Retighten cross tube to skid tube saddle bolts.		
	8.4. Torque forward and aft cross tube support fittings U-bolts to 80 to 100 inch-pounds (9.04 to 11.29 Nm).		
	8.5. Torque check ground handling wheel attachment eyebolts on skid tubes 40 to 58 foot-pounds (54 to 79 Nm). Eyebolts that rotate prior to reaching 40 foot-pounds (54 Nm) shall be removed and inspected for evidence of bending. No bending is permitted.		
BHT-412-SI-58	8.6. If gross weight towing kit provisions are installed, inspect fitting assembly for condition and security. Torque check attachment bolts 160 to 190 inch-pounds (18.1 to 21.4 Nm). If bolts are found loose, inspect for condition.		
412-A-53-00-00-00A-009A-A / 00897	9. Underside of fuselage:		
	9.1. Fuselage structure for damage, corrosion, and working rivets.		
	9.2. Exterior finish for condition and cleanliness.		
	9.3. Evidence of excessive fluid leakage.		
	9.4. Structure around landing gear for condition.		
	9.5. Forward crosstube support structure for working fasteners, cracks, corrosion, and excessive wear.		
	9.6. Aft crosstube support structure for working fasteners, cracks, corrosion, and excessive wear.		
	FUSELAGE AFT OF CABIN LEFT AND RIGHT SIDE		
	NOTE Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for engine inspection requirements.		
412-A-71-00-00-00A-009A-A / 01584	1. No. 1 (left) engine power section:		
	1.1. Chip detectors for debris.		
	1.2. Clean chip detectors.		
	1.3. Test chip detector electrical circuits.		
412-A-71-00-00-00A-009A-A / 01584	2. No. 2 (right) engine power section:		
	2.1. Chip detectors for debris.		
	2.2. Clean chip detectors.		
	2.3. Test chip detector electrical circuits.		
412-A-71-00-00-00A-009A-A / 01584	3. Reduction (combining) gearbox:		
	3.1. Chip detector for debris.		
	3.2. Clean chip detector.		
	3.3. Test chip detector electrical circuits.		
412-A-71-00-00-00A-009A-A / 01584	4. Engine electrical connections:		
	4.1. Power section 1 ignition leads for corrosion, chafing, and security.		
	4.2. Power section 2 ignition leads for corrosion, chafing, and security.		

Table 1. 300 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-71-00-00-00A-009A-A / 01584	5. Starter generator:		
	5.1. Power section 1 starter generator brushes for allowable wear.		
	5.2. Power section 2 starter generator brushes for allowable wear.		
	5.3. Power section 1 starter generator cooling ducts for obstruction, kinking, and security.		
	5.4. Power section 2 starter generator cooling ducts for obstruction, kinking, and security.		
	NOTE If applicable, refer to BHT-412-SI-87 for installation of adjustable seals or refer to 412-A-71-10-00-02A-271A-A / 01613 for seal adjustment.		
412-A-71-00-00-00A-009A-A / 01584 ASB 412-10-140	6. Inspect the left side and right side engine air inlet cowlings for gaps between the lower surface of the cowling firewall and the mating horizontal engine firewall.		
412-A-76-00-00-00A-009A-A / 01722	7. Engine fuel and power controls:		
	7.1. Engine control linkages for looseness, lost motion, chafing, damage, and security of attachment.		
	7.2. Bellcranks, mounts, and jackshafts for looseness, damage, and security of attachment.		
	7.3. Check controls for smooth movement throughout full range of travel.		
BHT-412-FM	7.4. Perform engine fuel control check (not applicable to PT6T-9 engine).		
412-A-26-00-00-00A-009A-A / 00291	8. Engine fire detection systems for condition and security.		
	TAILBOOM		
412-A-53-00-00-00A-009A-A / 00897	1. Tailboom structure:		
	1.1. Tailboom exterior structure for corrosion and damage.		
	1.2. Tailboom interior structure for corrosion and damage.		
	1.3. Inspect tailboom joints, splices, longerons, attach fittings, and attaching hardware for corrosion, damage, and cracks.		
ASB 412-90-49	1.4. Remove plug button at BS 99.00.		
	NOTE Paint fissures are common in the splice area, cracks will be evident by black powder emanating from cracked area, corrosion will leave evidence of white powder.		
	1.5. Using a borescope or other suitable means, inspect inside of longeron paying particular attention to cracking and/or corrosion in area of splice.		
	1.6. Report any cracks to Product Support Engineering.		
	1.7. Install plug button with sealant (C-308).		
	1.8. Ensure all drain holes are open.		

Table 1. 300 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	1.9. Vertical fin spar caps forward side and web from upper tailboom skin to approximately 4.0 inches (102 mm) below upper tailboom skin as follows: remove aft tailboom access door. Face aft and use bright light and small mirror to inspect area for cracks, especially near rivet holes.		
BHT-ALL-SPM, Standard Practices Manual	**NOTE** If necessary, clean area with cloth dampened with MEK (C-309) or equivalent. Ventilate area to prevent breathing fumes.		
412-A-67-00-00-00A-009A-A / 01319	2. Antitorque controls:		
	2.1. Tail rotor control tubes between tail rotor pedals and tail rotor gearbox for corrosion, wear, and damage.		
	2.2. Control tube bellcranks, supports, and attaching hardware for corrosion, security, wear, and damage.		
	2.3. Tail rotor hydraulic actuator for leaks and security of attachment, damage, corrosion, and bearings and linkages for looseness.		
ASB 412-15-166	2.4. Disconnect both ends of control tube 212-001-055-101 from bellcranks 212-001-705-001 and 212-001-759-101.		
	2.4.1. Without removing control tube 212-001-055-101, visually inspect control tube for cracks. No cracks permitted.		
	2.4.2. On bellcranks 212-001-705-001 and 212-001-759-101, inspect bearings DAS-14A1-512 for freedom of movement, bearings shall move freely in all directions and rotate freely. If a defective bearing is found, replace defective bearing in accordance with BHT-ALL-SPM, Standard Practices Manual.		
412-A-67-00-00-00A-009A-A / 01319	3. Aerodynamically actuated elevator:		
	3.1. Elevator control tube between forward support bracket and elevator horn for corrosion, wear, and damage.		
	3.2. Control tube support and attaching hardware for security, corrosion, wear, and damage.		
	3.3. Elevators and support structure for loose or missing rivets and cracks in the skin.		
	3.4. Trailing edge tabs and tip caps for condition and security.		
	3.5. Functionally check operation of tube as follows:  **DO NOT RELEASE ELEVATOR AND ALLOW THE SPRING TO DRIVE ELEVATOR BACK TO ORIGINAL POSITION.**		
	3.5.1. Move elevator to extreme nose down position and while holding allow elevator to slowly return to original position. Failure to return to original position, binding and/or lock-up are not acceptable.		
	3.5.2. If any of these conditions exist, replace tube.		
	3.6. Remove elevator control tube and measure spring breakout force for 90 to 100 pounds (400.32 to 444.80 N). Install elevator control tube.		

Table 1. 300 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	3.7. Functionally check elevator rigging for proper setting and travel.		
	3.8. Inspect and measure elevator trailing edge tab for 16° ±1/4° angle down from upper surface of elevator.		
412-A-52-00-00-00A-009A-A / 00774	4. Baggage compartment door for corrosion, damage, distortion, and positive locking mechanism.		
412-A-26-00-00-00A-009A-A / 00291	5. Baggage compartment smoke detector for condition and security.		
412-A-96-00-00-00A-009A-A / 01870	6. Inspect wire bundles and clamping for chafing, condition, and security.		
	NOTE *Step 7 pertains to helicopters S/N 33001 through 33213 and S/N 36001 through 36019 not modified by 412-570-001-103 or Pre BHT-412-SI-74 (412SP to 412HP Upgrade) or Pre BHT-412-SI-97 (412 Classic to 412HP Upgrade).*		
412-A-65-00-00-00A-009A-A / 01257	7. Tail rotor driveshaft:		
	7.1. Hanger assemblies for cracks at mounting ears, corrosion, and security of attachment.		
	7.2. Hanger supports for condition and security of attachment.		
	7.3. Driveshaft sections and attaching hardware for condition and security.		
	7.4. Coupling clamps for cracks in or near bolt lugs. Clamps for proper position.		
	NOTE *Step 8 pertains to helicopters S/N 36020 through 36999, S/N 37002 through 37999, S/N 38001 through 38999, S/N 39101 through 39999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade).*		
412-A-65-00-00-00A-009A-A / 01257	8. Tail rotor driveshaft:		
	8.1. Hanger assemblies for cracks at mounting ears, corrosion, and security of attachment.		
	8.2. Hanger supports for condition and security of attachment.		
	8.3. Driveshaft sections and attaching hardware for condition and security.		
	8.4. Disc pack couplings and attaching hardware for condition and security.		
	NOTE Step 9 and step 10 do not pertain to helicopters S/ N 37002 through 37999, S/N 38001 through 38999, and S/N 39101 through 39999.		
412-A-65-00-00-00A-009A-A / 01257	9. Intermediate gearbox:		
	9.1. Chip detector for debris.		
	9.2. Clean chip detector.		

Table 1. 300 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	9.3. Functionally test chip detector electrical circuit.		
412-A-12-00-00-00A-009A-A / 00125	9.4. Service gearbox as required.		
412-A-65-00-00-00A-009A-A / 01257	10. Tail rotor gearbox:		
	10.1. Chip detector for debris.		
	10.2. Clean chip detector.		
	10.3. Functionally test chip detector electrical circuit.		
	10.4. Service gearbox as required.		
412-A-64-00-00-00A-009A-A / 01225	11. Tail rotor hub and blade assembly:		
412-A-18-00-00-00A-009A-A / 00144	11.1. Dynamically balance tail rotor.		
	11.2. Torque check nuts on tail rotor blade retention bolts.		
	11.3. Torque check tail rotor retaining nut.		
	CABIN ROOF		
412-A-52-00-00-00A-009A-A / 00774	1. Cabin roof structure:		
412-A-53-00-00-00A-009A-A / 00897			
	1.1. Cabin roof structure, cowlings and fairings for damage, delamination, and general condition.		
	1.2. Cabin roof and cowling/fairing mounted antennas for condition and security.		
412-A-62-00-00-00A-009A-A / 00932	2. Main rotor hub and blade assembly:		
	2.1. Main rotor blades upper and lower surfaces for condition of bond lines and doublers. Leading edge for erosion. Blade surfaces for corrosion, cracks, damage, and voids.		
	2.2. Main rotor hub assembly for corrosion and damage.		
	2.3. Blade retention bolts for condition and security.		
	2.4. Torque check eight main rotor yoke assembly through bolt nuts to 70 to 80 foot-pounds (95 to 108 Nm).		
	2.5. Ensure bolts EWB0420 are installed (16 places) for simple pendulum absorber bracket. Torque check bolts. Apply 29 to 33 foot-pounds (39 to 45 Nm) in increasing torque direction.		
412-A-62-00-00-00A-009A-A / 00932	3. Main rotor controls:		
	3.1. Torque check swashplate link elastomeric rod end bearings (as applicable) attachment bolts 165 to 200 inch-pounds (18.65 to 22.59 Nm).		
	3.2. Torque check swashplate outer ring through bolts to 165 to 175 inch-pounds (18.65 to 19.77 Nm).		
	3.3. Torque check four hub and sleeve to rephasing lever attachment bolts to 24 to 34 foot-pounds (33 to 46 Nm) for 1/2 inch (13 mm) bolts or to 160 to 200 inch-pounds (18.08 to 22.59 Nm) for 3/8 inch bolts (if installed).		

Table 1. 300 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	3.4. Torque check four drive link to rephasing lever attachment bolts to 165 to 200 inch-pounds (18.65 to 22.59 Nm). 3.5. Torque check pitch link elastomeric rod end bearings (as applicable) attachment bolts to 165 to 200 inch-pounds (18.65 to 22.59 Nm). 3.6. Torque check swashplate support to mast plate bolts AN5H-6A to 130 to 140 inch-pounds (14.69 to 15.81 Nm).		
412-A-96-00-00-00A-009A-A / 01870	4. Emergency bus interconnect circuit breaker and diodes: **NOTE** Helicopters S/N 33001 through 33107, except those modified for CAA, remove screw, spacer, and washer installed in the emergency load switch guard. 4.1. Inspect EMERG BUS INTC circuit breaker for wear. Operate circuit breaker manually and check by feel. Worn or loose circuit breaker is not acceptable. Close circuit breaker. 4.2. Place all switches in OFF or NORM position and close all circuit breakers. 4.3. Connect battery. 4.4. Place BATTERY BUS 1 switch and BATTERY BUS 2 switch in the ON position. Position INV 1 and INV 2 switches in the ON position. Caution panel INVERTER 1 and INVERTER 2 segments should not be illuminated. 4.5. Check emergency bus power by pressing the FIRE EXT PRESS TO TEST button momentarily. Observe both FIRE PULL handle warning lights illuminate. 4.6. Position emergency bus switch to EMERG LOAD position. Check emergency bus power by pressing the FIRE EXT PRESS TO TEST button momentarily. Observe both FIRE PULL handles illuminate. Verify INVERTER 1 and INVERTER 2 segments are illuminated on the caution panel. **NOTE** S/N 33108 through 33213 and 36001 through 36999 — Check INVERTER 1 light segment on the caution panel is not illuminated and INVERTER 2 segment is illuminated. 4.7. Position BATTERY BUS 1 and BATTERY BUS 2 switches off. Check emergency bus power by pressing FIRE EXT PRESS TO TEST button momentarily. Observe FIRE PULL handles illuminate. 4.8. Position INVERTER 1 and INVERTER 2 switches OFF. Position emergency bus switch to NORMAL. Disconnect battery. 4.9. Install screw, spacer, and washer in the emergency load switch guard if removed.		
412-A-96-61-00-07A-340A-A / 01933	5. Perform functional check of power diodes.		

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Scheduled Inspections - 600 Hour/12 Month Inspection - Part B Procedure

Table 1. 600 Hour/12 Month Inspection - Part B

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME:_____		
	SIGNATURE:_____		
	NOTE For helicopters on the Part B inspection program, accomplish the following checks each 600 hours of flight operation or 12 calendar months, whichever occurs first.		
	GENERAL		
	Each listed inspection item or maintenance function is to be performed in accordance with the BHT-412-MM chapter specified.		
	PRELIMINARY REQUIREMENTS		
	1. All work accomplished during inspection shall be recorded in helicopter maintenance records.		
	2. Remove all fuselage and tailboom access panels, removable floor panels, fairings, and cowlings.		
	3. Ensure all applicable Special Inspections, Alert Service Bulletins, and Airworthiness Directives have been accomplished.		
	AFTER COMPLETION OF INSPECTION		
	1. Reinstall all panels, fairings, and cowlings removed during inspection.		
	2. Carry out a ground run and leak check prior to releasing the helicopter back to service.		
	FUSELAGE		
412-A-25-00-00-00A-009A-A / 00244	1. Emergency and safety equipment:		
	1.1. Emergency and safety equipment for inspection due dates.		
	1.2. Contents of first aid kit for missing or over age items.		
412-A-26-20-00-03A-280A-A / 00320	2. Perform weight check of cockpit and cabin portable fire extinguishers.		
412-A-52-10-00-00A-320A-A / 00785	3. Perform operational check of crew doors emergency release mechanisms.		
412-A-21-00-00-00A-009A-A / 00189	4. Bleed air heating and ventilation/defog system components:		
	4.1. Perform functional check of bleed air heating system and components.		
412-A-96-06-00-02A-320A-A / 02146	4.2. Perform functional check of defog blower.		
412-A-95-00-00-00A-009A-A / 01787	5. Instruments: (required at 12 months calendar time only)		
	5.1. Calibrate pilot, copilot compass systems, and standby magnetic compass.		

Table 1. 600 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	5.2. Drain pitot-static system of any accumulated moisture.		
	5.3. Test pitot-static system for leaks.		
412-A-96-03-01-24A-320A-A / 02078 or 412-A-96-03-01-24B-320A-A / 02080	6. Perform functional check of battery temperature sensor caution light system.		
412-A-67-00-00-00A-009A-A / 01319	7. Collective flights controls:		
	7.1. Inspect flight control systems bellcrank and control tubes for damage, chafing, corrosion, and security of attachment.		
	7.2. Inspect friction shoes and liners for condition.		
	7.3. Check collective lever friction adjuster for operation.		
	7.4. Check for proper collective minimum friction.		
	7.5. Check collective flight controls for smooth movement throughout full range of travel.		
	8. Cyclic flight controls:		
	8.1. Inspect flight control systems bellcrank and control tubes for damage, chafing, corrosion, and security of attachment.		
	8.2. Check cyclic stick friction adjuster for proper operation.		
	8.3. Check for proper cyclic minimum friction.		
	8.4. Check cyclic flight controls for smooth movement throughout full range of travel.		
	9. Anti-torque flight controls:		
	9.1. Flight control systems bellcrank and control tubes for damage, chafing, corrosion, and security of attachment.		
	9.2. Check for proper anti-torque minimum friction.		
	9.3. Check anti-torque flight controls for smooth movement throughout full range of travel.		
412-A-76-00-00-00A-009A-A / 01722 Pratt & Whitney Canada PT6T-3 Series Maintenance Manual	10. Engine: Perform maximum N_G topping check. **NOTE** Maximum N_G topping check is not applicable to Pratt & Whitney PT6T-9 engine.		
412-A-76-00-00-00A-009A-A / 01722	11. Engine controls:		
	11.1. Check collective lever throttle friction adjusters for proper operation.		
412-A-28-00-00-00A-009A-A / 00323	12. Inspect all exposed fuel lines and attachments for leakage, damage, and security.		
412-A-71-00-00-00A-009A-A / 01584			
412-A-29-00-00-00A-009A-A / 00610	13. Hydraulic system:		

Table 1. 600 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	NOTE Hydraulic filter elements 205-076-034-003 are non-cleanable. It is recommended they be replaced every 600 hours or 12 months. Filter elements 205-076-034-007 are cleanable. The 212-076-006-007 assembly must have -007 element only. The -003 and -007 elements are interchangeable on the assembly 212-076-006-105; intermixing of different element types is not permitted.		
	13.1. Remove and inspect hydraulic filter elements.		
	13.2. Discard or clean filter elements as applicable.		
	13.3. Install hydraulic filter elements.		
412-A-63-00-00-00A-009A-A / 00987	13.4. Install 0 to 500 PSI (0 to 3447 kPa) gauge in quick disconnect at rotor brake caliper. Slowly pull rotor brake handle through travel. Cylinder should generate 240 to 260 PSI (1655 to 1793 kPa) at any handle velocity and shall maintain 150 PSI (1034 kPa) minimum in over center position for 3 minutes. Adjust as necessary.		
	13.5. Inspect brake linings for wear (minimum thickness 0.150 inch (3.81 mm)).		
412-A-53-00-00-00A-009A-A / 00897	14. Fuselage cabin structure:		
	14.1. Fuselage structure for corrosion and damage to finish and sealant. Pay particular attention to edges of joints and seams and around fastener heads.		
	14.2. Ensure all drain holes are open.		
	14.3. Engine compartment floor and service deck, FS 155.06 to 241.22, for damage, delamination, and corrosion.		
	14.4. Fuselage bonded panels for damage and delamination.		
	NOTE To adequately inspect the following, it will be necessary to remove the auxiliary fuel tanks, if installed.		
	14.5. Pylon wall for damage, working fasteners, cracks, delamination, and corrosion.		
412-A-96-00-00-00A-009A-A / 01870	15. Electrical system:		
	15.1. Check inverters for security of mounting and connections.		
	15.2. Check generator control units for damage and security of mounting and terminals.		
	15.3. Inspect all exposed wire bundles, bundle supports, and connectors for damage, corrosion, and security. Pay particular attention to engine deck connectors and harnesses.		
	15.4. Inspect relays and main bus area for security of mounting and connections.		
	15.5. Inspect bus insulation for damage and condition.		

Table 1. 600 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-04-00-00-00A-288A-A / 00013 412-A-26-00-00-00A-009A-A / 00291 412-A-96-00-00-00A-009A-A / 01870	16. Fire extinguishing system:		
	16.1. Functionally check voltage of engine fire extinguishing circuits.		
	16.2. Replace engine fire extinguisher container firing cartridges in accordance with specified service life.		
ASB 412-12-152	16.3. Visually inspect the loop clamp installed on modified discharge tubes 212-060-921-001FM and -002FM for condition and security until Part III of referenced ASB is accomplished.		
412-A-28-00-00-00A-009A-A / 00323	17. Remove four access doors, located just forward of the aft crosstube, from the underside of the fuselage. Inspect fuel lines for evidence of leakage, damage, and security.		
412-A-62-00-00-00A-009A-A / 00932	18. Swashplate and support assembly - hub and sleeve assembly:		
	18.1. Disconnect drive and swashplate links from swashplate outer ring.		
	18.2. Disconnect pitch links from main rotor hub pitch horns.		
	18.3. Disconnect the collective pitch drive plate set.		
	18.4. Check swashplate duplex bearing and collective sleeve hub bearing set for roughness and ease of rotation.		
	18.5. Reconnect the collective pitch drive plate set.		
	NOTE It is not necessary to check the shim thickness if the original collective pitch drive plate set is being reinstalled.		
	18.6. Purge lubricate bearings.		
	18.7. Reconnect drive and swashplate links and tubes.		
	NOTE The following step is to be performed every 12 months.		
	19. Inspect all fuel system, oil system, and hydraulic system filler caps for proper functioning and sealing. Make sure the sealing O-rings within the filler caps are in good condition. Repair or replace the filler caps or replace sealing O-rings as required.		
	TAILBOOM		
412-A-53-00-00-00A-009A-A / 00897	1. Tailboom structure for corrosion and damage to finish and sealant. Pay particular attention to edges of joints and seams and around fastener heads.		
	NOTE For helicopters equipped with the Dual Tailboom Strakes and FastFin™ system Supplemental Type Certificate (STC) , refer to the applicable BLR Aerospace documents for inspection criteria.		

Table 1. 600 Hour/12 Month Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	BLR FastFin Kit: 2. Inspect upper and lower tail fairings for visible damage to the exterior skin and cracks at the attachment points. If paint is missing locally or has scratches, inspect the fairing for delamination and voids with a tap hammer. Inspect fairings for loose or missing attaching hardware.		
	BLR FastFin Kit: 3. Inspect upper and lower strakes for visible damage, deformation, cracks at the attachment points, and loose or missing rivets.		
412-A-26-10-00-03A-320A-A / 00307	4. Functionally check baggage compartment smoke detector.		
412-A-67-00-00-00A-009A-A / 01319	5. Aerodynamically actuated elevator:		
	5.1. Remove left and right elevators.		
	5.2. Clean elevator spars and inspect for corrosion.		
	5.3. Clean internal bore of elevator horn and inspect for corrosion.		
	5.4. With elevator control removed, attach a spring scale to hole in arm of horn where control tube is normally installed. Pull scale at right angle to arm, verify friction is 26 to 32 pounds (115.65 to 142.34 N). If friction is not within limits, adjust shim thickness.		
	5.5. Install left and right elevator.		

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Scheduled Inspections - 5000 Hour/5 Year Inspection - Part B

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-172	AR	
Grease	C-561	AR	

Procedure

Table 1. 5000 Hour/5 Year Inspection - Part B

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE For helicopters on the Part B inspection program, accomplish the following checks each 5000 hours flight time or 60 months, whichever occurs first. This inspection is meant to be a full visual inspection of the airframe primarily for environmental damage and corrosion.		
	GENERAL 1. Each listed inspection item or maintenance function is to be performed in accordance with the BHT-412-MM chapter specified or the BHT-412-CR&O.		
	PRELIMINARY REQUIREMENTS 1. All work accomplished during this inspection shall be recorded in the helicopter maintenance records. 2. Remove all panels, interior coverings, fairings, and cowlings required to satisfactorily carry out the inspection. 3. Remove the following items: 3.1. Main battery. 3.2. Electrical and avionics equipment from the nose area, as required, to allow a full visual inspection of the nose area to be performed. 3.3. Main rotor transmission. 3.4. Intermediate gearbox. 3.5. Tail rotor gearbox. 3.6. Tailboom (to allow a full inspection of the aft fuselage bulkhead and forward tailboom bulkhead). 3.7. Landing gear assembly (to allow a complete inspection of the landing gear attachment points and support structure).		
412-A-71-61-00-01A-520A-A / 01701	*3.8. Air management system ejectors.* 4. In addition to the 5000 Hour/5 Year Inspection - Part B items, do the following:		

Table 1. 5000 Hour/5 Year Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
SN: 33001-33213 or 36001-36999: 412-A-05-40-00-04A-281A-A / 00026 SN: 37002-37999: 412-MMS-EPI SN: 38001-38999 or 39101-39999: 412-MMS-EPX	4.1. Complete 25 Hour/30 Day Inspection - Part B.		
SN: 33001-33213 or 36001-36999: 412-A-05-40-00-05A-281A-A / 00027 SN: 37002-37999: 412-MMS-EPI SN: 38001-38999 or 39101-39999: 412-MMS-EPX	4.2. Complete 300 Hour/12 Month Inspection - Part B.		
SN: 33001-33213 or 36001-36999: 412-A-05-40-00-06A-281A-A / 00028 SN: 37002-37999: 412-MMS-EPI SN: 38001-38999 or 39101-39999: 412-MMS-EPX	4.3. Complete 600 Hour/12 Month Inspection - Part B.		
	AFTER COMPLETION OF INSPECTION		
	1. Install the following items:		
	1.1. Main battery.		
	1.2. Electrical and avionics equipment on the nose area.		
	1.3. Main rotor transmission.		
	1.4. Intermediate gearbox.		
	1.5. Tail rotor gearbox.		
	1.6. Tailboom.		
	1.7. Landing gear assembly.		
412-A-71-61-00-01A-720A-A / 01704	*1.8. Air management system ejectors.*		
	2. Install all panels, interior coverings, fairings, and cowlings removed during inspection.		
	3. Do a ground run and leak check before you return the helicopter to service.		
412-A-53-00-00-00A-009A-A / 00897	FORWARD FUSELAGE		
	1. Nose panels in area of pitot tube attachment, areas inside and outside for disbonding and cracking.		
	2. Pitot tubes attachment for looseness.		

Table 1. 5000 Hour/5 Year Inspection - Part B (continued)

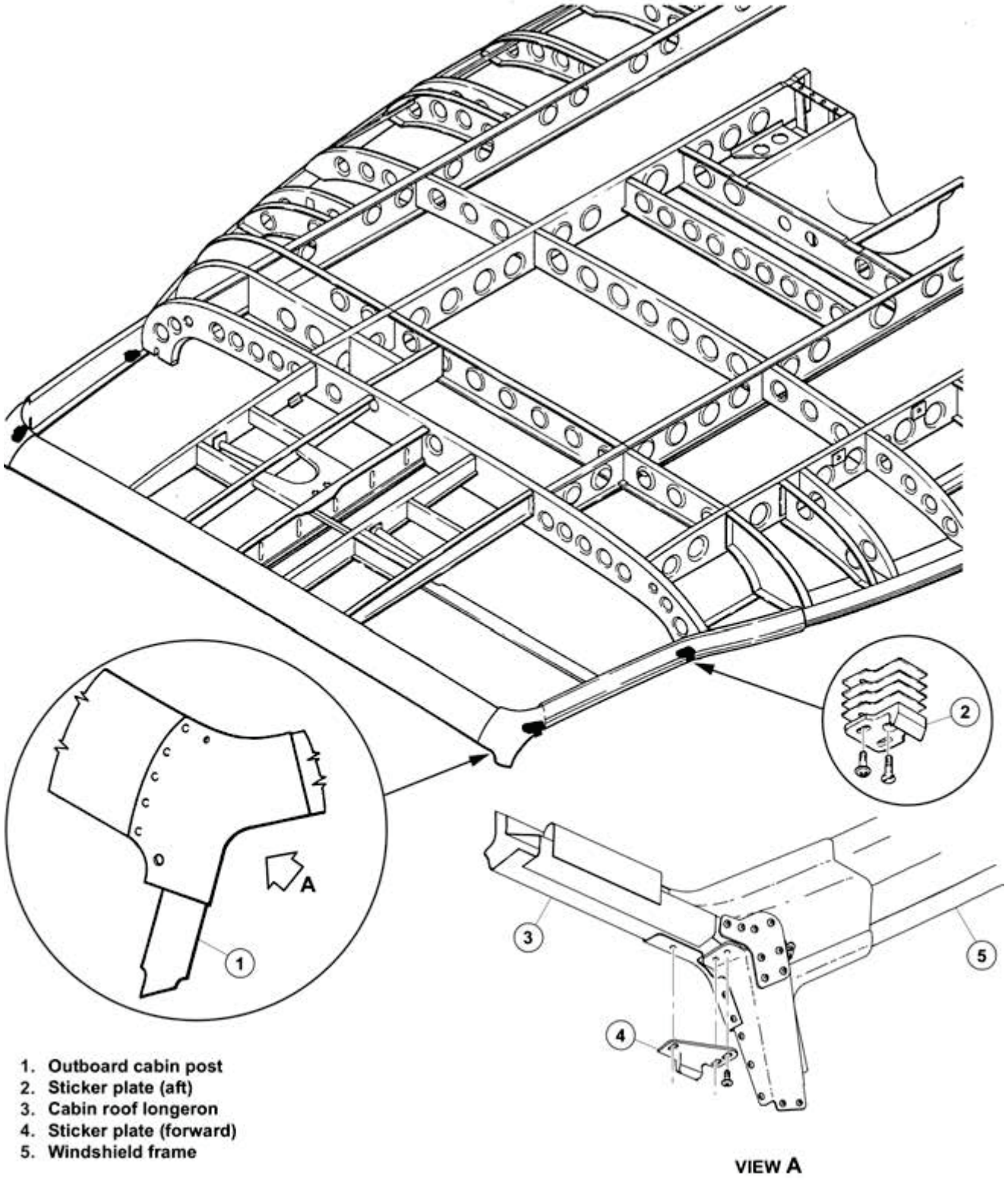
DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	3. Nose shelves for damage, corrosion, and disbonding.		
	4. Nose shelf attachment locations for damage, corrosion, and cracking.		
	5. Right and left BL 14.00 main beams, forward of STA 23.00 for damage, corrosion, and cracking.		
	6. Instrument panel attachment structure for damage, corrosion, and cracking, primarily in the door frame attachment area.		
	7. Instrument panel glareshield for chafing and cracking.		
	8. Crew door hinge attachment structure for damage and cracks.		
	9. Outboard cabin post (1, Figure 1), cabin roof longeron (2), and windshield frame (5) upper corner area for damage and cracks.		
	10. Striker plates (3 and 4) attachment areas for damage and cracks.		
	11. Striker plates (3 and 4) for excessive wear, damage, and corrosion.		
	12. Cabin floor area under crew doors for chafing and damage.		
	13. Crew door hinges for wear, damage, corrosion, and security of attachment.		
	14. Crew doors for proper fit and operation.		
	15. Crew seat rails for excessive wear and damage.		
	16. Center and outboard bulkhead, below WL 22.00 at FS 23.00, for cracks, deformation, and corrosion along structure joints.		
	17. Collective and cyclic control jackshafts support intercostals for cracks and corrosion.		
	18. Cyclic support for corrosion and cracks.		
	19. Collective, cyclic, and anti-torque control system bellcranks, levers, and support brackets for corrosion, cracks, and indication of wear at attaching points.		
	20. Crew seat support beams, at BL 14.00 and 30.00 between FS 23.00 and 74.30, for cracks and corrosion.		
	21. Forward crosstube attachment points and support structure at FS 63.33 and 74.30 for damage, corrosion, and cracks.		
	22. Aft crosstube attachment points and support structure at FS 155.06 and 166.00 for damage, corrosion, and cracks.		
	23. Crew and passenger cabin floor, FS 23.00 to 155.06, for corrosion and damage. Floor bonded panels for voids. Seat fasteners and cargo tie-down rings for condition. Area around pilot and copilot ICS foot switches for cracks.		
	NOTE In the following step, removal of the fuel cells is not required unless there is evidence of fuel leakage or damage to the fuel cells or structure is suspected.		
	24. All structure under floor panels and in forward fuel cell cavities including caps, angles, transverse bulkheads and bonded panels between FS 74.30 and 155.06 for damage, corrosion, cracks, and voids.		
	NOTE In the following step, removal of the fuel cells is not required unless there is evidence of fuel leakage or damage to the fuel cells or structure is suspected. Structure that is hidden by the fuel cells only needs to be inspected if the fuel cells are removed for some reason or structural damage is suspected.		

Table 1. 5000 Hour/5 Year Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	25. All structure under floor panels and in aft fuel cell cavities including caps, angles, and panels between FS 155.06 and 200.00 for damage, corrosion, cracks, and voids.		
	26. Bottom skin, FS 23.00 to 243.937, for damage. Joints for corrosion. Access panels and covers for condition and attachment.		
	27. Main transmission compartment interior, FS 129.00 to 166.00 and WL 7.44 to 76.00, for corrosion. Bulkheads and main beam panels for damage and corrosion. Attaching supports and brackets for condition and security.		
	28. Fluid lines in transmission compartment for evidence of leaks, condition, wear, and chafing. Clamps for condition (fluid contamination and swelling), and security of attachment.		
	29. Lift beam fitting for condition and working rivets, lift link attachment bolt hole bores for excessive wear, and bushings for looseness.		
	30. Pylon (transmission) support structure for wear and cracks. Pay particular attention to right and left horizontal supports, forward vertical supports, and aft vertical webs.		
	31. Aft cabin bulkheads, FS 166.00, for damage, upper aft flange for cracks, and bulkhead panel for damage and disbonding.		
	32. Electrical/avionics compartment, FS 166.00 to 243.937, doors for condition and security. Door frames for cracks and corrosion. Compartment interior for condition and joints for corrosion. Skin for damage and cracks.		
	33. Oil cooler compartment, FS 166.00 to 243.937, doors for condition and security. Main beam panels and bulkheads for condition and corrosion.		
	34. Engine deck below the reduction gearbox, FS 211.00 to 238.00 for damage, cracks, and corrosion.		
	35. Inspect the ejectors shim AAL-238-011 (if installed) for damage, cracks, delamination and proper bonding to the firewall. If any damage is found, replace with new shim (412-A-71-61-00-01A-720A-A / 01704).		
	36. Firewalls, FS 155.06 to 241.22, for damage, cracks, security of attachment, seals for condition, and seal attachment angles for cracks.		
	37. Cabin roof, FS 35.00 to 166.00 above WL 68.00, BL 50.00 left and right side, for damaged skin and joints for corrosion, bonded panels for disbonding and corrosion. Structure above overhead console (circuit breaker panels) for cracks and corrosion.		
	38. Cargo door tracks for wear, cracks, and corrosion.		
	39. Engine and transmission cowlings and fairings for evidence of chafing, cracks, corrosion, and seals for condition.		
	40. Bulkhead, FS 241.22, for damage, corrosion, and cracks. Tailboom attachment fittings for damage, cracks, and loose fasteners. Tailboom attachment bolt holes for damage and corrosion.		
ASB 412-21-187	41. Discard the upper left-hand attachment bolts.		
	42. Do a detailed visual inspection of the remaining three attachment bolts for mechanical or corrosion damage. Any corrosion including surface corrosion is cause for rejection. Damaged threads and detectable wear or mechanical damage on the bolt shank or bolt head radii are cause for rejection.		
	43. Apply a thin layer of grease (C-561) or grease (C-172) on the shanks of the bolts only. Both threads are to remain dry.		

Table 1. 5000 Hour/5 Year Inspection - Part B (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-53-00-00-00A-009A-A / 00897	44. Install tailboom after completion of inspection with a new upper left-hand bolt and serviceable bolts at the three other locations.		
412-A-53-00-00-00A-009A-A / 00897	TAILBOOM 1. Bulkhead, BS 17.42, for corrosion and cracks. Fuselage attachment bolt holes for damage and corrosion. Barrel nuts for damage and corrosion. 2. Tailboom and fin driveshaft covers for condition and attachment. 3. Tailboom and fin external skin for damage, corrosion, and cracks. 4. Tailboom access panels and covers for condition and attachment. Tail rotor driveshaft cover hinges for wear. 5. Tailboom baggage compartment door for condition, attachment, and door handle operation. 6. Tailboom baggage compartment interior for damage, panels for disbonding, and corrosion. Longerons above roof panel for corrosion and cracks. Longerons below floor panel for corrosion and cracks. 7. Tailboom interior bulkheads and stringers for cracks and corrosion. 8. Intermediate gearbox mount pad for damage, cracks, fretting, and corrosion. 9. Tail rotor gearbox support fitting assembly for damage, cracks, fretting, and corrosion. Pay particular attention to mounting holes and lugs. 10. Tailboom exterior bonded panels adjacent to baggage compartment for damage, delamination, and corrosion. 11. With a borescope or equivalent, inspect interior structure of vertical fin for cracks, corrosion, fretting, and damage. 12. Control tube fairleads (grommets and guides) for wear and security of attachment.		



1. Outboard cabin post
2. Sticker plate (aft)
3. Cabin roof longeron
4. Sticker plate (forward)
5. Windshield frame

ICN-412-A-054000-B-97499-00002-A-001-01

Figure 1. Cabin Frame Inspection - Part B (Sheet 1 of 1)

Special Inspections - Description

Perform applicable special inspection of helicopter:

- Daily or each 10 hours, whichever occurs first until 250 hours.
- Between 1 and 5 flight hours after main rotor hub installation.
- Between 1 and 5 flight hours after tailboom installation or attachment bolt replacement.
- Between 1 and 5 flight hours after tail rotor gearbox or intermediate gearbox installation.
- Between 1 and 25 flight hours after expandable blade bolt installation.
- Between 5 and 10 hours of flight after tail rotor hub and blade assembly installation.
- Each 25 hours for the next four inspections.
- Each 25 hours of tail rotor operation.
- *Each 25 hours tailboom attachment inspection.*
- 50 hours after installation of components.
- *Each 50 hours tailboom attachment inspection.*
- Each 100 hours tailboom vertical fin spar cap inspection.
- Each 100 hours of collective lever operation.
- Each 150 hours of starter generator (200SG119Q) operation.
- Each 300 hours or 6 months of expandable blade bolt operation.
- Each 600 hours of tail rotor driveshaft operation or 12 months, whichever occurs first.
- Each 600 hours or 12 months inspection of tailboom attachment bolts.
- Each 600 hours of main driveshaft operation or 12 months, whichever occurs first.
- Each 600 hours or 12 months inspection of magnetic brake assembly.
- Each 1000 hours of component operation.
- Each 12 months or 2500 landings of aft high crosstube operation.
- Each 24 months of control bolt operation.
- Each 24 months of main rotor mast operation.
- Each 2500 hours of main rotor hub assembly operation.
- Each 2500 hours of main rotor blade operation.
- Each 2500 or 3000 hours or 5 years of main rotor mast (412-040-366-109 and Subsequent) operation.
- Each 2500 hours of tail rotor drive system operation.
- Each 3000 hours or 5 years of main rotor mast (412-040-366-103 and -105) operation.
- Each 3000 hours of transmission (412-040-002) operation.
- Each 3000 hours of transmission (412-040-004/412-040-007/412-040-802) operation.
- 3600 hours total airframe time and each 300 hours or 12 months, whichever occurs first thereafter.
- Each 3000 hours or 5 years of transmission mount assembly (204-031-927) and friction damper assembly (204-031-920) operation.
- Each 2500 Hours or 10 years Inspection of Collective Stick Tube.

NOTE

Torque check is the term used when the specified torque or standard torque plus tare torque is applied to a fastener, in a tightening direction. The specified or standard torque plus tare torque would have to be previously recorded. However, if the specified or standard torque and tare were not recorded, use the minimum specified or standard torque, plus the tare torque listed in the BHT-ALL-SPM, Chapter 2, Standard Practices Manual.

Looseness may occur until components seat themselves and fasteners require tightening. This is not cause for disassembly, however the fastener must be torque checked again at the same scheduled interval set for the first torque check until the assembly is completely seated.

For additional information, refer to the BHT-ALL-SPM, Standard Practices Manual. If any confusion exists, or a clarification is desired, contact Product Support Engineering.

Use the following time frame criteria when accomplishing the required maintenance manual torque checks after installation of components.

MANUAL SPECIFIES	SUGGESTED TIME FRAME TO ACCOMPLISH
100 hours	90 to 110 hours
50 hours	40 to 60 hours
25 hours	20 to 30 hours
10 hours	5 to 15 hours

Special Inspections - Daily/10 Hour Inspection, Whichever Occurs First Until 250 Hours

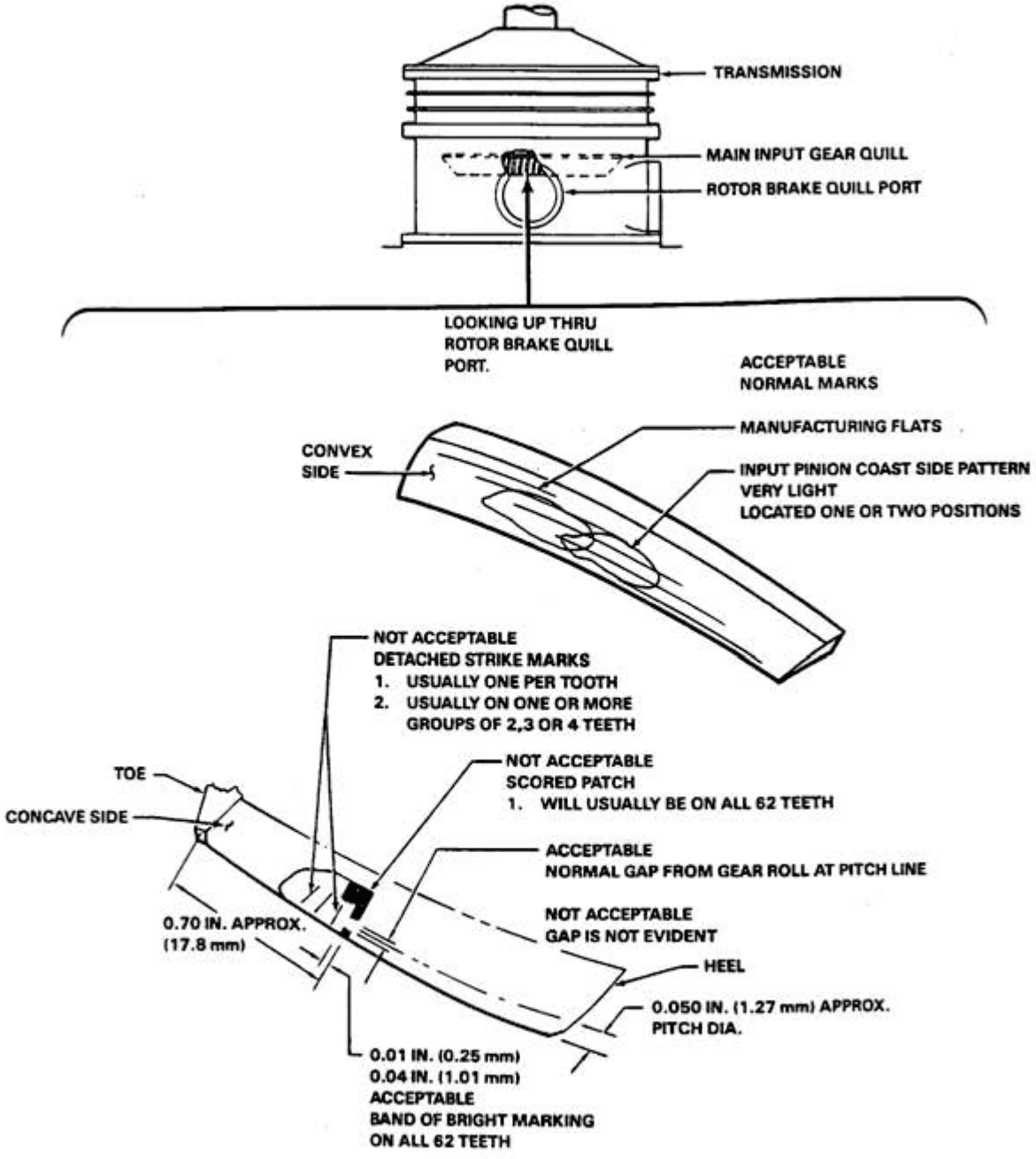
Procedure

Table 1. Daily/10 Hour Inspection, Whichever Occurs First Until 250 Hours

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish daily or each 10 hours, whichever occurs first until 250 hours, on transmissions with affected gears having 50 hours or more.		
	SPIRAL BEVEL GEAR (204-040-701-103)		
ASB 412-89-42	Affected helicopter S/N 33001 through 33138, 33150 through 33160, 33168 through 33175, 33177, 33178, 33179, 33181, 33182, 33185, 33186, and 33192.		
	Spare (uninstalled) spiral bevel gears 04-040-701-103, affected serial numbers:		
	A-3819, A-3821, A-3825, A-3826, A-3829, A-3833, A-3836, A-3834, A-3838, A-3840, A-3847, A-3848, A-3850, A-3855, A-3856, A-3857, A-3858, A-3861, A-3878, A-3880, A-3885, A-3886, A-3889, A-3891, A-3892, A-3893, A-3895, A-3896, A-3897, A-3899, A-3901, A-3911, A-3915, A-3916, A-3919, A-3920, A-4008, A-4014, A-4017, A-4019, A-4020, A-4021, A-4027, A-4029, A-4068, A-4069, A-4071, A-4075, A-4077, A-4079, A-4080, A-4081, A-4083, A-4084, A-4085, A-4087, A-4089, A-4090, A-4091, A-4092, A-4093, A-4094, A-4095, A-4096, A-4097, A-4098, A-4107, A-4108, A-4109, A-4147, A-4184, A-4187, A-4188, A-4191, A-4192, A-4193, A-4213, A-4233, A-4235, A-4236, A-4241, A-4243, A-4244, A-4245, A-4254, A-4266, A-4267, A-4274, A-4275, A-4280, A-4282, A-4288, A-4289, A-4290, A-4303, A-4319, A-4320, A-4325, A-4327, A-4328, A-4329, A-4332, A-4333, A-4334, A-4335, A-4336, A-4337, A-4358, A-4366, A-4368, A-4369, A-4370, A-4371, A-4372, A-4374, A-4376, A-4377, A-4378, A-4379, A-4380, A-4383, A-4385, A-4386, A-4387, A-4394, A-4395, A-4397, A-4400, A-4401, A-4403, A-4411, A-4417, A-4418, A-4428.		
	NOTE Spiral bevel gears with the suffix R after the serial number are acceptable for service.		
	Transmissions/helicopters delivered with an affected serial numbered spiral bevel gear 204-040-701-103		
	Bevel Gear: A-3843, A-3863, A-3870, A-3876, A-3898, A-3900, A-4026, A-4037, A-4229, A-4231		
	Transmission: A-114, A-122, A-124, A-125, A-127, A-128, A-129, A-132, A-133, A-140		
	Model 412: 33175, 33177, 33178, 33179, 33181, 33182, 33185, 33186, SPARE, 33192		

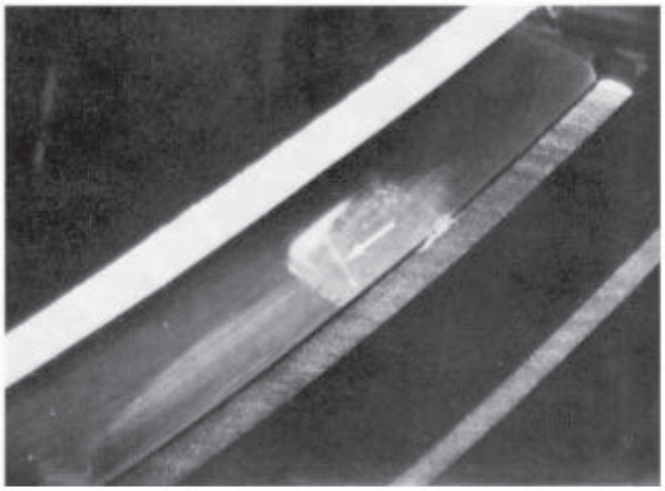
Table 1. Daily/10 Hour Inspection, Whichever Occurs First Until 250 Hours (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-63-00-00-00A-009A-A / 00987 BHT-412-CR&O	1. Remove and inspect transmission internal sump filter for metal contamination. If metal contamination is evident, notify Product Support Engineering of spiral bevel gear serial number, hours, and type contamination. 2. Every 50 hours (until 250 hours), visually inspect spiral bevel gear. 2.1. Remove quill pad cover (204-040-174-001) or rotor brake quill (if installed) from transmission. 2.2. Utilizing the port opening in the main gear case and a bright light and mirror, inspect all 62 teeth in spiral bevel gear 204-040-701-103. Carefully inspect each tooth for evidence of chipping or loss of material (Figure 1). 2.3. Remove and replace any gear not meeting inspection requirements set forth in step b. Notify Product Support Engineering of serial number and total time of any gear replaced. 2.4. Install quill pad cover (or rotor brake quill). 3. Return spare (uninstalled) affected serial numbered spiral bevel gears to Bell Helicopter Textron for inspection and re-identification.		

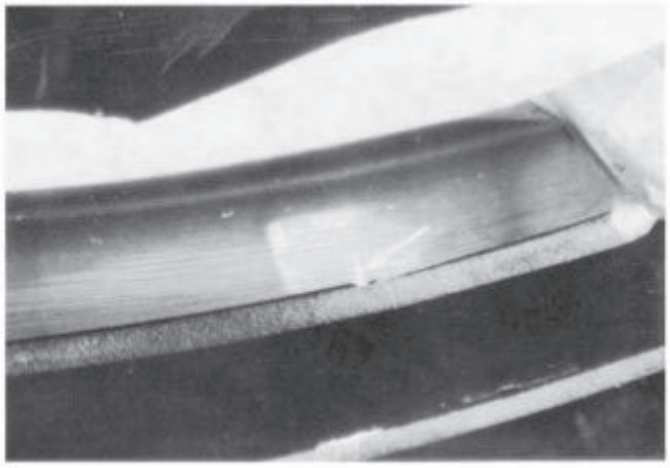


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Figure 1. Spiral Bevel Gear Inspection (Sheet 1 of 3)



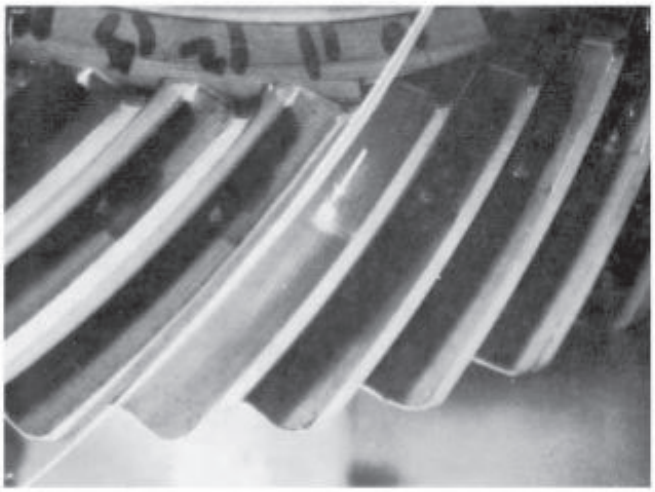
204-040-701 AND 412-040-701 SPIRAL BEVEL GEAR UNACCEPTABLE PATTERN



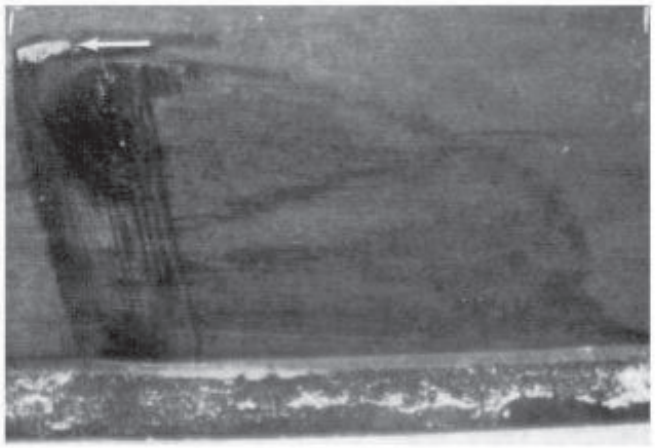
204-040-701 AND 412-040-701 SPIRAL BEVEL GEAR UNACCEPTABLE PATTERN

ICN-412-A-054700-B-97499-00002-A-001-01

Figure 1. Spiral Bevel Gear Inspection (Sheet 2 of 3)



204-040-701 AND 412-040-701 SPIRAL BEVEL GEAR UNACCEPTABLE PATTERN



204-040-701 AND 412-040-701 SPIRAL BEVEL GEAR UNACCEPTABLE PATTERN

ICN-412-A-054700-B-97499-00003-A-001-01

Figure 1. Spiral Bevel Gear Inspection (Sheet 3 of 3)

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Special Inspections - Between 1 and 5 Flight Hours After Main Rotor Hub Installation Procedure

Table 1. Between 1 and 5 Flight Hours After Main Rotor Hub Installation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____ **NOTE** Accomplish between 1 and 5 hours of flight after installation of main rotor hub.		
412-A-62-00-00-00A-009A-A / 00932	MAIN ROTOR HUB 1. Torque check eight main rotor hub/yoke through bolt nuts to 70 to 80 foot-pounds (95 to 108 Nm). **CAUTION** **SPECIAL TORQUE SEQUENCE REQUIRED FOR FOLLOWING STEPS.** 2. Torque check eight main rotor hub cap bolt assemblies to 250 to 260 inch-pounds (28.25 to 29.37 Nm). 3. Torque check lower cone bolts to 125 to 135 inch-pounds (14.13 to 15.25 Nm).		

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Special Inspections - Between 1 and 5 Flight Hours After Tailboom Installation or Attachment Bolt Replacement

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Torque seal	C-049	AR	

Procedure

Table 1. BETWEEN 1 AND 5 FLIGHT HOURS AFTER TAILBOOM INSTALLATION OR ATTACHMENT BOLT REPLACEMENT

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____ **NOTE** Accomplish between 1 and 5 hours of flight after installation of tailboom or replacement of attachment bolts. **NOTE** The below torque check must be repeated every 1 to 5 flight hours until torque has stabilized (three times maximum).		
412-A-53-00-00-00A-009A-A / 00897	TAILBOOM 1. Torque check the upper left bolt to 1300 to 1600 inch-pounds (147 to 181 Nm). 2. Torque check the upper right bolt to 1000 to 1200 inch-pounds (113 to 136 Nm). 3. Torque check both the bottom bolts to 400 to 430 inch-pounds (45.2 to 48.6 Nm). 4. If the torque has not stabilized after three torque checks, examine the bolt and replace the barrel nut. 5. Once the torque is stabilized, apply torque seal (C-049) witness mark to all four bolt heads.		

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Special inspections - Between 1 and 5 Flight Hours After Tail Rotor Gearbox or Intermediate Gearbox Installation

Procedure

Table 1. Between 1 and 5 Hours Of Flight After Each Installation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____ **NOTE** Accomplish between 1 and 5 hours of flight after each installation. **NOTE** The below torque check must be repeated every 1 to 5 flight hours until torque has stabilized.		
412-A-65-00-00-00A-009A-A / 01257	TAIL ROTOR GEARBOX Torque check tail rotor gearbox retaining nuts to 200 to 235 inch-pounds (22.60 to 26.55 Nm). Using a 0.005 inches (0.13 mm) or less feeler gauge, verify no gap exists between gearbox input quill shim and gearbox case shim.		
412-A-65-00-00-00A-009A-A / 01257	INTERMEDIATE GEARBOX Torque check intermediate gearbox retaining bolts to 50 to 70 inch-pounds (5.65 to 7.90 Nm).		

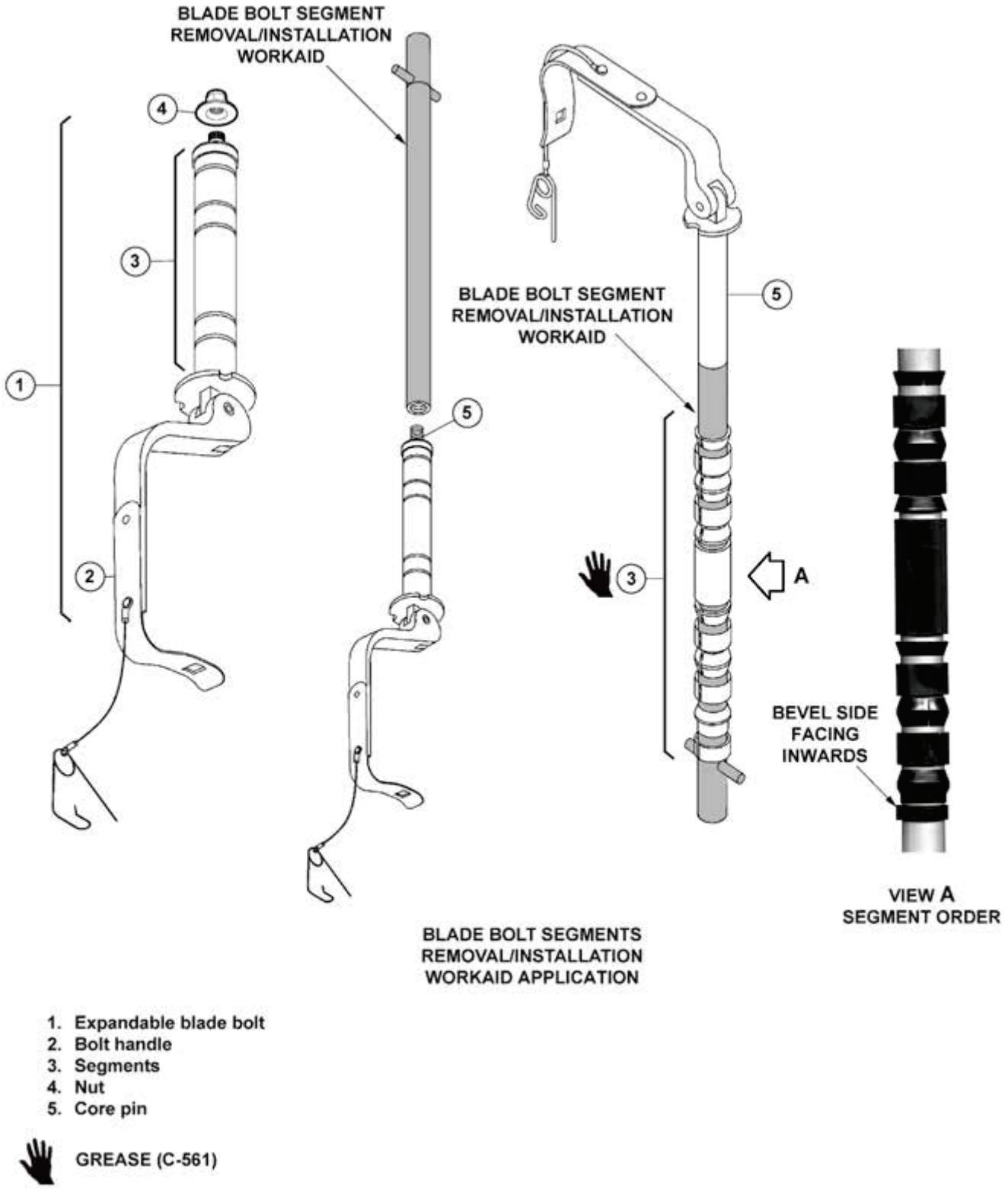
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Special Inspections - Between 1 and 25 Flight Hours After Expandable Blade Bolt Installation

Procedure

Table 1. Between 1 and 25 Flight Hours After Expandable Blade Bolt Installation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Verify expandable blade bolt handle (1, Figure 1) force at initial installation and again in 1 to 25 flight hours after installation.		
412-A-62-00-00-00A-009A-A / 00932	EXPANDABLE BLADE BOLT (412-010-137-103) 1. Remove safety lock pin from lower end of expandable blade bolt (1). 2. Raise bolt handle (2) to reduce the diameter of the bolt ring segments. 3. Push down on bolt handle (2). Verify that force required to push handle down is between 50 and 75 pounds (222 to 334 N). Refer to 412-A-62-10-00-00A-720A-A / 00946 for expandable blade bolt handle tension adjustment. 4. If adjustment is required, do an additional handle force check in 1 to 25 flight hours. In two consecutive inspections if the handle force of an individual bolt is more than the permitted limits 50 and 75 pounds (222 to 334 N) notify Product Support Engineering.		



ICN-412-A-054700-B-97499-00004-A-002-01

Figure 1. Expandable Blade Bolt - Special Inspection (Sheet 1 of 1)

Special Inspections - Between 5 and 10 Hours of Flight After Tail Rotor Hub and Blade Assembly Installation

Procedure

Table 1. Between 5 and 10 Hours Of Flight After Tail Rotor Hub and Blade Assembly Installation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE *Accomplish between 5 and 10 hours of flight after installation of tail rotor hub and blade assembly.*		
412-A-64-00-00-00A-009A-A / 01225	TAIL ROTOR HUB AND BLADE ASSEMBLY 1. Torque check tail rotor hub retaining nut to 900 inch-pounds (101.68 Nm).		
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Special Inspections - Each 25 Hours For the Next Four Inspections

Procedure

Table 1. Each 25 Hours for the Next Four Inspections

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish visual inspection of witness marks (only applicable to hanger bearing 204-040-623-003) each 25 flight hours for the next four inspections after installation or initial inspection of hanger bearings.		
ASB 412-95-84 Revision C	TAIL ROTOR HANGER BEARINGS WITNESS MARKS 1. Perform helicopter run-up for 5 minutes. Shut down helicopter and inspect tail rotor hanger bearings for movement of the dust shields, identified by breakage of the witness marks. Remove any bearings from service that display dust shield movement. 2. Hanger bearings not displaying movement at the run-up may remain in service, but must be inspected for dust shield movement at each of the next four mandatory 25 hour inspections, or any time the bearing is inspected between any of the 25 hour inspections. Bearings that do not display grease shield movement after the fourth 25 hour inspection may remain in service with no further inspection required. Record accomplishment of inspection in the helicopter log book. 3. Remove any hanger bearing from service whose witness marks indicates that grease shield has moved.		

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Special Inspections - Each 25 Hours of Tail Rotor Operation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cheesecloth	C-486	AR	
Abrasive cloth or paper	C-406	AR	
Abrasive pad	C-407	AR	

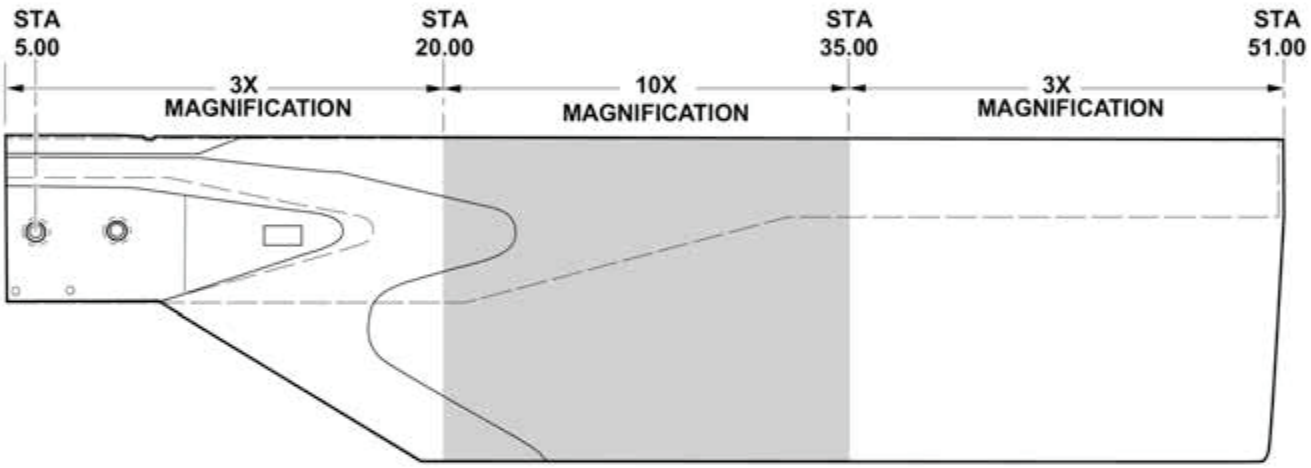
Procedure

Table 1. Each 25 Hours of Tail Rotor Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish inspection each 25 hours of flight operation.		
412-A-64-00-00-00A-009A-A / 01225 *ASB 412-96-89*	TAIL ROTOR STATIC STOP (212-011-713) Tail rotor static stop 212-011-713 for straightness across the outboard surface. If yielding has occurred, replace the tail rotor yoke and static stop. If any doubt exists as to the serviceability of these components, contact Product Support Engineering.		
412-A-64-00-00-00A-009A-A / 01225 ASB 412-13-155	TAIL ROTOR BLADE (212-010-750-ALL ONLY) **NOTE** Removal of the tail rotor blades is not required for this inspection. 1. Clean each tail rotor blade by hand using mild soap and cheesecloth (C-486) in a spanwise direction; dry thoroughly. 2. Carry out a detailed visual inspection of both sides of the entire blade with a 3X magnifying glass and a bright light. Inspect the tail rotor blade skin, leading edge spar, doublers, grip plates, and trailing edge for cracks, corrosion, nicks, dents, and scratches. Refer to 412-A-64-10-01-02A-280A-A / 01253 for damage limits. 3. Carry out a further detailed visual inspection, with a 10X magnifying glass, of the shaded area shown in Figure 1, between stations 20.00 and 35.00 of both sides of the blade, with a bright light.		

Table 1. Each 25 Hours of Tail Rotor Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	4. If any blistering, peeling, flaking, bubbling, or cracked paint is detected, remove the paint from the affected area by sanding in a spanwise direction only, with an abrasive cloth or paper (C-406) 240 grit or finer, and a final sanding with an abrasive cloth or paper (C-406) 400 grit or finer. After paint removal, visually inspect the affected area for damage with a 10X magnifying glass. Damage beyond the limits in 412-A-64-10-01-02A-280A-A / 01253 requires the blade be removed from service. 5. Only superficial corrosion that can be removed with aluminum wool or very fine grade abrasive pad (C-407) is permissible. Corrosion on the spar 0.003 inch (0.08 mm) deep or less is acceptable when polished out. Replace any blade that has corrosion on the spar greater than 0.003 inch (0.08 mm). 6. If a nick, scratch, or dent is found, visually inspect for crack using a 10X magnifying glass, or higher, and measure the depth of the damage. (A digital optical micrometer is one tool that can be used for this measurement.) 7. Repair damage and/or corrosion found on tail rotor blades in accordance with 412-A-64-10-01-02A-685A-A / 01255. 8. Replace any blade that has a crack. No cracks are permitted. 9. If paint was removed during the previous steps, carry out paint touch-up/refinishing (412-A-64-10-01-02A-257A-A / 01256).		



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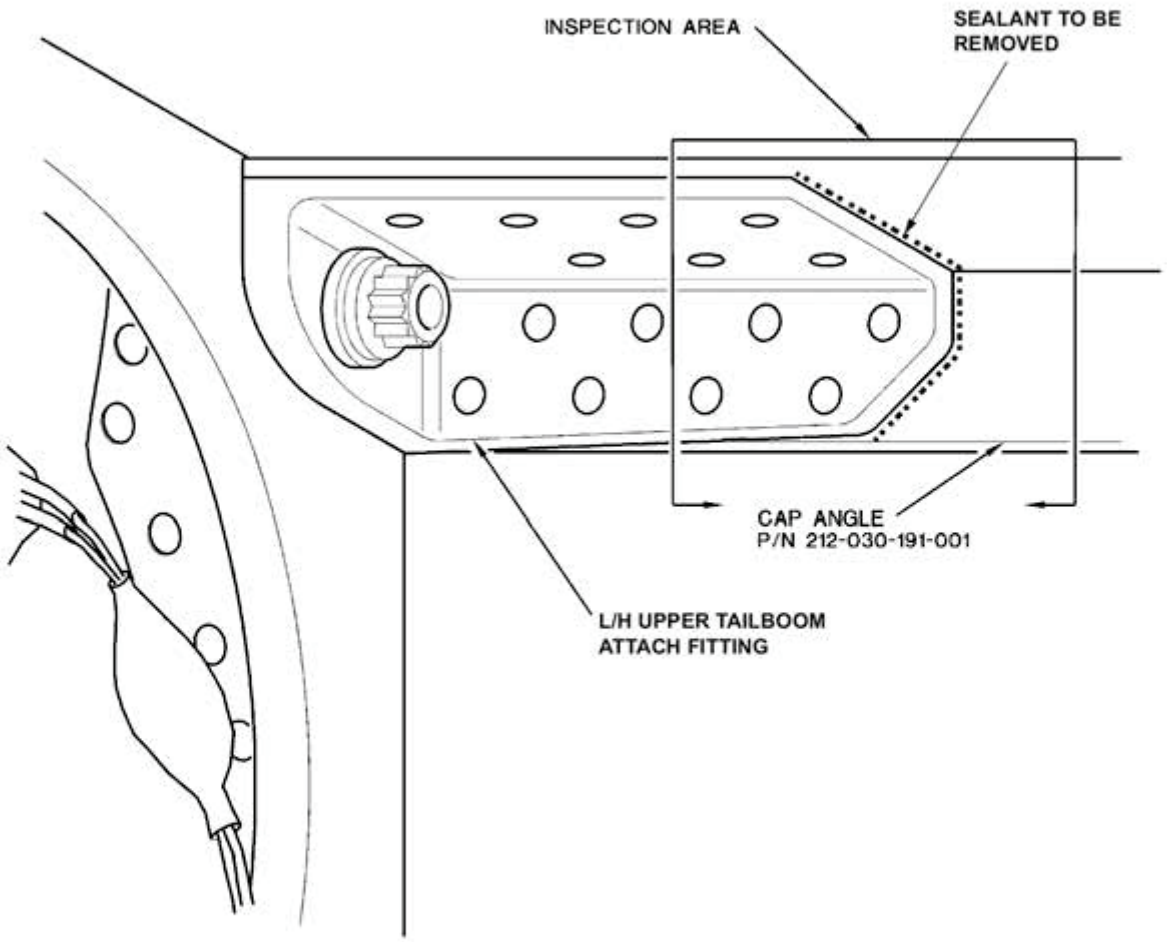
Figure 1. Tail Rotor Blade - Special Inspection (Sheet 1 of 1)

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Special Inspections - Each 25 Hours Tailboom Attachment Inspection Procedure

Table 1. ***Each 25 Hours Tailboom Attachment Inspection***

DATA REFERENCE	*INSPECTION TASK DESCRIPTION*	*INITIAL*	
		MECH	*OTHER*
	DATE: _____W.O. _____ *FACILITY: _____* *HELICOPTER S/N: _____* *REGISTRY NO.: _____* *TOTAL TIME:_____* *SIGNATURE:_____*		
	NOTE *Applicable to serial numbers 33001 through 33213, 34001 through 34036, 36001 through 36693.*		
	NOTE *Accomplish every 25 hours of helicopter operation.* *This inspection is mandated by 412-A-04-00-00-00A-288A-A / 00013 (Airworthiness Limitations Schedule), no extensions are permitted.*		
412-A-04-00-00-00A-288A-A / 00013 *ASB 412-24-197*	***FUSELAGE/TAILBOOM ATTACHMENT***  ***DO NOT DAMAGE ALUMINUM CAP ANGLE OR FITTING DURING SEALANT REMOVAL.*** 1. *If sealant is present, remove sealant from edge of fitting (Figure 1).* 2. *Inspect fuselage and tailboom attachment components and hardware for cracks with 10X magnifying glass and a bright light. Pay particular attention to interface between forward end of tailboom attachment fitting and cap angle, and main beam web area below cap angle (Figure 1).* 3. *Touch up primer if damaged during sealant removal. Paint touchup is optional. Do not reapply sealant.* 4. *If a crack is found, replace the cap angle before further flight. If the helicopter is at a location where the cap angle cannot be replaced, contact Product Support Engineering.*		



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Figure 1. Fuselage Tailboom Attachment Inspection (Sheet 1 of 1)

Special Inspections - 50 Hours After Installation of Components

Procedure

Table 1. 50 Hours After Installation of Components

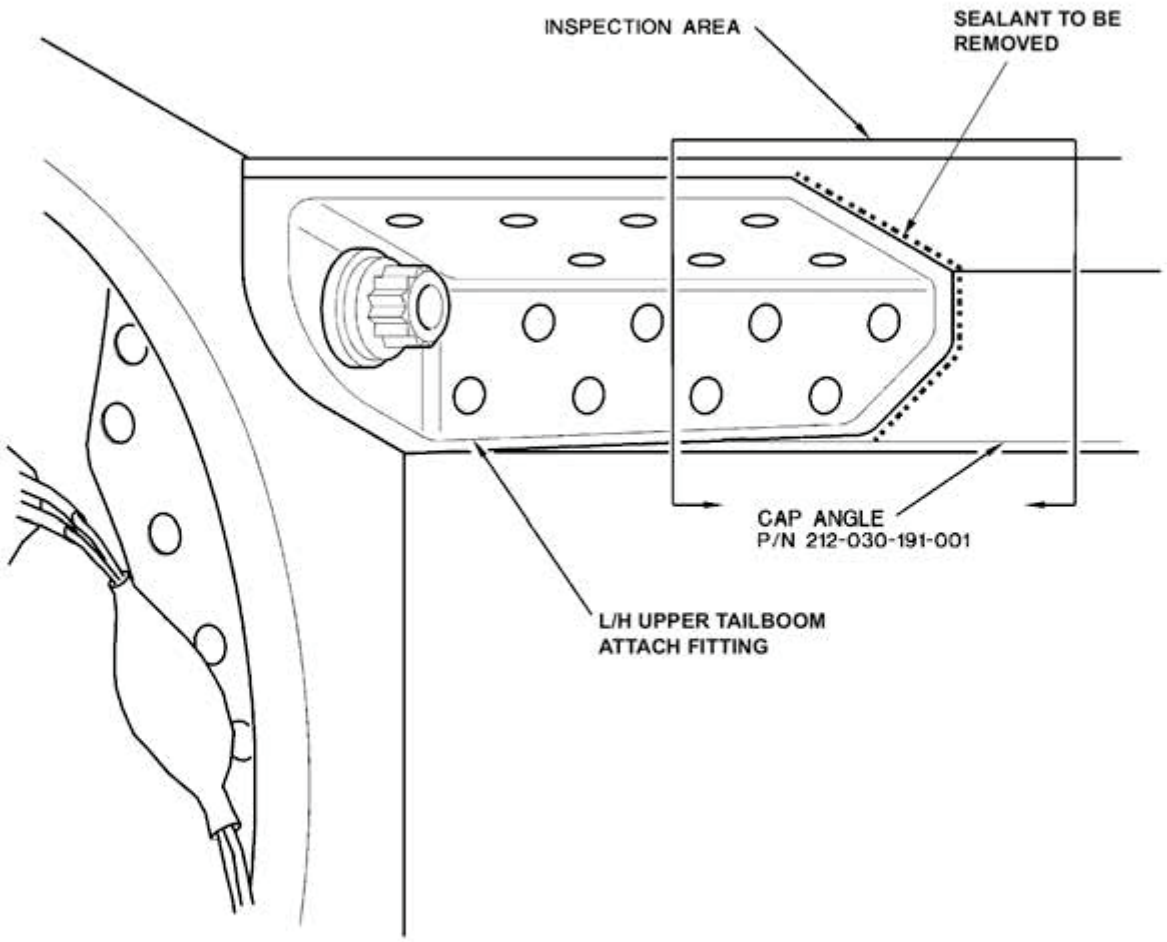
DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish 50 flight hours after each installation of components.		
412-A-53-00-00-00A-009A-A / 00897	TAILBOOM Torque check anti-torque bellcrank support (212-001-706-001, -101) attachment nuts to 75 to 95 inch-pounds (8.47 to 10.73 Nm).		
412-A-63-00-00-00A-009A-A / 00987	ROTOR BRAKE 1. Torque check brake assembly (caliper) attachment bolts to 80 to 100 inch-pounds (9.04 to 11.29 Nm). 2. Torque check brake disc attachment bolts to 50 to 70 inch-pounds (5.65 to 7.90 Nm).		

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Special Inspections - Each 50 Hours Tailboom Attachment Inspection Procedure

Table 1. Each 50 Hours Tailboom Attachment Inspection

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Applicable to Serial numbers 36694 through 36999 and 37002 through 37999.		
	NOTE Accomplish every 50 hours of helicopter operation. This inspection is mandated by 412-A-04-00-00-00A-288A-A / 00013 (Airworthiness Limitations Schedule), no extensions are permitted.		
412-A-04-00-00-00A-288A-A / 00013 ASB 412-24-197	**FUSELAGE/TAILBOOM ATTACHMENT**  **DO NOT DAMAGE ALUMINUM CAP ANGLE OR FITTING DURING SEALANT REMOVAL.** 1. If sealant is present, remove sealant from edge of fitting (Figure 1). 2. Inspect fuselage and tailboom attachment components and hardware for cracks with 10X magnifying glass and a bright light. Pay particular attention to interface between forward end of tailboom attachment fitting and cap angle, and main beam web area below cap angle (Figure 1). 3. Touch up primer if damaged during sealant removal. Paint touchup is optional. Do not reapply sealant. 4. If a crack is found, replace the cap angle before further flight. If the helicopter is at a location where the cap angle cannot be replaced, contact Product Support Engineering.		



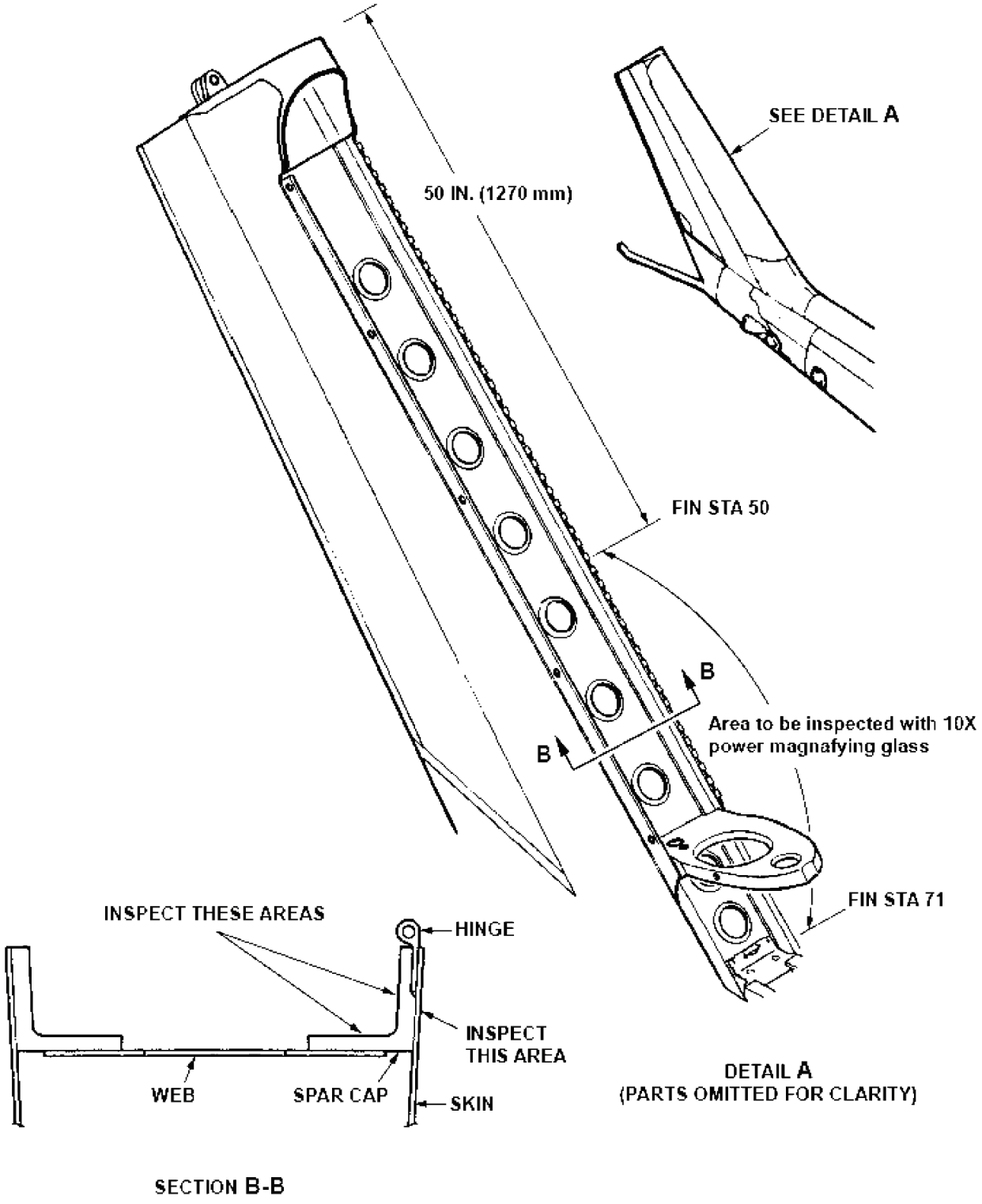
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Figure 1. Fuselage Tailboom Attachment Inspection (Sheet 1 of 1)

Special Inspections - Each 100 Hours Tailboom Vertical Fin Spar Cap Inspection Procedure

Table 1. Each 100 Hours Tailboom Vertical Fin Spar Cap Inspection

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____ **NOTE** Accomplish every 100 hours of helicopter operation. Applicable to tailboom vertical left hand fin spar cap 212-030-447-117.		
412-A-53-50-00-00A-280A-A / 00913 ASB 412-20-180	TAILBOOM VERTICAL LEFT HAND FIN SPAR CAP 212-030-447-117 1. Gain access to the spar cap angle by removing the 42-degree gearbox cover and opening the vertical fin driveshaft cover. 2. Clean exposed surfaces of spar cap to facilitate inspection. 3. Inspect both flanges of the left hand spar cap, between F.S. 50 and F.S. 71 using a 10x magnifying glass and a bright light, for cracks, loose rivets and other damage (Figure 1). 4. Inspect the exterior of the fin skin in the area in contact with the spar cap for cracks, loose rivets and/or distortion. 5. If no cracks or other damage is found, close the driveshaft cover and reinstall the 42-degree gearbox cover. 6. If a crack or other damage is found, contact Product Support Engineering before further flight: Bell Product Support Engineering Tel: 1-450-437-2862 / 1-800-363-8023 productsupport@bellflight.com.		



ICN-412-A-054700-B-97499-00933-A-001-01

Figure 1. Vertical Fin Spar Cap - Special Inspection (Sheet 1 of 1)

Special Inspections - Each 100 Hours of Collective Lever Operation

Preliminary requirements

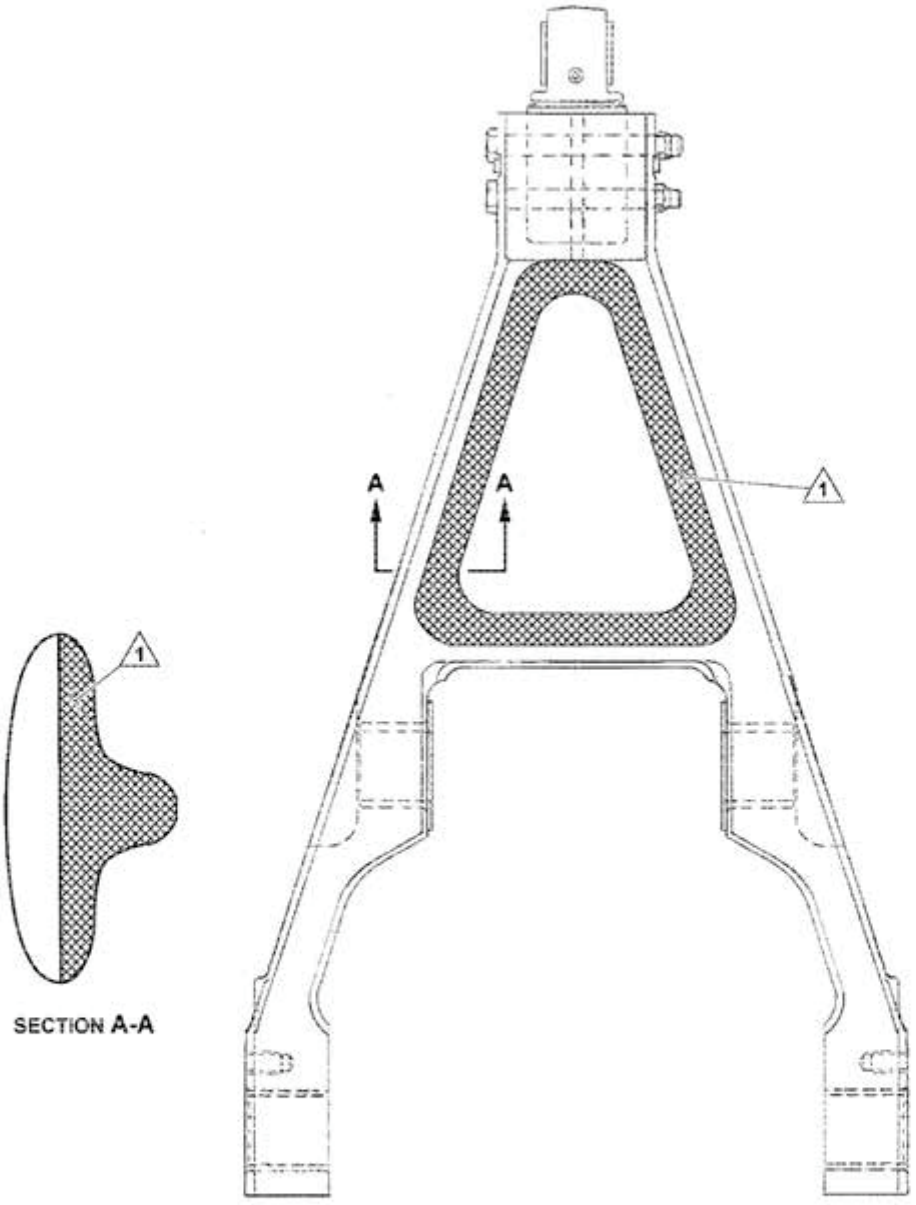
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cleaning compound	C-318	AR	

Procedure

Table 1. Each 100 Hours of Collective Lever Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
ASB 412-11-148	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____ **NOTE** Accomplish each 100 hours of collective lever 412-010-408-101 and -105 operation. **COLLECTIVE LEVER 412-010-408-101/-105** 1. Prepare helicopter for maintenance. 1.1. Gain access to the collective lever. 2. Use cleaning compound (C-318) or equivalent to thoroughly clean the collective lever in preparation for inspection. 3. Use a strong light source and a 10X magnifying glass to inspect the collective lever in the area shown in Figure 1. Due to difficult access, it is acceptable to inspect the lower side of the affected area with a strong light source and a mirror only. 4. Look for evidence of cracks in the paint. 5. If a crack is suspected, remove the collective lever from the swashplate assembly (412-A-62-30-00-08A-520A-A / 00981). 5.1. Perform a fluorescent penetrant inspection with local paint and primer removed in the suspected area (BHT-ALL-SPM, Chapter 6, Standard Practices Manual) 6. If a crack is found, replace the collective lever with a serviceable part and contact Product Support Engineering to report the defect. 7. If no cracks are found, refinish as required per the 412-CRO.		



NOTE

1 Area to be inspected with a 10X power magnifying glass. Due to difficult access, it is acceptable to inspect the lower side of the affected area with a strong light source and a mirror only.

ICN-412-A-054700-B-97499-00006-A-001-01

Figure 1. Collective Lever - Special Inspection (Sheet 1 of 1)

Special Inspections - Each 150 Hours of Starter Generator (200SG119Q) Operation

Procedure

Table 1. Each 150 Hours of Starter Generator (200SG119Q) Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____ **NOTE** Accomplish each 150 hours of starter generator 200SG119Q operation or more frequently if conditions warrant. Part number 200SG119Q refers to the starter generator without the QAD kit consisting of the mounting flange and ring clamp. This special inspection is not applicable to starter generator 209-060-221-001.		
412-A-71-01-00-11A-520A-A / 01694	1. Remove starter generator 200SG119Q.		
	2. Inspect starter generator brushes for wear.		
TM106 - Aircraft Parts Corp. Overhaul Instructions with Illustrated Parts Breakdown	3. If brushes are worn beyond allowable limits, refer to vendor manual and replace in accordance with manufacturer's instructions.		
	4. Install starter generator.		

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Special Inspections - Each 300 Hours or 6 Months of Expandable Blade Bolt Operation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Isopropyl alcohol	C-385	AR	
Abrasive cloth or paper	C-423	AR	
Solid film lubricant	C-021	AR	
Grease	C-561	AR	

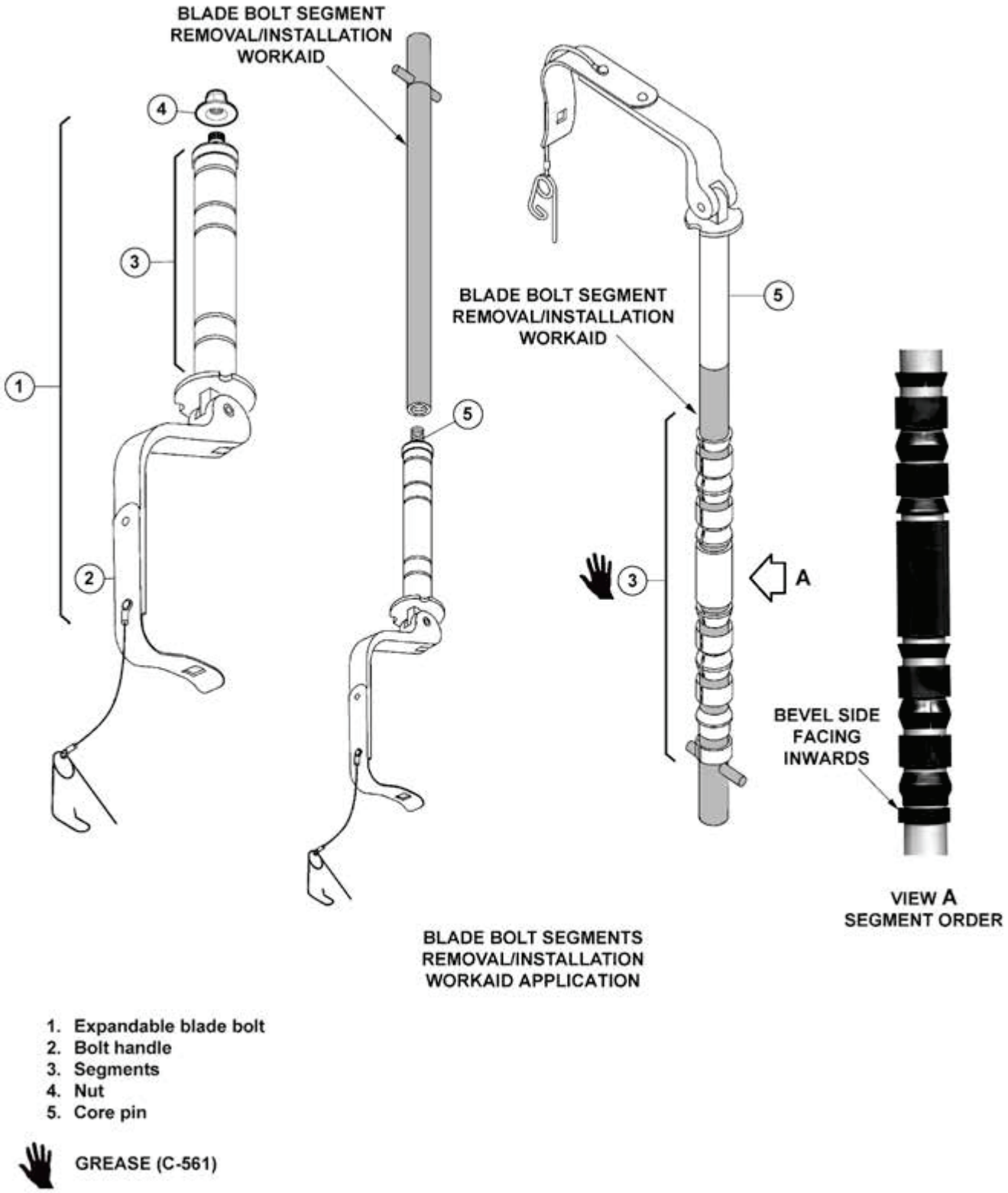
Procedure

Table 1. Each 300 Hours or 6 Months of Expandable Blade Bolt Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish each 300 hours or 6 months of main rotor expandable blade bolt operation.		
ASB 412-09-137	EXPANDABLE BLADE BOLT (412-010-137-103)		
	 **EXPANDABLE BOLT HANDLE SHALL NOT BE CLOSED WHEN THE BOLT IS NOT INSTALLED UNLESS THE TENSIONING NUT HAS BEEN LOOSENED. FAILURE TO FOLLOW THIS CAUTION COULD RESULT IN DAMAGING THE BOLT RING SEGMENTS.**		
412-A-62-10-00-00A-520A-A / 00935	1. Remove the main rotor expandable blade bolts (if installed) from the main rotor hub assembly.		
	 **PAY PARTICULAR ATTENTION NOT TO DISTURB THE RING SEGMENT ORDER AND ORIENTATION DURING THE DISASSEMBLY AND ASSEMBLY OF THE EXPANDABLE BOLT.**		
BHT-412-CR&O	2. Clean the expandable blade bolt (1, Figure 1) with a clean cloth moistened with aliphatic naphtha (C-305) or isopropyl alcohol (C-385) and wipe dry with a clean rag.		
	3. Make sure the nut (4) is facing up before you disassemble the expandable blade bolt (1).		
	4. Remove the nut (4) from the core pin (5) with the expandable blade bolt handle (2) in the open (as removed) position.		

Table 1. Each 300 Hours or 6 Months of Expandable Blade Bolt Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	NOTE For the blade bolt segment removal/installation workaid fabrication instructions, refer to the 412-CRO, Chapter 62. 5. Attach the blade bolt segment removal/installation workaid to the threaded end of the core pin (5) until firmly seated (finger tight only). 6. Hold or secure the bolt handle (2) and, while holding the segments (3), invert the expandable bolt assembly to allow the segments to slide freely from the core pin (5) to the blade bolt segment removal/installation workaid. 7. With aliphatic naphtha (C-305) or isopropyl alcohol (C-385), clean the segments (3) and the core pin (5) without removing them from the blade bolt segment removal/installation workaid. Separate the segments to remove all dirt and debris. 8. Thoroughly dry the segments (3) and the core pin (5) with clean compressed air. 9. Visually inspect the segments (3) and the expandable blade bolt (1) for cracks, fretting, corrosion, or mechanical damage. Refer to the 412-CRO, Chapter 62 for damage limits. **NOTE** Any bolts with damage beyond the limits of the 412-CRO, Chapter 62 must be removed from service. Repairable damage of the expandable blade bolt is limited to blending/polishing with 240-grit abrasive cloth or paper (C-423) followed by 320 and 400-grit abrasive cloth or paper (C-423). Apply solid film lubricant (C-021) to repaired areas and areas of burnished or missing solid film lubricant. 10. Apply a thin continuous layer of grease (C-561) to the core pin (5) and the segments (3) both internally and externally on all surfaces. 11. To begin reassembly, hold the end of the blade bolt segment removal/installation workaid and bolt handle (2). Invert the expandable blade bolt assembly to allow segments (3) to slide freely from the blade bolt segment removal/installation workaid back onto the core pin (5). Make sure the segments seat correctly against each other. 12. Remove the blade bolt segment removal/installation workaid and install the nut (4) on the core pin (5).		
412-A-62-10-00-00A-720A-A / 00946	13. Install the main rotor expandable blade bolts (1) in the main rotor hub assembly.		



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Figure 1. Expandable Blade Bolt - Special Inspection (Sheet 1 of 1)

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Special Inspections - Each 600 Hours of Tail Rotor Driveshaft Operation or 12 Months

Procedure

Table 1. Each 600 Hours of Tail Rotor Driveshaft Operation or 12 Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish each 600 hours of tail rotor driveshaft operation or 12 months, whichever occurs first (no Non Destructive Inspection (NDI) required).		
412-A-65-00-00-00A-009A-A / 01257	TAIL ROTOR DRIVESHAFT FLEXIBLE COUPLINGS		
	NOTE *Applicable to S/N 33001 through 33213 and 36001 through 36019 not modified by 412-570-001-103 or Pre BHT-412-SI-74 (412SP to 412HP Upgrade) or Pre BHT-412-SI-97 (412 Classic to 412HP Upgrade).*		
	1. Remove tail rotor driveshaft assemblies.		
	2. Remove four tail rotor hanger assemblies 212-040-600-ALL.		
	NOTE Disassemble flexible couplings only to the extent necessary to remove old grease and repack.		
412-CRO, Chapter 65	3. Clean, visually inspect, and repack tail rotor hanger assembly flexible couplings.		
	4. Install four tail rotor hanger assemblies 212-040-600-ALL.		
	5. Clean, visually inspect, and repack intermediate gearbox flexible couplings.		
	6. Clean, visually inspect, and repack tail rotor gearbox flexible coupling.		
	7. Clean, visually inspect, and repack transmission tail rotor driveshaft flexible coupling.		
412-A-65-00-00-00A-009A-A / 01257	8. Install tail rotor hanger assemblies.		
	9. Install tail rotor driveshaft assemblies.		
	TAIL ROTOR DRIVESHAFT FLEXIBLE COUPLINGS		
	NOTE *Applicable to S/N 36020 through 36999, S/N 37002 through 37999, S/N 38001 through 38999, S/N 39101 through 39999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade).*		

Table 1. Each 600 Hours of Tail Rotor Driveshaft Operation or 12 Months (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-65-10-00-00A-520A-A / 01259 BHT-412-CR&O, Chapter 65	1. Remove tail rotor coupling assembly 412-040-601-101 from transmission. 2. Remove tail rotor coupling assembly 412-040-601-105/-107 from hanger support assembly. **NOTE** Disassemble flexible couplings only to the extent necessary to remove old grease and repack. 3. Clean, visually inspect, and repack removed tail rotor coupling assemblies. 4. Install tail rotor coupling assembly 412-040-601-101 on transmission. 5. Install tail rotor coupling assembly 412-040-601-105/-107 on hanger support assembly. 6. Remove tail rotor driveshaft assembly between intermediate gearbox and tail rotor gearbox. **NOTE** Disassemble flexible coupling only to the extent necessary to remove old grease and repack. 7. Clean, visually inspect, and repack intermediate gearbox flexible coupling. 8. Install tail rotor driveshaft assembly.		

Special Inspections - Each 600 Hours or 12 Months Inspection of Tailboom Attachment Bolts Procedure

Table 1. EACH 600 HOURS OR 12 MONTHS INSPECTION OF TAILBOOM ATTACHMENT BOLTS

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish the following inspection each 600 hours or 12 months, whichever occurs first.		
412-A-53-00-00-00A-009A-A / 00897	TAILBOOM		
ASB 412-21-187			
	1. Carry out a torque check of tailboom attachment bolts.		
	2. If any bolt is found below minimum torque, carry out the following:		
	a. Replace the affected bolt and remove barrel nut. Examine the barrel nut for condition and for loss of tare torque, replace as necessary.		
412-A-05-47-00-00A-283B-A / 00033	b. Carry out a torque check of the tailboom attachment bolts between 1 to 5 flight hours after replacement.		

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Special Inspections - Each 600 Hours of Main Driveshaft Operation or 12 Months

Procedure

Table 1. Each 600 Hours of Main Driveshaft Operation or 12 Months

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish each 600 hours of main driveshaft operation or 12 months, whichever occurs first (no Non Destructive Inspection (NDI) required).		
	MAIN DRIVESHAFT 212-040-005-003/-007/-103, AND 412-040-005-101/-103		
412-A-63-00-00-00A-009A-A / 00987 BHT-412-CR&O, Chapter 63	1. Remove main driveshaft.		
	NOTE Only disassemble driveshaft to the extent necessary to remove old grease and repack.		
	2. Clean, visually inspect, and repack flexible couplings. 3. Install main driveshaft.		
	NOTE *Balancing of main driveshaft is required on helicopters S/N 36020 through 36999, S/N 37002 through 37999, S/N 38001 through 38999, S/N 39101 through 39999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade) anytime driveshaft is removed and reinstalled. Prior serial number helicopters, balancing of driveshaft is optional.*		
412-A-18-11-00-09A-271A-A / 00173	4. Balance driveshaft to below 0.2 IPS.		

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Special Inspections - Each 600 Hours or 12 Months Inspection Of Magnetic Brake Assembly Procedure

Table 1. EACH 600 HOURS OR 12 MONTHS INSPECTION OF MAGNETIC BRAKE ASSEMBLY

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish the following inspection each 600 hours or 12 months, whichever occurs first.		
	The following inspection is not applicable for the helicopter serial numbers 36026, 36052 and 36053 equipped with EFIS and 4-Axis AFCS Kits, 36087 through 36999 equipped with EFIS and 4-Axis AFCS kits (412-705-009, 412-705-051 and 412-705-024), 37002 through 37999 equipped with 4-Axis AFCS kit (412-705-042) and 38001 through 38999, 39101 through 39999 equipped with 4-Axis AFCS kit (412-706-151-101). MAGNETIC BRAKE ASSEMBLY 204-001-376-003		
ASB 412-17-173	1. Perform a detailed visual inspection of the magnetic brake assembly in accordance with ASB 412-17-173. 2. If the mechanical damage exceeds the limitation, contact Product Support Engineering.		

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Special Inspections - Each 1000 Hours of Component Operation

Procedure

Table 1. Each 1000 Hours of Component Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish each 1000 hours of component operation.		
412-A-29-00-00-00A-009A-A / 00610 TB 412-17-240	HYDRAULIC ACCUMULATORS **NOTE** The following inspection is not required if the intent of the TB 412-17-240 is accomplished. **NOTE** The following procedure only applies to helicopters S/N 33001 through 33213 and 36001 through 36627.		
BHT-412-CR&O, Chapter 29	1. Remove hydraulic accumulators. 2. Disassemble, clean, and visually inspect for corrosion and damage. 3. Replace any unserviceable parts. 4. Reassemble hydraulic accumulators.		
412-A-29-10-00-09A-720A-A / 00660	5. Install hydraulic accumulators.		
412-A-79-00-00-00A-009A-A / 01765	OIL COOLER BLOWERS		
BHT-412-CR&O, Chapter 79	1. Remove oil cooler blowers. 2. Disassemble, clean, and visually inspect for corrosion and damage. 3. Replace bearings and any other unserviceable parts. 4. Reassemble oil cooler blowers. 5. Install oil cooler blowers.		

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Special Inspections - Each 12 Months or 2500 Landings of Aft High Crosstube Operation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Paint remover	C-436	AR	
Abrasive pad	C-407	AR	
Corrosion preventive compound	C-125	AR	
Barrier material	C-427	AR	
Tape	C-410	AR	
Epoxy primer coating	C-246	AR	
Sealant	C-308	AR	

Procedure

Table 1. Each 12 Months or 2500 Landings of Aft High Crosstube Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish each 12 Months or 2500 landings, whichever occurs first.		
	HIGH AFT CROSSTUBE ASSEMBLIES (412-050-011-ALL AND 412-050-045-ALL) DEFLECTION CHECK		
412-A-32-00-00-00A-009A-A / 00694 ASB 412-09-135	1. Hoist or jack the helicopter until the weight is removed from the landing gear assembly.		
	 **ENSURE HELICOPTER IS STABILIZED ON JACKS BEFORE PROCEEDING TO ACQUIRING MEASUREMENTS.**		
412-A-32-00-00-00A-009A-A / 00694	2. Confirm aft crosstube is within deflections limits in accordance with 412-A-32-01-06-00A-280A-A / 00726.		
	3. Replace aft crosstubes that exceed the required dimensional tolerances.		
	HIGH AFT CROSSTUBE ASSEMBLIES (412-050-011-ALL AND 412-050-045-ALL) FLUORESCENT PENETRANT INSPECTION		
BHT-ALL-SPM, Standard Practices Manual	**NOTE** The Fluorescent Penetrant Inspection shall be accomplished by persons meeting the personnel qualification and certification requirements of BHT-ALL-SPM, Chapter 6, Standard Practices Manual. Bell Helicopter trained Level 1 Special individuals are qualified to perform this inspection.		

Table 1. Each 12 Months or 2500 Landings of Aft High Crosstube Operation (continued)

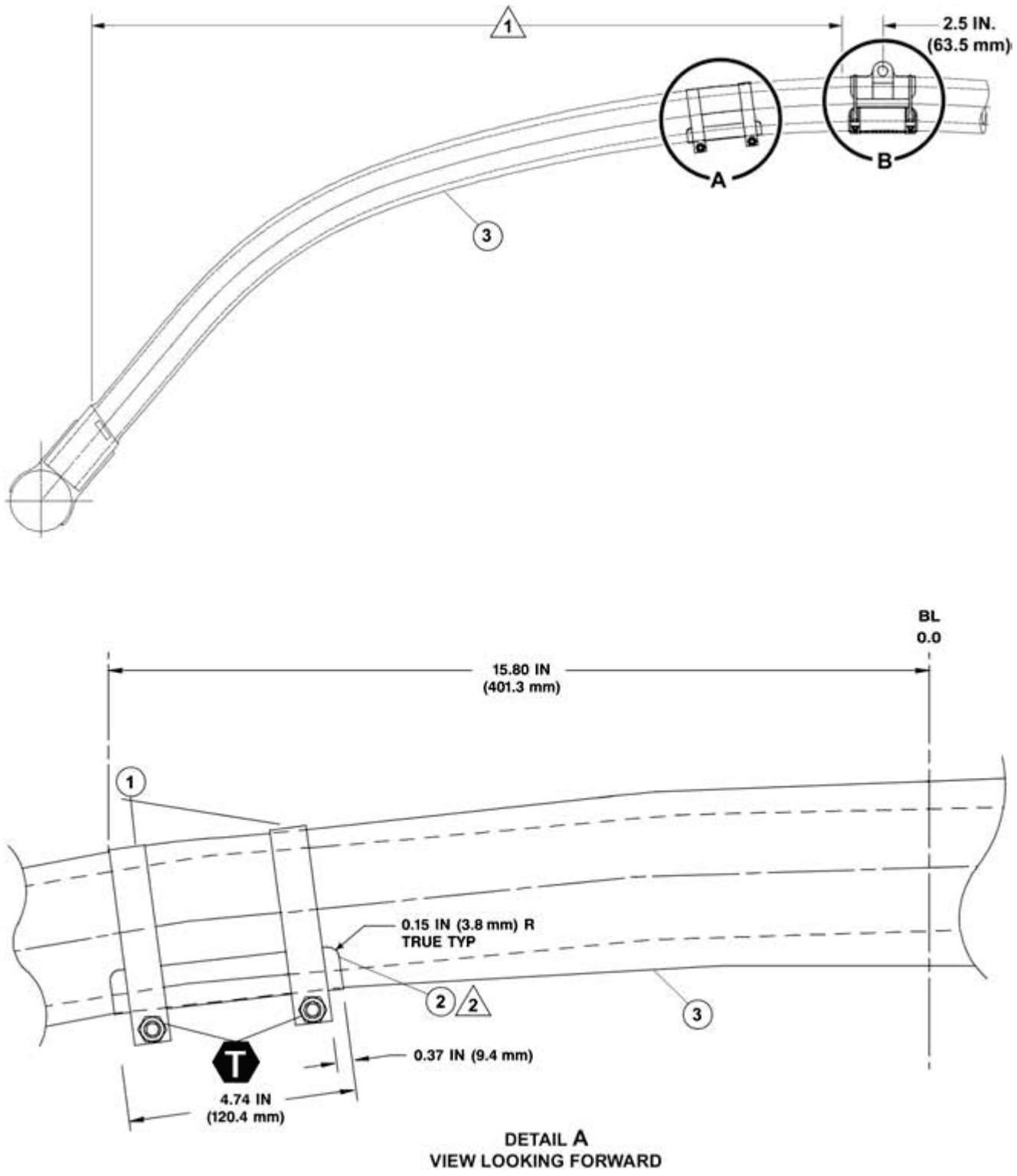
DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-32-00-00-00A-009A-A / 00694 ASB 412-09-135	1. Remove the landing gear assembly. 2. Remove the aft crosstube from the landing gear assembly. 3. Remove the support beam assembly from the high aft crosstube assembly. 4. Remove the hardware retaining the left and right clamp assemblies (1, Figure 1) on the crosstube. **CAUTION** **CROSSTUBE TEMPERATURE NOT TO EXCEED 200ºF (93.3ºC). EXCESSIVE HEAT WILL DAMAGE CROSSTUBE.** 5. Use a heat gun to soften sealant bonding the clamp assembly (1) to the aft crosstube (3). 6. Separate the clamp assembly (1) from the aft crosstube (3) and abrasion strip (2). **NOTE** Removal of the abrasion strips (2) from the aft crosstube (3) is not necessary to accomplish the Fluorescent Penetrant Inspection.		
412-A-32-00-00-00A-009A-A / 00694	7. Inspect the abrasion strips (2) for de-bonding. If de-bonding is found refer to 412-A-32-01-08-01A-280A-A / 00738 for repair instructions. **NOTE** Complete edge sealing is required to prevent contamination of the bond between the crosstube and upper and lower supports (4 and 6). 8. Inspect the sealant bead around the periphery of the upper and lower supports (4 and 6) for condition. If the sealant bead has been damaged, investigate further.		
412-A-32-00-00-00A-009A-A / 00694	9. Inspect crosstube lower support (6) and abrasion strip (5) for signs of looseness or debonding. If looseness or debonding is found, refer to 412-A-32-01-08-01A-280A-A / 00738 for repair instructions. **NOTE** The upper support (4) on the aft crosstube (3) is bonded with structural adhesive and is not designed to be removed.		
412-A-32-00-00-00A-009A-A / 00694	10. Inspect the upper support (4) for signs of looseness or debonding. If looseness or debonding is found refer to 412-A-32-01-08-01A-280A-A / 00738 for repair instructions. **CAUTION** **PROTECT ABRASION STRIPS AND UPPER AND LOWER SUPPORTS FROM PAINT REMOVER CONTAMINATION.** 11. Prepare the aft crosstube for the inspection by removing the paint on the crosstube in the area indicated in Figure 1 with the exception of the abrasion strip areas and the upper and lower supports.		

Table 1. Each 12 Months or 2500 Landings of Aft High Crosstube Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
BHT-ALL-SPM, Standard Practices Manual	12. Apply paint remover (C-436) to the crosstube with a brush following manufacturer's directions and the BHT-ALL-SPM, Standard Practices Manual.  **A PLASTIC SCRAPER OR abrasive pad (C-407) MAY BE USED TO GENTLY ASSIST PAINT REMOVAL. USE THE PLASTIC SCRAPER OR ABRASIVE PAD IN A LONGITUDINAL DIRECTION ONLY, ENSURING NOT TO ABRADE THE CROSSTUBE METAL SURFACE.** 13. Reapply paint remover as required to completely remove paint and primer. 14. Using potable water rinse the crosstube surface until clean and dry thoroughly. 15. Ensure the crosstube is free of paint and primer residue and bare metal is clearly visible in the required area.		
BHT-ALL-SPM, Standard Practices Manual	16. If the fluorescent penetrant inspection is not going to be accomplished immediately then protect the crosstube by applying corrosion preventive compound (C-125). Wrap exposed area in barrier material (C-427) and secure with tape (C-410).  **USE CARE TO PREVENT CONTAMINATION OF THE ABRASION STRIPS AND UPPER AND LOWER SUPPORTS FROM THE FLUORESCENT PENETRANT CHEMICALS.**		
BHT-ALL-SPM, Standard Practices Manual	**NOTE** The crosstube Fluorescent Penetrant Inspection can be accomplished by any person meeting the personnel Qualification/Certification Requirements of the BHT-ALL-SPM, Chapter 6, Standard Practices Manual. A Bell Helicopter trained Level 1 Special individual is qualified. 17. Inspect the aft crosstube (3, Figure 1) with the exception of the areas under the abrasion strips (2 and 5) and upper support (4) as follows:		
	NOTE All penetrant inspection materials utilized shall be compatible, and shall meet requirements of SAE AMS 2644. The use of visible dye systems is prohibited.		
BHT-ALL-SPM, Standard Practices Manual	17.1. Use fluorescent penetrant inspection method per ASTM E1417, Type 1 Method B, C, or D using a penetrant sensitivity level of 2 or 3 (BHT-ALL-SPM, Chapter 6, Standard Practices Manual).		
412-A-32-00-00-00A-009A-A / 00694	17.2. Inspect crosstube for damage per limits specified in the 412-A-32-01-05-00A-685A-A / 00722 to determine allowable damage and cleanup. 17.3. No cracks are permitted.		
412-A-32-01-05-00A-685A-A / 00722	17.4. Repair all damage in accordance with 412-A-32-01-05-00A-685A-A / 00722.		
BHT-ALL-SPM, Standard Practices Manual	18. Apply epoxy primer coating (C-246) to areas where paint has been removed for inspection as per the BHT-ALL-SPM, Standard Practices Manual.		

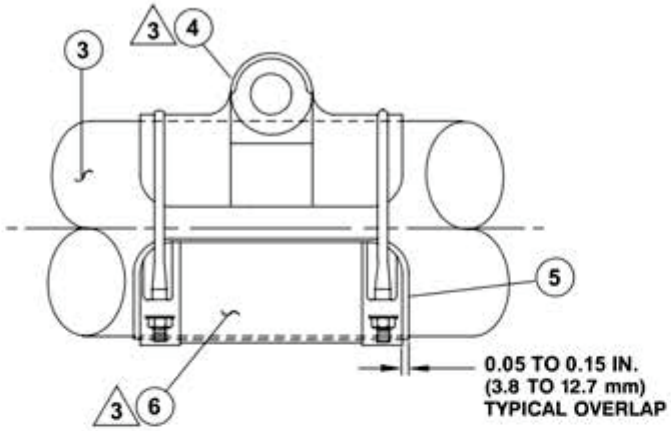
Table 1. Each 12 Months or 2500 Landings of Aft High Crosstube Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	 **DO NOT SAND CROSSTUBE IN A CIRCUMFERENTIAL DIRECTION. SAND CROSSTUBE IN A LONGITUDINAL DIRECTION ONLY.** 19. Taking care not to expose bare material, lightly abrade the primer on the crosstube and abrasion strip where the clamp assembly (1) is installed.  **CONFIRM BAND CLAMP ASSEMBLY POSITIONING PRIOR TO INSTALLATION. DO NOT POSITION BAND CLAMP OVERLAP AT CENTER OF RETENTION CAP RUBBER BUMPER. OVER-ROTATION OF THE CLAMP ASSEMBLY COULD RESULT IN DAMAGE TO THE AIRFRAME FROM THE CLAMP "T" BOLTS.** 20. Remove protective backing film (to expose adhesive film) from clamp assembly (1).		
BHT-ALL-SPM, Standard Practices Manual and 412-A-32-00-00-00A-009A-A / 00694	21. Install clamp assembly (1) using sealant (C-308) as per 412-A-32-01-08-01A-720A-A / 00739. Ensure complete sealant coverage of the inner clamp surface, avoiding air pockets. 22. Torque clamp assembly (1) nuts 50 to 70 inch-pounds (5.6 to 7.9 Nm). **NOTE** The "T" bolt should have approximately 0.5 inch (13 mm) of threads visible after installation.		
BHT-ALL-SPM, Standard Practices Manual	23. If required, trim the "T" bolts on the clamp assembly (1) to provide adequate clearance with fuselage tunnel; deburr as required. Apply epoxy primer coating (C-246) and paint bare metal on "T" bolt for corrosion protection.		
BHT-ALL-SPM, Standard Practices Manual	24. Repaint crosstube (BHT-ALL-SPM, Chapter 4, Standard Practices Manual).		
412-A-32-00-00-00A-009A-A / 00694	25. Install the support beam assembly onto the aft crosstube. 26. Install the aft crosstube onto the landing gear assembly. 27. Install the landing gear assembly.		



ICN-412-A-054700-B-97499-00007-A-001-01

Figure 1. Aft High Crosstube - Special Inspection (Sheet 1 of 2)



DETAIL B
VIEW LOOKING FORWARD

1. Clamp assembly
2. Abrasion strip
3. Aft crosstube
4. Upper support
5. Larson L101 abrasion strip
6. Lower support

T 50 TO 70 IN-LBS
(5.6 TO 7.9 Nm)

NOTES

1. Remove paint/primer from the aft crosstube (3) on the left and right-hand sides.
2. Protect abrasion strip (if installed) during paint removal process.
3. Protect upper and lower supports (4 and 6) during paint removal process.

ICN-412-A-054700-B-97499-00008-A-001-01

Figure 1. Aft High Crosstube - Special Inspection (Sheet 2 of 2)

Special Inspections - Each 24 Months of Control Bolt Operation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
MEK	C-309	AR	
Corrosion preventive compound	C-104	AR	
Corrosion preventive compound	C-101	AR	

Procedure

Table 1. Each 24 Months of Control Bolt Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION		INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____			
	FACILITY: _____			
	HELICOPTER S/N: _____			
	REGISTRY NO.: _____			
	TOTAL TIME:_____			
	SIGNATURE:_____			
	NOTE Accomplish each 24 months of control bolt operation.			
	FLIGHT CONTROL SYSTEM BOLTS			
	NOTE Refer to 412-A-04-00-00-00A-288A-A / 00013 for retirement life of flight control bolts.			
412-A-29-00-00-00A-009A-A / 00610 412-A-62-00-00-00A-009A-A / 00932 412-A-63-00-00-00A-009A-A / 00987 412-A-67-00-00-00A-009A-A / 01319	1. Remove the following flight control bolts:			
	BOLT PART NUMBER	**LOCATION**		
	20-057-5-24D	Swashplate inner ring to L/H and R/H cyclic boost		
	20-057-5-24D	Collective lever to collective boost		
	20-057-5-24D	Boost tube to universal		
	20-057-5-24D	Universal to boost tube		
	20-057-5-28D	Pitch horn to pitch link (2)		
	20-057-6-36D	Pitch link to rephasing lever		
	(or 20-057-6-52D)			
	20-057-6-36D	Drive Link to rephasing lever (1)		

Table 1. Each 24 Months of Control Bolt Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION		INITIAL	
			MECH	OTHER
	20-057-6-36D	Swashplate link to rotating ring (1)		
	(or 20-057-6-38D)			
	20-057-6-52D	Swashplate link to rephasing lever (1)		
	20-057-8-84D	Rephasing lever to hub		
	20-057-8-92D	Collective lever to swashplate support		
	212-001-323-001	Hydraulic cylinder to lower support		
BHT-ALL-SPM, Standard Practices Manual	2. Clean bolts with cloth dampened with MEK (C-309) or equivalent.			
	3. Inspect bolts for thread damage, shank wear, and corrosion. Replace any bolt that has damaged threads, detectable shank wear, or exhibits corrosion pitting.			
	4. Apply corrosion preventive compound (C-104) to bolt shanks only and install. Do not apply to the bolt threads.			
	5. Operations in extreme environmental conditions, that erode hard film corrosion preventive compound (C-101), may warrant periodic touch-up of control bolts external surfaces. Prior to touch-up, visually check exposed surfaces of hardware for evidence of corrosion and fretting. Remove, clean, and inspect bolts as conditions warrant.			
	6. After bolts have been installed and nuts torqued and lockwired, coat bolt heads and nuts with corrosion preventive compound (C-101).			
	NOTES:			
TB 412-86-52	1	If dual drive links were not installed (S/N 33001 through 33107), the following quantities apply: - Drive link to rephasing lever 20-057-36D - Swashplate link to rephasing lever, 20-057-6-38D - Swashplate link to rotating ring 20-057-6-36D.		
	2	As per ASB 412-98-92, bolt 20-057-5-25D is replaced by 20-057-5-28D.		

Special Inspections - Each 24 Months of Main Rotor Mast Operation

Procedure

Table 1. Each 24 Months of Main Rotor Mast Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish every 24 months of main rotor mast operation when operating in a non-corrosive environment. If helicopter is operated in a corrosive environment, it is recommended this inspection be accomplished every 12 months.		
	MAIN ROTOR MAST ASSEMBLY (412-040-366-SERIES AND 412-040-368-SERIES)		
412-A-62-20-00-02A-520A-A / 00948	1. Remove main rotor hub assembly.		
BHT-412-CR&O, Chapter 63	2. Clean then visually inspect main rotor mast upper splines for corrosion.		
412-A-62-20-00-02A-720A-A / 00952	3. Install main rotor hub assembly.		

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Special Inspections - Each 2500 Hours of Main Rotor Hub Assembly Operation

Procedure

Table 1. Each 2500 Hours of Main Rotor Hub Assembly Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish each 2500 hours of main rotor hub operation.		
	MAIN ROTOR HUB ASSEMBLY (412-010-100-SERIES) AND 412-010-502-SERIES		
412-A-62-20-00-02A-520A-A / 00948 BHT-412-CR&O, Chapter 62	1. Disassemble main rotor hub. Visually inspect eight drive pins 412-010-166.		
	NOTE Pitch horn to spindle attachment bolts do not need to be removed unless there is evidence of looseness or external corrosion on the bolt head.		
TB 412-92-112	2. Visually inspect yoke flexures, spindles, pitch horns, upper and lower cone seats, and spline plate for condition. Inspect and replace (if required) buffer pads on yoke. Inspect yoke bushings for condition.		
	NOTE Some yoke assemblies 412-010-101-123, -127 were delivered with oversized bushings.		
	3. Visually inspect all elastomeric bearings for condition, deterioration, and separation.		
	NOTE The following recommendations are not mandatory, but are intended to aid the operator to reduce unscheduled maintenance. Recommend replacement of damper yoke sets at each 2500 hour special inspection and replacement of damper bearings at 5000 hour interval (each second 2500 hour special inspection).		
	4. Visually inspect droop restraint system components for condition and security. Inspect and replace droop restraint bolts if any thread damage, shank wear, distortion, or corrosion is detected.		
	5. Visually inspect simple pendulum absorbers components for condition and security. Inspect and replace simple pendulum absorbers bolts if any thread damage, shank wear, distortion, or corrosion is detected.		
	6. Inspect main rotor mast nut and blade bolts for condition.		
412-A-04-00-00-00A-288A-A / 00013	7. Check for retirement life of components.		

Table 1. Each 2500 Hours of Main Rotor Hub Assembly Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
BHT-412-CR&O, Chapter 62 412-A-62-20-00-02A-720A-A / 00952	8. Treat and/or touch-up all damaged painted surfaces as required. 9. Reassemble main rotor hub.		

Special Inspections - Each 2500 Hours of Main Rotor Blade Operation

Procedure

Table 1. Each 2500 Hours of Main Rotor Blade Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish every 2500 hours of main rotor blade operation.		
	MAIN ROTOR BLADES (412-015-200/-300 SERIES)		
412-A-62-10-00-00A-520A-A / 00935	1. Remove each main rotor blade.		
BHT-412-CR&O, Chapter 62	2. Inspect for condition in accordance with main rotor blade normal inspection criteria.		

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Special Inspections - Each 2500 or 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-109 and Subsequent) Operation

Procedure

Table 1. Each 2500 or 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-109 and Subsequent) Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE *Accomplish the inspection each 2500 hours or 5 years of main rotor mast (412-040-366-109 and subsequent) operation, whichever occurs first, on helicopters S/N 36020 through 36670 and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade) .*		
	NOTE *Accomplish the inspection each 3000 hours or 5 years of main rotor mast (412-040-366-109 and subsequent) operation, whichever occurs first, if the intent of TB 412-14-231 has been complied with on helicopters S/N 36020 through 36670 and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade) .*		
	NOTE *Accomplish the inspection each 3000 hours or 5 years of main rotor mast (412-040-366-127/-129 and subsequent) operation, whichever occurs first, on helicopters S/N 36671 through 36999 and S/ N 37002 through 37999.*		
412-A-63-00-00-00A-009A-A / 00987 BHT-412-CR&O, Chapter 63	**MAIN ROTOR MAST (412-040-366-109, AND SUBSEQUENT)**		
	 **REMOVE MONOPOLE SENSOR BEFORE REMOVING THE MAST.**		

Table 1. Each 2500 or 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-109 and Subsequent) Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	NOTE Prior to removing mast assembly from transmission, check chip detector located in top case for evidence of debris. Presence of ferrous foreign material on chip detector will require additional inspection to determine origin. The 412-040-366-109 (Pre TB 412-01-179) mast is not field serviceable and must be returned to BHTI for the following inspections. The mast assembly must be returned with torquemeter tube, plug, and lower torquemeter rotor in place. Also note the number of the monopole sensor 412-375-004-105 or -107 used with the mast. 1. Remove main rotor mast assembly from the transmission.  **DO NOT STAND OR LAY THE MAST HORIZONTALLY WITH THE MAST BEARING RETAINING PLATE SUPPORTING WEIGHT OF THE MAST.** 2. Place main rotor mast assembly in horizontal position on the padded support, such as V blocks. 3. Remove lock ring, mast bearing retaining plate, and shim. 4. Visually examine the seal for signs of overheating, cracking, wear, or damage. 5. Check mast bearing for smooth rolling and freedom from binding. 6. Visually examine the lower mast bearing inner race for scratches, dents, corrosion, pitting, spalling, roller skidding, or scoring. 7. Visually examine the splines on the mast for dents, wear, or damage.  **CAREFULLY REMOVE AND INSTALL THE TORQUEMETER TUBE ASSEMBLY IN THE MAST TO PREVENT DAMAGE TO THE CADMIUM PLATING OF THE MAST INNER DIAMETER AND TO AVOID DAMAGE TO THE SPLINE TEETH AT THE LOWER END OF THE TORQUE TUBE.** 8. Remove the torquemeter tube assembly and plug at the upper end of the mast from the mast inner diameter. **NOTE** Mark an index on the torque multiplier wrench to determine if the upper mast bearing nut rotation occurs. 9. Check the torque on the upper mast bearing nut 412-040-104-101 or 412-040-104-103 by applying torque in increasing torque direction. Do not torque more than 1400 foot-pounds (1898 Nm) for -101 nut or 900 foot-pounds (1220 Nm) for -103 nut.		

Table 1. Each 2500 or 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-109 and Subsequent) Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	10. If rotation is evidenced at torque less than the values above, remove the nut and mast bearing. Examine bearing inside diameter and mast journal for evidence of rotation and/or fretting. Visually examine the inside diameter and outside diameter of the mast for evidence of corrosion and/or mechanical damage. Pay attention for corrosion at the journal, the annular groove and the (bearing nut) threads above the bearing. Record that the reason for the disassembly was low nut torque. Replace any discrepant parts. **NOTE** Step 11 is applicable to Post-TB 412-14-231 mast assembly and step 12 is applicable to Pre-TB 412-14-231 mast assembly only. *11. For Post-TB 412-14-231 mast configuration and mast assembly 412-040-366-127/-129 and subsequent, if there was no evidence of nut rotation at step 10, visually examine the inside diameter and outside diameter of the mast for evidence of corrosion and/or mechanical damage. Continue with step 13.* 12. For Pre-TB 412-14-231 mast configuration, if there was no evidence of nut rotation at step 10, remove the nut and inspect for corrosion and/or mechanical damage. It is not necessary to remove the mast bearing. Visually examine the inside diameter and outside diameter of the mast for evidence of corrosion and/ or mechanical damage. Pay attention for corrosion on the journal above the bearing, the visible portions of the bearing, the annular groove and the (bearing nut) threads above the bearing. Replace any discrepant parts. 13. Assemble main rotor mast assembly. 14. Install the torquemeter tube assembly and plug. 14.1. For mast assemblies 412-040-366-109, Bell Helicopter will align the spline teeth in the torquemeter tube assembly relative to those in the lower torquemeter rotor. The alignment procedure used for the spline teeth depends on the serial number of the mast assembly and the part number of the monopole sensor used with the mast assembly. 14.2. For mast assemblies 412-040-366-111 and subsequent, align the spline teeth in the torquemeter tube assembly relative to the spline teeth in the lower torquemeter rotor using tool 412-240-028-101. 15. Install main rotor mast assembly.		

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Special Inspections - Each 2500 Hours of Tail Rotor Drive System Operation

Procedure

Table 1. Each 2500 Hours of Tail Rotor Drive System Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE *Accomplish each 2500 hours of tail rotor drive system operation on helicopters S/N 36020 through 36999, S/N 37002 through 37999, S/N 38001 through 38999, S/N 39101 through 39999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade).*		
412-A-65-00-00-00A-009A-A / 01257	**TAIL ROTOR DRIVESHAFT** 1. Check torque of nuts at laminated disc couplings by applying torque. Do not exceed 280 inch-pounds (31.64 Nm). If nut rotates under torque, inspect attachment hardware, driveshafts, disc packs, couplings, and adapters, as applicable, for serviceability. Replace as required. 2. Visually examine laminated disc assemblies for separation, deflection, distortion, cracks, missing sections, and correct installation of bolts and washers.		
	 **INDEXING FLATS OF 0.015 INCH (0.38 MM) (TWO PLACES 180° APART) ON DISC ASSEMBLIES ARE PREFERENTIALLY ORIENTED AND SHOULD NOT BE DISTURBED IN HANDLING. IF ORIENTATION IS INADVERTENTLY DISTURBED, REINDEX DISCS BY ALTERNATING INDEX FLATS ON EACH DISC 60° FROM ADJACENT DISC.**		
412-A-65-10-00-00A-520A-A / 01259	3. Remove driveshafts from helicopter. 4. Visually examine driveshafts for nicks and dents. Examine flange mating surfaces and pilot diameters for evidence of fretting, wear, and/or corrosion. 5. Visually examine coupling 412-340-610 laminations for evidence of wear and or corrosion. 6. After intermediate gearbox, tail rotor gearbox, and hanger assemblies have been inspected, install tail rotor driveshafts.		
412-A-65-00-00-00A-009A-A / 01257 BHT-412-CR&O, Chapter 65	**INTERMEDIATE GEARBOX** 1. Visually inspect exterior surfaces of intermediate gearbox for evidence of seeping, leaking, corrosion, and/or damage.		

Table 1. Each 2500 Hours of Tail Rotor Drive System Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
BHT-412-CR&O, Chapter 65 412-A-65-00-00-00A-009A-A / 01257	2. Inspect output quill assembly as follows:  **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.** 2.1. Visually examine exterior surface and TEMP-PLATE overtemperature indicators of outer coupling for discoloration and evidence of overheating. A change in color of TEMP-PLATES (from white to black) indicates a possible overheat condition and/or component degradation. The cause of TEMP-PLATES and/or exterior surface discoloration shall be determined and corrected prior to continued operation. **NOTE** Refer to BHT-412-CR&O for disassembly, cleaning, assembly, inspection procedures, and wear and damage limits. 2.2. Remove retaining ring, end plate, and packing. Remove retaining ring, retainer, and packing. 2.3. Check torque of bolt 204-040-612 by applying torque. Do not exceed 100 foot-pounds (136 Nm). If bolt turns under torque, inspect for serviceability after removal. Replace as required. After torque verification, remove bolt 204-040-612. 2.4. Remove outer coupling and inner crown tooth coupling. 2.5. Remove inner coupling from outer coupling. Remove retainer 214-040-633/427-044-121, seal 214-040-812/412-340-004, and housing 214-040-634/427-044-113 from female coupling as an assembly. Using an arbor press, ensure retainer 214-040-633/427-044-121 is fully seated in housing 214-040-634/427-044-113. 2.6. Remove all old grease from inner outer coupling teeth using clean, lint-free cloth. Clean teeth thoroughly. 2.7. Visually inspect internal spline teeth of male coupling and mating spline of gear for evidence of fretting and/or corrosion. Visually inspect teeth of inner and outer coupling for evidence of wear, pitting, dings, or chipped teeth. 2.8. Inspect seal 214-040-812/412-340-004, in place for distortion, cuts, tears, and deterioration. 2.9. Visually examine wear sleeve of male coupling for evidence of wear and/or overheating. 2.10. Inspect seal housing for damage that could cause leakage of lubricant. 2.11. Inspect end plate for damage that could cause leakage of lubricant.		
BHT-412-CR&O, Chapter 65 412-A-65-00-00-00A-009A-A / 01257	3. Assemble and lubricate intermediate gearbox output quill coupling. **HANGER ASSEMBLIES** 1. Visually examine exterior surfaces of hanger assembly and adjacent area surfaces for evidence of grease leakage.		

Table 1. Each 2500 Hours of Tail Rotor Drive System Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	NOTE Refer to the BHT-412-CR&O for disassembly, cleaning, assembly, inspection procedures, and wear and damage limits.		
BHT-412-CR&O, Chapter 65	2. Clean external surfaces of hanger assembly.		
	3. Visually inspect hanger assembly as follows:		
	3.1. Inspect hanger for evidence of wear, dents, enlarged mounting slots, fretting, or corrosion.		
	 **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.**		
	3.2. Inspect hanger exterior surfaces for discoloration and overheat condition. A change in color indicates a possible overheat condition and/or component degradation. The cause of exterior surface discoloration shall be determined and corrected prior to continued operation.		
	3.3. Inspect shaft and flange for evidence of wear, nicks, dents, enlarged holes, and cracks.		
	3.4. Inspect bearing for grease leakage.		
	3.5. Inspect bearing in hanger for smoothness of rotation and binding by manually rotating bearing.		
	4. Lubricate hanger assembly.		
	COUPLING ASSEMBLY (412-040-601-101/-109)		
	1. Visually examine exterior surfaces of coupling assembly and adjacent area surfaces for evidence of grease leakage.		
412-A-65-00-00-00A-009A-A / 01257	2. Remove coupling assembly 412-040-601-101/-109 from helicopter.		
	NOTE Refer to the BHT-412-CR&O for disassembly, cleaning assembly, inspection procedures, and wear and damage limits.		
BHT-412-CR&O, Chapter 65	3. Clean external surfaces of coupling assembly.		
	4. Visually inspect coupling assembly, as follows:		
	 **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.**		
	4.1. Inspect coupling TEMP-PLATE overtemperature indicator and exterior surfaces for discoloration and overheat condition. A change in color of TEMP-PLATES (from white to black) indicates a possible overheat condition and possible component degradation. The cause of indicator or exterior surface discoloration shall be determined and corrected prior to continued operation.		
	4.2. Inspect flange and outer coupling for nicks, dents, enlarged holes, and cracks.		
BHT-412-CR&O, Chapter 65	5. Disassemble coupling assembly for internal inspection as follows:		

Table 1. Each 2500 Hours of Tail Rotor Drive System Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	5.1. While holding end plate against spring, remove retaining ring. Remove end plate and spring. Visually examine end plate, spring, and retaining ring for evidence of wear and/or corrosion.		
	5.2. Remove cotter pin. Check torque on nut by applying torque. Do not exceed 50 inch-pounds (5.65 Nm). If nut rotates under torque, inspect attachment hardware for serviceability. Replace as required.		
	5.3. Remove bolt, nut, and plates. Visually examine faying surfaces of plates for evidence of wear and/or corrosion. Remove inner and outer couplings and sleeve from shaft.		
	5.4. Remove inner coupling from outer coupling. Remove retaining ring. Remove retainer, seal, and housing from female coupling as an assembly. Using an arbor press, ensure retainer is fully seated in housing.		
	5.5. Remove sleeve from inner coupling and inspect for evidence of wear and/or corrosion.		
	5.6. Remove all old grease from inner and outer coupling teeth using a clean, lint-free cloth. Clean teeth thoroughly.		
	5.7. Visually inspect splines of shaft and inner coupling for evidence of fretting, wear, and/or corrosion. Visually inspect mating surfaces of male coupling and sleeve for evidence of wear and/or corrosion. Visually inspect teeth of inner and outer couplings for evidence of wear, pitting, dings, dents, or chipped teeth.		
	5.8. Inspect seal 214-040-812/412-340-004, in place for distortion, cuts, tears, and deterioration.		
	5.9. Inspect seal housing for damage that could cause leakage of lubricant.		
	5.10. Inspect end plate for damage that could cause leakage of lubricant.		
BHT-412-CR&O, Chapter 65	6. Assemble and lubricate coupling assembly.		
	7. Replace TEMP-PLATE overtemperature indicators if discolored or damaged.		
412-A-65-00-00-00A-009A-A / 01257	8. Install coupling assembly on helicopter.		
	COUPLING ASSEMBLY (412-040-601-105/-107/-111)		
412-A-65-00-00-00A-009A-A / 01257	1. Visually examine exterior surfaces of coupling assembly and adjacent area surfaces for evidence of grease leakage.		
BHT-412-CR&O, Chapter 65	2. Remove coupling assembly from helicopter.		
	NOTE Refer to the BHT-412-CR&O for disassembly, cleaning, assembly, inspection procedures, and wear and damage limits.		
BHT-412-CR&O, Chapter 65	3. Clean external surfaces of coupling assembly.		
BHT-412-CR&O, Chapter 65	4. Visually inspect coupling assembly as follows:		
	4.1. Inspect hanger for evidence of wear, dents, enlarged mounting slots, fretting, or corrosion.		

Table 1. Each 2500 Hours of Tail Rotor Drive System Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	 **CAUSE OF ANY DISCOLORATION SHALL BE DETERMINED AND CORRECTED PRIOR TO NEXT FLIGHT.** 4.2. Inspect hanger and outer coupling TEMP-PLATES overtemperature indicators and exterior surfaces for discoloration and overheat condition. A change in color of TEMP-PLATES (from white to black) indicates a possible overheat condition and/or possible component degradation. The cause of discoloration shall be determined and corrected prior to continued operation. 4.3. Inspect shaft and outer coupling for nicks, dents, enlarged holes, and cracks. 4.4. Inspect bearing for grease leakage and dislocation of bearing seals. 4.5. Inspect bearing in hanger for smoothness of rotation and binding by manually rotating bearing.		
BHT-412-CR&O, Chapter 65	5. Disassemble coupling assembly for internal inspection as follows: 5.1. While holding end plate against spring, remove retaining ring. Remove end plate and spring. Visually examine end plate spring and retaining ring for evidence of wear and/or corrosion. 5.2. Remove cotter pin. Check torque on nut by applying torque. Do not exceed 50 inch-pounds (5.65 Nm). If nut rotates under torque, inspect attachment hardware for serviceability. Replace as required. 5.3. Remove bolt, nut, and plates. Visually examine faying surfaces of plates for evidence of wear and/or corrosion. Remove inner and outer couplings and sleeve from shaft. 5.4. Remove inner coupling from outer coupling. Remove retaining ring. Remove retainer, seal, and housing from female coupling as an assembly. Using an arbor press, ensure retainer is fully seated in housing. 5.5. Remove sleeve from inner coupling and inspect for evidence of wear and/or corrosion.		
	 **DO NOT USE SOLVENT TO CLEAN COUPLING TEETH. SOLVENT MAY LEAVE A RESIDUE, WHICH PREVENTS LUBRICANTS FROM ADHERING TO COUPLING.** 5.6. Remove all old grease from inner and outer coupling teeth using a clean, lint-free cloth. Clean teeth thoroughly. 5.7. Visually inspect splines of shaft and inner coupling for evidence of fretting, wear, and/or corrosion. Visually inspect mating surfaces of male coupling and sleeve for evidence of wear and/or corrosion. Visually inspect teeth of inner and outer coupling for evidence of wear, pitting, dents, or chipped teeth. 5.8. Inspect end plate for damage that could cause leakage of lubricant. 5.9. Inspect seal, in place, for distortions, cuts, tears, and deterioration. 5.10. Inspect seal housing for damage that could cause leakage of lubricant.		
BHT-412-CR&O	6. Assemble and lubricate bearing and coupling assembly.		

Table 1. Each 2500 Hours of Tail Rotor Drive System Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	7. Replace TEMP-PLATE overtemperature indicators if discolored or damaged.		
412-A-65-00-00-00A-009A-A / 01257	8. Install assembly on helicopter.		
	TAIL ROTOR GEARBOX		
	NOTE Refer to the BHT-412-CR&O for disassembly, cleaning, assembly, inspection procedures, and wear and damage limits.		
412-A-65-00-00-00A-009A-A / 01257 BHT-412-CR&O, Chapter 65	1. Visually inspect exterior surface of gearbox and adjacent area for evidence of leaking, seepage, corrosion, and damage.		
	2. Remove input quill adapter spring lock and retainer. Check torque of bolt using increasing torque. Do not exceed 80 foot-pounds (108 Nm). If bolt turns under torque, inspect for serviceability.		
	3. Install input quill adapter retainer and spring lock.		

Special Inspections - Each 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-103 and -105) Operation

Procedure

Table 1. Each 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-103 and -105) Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE *Accomplish each 3000 hours or 5 years of main rotor mast (412-040-366-103 and -105) operation on helicopters S/N 33001 through 33213 and S/N 36001 through 36019 not modified by 412-570-001-103 or BHT-412-SI-74 (412SP to 412HP Upgrade) or BHT-412-SI-97 (412 Classic to 412HP Upgrade).*		
	MAIN ROTOR MAST (412-040-366-103 and 412-040-366-105)		
	NOTE Prior to removing mast assembly from transmission, check chip detector, located in top case, for evidence of debris. Presence of ferrous foreign material on chip detector will require additional inspection to determine origin.		
412-A-63-21-01-00A-520A-A / 01006 BHT-412-CR&O, Chapter 63	1. Remove main rotor mast assembly from the transmission.		
	 **DO NOT STAND OR LAY THE MAST HORIZONTALLY WITH THE MAST BEARING RETAINING PLATE SUPPORTING WEIGHT OF THE MAST.**		
	2. Place main rotor mast assembly in horizontal position on padded support, such as V blocks.		
	3. Remove lock ring, mast bearing retaining plate, and shim.		
	4. Visually examine the seal for signs of overheating, cracking, wear, or damage.		
	5. Check mast bearing for smooth rolling and freedom from binding.		
	6. Visually examine the lower mast bearing inner race for scratches, dents, corrosion, pitting, spalling, roller skidding, or scoring.		
	7. Visually examine splines on mast for dents, wear, or damage.		
	NOTE Mark an index on the torque multiplier wrench to determine if the upper mast bearing nut rotation occurs.		

Table 1. Each 3000 Hours or 5 Years of Main Rotor Mast (412-040-366-103 and -105) Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	8. Check the torque on the upper mast bearing nut 412-040-104-101 or 412-040-104-103 by applying torque in increasing torque direction. Do not torque more than 1400 foot-pounds (1898 Nm) for -101 nut or 900 foot-pounds (1220 Nm) for -103 nut. 9. If rotation is evidenced at torque less than the values above, remove the nut and mast bearing. Examine bearing inside diameter and mast journal for evidence of rotation and/or fretting. Visually examine the inside diameter and outside diameter of the mast for evidence of corrosion and/or mechanical damage. Pay attention for corrosion at the journal, the annular groove and the (bearing nut) threads above the bearing. Record that the reason for the disassembly was low nut torque. Replace any discrepant parts. **NOTE** Step 10 is applicable to Post-TB 412-14-231 mast assembly and step 11 is applicable to Pre-TB 412-14-231 mast assembly only. 10. For Post-TB 412-14-231 mast configuration, if there was no evidence of nut rotation at step 9, visually examine the inside diameter and outside diameter of the mast for evidence of corrosion and/or mechanical damage. Continue with step 12. 11. For Pre-TB 412-14-231 mast configuration, if there was no evidence of nut rotation at step 9, remove the nut and inspect for corrosion and/or mechanical damage. It is not necessary to remove the mast bearing. Visually examine the inside diameter and outside diameter of the mast for evidence of corrosion and/ or mechanical damage. Pay attention for corrosion on the journal above the bearing, the visible portions of the bearing, the annular groove and the (bearing nut) threads above the bearing. Replace any discrepant parts. 12. Assemble main rotor mast assembly.		

Special Inspections - Each 3000 Hours of Transmission (412-040-002) Operation

Procedure

Table 1. Each 3000 Hours of Transmission (412-040-002) Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE *Accomplish each 3000 hours of transmission (412-040-002) operation on helicopters S/N 33001 through 33213 and S/N 36001 through 36019 not modified by 412-570-001-103 or BHT-412-SI-74 (412SP to 412HP Upgrade) or BHT-412-SI-97 (412 Classic to 412HP Upgrade).*		
	TRANSMISSION (412-040-002)		
BHT-412-CR&O, Chapter 63	1. Disassemble transmission sufficiently to remove main input driven gear quill.		
	2. Remove gear support case and debris collector to gain access to spiral bevel gear retaining bolts.		
	2.1. Using torque wrench, torque check each of the 32 bevel gear retaining bolts for minimum torque of 300 inch-pounds (33.89 Nm).		
	2.2. If torque value of any one retaining bolt is less than 300 inch-pounds (33.89 Nm), remove bevel gear and inspect mating surfaces of gear and shaft for fretting damage.		
	2.3. The maximum acceptable depth of pitting is 0.0005 inch (0.013 mm). Depth may be measured by using a dial indicator with a needle pointed probe. Pitting of measurable depth is acceptable only in the area on gear or shaft surface outside of diameter of bolt holes. It is not acceptable within 0.10 inch (2.5 mm) of edge of a bolt hole. Damage in excess of these limits is cause for rejection of part.		
BHT-412-CR&O, Chapter 63	3. Inspect upper flange surface and pilot diameter of ring gear and mating surfaces of top case fretting and wear.		
BHT-412-CR&O, Chapter 63	4. Inspect lower flange surface and pilot diameter of ring gear and mating surfaces of bevel gear support case for fretting and wear.		
	5. Visually check index marks on input pinion and inner race of bearing set for alignment. Inspection may be accomplished with input quill installed by viewing pinion shaft and bearing inner race between pinion teeth and main case.		
	NOTE If index marks indicate rotational movement between pinion and inner race of bearing set, remove quill and bearing set, and inspect pinion bearing journal for signs of fretting and bearing inner race spinning. Visually inspect oil holes in input quill sleeve to ensure no foreign material is present. Inspect detail parts to determine cause for bearing inner race rotation and replace parts as required.		
BHT-412-CR&O, Chapter 63	6. Inspect bevel gear and main input pinion for general condition and wear pattern.		

Table 1. Each 3000 Hours of Transmission (412-040-002) Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
BHT-412-CR&O, Chapter 63	7. Inspect planetary ring gear, sun gears, and planetary pinions for general condition and wear pattern. Inspect all parts of upper and lower planetary visually for excessive wear and damage. Dimensionally check parts showing evidence of wear or physical damage. 8. Remove tail rotor drive quill to gain access to accessory drive and sump gears. Visually inspect accessory case input quill gear and tail rotor drive quill gear for general condition and wear pattern. 9. Inspect transmission support case lateral mounts as follows: 9.1. Remove screws and washers from transmission support case lateral mounts, top and bottom. 9.2. Visually inspect screws for corrosion and thread damage. 9.3. Visually inspect upper washers 204-040-236 and lower washers 204-040-736 for corrosion and/or pitting. 9.4. Visually inspect installed bushings for corrosion and/or pitting. 9.5. Visually inspect transmission support case lateral mount surfaces and threaded holes for mechanical and corrosion damage. 10. Perform magnetic particle inspection on lower planetary spider 204-040-785-003. **CAUTION** **USE OF ANY FORM OF LUBRICANT TO FACILITATE INSTALLATION OF GEAR ON SHAFT WILL RESULT IN TORQUE LOSS OF ATTACHMENT BOLTS AND SUBSEQUENT FRETTING DAMAGE.** **NOTE** During assembly of spiral bevel gear to shaft, ensure mating surfaces are dry.		
BHT-412-CR&O, Chapter 63	11. Assemble transmission.		

Special Inspections - Each 3000 Hours of Transmission (412-040-004/412-040-007/412-040-802) Operation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
MEK	C-309	AR	
Drycleaning solvent	C-304	AR	

Procedure

Table 1. Each 3000 Hours of Transmission (412-040-004/412-040-007/412-040-802) Operation

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE *Accomplish each 3000 hours of transmission 412-040-004/412-040-007/412-040-802 operation on helicopters S/N 36020 through 36999, S/N 37002 through 37999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade).*		
412-A-63-00-00-00A-009A-A / 00987 BHT-412-CR&O, Chapter 63	**MAIN TRANSMISSION (412-040-004/412-040-007/412-040-802)**		
	NOTE Refer to the BHT-412-CR&O for disassembly, cleaning, assembly, inspection procedures, wear, and damage limits.		
412-A-63-00-00-00A-009A-A / 00987 BHT-412-SI-12 BHT-412-CR&O, Chapter 63	1. Visually examine exterior surfaces of rotor brake quill for evidence of oil seeping and/or leaking. Visually examine area of shaft seal for evidence of seal leakage.		
	NOTE Do not remove brake disc from drive flange. To facilitate removal of rotor brake quill from transmission, remove nut retaining drive flange and remove disc and flange as an assembly.		
412-A-63-21-02-08A-520A-A / 01177	2. Remove rotor brake quill from transmission. Refer 412-A-05-11-00-00A-042A-A / 00070 for Component Overhaul Schedule.		
412-A-63-21-02-08A-720A-A / 01181	3. After input gear quill has been assembled into main case, install rotor brake quill in transmission.		

Table 1. Each 3000 Hours of Transmission (412-040-004/412-040-007/412-040-802) Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-63-00-00-00A-009A-A / 00987 BHT-412-CR&O, Chapter 63	4. Visually inspect outside of transmission for evidence of corrosion, damage, and oil leaks (including leaking at seals at input, mast, tail rotor output, and accessory quills) before disassembling transmission.		
	5. Drain and strain transmission lubricant. Bag any foreign material found and deliver with transmission for evaluation during this inspection.		
	6. Remove mast chip detector from top case, planetary detector from main case, full flow detector from sump case, and chip detector from bottom of sump case. Visually inspect detectors for cut packings, burrs, bent tubes, and foreign material. Clean detectors as required with MEK (C-309).		
	7. Any foreign material found on detectors or in lubricant shall be evaluated. Inspection may require additional investigation to identify course(s) of foreign material and initiate appropriate action.		
BHT-412-CR&O, Chapter 63	8. Disassemble main transmission to accomplish the following:		
	8.1. Remove and inspect the following jet assemblies from transmission and check for evidence of clogging.		
	8.1.1. 204-040-262 - Jet No. 2		
	8.1.2. 204-040-326 - Jet No. 3		
	8.1.3. 204-040-288 - Jet No. 5		
	8.1.4. 205-040-135 (2 each) - Jet located in ring gear case		
	8.2. Remove top case. Inspect mast assembly mating surface, ring gear mating surface, and pilot diameters for fretting and wear.		
	9. Planetary component removal:		
	9.1. Remove upper planetary assembly 204-040-360 and adapter 412-040-107 as an assembly.		
	9.2. Remove retaining ring M27426-1220C and adapter 412-040-107 from planetary assembly 204-040-360. Do not disassemble planetary assembly.		
	9.3. Remove liner 204-040-337, bearing 204-040-135, and extension 412-040-109 as an assembly. Do not disassemble.		
	9.4. Remove ring gear assembly 205-040-231.		
	9.5. Remove upper sun gear 412-040-230, deflector 204-040-378, and retaining ring RR687L as an assembly.		
	9.6. Remove lower planetary assembly 204-040-784. Do not disassemble.		
	9.7. Remove liner 204-040-338, bearing 204-040-135, and liner 204-040-257 as an assembly. Do not disassemble.		
	9.8. Remove lower sun gear 205-040-229.		
BHT-412-CR&O, Chapter 63	10. Planetary component inspection:		
	10.1. Visually inspect all parts of planetary assembly for evidence of wear and/or damage.		
	10.2. Visually inspect gear tooth contact patterns on the eight pinion assemblies in upper planetary assembly. It is normal for ring gear meshing side of planet pinions to have a double wear pattern. Check bearings for smoothness and freedom from binding by rotating pinion assemblies. Visually inspect all lockwire provisions.		
	10.3. Visually inspect external and internal spline teeth of adapter 412-040-107 for excessive wear and general condition.		

Table 1. Each 3000 Hours of Transmission (412-040-004/412-040-007/412-040-802) Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	10.4. Check planetary support liners and bearings 204-040-135 for smoothness and freedom of rotation by manually turning. 10.5. Visually inspect upper flange surface and upper pilot diameter and lower flange surface and lower pilot diameter of ring gear 205-040-231 for fretting and wear. Visually inspect upper and lower gear teeth contact patterns for evidence of pitting, scoring, wear, or damage. 10.6. Visually inspect upper sun gear 412-040-230 teeth for wear, damage, and contact pattern.  **DO NOT USE SOLVENT TO CLEAN COUPLING TEETH. SOLVENT MAY LEAVE A RESIDUE, WHICH PREVENTS LUBRICANTS FROM ADHERING TO COUPLING.** 10.7. Visually inspect gear tooth contact pattern on the four pinion assemblies in lower planetary assembly. It is normal for ring gear meshing side of planet pinions to have a double wear pattern. Check bearings for smoothness and freedom from binding by rotating the four pinion assemblies. Visually inspect all lockwire provisions. 10.8. Visually inspect lower sun gear 205-040-229 teeth for wear, damage, and contact pattern. 11. Check backlash in three places approximately 120° apart between gear 412-040-701 and pinion 412-040-700 using backlash tool 412-240-006. Record backlash on inspection log. 12. Visually inspect manifold assembly 204-040-393. Remove, clean, and inspect jet No. 7 for evidence of clogging. 13. Remove quill assembly 412-040-362 from main case. Do not disassemble. Visually inspect mating flanges and pilot diameters for corrosion, fretting, and wear. Visually inspect all lockwire provisions. Check duplex bearing for smoothness and freedom of rotation by manually turning gear 412-040-701. Visually check index marks on gear and inner race of bearing set 205-040-245 for alignment. 13.1. If the index marks indicate movement between gear and inner race of bearing set, remove gear 412-040-701, plate 204-040-357, shim 204-040-348, bearing set 205-040-245, and shim 204-040-350. 13.2. Visually inspect inside diameter of bearing set 205-040-245 for signs of spinning. Visually inspect bearing journal on gear 412-040-701 for sign of fretting. Inspect detail components to determine cause for inner race rotation. 14. Visually inspect drive and coast side of gear 412-040-701 teeth for wear, scoring, pitting, and contact pattern. Visually inspect upper and lower spline teeth for excessive wear and/or damage. 15. Visually inspect roller bearing race on lower end of gear for evidence of skidding, scoring, pitting, or damage. 16. Visually inspect splines on the upper and lower end of gear for damage and/or wear. 17. Visually inspect drive and coast side of pinion 412-040-700 teeth for excessive wear, scoring, pitting, and contact pattern. Visually inspect vibro-etched index marks on pinion 412-040-700 and bearing set 214-040-118 for alignment. Inspection may be accomplished with input quill installed by viewing pinion shaft and bearing inner race between teeth and main case.		

Table 1. Each 3000 Hours of Transmission (412-040-004/412-040-007/412-040-802) Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	17.1. If index marks indicate rotational movement between pinion and inner race of bearing set, quill 412-040-263 shall be removed from main case. Disassemble quill to remove pinion 412-040-700 and bearing set 214-040-118.		
	17.2. Visually inspect pinion bearing journal for signs of fretting and bearing inner race spinning. Clean sleeve 412-040-137 with drycleaning solvent (C-304). Visually verify oil holes in sleeve are free of any foreign material. Inspect detail components to determine cause for inner race rotation.		
BHT-412-CR&O, Chapter 63	18. Visually inspect teeth of pinion 412-040-302 in quill assembly 204-040-379 for wear, scoring, pitting, and gear tooth contact pattern. Inspect internal spline of pinion for wear and general condition.		
	19. Check bearing set 204-040-424 for smoothness and freedom from binding by rotating pinion 412-040-302 by hand.		
	20. Remove main case assembly. Visually inspect upper and lower flanges and pilot diameters for evidence of fretting and wear.		
	21. Visually inspect the rollers of bearing 204-040-271 for evidence of skidding, scoring, and end damage. Do not remove bearing from case.		
	22. Visually inspect the rollers of 205-040-249 bearing for any damage. Do not remove from main case.		
BHT-412-CR&O, Chapter 63	23. Inspect 212-040-208 quill assembly:		
	23.1. Remove drive and sump assembly 212-040-365 from support case assembly 212-040-054.		
	23.2. Remove adapter 212-040-206. Inspect both splines for evidence of damage and/or wear.		
	23.3. Check installation of cotter pin six places.		
	23.4. Visually inspect for proper installation of retainer.		
	23.5. Visually inspect gear 212-040-201 teeth and pinion 212-040-202 teeth for evidence for wear, scoring, and pitting.		
	23.6. Manually turn pinion and check bearings for smoothness of rotation and freedom from binding.		
BHT-412-CR&O, Chapter 63	24. Inspect components in case assembly 212-040-054:		
	24.1. Visually inspect mating flange and pilot diameter of support case at main case joint for evidence of corrosion, fretting, and/or wear.		
	24.2. Remove lift link fitting.		
	24.3. Visually inspect lift link fitting pads for evidence of fretting and wear.		
	24.4. Visually inspect bearing rollers for evidence of damage.		
	24.5. Remove screws and washers from transmission support case lateral mounts, top and bottom.		
	24.6. Visually inspect screws for corrosion and thread damage.		
	24.7. Visually inspect upper washers 204-040-236 and lower washers 204-040-736 for corrosion and/or pitting.		
	24.8. Visually inspect installed bushings for corrosion and/or pitting.		
	24.9. Visually inspect transmission support case lateral mount surfaces and threaded holes for mechanical and corrosion damage.		
BHT-412-CR&O, Chapter 63	25. Inspect components in drive sump assembly 212-040-365:		

Table 1. Each 3000 Hours of Transmission (412-040-004/412-040-007/412-040-802) Operation (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	25.1. Inspect mating surface and pilot diameter for evidence of fretting and wear.		
	25.2. Remove and inspect sump output quill assembly. Do not disassemble.		
	25.3. Visually inspect pinion 212-040-151 teeth for evidence of wear, scoring, and pitting. Inspect tooth contact pattern.		
	25.4. Check freedom from binding and smoothness of roll of bearings by rotating pinion gear by hand.		
	25.5. Visually inspect tail rotor input gear teeth for evidence of wear, scoring, and pitting through the output quill bore. Inspect gear tooth contact pattern. Do not remove input quill.		
	25.6. Check freedom from binding and smoothness of roll of bearings by rotating pinion by hand.		
	25.7. Remove oil pump and visually examine spline for general condition. Rotate pump by hand and check for freedom of rotation.		
	25.8. Remove accessory quill (hydraulic and tach drive). Visually examine pilot diameters and mating flanges for evidence of wear and/or corrosion.		
	25.9. Visually inspect hydraulic pump and tachometer drive gear for excessive wear, scoring, and pitting. Inspect tooth contact pattern.		
	25.10. Check freedom from binding and smoothness of roll of bearings by manually rotating gear 212-040-149.		
	25.11. Remove, inspect, and clean, chip detector 212-040-122 and screen assembly 204-040-237 for foreign material and No. 4 oil jet for evidence of clogging.		
	25.12. Visually examine oil pump shaft, through pump bore in case, for general condition of lower spines.		
BHT-412-CR&O, Chapter 63	26. Assemble transmission.		

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Special Inspections - 3600 Hours Total Airframe Time and Each 300 Hours/12 Months Inspection Procedure

Table 1. 3600 Hours Total Airframe Time and Each 300 Hours/12 Months Inspection

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____ **NOTE** For helicopters S/N 33001 through 33179 with main beam cap 205-030-186-005 installed, accomplish at 3600 hours total airframe time and each 300 hours or 12 months thereafter, whichever occurs first. If main beam cap 205-030-186-005 has been replaced, use replacement cap total time in lieu of helicopter total time. This inspection is not required if main beam cap 205-030-186-005 has been replaced with main beam cap 205-030-186-101.		
TB 412-88-66	**MAIN BEAM CAP** Inspect main beam cap 205-030-186-005 as follows: 1. Remove components as necessary to gain access to main beam cap between FS 164 and 168. 2. Clean exposed area of main beam cap. 3. Inspect cap for crack(s) using a bright light and a 3X magnifying glass. 4. Log inspection in helicopter log book. 5. Replace components that were removed to gain access.		

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Special Inspections - Each 3000 Hours or 5 Years Of Transmission Mount Assembly (204-031-927) And Friction Damper Assembly (204-031-920) Operation

Procedure

Table 1. EACH 3000 HOURS OR 5 YEARS OF TRANSMISSION MOUNT ASSEMBLY (204-031-927) AND FRICTION DAMPER ASSEMBLY (204-031-920) OPERATION

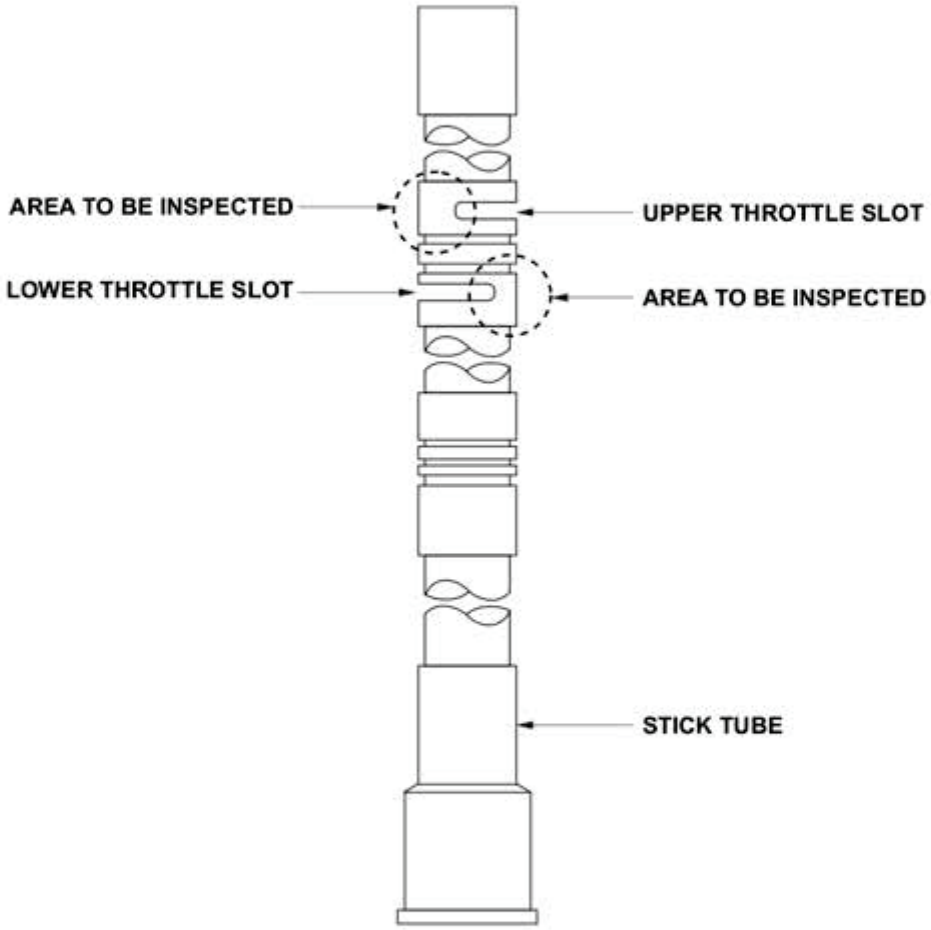
DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish each 3000 hours or 5 years of transmission mount assembly (204-031-927) and friction damper assembly (204-031-920) operation, whichever occurs first. When you replace one or more transmission mounts, it is recommended to replace all four transmission mounts at the same time. This will be only applicable to the transmission mounts with more than 3000 hours/5 years time in service.		
412-A-63-00-00-00A-009A-A / 00987	TRANSMISSION MOUNT ASSEMBLY 1. Remove mount assembly.		
412-A-63-21-04-00A-280A-A / 01188	2. Do a detailed inspection of mount assembly.		
412-A-63-00-00-00A-009A-A / 00987	TRANSMISSION AFT FRICTION DAMPER ASSEMBLY 1. Remove friction damper assembly.		
BHT-412-CR&O, Chapter 63	2. Do a detailed inspection of friction damper assembly.		
BHT-412-CR&O, Chapter 63	3. Test friction damper assembly.		

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Special Inspections - Each 2500 Hours or 10 Years Inspection Of Collective Stick Tube Procedure

Table 1. ***Each 2500 Hours or 10 Years Of Collective Stick Tube Inspection***

DATA REFERENCE	*INSPECTION TASK DESCRIPTION*	*INITIAL*	
		MECH	*OTHER*
	DATE: _____W.O. _____ *FACILITY: _____* *HELICOPTER S/N: _____* *REGISTRY NO.: _____* *TOTAL TIME:_____* *SIGNATURE:_____* **NOTE** *Accomplish the inspection of pilot collective stick tube (212-001-181-001) and copilot collective stick tube (212-001-170-001) each 2500 hours or 10 years, whichever occurs first.*		
412-A-67-00-00-00A-009A-A / 01319 BHT-412-CR&O, Chapter 67 *BHT-412-SI-44* *TB 412-87-60*	*FLIGHT CONTROLS* *1. Disassemble pilot and copilot (if installed) collective stick(s) to the extent necessary to remove throttle twist grips.* *2. Inspect the collective stick tube for cracks by Fluorescent Penetrant Inspection (FPI) method (BHT-ALL-SPM, Chapter 6). Pay particular attention to the base of the throttle slots (Refer to Figure 1) that form the twist grip stops.* *3. If a crack is found, replace stick tube.* *4. Assemble the pilot and copilot (if applicable) collective stick(s).* *5. Check throttle controls for proper operation.* *6. Check collective controls for proper operation.*		



ICN-412-A-054700-B-97499-00839-A-001-01

Figure 1. Collective Stick Tube - Special Inspection (Sheet 1 of 1)

Conditional Inspection - Description

Perform applicable conditional inspection of helicopter after hard landing, after blade strike or other rotating system torque spike, overspeed, overtorque, compressor stall or surge, lightning strike, and after engine combining gearbox clutch non-engagement, misengagement, or in-flight slippage.

If overhaul evaluation is specified, complete both the conditional inspection and normal inspection procedures (as applicable) for that component listed in BHT-412-CR&O.

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Conditional Inspection - After Hard Landing

Procedure

Table 1. After Hard Landing

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish after hard landing. Evaluate components removed from a helicopter following a hard landing as an interrelated group. Make entries in component records of each component removed to cross reference part and serial numbers of other drive system components removed for evaluation.		
	Hard landing is defined as any accident or incident in which ground impact of helicopter results in yielding or cracking of mounting lugs of transmission support case or noticeable yielding or cracking of fuselage pylon support structure, landing gear, or tailboom attachment structure. This definition is confined only to those accidents not involving sudden stoppage of main rotor or tail rotor.		
	AFTER HARD LANDING		
	1. If a hard landing is suspected, accomplish the following:		
	1.1. Inspect main and tail rotor blades for evidence of strike damage. If such evidence is found on either rotor, perform Sudden Stoppage (power on or off).		
	1.2. Visually inspect underside of fuselage and tailboom for evidence of ground contact.		
412-A-32-00-00-00A-009A-A / 00694	1.3. Perform landing gear deflection check.		
	1.3.1. If crosstubes have yielded, remove landing gear and inspect support and attaching structure for signs of yielding or other damage. Replace yielded crosstube and aft crosstube support beam pivot bolt.		
	1.3.2. If supports and attaching structure are not damaged, replace damaged landing gear components.		
	1.4. Remove mast adapter lower cone and inspect lower cone seat for damage.		
	1.5. Inspect area around pylon mounts for loose rivets or other damage.		
	1.6. Inspect tailboom and fuselage attachment for loose rivets, cracks, or other damage.		
	1.7. If no damage other than yielded landing gear crosstubes has been found at this point, a true hard landing probably did not occur. Complete a 25 hour inspection and return helicopter to flight status provided no further evidence of damage is found.		
	1.8. If damage is more extensive than landing gear crosstube yielding, a hard landing has occurred. Comply with requirements of step 2 through step 4.		
	2. If a hard landing has occurred, the following steps shall be accomplished:		
BHT-412-CR&O, Chapter 63	2.1. Remove and perform an overhaul evaluation inspection of following components:		

Table 1. After Hard Landing (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTH-ER
ASB 412-96-89	2.1.1. Mast assembly.  **MAST ASSEMBLY 412-040-366-109 IS NOT FIELD SERVICEABLE. IT MUST BE RETURNED TO BHTI IF ANY INSPECTIONS REQUIRING DISASSEMBLY ARE REQUIRED.** **NOTE** If there is any yielding or deformation in the main rotor attachment area or any other obvious damage, mast is unserviceable and non-repairable. 2.1.2. Transmission. 2.1.3. Main driveshaft. 2.2. Perform a thorough visual inspection of the following components that may be kept in service if no discrepancy or obvious damage is found. Replace any damaged component: 2.2.1. Main rotor blades. 2.2.2. Main rotor hub. 2.2.3. Tail rotor blades. 2.2.4. Tail rotor hub. **NOTE** Inspect tail rotor static stops for yielding as noted in 412-A-64-00-00-00A-280A-A / 01228 to determine if tail rotor yoke assembly may have been exposed to bending. 2.2.5. Tail rotor static stop 212-011-713. 2.2.6. Tail rotor yoke 212-011-702. 2.2.7. Intermediate gearbox. 2.2.8. Tail rotor gearbox 2.2.9. Tail rotor driveshafts 2.2.10. Tail rotor driveshaft hangers 2.2.11. Swashplate and support assembly 2.2.12. Hub and sleeve assembly and collective lever 2.2.13. Helicopter structure directly supporting damaged components identified in above inspections 2.3. Check all cowling and doors for proper fit and alignment. Remove cowling and inspect all attachment fittings. **NOTE** If significant damage has been found in any area of the airframe, expand inspection in those areas until no further damage is found. 2.4. Using a 10X magnifying glass, make a complete inspection of pylon support structure for loose or sheared rivets, cracked brackets, buckled or cracked support angles, and webs. Pay particular attention to pylon mounts attaching points.		

Table 1. After Hard Landing (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTH-ER
	2.5. Make a complete inspection of lift link, lift link attachment fittings, and lift beam for cracks and other evidence of damage. Remove lift link and replace with like serviceable item, if damaged.		
	2.6. Remove both pylon dampers. Disassemble and check for internal yielding. Assemble and install if no evidence of damage. If any damage is found, replace with like serviceable item.		
	2.7. Install serviceable mast, transmission assembly, and main driveshaft assembly. Install removed pylon control components.		
	2.8. Check all engine mount fittings and bolts for damage and looseness.		
	2.9. Inspect engine firewalls for evidence of warping, crushing, or other damage.		
412-A-53-00-00-00A-009A-A / 00897	2.10. Make a complete inspection of area where tailboom is attached to forward fuselage section. This includes both sets of attachment fittings and longerons, beam caps, skins, webs, bulkhead flanges, and other structural members. Check torque on attachment bolts to determine if yielding has occurred.		
412-A-67-00-00-00A-009A-A / 01319	2.11. Completely inspect flight control system from pilot controls to rotor head for bent or damaged tubes, bellcranks, bellcrank supports, and for damaged control system bearings. Pay particular attention to pylon controls, lower cylinder attachment support fitting, and adjacent airframe structure.		
412-A-29-00-00-00A-009A-A / 00610	2.12. Pressurize hydraulic system and check for leaks, interference, binding, and satisfactory operation.		
	2.13. Inspect fuel, oil, and pneumatic system for damage. Perform engine ground run and visually check fuel, oil, and pneumatic lines for leaks.		
Pratt & Whitney Canada Engine Maintenance Manual	3. Inspect engine in accordance with applicable Pratt & Whitney Canada Engine Maintenance Manual.		
	4. If no significant damage is found, further inspection is unnecessary.		

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Conditional Inspection - Sudden Stoppage - Power ON or OFF

Procedure

Table 1. Sudden Stoppage - Power ON or OFF

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____ **NOTE** If the information contained in the following conditional inspection does not fully reflect the occurrence or if additional information or clarification is required, please contact Product Support Engineering for assistance. **NOTE** Sudden stoppage conditional inspection is to be accomplished after main or tail rotor blade strike or any drive system incident which inhibits free rotation of the drive system. Sudden stoppage is defined as any rapid deceleration of the drive system whether caused by seizure within the helicopter transmission or by contact of the main or tail rotor blades with the ground, water, snow, dense vegetation, or other object of sufficient inertia to cause rapid deceleration. Tail rotor blade damage, when caused by striking some object sufficiently to require blade replacement, is considered sudden stoppage. If the sudden stoppage conditional inspection is the result of crash damage, all rotating system components shall be considered unserviceable and non-repairable. Crash damage is any damage sustained beyond the scope of that identified in conditional inspection (412-A-05-50-00-00A-042A-A / 00062). A blade strike is due to contact of main or tail rotor blades with the ground, water, snow, dense vegetation or other object sufficient to cause blade damage. However, not all blade strikes will meet the sudden stoppage definition. A blade strike which causes damage, dents or tear, to a main rotor or tail rotor blade skin behind the spar due to impact with a small foreign object (e.g. small stone or low mass item) that did not contact the leading edge and would not cause a sudden deceleration of the drive system is not considered a sudden stoppage, but may require the affected blade(s) to be repaired. A rotating system torque spike is defined as any other incident which causes a severe torque spike in the drive system. These incidents include, but are not limited to foreign object ingestion through the gear mesh in any gearbox. **NOTE** If overhaul evaluation is specified, complete both the conditional inspection and normal inspection procedures (as applicable) for that component listed in BHT-412-CR&O.		

Table 1. Sudden Stoppage - Power ON or OFF (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	NOTE Components removed from a helicopter for evaluation following a sudden stoppage shall be evaluated as an interrelated group. Removal records accompanying each component shall cross reference part and serial numbers of other drive system components removed for evaluation. When any of these sudden stoppage incidents are reported, do the steps that follow:		
412-A-62-10-00-00A-280A-A / 00937	1. If the sudden stoppage was caused by a main rotor strike do the inspection that follows. 1.1. Main Rotor Blades: **NOTE** In all cases, ensure damage limits are adhered to. 1.1.1. Visually inspect main rotor blades for damage such as cracks, bond voids and delamination. 1.1.2. If a blade is damaged sufficiently to require replacement, return it to an authorized repair facility. Make an entry in the component record to show reason for removal is a sudden stoppage. Refer to BHT-412-CR&O, Chapter 62 for repair limits. 1.1.3. Blades may remain in service if no damage was found or if damage was within acceptable damage limits. 1.2. Main Rotor Hub and Pylon Controls. 1.2.1. If main rotor blades were not damaged sufficiently to require blade replacement, perform a visual inspection of the following components. If no evidence of damage is found, the inspected components may be retained in service. 1.2.1.1. Main rotor hub assembly. 1.2.1.2. Main rotor pitch links. 1.2.1.3. Main rotor swashplate and support. 1.2.1.4. Main rotor hub and sleeve. 1.2.1.5. Collective lever. 1.2.2. If the main rotor blade(s) were damaged as a result of a sudden stoppage the following shall be removed and an overhaul evaluation performed. 1.2.2.1. Main rotor hub assembly. 1.2.2.2. Main rotor pitch links. 1.2.2.3. Main rotor swashplate and support. 1.2.2.4. Main rotor hub and sleeve. 1.2.2.5. Collective lever. Remove and discard all rotating control bolts. Make an entry in the component records to show reason for removal is a sudden stoppage. 1.3. Main Rotor Driveshaft 1.3.1. Remove and disassemble main driveshaft. 1.3.2. Visually inspect components for damage in accordance with BHT-412-CR&O, Chapter 63. If no visual evidence of damage is detected, reassemble per BHT-412-CR&O, Chapter 63.		

Table 1. Sudden Stoppage - Power ON or OFF (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	1.3.3. If damage is detected, make an entry in the component records to show the reason for removal is a sudden stoppage.		
	1.3.4. Perform Main Driveshaft - Conditional inspection in accordance with BHT-412-CR&O, Chapter 63 or BHT-412-CR&O, Chapter 63.		
	1.4. Transmission and mast assembly		
	1.4.1. If a sudden stoppage incident was confirmed as a Result of impact damage to a main rotor blade leading edge exceeding limits specified in 412-A-62-10-00-00A-685A-A / 00938, remove the mast assembly and main rotor transmission assembly. Make an entry in the component records to show reason for removal is a sudden stoppage.		
	1.4.1.1. If the main rotor mast assembly is visibly bent, twisted or distorted, the mast assembly shall be scrapped and the main rotor transmission assembly and rotor brake quill assembly (if installed) shall have an overhaul evaluation performed. The lower planetary spider and lower mast bearing shall be scrapped.		
	1.4.1.2. If the main rotor mast assembly is not visibly bent, twisted or distorted then the main rotor mast, main rotor transmission assembly and rotor brake quill assembly (if installed) shall have an overhaul evaluation performed. Make an entry in the component records to show reason for removal is a sudden stoppage.		
	1.4.2. If a sudden stoppage incident was confirmed but the blade damage is confined to skin compression or wrinkling behind the spar, perform the following inspection:		
	1.4.2.1. Remove main rotor blades, main rotor hub, main rotor mast swashplate and support, hub and sleeve assembly from helicopter. Remove main rotor transmission if required.		
	1.4.2.2. Inspect mast assembly for distortion per BHT-412-CR&O, Chapter 63. Provided the mast assembly does not exceed distortion limits, continue with step 1.4.2.3. If distortion exceeds limits, the mast assembly shall be considered scrap and transmission assembly shall have an overhaul evaluation performed. Make an entry in the component records to show the reason for removal is sudden stoppage.		
	1.4.2.3. Remove transmission top case, mast adapter, upper planetary assembly, upper sun gear, lower planetary assembly, and lower sun gear. Visually inspect all removed components as well as ring gear case for damage resulting from sudden stoppage. If damage is found the main rotor transmission and rotor brake quill (if installed) shall have an overhaul evaluation performed. Make an entry in component records to show reason for removal is a sudden stoppage.		
	1.4.2.4. If evaluation reveals no damage exceeding limits in BHT-412-CR&O, Chapter 63, reassemble transmission and install main rotor mast, swashplate and support, hub and sleeve, main rotor hub and main rotor blades. Annotate component records as required.		
	2. If the sudden stoppage was caused by a tail rotor strike do the inspection that follows.		
	2.1. Tail Rotor Hub and Blade assembly.		
	NOTE In all cases ensure damage limits are adhered to.		
ASB 412-96-89	2.1.1. Inspect tail rotor static stop for yielding as noted in 412-A-64-00-00-00A-280A-A / 01228 to determine if tail rotor yoke assembly may have been exposed to bending. Check for straightness across static stop P/N 212-011-713 outboard surface. If yielding has occurred, scrap the tail rotor yoke and static stop.		

Table 1. Sudden Stoppage - Power ON or OFF (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.1.2. If a sudden stoppage incident was confirmed as a result of impact damage to the leading edge exceeding limits specified in 412-A-64-00-00-00A-280B-A / 01230, the tail rotor hub and blade assembly shall be considered scrap and the following components shall be overhauled. Make an entry in the component records to show reason for removal is sudden stoppage. 2.1.2.1. Tail rotor gearbox assembly. - If during the overhaul; cracks, sheared or bend studs or evidence of case distortion are found the gearbox is unserviceability and nonrepairable and shall be scrapped. 2.1.2.2. Intermediate gearbox assembly. - If during overhaul; cracks, case distortion or broken lugs are found, the gearbox is unserviceable and non-repairable and shall be scrapped. 2.1.2.3. Tail rotor driveshaft hanger assemblies 2.1.2.4. Tail rotor rotating controls. 2.1.3. If a sudden stoppage incident was confirmed but blade damage is confined to skin compression, wrinkling directly behind the spar, perform the following: 2.1.3.1. Overhaul tail rotor hub assembly. Scrap or repair tail rotor blades. 2.1.3.2. Perform a close visual inspection of the tail rotor rotating controls. If no evidence of damage is found, the tail rotor rotating controls may remain in service. Replace all tail rotor rotating controls bolts. If damage to the tail rotor rotating controls is found, perform overhaul of the tail rotor rotating controls. Make an entry in the component record to show that reason for the removal is sudden stoppage. 2.1.3.3. Remove output shaft case (sleeve assy) from tail rotor gearbox. Visually inspect input pinion and bevel gear for indication of sudden stoppage. Any indication or damage found will require overhaul of tail rotor gearbox. Make an entry in the component records to show reason for removal is a sudden stoppage. 2.1.3.4. Remove intermediate gearbox input and output quill. Visually inspect pinion for indication of a sudden stoppage. Any indication or damage found will require overhaul of the intermediate gearbox. Make an entry in the component records to show reason for removal is a sudden stoppage. 2.1.4. If the inspection is being conducted as a result of a tail rotor driveshaft, intermediate gearbox or tail rotor gearbox assembly induced torque spike, an overhaul evaluation of the tail rotor hub and blade assembly shall be performed. Make and entry in the component record to show reason for removal is a sudden stoppage. 2.1.5. If the inspection is being conducted as a result of the Main rotor strike or transmission induced torque spike, the tail rotor hub and blade assembly may remain in service provided there is no visible damage. If visible damage is noted on tail rotor hub and blade assembly, an overhaul evaluation of the tail rotor hub and blade assembly shall be performed. Make an entry in component record to show reason for removal is a sudden stoppage. **NOTE** If you replace driveshafts or hanger bearing assemblies because of the results in step 2.1.3.2 through 2.1.3.3, do a detailed visual inspection of the hanger bearing support structure for cracks, loose or sheared rivets and elongated holes.		

Table 1. Sudden Stoppage - Power ON or OFF (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.2. Tail rotor driveshaft hanger assemblies:		
	2.2.1. Remove hanger assemblies from helicopter and disassemble as required to allow visual inspection for indications of sudden stoppage. Any indication or damage found will require overhaul of the hanger assemblies. Make an entry in component records to show reason for removal is a sudden stoppage.		
	2.2.2. If tail rotor driveshaft has been damaged beyond the limits specified in 412-A-65-10-00-00A-280A-A / 01261 due to contact with main rotor blade or any other similar circumstances other than torsional overload, overhaul the hanger assembly that the driveshaft was attached to. If a tail rotor driveshaft is visibly twisted, all hanger assemblies and shafts shall be overhauled. Make an entry in the components records to show reason for removal is a sudden stoppage.		
	2.2.3. If any driveshafts fails because of torsional overload, remove and scrap all hanger assemblies and driveshaft.		
	2.3. Tail rotor driveshafts:		
	2.3.1. Removal tail rotor driveshafts and inspect them for the following conditions. If one or more conditions listed in step 2.3.1.1 through step 2.3.1.4 below are noted, all hanger assemblies and driveshafts shall be overhauled. Make an entry in the component record to show that reason for the removal is sudden stoppage.		
	2.3.1.1. Inspect driveshaft curvic faces for distortion. (412 and 412SP only).		
	2.3.1.2. Inspect driveshaft for bends, twisting, out of roundness exceeding limits specified in 412-A-65-10-00-00A-280A-A / 01261.		
	2.3.1.3. Inspect driveshaft for cracks.		
	2.3.1.4. Inspect driveshaft for loose or shear rivets.		
	2.3.2. Disc pack couplings - Any deformation that results in gaps between laminates that are greater than 0.015 inch (0.381 mm), scrap the disk pack coupling. (Not applicable to 412 and 412SP).		
	2.3.3. Disc pack coupling bolts - Distortion or cracks in any bolt, scrap bolt. (Not applicable to 412 and 412SP).		
	2.3.4. Tail rotor drive adapters - Distortion beyond the limits specific in 412-A-65-10-00-00A-280A-A / 01261, scrap adapter.		
	2.3.5. If no damage that would render the tail rotor driveshaft unserviceable is noted, it may be returned to service.		
	2.4. Inspect tail rotor drive system structural attachment as follows:		
	2.4.1. Inspect the tail rotor gearbox, intermediate gearbox attachment for cracks, sheared rivets or damage.		
	2.4.2. Inspect tail rotor driveshaft support structure for cracks, shear rivets or damage.		
	2.5. Transmission sump assembly:		
	NOTE If transmission assembly has already been removed for overhaul omit step 2.5. If no evidence of damage was found in step 2.2 and step 2.3 above, omit step 2.5.1 through step 2.5.3 below.		
	2.5.1. If damage was found on driveshaft hanger assemblies, tail rotor driveshaft, intermediate gearbox assembly or tail rotor gearbox assembly, remove tail rotor output quill assembly from transmission sump assembly.		

Table 1. Sudden Stoppage - Power ON or OFF (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.5.2. Inspect output quill pinion for unusual load patterns on both sides of teeth. Inspect sump case internally for gear contact damage. If no damage is found, install quill assembly. Transmission assembly may be retained in service. 2.5.3. If inspection of the tail rotor output quill assembly reveals discrepancies, or gear contact with sump case is evident, remove transmission assembly and an overhaul evaluation of the sump case assembly shall be performed. Make an entry in component record to show reason for removal was blade strike or torque spike. 3. Install or replace, as applicable, all removed components with like serviceable components. 4. Refer to applicable Pratt & Whiney Canada Engine Maintenance Manual for sudden stoppage inspection requirements for the engine.		

Conditional Inspection - After Overspeed

Procedure

Table 1. After Overspeed

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME:_____		
	SIGNATURE:_____		
	NOTE Accomplish after overspeed. Components removed from a helicopter for evaluation following an overspeed shall be evaluated as an interrelated group. Removal records accompanying each component shall cross reference part and serial numbers of other drive system components removed for evaluation.		
	AFTER OVERSPEED		
	Overspeed is defined as any incident in which 110% main rotor rpm is exceeded and/or engine overspeed limits in engine manual are exceeded.		
	1. Perform overspeed inspection as follows:		
412-A-62-00-00-00A-009A-A / 00932	1.1. Main rotor hub assembly:		
	1.1.1. Remove main rotor hub. Remove main rotor blades.		
BHT-412-CR&O, Chapter 63	1.1.2. Perform an overhaul conditional evaluation inspection. Make an entry in component record to show reason for removal was overspeed.		
BHT-412-CR&O, Chapter 62	1.1.3. Inspect main rotor blade retention bolts.		
	1.2. Main rotor blades:		
	1.2.1. Inspect main rotor blades skin for wrinkles and deformation.		
	1.2.2. If no discrepancies are found, main rotor blades may be retained in service.		
	1.2.3. If discrepancies are found, return all blades to an authorized blade repair station. Make an entry in component record to show reason for removal was overspeed. If known, list percent of overspeed and duration.		
412-A-64-00-00-00A-009A-A / 01225	1.3. Tail rotor hub and blades:		
	1.3.1. Remove tail rotor hub and blade assembly.		
	1.3.2. Remove tail rotor blades.		
	1.3.3. Replace tail rotor blade retention bolts.		
	1.4. Tail rotor blades:		
BHT-412-CR&O, Chapter 64	1.4.1. Perform Major Overspeed Inspection of tail rotor blades.		
	1.4.2. Make an entry in component record to show reason for removal was overspeed.		
	1.5. Perform a close inspection of the following components. If no visual damage is found, components may be retained in service.		

Table 1. After Overspeed (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	1.5.1. Main transmission.		
	1.5.2. Intermediate gearbox. Check gearbox for security and retorque retaining bolts.		
	1.5.3. Tail rotor gearbox. Check gearbox for security and retorque retaining nuts.		
	1.5.4. Main rotor mast		
	1.5.5. Main input driveshaft		
	1.5.6. Tail rotor driveshafts		
	1.5.7. Tail rotor driveshafts hangers		
	1.5.8. Swashplate		
	1.5.9. Drive hub and sleeve		
	1.5.10. Tail rotor hub		
	1.6. Install a serviceable main rotor hub and blade assembly. Install serviceable tail rotor blades. Balance and install tail rotor hub and blade assembly.		
Pratt & Whitney Canada Engine Maintenance Manual	1.7. Refer to applicable Pratt & Whitney Canada Engine Maintenance Manual for engine overspeed and inspection requirements.		

Conditional Inspection - After Overtorque

Procedure

Table 1. AFTER OVERTORQUE

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish after overtorque. Components removed from a helicopter for evaluation following an overtorque shall be evaluated as an interrelated group. Make entries in component records to cross reference part and serial numbers of other drive system components removed for evaluation.		
	AFTER OVERTORQUE Overtorque is defined as any incident in which torsional loads are introduced into the helicopter dynamic system in excess of established limits.		
Pratt & Whitney Canada Engine Maintenance Manuals	1. Refer to applicable Pratt & Whitney Canada Engine Maintenance Manual for engine overtorque inspection limits.		
	NOTE To assist with determining engine overtorque inspection requirements, the following table provides a comparison between Bell Helicopter cockpit torque gauge % values and Pratt & Whitney Maintenance Manual FT LB values.		

Table 1. AFTER OVERTORQUE (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION						INITIAL	
							MECH	OTHER
	PT6T-3B/-3BF (1)		PT6T-3B/-3BE/-3BF/-3BG (2)		PT6T-3D/-3DE/-3DF (3)			
	BELL % TQ	P&WC FT LB	BELL % TQ	P&WC FT LB	BELL % TQ	P&WC FT LB		
	85.3	915	82.1	915	98.7	1100		
	81.6	875	78.5	875	96.5	1075		
	76	815	73.2	815	94.2	1050		
	66.8	738	66.2	738	89.8	1000		
					85.3	950		
					81.0	902		
					77.1	859		
					76.3	850		
					73.2	815		
					71.8	800		
					67.3	750		
	NOTES: 1 412 S/N 33001 through 33107 Pre TB 412-84-44. 2 412 S/N 33001 through 33107 Post TB 412-84-44 and all 412SP S/N 33108 through 33213, 36001 through 36019, and 412HP S/N 36020 through 36086 with PT6T-3B series engines installed. 3 412SP S/N 33108 through 33213, 36001 through 36019, and 412HP S/N 36020 through 36086 with PT6T-3D series engines installed and 412EP S/N 36087 through 36999.							
	NOTE Steps 2 to 5 are applicable for airspeeds below 105 Knots and a torque indication above 100%. 2. When torque has exceeded 100% for 3 seconds or less, but has not exceeded 104% (105% mast torque for helicopters modified by 412-570-001-103 and S/N 36020 through 36999), with transmission lower spider 204-040-785-003 or 412-040-785-103 installed, then no further maintenance is required. **NOTE** If transmission lower spider 412-040-785-101 is installed, when torque has exceeded 100%, but has not exceeded 104% (105% mast torque for helicopters modified by 412-570-001-103, and S/N 36020 through 36058), perform a thorough visual inspection of the components listed in the following step. If inspection does not reveal any discrepancies or obvious damage, components may be retained in service.							

Table 1. AFTER OVERTORQUE (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
412-A-62-00-00-00A-009A-A / 00932 412-A-63-00-00-00A-009A-A / 00987 412-A-64-00-00-00A-009A-A / 01225 412-A-65-00-00-00A-009A-A / 01257	3. When torque has exceeded 100% for more than 3 seconds, but has not exceeded 104% (105% mast torque for helicopters modified by 412-570-001-103 and S/N 36020 through 36999), perform a thorough visual inspection of the following components. If inspection does not reveal any discrepancies or obvious damage, components may be retained in service.		
	3.1. Tail rotor blades.		
	3.2. Main rotor hub.		
	3.3. Tail rotor hub.		
	3.4. Intermediate gearbox. Check gearbox for security and torque check retaining bolts.		
	3.5. Tail rotor gearbox. Check gearbox for security and torque check retaining nuts.		
	3.6. Tail rotor driveshafts.		
	3.7. Tail rotor driveshaft hangers.		
	3.8. Swashplate.		
	3.9. Hub and sleeve assembly.		
	3.10. Main driveshaft.		
	3.11. Mast.		
	3.12. Transmission.		
	4. When overtorque exceeds 104%, but does not exceed 112% (105 and 110% mast torque respectively for helicopters S/N 36020 through 36999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade), do as follows:		
	4.1. Perform thorough visual inspection of components listed in step 3.		
412-A-62-00-00-00A-009A-A / 00932	4.2. Inspect main transmission chip detector(s).		
412-A-62-00-00-00A-009A-A / 00932	4.3. Inspect main transmission internal filter or full flow debris monitor (as applicable).		
	4.3.1. If metal particles are found, indicating internal failure, remove transmission for overhaul evaluation. Make an entry in the component record explaining reason for removal was overtorque. Remove all transmission system oil lines. Flush and reinstall oil lines. Replace external oil filter element. Remove and scrap oil cooler. Install new oil cooler.		
	4.3.2. If chip detector(s) and internal filter or full flow debris monitor (as applicable) appear normal and there is no evidence of internal failure, return helicopter to service. Operate normally for 3 to 10 hours and then check chip detector(s) and internal filter or full flow debris monitor (as applicable). If no metal particles are found, indicating internal failure, normal scheduled inspection intervals may then be followed. If metal particles are present prior to or at the 3 to 10-hour check or if there is any evidence of internal failure, remove transmission for overhaul evaluation. Make an entry in the component record explaining reason for removal was overtorque. Remove all transmission system oil lines. Flush and reinstall oil lines. Replace external oil filter element. Remove and scrap oil cooler. Install new oil cooler.		

Table 1. AFTER OVERTORQUE (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	5. When overtorque has exceeded 112% (110% mast torque for helicopters S/N 36020 through 36999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade):		
BHT-412-CR&O	5.1. Return the following components to an overhaul facility for overhaul evaluation. **NOTE** Component removal record of dynamic components shall reflect over-torque as reason for removal. 5.1.1. Transmission. 5.1.2. Main driveshaft. 5.1.3. Main rotor hub. 5.1.4. Mast. 5.2. Perform thorough visual inspection of other components outlined in step 3. **NOTE** Steps 6 to 9 are applicable to helicopters modified by 412-570-001-103 and S/N 36020 through 36999 for airspeeds above 105 Knots with a mast torque indication above 81%. 6. With an airspeed greater than 105 Knots if mast torque has exceeded 81% for 3 seconds or less but has not exceeded 84%. No action is required.		
412-A-62-00-00-00A-009A-A / 00932 412-A-63-00-00-00A-009A-A / 00987 412-A-64-00-00-00A-009A-A / 01225	7. With an airspeed greater than 105 Knots if mast torque has exceeded 81% for more than 3 seconds but has not exceeded 84% perform a thorough visual inspection of the following components. If inspection does not reveal any discrepancies or obvious damage, components may be retained in service. 7.1. Main rotor hub 7.2. Main driveshaft 7.3. Mast 7.4. Transmission 7.5. Tail rotor blades 7.6. Tail rotor hub 7.7. Swashplate 7.8. Hub and sleeve assembly		
412-A-63-00-00-00A-009A-A / 00987	8. With an airspeed greater than 105 Knots if mast torque has exceeded 84% but has not exceeded 100% for more than 3 seconds or 103%, do as follows: 8.1. Perform thorough visual inspection of components listed in step 7. 8.2. Inspect main transmission chip detectors. 8.3. Inspect main transmission full flow debris monitor. 8.3.1. If metal particles are found, refer to 412-A-63-00-00-01A-280A-A / 00989 and 412-A-63-00-00-01A-310A-A / 00990 for oil contamination procedures.		

Table 1. AFTER OVERTORQUE (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	8.3.2. If chip detectors and full flow debris monitor appear normal and there is no evidence of internal failure, return helicopter to service. Operate normally for 3 to 10 hours and then check chip detectors and full flow debris monitor. If no metal particles are found, normal scheduled inspection intervals may then be followed. If metal particles are present prior to or at the 3 to 10-hour check refer to 412-A-63-00-00-01A-280A-A / 00989 and 412-A-63-00-00-01A-310A-A / 00990 for oil contamination procedures.		
	9. With an airspeed greater than 105 Knots if mast torque has exceeded 100% for more than 3 seconds or 103%, do as follows:		
	NOTE If overhaul evaluation is specified, complete both the conditional inspection and normal inspection procedures (as applicable) for that component listed in BHT-412-CR&O.		
BHT-412-CR&O	9.1. Return the following components to an overhaul facility for overhaul evaluation.		
	NOTE Component removal record of dynamic components shall reflect overtorque as reason for removal.		
	9.1.1. Transmission		
	9.1.2. Main driveshaft		
	9.1.3. Main rotor hub		
	9.1.4. Mast.		
412-A-62-00-00-00A-009A-A / 00932 412-A-63-00-00-00A-009A-A / 00987 412-A-64-00-00-00A-009A-A / 01225	9.2. Perform a thorough visual inspection of the following components.		
	9.2.1. Tail rotor blades		
	9.2.2. Tail rotor hub		
	9.2.3. Intermediate gearbox. Check gearbox for security and torque check retaining bolts.		
	9.2.4. Tail rotor gearbox. Check gearbox for security and torque check retaining nuts.		
	9.2.5. Tail rotor driveshafts		
	9.2.6. Tail rotor driveshaft hangers		
	9.2.7. Swashplate		
	9.2.8. Hub and sleeve assembly.		

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Conditional Inspection - After Compressor Stall or Surge

Procedure

Table 1. After Compressor Stall or Surge

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish after engine compressor stall or surge. Discuss circumstances of reported compressor stall with pilot. Determine N_1 speed at which reported stall occurred. Check helicopter and engine logs for any pertinent history.		
	AFTER ENGINE COMPRESSOR STALL OR SURGE Engine compressor stall or surge is characterized by a sharp rumble or a series of loud sharp reports, severe engine vibration, and a rapid rise in Inter-Turbine Temperature (ITT) , depending on severity of surge. When a surge has been reported, progressively perform step 1 through step 3, as dictated by discrepant conditions. Components removed from a helicopter for evaluation following a compressor stall or surge shall be evaluated as an interrelated group. Removal records accompanying each component shall cross reference part and serial numbers of other drive system components removed for evaluation.		
	NOTE If overhaul evaluation is specified, complete both the conditional inspection and normal inspection procedures (as applicable) for that component listed in BHT-412-CR&O.		
	1. Power plant:		
	1.1. Examine inlet screen for blockage.		
Pratt & Whitney Canada Engine Maintenance Manual	1.2. Inspect engine compressor region for salt, dust, oil, or other contaminants. If contaminants are found, clean and perform a power check in accordance with applicable Pratt & Whitney Canada Engine Maintenance Manual.		
	1.3. Inspect for visible Foreign Object Damage (FOD) to visible compressor blades.		
	1.4. If compressor stall (surge) occurred during acceleration, refer to applicable Pratt & Whitney Canada Engine Maintenance Manual for inspection procedures.		
	1.5. Perform test on pneumatic sense lines.		
	1.6. If step 1.1 through step 1.5 do not reveal cause of surge, refer to applicable Pratt & Whitney Canada Engine Maintenance Manual for inspection requirements.		
412-A-63-00-00-00A-009A-A / 00987 412-A-65-00-00-00A-009A-A / 01257	2. Power train:		

Table 1. After Compressor Stall or Surge (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.1. If compressor stall occurs below 85% N_1 speed, only step 2.2 and step 2.3 are required, unless otherwise instructed in step 2.3.		
	2.2. Remove magnetic chip detectors from transmission, intermediate gearbox, and tail rotor gearbox and inspect for metal particles.		
	2.3. If no evidence of damage is found on tailboom fin and no indication of metal particles are found on chip detectors, clean chip detectors and reinstall. Return helicopter to flight status and repeat chip detector inspection after 5 to 10 operating hours. If positive indication of damage is found on tailboom fin or metal chips are found on chip detectors, on initial or 5 to 10 hour inspection, comply with step 2.5 through step 2.8.		
	2.4. If compressor stall occurs at 85% N_1 or above, comply with step 2.2, step 2.3, and step 2.5 through step 2.8, in progression.		
	2.5. Remove and inspect tail rotor driveshafts.		
	2.6. Remove input and output drive quill from intermediate gearbox and inspect gear teeth on pinion and gear for damage with a 10X magnifying glass. If no evidence of scoring or scuffing is found and there is no mechanical damage that would render gearbox unserviceable, reassemble and perform step 3. If gear teeth are scuffed or scored or gearbox has sustained other damage, gearbox shall be removed and an overhaul evaluation performed (Make an entry in the component record to show the reason for removal was compressor stall or surge) and step 2.7 and step 2.8 shall be accomplished.		
	2.7. Remove tail rotor gearbox from helicopter and remove input quill. Inspect gear teeth on pinion and gear for damage with a 10X magnifying glass. If no evidence of scoring or scuffing is found and there is no other damage that would render gearbox unserviceable, it may be reassembled and reinstalled for continued use. If gear teeth are scored or scuffed or there is other damage that would render gearbox unserviceable, gearbox shall be removed and an overhaul evaluation performed. Make an entry in the component record to show the reason for removal was compressor stall or surge.		
	2.8. Remove tail rotor drive quill from transmission and inspect gear teeth for damage with a 10X magnifying glass. If there is no indication of scoring or scuffing and there is no other damage that would render transmission unserviceable, it is suitable for continued use. If gear teeth are scored or scuffed or there is other damage that would render the transmission unserviceable gearbox shall be removed and an overhaul evaluation performed. Make an entry in the component record to show the reason for removal was compressor stall or surge. Comply with step 2.9.		
	2.9. If the transmission is sent for an overhaul evaluation, the procedures outlined in step 2.9.1 through step 2.9.9 shall be performed.		
	2.9.1. Remove tail rotor driveshafts and inspect for the following conditions. Replace any tail rotor driveshafts exceeding limits specified in 412-A-65-10-00-00A-280A-A / 01261. If one or more of conditions listed in step 2.9.1.1 through step 2.9.1.4 are noted, all hanger assemblies shall be sent for overhaul evaluation. Note on removal tag that reason for removal was compressor stall or surge.		
	2.9.1.1. Curvic faces distorted (412 and 412SP only).		
	2.9.1.2. Tube bent, twisted, or out of roundness.		
	2.9.1.3. Cracks.		
	2.9.1.4. Loose or sheared rivets.		

Table 1. After Compressor Stall or Surge (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.9.2. Disc pack couplings (412HP and 412EP/412EPI/412EPX only) - Any deformation of disc pack that results in gaps between laminates that are greater than 0.015 inch (0.38 mm) will require replacement. Not applicable to 412 and 412SP. 2.9.3. Disc pack coupling bolts - Distortion or cracks in any bolt will require replacement. Not applicable to 412 and 412SP. 2.9.4. Tail rotor drive adapters - Distortion beyond the limits specified in 412-A-65-10-00-00A-280A-A / 01261 will require replacement. 2.9.5. Tail rotor hub and blade assembly - perform an overhaul evaluation. Make an entry in component record showing reason for removal was compressor stall. 2.9.6. Remove main rotor blades retention bolts. Check 412-010-124 type bolts for deformation and perform magnetic particle inspection on bolts. Inspect 412-010-137-103 expandable bolts for damage. If satisfactory, return to service. 2.9.7. Remove main rotor damper bridge assemblies and check for deformation of bushings and bushing holes. 2.9.8. Perform close visual inspection of all other main rotor components. 2.9.9. If any discrepancies are noted as a result of inspection in step 2.9.6 through step 2.9.8, remove main rotor hub assembly, main rotor blades and mast assembly. Removed assemblies shall have an overhaul evaluation performed. Make an entry in component records showing reason for removal was compressor stall. 3. Airframe: 3.1. Check tailboom fin for evidence of damaged skin panels and/or structure and rivets for looseness and/or sheared heads. If inspection shows no indications of damage, return helicopter to flight status. If positive evidence of damage is found, comply with following step 3.2 through step 3.5. 3.2. Remove skin from tailboom fin adjacent to tail rotor gearbox mounting. Inspect all support structures in this area and repair as required. Install new skin. 3.3. Make close visual inspection of complete tailboom structure for distortion, buckles, skin cracks, and sheared or loose rivets. Pay particular attention to tailboom attachment points at FS 241.43 to 243.9, adjacent fuselage to tailboom intermediate gearbox support structure. 3.4. Make close visual inspection of main pylon support and engine mount attachment structure for distortion, buckles, cracks, sheared, or loose rivets, etc. 3.5. If discrepancies found during inspection in step 3.2 through step 3.4 cannot be repaired by standard procedures, replace discrepant assembly. 4. Install serviceable components.		

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Conditional Inspection - After Lightning Strikes

Procedure

Table 1. After Lightning Strikes

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____		
	FACILITY: _____		
	HELICOPTER S/N: _____		
	REGISTRY NO.: _____		
	TOTAL TIME:_____		
	SIGNATURE:_____		
	NOTE Accomplish after lightning strikes. In all instances below, if significant damage has been found in any area, inspection shall be expanded in those areas until it extends beyond the zone of damage. Any damage found anywhere on helicopter shall be recorded in detail and copies of these records shall be provided along with any component returned for overhaul to assist overhauling facility in evaluating component.		
	AFTER LIGHTNING STRIKES When helicopter is suspected of receiving a lightning strike, accomplish the following:		
	1. Visually inspect all external surfaces of helicopter with particular attention to main rotor hub and blades, transmission and mast assembly, tail rotor blades and hub, tail rotor gearbox, and vertical fin. Check electrical instruments and systems. Verify magnetic compass accuracy.		
	2. If visual indications of damage are present, proceed as follows:		
	2.1. Remove and visually inspect main rotor blades. If blades show any of the following indications, send the blades to a Bell Helicopter Textron approved blade repair facility for evaluation. State lightning strike as reason for removal.		
	2.1.1. Inspect blades for signs of burns. Burn marks can be very minute.		
	2.1.2. Inspect blades for debond in all bonded areas.		
	2.2. Remove main rotor hub for overhaul. State lightning strike reason for removal. Inspect main rotor hub and rotating controls for indication of arcing or burning.		
	2.3. Remove main driveshaft for inspection.		
BHT-412-CR&O, Chapter 63	2.3.1. Disassemble to extent required for cleaning and repacking coupling.		
	2.3.2. Clean coupling.		
	2.3.3. Visually inspect couplings for any evidence of arc burning or pitting. Pay special attention to tips, roots, and profiles of male and female coupling teeth. Any evidence of arc burning or pitting is cause for rejection.		
	2.3.4. Repack coupling.		
	2.4. Remove tail rotor output coupling for inspection.		
BHT-412-CR&O, Chapter 65	2.4.1. Disassemble to extent required for cleaning and repacking coupling.		
	2.4.2. Clean coupling.		
	2.4.3. Visually inspect couplings for any evidence of arc burning or pitting. Pay special attention to tips, roots, and profiles of male and female coupling teeth. Any evidence of arc burning or pitting is cause for rejection.		

Table 1. After Lightning Strikes (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.4.4. Repack coupling.		
	2.5. If main rotor blades, main rotor hub, main driveshafts, or tail rotor output coupling exhibit evidence of damage that can be attributed to a lightning strike, remove transmission and mast assembly for overhaul. State lightning strike as reason for removal. If no evidence of damage is noted on the above mentioned components, inspect main transmission as follows:		
412-A-63-00-00-00A-009A-A / 00987	2.5.1. Remove and inspect all transmission chip detectors.		
	2.5.2. If any evidence of arc burning or pitting is noted or excessive debris is found on chip detectors, remove transmission and mast assembly for overhaul. State lightning strike as reason for removal.		
	2.5.3. If no evidence of arc burning or pitting is noted, ground run light on skids for 1 hour. Reinspect chip detectors and remove and inspect oil filter. Repeat these inspections after accumulating 5 flight hours but prior to 10 flight hours.		
412-A-64-00-00-00A-009A-A / 01225	2.6. Inspect tail rotor blades and hub. If indications of burns or debonding are present, send the blades to a Bell Helicopter Textron approved blade repair facility for evaluation. State lightning strike as reason for removal. Remove tail rotor hub for overhaul.		
412-A-65-00-00-00A-009A-A / 01257	2.7. Remove and inspect tail rotor gearbox and intermediate gearbox chip detectors.		
	2.8. If tail rotor blades, tail rotor hub, or tail rotor output coupling exhibit evidence of damage that can be attributed to a lightning strike or excessive debris is found on chip detectors, remove tail rotor gearbox, intermediate gearbox, and tail rotor driveshaft hangers for overhaul. State lightning strike as reason for removal. Additionally, tail rotor driveshaft tubes, disc pack couplings, and attaching hardware shall be visually inspected for evidence of arc burns or pitting. Any evidence of arc burning or pitting is cause for rejection.		
	2.9. If no evidence of arc burning or pitting is noted, ground run light on skids for one hour. Reinspect chip detector. Repeat this inspection after accumulating 5 flight hours but prior to 10 flight hours.		
	3. If no visual indications of damage are present, proceed as follows:		
412-A-63-00-00-00A-009A-A / 00987	3.1. Remove and inspect all transmission chip detectors.		
	3.1.1. If excessive debris is found on chip detectors, remove transmission and mast assembly for overhaul.		
BHT-412-CR&O, Chapter 63	3.1.2. If little or no debris is found, ground run light on skids for one hour. Reinspect chip detectors.		
	3.2. Remove and inspect oil filter.		
	3.3. Repeat these inspections after accumulating 5 flight hours but prior to 10 flight hours.		
	4. Prior to first flight after a suspected or confirmed lightning strike, verify proper function of all drive system component chip detectors as follows:		
	4.1. Remove electrical connector from chip detector.		
	4.2. Remove chip detector and reinstall connector.		
	4.3. With electrical power applied to helicopter systems, bridge chip detector gap with a clean screwdriver or other clean conductive object.		
	4.4. Verify proper indication on caution panel. Verify illumination of MASTER CAUTION light.		

Table 1. After Lightning Strikes (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	4.5. Perform repairs as required. 5. When it has been established lightning has struck helicopter, inspections of electrical and instrument systems that follow are mandatory to ensure safety of flight: 5.1. Inspect all electrical wiring, bundles, and connectors for burning or electrical arcing. Unplug all connectors and inspect pins and housing for electrical arcing or burning. Inspect interior of all circuit breaker panels for burning or electrical arcing. Replace any wires, connectors, or circuit breakers found to be damaged. 5.2. Inspect main rotor blades and control links, transmission system, driveshafts, gearboxes, and tailboom structure for magnetization. Using a magnetometer with a range no larger than ±5 gauss, place arrow or red dot (depending on magnetometer model) within 0.5 inch (13 mm) of item being checked with arrow pointing directly at item. If components of any item have a reading greater than one gauss, those items shall be degaussed. **NOTE** Do not test chip detectors for magnetization. If transmission or gearbox magnetic readings are greater than one gauss near chip detector, remove chip detector from housing and repeat test. 5.3. Remove and bench test DC control units. Operationally check starter generators and inverters for proper operation. Visually inspect starter generators and inverters for burns or electrical arcing. If damaged, remove for internal inspection and bench test. 5.4. Perform operational check of bussing system as dictated in the 412-A-96-00-00-00A-009A-A / 01870. 5.5. Inspect transmission and tail rotor gearbox chip detectors for proper operation. Remove chip detectors found inoperative. 5.6. Perform operational check of interior and exterior lighting system. Replace lamps, bulbs, and lighting assemblies as required. 5.7. Perform operational check on all instruments. Remove and repair/replace instruments and sensors found to be defective. 5.8. Perform operational check on all caution messages for proper operation. 6. When it has been established lightning has struck helicopter, the structural inspections that follow are mandatory to ensure safety of flight: **NOTE** Arcing damage on metal components of airframe structure, when cleaned out to twice visible depth, shall be treated as mechanical damage, and damage limits in the 412-A-53-00-00-00A-009A-A / 00897 establish repairability and/or scrap of component. Any other structural damage (tears, voids, rupture, etc., directly or indirectly related to lightning strike) shall also be treated as mechanical damage. 6.1. Check honeycomb sandwich panels in suspect areas for voids or debond. If damage is apparent, proceed with normal maintenance procedures.		
412-A-96-00-00-00A-009A-A / 01870			

Table 1. After Lightning Strikes (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	6.2. Check fixed controls and support system components for possible arcing damage. Bearings in rod ends, bellcranks, and supports are most susceptible to arcing damage; check bearings for smooth rotation. Visibly inspect attaching hardware of support for signs of lightning damage (damaged finish and/or burns). If damage is evident, remove supports and inspect mounting holes and mating surfaces for arcing damage. Arcing damage shall be blended out to twice its visible depth. Repaired damages shall not exceed mechanical damage limits.		
412-A-32-00-00-00A-009A-A / 00694	6.3. When it is apparent lightning has been grounded through skid landing gear, remove entire landing gear assembly and inspect crosstubes and airframe support fittings for possible arcing damage; specifically, inspect attaching holes and mating surfaces of crosstubes directly beneath landing gear bearing/retaining supports. Clean out arcing damage to twice visible depth. Damage, after cleanup, shall not exceed allowable mechanical damage limits.		
	6.4. Any airframe metal parts not specifically identified above but are suspect, shall be noted in maintenance log and shall be reinspected prior to next 100 hours of flight.		

Conditional Inspection - After Engine Combining Gearbox Clutch Non-Engagement, Misengagement, or In-flight Slippage

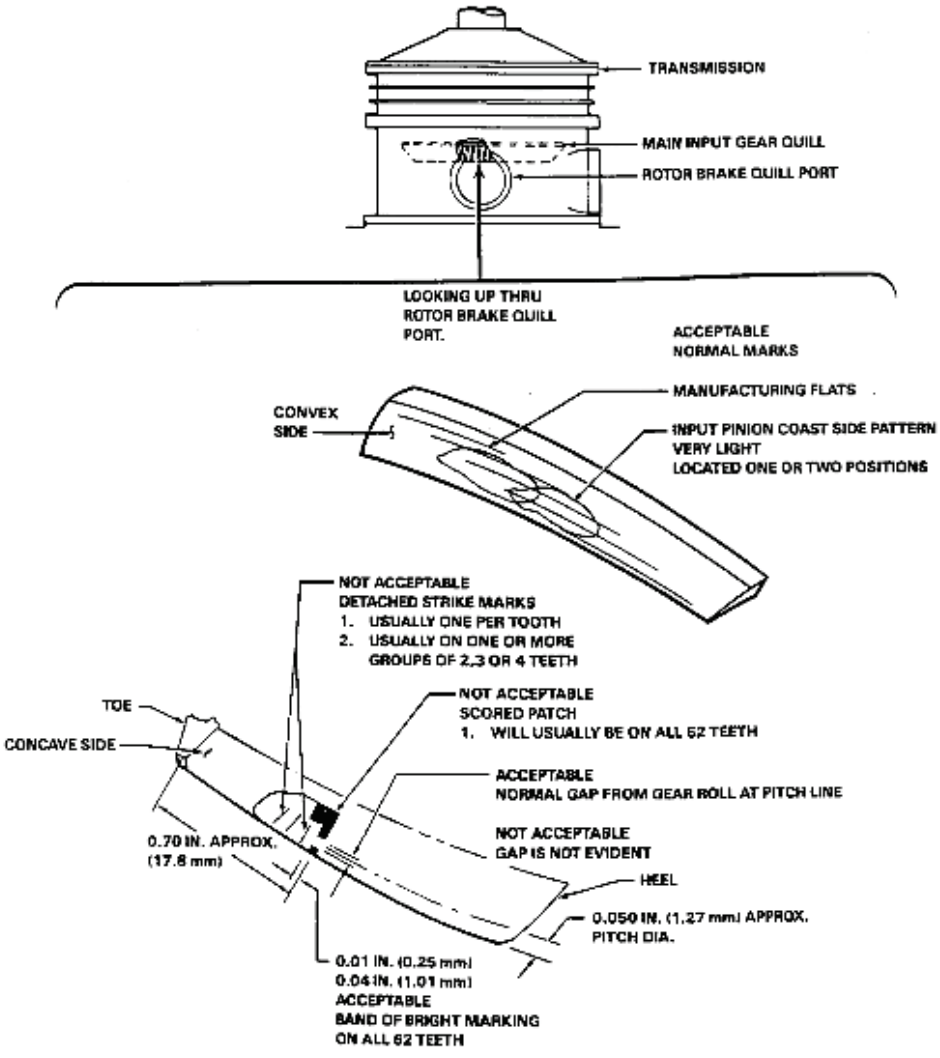
Procedure

Table 1. After Engine Combining Gearbox Clutch Non-Engagement, Misengagement, or In-flight Slippage

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL MECH	INITIAL OTHER
	DATE: _____W.O. _____ FACILITY: _____ HELICOPTER S/N: _____ REGISTRY NO.: _____ TOTAL TIME:_____ SIGNATURE:_____		
	NOTE Accomplish after engine combining gearbox clutch nonengagement, misengagement, or in-flight slippage. A nonengaged engine is indicated by near zero torque, higher N_2, and much cooler ITT as compared to engaged engine.		
	AFTER ENGINE COMBINING GEARBOX CLUTCH NONENGAGEMENT, MISENGAGEMENT, OR IN-FLIGHT SLIPPAGE		
Pratt & Whitney Canada Engine Maintenance Manual	1. Non-engagement - perform inspection in accordance with applicable Pratt & Whitney Canada Engine Maintenance Manual.		
	2. Misengagement.		
	NOTE If a sprag clutch has failed to engage, and/or subsequently engages with or without audible or physical (helicopter jolt) indications, proceed as follows:		
Pratt & Whitney Canada Engine Maintenance Manual	2.1. Combining gearbox shall be removed and returned to Pratt & Whitney Canada for investigation.		
	2.2. Inspect transmission spiral bevel gear as follows:		
BHT-412-CR&O, Chapter 63	2.2.1. Gain access to left side of transmission and, if installed, remove rotor brake and quill assembly or rotor brake quill port cover.		
	2.2.2. Utilizing rotor brake quill port opening, inspect all sixty two teeth of main spiral bevel gear convex side for unsatisfactory conditions such as scoring, hard lines, nicks, dents, chipping, metal deformation, etc. Inspect all 62 teeth on spiral bevel gear concave side (Figure 1).		
	NOTE Inspection mirrors and a suitable light are required. The rotor must be turned slowly to permit inspection of all gear teeth.		
BHT-412-CR&O, Chapter 63	2.2.3. Remove and replace any gear not meeting inspection requirements set forth in step 2.2.2. Overhaul rotor brake quill.		
	NOTE Components removed from a helicopter for evaluation following combining gearbox clutch nonengagement, misengagement, or in-flight slippage shall be evaluated as an interrelated group. Removal records accompanying each component shall cross reference part and serial numbers of other drive system components removed for evaluation.		
	2.2.4. Install rotor brake and quill.		

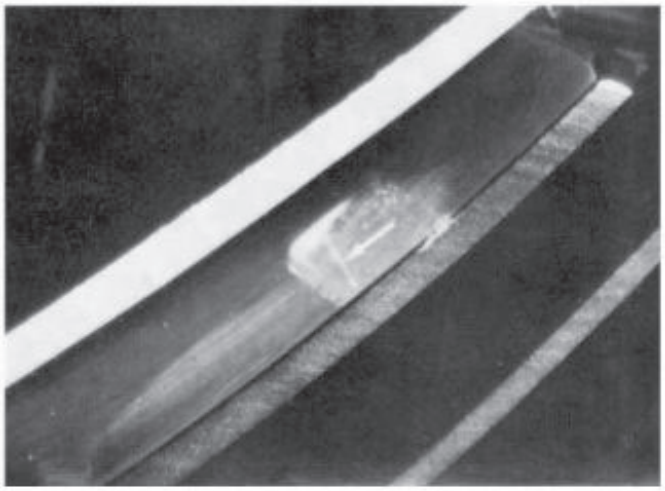
Table 1. After Engine Combining Gearbox Clutch Non-Engagement, Misengagement, or In-flight Slippage (continued)

DATA REFERENCE	INSPECTION TASK DESCRIPTION	INITIAL	
		MECH	OTHER
	2.2.5. Install cowling and inspection panels.		
	2.2.6. Perform a 15 minute ground run to determine if any oil leaks are present at rotor brake quill (or quill port cover) assembly.		
	2.3. Perform inspection in accordance with compressor stall or surge inspection, except inspect drive train as follows:		
	2.3.1. Visually inspect main rotor mast and controls for damage. If signs of yielding are evident, transmission and mast assembly and any other damaged components shall be replaced.		
412-A-65-10-00-00A-280A-A / 01261	2.3.2. Visually inspect tail rotor driveshaft for damage.		
	2.3.3. Remove output quill from tail rotor gearbox and inspect both sides of gear teeth for scoring, scuffing, or other damage or marks indicating excessive load.		
	NOTE If a tail rotor driveshaft has failed or yielded, all tail rotor driveshafts, hanger assemblies, intermediate gearbox, and tail rotor gearbox shall be replaced and tail rotor drive quill in transmission shall be removed and inspected for damage in accordance with compressor stall inspection. If scoring, scuffing, or other damage or marks indicating excessive load are found in intermediate gearbox or tail rotor gearbox, but tail rotor driveshafts are serviceable, gearbox(es) that are not serviceable shall be replaced.		
	3. In-flight clutch slippage.		
	NOTE If clutch slippage occurs, proceed as follows:		
Pratt & Whitney Canada Engine Maintenance Manual	3.1. Combining gearbox shall be removed and returned to Pratt & Whitney Canada for investigation.		
	3.2. Inspect spiral bevel gear in accordance with step 2.2.2. of step 2.2. If spiral bevel gear is damaged, perform misengagement inspection given in step 2.3. of step 2.		

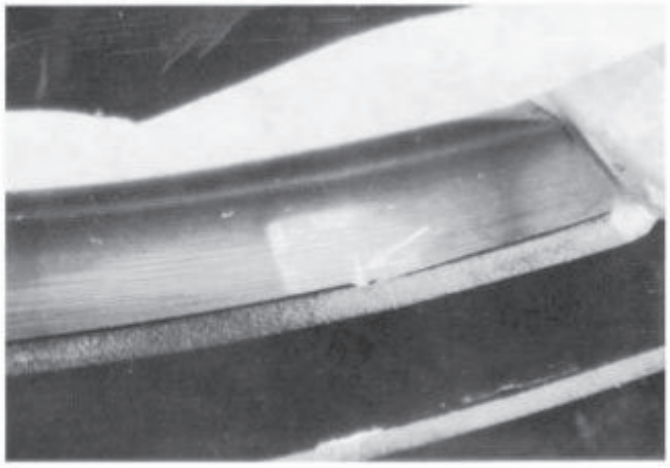


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Figure 1. Spiral Bevel Gear Inspection (Sheet 1 of 3)



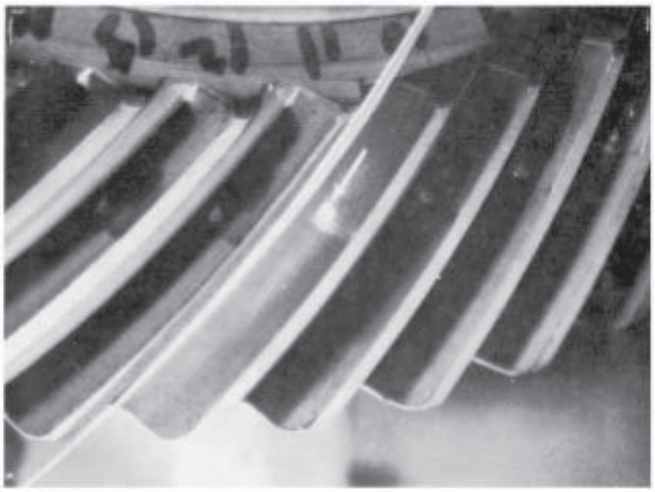
204-040-701 AND 412-040-701 SPIRAL BEVEL GEAR UNACCEPTABLE PATTERN



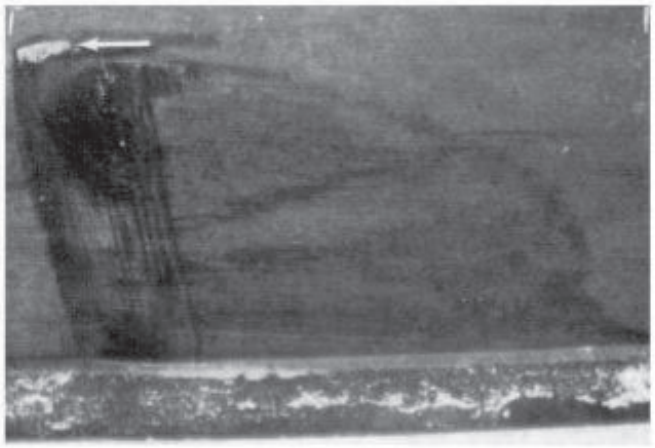
204-040-701 AND 412-040-701 SPIRAL BEVEL GEAR UNACCEPTABLE PATTERN

ICN-412-A-054700-B-97499-00002-A-001-01

Figure 1. Spiral Bevel Gear Inspection (Sheet 2 of 3)



204-040-701 AND 412-040-701 SPIRAL BEVEL GEAR UNACCEPTABLE PATTERN



204-040-701 AND 412-040-701 SPIRAL BEVEL GEAR UNACCEPTABLE PATTERN

ICN-412-A-054700-B-97499-00003-A-001-01

Figure 1. Spiral Bevel Gear Inspection (Sheet 3 of 3)

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Component Overhaul Schedule - Description

The Component Overhaul Schedule (Table 1) provides the time interval between overhaul for each applicable helicopter component.



DO NOT APPLY TOLERANCES TO PARTS WITH A LIMITED AIRWORTHINESS LIFE (412-A-04-00-00-00A-288A-A / 00013).



DO NOT EXCEED RETIREMENT LIFE FOR CRITICAL COMPONENTS. REFER TO AIRWORTHINESS LIMITATIONS SCHEDULE (412-A-04-00-00-00A-288A-A / 00013).

SOME PARTS INSTALLED AS ORIGINAL EQUIPMENT ON MILITARY HELICOPTERS MAY HAVE A LOWER AIRWORTHINESS LIFE AND/OR OVERHAUL SCHEDULE THAN WHEN USED ON A COMMERCIAL HELICOPTER. CONSEQUENTLY, PARTS THAT HAVE BEEN USED ON MILITARY HELICOPTERS SHOULD NOT BE USED ON COMMERCIAL HELICOPTERS.



OVERHAUL SCHEDULE FOR SOME KIT COMPONENTS AND/OR PARTS IS NOT COVERED IN THIS SCHEDULE. REFER TO APPLICABLE SERVICE INSTRUCTIONS FOR KIT COMPONENTS SCHEDULE.

NOTE

Refer to 412-A-05-00-00-04A-042A-A / 00020 for information on inspection and overhaul tolerance.

NOTE

Neither assignment of a time period for overhaul of a component or failure to assign a time period for overhaul of a component constitutes a warranty of any kind. The only warranty applicable to helicopter and any component is that warranty included in the Purchase Agreement for the helicopter or component.

The overhaul interval specified for any given part number contained in this Component Overhaul Schedule applies to all successive dash numbers (or suffixes) for that item unless otherwise specified.

Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual or Service Bulletin for engine and related component overhaul intervals.

Table 1. Component Overhaul Schedule

NOMENCLATURE	PART NUMBER (1)	OVERHAUL INTERVAL (HOURS)
	ROTORS	
Swashplate and Support Assembly	412-010-402-101	2500 hours
Collective Lever Assembly	412-010-408-ALL	2500 hours
Collective Lever Assembly	412-010-464-101	2500 hours
Hub and Sleeve Assembly	412-010-401-105	2500 hours
Main Rotor Pitch Link Assembly	412-010-425-ALL	2500 hours
Main Rotor Hub Assembly	412-010-100 Series	(2)
Tail Rotor Hub Assembly	212-011-701-003	(10) 2500 hours
Main Rotor Blades	412-015-200 & -300 Series	(2)
	POWER TRAIN	
Rotor Brake Quill	412-040-123-101	3000 hours
Rotor Brake Quill	412-040-123-103	3000 hours
Rotor Brake Quill	412-540-014-101	(12) 3000 hours

Table 1. Component Overhaul Schedule (continued)

NOMENCLATURE	PART NUMBER (1)	OVERHAUL INTERVAL (HOURS)	
Driveshaft Assembly	212-040-005-103		3000 hours
Driveshaft Assembly	412-040-005-101		3000 hours
Transmission	412-040-002-103	(6) (18)	6000 hours
Transmission	412-040-004-103	(6) (13) (14)	6000 hours
Transmission	412-040-007-101	(6) (13)	6000 hours
Quill Assembly, Auxiliary Equipment	412-040-703-103	(15)	1000 hours
Intermediate Gearbox Assembly	412-040-006-101	(7) (17)	5000 hours
Intermediate Gearbox Assembly	212-040-003-023		5000 hours
Tail Rotor Gearbox Assembly	212-040-004-103	(7) (17)	5000 hours
Tail Rotor Gearbox Assembly	212-040-004-009		5000 hours
Tail Rotor Driveshaft Hangers	212-040-600-005		3000 hours
Tail Rotor Driveshaft Hangers	412-040-600-101	(17)	5000 hours
Tail Rotor Driveshaft Coupling	412-040-601-101	(17)	5000 hours
Tail Rotor Driveshaft Coupling	412-040-601-103	(17)	5000 hours
Mast Assembly	412-040-366-103/-105	(9) (19) (24)	5000 hours
Mast Assembly	412-040-366-109	(4) (8) (9) (16) (23) (24)	5000 hours
Mast Assembly	412-040-366-111/-113	(4) (9) (16) (23) (24)	5000 hours
Mast Assembly	412-040-366-125	(9) (19)	6000 hours
Mast Assembly	*412-040-366-127/-129*	(9) (16) (22)	6000 hours
HYDRAULIC			
Flight Control Hydraulic Cylinders (Cyclic and Collective)	212-076-005-101		2500 hours
POWER PLANT			
Engine Combining (Reduction) Gearbox	PT6T-3B	(3) (21)	3500 hours
Engine Combining (Reduction) Gearbox	PT6T-3BE	(7) (20) (21)	4000 hours
Engine Combining (Reduction) Gearbox	PT6T-3D	(7) (20) (21)	4000 hours
Engine Combining (Reduction) Gearbox	PT6T-3DE	(7) (20) (21)	4000 hours
Engine Combining (Reduction) Gearbox	PT6T-3DF	(7) (20) (21)	4000 hours
Starter Generator	209-060-221-001		1000 hours
Starter Generator	200SG119Q		1000 hours
Fire Extinguisher Container	209-062-908-001	(11)	5 years

NOTES:

1 Operating time specified for overhaul of any given part number in this schedule applies to all successive dash numbers (or suffixes), unless otherwise specified.

2 On condition. Special Inspection required each 2500 hours (412-A-05-47-00-17A-283A-A / 00052 and 412-A-05-47-00-18A-283A-A / 00053).

3 The overhaul interval for engine combining (reduction) gearboxes modified in accordance with TB 412-89-80is 4000 hours.

Table 1. Component Overhaul Schedule (continued)

NOMENCLATURE	PART NUMBER (1)	OVERHAUL INTERVAL (HOURS)

4 *Special Inspection required each 2500 hours or 5 years on helicopters S/N 36020 through 36999 and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade) (412-A-05-47-00-19A-283A-A / 00054).*

5 Refer to (19)

6 All transmission quills, with the exception of rotor brake quill, shall be overhauled at time of transmission overhaul.

7 Incorporation of Increased Maximum Continuous Power Kit (BHT-412-SI-38) P/N 412-706-029-ALL reduces the Time Between Overhaul (TBO) interval to 2500 hours.

8 Mast assembly 412-040-366-109 is not field serviceable. It must be returned to Bell Helicopter Textron for any maintenance requiring disassembly.

9 For each hoist operation performed within penalty CG region, 4 additional flight hours must be logged against main rotor mast, yoke, and lower cone seat. Refer to 412-A-04-00-00-00A-288A-A / 00013 for penalty region chart.

10 The following items are to be overhauled at the same time as the tail rotor hub assembly:

Idler assembly 209-011-711-ALL

Lever assembly 209-011-712-ALL

Nut, tail rotor retention 212-010-706-ALL

Crosshead 212-010-775-ALL

Link assembly 209-011-713-ALL

11 Hydrostatic test in accordance with specification DOT-4 DA, DOT-4 DS700, or DOT-SP-7945, as marked on the reservoir, every 5 years or prior to refill after leakage or discharge. Extensions to this item are not permitted.

12 Refer to TB 412-03-191.

13 *Special inspection required each 3000 hours on helicopters S/N 36020 through 36999, S/N 37002 through 37999 and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade) (412-A-05-47-00-23A-283A-A / 00058).*

14 Incorporation of Increased Maximum Continuous Power Kit (BHT-412-SI-38) P/N 412-706-029-ALL reduces the Time Between Overhaul (TBO) interval to 3000 hours.

15 Refer to BHT-412-SI-36 for maintenance information.

16 Visually inspect (BHT-412-CR&O, Chapter 63) tube assembly 412-040-510-103 at mast assembly overhaul.

17 *Special inspection required each 2500 hours on helicopters S/N 36020 through 36999, S/N 37002 through 37999 and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade) (412-A-05-47-00-20A-283A-A / 00055).*

18 *Special inspection required each 3000 hours on helicopters prior to S/N 36020 and helicopters not modified by 412-570-001-103 or BHT-412-SI-74 (412SP to 412HP Upgrade) or BHT-412-SI-97 (412 Classic to 412HP Upgrade) (412-A-05-47-00-22A-283A-A / 00057).*

19 *Special inspection required each 3000 hours or 5 years on helicopters prior to S/N 36020 and helicopters not modified by 412-570-001-103 or BHT-412-SI-74 (412SP to 412HP Upgrade) or BHT-412-SI-97 (412 Classic to 412HP Upgrade) (412-A-05-47-00-21A-283A-A / 00056).*

20 Refer to Pratt & Whitney Canada PT6T-3D Series Maintenance Manual for instructions on 2500 hour clutch inspection requirements.

21 The inspection and overhaul tolerance permitted by Bell only applies to the reduction gearbox (RGB). For all other engine components, contact Pratt & Whitney Canada for any deviation in maintenance requirements.

22 *Special inspection required each 3000 hours or 5 years on helicopters S/N 36020 through 36999, S/N 37002 through 37999 and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade) and Post TB 412-14-231 (412-A-05-47-00-19A-283A-A / 00054).*

23 *If the intent of TB 412-14-231 has been accomplished, special inspection (412-A-05-47-00-19A-283A-A / 00054) is required each 3000 hours or 5 years on helicopters S/N 36020 through 36999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412 Classic to 412HP Upgrade).*

24 If the intent of TB 412-14-231 has been accomplished, then the overhaul interval is 6000 hours.

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CHAPTER 06
DIMENSIONS AND CHARTS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

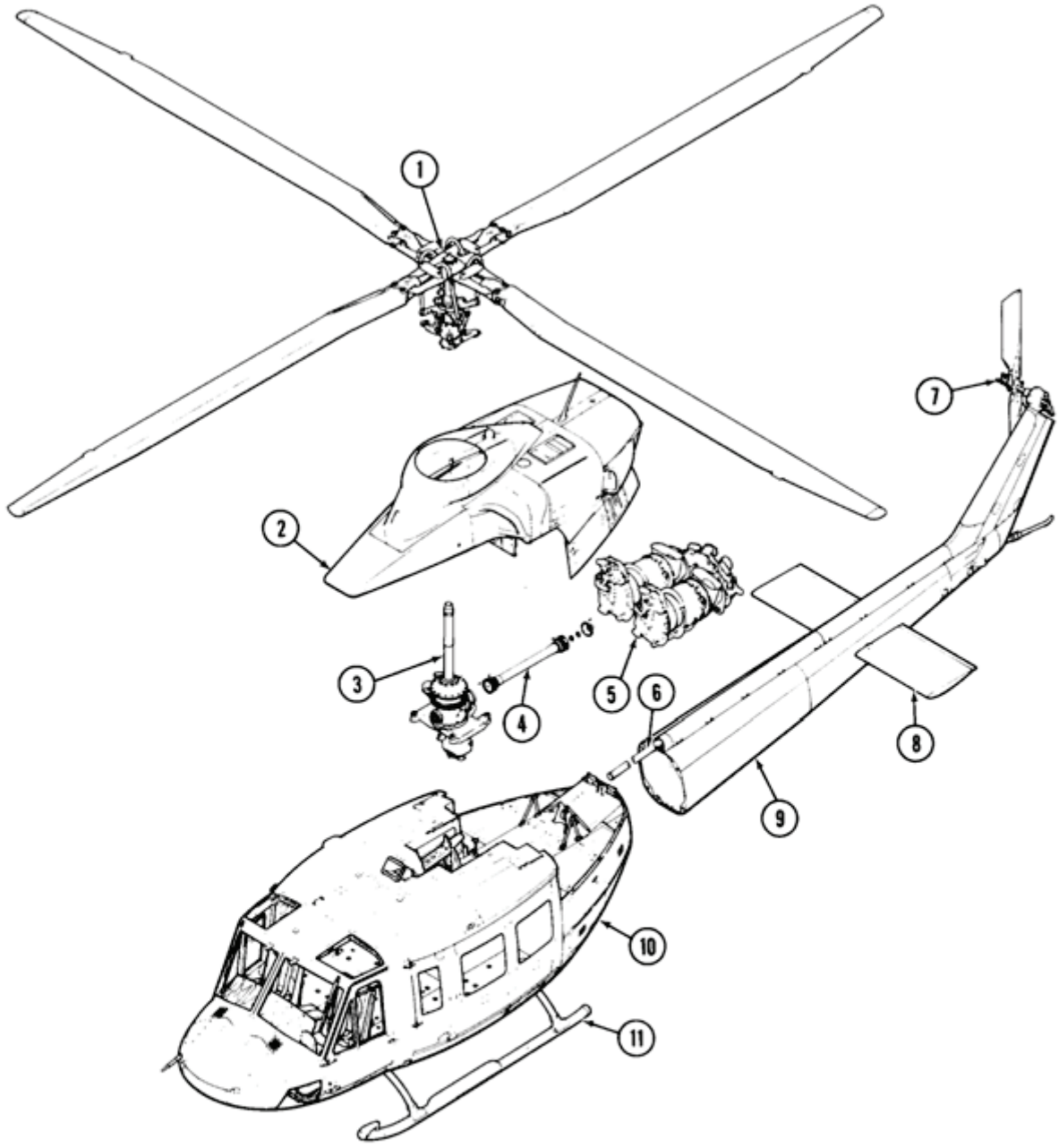
C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Airframe Principal Dimensions - Description	DMC-412-A-06-10-00-00A-028A-A / 00072	2023-05-31	ALL
Station Lines, Buttock Lines, and Water lines - Description	DMC-412-A-06-20-00-00A-042A-A / 00073	2023-05-31	ALL

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Airframe Principal Dimensions - Description

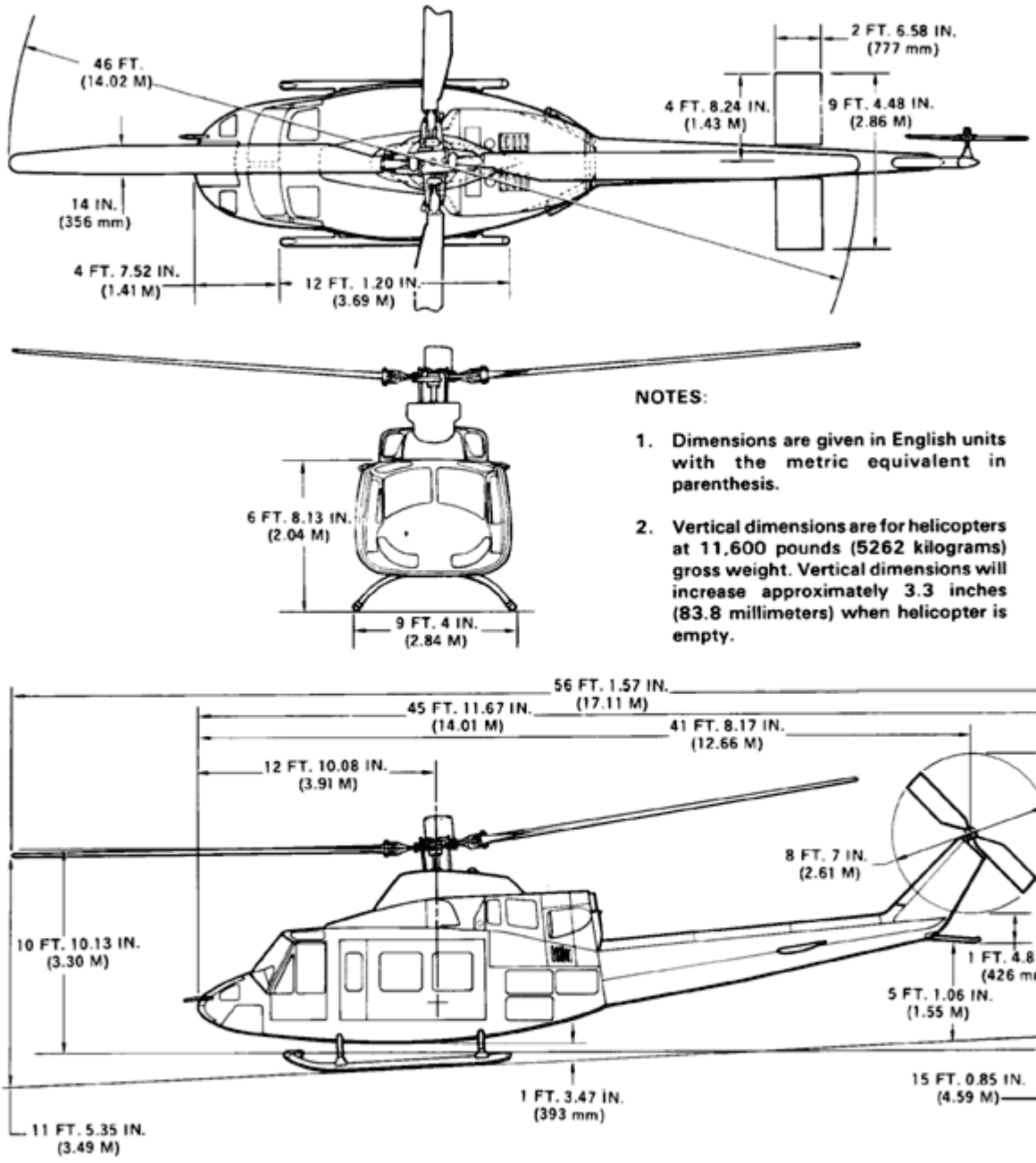
Figure 1 depicts the major components of the helicopter. Figure 2 depicts principal dimensions of helicopter airframe. Due to variations in loading and landing gear deflection, all height dimensions are approximate. With ground handling wheels installed and fully extended, height is increased by 5.0 in. (12.7 cm).



1. Main rotor hub and blades
2. Transmission and engine cowl
3. Transmission and mast
4. Main driveshaft
5. Engine
6. Tail rotor driveshafts
7. Tail rotor hub and blades
8. Aerodynamically actuated elevator
9. Tailboom
10. Forward fuselage
11. Landing gear

ICN-412-A-061000-B-97499-00001-A-001-01

Figure 1. Major Components Location (Sheet 1 of 1)



ICN-412-A-061000-B-97499-00002-A-001-01

Figure 2. Airframe Principal Dimensions (Sheet 1 of 1)

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Station Lines, Buttock Lines, and Water lines - Description

GENERAL. Station lines, buttock lines, and water lines are used to determine locations on, and within, the helicopter. All three are measured in inches from known points therefore these lines will not be expressed in metric equivalents within this manual.

NOTE

The nose of the helicopter is station -20.0. Station 0 is 20.0 inches aft of the most forward part of the nose.

STATION (STA) LINES: Stations are vertical planes perpendicular to, and measured along, the longitudinal axis of helicopter. Several station lines are shown on Figure 1 at recognizable locations on the airframe. Other station locations can be measured from these lines. Tailboom stations, stations within the baggage compartment stations and stations along the vertical fin are illustrated in the same manner. These stations are perpendicular to the centerline of tailboom and fin as applicable because these components are mounted at an angle to the horizontal plane of fuselage.

WATER LINE (WL) : Water lines are horizontal planes perpendicular to, and measured along, the vertical axis of the helicopter. Water line 0 is 7.44 in. (18.9 cm) below the lowest point of fuselage as shown on Figure 1. Water lines can be used to measure locations as described for station lines in Paragraph 2.

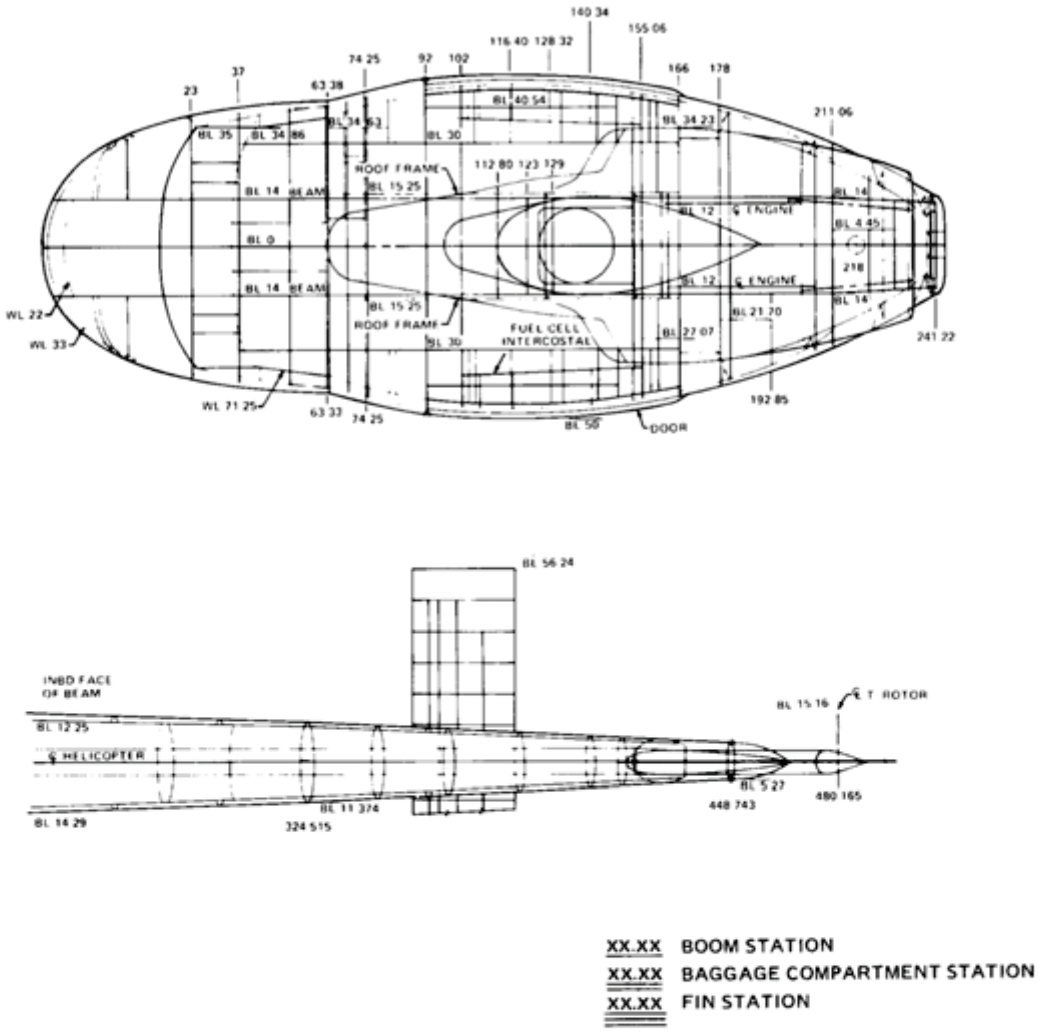
BUTTOCK LINES (BL) : Buttock lines are vertical planes perpendicular to, and measured to, the left and right along the lateral axis of the helicopter. Buttock line 0 is the plane of the vertical centerline of the helicopter. Buttock lines can be used to measure locations as described for station lines in Paragraph 2.

TAILBOOM AND BAGGAGE COMPARTMENT STATION LINES: Baggage compartment station lines are measured from tailboom fuselage attach point to aft end of baggage compartment. These stations are numbered 17.42 through 101.38. Tailboom station lines are measured from aft end of baggage compartment to center of intermediate gearbox. These are numbered 101.38 through 206.17.

ELEVATOR STATION (ES) LINES: Elevator station lines are buttock lines extended through the elevator to the elevator outboard tips. These lines are referred to as Elevator Stations (ES) . Fuselage Stations (FS) and Bottom Stations (BS) also apply to the elevator.

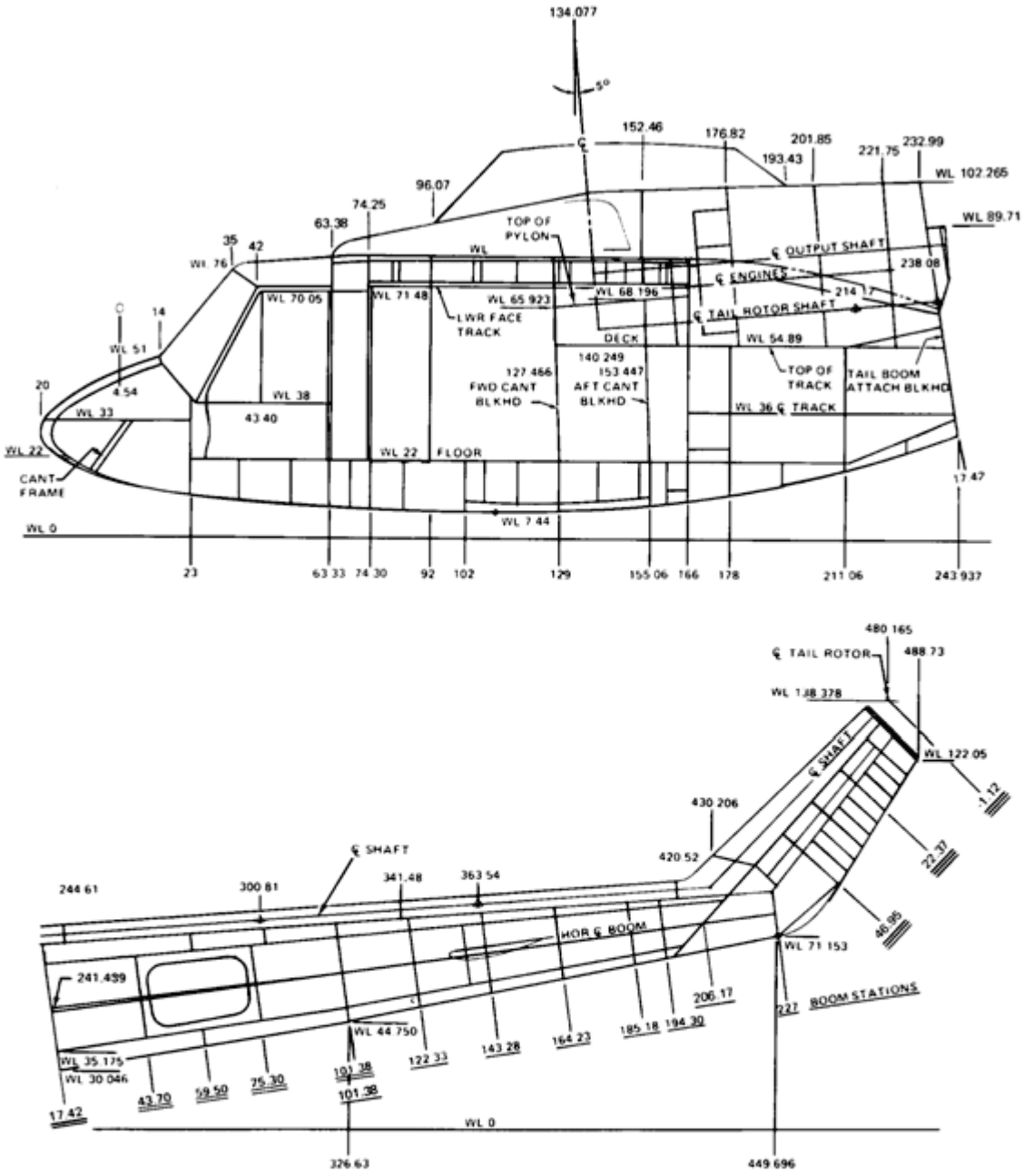
VERTICAL FIN STATION (VFS) LINES: Vertical fin stations are parallel vertical lines perpendicular to the centerline of tail rotor shaft, below leading edge of vertical fin. These stations are measured from Vertical Fin Station (VFS) 0 which is 1.12 in (2.85 mm) below centerline of tail rotor which is at vertical fin station -1.12. These lines are referred to as Vertical Fin Station (VFS) s. Fuselage Stations (FS) also apply to vertical fin.

MAIN AND TAIL ROTOR BLADE STATION LINES. Main and tail rotor blade station lines are measured from the center of hub to tip of blade.



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Figure 1. Airframe Reference Lines (Sheet 1 of 2)



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Figure 1. Airframe Reference Lines (Sheet 2 of 2)

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CHAPTER 07
LIFTING AND JACKING

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Complete Helicopter (Main Rotor Hub Installed) - Lifting	DMC-412-A-07-00-00-00A-171A-A / 00075	2023-05-31	ALL
Complete Helicopter (Mast Installed) - Lifting	DMC-412-A-07-00-00-01A-171A-A / 00076	2023-05-31	ALL
Complete Helicopter (Mast Removed) - Lifting	DMC-412-A-07-00-00-02A-171A-A / 00077	2023-05-31	ALL
Complete Helicopter (Transmission Removed) - Lifting	DMC-412-A-07-00-00-03A-171A-A / 00078	2023-05-31	ALL
Tailboom Only - Lifting	DMC-412-A-07-00-00-04A-171A-A / 00079	2023-05-31	ALL
Helicopter - Jacking	DMC-412-A-07-10-00-00A-172A-A / 00080	2023-05-31	ALL

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Complete Helicopter (Main Rotor Hub Installed) - Lifting

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Clevis	204-011-178-001 or equivalent	1	

Procedure



IF MAIN ROTOR HUB HAS BEEN REMOVED, DO NOT LIFT HELICOPTER WITH MAST CAP (412-010-160) INSTALLED. USE clevis (204-011-178-001 or equivalent) WITH MAST NUT 204-010-481-001.

DO NOT USE clevis (204-011-178-001 or equivalent) TO LIFT HELICOPTER WITH GROSS WEIGHT IN EXCESS OF 10,000 POUNDS.

1. Attach hoisting clevis (204-011-178-001 or equivalent) to eye on bolted down mast cap (412-010-160). Connect suitable hoist and take up slack.

2. Station a person at tail skid to steady helicopter when hoisting. If lifting beyond reach from ground, use a rope to steady helicopter.

3. Hoist slowly with a constant lifting force.

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Complete Helicopter (Mast Installed) - Lifting

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Mast Nut	204-010-481-001	1	
Clevis	204-011-178-001 or equivalent	1	

Procedure



DO NOT LIFT HELICOPTER WITH MAST CAP (412-010-160) INSTALLED. USE clevis (204-011-178-001 or equivalent) WITH mast nut (204-010-481-001).

DO NOT USE clevis (204-011-178-001 or equivalent) TO LIFT HELICOPTER WITH GROSS WEIGHT IN EXCESS OF 10,000 POUNDS.

1. Attach hoisting clevis (204-011-178-001 or equivalent) to eye provided on mast nut (204-010-481-001) at top of main rotor mast. Connect suitable hoist and tack up slack.

2. Station a person at tail skid to steady helicopter when hoisting. If lifting beyond reach from ground, use a rope to steady helicopter.

3. Hoist slowly with a constant lifting force.

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Complete Helicopter (Mast Removed) - Lifting

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Lift Plate and Cover	T103211-101	1	
Clevis	204-011-178-001	1	

Procedure

1. Install lift plate and cover (T103211-101) (or equivalent) on transmission.

2. Attach hoisting clevis (204-011-178-001) (or equivalent) to lifting eye.

3. Station a person at tail skid to steady helicopter when hoisting. If lifting beyond reach from ground, use a rope to steady helicopter.

4. Hoist slowly with a constant lifting force.

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Complete Helicopter (Transmission Removed) - Lifting

Procedure

1. Attach a suitable hoist to pylon lift link (412-A-63-00-00-00A-009A-A / 00987).
2. Station a person at tail skid to steady helicopter when hoisting. If lifting beyond reach from ground, use a rope to steady helicopter.
3. Hoist slowly with a constant lifting force.

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Tailboom Only - Lifting

Procedure

1. Hoist tailboom as a separate component with sling positioned at tailboom CG. Use sling manufactured from belting material. Position a person at each end of tailboom to steady tailboom in sling.

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Helicopter - Jacking

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hydraulic Jacks	Commercial	4	Tronair P/N 02-0511-0132 or equivalent
Hydraulic Jacks	Commercial	4	Tronair P/N 02-0517C0140 or equivalent, for high skid gear

Procedure



DO NOT ALLOW PERSONNEL TO CRAWL INTO OR ONTO HELICOPTER WHILE RAISING OR WHILE HELICOPTER IS SUPPORTED ON JACKS. ROPE OFF AREA AND DISPLAY SIGNS STATING “THIS HELICOPTER ON JACKS”.



DO NOT PLACE HELICOPTER ON JACKS IF WIND VELOCITY EXCEEDS 20 KNOTS. HEAD HELICOPTER INTO THE PREVAILING WIND.

PROVIDE SUITABLE SURFACE FOR JACKS TO REST ON (LEVEL, HARD SURFACE OR EQUIVALENT).

1. Raise helicopter as follows:

1.1. Tie down main rotor blades (412-A-10-00-00-00A-042A-A / 00110).

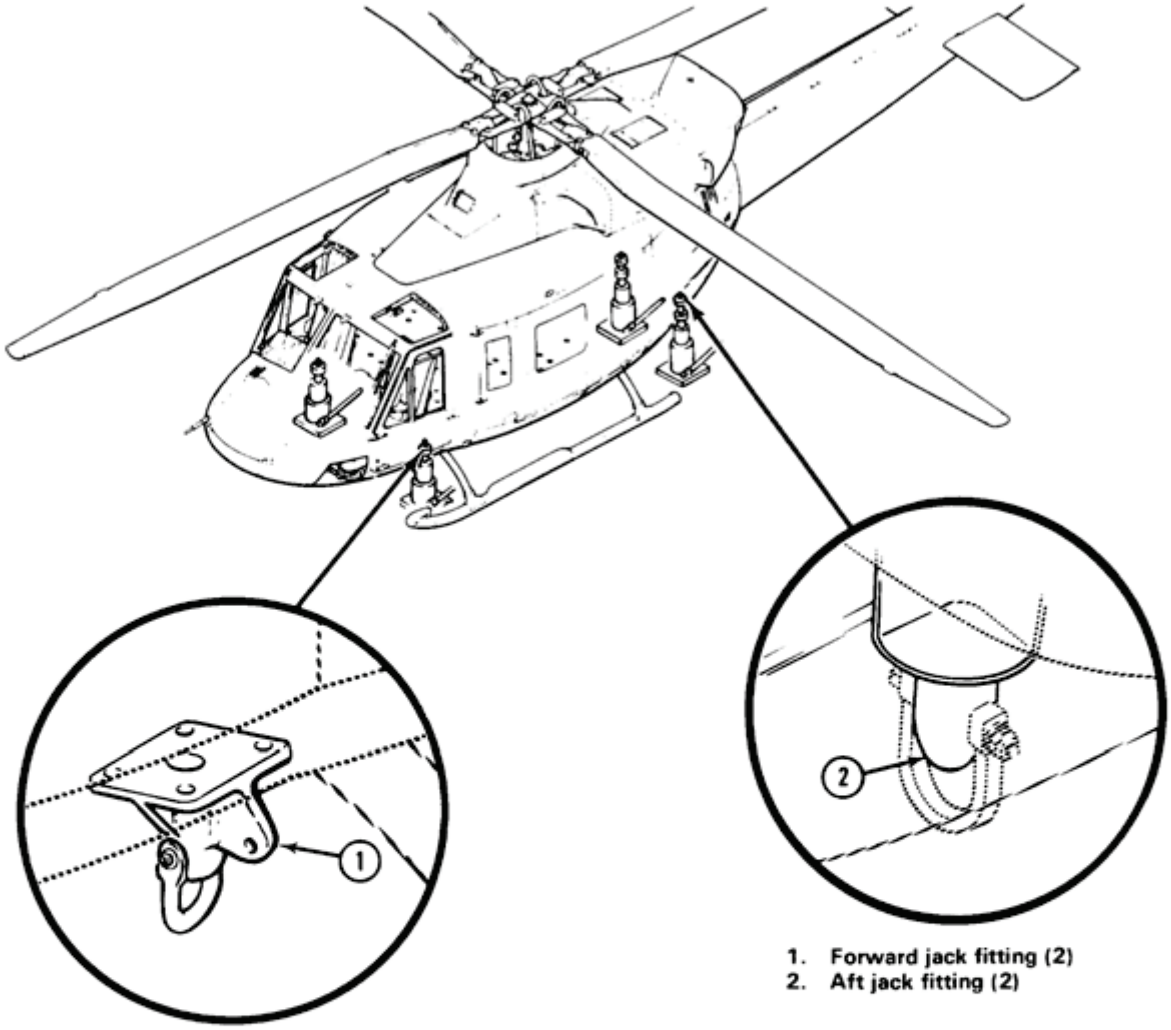
1.2. Position a jack under each jack fitting (preferred method) or under each forward jack fitting and one aft jack fitting (alternate method) and raise helicopter clear of floor (Figure 1).

1.3. Operate jack handles slowly and evenly at all positions, being careful to keep helicopter level, as it is raised to desired height.

2. Lower helicopter as follows:

2.1. Slowly and carefully lower all jacks simultaneously.

2.2. Remove jacks and other equipment from area.



1. Forward jack fitting (2)
2. Aft jack fitting (2)

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Figure 1. Jacking (Sheet 1 of 1)

CHAPTER 08
WEIGHT AND BALANCE

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Weight and Balance - Description	DMC-412-A-08-40-00-00A-028A-A / 00082	2023-05-31	ALL
Weight and Balance - General	DMC-412-A-08-40-00-00A-040A-A / 00083	2023-05-31	ALL
Weight and Balance - Terminology	DMC-412-A-08-40-00-00A-018A-A / 00084	2023-05-31	ALL
Weight and Balance - Leveling	DMC-412-A-08-40-00-00A-169A-A / 00085	2023-05-31	ALL
Weight and Balance - Weighing	DMC-412-A-08-40-00-01A-169A-A / 00086	2023-05-31	ALL
Weighing - Preparation	DMC-412-A-08-40-00-01A-160A-A / 00087	2023-05-31	ALL
Weighing - Procedure	DMC-412-A-08-40-00-01A-160B-A / 00088	2023-05-31	ALL
Weight and Balance - Calculations	DMC-412-A-08-40-00-02A-169A-A / 00089	2023-05-31	ALL
Center of Gravity - Calculations	DMC-412-A-08-40-00-03A-169A-A / 00090	2023-05-31	ALL
Longitudinal CG - Calculations	DMC-412-A-08-40-00-04A-169A-A / 00091	2023-05-31	ALL
Lateral CG - Calculations	DMC-412-A-08-40-00-05A-169A-A / 00092	2023-05-31	ALL
Empty Weight - Calculations	DMC-412-A-08-40-00-06A-169A-A / 00093	2023-05-31	ALL
Ballast Requirements - Using the Gross Weight CG Envelope Chart - Calculations	DMC-412-A-08-40-00-07A-169A-A / 00094	2023-05-31	ALL
Weight and Balance - Actual Weight Record (Example)	DMC-412-A-08-40-00-08A-169A-A / 00095	2023-05-31	ALL
Weight and Balance - Ballast Removal	DMC-412-A-08-40-00-00A-520A-A / 00096	2023-05-31	ALL
Weight and Balance - Ballast Installation	DMC-412-A-08-40-00-00A-720A-A / 00097	2023-05-31	ALL
Tail Ballast (Weight) - Removal	DMC-412-A-08-40-20-00A-520A-A / 00098	2023-05-31	ALL
Tail Ballast (Weight) - Installation	DMC-412-A-08-40-20-00A-720A-A / 00099	2023-05-31	ALL
Tail Ballast (Lead Shot) Alternate Method - Removal	DMC-412-A-08-40-20-01A-520A-A / 00100	2023-05-31	ALL
Tail Ballast (Lead Shot) Alternate Method - Installation	DMC-412-A-08-40-20-01A-720A-A / 00101	2023-05-31	ALL

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Weight and Balance - Description

This chapter gives the procedures to find the actual Empty Weight and the Center of Gravity (CG) of a Model 412 helicopter.

It also provides instructions to determine if any changes (i.e., ballast) are necessary to assist in keeping the helicopter within the Gross Weight CG Envelope (BHT-412-FM) during operation. Instructions for the installation or removal of ballast are also provided.

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Weight and Balance - General

Procedure

1. The Center of Gravity (CG) is the balance point of a body. It is used in weight and balance calculations and directly affects the payload capabilities and handling characteristics of a helicopter.



MAKE SURE THE CENTER OF GRAVITY IS ALWAYS WITHIN THE OPERATIONAL LIMITS, OR MANEUVERABILITY OF THE HELICOPTER MAY BE REDUCED. THIS COULD RESULT IN UNSAFE FLIGHT CONDITIONS THAT MAY CAUSE DEATH OR SERIOUS INJURIES TO PERSONS.

2. It is the pilot's responsibility to make sure that each flight is accomplished within the limits of the Gross Weight CG Envelope given in the Flight Manual.

3. If the helicopter CG is not within the certified limits (i.e., the Gross Weight CG Envelope), the helicopter will assume an unfavorable attitude. This condition reduces the cyclic control range available for operations and introduces high loads into the rotor system, causing excessive wear of components.

NOTE

For additional weight and balance information, operators in territorial United States should refer to FAR 29. Operators outside territorial United States should refer to information issued by the licensing authority.

4. Because this condition can be dangerous and is undesirable, it is important to keep the helicopter operational CG within the Gross Weight CG Envelope. Do this as follows:

4.1. Change the location of the helicopter Empty Weight CG (i.e., relocate equipment or add/remove fixed ballast) as required.

4.2. Calculate the combination of Useful Load items that are permitted for each flight.

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Weight and Balance - Terminology

Refer to Chapter 1 (412-A-00-00-00-00A-042A-A / 00009) for definitions of the technical terms used in this chapter.

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Weight and Balance - Leveling

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Plumb Bob	Commercial	1	
Level	Commercial	1	(Alternate)

Procedure

NOTE

The following information may be used for any maintenance task that requires the helicopter to be leveled.

NOTE

The leveling plate (2, Figure 1) is marked in increments of 1/4°. The leveling plate is located on the cabin floor just inside the left passenger cargo door. A slotted hole is located in the upper door frame above the leveling plate to support the plumb bob (1).

1. To level the helicopter longitudinally and laterally, do the following steps:

1.1. Put the helicopter on a hard and level surface.

1.2. Support the helicopter as necessary (412-A-07-10-00-00A-172A-A / 00080).

1.3. Hang the plumb bob (1) in the slotted opening in the door frame with the point of the plumb bob just above the leveling plate (2).

1.4. If you are leveling the helicopter for weighing, install the weighing devices (412-A-08-40-00-01A-160B-A / 00088).



THE HELICOPTER MUST BE ON A HARD AND LEVEL SURFACE BEFORE LIFTING WITH JACKS. OTHERWISE, IT MAY FALL OFF THE JACKS, GET DAMAGED, AND/OR CAUSE INJURY TO PERSONNEL.



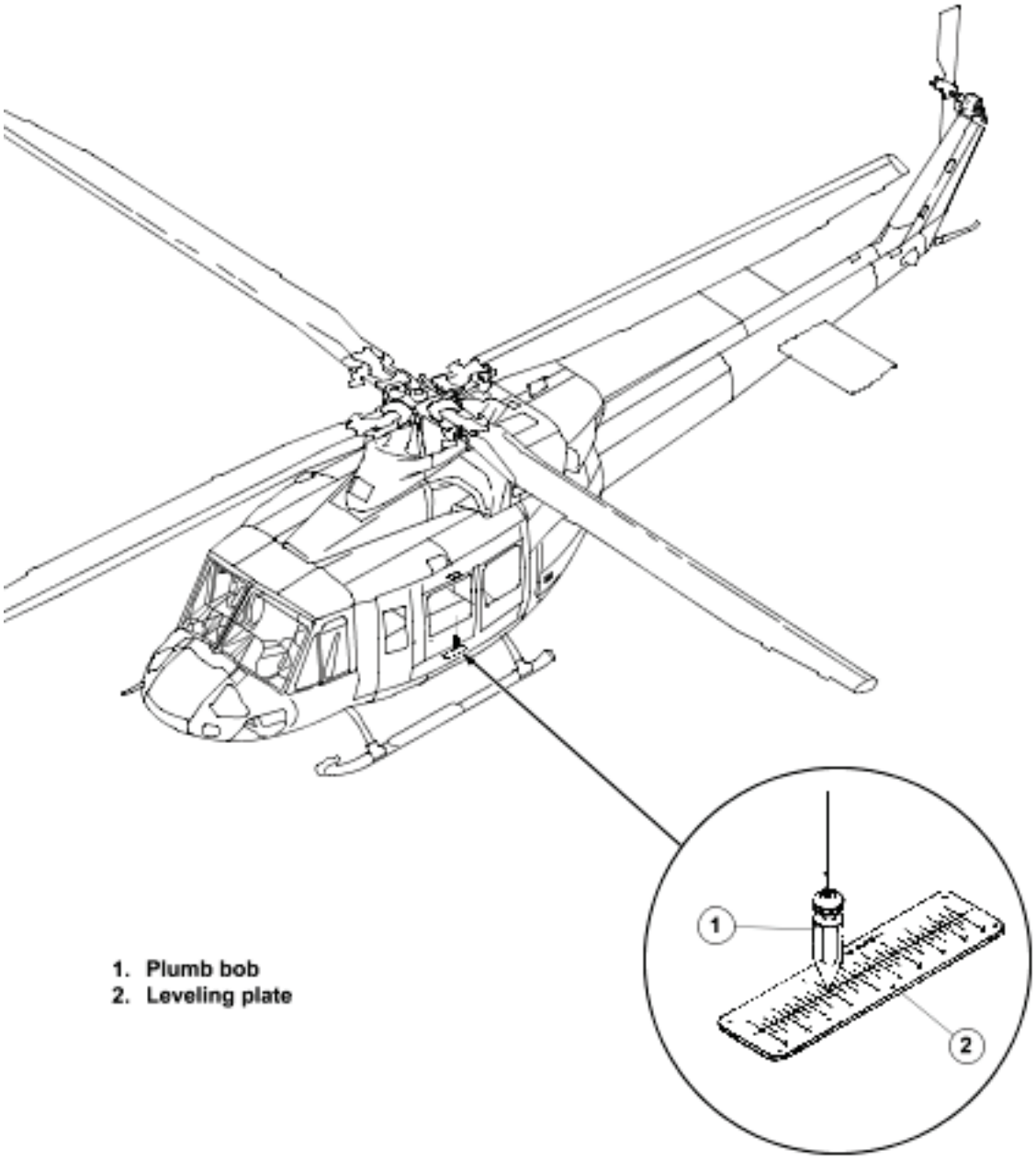
DO NOT ALLOW PERSONNEL IN OR ON THE HELICOPTER WHEN IT IS BEING LIFTED. THE HELICOPTER MAY NOT BE STABLE. FAILURE TO OBEY MAY RESULT IN INJURY TO PERSONNEL AND DAMAGE TO THE HELICOPTER.

1.5. Lift the helicopter with jacks (412-A-07-10-00-00A-172A-A / 00080).



IF YOU ARE LEVELING THE HELICOPTER FOR WEIGHING, MAKE SURE THAT THE HELICOPTER IS CLEAR OF THE FLOOR AND NOT TOUCHING ANY OTHER OBJECT. OTHERWISE, THE SCALE READINGS WILL BE INACCURATE.

1.6. Adjust the jacks until the protractor or level indicates that the helicopter is level, both longitudinally and laterally.



1. Plumb bob
2. Leveling plate

ICN-412-A-084000-B-97499-00001-A-001-01

Figure 1. Leveling (Sheet 1 of 1)

Weight and Balance - Weighing

Procedure

1. It is recommended to complete a checklist that will be used as a record of the weighed configuration, indicating differences (if any) to the Empty Weight configuration, fluid quantities, and any observations regarding the weighing. For an example of a completed weighing checklist, refer to Table 1 and for a blank weighing checklist form, refer to Table 2.

Table 1. Weight and Balance - Weighing Checklist (Example)

Helicopter Serial Number: _____ **Weighing Kit Name:** _____

Date Weighed: _____ **Calibration Expiry Date:** _____

ITEM	DESCRIPTION	YES	NO	ITEM	DESCRIPTION	YES	NO
BASIC HELICOPTER CONFIGURATION				**BALLAST**			
Fuel	Empty (cells drained)	x		Nose			
	Undrainable in system	x		Tailboom			
Oil	Engines full	x		**OPTIONAL KITS**			
	Transmission full	x					
	Engine and transmission topped-up post ground run	x					
	Tail rotor gearbox full	x					
Hydraulic fluid	Full	x					
Battery fitted		x					
Fire extinguisher		x					
Interior installed		x					
Headset(s)	Quantity installed _6_	x					
First aid kit		x					
Finish paint		x					
Baggage compartment	Empty	x					
Ground handling wheels	Removed	x					
Helicopter level	Forward and aft	x					
	Lateral	x					
Cowlings, doors, and panels	Closed/installed	x					

Parts Added: _____

Parts Omitted: _____

Notes: _____

Table 2. Actual Weight Record - As-weighed Weighing Results

DATE WEIGHED:_____ SERIAL NUMBER: _____

WEIGHING KIT:_____ CALIBRATION EXPIRY: _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**A**)						
Forward Right (**B**)						
Aft Left (**C**)						
Aft Right (**D**)						
As-Weighed Weight						

SECOND WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**E**)						
Forward Right (**F**)						
Aft Left (**G**)						
Aft Right (**H**)						
As-Weighed Weight						

CONFIRMATION OF WEIGHING RESULTS

SCALE LOCATION (SCALE LETTER)	NET WEIGHTS LB (KG)	SCALE LOCATION (SCALE LETTER)	NET WEIGHTS NET WEIGHTS LB (KG)
Forward Left (**A**)		Forward Left (**E**)	
Forward Right (**B**)		Forward Right (**F**)	
Aft Left (**C**)		Aft Left (**G**)	
Aft Right (**D**)		Aft Right (**H**)	
Total first net Weight (I)		**Total second net Weight (J)**	

TOTAL AS-WEIGHED WEIGHT DIFFERENCE (I-J) (MUST BE < 20 POUNDS (9 KG)) = _____POUNDS (_____KG)

FORWARD VERSUS AFT WEIGHT DIFFERENCE (A+B-C-D-E-F+G+H) (MUST BE < 10 POUNDS (4.5 KG)) = _____POUNDS (_____KG)

NOTES:

1. As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

2. If total helicopter As-weighed Weight difference is less than 20 pounds (9 kg), and forward versus aft Net Weight difference is less than 10 pounds (4.5 kg), the weighing results are acceptable.

Weighing - Preparation

Procedure

1. Make sure that the helicopter configuration is as close to the Empty Weight as possible. Complete the weighing checklist (Table 1). Refer to Table 2 as an example.
2. Remove excessive dirt, grease, and moisture from the helicopter.
3. Remove all items that are not part of the Empty Weight configuration.
4. Add all missing items required for Empty Weight configuration in their appropriate locations.
5. Make sure that the baggage compartment is empty.
6. Make sure that the engine oil, transmission oil, tail rotor gearbox oil, and hydraulic fluid are at the correct (full) level (412-A-12-00-00-00A-040A-A / 00126).
7. Drain the fuel system (412-A-12-10-01-00A-221A-A / 00131).
8. Proceed to 412-A-08-40-00-01A-160B-A / 00088.

Table 1. Actual Weight Record - As-weighed Weighing Results

DATE WEIGHED:_____ **SERIAL NUMBER:** _____

WEIGHING KIT:_____ **CALIBRATION EXPIRY:** _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**A**) Forward Right (**B**) Aft Left (**C**) Aft Right (**D**)						
As-Weighed Weight						

SECOND WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**E**) Forward Right (**F**) Aft Left (**G**) Aft Right (**H**)						
As-Weighed Weight						

CONFIRMATION OF WEIGHING RESULTS

SCALE LOCATION (SCALE LETTER)	NET WEIGHTS LB (KG)	SCALE LOCATION (SCALE LETTER)	NET WEIGHTS NET WEIGHTS LB (KG)
Forward Left (**A**) Forward Right (**B**)		Forward Left (**E**) Forward Right (**F**)	

Table 1. Actual Weight Record - As-weighed Weighing Results (continued)

DATE WEIGHED:_____ SERIAL NUMBER: _____

WEIGHING KIT:_____ CALIBRATION EXPIRY: _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Aft Left (**C**)			Aft Left (**G**)			
Aft Right (**D**)			Aft Right (**H**)			
Total first net Weight (I)			**Total second net Weight (J)**			

TOTAL AS-WEIGHED WEIGHT DIFFERENCE (I-J) (MUST BE < 20 POUNDS (9 KG)) = _____POUNDS (_____KG)

FORWARD VERSUS AFT WEIGHT DIFFERENCE (A+B-C-D-E-F+G+H) (MUST BE < 10 POUNDS (4.5 KG)) = _____POUNDS (_____KG)

NOTES:

1. As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.
2. If total helicopter As-weighed Weight difference is less than 20 pounds (9 kg), and forward versus aft Net Weight difference is less than 10 pounds (4.5 kg), the weighing results are acceptable.

Table 2. Weight and Balance - Weighing Checklist (Example)

Helicopter Serial Number: _____ Weighing Kit Name: _____

Date Weighed: _____ Calibration Expiry Date: _____

ITEM	DESCRIPTION	YES	NO	ITEM	DESCRIPTION	YES	NO
BASIC HELICOPTER CONFIGURATION				**BALLAST**			
Fuel	Empty (cells drained)	x		Nose			
	Undrainable in system	x		Tailboom			
Oil	Engines full	x		**OPTIONAL KITS**			
	Transmission full	x					
	Engine and transmission topped-up post ground run	x					
	Tail rotor gearbox full	x					
Hydraulic fluid	Full	x					
Battery fitted		x					
Fire extinguisher		x					
Interior installed		x					
Headset(s)	Quantity installed _6_	x					
First aid kit		x					
Finish paint		x					
Baggage compartment	Empty	x					
Ground handling wheels	Removed	x					

Table 2. Weight and Balance - Weighing Checklist (Example) (continued)

Helicopter Serial Number: _____				**Weighing Kit Name:** _____		
Date Weighed: _____				**Calibration Expiry Date:** _____		
Helicopter level	Forward and aft	x				
	Lateral	x				
Cowlings, doors, and panels	Closed/installed	x				
Parts Added: _____			Parts Omitted: _____			
Notes: _____						

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Weighing - Procedure

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Mechanical Scales	Range 0 to 5000 pounds (0 to 2268 kg)	3	Precision: 1.1% of applied load
Electron Platform Scales	Range 0 to 5000 pounds (0 to 2268 kg)	3	Precision: 1.1% of applied load
Electron Load Cells	Range 0 to 5000 pounds (0 to 2268 kg)	3	Precision: 1.1% of applied load

Procedure



DO NOT WEIGH THE HELICOPTER OUTDOORS OR IN AN OPEN BUILDING. WIND, FLAPPING ROTORS, AND BODY SWAY MAY SERIOUSLY AFFECT THE ACCURACY OF THE SCALE READINGS.

NOTE

For blank forms to complete your weight and balance calculations and to record all of the weighing results, refer to Table 1 through Table 5.

1. Put the helicopter on a hard and level surface, in a closed hangar.

NOTE

Use the two forward jack fittings and only one of the aft jack fittings to weigh the helicopter.

2. Install the weighing devices as follows:

2.1. If you use electronic load cells, install them on the jacks used to lift the helicopter. Make sure that the load cells and their adapters are tightened to the pads of the jacks.

2.2. If you use electron platform scales (Range 0 to 5000 pounds (0 to 2268 kg)), put each scale in position below the jacking point, and align the center of the scale with the jack fitting. Make sure of the following:

2.2.1. Do not allow the cables that connect the scales to the control unit to cross each other.

2.2.2. Do not allow any weight on the cables.

2.3. If you use mechanical scales (Range 0 to 5000 pounds (0 to 2268 kg)), put each scale in position below the jacking point and align the center of the scale with the jack fitting.

2.4. If applicable, when you use electron platform scales (Range 0 to 5000 pounds (0 to 2268 kg)), do the recommended warmup procedure in accordance with the manufacturer's instructions. Adjust each scale to read zero with no load.



DO NOT ALLOW PERSONNEL IN OR ON THE HELICOPTER WHEN IT IS BEING LIFTED. THE HELICOPTER MAY NOT BE STABLE. FAILURE TO OBEY MAY RESULT IN INJURY TO PERSONNEL AND DAMAGE TO THE HELICOPTER.

NOTE

For information on the characteristics of the jacks to be used, refer to 412-A-07-10-00-00A-172A-A / 00080.

3. With the jacks and weighing devices in position, raise and adjust the jacks to make the helicopter level in the longitudinal and lateral directions. Refer to 412-A-08-40-00-00A-169A-A / 00085 for the leveling procedure.

4. Make sure no side loads are imposed on the jacks, scales, or load cells.

CAUTION

IF YOU ARE LEVELING THE HELICOPTER FOR WEIGHING, MAKE SURE THE HELICOPTER IS CLEAR OF THE FLOOR AND NOT TOUCHING ANY OTHER OBJECT. OTHERWISE, THE SCALE READINGS WILL BE INACCURATE.

5. Refer to the scale manufacturer's instructions to determine the weight registered.

NOTE

To record the measurements and do the calculations for the following steps, refer to Table 6 for the example and Table 1 for the blank form.

NOTE

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

6. Record the scale readings for each scale in Table 7. Refer to Table 6 for an example.

7. Lower the helicopter (412-A-07-10-00-00A-172A-A / 00080), but do not remove the jacking/weighing equipment at this time.

8. Make sure there is no contact between the helicopter and the jacking/weighing equipment.

9. Record the Tare Weight for each scale in Table 7. Refer to Table 6 for an example.

10. Calculate the Net Weight for each scale in Table 7 as follows:

Net Weight = Scale Reading - Tare Weight

11. Add the three Net Weights that you calculated to get the As-weighed Weight of the helicopter. Record the value in Table 7. Refer to Table 6 for an example.

12. To make sure the scale readings are accurate, do a second weighing as follows:

NOTE

Make sure the same aft jack fitting used for the first weighing is used for the second weighing. Do not use the other aft jack fitting.

12.1. Rotate the mechanical scales (Range 0 to 5000 pounds (0 to 2268 kg)), electron platform scales (Range 0 to 5000 pounds (0 to 2268 kg)), orelectron load cells (Range 0 to 5000 pounds (0 to 2268 kg)), one jack point position clockwise. When you rotate the forward right side scale to the aft jack point, use the same aft jack point used for the first weighing.

12.2. Adjust each mechanical scales (Range 0 to 5000 pounds (0 to 2268 kg)), electron platform scales (Range 0 to 5000 pounds (0 to 2268 kg)), orelectron load cells (Range 0 to 5000 pounds (0 to 2268 kg)), reading to zero as applicable.

12.3. Repeat Step 3. through Step 11..

12.4. Make sure the results of the two consecutive weighings meet the following conditions. Refer to Table 6 for an example.

12.5. Difference between first and second As-weighed Weight must be less than 20 pounds (9 kg).

12.6. Calculate the difference between the forward and aft Net Weights for both the first and second weighings. Values must be less than 10 pounds (4.5 kg) in difference.

12.7. If the two consecutive weighings do not meet the previous conditions, apply corrective actions as necessary. Repeat the weighing procedure until acceptable results are obtained.

13. Lower the helicopter and remove the jacks and weighing devices (412-A-07-10-00-00A-172A-A / 00080).

14. Proceed to 412-A-08-40-00-04A-169A-A / 00091.

Table 1. Actual Weight Record - As-weighed Weight, Longitudinal CG, and Lateral CG Calculations

SCALE LOCATION	NET WEIGHT	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
Forward Left		61.69 (1567)		-30.0 (-762)	
Forward Right		61.69 (1567)		30.0 (762)	
Aft Left		211.58 (5374)		-14.53 (-369)	
Aft Right		211.58 (5374)		14.53 (369)	
TOTAL					
CG CALCULATION					
AS-WEIGHED WEIGHT AND CG					

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

Table 2. Actual Weight Record - Deriving Unballasted Empty Weight and Center of Gravity Calculations

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
AS-WEIGHED (NET WEIGHT)					
REMOVE:					
ADD:					
TOTAL UNBALLASTED EMPTY WEIGHT, MOMENT, AND CG					

Table 3. Actual Weight Record - Final Empty Weight and Fixed Ballast Using Empty Weight/CG Chart Calculations

Using U.S. or Metric System units:

$$\text{Required Ballast} = \frac{\text{Empty Weight x (Required CG - Empty Weight CG)}}{\text{(Ballast Location CG - Required CG)}}$$

$$\text{Required Ballast} = \frac{\text{x (-)}}{\text{-}}$$

Required Ballast = _____ pounds (_____ kg) at FS _____ (_____ mm)

(Per Table 4, can add _____ pounds (_____ kg) at FS _____.)

Table 3. Actual Weight Record - Final Empty Weight and Fixed Ballast Using Empty Weight/CG Chart Calculations (continued)

Add these values to the Empty Weight derivation table and transfer the new Empty Weight to the Most Forward and Most Aft CG cases.

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (NM)	IN. (MM)	IN-LB (NM)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD:					
EMPTY WEIGHT					

Table 4. Actual Weight Record (U.S.)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

_____

SIGNATURE: _____

WEIGHING RESULTS

[Units = pounds (lb), inches (in), U.S. gallons (gal)]

SCALE LOCATION	READING	TARE	NET
Forward Jack Point LH [FS 61.69, BL -30]			
Forward Jack Point RH [FS 61.69, BL 30]			
Aft Jack Point LH [FS 211.58, BL -14.53]			
Aft Jack Point RH [FS 211.58, BL 14.53]			
TOTAL (AS-WEIGHED NET WEIGHT)			

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{Fwd LH weight} + \text{Fwd RH weight})] + [\text{Aft FS} \times (\text{Aft LH weight} + \text{Aft RH weight})]}{\text{Total weight}}$$

$$CG = \frac{[61.69 \times (___ + ___)] + [211.58 \times (___ + ___)]}{(__ + ___ + ___ + __)} = _____ = _____ \text{inches}$$

LATERAL CG AS WEIGHED

Table 4. Actual Weight Record (U.S.) (continued)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

_____

SIGNATURE: _____

$$CG = \frac{[\text{Fwd FS} \times (\text{Fwd LH weight} + \text{Fwd RH weight})] + [\text{Aft FS} \times (\text{Aft LH weight} + \text{Aft RH weight})]}{\text{Total weight}}$$

$$CG = \frac{(-30 \times __) + (30 \times __) + (-14.53 \times __) + (14.53 \times __)}{(__ + __ + __ + __)} = \frac{_____}{_____} = _____ \text{inches}$$

EMPTY WEIGHT DERIVATION

ITEM	NET WEIGHT (LB)	LONGITUDINAL ARM (IN.)	LONGITUDINAL MOMENT (IN-LB)	LATERAL ARM (IN.)	LATERAL MOMENT (IN-LB)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD:					
Fixed ballast:					
EMPTY WEIGHT					

MOST FORWARD CG

ITEM		WEIGHT (LB)	CENTER OF GRAVITY AND MOMENTS: LONGITUDINAL ARM (IN.)	LONGITUDINAL MOMENT (IN-LB)	LATERAL ARM (IN.)	LATERAL MOMENT (IN-LB)
EMPTY WEIGHT						
	+ Pilot					
	+ Passengers (4) Fwd					
	+ Passengers (5) Mid					
	+ Passengers (2) Facing Outboard, Fwd Seat					
	+ Engine Oil					
	+ Most Fwd Fuel ( ) U.S. gallons					
TOTAL						

MOST AFT CG

Table 5. Actual Weight Record (Metric)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

SIGNATURE: _____

WEIGHING RESULTS

[Units = kilograms (kg), millimeters (mm)]

SCALE LOCATION	READING	TARE	NET
Forward Jack Point LH [FS 1567, BL -762]			
Forward Jack Point RH [FS 1567, BL 762]			
Aft Jack Point LH [FS 5374, BL -369]			
Aft Jack Point RH [FS 5374, BL 369]			
TOTAL (AS-WEIGHED NET WEIGHT)			

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS x (Fwd LH weight + Fwd RH weight)}] + [\text{Aft FS x (Aft LH weight + Aft RH weight)}]}{\text{Total weight}}$$

$$CG = \frac{[1567 \times (__ + __)] + [5374 \times (__ + __)]}{(__ + __ + __ + __)} = _____ = _____ mm$$

LATERAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS x (Fwd LH weight + Fwd RH weight)}] + [\text{Aft FS x (Aft LH weight + Aft RH weight)}]}{\text{Total weight}}$$

$$CG = \frac{(-762 \times __) + (762 \times __) + (-369) + (369 \times __)}{(__ + __ + __ + __)} = _____ = _____ mm$$

EMPTY WEIGHT DERIVATION

ITEM	NET WEIGHT (KG)	LONGITUDINAL ARM (MM)	LONGITUDINAL MOMENT (KG•MM)	LATERAL ARM (MM)	LATERAL MOMENT (KG•MM)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD: Fixed ballast:					
EMPTY WEIGHT					

MOST FORWARD CG

Table 5. Actual Weight Record (Metric) (continued)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

SIGNATURE: _____

ITEM	WEIGHT (KG)	CENTER OF GRAVITY AND MOMENTS			
		LONGITUDINAL		LATERAL	
		ARM (MM)	MOMENT (KG•MM)	ARM (MM)	MOMENT (KG•MM)
EMPTY WEIGHT					
+ Pilot					
+ Passengers (4) Fwd					
+ Passengers (5) Mid					
+ Passengers (2) Facing Outboard, Fwd Seat					
+ Engine Oil					
+ Most Fwd Fuel ( ) U.S. gallons					
TOTAL					

MOST AFT CG

Table 6. Weight and Balance - As-weighed Weighing Results (Example)

SERIAL NUMBER:_____ **WEIGHING KIT:** _____

DATE WEIGHED:_____ **CALIBRATION EXPIRY:** _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**A**)	776.0	352.0	10.0	4.5	766.0	947.5
Forward Right (**B**)	2709.0	1228.8	0.0	0.0	2709.0	1228.8
Aft Left (**C**)	3963.0	1797.6	6.0	2.7	3957.0	1794.9
Aft Right (**D**)	0.0	0.0	0.0	0.0	0.0	0.0
As-Weighed Weight	1448.0	3378.4	16.0	7.2	7432.0	3371.2

SECOND WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**E**)	774.0	351.1	10.0	4.5	764.0	346.6

Table 6. Weight and Balance - As-weighed Weighing Results (Example) (continued)

SERIAL NUMBER:_____ WEIGHING KIT: _____

DATE WEIGHED:_____ CALIBRATION EXPIRY: _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Right (**F**)	2712.0	1230.2	0.0	0.0	2712.0	1230.2
Aft Left (**G**)	3960.0	1796.2	0.0	0.0	3960.0	1796.2
Aft Right (**H**)	0.0	0.0	0.0	0.0	0.0	0.0
As-Weighed Weight	7446.0	3377.5	10.0	4.5	7436.0	3373.0

CONFIRMATION OF WEIGHING RESULTS

SCALE LOCATION (SCALE LETTER)	NET WEIGHTS		SCALE LOCATION (SCALE LETTER)	Net weights	
	LB	KG		LB	KG
Forward Left (**A**)	766.0	347.5	Forward Left (**E**)	764.0	346.6
Forward Right (**B**)	2709.0	1228.8	Forward Right (**F**)	2712.0	1230.2
Aft Left (**C**)	3957.0	1794.9	Aft Left (**G**)	3960.0	1796.2
Aft Right (**D**)	0.0	0.0	Aft Right (**H**)	0.0	0.0
Total first net Weight (I)	7432.0	3371.2	**Total second net Weight (J)**	7436.0	3373.0

TOTAL AS-WEIGHED WEIGHT DIFFERENCE (I-J) (MUST BE < 20 POUNDS (9 KG)) = 4 POUNDS (1.8 KG)

FORWARD VERSUS AFT WEIGHT DIFFERENCE (A+B-C-D-E-F+G+H) (MUST BE < 10 POUNDS (4.5KG)) = 4 POUNDS (1.9 KG)

NOTES:

1. As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

2. If total helicopter As-weighed Weight difference is less than 20 pounds (9 kg), and forward versus aft Net Weight difference is less than 10 pounds (4.5 kg), the weighing results are acceptable.

Table 7. Actual Weight Record - As-weighed Weighing Results

DATE WEIGHED:_____ SERIAL NUMBER: _____

WEIGHING KIT:_____ CALIBRATION EXPIRY: _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**A**)						
Forward Right (**B**)						
Aft Left (**C**)						
Aft Right (**D**)						
As-Weighed Weight						

Table 7. Actual Weight Record - As-weighed Weighing Results (continued)

DATE WEIGHED:_____ SERIAL NUMBER: _____

WEIGHING KIT:_____ CALIBRATION EXPIRY: _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG

SECOND WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**E**)						
Forward Right (**F**)						
Aft Left (**G**)						
Aft Right (**H**)						
As-Weighed Weight						

CONFIRMATION OF WEIGHING RESULTS

SCALE LOCATION (SCALE LETTER)	NET WEIGHTS LB (KG)	SCALE LOCATION (SCALE LETTER)	NET WEIGHTS NET WEIGHTS LB (KG)
Forward Left (**A**)		Forward Left (**E**)	
Forward Right (**B**)		Forward Right (**F**)	
Aft Left (**C**)		Aft Left (**G**)	
Aft Right (**D**)		Aft Right (**H**)	
Total first net Weight (I)		**Total second net Weight (J)**	

TOTAL AS-WEIGHED WEIGHT DIFFERENCE (I-J) (MUST BE < 20 POUNDS (9 KG)) = _____POUNDS (_____KG)

FORWARD VERSUS AFT WEIGHT DIFFERENCE (A+B-C-D-E-F+G+H) (MUST BE < 10 POUNDS (4.5 KG)) = _____POUNDS (_____KG)

NOTES:

1. As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

2. If total helicopter As-weighed Weight difference is less than 20 pounds (9 kg), and forward versus aft Net Weight difference is less than 10 pounds (4.5 kg), the weighing results are acceptable.

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Weight and Balance - Calculations

Procedure

1. The information in 412-A-08-40-00-03A-169A-A / 00090, 412-A-08-40-00-04A-169A-A / 00091, 412-A-08-40-00-05A-169A-A / 00092, 412-A-08-40-00-06A-169A-A / 00093 and 412-A-08-40-00-07A-169A-A / 00094, explains how the weighing results are used to make sure the helicopter weight and balance complies with the limitations described in the BHT-412-FM, Section 1.

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Center of Gravity - Calculations

Procedure

1. Arms are measured from the datum to the point at which the weight of an item acts. Multiplying the weight by the arm gives the item's moment. Adding the moments of all items and dividing by their total weight gives an equivalent arm, or Center of Gravity (CG) , for all of the items.

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Longitudinal CG - Calculations

Procedure

NOTE

The Fuselage Station (FS) 0.0 is located 20 inches (508 mm) inside the nose of the helicopter.

1. The distance from FS 0.0 to the centers of the forward jack fittings is called the forward arm. The forward arm is 61.69 inches (1567 mm), as shown in Figure 1.

2. The distance from FS 0.0 to the center of the aft jack fitting is called the aft arm. The aft arm is 211.58 inches (5374 mm), as shown in Figure 1.

NOTE

Refer to Table 1 for an example of the calculations and Table 2 for a blank calculation form.

2.1. Multiply the Net Weight for each scale by its arm. The end results are called the longitudinal moments. Record the results in Table 2. Refer to Table 1 for an example.

2.2. Add the longitudinal moments and divide this total by the As-weighed Weight. The result is the helicopter As-weighed Longitudinal CG in inches (mm), aft of the datum. Record the result in Table 2. Refer to Table 1 for an example.

2.3. Proceed to 412-A-08-40-00-05A-169A-A / 00092.

Table 1. Calculations - As-weighed Weight, Longitudinal CG, and Lateral CG (Example)

Scale Location	Net Weight	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
Forward Left	766.0 (347.5)	61.69 (1567)	47,255 (544,533)	-30.0 (-762)	-22,980 (-264,795)
Forward Right	2709.0 (1228.8)	61.69 (1567)	167,118 (1,925,407)	30.0 (762)	81,270 (936,346)
Aft Left	3957.0 (1794.9)	211.58 (5374)	837,222 (9,645,793)	-14.53 (-369)	-57,495 (-662,318)
Aft Right	0.0 (0.0)	211.58 (5374)	0 (0)	14.53 (369)	0 (0)
As-Weighed Weight and CG	7432.0 (3371.2)	141.5 (3594)	1,051,595 (12,115,732)	0.1 (3)	795 (9233)

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

Table 2. Actual Weight Record - As-weighed Weight, Longitudinal CG, and Lateral CG Calculations

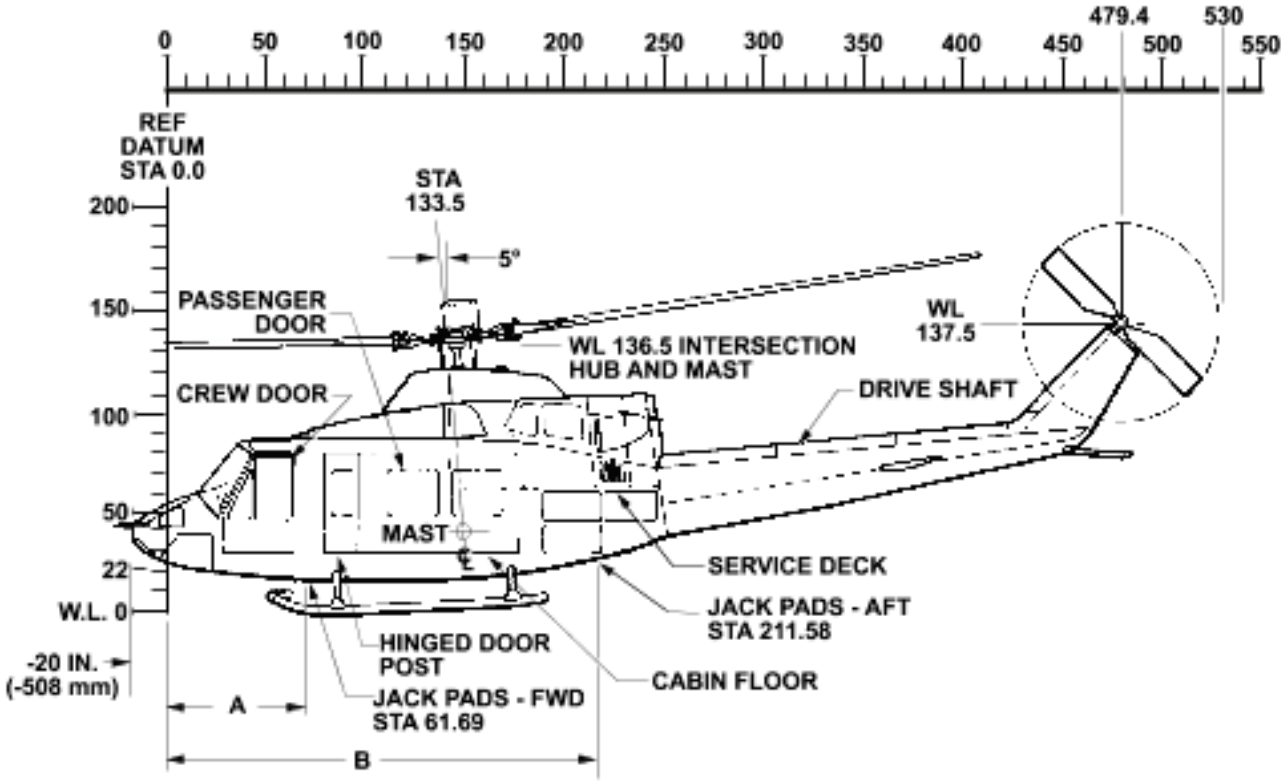
SCALE LOCATION	NET WEIGHT	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
Forward Left		61.69 (1567)		-30.0 (-762)	
Forward Right		61.69 (1567)		30.0 (762)	

Table 2. Actual Weight Record - As-weighed Weight, Longitudinal CG, and Lateral CG Calculations (continued)

SCALE LOCATION	NET WEIGHT	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
Aft Left		211.58 (5374)		-14.53 (-369)	
Aft Right		211.58 (5374)		14.53 (369)	
TOTAL					
CG CALCULATION					
AS-WEIGHED WEIGHT AND CG					

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.



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Figure 1. Weight and Balance - Station Diagram (Sheet 1 of 1)

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Lateral CG - Calculations

Procedure

1. The helicopter centerline is Buttock Line (BL) 0.0. The arms to the left of the centerline (looking forward) are negative (LBL) and arms to the right are positive (RBL), as shown in Figure 1.

2. The forward jack fittings are located at LBL -30 inches (-762 mm) and RBL 30 inches (762 mm) from the centerline.

3. The aft jack fittings are located at LBL -14.53 inches (-369 mm) and RBL 14.53 inches (369 mm) from the centerline.

NOTE

Refer to Table 1 for an example of the calculations and Table 2 for a blank calculation form.

3.1. Multiply the Net Weight for each scale by its arm, including the appropriate sign. The results are called the lateral moments. Record the results in Table 2. Refer to Table 1 for an example.

3.2. Add the lateral moments and divide this total by the As-weighed Weight. The result is the helicopter As-weighed Lateral CG in inches (mm) left or right of BL 0.0, depending on the sign. Record the result in Table 2. Refer to Table 1 for an example.

3.3. Proceed to 412-A-08-40-00-06A-169A-A / 00093.

Table 1. Calculations - As-weighed Weight, Longitudinal CG, and Lateral CG (Example)

Scale Location	Net Weight	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
Forward Left	766.0 (347.5)	61.69 (1567)	47,255 (544,533)	-30.0 (-762)	-22,980 (-264,795)
Forward Right	2709.0 (1228.8)	61.69 (1567)	167,118 (1,925,407)	30.0 (762)	81,270 (936,346)
Aft Left	3957.0 (1794.9)	211.58 (5374)	837,222 (9,645,793)	-14.53 (-369)	-57,495 (-662,318)
Aft Right	0.0 (0.0)	211.58 (5374)	0 (0)	14.53 (369)	0 (0)
As-Weighed Weight and CG	7432.0 (3371.2)	141.5 (3594)	1,051,595 (12,115,732)	0.1 (3)	795 (9233)

NOTE:

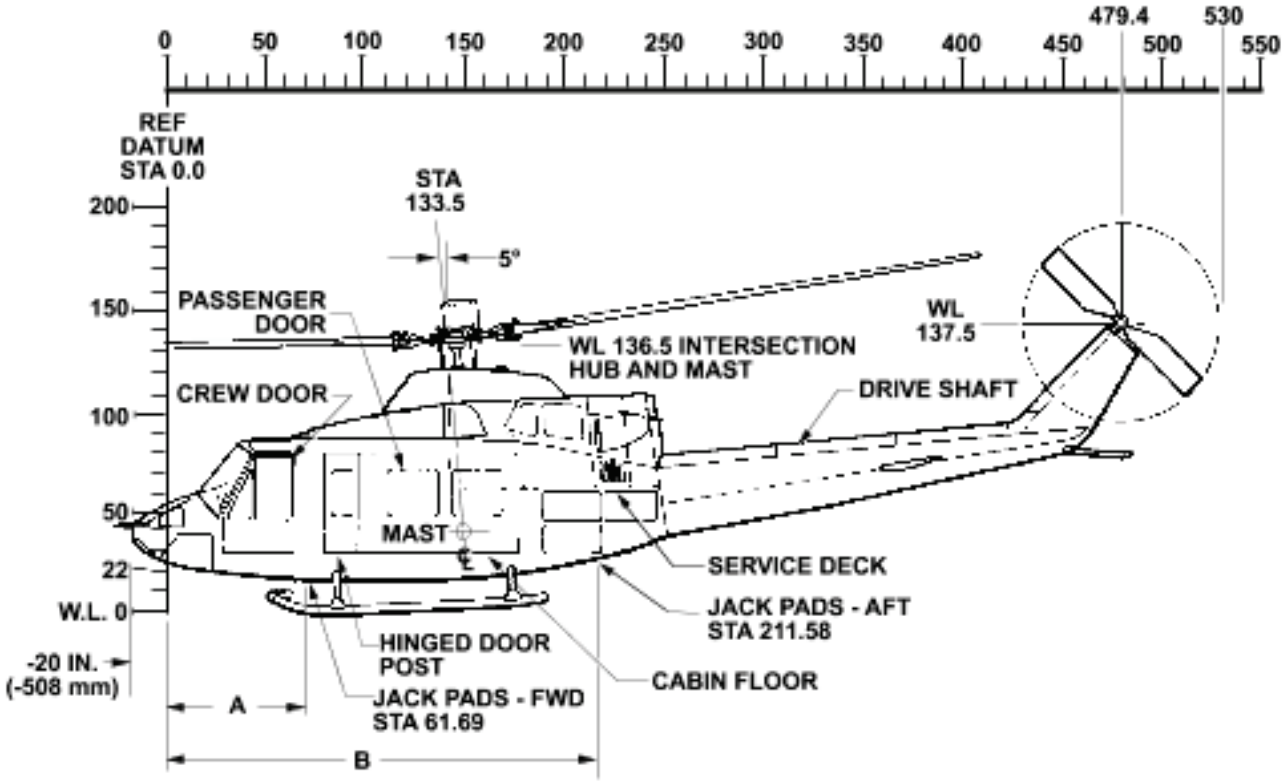
As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

Table 2. Actual Weight Record - Deriving Unballasted Empty Weight and Center of Gravity Calculations

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
AS-WEIGHED (NET WEIGHT)					
REMOVE:					

Table 2. Actual Weight Record - Deriving Unballasted Empty Weight and Center of Gravity Calculations (continued)

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
ADD:					
TOTAL UNBALLASTED EMPTY WEIGHT, MOMENT, AND CG					



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Figure 1. Weight and Balance - Station Diagram (Sheet 1 of 1)

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Empty Weight - Calculations

Procedure

1. The basis of all the operational balance calculations is the Empty Weight configuration. The following information gives the steps required to derive the Unballasted Empty Weight configuration from the As-weighed Weight. The fixed ballast required, if any, to keep the helicopter within the Gross Weight Center of Gravity (CG) limits, can then be calculated as described in 412-A-08-40-00-07A-169A-A / 00094.

NOTE

Refer to Table 1 for an example of the calculations and Table 2 for a blank calculation form.

NOTE

If required, Table 3 lists the densities, weights, CGs, and moments of different fuels and oils that must be used for weight and balance calculations. The weight and balance data for the fuels and oils are provided for a standard International Standard Atmosphere (ISA) day.

If a helicopter has to be weighed with full tanks (it is not recommended), a specific gravity check must be carried out to find the correct weight of the fuel on board.

1.1. Start with the As-weighed Weight, CG, and moment values previously calculated in 412-A-08-40-00-01A-160B-A / 00088 and recorded in Table 2. Do the following steps:

NOTE

If the helicopter was weighed with fixed ballast installed, then it must be subtracted at this point.

1.1.1. Subtract all non-Empty Weight items that were on the helicopter when it was weighed. Record the items, weights, and calculated results in Table 4. Refer to Table 1 for an example.

NOTE

When the fuel system has been drained (412-A-08-40-00-01A-160A-A / 00087), add the residual fuel value (Table 3).

1.1.2. Add all Empty Weight items that were not on the helicopter when it was weighed. Record the items, weights, and calculated results in Table 4. Refer to Table 1 for an example.

1.1.3. Divide the total moment by the total weight to get the Unballasted Empty Weight CG. Record the results in Table 4, as shown in the Table 1 example.

1.2. Proceed to 412-A-08-40-00-07A-169A-A / 00094.

Table 1. Calculations - Deriving Unballasted Empty Weight and Center of Gravity (Example)

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
AS-WEIGHED (NET WEIGHT)	7432.0 (3371.2)	141.5 (3594)	1,051,595 (12,115,732)	0.1 (3)	795 (9233)
REMOVE:					
Usable Oil	-24.5 (-11.1)	169.1 (4295)	-4143 (-47,732)	0.0 (0)	0 (0)
Ballast - Battery Block	-76.0 (-34.5)	-5.7 (-145)	433 (4991)	7.9 (201)	-600 (-6917)
Ballast - Station -4.7	-74.0 (-33.6)	-4.7 (-119)	348 (4007)	0.0 (0)	0 (0)
Ballast - Station -4.4	-64.0 (-29.0)	-4.4 (-112)	282 (3244)	0.1 (3)	-6 (-74)

Table 1. Calculations - Deriving Unballasted Empty Weight and Center of Gravity (Example) (continued)

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
Ballast - Station -4.1	-15.0 (-6.8)	-4.1 (-104)	62 (709)	0.2 (5)	-3 (-35)
ADD:					
Residual Fuel	28.1 (12.7)	126.0 (3200)	3541 (40,793)	0.0 (0)	0 (0)
UNBALLASTED EMPTY WEIGHT	7206.6 (3269.0)	146.0 (3708)	1,052,117 (12,121,743)	0.0 (1)	185 (2207)

Table 2. Actual Weight Record - Deriving Unballasted Empty Weight and Center of Gravity Calculations

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
AS-WEIGHED (NET WEIGHT)					
REMOVE:					
ADD:					
TOTAL UNBALLASTED EMPTY WEIGHT, MOMENT, AND CG					

Table 3. Weight and Balance Data for Fuels and Oils

FUEL	ENGLISH			METRIC		
	WEIGHT	CG	MOMENT	WEIGHT	CG	MOMENT
	LB	IN.	IN-LB	KG	MM	KG•MM
TOTAL UNUSABLE FUEL						
SN: 33001-33107:						
Jet A (JP-5)	28.3	142.8	4041	12.8	3627	46557
Jet B (JP-4)	27.1	142.8	3863	12.3	3627	44503
SN: 33108-33213 or 36001-36999 or 37002-37999:						
Jet A (JP-5)	47.6	128.0	6093	21.6	3251	70222
Jet B (JP-4)	45.5	128.0	5284	20.6	3251	66971
UNDRAINABLE FUEL						
SN: 33001-33107:						

Table 3. Weight and Balance Data for Fuels and Oils (continued)

FUEL	ENGLISH			METRIC		
	WEIGHT	CG	MOMENT	WEIGHT	CG	MOMENT
	LB	IN.	IN-LB	KG	MM	KG•MM
Jet A (JP-5)	15.3	163.4	2499	6.9	4150	28803
Jet B (JP-4)	14.6	163.4	2390	6.6	4150	27533
SN: 33108-33213 or 36001-36999 or 37002-37999:						
Jet A (JP-5)	19.5	130.9	2553	8.8	3325	29260
Jet B (JP-4)	18.6	130.9	2440	8.5	3325	28263
RESIDUAL FUEL						
SN: 33001-33107:						
Jet A (JP-5)	13.0	118.6	1542	5.9	3012	17763
Jet B (JP-4)	12.4	118.6	1474	5.6	3012	16980
SN: 33108-33213 or 36001-36999 or 37002-37999:						
Jet A (JP-5)	28.1	126.0	3541	12.7	3201	40653
Jet B (JP-4)	26.9	126.0	3384	12.2	3200	39040
OIL	**ENGLISH**			**METRIC**		
	WEIGHT	CG	MOMENT	WEIGHT	CG	MOMENT
	LB	IN.	IN-LB	KG	MM	KG•MM
USABLE ENGINE OIL						
DOD-PRF-85734	26.7	169.1	4515	12.1	4295	51970
MIL-PRF-7808	24.5	169.1	4143	11.1	4295	47675
MIL-PRF-23699	26.7	169.1	4515	12.1	4295	51970

NOTES:

Densities reflect 15°C (59°F).

Density of fuels:

Jet A (JP-5): 6.8 pounds per gallon (0.815 kg/L)

Jet B (JP-4): 6.5 pounds per gallon (0.779 kg/L)

Density of oils:

DOD-PRF-85734: 8.4 pounds per gallon (1.007 kg/L)

MIL-PRF-7808: 7.7 pounds per gallon (0.923 kg/L)

MIL-PRF-23699: 8.4 pounds per gallon (1.007 kg/L)

Table 4. Actual Weight Record - Final Empty Weight and Fixed Ballast Using Empty Weight/CG Chart Calculations

Using U.S. or Metric System units:

$$\text{Required Ballast} = \frac{\text{Empty Weight x (Required CG - Empty Weight CG)}}{\text{(Ballast Location CG - Required CG)}}$$

$$\text{Required Ballast} = \frac{\text{x (-)}}{\text{-}}$$

Required Ballast = _____ pounds (_____ kg) at FS _____ (_____ mm)

(Per Table 5, can add _____ pounds (_____ kg) at FS _____.)

Add these values to the Empty Weight derivation table and transfer the new Empty Weight to the Most Forward and Most Aft CG cases.

Table 4. Actual Weight Record - Final Empty Weight and Fixed Ballast Using Empty Weight/CG Chart Calculations (continued)

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (NM)	IN. (MM)	IN-LB (NM)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD:					
EMPTY WEIGHT					

Table 5. Ballast Requirements - Maximum Fixed Ballast Weight and Location

LOCATION	PART NUMBER	MAXIMUM		FUSELAGE STATION (FS)		BUTTOCK LINE (BL)	
		LB	KG	IN.	MM	IN.	MM
Battery Block (1)	212-030-173-101 (x1)	76.0	34.5	-5.7	-145	7.9	201
	212-030-173-103 (x1) (8)	52	23.6	-5.9	-150	7.9	201
SN: 33001-33213 or 36001-36036: Nose Shelf (2)	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-011 (x4)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-011 (x4)	16	7.3	-4.1	-104	20.8	528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	20.8	528
	212-030-245-015 (x1)	19	8.6	-4.7	-119	-20.5	-521
	212-030-245-015 (x1)	19	8.6	-4.7	-119	20.5	521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521

Table 5. Ballast Requirements - Maximum Fixed Ballast Weight and Location (continued)

LOCATION	PART NUMBER	MAXIMUM		FUSELAGE STATION (FS)		BUTTOCK LINE (BL)	
		LB	KG	IN.	MM	IN.	MM
SN: 36037-36382: Nose Shelf (2)	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-121 (x3) (5)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-151 (x3) (5)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-123 (x3) (6)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-153 (x3) (6)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-011 (x4)	16	7.3	-4.1	-104	20.8	528
	212-030-245-125 (x4)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	20.8	528
	212-030-245-127 (x2)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-143 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-145 (x1)	17.6	8	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-131 (x1)	18	8.2	-4.7	-119	-20.5	-521
SN: 36383-36999 or 37002-37999: Nose Shelf (2)	212-030-245-147 (x3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-151 (x3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-149 (x3)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-153 (x3)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-011 (x4)	16.0	7.3	-4.1	-104	20.8	528
	212-030-245-125 (x4)	16.0	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16.0	7.3	-4.1	-104	20.8	528
	212-030-245-127 (x2)	16.0	7.3	-4.1	-104	-20.8	-528
	212-030-245-143 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-145 (x1)	17.6	8.0	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-131 (x1)	18.0	8.2	-4.7	-119	-20.5	-521
Tailboom (7)	204-071-566-001 (4)	25.2	11.4	468.0	11,887	0	0
	204-071-566-001 (6)	37.8	17.1	441.0	11,201	0	0
	204-071-566-003 (4)	25.2	11.4	441.0	11,201	0	0
	No. 8 Lead Shot (7)	15.0	6.8	457.0	11,608	0	0
		Completely fill tube (do not pack)					

Table 5. Ballast Requirements - Maximum Fixed Ballast Weight and Location (continued)

LOCATION	PART NUMBER	MAXIMUM		FUSELAGE STATION (FS)		BUTTOCK LINE (BL)	
		LB	KG	IN.	MM	IN.	MM

NOTES:

1 Actual weight is stamped on 212-030-173 block. Battery block ballast not included in 164 pounds (74.4 kg) maximum weight.

2 Maximum nose shelf ballast is 82 pounds (37.2 kg) per side. Add ballast symmetrically.

3 212-030-245-147 replaces -007 for future procurement.

4 212-030-245-149 replaces -009 for future procurement.

5 212-030-245-151 replaces -121 for future procurement.

6 212-030-245-153 replaces -123 for future procurement.

7 Maximum tailboom ballast is 50 pounds (22.7 kg).

8 Use the 212-030-173-103 battery block ballast when 53 Amp-hour Lead Acid Battery Kit (412-706-131) is installed. Weigh battery and ballast underneath the battery before installation. The total weight of both parts must not exceed 155 pounds (70 kg).

Ballast Requirements - Using the Gross Weight CG Envelope Chart - Calculations

Procedure

NOTE

There is no provision to ballast the helicopter for lateral CG.

1. The ballast requirements and, therefore, the final Empty Weight and Empty Weight Center of Gravity (CG) , can be found by using the Gross Weight CG limitations.

NOTE

Use the following procedure to ballast the helicopter to accommodate the most critical forward and aft balance loading conditions for your specific operation. These critical points consist of useful load combinations (crew, passengers, baggage, cargo, fuel), which bring the resulting helicopter weight and CG closest within (or furthest outside, if exceeding limits) of the Gross Weight CG Envelope. This will help the pilot to calculate and make sure that each flight is accomplished within the limits of the Gross Weight CG Envelope (BHT-412-FM, Section 1).

1.1. Determine the Most Forward Gross Weight and CG as follows:

1.1.1. Calculate the combination of Useful Load items that, when added to the Unballasted Empty Weight, will result in the helicopter CG closest within (or furthest outside, if exceeding limits) the Gross Weight CG Envelope forward limit as follows:

1.1.1.1. Include the mission crew and fuel weight, and CG, as determined from the fuel loading tables (Applicable Flight Manual Supplement, or STC manufacturer documents) specific to the helicopter configuration, which will place the helicopter CG closest to the forward limit on the Gross Weight CG Envelope.

Applicable: SN: 33001-33213 or 36001-36019:

1.1.1.1.1. BHT-412-MD-1, Section 1, Manufacturer's Data (S/N 33001 - 33107) or BHT-412-MD-2, Section 1, Manufacturer's Data (S/N 33108 - 33213, 34001 - 34024 and 36001 - 36019).

Applicable: SN: 36020-36999 or 37002-37999:

1.1.1.1.2. BHT-412-FM-1, Section 5, Flight Manual (S/N 33001 - 33107) or BHT-412-FM-2, Section 5, Flight Manual (S/N 33108 - 33213, 34001 - 34024 and 36001 - 36019).

1.1.1.2. Include items that have a CG forward of the helicopter weight empty CG.

1.1.1.3. Exclude items that have a CG aft of the helicopter weight empty CG, unless their inclusion brings the resulting gross weight CG closer to the forward limit.

1.1.2. Add the combination of Useful Load items to the Unballasted Empty Weight. The result is the critical mission Most Forward Gross Weight and CG.

1.2. Determine the Most Aft Gross Weight and CG as follows:

1.2.1. Calculate the combination of Useful Load items which, when added to the Unballasted Empty Weight, will result in the helicopter CG closest within (or furthest outside, if exceeding limits) the Gross Weight CG Envelope aft limit as follows:

1.2.1.1. Include the minimum crew weight and fuel as determined from the fuel loading tables (Applicable Flight Manual Supplement, or STC manufacturer documents) specific to the helicopter configuration, which will place the helicopter CG closest within (or furthest outside, if exceeding limits) the aft limit on the Gross Weight CG Envelope.

Applicable: SN: 33001-33213 or 36001-36019:

1.2.1.1.1. BHT-412-MD-1, Section 1, Manufacturer's Data (S/N 33001 - 33107) or BHT-412-MD-2, Section 1, Manufacturer's Data (S/N 33108 - 33213, 34001 - 34024 and 36001 - 36019).

Applicable: SN: 36020-36999 or 37002-37999:

1.2.1.1.2. BHT-412-FM-1, Section 5, Flight Manual (S/N 33001 - 33107) or BHT-412-FM-2, Section 5, Flight Manual (S/N 33108 - 33213, 34001 - 34024 and 36001 - 36019).

1.2.1.2. Exclude items that have a CG forward of the helicopter weight empty CG, unless their inclusion brings the resulting gross weight CG closest within (or furthest outside, if exceeding limits) the aft limit.

1.2.2. Add the combination of Useful Load items to the Unballasted Empty Weight. The result is the critical mission Most Aft Gross Weight and CG.

1.3. Plot the Most Forward and Most Aft Gross Weights and CGs on the Gross Weight CG Envelope chart (BHT-412-FM, Section 1).

1.4. If the Most Forward and Most Aft Gross Weight CGs are both inside the Gross Weight CG Envelope chart (BHT-412-FM, Section 1), the helicopter does not require any fixed ballast. The final Empty Weight is the same as the Unballasted Empty Weight.



IF THE CALCULATION OF THE UNBALLASTED EMPTY WEIGHT SUBTRACTED THE FIXED BALLAST THAT WAS INSTALLED ON THE HELICOPTER WHEN IT WAS WEIGHED, MAKE SURE IT IS PHYSICALLY REMOVED FROM THE HELICOPTER AT THIS POINT.

NOTE

Table 1 shows the maximum fixed ballast that can be installed at each location.

1.5. If either the Most Forward or the Most Aft Gross Weight CGs are outside the Gross Weight CG Envelope chart (BHT-412-FM, Section 1), ballast the helicopter to move the appropriate case inside as follows:

1.5.1. If the Most Forward Gross Weight CG is forward of the Gross Weight CG Envelope forward limit (BHT-412-FM, Section 1), fixed ballast must be added in a location aft of the CG.

1.5.2. If the Most Aft Gross Weight CG is aft of the Gross Weight CG Envelope aft limit (BHT-412-FM, Section 1), fixed ballast must be added in a location forward of the CG.

1.5.3. Determine the requirement for the fixed ballast.



WHEN YOU ADD FIXED BALLAST TO MOVE THE MOST FORWARD (OR MOST AFT) CG INSIDE THE GROSS WEIGHT CG ENVELOPE (BHT-412-FM, SECTION 1), RECALCULATE THE CRITICAL MOST AFT AND MOST FORWARD CG TO MAKE SURE THESE DID NOT MOVE OUTSIDE THE ENVELOPE.

NOTE

If the Most Aft (or Most Forward) CG moved outside the Gross Weight CG Envelope (BHT-412-FM, Section 1), the combination of Useful Load items or the Empty Weight configuration must be changed, as described in the following steps.

1.6. If both the Most Forward and the Most Aft Gross Weight CGs cannot be ballasted within the Gross Weight CG Envelope (BHT-412-FM, Section 1), or if the required fixed ballast is more than the maximum permitted, do one of the following:

- Adjust the combination of Useful Load items to decrease the required fixed ballast to the maximum permitted or less.
- Change the Empty Weight configuration to move the Empty Weight CG closer to the flight limit until the required fixed ballast is equal to or less than the maximum permitted.

NOTE

Limiting the combinations of Useful Load items can replace the effect of fixed ballast installations and minimize the Empty Weight.

However, consideration must also be given to the type of helicopter operations to be flown and the effect the limitations will have on them.

1.7. After determining the correct fixed ballast, calculate the final Empty Weight in Table 2. Use Table 3 as an example.

1.8. Enter these values and the Useful Load limitations (if any), as well as associated data from your weight and balance calculations into the helicopter Actual Weight Record (Table 4 (U.S.) or Table 5 (Metric)).



IF THE CALCULATION OF THE UNBALLASTED EMPTY WEIGHT SUBTRACTED THE FIXED BALLAST THAT WAS INSTALLED ON THE HELICOPTER WHEN IT WAS WEIGHED, MAKE SURE IT IS PHYSICALLY REMOVED FROM THE HELICOPTER AT THIS POINT, UNLESS IT WILL BE PART OF THE FINAL EMPTY WEIGHT FIXED BALLAST REQUIREMENT.

1.9. Install and secure the required fixed ballast (412-A-08-40-00-00A-720A-A / 00097).

Table 1. Ballast Requirements - Maximum Fixed Ballast Weight and Location

LOCATION	PART NUMBER	MAXIMUM		FUSELAGE STATION (FS)		BUTTOCK LINE (BL)	
		LB	KG	IN.	MM	IN.	MM
Battery Block (1)	212-030-173-101 (x1)	76.0	34.5	-5.7	-145	7.9	201
	212-030-173-103 (x1) (8)	52	23.6	-5.9	-150	7.9	201
SN: 33001-33213 or 36001-36036: Nose Shelf (2)	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-011 (x4)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-011 (x4)	16	7.3	-4.1	-104	20.8	528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	20.8	528
	212-030-245-015 (x1)	19	8.6	-4.7	-119	-20.5	-521
	212-030-245-015 (x1)	19	8.6	-4.7	-119	20.5	521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521
SN: 36037-36382: Nose Shelf (2)	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-121 (x3) (5)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-151 (x3) (5)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-123 (x3) (6)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-153 (x3) (6)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-011 (x4)	16	7.3	-4.1	-104	20.8	528
	212-030-245-125 (x4)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	20.8	528
	212-030-245-127 (x2)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-143 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-145 (x1)	17.6	8	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-131 (x1)	18	8.2	-4.7	-119	-20.5	-521

Table 1. Ballast Requirements - Maximum Fixed Ballast Weight and Location (continued)

LOCATION	PART NUMBER	MAXIMUM		FUSELAGE STATION (FS)		BUTTOCK LINE (BL)	
		LB	KG	IN.	MM	IN.	MM
SN: 36383-36999 or 37002-37999: Nose Shelf (2)	212-030-245-147 (x3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-151 (x3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-149 (x3)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-153 (x3)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-011 (x4)	16.0	7.3	-4.1	-104	20.8	528
	212-030-245-125 (x4)	16.0	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16.0	7.3	-4.1	-104	20.8	528
	212-030-245-127 (x2)	16.0	7.3	-4.1	-104	-20.8	-528
	212-030-245-143 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-145 (x1)	17.6	8.0	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-131 (x1)	18.0	8.2	-4.7	-119	-20.5	-521
Tailboom (7)	204-071-566-001 (4)	25.2	11.4	468.0	11,887	0	0
	204-071-566-001 (6)	37.8	17.1	441.0	11,201	0	0
	204-071-566-003 (4)	25.2	11.4	441.0	11,201	0	0
	No. 8 Lead Shot (7)	15.0	6.8	457.0	11,608	0	0
		Completely fill tube (do not pack)					

NOTES:

1 Actual weight is stamped on 212-030-173 block. Battery block ballast not included in 164 pounds (74.4 kg) maximum weight.

2 Maximum nose shelf ballast is 82 pounds (37.2 kg) per side. Add ballast symmetrically.

3 212-030-245-147 replaces -007 for future procurement.

4 212-030-245-149 replaces -009 for future procurement.

5 212-030-245-151 replaces -121 for future procurement.

6 212-030-245-153 replaces -123 for future procurement.

7 Maximum tailboom ballast is 50 pounds (22.7 kg).

8 Use the 212-030-173-103 battery block ballast when 53 Amp-hour Lead Acid Battery Kit (412-706-131) is installed. Weigh battery and ballast underneath the battery before installation. The total weight of both parts must not exceed 155 pounds (70 kg).

Table 2. Actual Weight Record - Final Empty Weight and Fixed Ballast Using Empty Weight/CG Chart Calculations

Using U.S. or Metric System units:

$$\text{Required Ballast} = \frac{\text{Empty Weight x (Required CG - Empty Weight CG)}}{\text{(Ballast Location CG - Required CG)}}$$

$$\text{Required Ballast} = \frac{\text{x (-)}}{\text{-}}$$

Required Ballast = _____ pounds (_____ kg) at FS _____ (_____ mm)

(Per Table 5, can add _____ pounds (_____ kg) at FS _____.)

Add these values to the Empty Weight derivation table and transfer the new Empty Weight to the Most Forward and Most Aft CG cases.

Table 2. Actual Weight Record - Final Empty Weight and Fixed Ballast Using Empty Weight/CG Chart Calculations (continued)

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (NM)	IN. (MM)	IN-LB (NM)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD:					
EMPTY WEIGHT					

Table 3. Weight and Balance - Actual Weight Record (U.S.) (Example)

1. Perform the weighing as described in 412-A-08-40-00-01A-160B-A / 00088. Enter the As-weighed weighing results for each scale and subtract the Tare Weight (if any). Refer to the Table 6 example. Calculate the As-weighed net weight and Center of Gravity. Refer to the Table 7 example.

WEIGHING RESULTS

[Units = pounds (lb), inches (in), U.S. gallons (gal)]

SCALE LOCATION	READING (LB)	TARE (LB)	NET (LB)
Forward Jack Point LH [FS 61.69, BL -30.0]	776	10.0	766.0
Forward Jack Point RH [FS 61.69, BL 30.0]	2709	0.0	2709.0
Aft Jack Point LH [FS 211.58, BL -14.53]	3963.0	6.0	3957.0
Aft Jack Point RH [FS 211.58, BL 14.53]	0.0	0.0	0.0
TOTAL	**7449.0**	**17.0**	**7432.0**

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{LH weight} + \text{RH weight})] + [\text{Aft FS} \times \text{Aft weight})]}{\text{Total weight}}$$

$$CG = \frac{[61.69 \times (766.0 + 2709.0)] + [211.58 \times 3957.0]}{(766.0 + 2709.0 + 3957.0)} = \frac{1{,}051{,}595}{7432.0} = 141.5 \text{ inches}$$

LATERAL CG AS WEIGHED

$$CG = \frac{(\text{Fwd LBL} \times \text{LH weight}) + (\text{Fwd RBL} \times \text{RH weight}) + (\text{Aft BL} \times \text{Aft weight})}{\text{Total weight}}$$

$$CG = \frac{(-30.0 \times 766.0) + (30.0 \times 2709.0) + (-14.53 \times 3957.0)}{(766.0 + 2709.0 + 3957.0)} = \frac{795}{7432.0} = 0.1 \text{ inch}$$

Table 3. Weight and Balance - Actual Weight Record (U.S.) (Example) (continued)

2. Remove the weight and moment for those items installed that are not part of the Empty Weight. Add the weight and moment for those items that are part of the Empty Weight, but were not installed. The result is the Unballasted Empty Weight. Refer to the Table 8 example.

EMPTY WEIGHT DERIVATION

ITEM	WEIGHT (LB)	LONGITUDINAL CG (IN.)	LONGITUDINAL MOMENT (IN-LB)	LATERAL CG (IN.)	LATERAL MOMENT (IN-LB)
AS WEIGHED (NET WEIGHT)	7432.0	141.5	1,051,595	0.1	795
REMOVE:					
Unusable Oil	-24.5	169.1	-4143	0.0	0
Ballast at Battery Block	-76.0	-5.7	433	7.9	-600
Ballast at FS -4.7	-74.0	-4.7	348	0.0	0
Ballast at FS -4.4	-64.0	-4.4	282	0.1	-6
Ballast at FS -4.1	-15.0	-4.1	62	0.2	-3
ADD:					
Residual Fuel	28.1	126.0	3541	0.0	0
UNBALLASTED EMPTY WEIGHT	**7206.6**	**146.0**	**1,052,117**	**0.0**	**185**

3. Determine the Most Forward and Most Aft Useful Load weights and moments and add these to the Unballasted Empty Weight to give the most Forward and Most Aft Gross Weight and Center of Gravity.

MOST FORWARD CG

ITEM	WEIGHT (LB)	CENTER OF GRAVITY AND MOMENTS: LONGITUDINAL CG (IN.)	LONGITUDINAL MOMENT (IN-LB)	LATERAL CG (IN.)	LATERAL MOMENT (IN-LB)
EMPTY WEIGHT	7206.6	146.0	1,052,117	0.0	185
+ Pilot and Copilot	340.0	47.0	15,980	0.4	136
+ Passengers (4) Fwd	680.0	85.7	58,276	0.0	0
+ Passengers (5) Mid	850.0	114.7	97,495	0.0	0
+ Passengers (2), Facing Outboard, Fwd Seat	340.0	139.7	47,498	0.0	0
Engine Oil	24.5	169.1	4143	0.0	0
+ Most Fwd Fuel (Jet A) 58.3 U.S. gallons	396.7	139.9	55,540	0.0	0.0
TOTAL	**9837.8**	**135.4**	**1,331,049**	**0.0**	**321**

MOST AFT CG

Table 4. Actual Weight Record (U.S.)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

_____

SIGNATURE: _____

WEIGHING RESULTS

[Units = pounds (lb), inches (in), U.S. gallons (gal)]

SCALE LOCATION	READING	TARE	NET
Forward Jack Point LH [FS 61.69, BL -30]			
Forward Jack Point RH [FS 61.69, BL 30]			
Aft Jack Point LH [FS 211.58, BL -14.53]			
Aft Jack Point RH [FS 211.58, BL 14.53]			
TOTAL (AS-WEIGHED NET WEIGHT)			

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{Fwd LH weight} + \text{Fwd RH weight})] + [\text{Aft FS} \times (\text{Aft LH weight} + \text{Aft RH weight})]}{\text{Total weight}}$$

$$CG = \frac{[61.69 \times (_+_)] + [211.58 \times (_+_)]}{(__+__+__+__)} = \frac{\qquad}{\qquad} = _____\text{inches}$$

LATERAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{Fwd LH weight} + \text{Fwd RH weight})] + [\text{Aft FS} \times (\text{Aft LH weight} + \text{Aft RH weight})]}{\text{Total weight}}$$

$$CG = \frac{(-30 \times _) + (30 \times _) + (-14.53 \times _) + (14.53 \times _)}{(__+__+__+__)} = \frac{\qquad}{\qquad} = _____\text{inches}$$

EMPTY WEIGHT DERIVATION

ITEM	NET WEIGHT (LB)	LONGITUDINAL ARM (IN.)	LONGITUDINAL MOMENT (IN-LB)	LATERAL ARM (IN.)	LATERAL MOMENT (IN-LB)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD:					
Fixed ballast:					
EMPTY WEIGHT					

MOST FORWARD CG

Table 4. Actual Weight Record (U.S.) (continued)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

_____

SIGNATURE: _____

ITEM	WEIGHT	CENTER OF GRAVITY AND MOMENTS			
		LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	(LB)	(IN.)	(IN-LB)	(IN.)	(IN-LB)
EMPTY WEIGHT					
+ Pilot					
+ Passengers (4) Fwd					
+ Passengers (5) Mid					
+ Passengers (2) Facing Outboard, Fwd Seat					
+ Engine Oil					
+ Most Fwd Fuel ( ) U.S. gallons					
TOTAL					

MOST AFT CG

Table 5. Actual Weight Record (Metric)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

SIGNATURE: _____

WEIGHING RESULTS

[Units = kilograms (kg), millimeters (mm)]

SCALE LOCATION	READING	TARE	NET
Forward Jack Point LH [FS 1567, BL -762]			
Forward Jack Point RH [FS 1567, BL 762]			
Aft Jack Point LH [FS 5374, BL -369]			
Aft Jack Point RH [FS 5374, BL 369]			
TOTAL (AS-WEIGHED NET WEIGHT)			

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

Table 5. Actual Weight Record (Metric) (continued)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

SIGNATURE: _____

$$CG = \frac{[\text{Fwd FS x (Fwd LH weight + Fwd RH weight)}] + [\text{Aft FS x (Aft LH weight + Aft RH weight)}]}{\text{Total weight}}$$

$$CG = \frac{[1567 \times (\ +\ )] + [5374 \times (\ +\ )]}{(\ +\ +\ +\ )} = \frac{\qquad}{\qquad} = _____\text{mm}$$

LATERAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS x (Fwd LH weight + Fwd RH weight)}] + [\text{Aft FS x (Aft LH weight + Aft RH weight)}]}{\text{Total weight}}$$

$$CG = \frac{(-762 \times\ ) + (762 \times\ ) + (-369\ ) + (369 \times\ )}{(\ +\ +\ +\ )} = \frac{\qquad}{\qquad} = _____\text{mm}$$

EMPTY WEIGHT DERIVATION

ITEM	NET WEIGHT (KG)	LONGITUDINAL ARM (MM)	LONGITUDINAL MOMENT (KG•MM)	LATERAL ARM (MM)	LATERAL MOMENT (KG•MM)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD: Fixed ballast:					
EMPTY WEIGHT					

MOST FORWARD CG

ITEM	WEIGHT (KG)	CENTER OF GRAVITY AND MOMENTS — LONGITUDINAL ARM (MM)	LONGITUDINAL MOMENT (KG•MM)	LATERAL ARM (MM)	LATERAL MOMENT (KG•MM)
EMPTY WEIGHT					
+ Pilot					
+ Passengers (4) Fwd					
+ Passengers (5) Mid					
+ Passengers (2) Facing Outboard, Fwd Seat					
+ Engine Oil					

Table 5. Actual Weight Record (Metric) (continued)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

SIGNATURE: _____

+ Most Fwd Fuel ( ) U.S. gallons					
TOTAL					

MOST AFT CG

Table 6. Weight and Balance - As-weighed Weighing Results (Example)

SERIAL NUMBER:_____ **WEIGHING KIT:** _____

DATE WEIGHED:_____ **CALIBRATION EXPIRY:** _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**A**)	776.0	352.0	10.0	4.5	766.0	947.5
Forward Right (**B**)	2709.0	1228.8	0.0	0.0	2709.0	1228.8
Aft Left (**C**)	3963.0	1797.6	6.0	2.7	3957.0	1794.9
Aft Right (**D**)	0.0	0.0	0.0	0.0	0.0	0.0
AS-WEIGHED WEIGHT	1448.0	3378.4	16.0	7.2	7432.0	3371.2

SECOND WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**E**)	774.0	351.1	10.0	4.5	764.0	346.6
Forward Right (**F**)	2712.0	1230.2	0.0	0.0	2712.0	1230.2
Aft Left (**G**)	3960.0	1796.2	0.0	0.0	3960.0	1796.2
Aft Right (**H**)	0.0	0.0	0.0	0.0	0.0	0.0
AS-WEIGHED WEIGHT	7446.0	3377.5	10.0	4.5	7436.0	3373.0

CONFIRMATION OF WEIGHING RESULTS

SCALE LOCATION (SCALE LETTER)	NET WEIGHTS		SCALE LOCATION (SCALE LETTER)	NET WEIGHTS	
	LB	KG		LB	KG
Forward Left (**A**)	766.0	347.5	Forward Left (**E**)	764.0	346.6
Forward Right (**B**)	2709.0	1228.8	Forward Right (**F**)	2712.0	1230.2
Aft Left (**C**)	3957.0	1794.9	Aft Left (**G**)	3960.0	1796.2

Table 6. Weight and Balance - As-weighed Weighing Results (Example) (continued)

SERIAL NUMBER:_____ WEIGHING KIT: _____

DATE WEIGHED:_____ CALIBRATION EXPIRY: _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Aft Right (**D**)	0.0	0.0	Aft Right (**H**)		0.0	0.0
TOTAL FIRST NET WEIGHT (I)	7432.0	3371.2	**TOTAL SECOND NET WEIGHT (J)**		7436.0	3373.0

TOTAL AS-WEIGHED WEIGHT DIFFERENCE (I-J) (MUST BE < 20 POUNDS (9 KG)) = 4 POUNDS (1.8 KG)

FORWARD VERSUS AFT WEIGHT DIFFERENCE (A+B-C-D-E-F+G+H) (MUST BE < 10 POUNDS (4.5KG)) = 4 POUNDS (1.9 KG)

NOTES:

1. As only one aft jack point weighing position is used, only enter data in the applicable aft left or aft right location within the table.
2. If total helicopter As-weighed Weight difference is less than 20 pounds (9 kg), and forward versus aft Net Weight difference is less than 10 pounds (4.5 kg), the weighing results are acceptable.

Table 7. Calculations - As-Weighed Weight, Longitudinal CG, and Lateral CG (Example)

SCALE LOCATION	NET WEIGHT	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
Forward Left	766.0 (347.5)	61.69 (1567)	47,255 (544,533)	-30.0 (-762)	-22,980 (-264,795)
Forward Right	2709.0 (1228.8)	61.69 (1567)	167,118 (1,925,407)	30.0 (762)	81,270 (936,346)
Aft Left	3957.0 (1794.9)	211.58 (5374)	837,222 (9,645,793)	-14.53 (-369)	-57,495 (-662,318)
Aft Right	0.0 (0.0)	211.58 (5374)	0 (0)	14.53 (369)	0 (0)
AS-WEIGHED WEIGHT AND CG	7432.0 (3371.2)	141.5 (3594)	1,051,595 (12,115,732)	0.1 (3)	795 (9233)

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

Table 8. Calculations - Deriving Unballasted Empty Weight and Center of Gravity (Example)

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
AS-WEIGHED (NET WEIGHT)	7432.0 (3371.2)	141.5 (3594)	1,051,595 (12,115,732)	0.1 (3)	795 (9233)

Table 8. Calculations - Deriving Unballasted Empty Weight and Center of Gravity (Example) (continued)

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
REMOVE:					
Usable Oil	-24.5 (-11.1)	169.1 (4295)	-4143 (-47,732)	0.0 (0)	0 (0)
Ballast - Battery Block	-76.0 (-34.5)	-5.7 (-145)	433 (4991)	7.9 (201)	-600 (-6917)
Ballast - Station -4.7	-74.0 (-33.6)	-4.7 (-119)	348 (4007)	0.0 (0)	0 (0)
Ballast - Station -4.4	-64.0 (-29.0)	-4.4 (-112)	282 (3244)	0.1 (3)	-6 (-74)
Ballast - Station -4.1	-15.0 (-6.8)	-4.1 (-104)	62 (709)	0.2 (5)	-3 (-35)
ADD:					
Residual Fuel	28.1 (12.7)	126.0 (3200)	3541 (40,793)	0.0 (0)	0 (0)
UNBALLASTED EMPTY WEIGHT	7206.6 (3269.0)	146.0 (3708)	1,052,117 (12,121,743)	0.0 (1)	185 (2207)

Weight and Balance - Actual Weight Record (Example)

Procedure

NOTE

The examples in Table 1 (U.S.) and Table 2 (Metric) make use of the values already used in this chapter.

1. The Table 1 (U.S.) and Table 2 (Metric) examples show the calculations that are necessary to complete an Actual Weight Record after weighing the helicopter. This may be after a configuration change or as required by local aviation authorities.

2. Blank forms of the Actual Weight Record are provided in Table 3 (U.S.) and Table 4 (Metric).

Table 1. Weight and Balance - Actual Weight Record (U.S.) (Example)

1. Perform the weighing as described in 412-A-08-40-00-01A-160B-A / 00088. Enter the As-weighed weighing results for each scale and subtract the Tare Weight (if any). Refer to the Table 5 example. Calculate the As-weighed net weight and Center of Gravity. Refer to the Table 6 example.

WEIGHING RESULTS

[Units = pounds (lb), inches (in), U.S. gallons (gal)]

SCALE LOCATION	READING (LB)	TARE (LB)	NET (LB)
Forward Jack Point LH [FS 61.69, BL -30.0]	776	10.0	766.0
Forward Jack Point RH [FS 61.69, BL 30.0]	2709	0.0	2709.0
Aft Jack Point LH [FS 211.58, BL -14.53]	3963.0	6.0	3957.0
Aft Jack Point RH [FS 211.58, BL 14.53]	0.0	0.0	0.0
TOTAL	**7449.0**	**17.0**	**7432.0**

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{LH weight} + \text{RH weight})] + [\text{Aft FS} \times \text{Aft weight}]}{\text{Total weight}}$$

$$CG = \frac{[61.69 \times (766.0 + 2709.0)] + [211.58 \times 3957.0]}{(766.0 + 2709.0 + 3957.0)} = \frac{1{,}051{,}595}{7432.0} = 141.5 \text{ inches}$$

LATERAL CG AS WEIGHED

$$CG = \frac{(\text{Fwd LBL} \times \text{LH weight}) + (\text{Fwd RBL} \times \text{RH weight}) + (\text{Aft BL} \times \text{Aft weight})}{\text{Total weight}}$$

$$CG = \frac{(-30.0 \times 766.0) + (30.0 \times 2709.0) + (-14.53 \times 3957.0)}{(766.0 + 2709.0 + 3957.0)} = \frac{795}{7432.0} = 0.1 \text{ inch}$$

2. Remove the weight and moment for those items installed that are not part of the Empty Weight. Add the weight and moment for those items that are part of the Empty Weight, but were not installed. The result is the Unballasted Empty Weight. Refer to the Table 7 example.

EMPTY WEIGHT DERIVATION

ITEM	WEIGHT (LB)	LONGITUDINAL		LATERAL	
		CG (IN.)	MOMENT (IN-LB)	CG (IN.)	MOMENT (IN-LB)
AS WEIGHED (NET WEIGHT)	7432.0	141.5	1,051,595	0.1	795
REMOVE:					

Table 1. Weight and Balance - Actual Weight Record (U.S.) (Example) (continued)

Unusable Oil	-24.5	169.1	-4143	0.0	0
Ballast at Battery Block	-76.0	-5.7	433	7.9	-600
Ballast at FS -4.7	-74.0	-4.7	348	0.0	0
Ballast at FS -4.4	-64.0	-4.4	282	0.1	-6
Ballast at FS -4.1	-15.0	-4.1	62	0.2	-3
ADD:					
Residual Fuel	28.1	126.0	3541	0.0	0
UNBALLASTED EMPTY WEIGHT	**7206.6**	**146.0**	**1,052,117**	**0.0**	**185**

3. Determine the Most Forward and Most Aft Useful Load weights and moments and add these to the Unballasted Empty Weight to give the most Forward and Most Aft Gross Weight and Center of Gravity.

MOST FORWARD CG

ITEM	WEIGHT	CENTER OF GRAVITY AND MOMENTS			
		LONGITUDINAL		LATERAL	
		CG	MOMENT	CG	MOMENT
	(LB)	(IN.)	(IN-LB)	(IN.)	(IN-LB)
EMPTY WEIGHT	7206.6	146.0	1,052,117	0.0	185
+ Pilot and Copilot	340.0	47.0	15,980	0.4	136
+ Passengers (4) Fwd	680.0	85.7	58,276	0.0	0
+ Passengers (5) Mid	850.0	114.7	97,495	0.0	0
+ Passengers (2), Facing Outboard, Fwd Seat	340.0	139.7	47,498	0.0	0
Engine Oil	24.5	169.1	4143	0.0	0
+ Most Fwd Fuel (Jet A) 58.3 U.S. gallons	396.7	139.9	55,540	0.0	0.0
TOTAL	**9837.8**	**135.4**	**1,331,049**	**0.0**	**321**

MOST AFT CG

Table 2. Weight and Balance - Actual Weight Record (Metric) (Example)

1. Perform the weighing as described in 412-A-08-40-00-01A-160B-A / 00088. Enter the As-weighed weighing results for each scale and subtract the Tare Weight (if any). Refer to the Table 5 example. Calculate the As-weighed net weight and Center of Gravity. Refer to the Table 6 example.

WEIGHING RESULTS

[Units = kilograms (kg), millimeters (mm)]

SCALE LOCATION	READING	TARE	NET
	(KG)	(KG)	(KG)
Forward Jack Point LH [FS 1567, BL -762]	352	4.5	347.5
Forward Jack Point RH [FS 1567, BL 762]	1228.8	0.0	1228.8
Aft Jack Point LH [FS 5374, BL -369]	1797.6	2.7	1794.9
Aft Jack Point RH [FS 5374, BL 369]	0.0	0.0	0.0
TOTAL	**3378.4**	**7.2**	**3371.2**

NOTE:

Table 2. Weight and Balance - Actual Weight Record (Metric) (Example) (continued)

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{LH weight} + \text{RH weight})] + [\text{Aft FS} \times \text{Aft weight})]}{\text{Total weight}}$$

$$CG = \frac{[1567 \times (347.5 + 1228.8)] + [5374 \times 1794.9]}{(347.5 + 1228.8 + 1794.9)} = \frac{12{,}115{,}855}{3371.2} = 3594 \text{ mm}$$

LATERAL CG AS WEIGHED

$$CG = \frac{(\text{Fwd LBL} \times \text{LH weight}) + (\text{Fwd RBL} \times \text{RH weight}) + (\text{Aft BL} \times \text{Aft weight})}{\text{Total weight}}$$

$$CG = \frac{(-762 \times 347.5) + (762 \times 1228.8) + (-369 \times 1794.9)}{(347.5 + 1228.8 + 1794.9)} = \frac{9233}{3371.2} = 3 \text{ mm}$$

2. Remove the weight and moment for those items installed that are not part of the Empty Weight. Add the weight and moment for those items that are part of the Empty Weight, but were not installed. The result is the Unballasted Empty Weight. Refer to the Table 7 example.

EMPTY WEIGHT DERIVATION

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		CG	MOMENT	CG	MOMENT
	(KG)	(MM)	(KG•MM)	(MM)	(KG•MM)
AS WEIGHED (NET WEIGHT)	3371.2	3594	12,115,855	3	9233
REMOVE:					
Unusable Oil	-11.1	4295	-47676	0	0
Ballast at Battery Block	-34.5	-145	4996	201	-6924
Ballast at FS -119	-33.6	-119	4012	0	0
Ballast at FS -112	-29.0	-112	3242	3	-73
Ballast at FS -104	-6.8	-104	708	5	-35
ADD:					
Residual Fuel	12.7	3200	40,645	0	0
UNBALLASTED EMPTY WEIGHT	**3268.9**	**3708**	**12,121,782**	**1**	**2201**

3. Determine the Most Forward and Most Aft Useful Load weights and moments and add these to the Unballasted Empty Weight to give the most Forward and Most Aft Gross Weight and Center of Gravity.

MOST FORWARD CG

ITEM	WEIGHT	CENTER OF GRAVITY AND MOMENTS			
		LONGITUDINAL		LATERAL	
		CG	MOMENT	CG	MOMENT
	(KG)	(MM)	(KG-MM)	(MM)	(KG-MM)
EMPTY WEIGHT	3268.9	3708	12,121,782	1	2201
+ Pilot and Copilot	154.2	1194	184,110	10	1567

Table 2. Weight and Balance - Actual Weight Record (Metric) (Example) (continued)

+ Passengers (4) Fwd	308.4	2177	671,413	0	0
+ Passengers (5) Mid	385.6	2913	1,123,265	0	0
+ Passengers (2) Facing Outboard, Fwd Seat	154.2	3548	547,237	0	0
+ Engine Oil	11.1	4295	47,732	0	0
Most Fwd Fuel (Jet A) 220.8 liters	180.0	3553	639,485	0	0.0
TOTAL	**4462.4**	**3436**	**15,335,023**	**1**	**3768**

MOST AFT CG

Table 3. Actual Weight Record (U.S.)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

_____

SIGNATURE: _____

WEIGHING RESULTS

[Units = pounds (lb), inches (in), U.S. gallons (gal)]

SCALE LOCATION	READING	TARE	NET
Forward Jack Point LH [FS 61.69, BL -30]			
Forward Jack Point RH [FS 61.69, BL 30]			
Aft Jack Point LH [FS 211.58, BL -14.53]			
Aft Jack Point RH [FS 211.58, BL 14.53]			
TOTAL (AS-WEIGHED NET WEIGHT)			

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{Fwd LH weight} + \text{Fwd RH weight})] + [\text{Aft FS} \times (\text{Aft LH weight} + \text{Aft RH weight})]}{\text{Total weight}}$$

$$CG = \frac{[61.69 \times (__+__)] + [211.58 \times (__+__)]}{(__+__+__+__)} = \frac{_____}{_____} = _____ \text{inches}$$

LATERAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{Fwd LH weight} + \text{Fwd RH weight})] + [\text{Aft FS} \times (\text{Aft LH weight} + \text{Aft RH weight})]}{\text{Total weight}}$$

Table 3. Actual Weight Record (U.S.) (continued)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

_____

SIGNATURE: _____

CG = (-30 x__) + (30 x__) + (-14.53 x__) + (14.53 x__) / (+ + +) = _____ = _____inches

EMPTY WEIGHT DERIVATION

ITEM	NET WEIGHT (LB)	LONGITUDINAL ARM (IN.)	LONGITUDINAL MOMENT (IN-LB)	LATERAL ARM (IN.)	LATERAL MOMENT (IN-LB)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD: Fixed ballast:					
EMPTY WEIGHT					

MOST FORWARD CG

ITEM	WEIGHT (LB)	CENTER OF GRAVITY AND MOMENTS — LONGITUDINAL ARM (IN.)	LONGITUDINAL MOMENT (IN-LB)	LATERAL ARM (IN.)	LATERAL MOMENT (IN-LB)
EMPTY WEIGHT					
+ Pilot					
+ Passengers (4) Fwd					
+ Passengers (5) Mid					
+ Passengers (2) Facing Outboard, Fwd Seat					
+ Engine Oil					
+ Most Fwd Fuel ( ) U.S. gallons					
TOTAL MOST AFT CG					

Table 4. Actual Weight Record (Metric)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

SIGNATURE: _____

WEIGHING RESULTS

[Units = kilograms (kg), millimeters (mm)]

SCALE LOCATION	READING	TARE	NET
Forward Jack Point LH [FS 1567, BL -762]			
Forward Jack Point RH [FS 1567, BL 762]			
Aft Jack Point LH [FS 5374, BL -369]			
Aft Jack Point RH [FS 5374, BL 369]			
TOTAL (AS-WEIGHED NET WEIGHT)			

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

LONGITUDINAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{Fwd LH weight} + \text{Fwd RH weight})] + [\text{Aft FS} \times (\text{Aft LH weight} + \text{Aft RH weight})]}{\text{Total weight}}$$

$$CG = \frac{[1567 \times (__+__)] + [5374 \times (\ +__)]}{(__+__+__+__)} = _____ = _____\text{mm}$$

LATERAL CG AS WEIGHED

$$CG = \frac{[\text{Fwd FS} \times (\text{Fwd LH weight} + \text{Fwd RH weight})] + [\text{Aft FS} \times (\text{Aft LH weight} + \text{Aft RH weight})]}{\text{Total weight}}$$

$$CG = \frac{(-762 \times __) + (762 \times __) + (-369) + (369 \times __)}{(__+__+__+__)} = _____ = _____\text{mm}$$

EMPTY WEIGHT DERIVATION

ITEM	NET WEIGHT (KG)	LONGITUDINAL ARM (MM)	LONGITUDINAL MOMENT (KG•MM)	LATERAL ARM (MM)	LATERAL MOMENT (KG•MM)
AS WEIGHED (NET WEIGHT)					
REMOVE:					
ADD: Fixed ballast:					
EMPTY WEIGHT					

MOST FORWARD CG

ITEM	WEIGHT (KG)	CENTER OF GRAVITY AND MOMENTS: LONGITUDINAL ARM (MM)	LONGITUDINAL MOMENT (KG•MM)	LATERAL ARM (MM)	LATERAL MOMENT (KG•MM)

Table 4. Actual Weight Record (Metric) (continued)

DATE: _____ **W.O.** _____

COMPANY:_____

HELICOPTER S/N:_____

REGISTRATION NO.:_____

TOTAL TIME:_____

SIGNATURE: _____

EMPTY WEIGHT						
	+ Pilot					
	+ Passengers (4) Fwd					
	+ Passengers (5) Mid					
	+ Passengers (2) Facing Outboard, Fwd Seat					
	+ Engine Oil					
	+ Most Fwd Fuel ( ) U.S. gallons					
TOTAL						

MOST AFT CG

Table 5. Weight and Balance - As-weighed Weighing Results (Example)

SERIAL NUMBER:_____ **WEIGHING KIT:** _____

DATE WEIGHED:_____ **CALIBRATION EXPIRY:** _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**A**)	776.0	352.0	10.0	4.5	766.0	947.5
Forward Right (**B**)	2709.0	1228.8	0.0	0.0	2709.0	1228.8
Aft Left (**C**)	3963.0	1797.6	6.0	2.7	3957.0	1794.9
Aft Right (**D**)	0.0	0.0	0.0	0.0	0.0	0.0
As-Weighed Weight	1448.0	3378.4	16.0	7.2	7432.0	3371.2

SECOND WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
Forward Left (**E**)	774.0	351.1	10.0	4.5	764.0	346.6
Forward Right (**F**)	2712.0	1230.2	0.0	0.0	2712.0	1230.2
Aft Left (**G**)	3960.0	1796.2	0.0	0.0	3960.0	1796.2
Aft Right (**H**)	0.0	0.0	0.0	0.0	0.0	0.0

Table 5. Weight and Balance - As-weighed Weighing Results (Example) (continued)

SERIAL NUMBER:_____ WEIGHING KIT: _____

DATE WEIGHED:_____ CALIBRATION EXPIRY: _____

FIRST WEIGHING RESULT

SCALE LOCATION (SCALE LETTER)	SCALE READING		TARE WEIGHT		NET WEIGHT	
	LB	KG	LB	KG	LB	KG
As-Weighed Weight	7446.0	3377.5	10.0	4.5	7436.0	3373.0

CONFIRMATION OF WEIGHING RESULTS

SCALE LOCATION (SCALE LETTER)	NET WEIGHTS		SCALE LOCATION (SCALE LETTER)	Net weights	
	LB	KG		LB	KG
Forward Left (**A**)	766.0	347.5	Forward Left (**E**)	764.0	346.6
Forward Right (**B**)	2709.0	1228.8	Forward Right (**F**)	2712.0	1230.2
Aft Left (**C**)	3957.0	1794.9	Aft Left (**G**)	3960.0	1796.2
Aft Right (**D**)	0.0	0.0	Aft Right (**H**)	0.0	0.0
Total first net Weight (I)	7432.0	3371.2	**Total second net Weight (J)**	7436.0	3373.0

TOTAL AS-WEIGHED WEIGHT DIFFERENCE (I-J) (MUST BE < 20 POUNDS (9 KG)) = 4 POUNDS (1.8 KG)

FORWARD VERSUS AFT WEIGHT DIFFERENCE (A+B-C-D-E-F+G+H) (MUST BE < 10 POUNDS (4.5KG)) = 4 POUNDS (1.9 KG)

NOTES:

1. As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.
2. If total helicopter As-weighed Weight difference is less than 20 pounds (9 kg), and forward versus aft Net Weight difference is less than 10 pounds (4.5 kg), the weighing results are acceptable.

Table 6. Calculations - As-weighed Weight, Longitudinal CG, and Lateral CG (Example)

Scale Location	Net Weight	LONGITUDINAL		LATERAL	
		ARM	MOMENT	ARM	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
Forward Left	766.0	61.69	47,255	-30.0	-22,980
	(347.5)	(1567)	(544,533)	(-762)	(-264,795)
Forward Right	2709.0	61.69	167,118	30.0	81,270
	(1228.8)	(1567)	(1,925,407)	(762)	(936,346)
Aft Left	3957.0	211.58	837,222	-14.53	-57,495
	(1794.9)	(5374)	(9,645,793)	(-369)	(-662,318)
Aft Right	0.0	211.58	0	14.53	0
	(0.0)	(5374)	(0)	(369)	(0)
As-Weighed Weight and CG	7432.0	141.5	1,051,595	0.1	795
	(3371.2)	(3594)	(12,115,732)	(3)	(9233)

NOTE:

As only one aft jack point weighing position is used, only enter data in the applicable Aft Left or Aft Right location within the table.

Table 7. Calculations - Deriving Unballasted Empty Weight and Center of Gravity (Example)

ITEM	WEIGHT	LONGITUDINAL		LATERAL	
		ARM (CG)	MOMENT	ARM (CG)	MOMENT
	LB (KG)	IN. (MM)	IN-LB (KG•MM)	IN. (MM)	IN-LB (KG•MM)
AS-WEIGHED (NET WEIGHT)	7432.0 (3371.2)	141.5 (3594)	1,051,595 (12,115,732)	0.1 (3)	795 (9233)
REMOVE:					
Usable Oil	-24.5 (-11.1)	169.1 (4295)	-4143 (-47,732)	0.0 (0)	0 (0)
Ballast - Battery Block	-76.0 (-34.5)	-5.7 (-145)	433 (4991)	7.9 (201)	-600 (-6917)
Ballast - Station -4.7	-74.0 (-33.6)	-4.7 (-119)	348 (4007)	0.0 (0)	0 (0)
Ballast - Station -4.4	-64.0 (-29.0)	-4.4 (-112)	282 (3244)	0.1 (3)	-6 (-74)
Ballast - Station -4.1	-15.0 (-6.8)	-4.1 (-104)	62 (709)	0.2 (5)	-3 (-35)
ADD:					
Residual Fuel	28.1 (12.7)	126.0 (3200)	3541 (40,793)	0.0 (0)	0 (0)
UNBALLASTED EMPTY WEIGHT	7206.6 (3269.0)	146.0 (3708)	1,052,117 (12,121,743)	0.0 (1)	185 (2207)

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Weight and Balance - Ballast Removal

Procedure

1. Remove the ballast (12, Figure 1) as follows:

1.1. Remove the battery (412-A-96-61-00-02A-520A-A / 01913).

1.2. Remove the nuts (11 and 14), aluminum washers (10 and 13), and eyebolts (3 and 6).

1.3. Remove the bolts (1 and 7), aluminum washers (2 and 8), and brackets (5 and 9).

1.4. Remove the pan (4) and the ballast (12).

NOTE

Remove the ballast in equal amounts from both sides of the nose shelf.

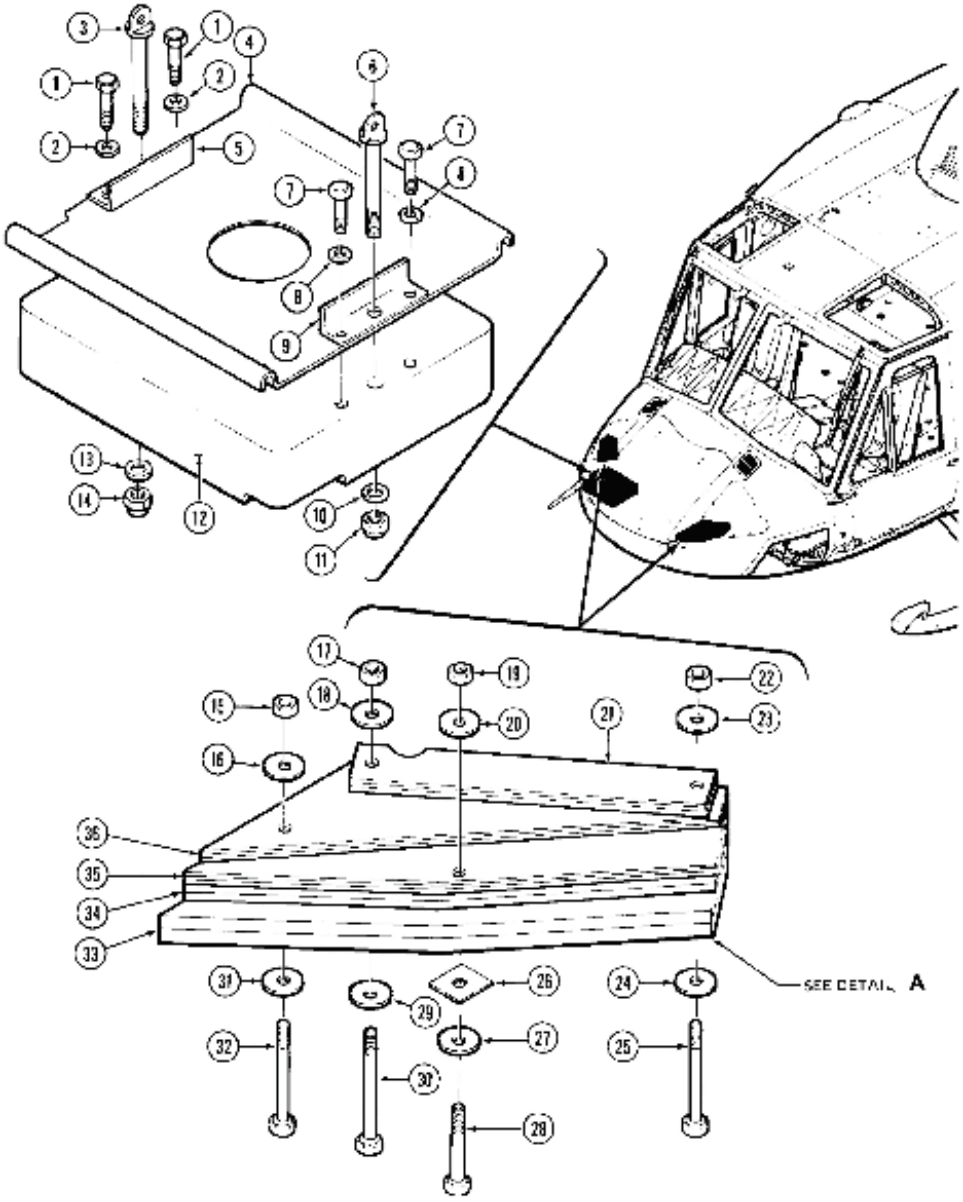
2. Remove the ballasts (21, 33, 34, 35, and 36) as follows:

2.1. Remove the nuts (22), steel washers (23), bolts (25), and steel washers (24).

2.2. Remove the nuts (17), steel washers (18), bolts (30), and steel washers (29). Remove the ballast (21).

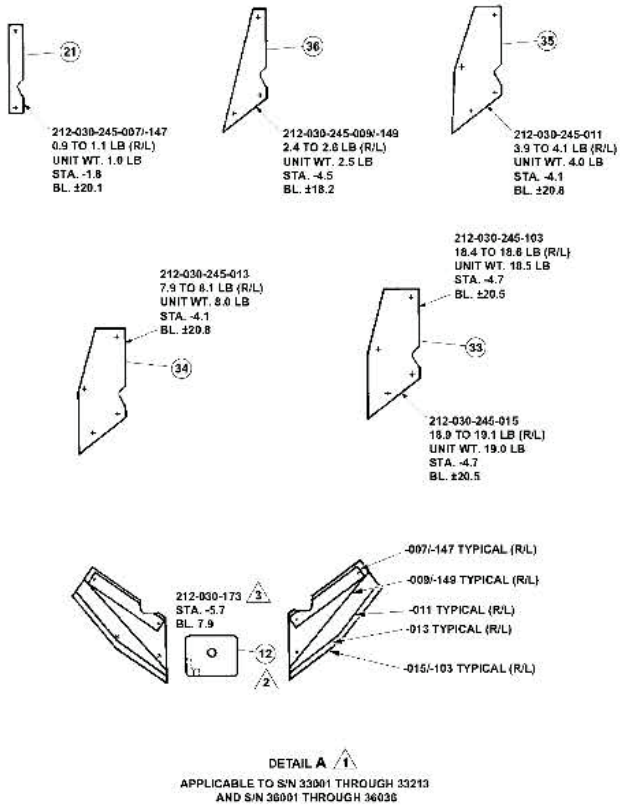
2.3. Remove the nuts (15), steel washers (16), bolts (32), and steel washers (31). Remove the ballast (36).

2.4. Remove the nuts (19), steel washers (20), blocks (26), steel washers (27), and bolts (28). Remove the ballasts (33, 34, and 35).



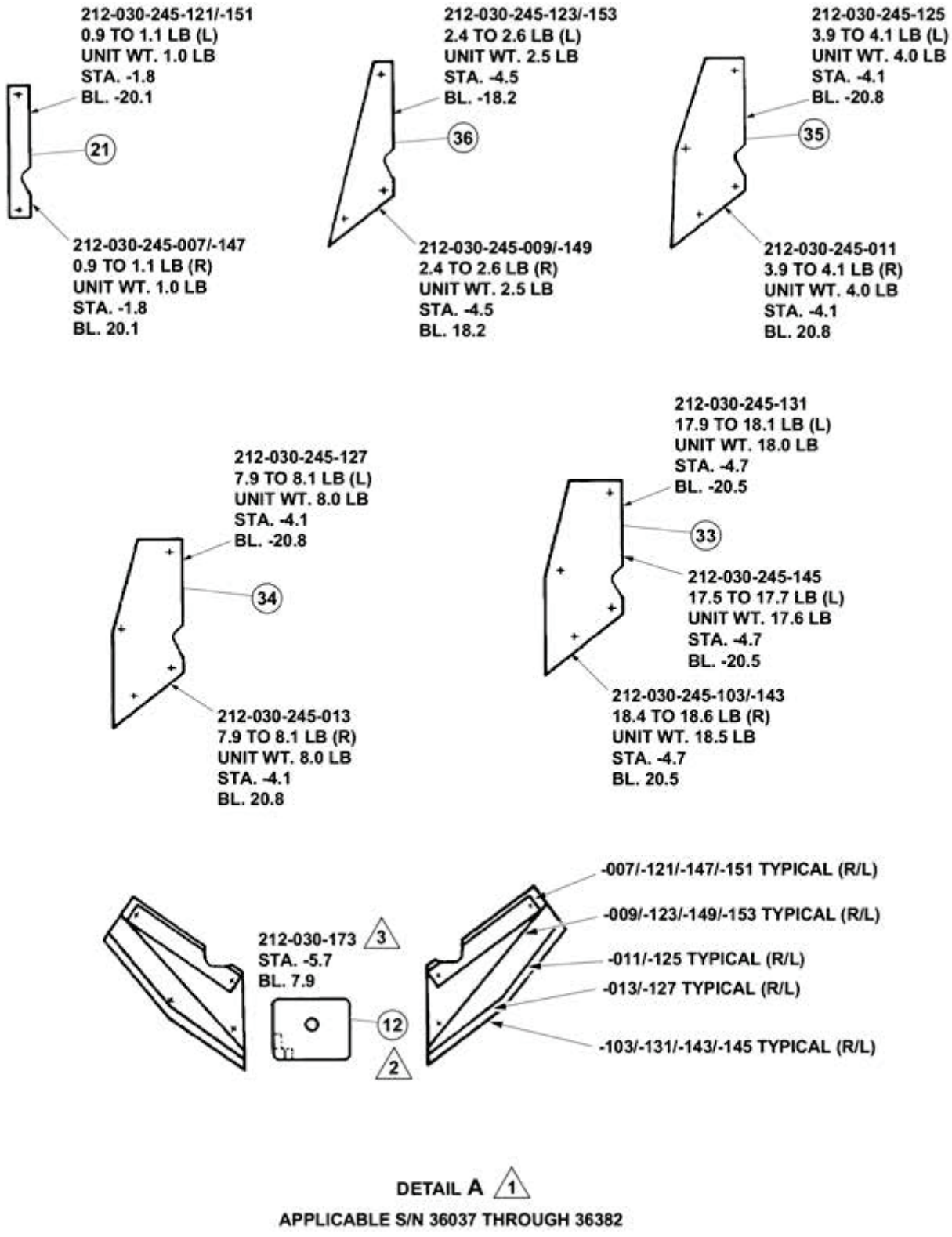
ICN-412-A-084000-B-97499-00003-A-001-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 1 of 5)



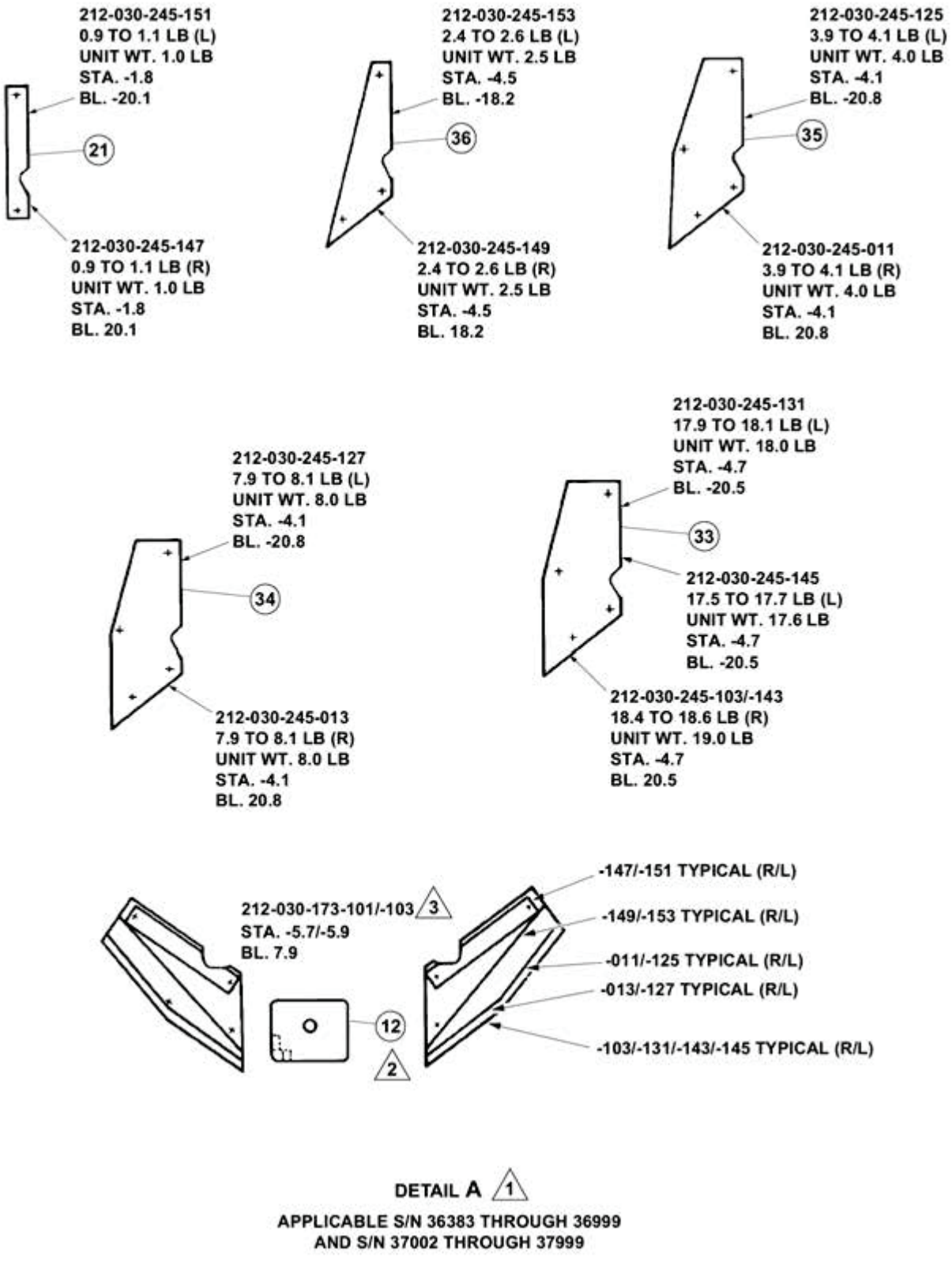
ICN-412-A-084000-B-97499-00004-A-001-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 2 of 5)



ICN-412-A-084000-B-97499-00005-A-002-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 3 of 5)



ICN-412-A-084000-B-97499-00006-A-002-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 4 of 5)

1. Bolt
2. Aluminum washer
3. Eyebolt
4. Pan
5. Bracket
6. Eyebolt
7. Bolt
8. Aluminum washer
9. Bracket
10. Aluminum washer
11. Nut
12. Ballast
13. Aluminum washer
14. Nut
15. Nut
16. Steel washer
17. Nut
18. Steel washer
19. Nut
20. Steel washer
21. Ballast
22. Nut
23. Steel washer
24. Steel washer
25. Bolt
26. Block
27. Steel washer
28. Bolt
29. Steel washer
30. Bolt
31. Steel washer
32. Bolt
33. Ballast
34. Ballast
35. Ballast
36. Ballast

NOTES

1. Maximum ballast is 82 pounds (37.2 kg) per side. Add ballast symmetrically.
2. Ballast (12) is not included in 164-pound (74.4-kg) maximum hose shelf ballast.
3. Actual weight is stamped on each 212-030-173 battery block ballast. Use 212-030-173-101 ballast at FS -5.7. When 53 Amp-hour Lead Acid battery kit (412-706-131) is installed, use the 212-030-173-103 battery block ballast at FS -5.9. Weigh battery and ballast underneath the battery before installation. The total of both parts weight must not exceed 155 lbs (70 kg). Make sure that the cutout in the ballast is facing down.

ICN-412-A-084000-B-97499-00007-A-001-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 5 of 5)

Weight and Balance - Ballast Installation

Procedure

1. Refer to 412-A-08-40-00-07A-169A-A / 00094 for the procedure to determine the amount of ballast and the location where the ballast is installed. If the ballast is installed or removed from the tail skid, refer to 412-A-32-20-00-00A-520A-A / 00747. To install or remove the ballast in the nose compartment, do the following steps:

2. Install the ballast (12, Figure 1) as follows:

2.1. Remove the battery (412-A-96-61-00-02A-520A-A / 01913).

2.2. Remove the nuts (11 and 14), aluminum washers (10 and 13), and eyebolts (3 and 6).

2.3. Remove the bolts (1 and 7), aluminum washers (2 and 8), and brackets (5 and 9).

2.4. Remove the pan (4).

2.5. If the 53 Amp-hour Lead Acid Battery Kit (412-706-131) is installed, make sure that the total weight of both parts does not exceed 155 pounds (70 kg) and the cutout in the ballast (12) faces down.

2.6. Position the ballast (12) on the shelf.

2.7. Position the pan (4) on the ballast (12).

2.8. Position the brackets (5 and 9) on the pan (4).

2.9. Install the eyebolts (3 and 6), aluminum washers (10 and 13), and nuts (11 and 14).

2.10. Install the bolts (1 and 7) with the aluminum washers (2 and 8).

2.11. Install the battery (412-A-96-61-00-02A-720A-A / 01915).

NOTE

Add the ballast in equal amounts to both sides of the nose shelf.

3. Install the ballasts (21, 33, 34, 35 and 36) as follows:

3.1. Remove the nuts (22), steel washers (23), bolts (25), and steel washers (24).

3.2. Remove the nuts (17), steel washers (18), bolts (30), and steel washers (29).

3.3. Remove the nuts (15), steel washers (16), bolts (32), and steel washers (31).

3.4. Remove the nuts (19), steel washers (20), blocks (26), steel washers (27), and bolts (28).

3.5. Position the required ballast on the shelf.

NOTE

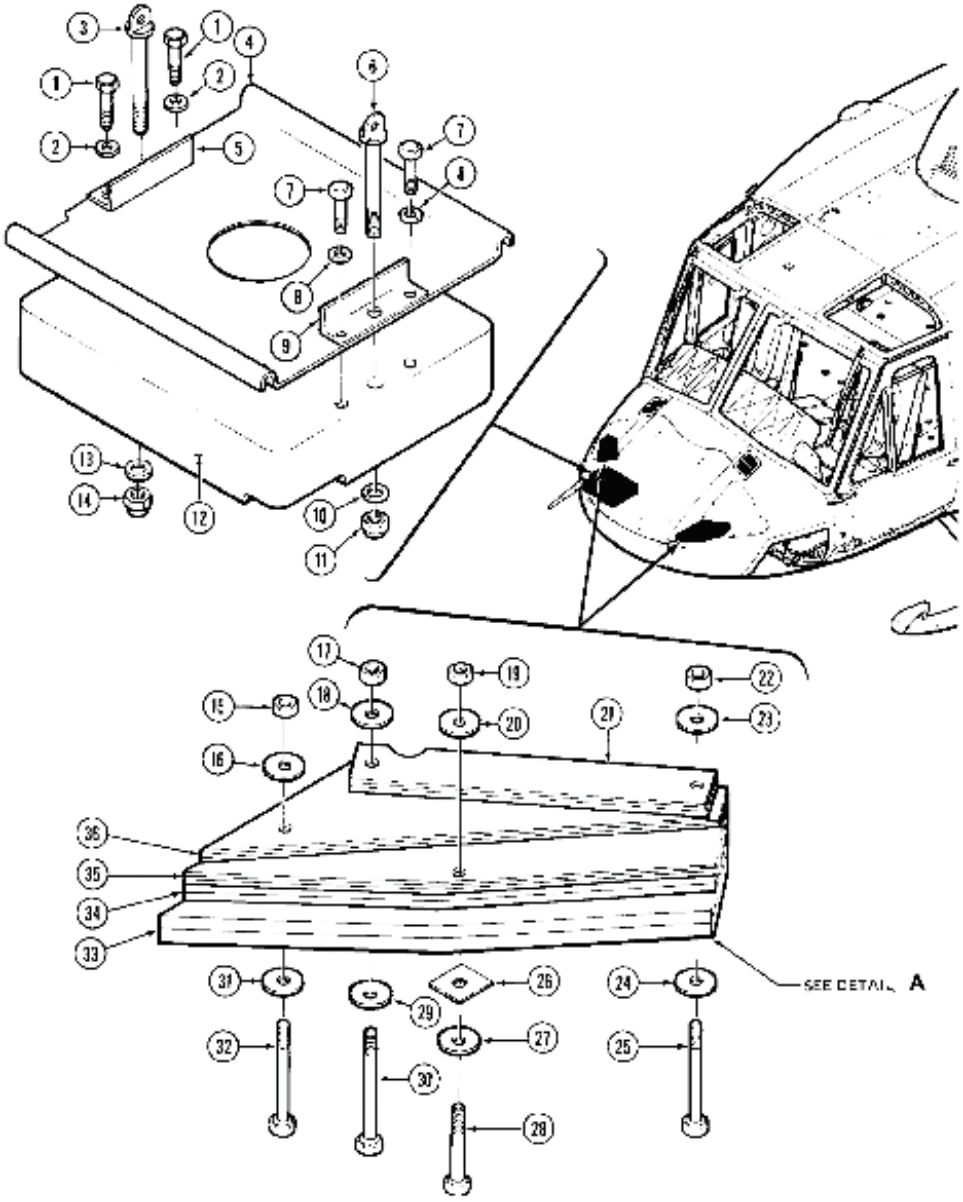
For ballast installations using less than the maximum configuration, some installation hardware may not be required.

3.6. Install the bolts (25), steel washers (24 and 23), and nuts (22).

3.7. Install the bolts (30), steel washers (29 and 18), and nuts (17).

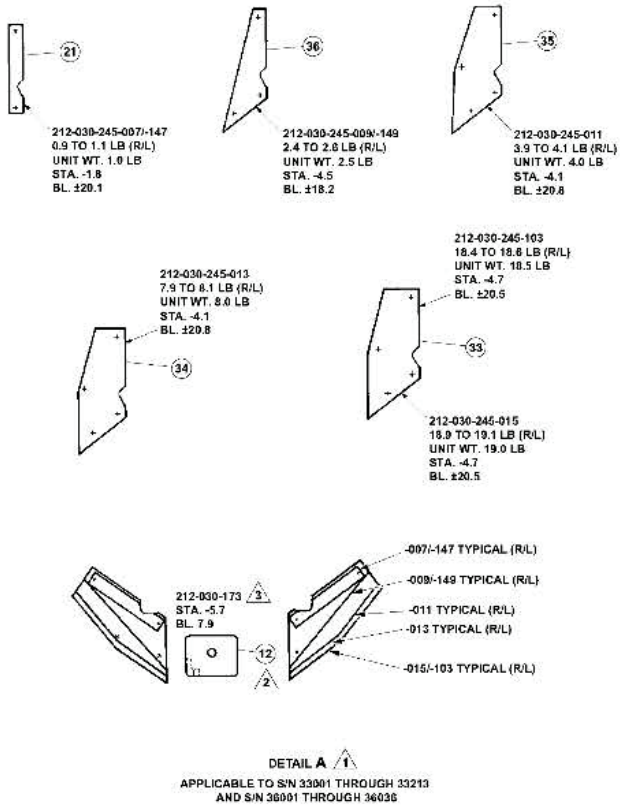
3.8. Install the bolts (32), steel washers (31 and 16), and nuts (15).

3.9. Install the bolts (28), steel washers (27), blocks (26), steel washers (20), and nuts (19).



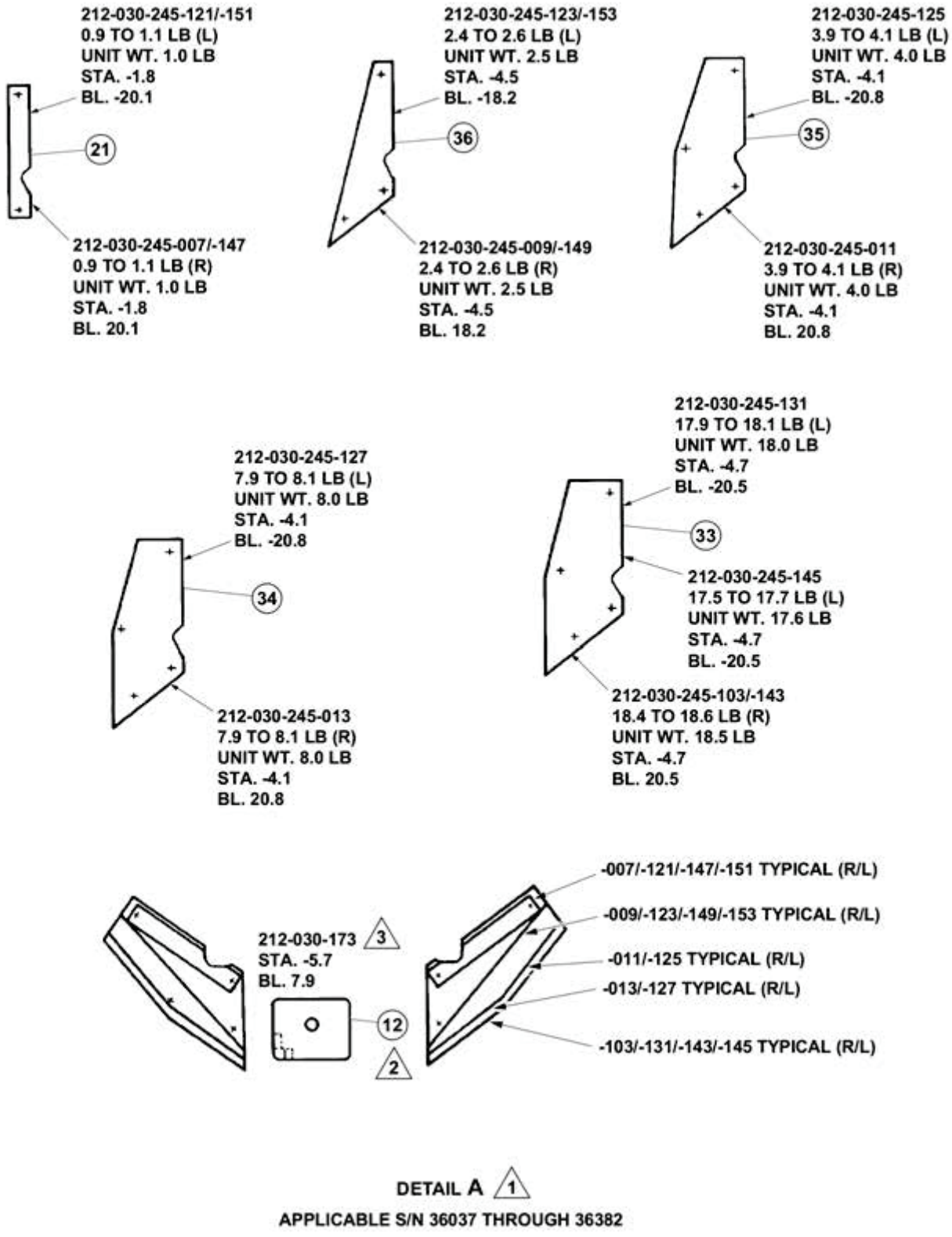
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Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 1 of 5)



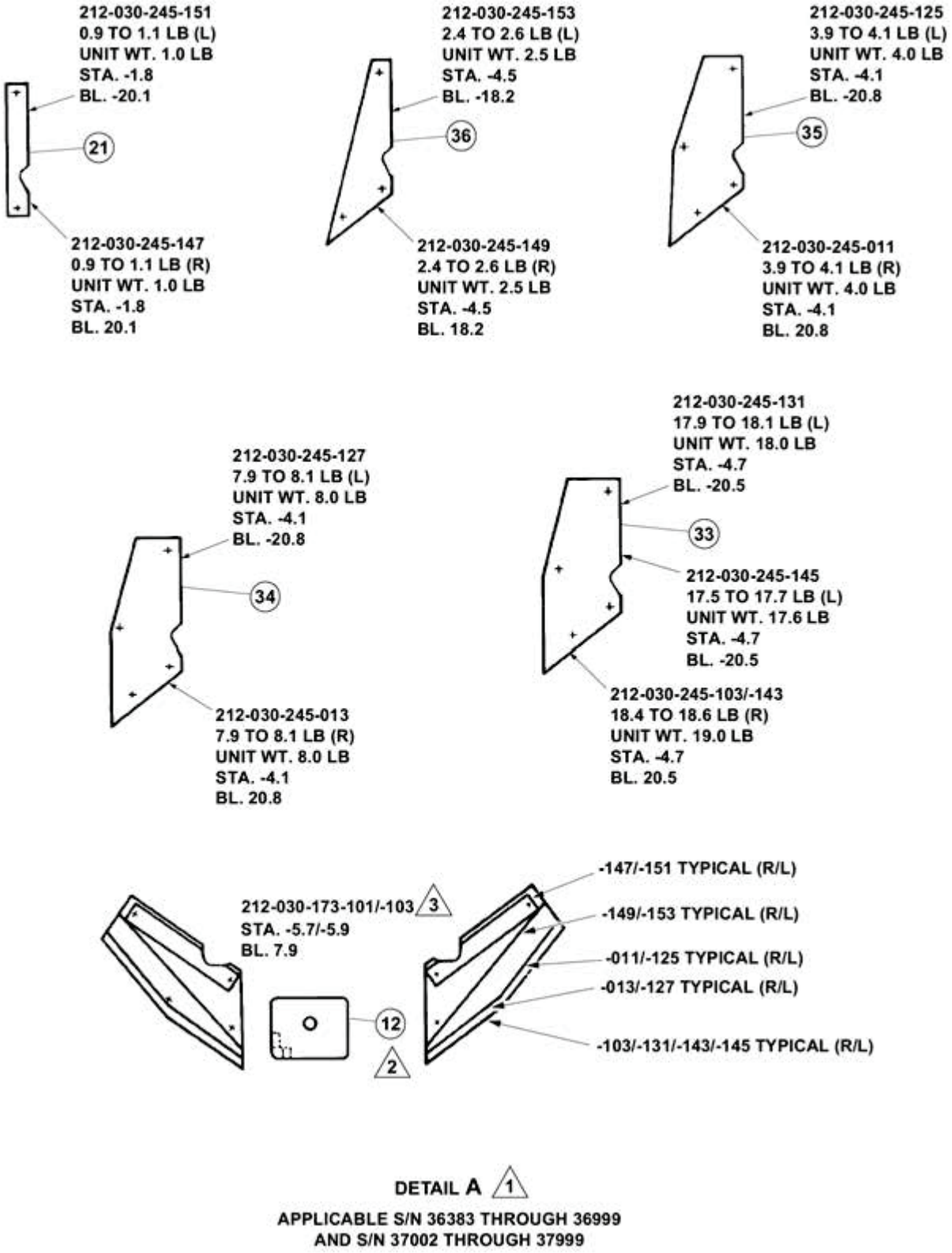
ICN-412-A-084000-B-97499-00004-A-001-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 2 of 5)



ICN-412-A-084000-B-97499-00005-A-002-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 3 of 5)



ICN-412-A-084000-B-97499-00006-A-002-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 4 of 5)

1. Bolt
2. Aluminum washer
3. Eyebolt
4. Pan
5. Bracket
6. Eyebolt
7. Bolt
8. Aluminum washer
9. Bracket
10. Aluminum washer
11. Nut
12. Ballast
13. Aluminum washer
14. Nut
15. Nut
16. Steel washer
17. Nut
18. Steel washer
19. Nut
20. Steel washer
21. Ballast
22. Nut
23. Steel washer
24. Steel washer
25. Bolt
26. Block
27. Steel washer
28. Bolt
29. Steel washer
30. Bolt
31. Steel washer
32. Bolt
33. Ballast
34. Ballast
35. Ballast
36. Ballast

NOTES

1. Maximum ballast is 82 pounds (37.2 kg) per side. Add ballast symmetrically.
2. Ballast (12) is not included in 164-pound (74.4-kg) maximum hose shelf ballast.
3. Actual weight is stamped on each 212-030-173 battery block ballast. Use 212-030-173-101 ballast at FS -5.7. When 53 Amp-hour Lead Acid battery kit (412-706-131) is installed, use the 212-030-173-103 battery block ballast at FS -5.9. Weigh battery and ballast underneath the battery before installation. The total of both parts weight must not exceed 155 lbs (70 kg). Make sure that the cutout in the ballast is facing down.

ICN-412-A-084000-B-97499-00007-A-001-01

Figure 1. Nose and Cabin Ballast (Weight) Installation (Sheet 5 of 5)

Tail Ballast (Weight) - Removal

Procedure

NOTE

If tail skid removal is required, refer to the 412-A-32-20-00-00A-520A-A / 00747.

1. Determine the amount of ballast to be removed (412-A-08-40-00-07A-169A-A / 00094).

2. To remove both light fairings (1, Figure 1) (if required), remove the screws (2) and disconnect the electrical connectors.

NOTE

Washer halves (7 and 11) are used only on the end of the ballast stack as a filler for the clamp halves (5 and 10).

3. Remove the tail skid ballast as follows:

3.1. Remove the nuts (9) and the bolts (4) from the clamp halves (5 and 10).

3.2. Remove the clamp halves (5 and 10) and the washers halves (7 and 11).

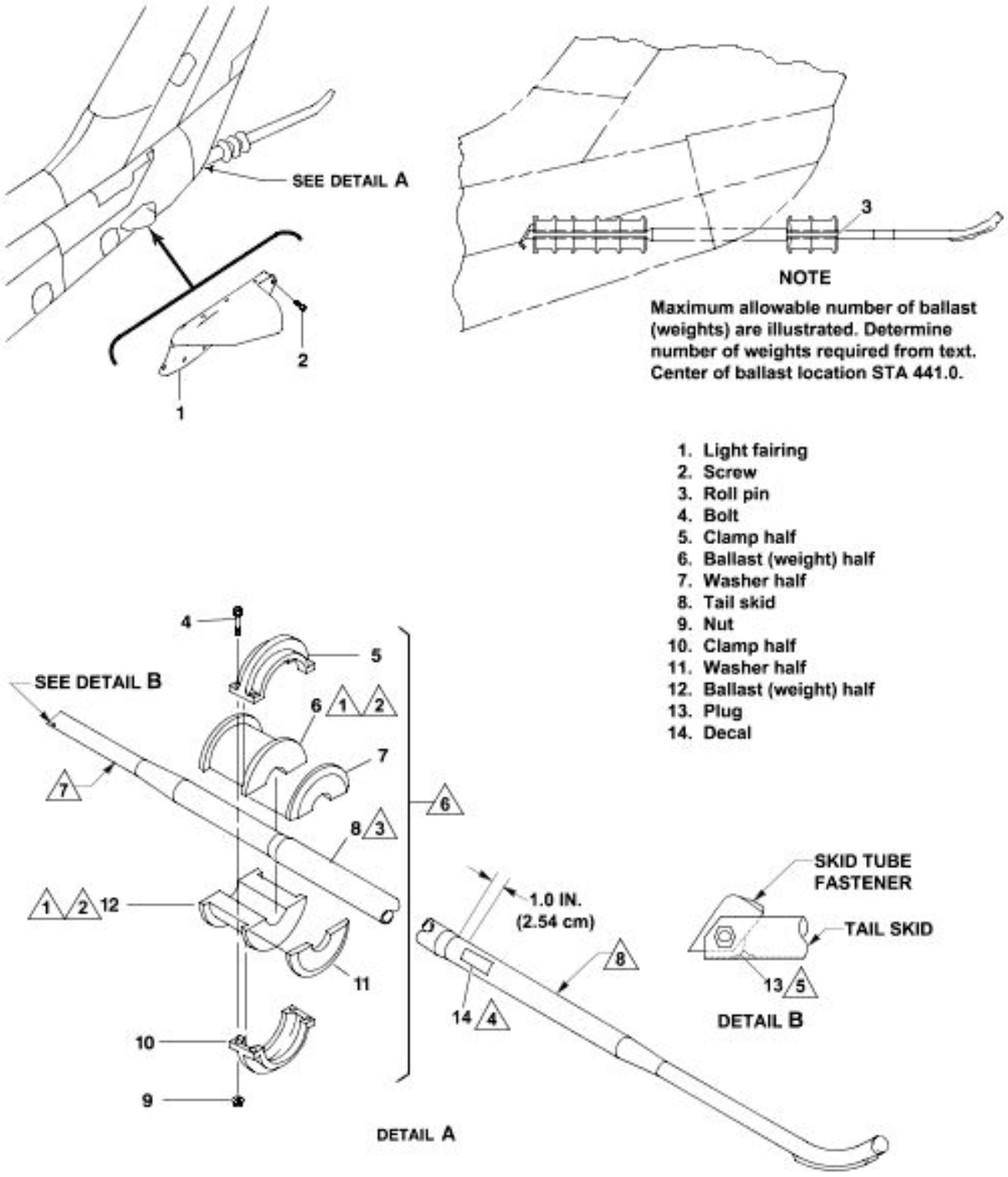
3.3. Remove the ballast halves (6 and 12).

3.4. Install the washer halves (7 and 11) and the clamp halves (5 and 10) with the bolts (4) and the nuts (9) on the end of the remaining ballast stack. Tighten the nuts.

3.5. Torque the nuts (9) 30 to 35 inch-pounds (3.4 to 4.0 Nm).

3.6. Gently tap the clamp halves (5 and 10) with a soft mallet and check the torque. Repeat until torque stabilizes.

4. Connect the electrical connectors and install the light fairings (1) with the screws (2), if removed. Tighten the screws.



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Figure 1. Tail Ballast (Weight) Installation (Sheet 1 of 2)

NOTES

1. Two ballast (weight) halves make a set.
2. Nominal weight of each ballast (weight) half is 6.3 pounds (2.86 kg). Actual weight is stamped on each ballast (weight) half.
3. Alternate ballast, No. 8 lead shot or equivalent installed inside tail skid.
4. "BALLAST INSIDE" decal installed when alternate ballast installed.
5. Install plug (13) with zinc chromate primer (C-201).
6. 204-071-566-001 ballast installation, typical three places.
7. 204-071-566-003 ballast.
8. 204-071-566-005 ballast.

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Figure 1. Tail Ballast (Weight) Installation (Sheet 2 of 2)

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Tail Ballast (Weight) - Installation

Procedure

NOTE

If tail skid removal is required, refer to the 412-A-32-20-00-00A-520A-A / 00747.

1. Determine the amount of ballast required (412-A-08-40-00-07A-169A-A / 00094). Ballast may be added to a maximum of 50 pounds (22.68 kg) on and in the tail skid.

2. To remove both light fairings (1, Figure 1) remove the screws (2) and disconnect the electrical connectors.

NOTE

Washer halves (7 and 11) are used only on the end of the ballast stack as a filler for the clamp halves (5 and 10).

3. Install the tail skid ballast as follows:

3.1. Put the ballast (weight) halves (6 and 12), washer halves (7 and 11), and clamp halves (5 and 10) in position on the tail skid (8).

3.2. Install the bolts (4) and the nuts (9).

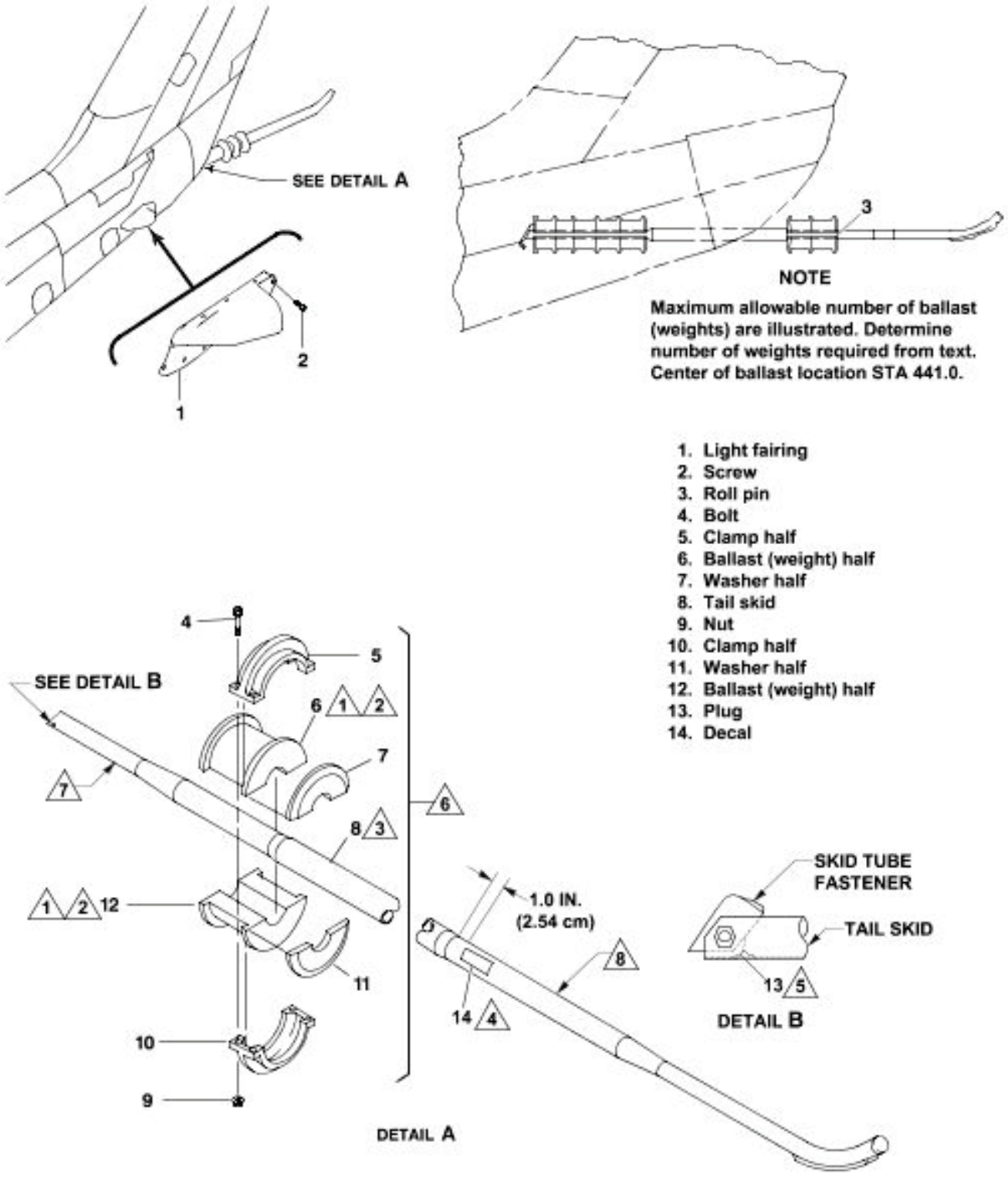
3.3. Torque the nuts (9) 30 to 35 inch-pounds (3.4 to 4.0 Nm).

3.4. Gently tap the clamp halves (5 and 10) with a soft mallet and check the torque. Repeat until torque stabilizes.

3.5. Install additional ballast as required in the same manner.

3.6. Inspect the roll pin (3) for secure installation.

4. Connect the electrical connectors and install the light fairings (1) with the screws (2).



ICN-412-A-084020-B-97499-00001-A-001-01

Figure 1. Tail Ballast (Weight) Installation (Sheet 1 of 2)

NOTES

1. Two ballast (weight) halves make a set.
2. Nominal weight of each ballast (weight) half is 6.3 pounds (2.86 kg). Actual weight is stamped on each ballast (weight) half.
3. Alternate ballast, No. 8 lead shot or equivalent installed inside tail skid.
4. "BALLAST INSIDE" decal installed when alternate ballast installed.
5. Install plug (13) with zinc chromate primer (C-201).
6. 204-071-566-001 ballast installation, typical three places.
7. 204-071-566-003 ballast.
8. 204-071-566-005 ballast.

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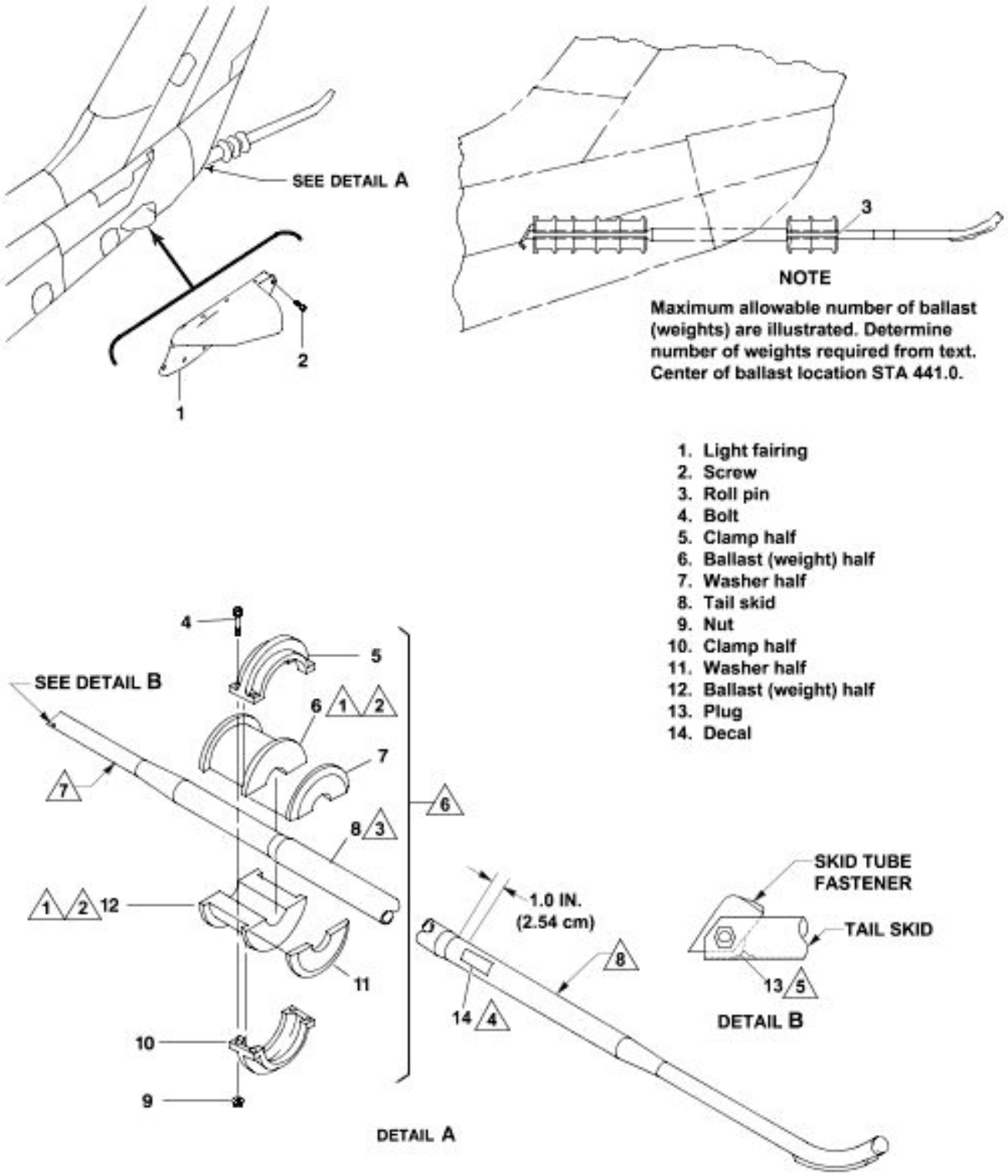
Figure 1. Tail Ballast (Weight) Installation (Sheet 2 of 2)

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Tail Ballast (Lead Shot) Alternate Method - Removal

Procedure

1. Remove the tail skid (412-A-32-20-00-00A-520A-A / 00747).
2. Remove all ballast inside the tail skid.
3. Use a heat gun to remove adhesive from around the plug (13, Figure 1) in the end of the tail skid (8).
4. Remove the plug (13).
5. Remove all the lead shot or equivalent ballast from inside the tail skid (8).
6. Remove the decal (14) when an alternate ballast is to be replaced by primary method ballast weights (6 and 12).



ICN-412-A-084020-B-97499-00001-A-001-01

Figure 1. Tail Ballast (Weight) Installation (Sheet 1 of 2)

NOTES

1. Two ballast (weight) halves make a set.
2. Nominal weight of each ballast (weight) half is 6.3 pounds (2.86 kg). Actual weight is stamped on each ballast (weight) half.
3. Alternate ballast, No. 8 lead shot or equivalent installed inside tail skid.
4. "BALLAST INSIDE" decal installed when alternate ballast installed.
5. Install plug (13) with zinc chromate primer (C-201).
6. 204-071-566-001 ballast installation, typical three places.
7. 204-071-566-003 ballast.
8. 204-071-566-005 ballast.

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Figure 1. Tail Ballast (Weight) Installation (Sheet 2 of 2)

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Tail Ballast (Lead Shot) Alternate Method - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Zinc chromate primer	C-201	AR	

Procedure

1. Determine ballast requirement (412-A-08-40-00-07A-169A-A / 00094).

NOTE

When ballast is installed inside tail skid, tail skid shall be filled completely full, approximately 15 pounds (6.8 kg).

NOTE

Do not pack lead shot into the tail skid.

2. Add required No. 8 lead shot, or equivalent ballast, inside the tail skid (8, Figure 1). Ballast may be added to a maximum of 50 pounds (22.68 kg) on and in the tail skid. With the lead shot ballast installed, a maximum of 35 pounds (15.9 kg) may be added on tail skid (Table 1).

3. Install the plug (13) into the end of the tail skid (8) with zinc chromate primer (C-201).

4. Make sure that the fastener hole in the plug (13) aligns with the hole in the tail skid (8).

5. Install the tail skid (8) (412-A-32-20-00-00A-720A-A / 00749).

6. Apply the decal (14) to the skid tube (8) (412-A-11-00-00-00A-067A-A / 00124).

Table 1. Ballast Requirements - Maximum Fixed Ballast Weight and Location

LOCATION	PART NUMBER	MAXIMUM		FUSELAGE STATION (FS)		BUTTOCK LINE (BL)	
		LB	KG	IN.	MM	IN.	MM
Battery Block (1)	212-030-173-101 (x1)	76.0	34.5	-5.7	-145	7.9	201
	212-030-173-103 (x1) (8)	52	23.6	-5.9	-150	7.9	201
SN: 33001-33213 or 36001-36036: Nose Shelf (2)	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-011 (x4)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-011 (x4)	16	7.3	-4.1	-104	20.8	528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	20.8	528
	212-030-245-015 (x1)	19	8.6	-4.7	-119	-20.5	-521
	212-030-245-015 (x1)	19	8.6	-4.7	-119	20.5	521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521

Table 1. Ballast Requirements - Maximum Fixed Ballast Weight and Location (continued)

LOCATION	PART NUMBER	MAXIMUM		FUSELAGE STATION (FS)		BUTTOCK LINE (BL)	
		LB	KG	IN.	MM	IN.	MM
SN: 36037-36382: Nose Shelf (2)	212-030-245-007 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-147 (x3) (3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-121 (x3) (5)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-151 (x3) (5)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-009 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-149 (x3) (4)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-123 (x3) (6)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-153 (x3) (6)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-011 (x4)	16	7.3	-4.1	-104	20.8	528
	212-030-245-125 (x4)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16	7.3	-4.1	-104	20.8	528
	212-030-245-127 (x2)	16	7.3	-4.1	-104	-20.8	-528
	212-030-245-143 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-145 (x1)	17.6	8	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-131 (x1)	18	8.2	-4.7	-119	-20.5	-521
SN: 36383-36999 or 37002-37999: Nose Shelf (2)	212-030-245-147 (x3)	3	1.4	-1.8	-46	20.1	511
	212-030-245-151 (x3)	3	1.4	-1.8	-46	-20.1	-511
	212-030-245-149 (x3)	7.5	3.4	-4.5	-114	18.2	462
	212-030-245-153 (x3)	7.5	3.4	-4.5	-114	-18.2	-462
	212-030-245-011 (x4)	16.0	7.3	-4.1	-104	20.8	528
	212-030-245-125 (x4)	16.0	7.3	-4.1	-104	-20.8	-528
	212-030-245-013 (x2)	16.0	7.3	-4.1	-104	20.8	528
	212-030-245-127 (x2)	16.0	7.3	-4.1	-104	-20.8	-528
	212-030-245-143 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-145 (x1)	17.6	8.0	-4.7	-119	-20.5	-521
	212-030-245-103 (x1)	18.5	8.4	-4.7	-119	20.5	521
	212-030-245-131 (x1)	18.0	8.2	-4.7	-119	-20.5	-521
Tailboom (7)	204-071-566-001 (4)	25.2	11.4	468.0	11,887	0	0
	204-071-566-001 (6)	37.8	17.1	441.0	11,201	0	0
	204-071-566-003 (4)	25.2	11.4	441.0	11,201	0	0
	No. 8 Lead Shot (7)	15.0	6.8	457.0	11,608	0	0
		Completely fill tube (do not pack)					

Table 1. Ballast Requirements - Maximum Fixed Ballast Weight and Location (continued)

LOCATION	PART NUMBER	MAXIMUM		FUSELAGE STATION (FS)		BUTTOCK LINE (BL)	
		LB	KG	IN.	MM	IN.	MM

NOTES:

1 Actual weight is stamped on 212-030-173 block. Battery block ballast not included in 164 pounds (74.4 kg) maximum weight.

2 Maximum nose shelf ballast is 82 pounds (37.2 kg) per side. Add ballast symmetrically.

3 212-030-245-147 replaces -007 for future procurement.

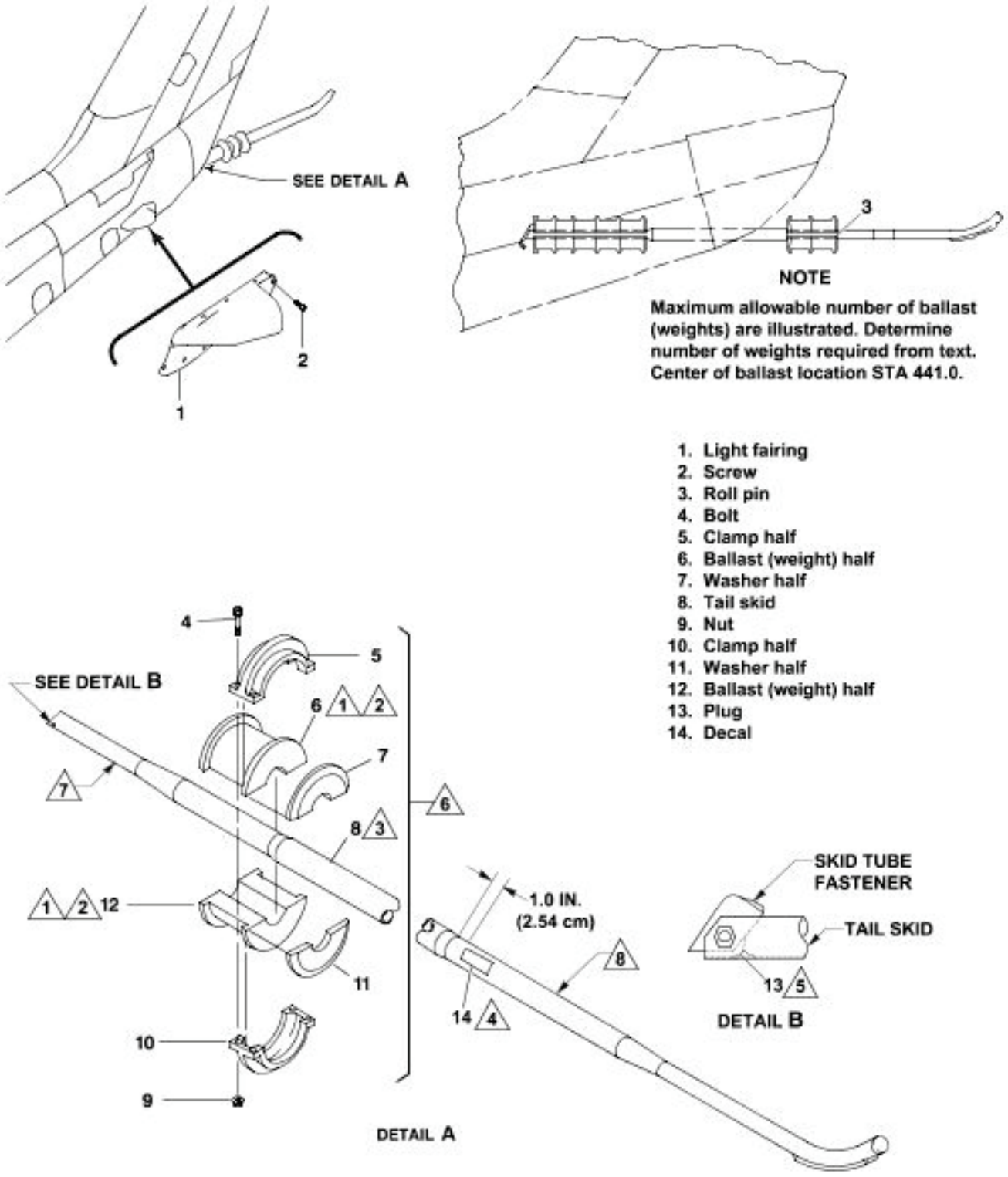
4 212-030-245-149 replaces -009 for future procurement.

5 212-030-245-151 replaces -121 for future procurement.

6 212-030-245-153 replaces -123 for future procurement.

7 Maximum tailboom ballast is 50 pounds (22.7 kg).

8 Use the 212-030-173-103 battery block ballast when 53 Amp-hour Lead Acid Battery Kit (412-706-131) is installed. Weigh battery and ballast underneath the battery before installation. The total weight of both parts must not exceed 155 pounds (70 kg).



ICN-412-A-084020-B-97499-00001-A-001-01

Figure 1. Tail Ballast (Weight) Installation (Sheet 1 of 2)

NOTES

1. Two ballast (weight) halves make a set.
2. Nominal weight of each ballast (weight) half is 6.3 pounds (2.86 kg). Actual weight is stamped on each ballast (weight) half.
3. Alternate ballast, No. 8 lead shot or equivalent installed inside tail skid.
4. "BALLAST INSIDE" decal installed when alternate ballast installed.
5. Install plug (13) with zinc chromate primer (C-201).
6. 204-071-566-001 ballast installation, typical three places.
7. 204-071-566-003 ballast.
8. 204-071-566-005 ballast.

ICN-412-A-084020-B-97499-00002-A-001-01

Figure 1. Tail Ballast (Weight) Installation (Sheet 2 of 2)

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CHAPTER 09
TOWING

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Towing - Description	DMC-412-A-09-00-00-00A-028A-A / 00103	2023-05-31	ALL
Towing - Preparation	DMC-412-A-09-00-00-00A-121A-A / 00104	2023-05-31	ALL
Ground Handling Wheels - Installation	DMC-412-A-09-00-00-00A-720A-A / 00105	2023-05-31	ALL
Helicopter - Towing	DMC-412-A-09-00-00-00A-174A-A / 00106	2023-05-31	ALL
Ground Handling Wheels - Removal	DMC-412-A-09-00-00-00A-520A-A / 00107	2023-05-31	ALL

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Towing - Description

NOTE

When towing the helicopter over the weights indicated in Table 1 Gross Weight Requiring Strapping, refer to BHT-412-SI-58.

NOTE

Puller assembly contained in BHT-412-SI-58 is equipped with a load limiter similar to a break click type torque wrench. When using this puller assembly and especially when using alternate methods of strapping, review the draw-up limits of Table 2 Aft Crosstube Pretension Values.

Table 1. Gross Weight Requiring Strapping

GEAR TYPE	AFT CROSSTUBE	TUBE DIA X WALL (INCHES)	STRAP AT: GROSS WEIGHT (POUNDS)
STD	BHT crosstubes and subsequent approved replacements	3.5 X 0.625	9100
HIGH	BHT crosstubes and subsequent approved replacements	3.5 X 0.625	8900

Table 2. Aft Crosstube Pretension Values

GEAR TYPE	AFT CROSSTUBE	DIA X WALL (INCHES)	COLUMN 1	COLUMN 2
			DRAW-UP (INCHES) (1)	LIMITING DRAW-UP
STD	BHT crosstubes and subsequent approved replacements	3.5 X 0.625	3.5	102 inches between skid tube
HIGH	BHT crosstubes and subsequent approved replacements	3.5 X 0.625	6.5	112.5 inches between skid tube

NOTES:

1 Do not draw up beyond limits in Column 2.

2 Based on a 3/8” draw-up cable.

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Towing - Preparation

Preliminary requirements

Support equipment

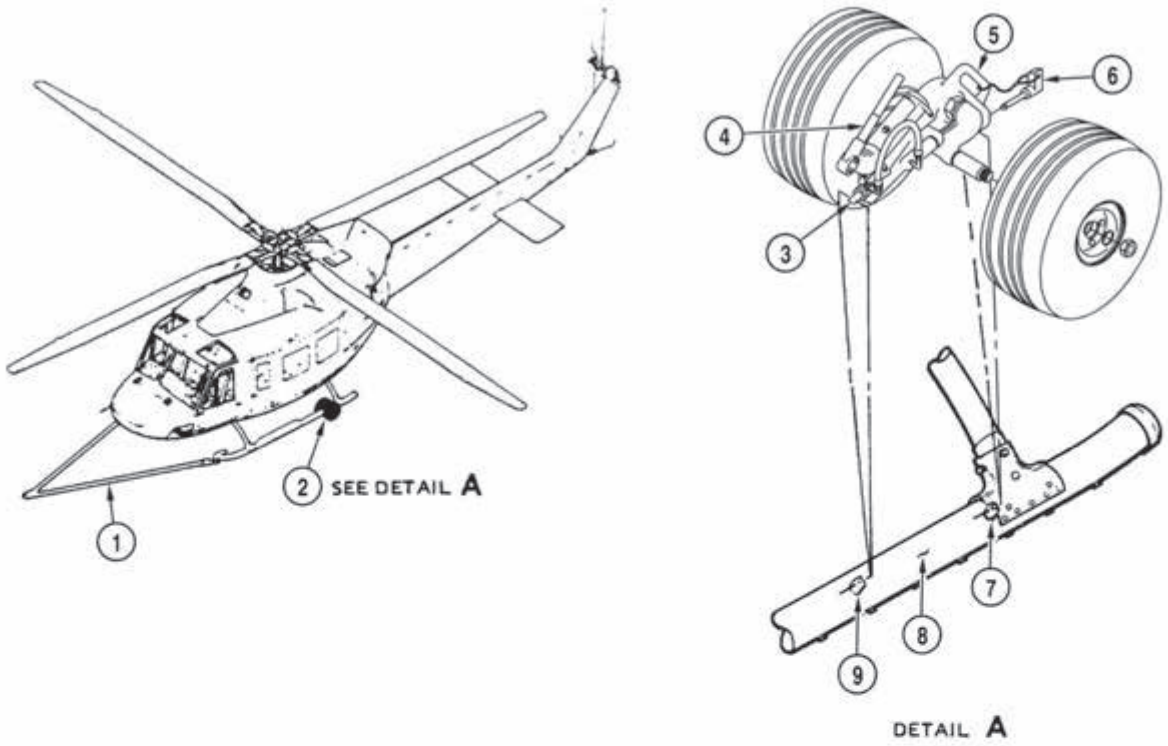
Nomenclature	Identification No.	Qty	Remarks
Tow Bar	T101808-101	1	

Procedure

1. Refer to 412-A-52-40-00-02A-720A-A / 00840 for installation requirements for access doors, panels, fairings, and covers for towing.

2. Tie down main rotor blades (412-A-10-00-00-00A-042A-A / 00110).

3. Attach tow bar (1, Figure 1) to skid gear tow ring and tractor pintle hook. Install safety pin in pintle hook.



1. Tow bar
2. Ground handling wheels
3. Fixed pin
4. Handle
5. Cradle assembly
6. Spring-loaded pin
7. Eyebolt
8. Skid tube
9. Eyebolt

ICN-412-A-090000-B-97499-00001-A-001-01

Figure 1. Tow Bar and Ground Handling Wheels - Installation (Sheet 1 of 1)

Ground Handling Wheels - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Ground Handling Wheels	214-706-104-107	1	or Equivalent

Procedure

NOTE

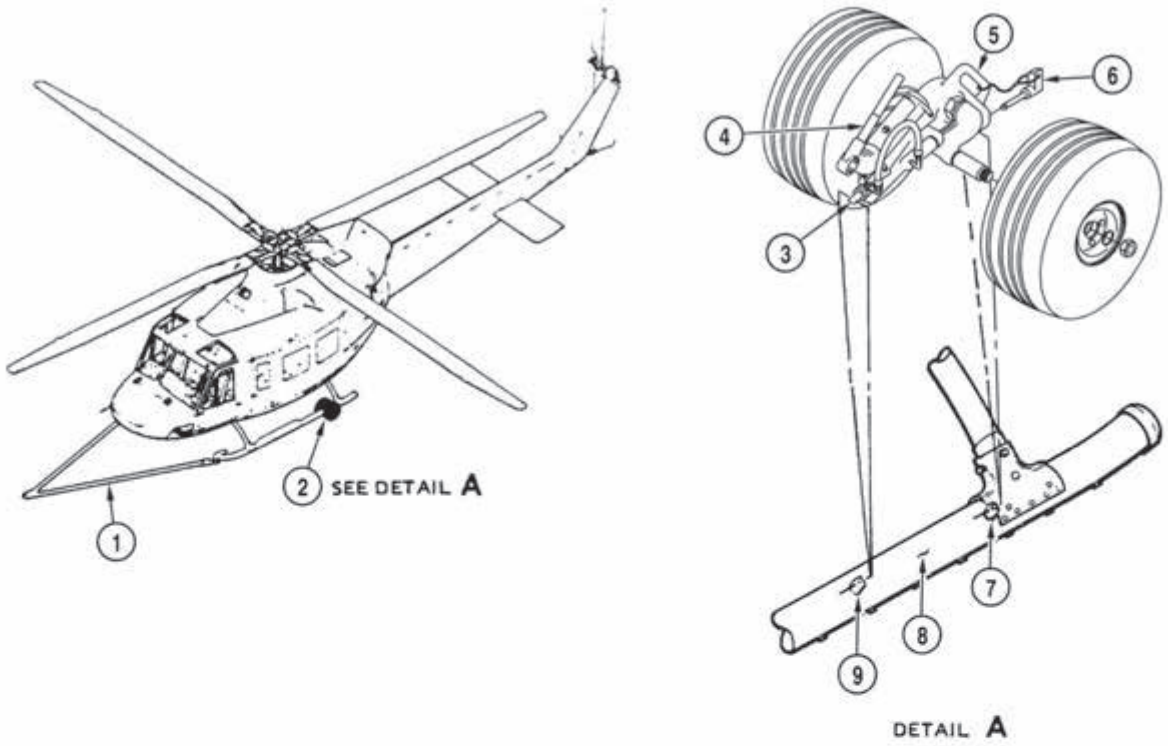
ground handling wheels (214-706-104-107) (2, Figure 1) can be installed with fixed pin (3) facing either forward or aft. Also, ground handling wheels (214-706-104-107) can be installed with fixed pin facing forward on one side of helicopter and fixed pin facing aft on opposite side of helicopter. Helicopter loading (center of gravity) dictates position in which ground handling wheels (214-706-104-107) should be installed. When ground handling wheels (214-706-104-107) are lowered, they should be on or near helicopter center of gravity. If helicopter center of gravity is aft, install both ground handling wheels (214-706-104-107) with fixed pins facing forward.

1. Position ground handling wheels (214-706-104-107) (2, Figure 1) over skid tube (8).
2. Insert fixed pin (3) in eyebolt (9).
3. Align cradle assembly (5) with eyebolt (7) and fully insert spring-loaded pin (6) through cradle assembly and eyebolt to install ground handling wheels (214-706-104-107) (2).
4. Install ground handling wheels (214-706-104-107) (2) on opposite skid tube (8) in the same manner.



DO NOT LEAVE HELICOPTER UNATTENDED WITH ground handling wheels (214-706-104-107) IN EXTENDED POSITION.

5. Station a person at tail skid to maintain helicopter in level position.
6. Operate handle (4) to extend ground handling wheels (214-706-104-107) (2) and raise skid gear clear of ground.
7. Disconnect ground wire.



1. Tow bar
2. Ground handling wheels
3. Fixed pin
4. Handle
5. Cradle assembly
6. Spring-loaded pin
7. Eyebolt
8. Skid tube
9. Eyebolt

ICN-412-A-090000-B-97499-00001-A-001-01

Figure 1. Tow Bar and Ground Handling Wheels - Installation (Sheet 1 of 1)

Helicopter - Towing

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Droop Restraint Tube Assembly	T101261-101	1	

Procedure



TOWING HELICOPTER WITH BLADES FOLDED SHALL BE ACCOMPLISHED WITH MAIN ROTOR droop restraint tube assembly (T101261-101) INSTALLED OR WITH BELL HELICOPTER APPROVED EQUIVALENT.



DO NOT START ENGINE POWER SECTION(S) WITH CROSSTUBES STRAPPED.

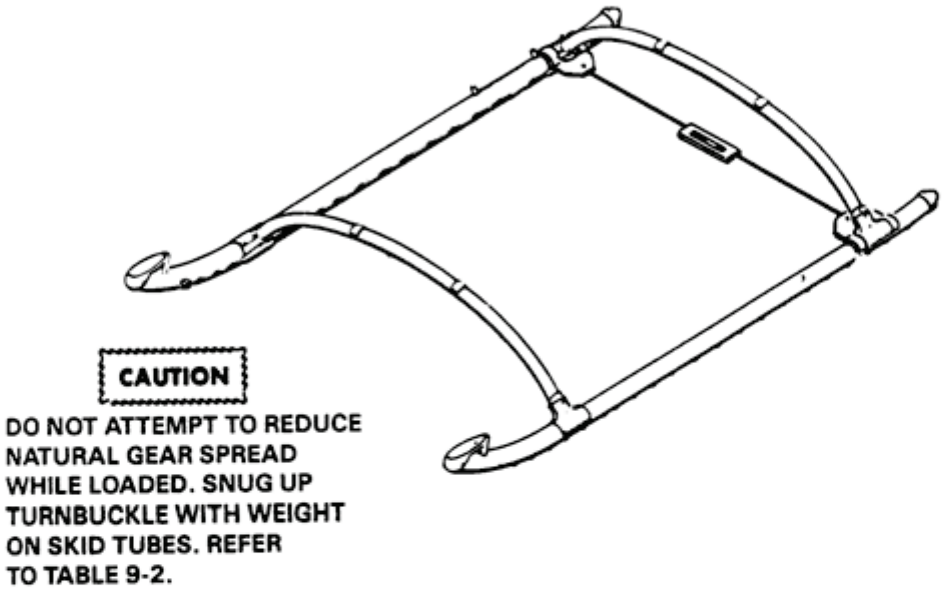


TOWING HELICOPTER ON UNPREPARED SURFACES OR ACROSS HANGAR TRACKS, ETC., EXCEEDING GROSS WEIGHTS SHOWN IN Table 1 MAY CAUSE A PERMANENT SET IN AFT CROSSTUBE. IN AN EMERGENCY, HELICOPTER MAY BE TOWED UP TO MAXIMUM GROSS WEIGHT OF 9500 POUNDS BY STRAPPING CROSSTUBES TOGETHER. REFER TO Figure 1.

1. Station a person at tail skid to maintain helicopter in level position during towing operations.
2. Clear towing area of auxiliary support equipment (work stands, power units, etc).
3. Tow or push slowly, balancing helicopter with tail skid.

Table 1. Gross Weight Requiring Strapping

GEAR TYPE	AFT CROSSTUBE	TUBE DIA X WALL (INCHES)	STRAP AT: GROSS WEIGHT (POUNDS)
STD	BHT crosstubes and subsequent approved replacements	3.5 X 0.625	9100
HIGH	BHT crosstubes and subsequent approved replacements	3.5 X 0.625	8900



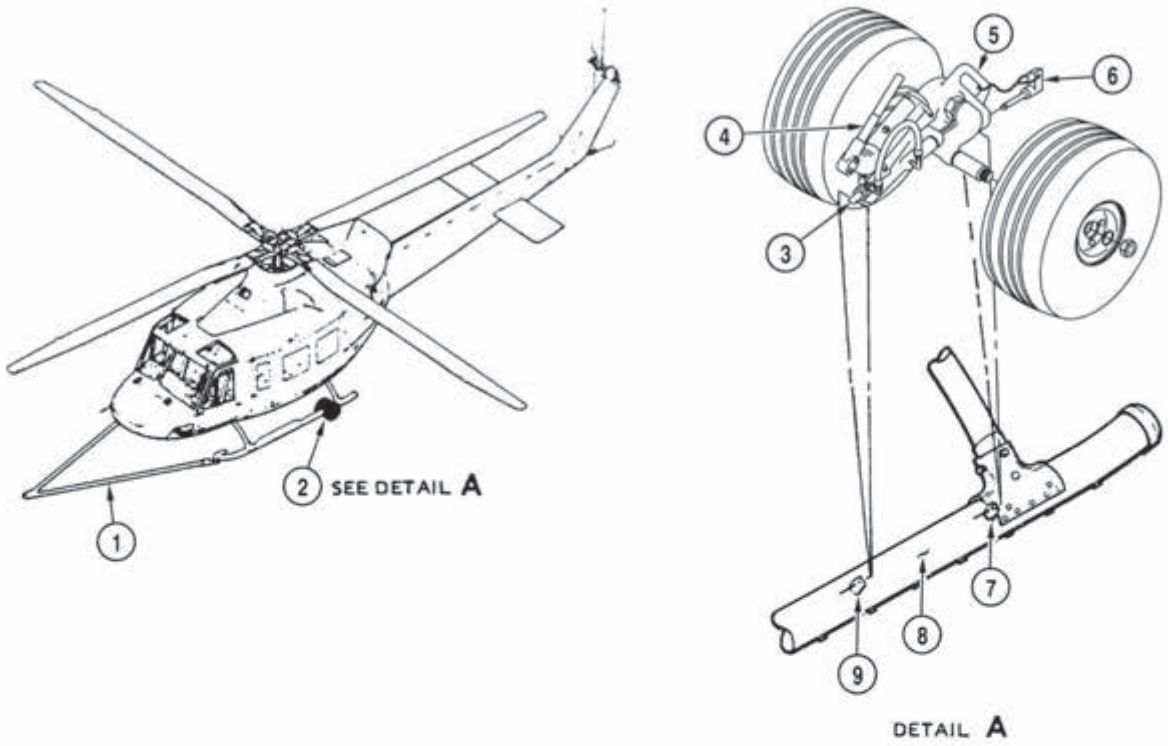
ICN-412-A-090000-B-97499-00002-A-001-01

Figure 1. Crosstube Strap - Installation (Sheet 1 of 1)

Ground Handling Wheels - Removal

Procedure

1. Attach ground wire to helicopter.
2. Retract ground handling wheels (2, Figure 1) individually to lower helicopter to skid tube (8) by turning pump valve "T" handle.
3. Pull spring-loaded pin (6) to release cradle assembly (5) from skid tube (8).
4. Move ground handling wheels (2) aft (or forward as applicable) to disengage fixed pin (3) from eyebolt. Remove ground handling wheels from skid tube (8).



1. Tow bar
2. Ground handling wheels
3. Fixed pin
4. Handle
5. Cradle assembly
6. Spring-loaded pin
7. Eyebolt
8. Skid tube
9. Eyebolt

ICN-412-A-090000-B-97499-00001-A-001-01

Figure 1. Tow Bar and Ground Handling Wheels - Installation (Sheet 1 of 1)

CHAPTER 10
PARKING AND MOORING

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Parking and Mooring - Conditions	DMC-412-A-10-00-00-00A-170A-A / 00109	2023-05-31	ALL
Main Rotor Blade Tie-down Provisions - Description	DMC-412-A-10-00-00-00A-042A-A / 00110	2023-05-31	ALL
Main Rotor Non-folded Blade Tie-down - Description	DMC-412-A-10-00-00-01A-042A-A / 00111	2023-05-31	ALL
Main Rotor Non-folded Blade Tie-down - Removal	DMC-412-A-10-00-00-01A-520A-A / 00112	2023-05-31	ALL
Main Rotor Non-folded Blade Tie-down - Inspection	DMC-412-A-10-00-00-01A-280A-A / 00113	2023-05-31	ALL
Main Rotor Non-folded Blade Tie-down - Installation	DMC-412-A-10-00-00-01A-720A-A / 00114	2023-05-31	ALL
Folding/Unfolding (Using Droop Restraint Tube Assembly) - Description	DMC-412-A-10-00-00-02A-042A-A / 00115	2023-05-31	ALL
Folding/Unfolding (Using Droop Restraint Tube Assembly) - Folding	DMC-412-A-10-00-00-02A-272A-A / 00116	2023-05-31	ALL
Folding/Unfolding (Using Droop Restraint Tube Assembly) - Unfolding	DMC-412-A-10-00-00-02A-272B-A / 00117	2023-05-31	ALL
Folding/Unfolding (Using Blade Folding Racks B67-43001-02) - Description	DMC-412-A-10-00-00-03A-042A-A / 00118	2023-05-31	ALL
Folding/Unfolding (Using Blade Folding Racks B67-43001-02) - Folding	DMC-412-A-10-00-00-03A-272A-A / 00119	2023-05-31	ALL
Folding/Unfolding (Using Blade Folding Racks B67-43001-02) - Unfolding	DMC-412-A-10-00-00-03A-272B-A / 00120	2023-05-31	ALL
Helicopter - Storage	DMC-412-A-10-30-00-00A-810A-A / 00121	2023-05-31	ALL

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Parking and Mooring - Conditions

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Strap Assembly	204-070-450-019	1	
Cover Assembly	204-070-470-001	1	
Cover Assembly	212-070-111-001	1	
Shield Assembly	212-070-112-001	1	
Shield Assembly	212-070-112-002	1	
Sock Assembly	412-782-113-103	1	
Wand Assembly	412-782-118-101	1	

Procedure

1. Parking and Mooring - Normal Conditions

NOTE

Use this procedure when forecast wind velocity is less than 20 knots.

1.1. Park helicopter on a level surface.

1.2. Retract or remove ground handling wheels (6, Figure 1) (412-A-09-00-00-00A-520A-A / 00107) to allow helicopter to rest on skid landing gear.

1.3. Install engine ejector cover assemblies (2).

1.4. Install engine inlet shield assemblies (4).

1.5. Install pitot tube cover assembly (9).

1.6. Install main rotor blade tie-down (412-A-10-00-00-01A-720A-A / 00114).



DO NOT EXCEED LOAD AND/OR DEFLECTION LIMITS DURING TIE-DOWN PROCEDURES FOR TAIL ROTOR; MAXIMUM LOAD ALLOWABLE AT BLADE TIP IS 50 POUNDS.

1.7. Install tail rotor tie-down strap (3) to tail rotor blade and footman's loop (not shown) located on the right side of the tailboom fin.

1.8. Disconnect battery.

1.9. Tighten friction locks on flight controls.

1.10. Close all doors and windows.

2. Parking and Mooring - Turbulent Conditions

NOTE

Use this procedure when winds are 20 to 45 knots.

2.1. Park helicopter on a level surface with helicopter headed into direction of highest forecast winds.

2.2. Accomplish Step 1.2. through Step 1.10..

3. Parking and Mooring - High Wind Conditions

NOTE

Use this procedure when forecast wind velocity is 45 to 65 knots. If forecast wind velocity is in excess of 65 knots, evacuate helicopter to a safe area if possible.

3.1. If a paved ramp with suitably spaced tie-down rings is available, park helicopter on skid landing gear headed in direction from which highest forecast winds are expected. Secure mooring shackles (4 and 5, Figure 2) to ramp tie-down rings with rope, cable, or manufactured tie-downs. If rope diameter is too large for mooring shackles, attach larger clevis to each mooring shackle.

3.2. If suitably-spaced ramp tie-downs are not available, park helicopter on an unpaved parking area headed in direction from which highest forecast winds are expected and retract ground handling wheels. Use mooring anchor or make “dead man” anchors. Moor helicopters as described in Step 3.1..

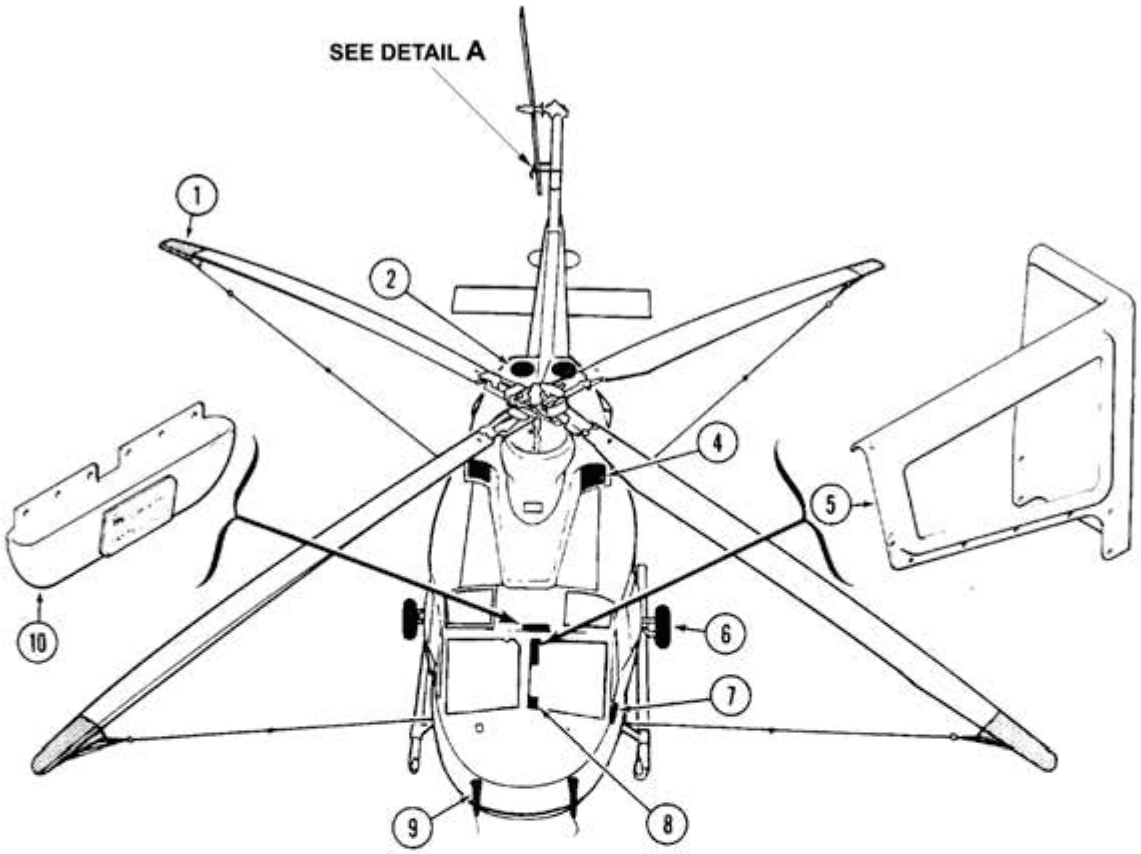
3.3. Fill fuel tank with prescribed fuel (412-A-12-10-01-00A-210A-A / 00130).

3.4. If storage space and time is available, remove main rotor blades and store. If blades cannot be removed, fold main rotor blades. Install main rotor droop restraint tube assembly (412-A-10-00-00-02A-272B-A / 00117). If main rotor blades cannot be removed or folded, install main rotor blade tie-down (412-A-10-00-00-01A-720A-A / 00114).

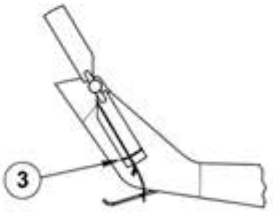
3.5. Accomplish Step 1.3. through Step 1.5. and Step 1.7. through Step 1.10..

3.6. Secure all ground handling equipment and other objects which might be blown by wind.

3.7. After winds subside, inspect helicopter carefully for damage which may have been inflicted by flying objects.



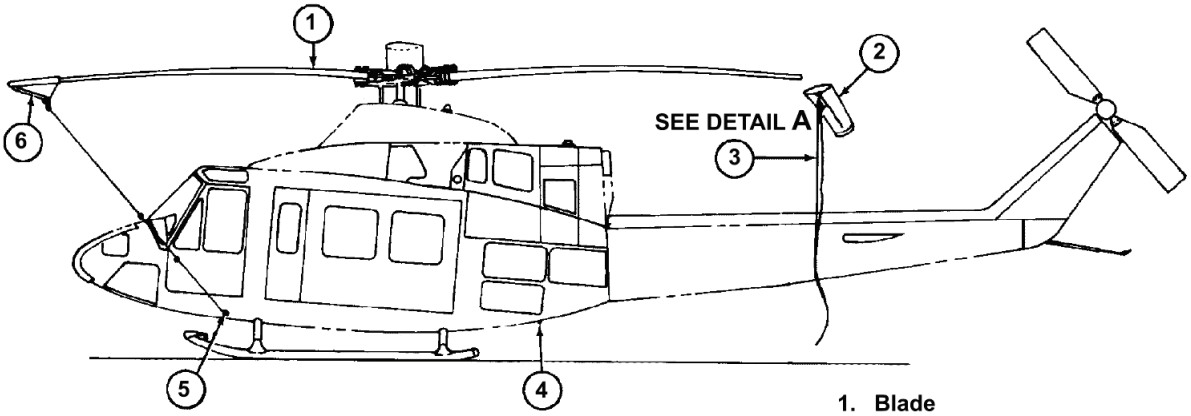
1. Sock assemblies
2. Engine ejector cover assemblies
3. Tail rotor tie-down strap
4. Engine inlet shield assemblies
5. Pilot console partition assembly
6. Ground handling wheels
7. Collective control cover
8. First aid kit
9. Pitot tube cover assembly
10. Overhead console guard assembly



DETAIL A

ICN-412-A-100000-B-97499-00001-A-001-01

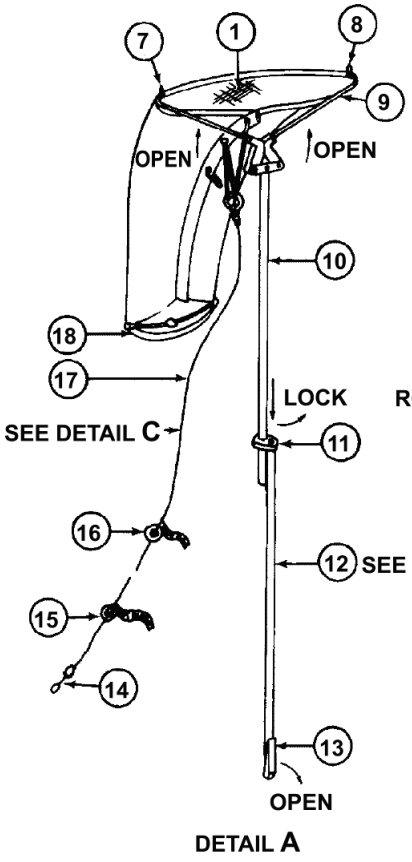
Figure 1. Loose Equipment (Sheet 1 of 1)



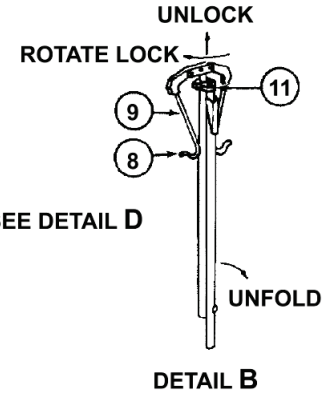
NOTE

The main rotor nonfolding blade tie-down consists of four tie-down sock assemblies (2) P/N 412-782-113-103 and one wand assembly (3) P/N 412-782-118-101.

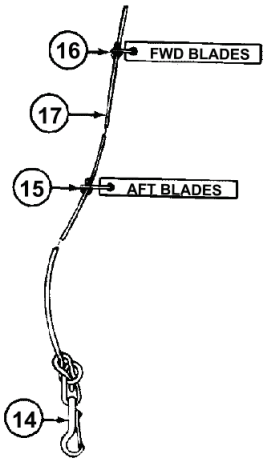
1. Blade
2. Sock assembly (not installed)
3. Wand assembly
4. Aft mooring shackle
5. Forward mooring shackle
6. Sock assembly (installed)
7. Grommet
8. Spreader arm guide
9. Spreader arm
10. Upper tube
11. Lock
12. Lower tube
13. Sock removal hook
14. Snap hook
15. Tie-down ring and AFT BLADE flag
16. Tie-down ring and FWD BLADE flag
17. Tie-down line
18. Sock removal loop



DETAIL A



DETAIL B



DETAIL C

ICN-412-A-100000-B-97499-00002-A-001-01

Figure 2. Main Rotor Non-folded Blade Tie-down (Sheet 1 of 1)

Main Rotor Blade Tie-down Provisions - Description

Main rotor blades may be tied down in non-folded position or in folded position.

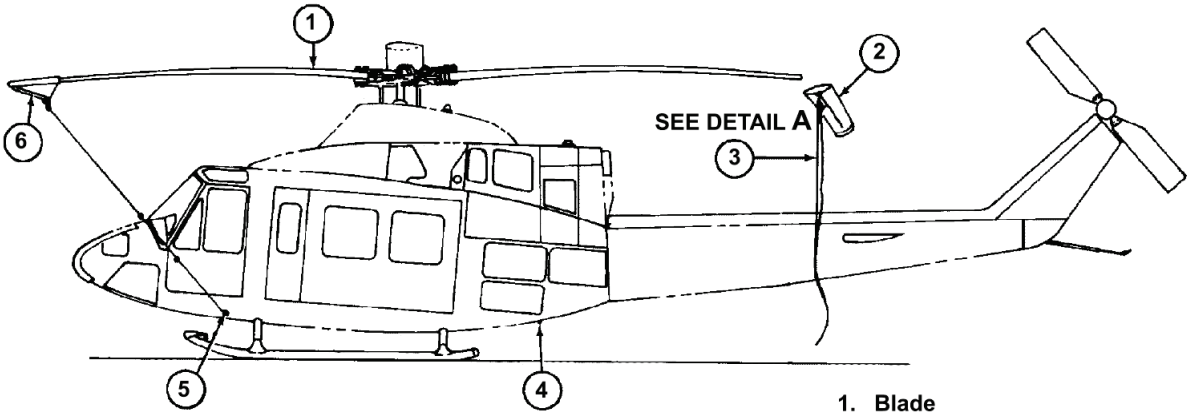
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Main Rotor Non-folded Blade Tie-down - Description

The main rotor tie-down kit for non-folded blades consists of four sock assemblies (2, Figure 1) and one wand assembly (3). The nylon mesh material sock assemblies include a sock removal loop (18) and tie-down line (17). Each tie-down line includes a tie-down ring and AFT BLADE flag (15), a tie-down ring and FWD BLADE flag (16), and snap hook (14).

The wand assembly (3) is used to install the socks on rotor blade tips and to remove socks from blade tips.

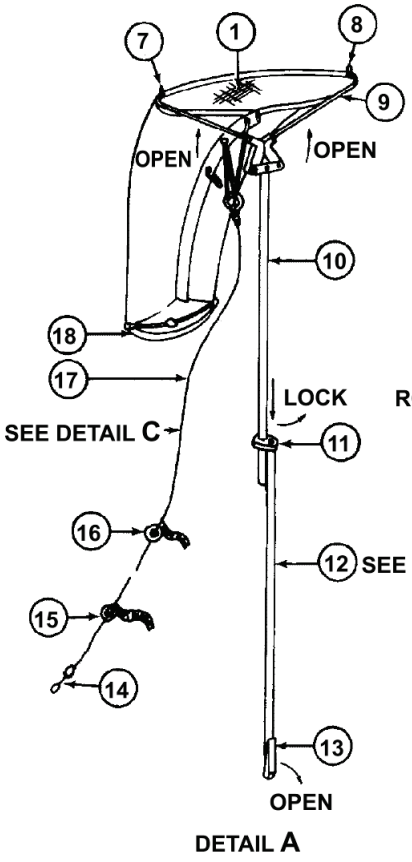
The tie-down rings and flags (15 and 16) may be positioned on tie-down line (17) so proper tension will be applied when sock assembly (6) is installed without tying knots in line each time a sock assembly is installed.



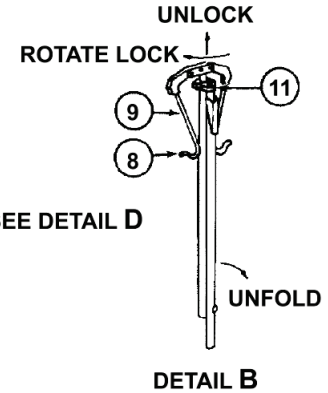
NOTE

The main rotor nonfolding blade tie-down consists of four tie-down sock assemblies (2) P/N 412-782-113-103 and one wand assembly (3) P/N 412-782-118-101.

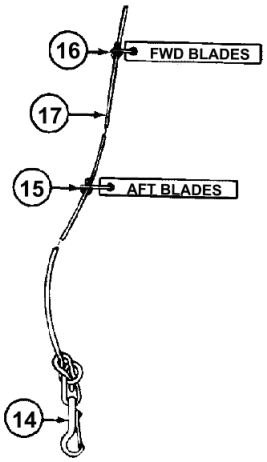
1. Blade
2. Sock assembly (not installed)
3. Wand assembly
4. Aft mooring shackle
5. Forward mooring shackle
6. Sock assembly (installed)
7. Grommet
8. Spreader arm guide
9. Spreader arm
10. Upper tube
11. Lock
12. Lower tube
13. Sock removal hook
14. Snap hook
15. Tie-down ring and AFT BLADE flag
16. Tie-down ring and FWD BLADE flag
17. Tie-down line
18. Sock removal loop



DETAIL A



DETAIL B



DETAIL C

ICN-412-A-100000-B-97499-00002-A-001-01

Figure 1. Main Rotor Non-folded Blade Tie-down (Sheet 1 of 1)

Main Rotor Non-folded Blade Tie-down - Removal

Procedure

1. Unfold wand assembly (3, Detail A, Figure 1).

2. Open sock removal hook (13).

3. Remove tie-down line from forward mooring shackle (5).

4. Remove sock (6) from blade (1).

5. Remove other three socks in same manner.

6. Stow wand assembly (3) as follows:

6.1. Unlock lock (11) and slide to top of upper tube (10).

6.2. Rotate lock (11) clockwise 90°.

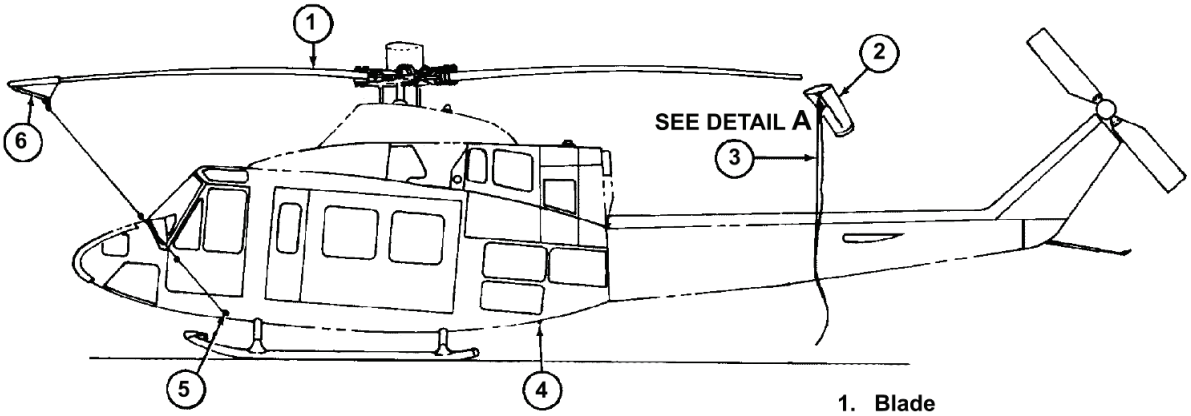
6.3. Fold lower tube (12) 180° parallel with upper tube (10) (Detail B).

6.4. Rotate lock (11) counterclockwise 90°.

6.5. Slide lock (11) down and engage end of lower tube (12).

6.6. Close spreader arms (9).

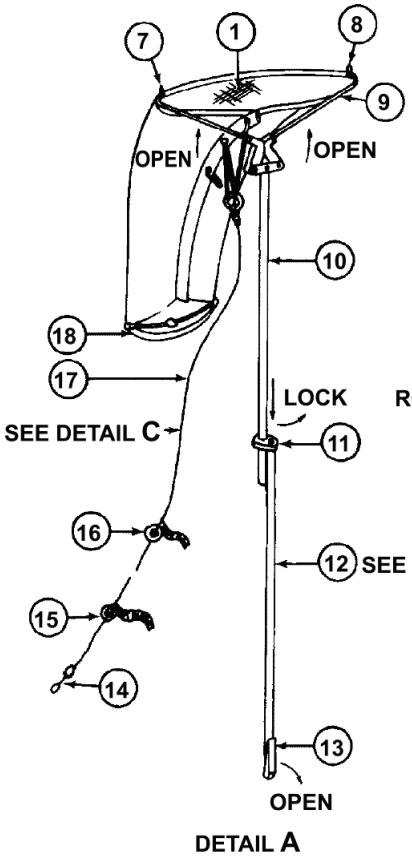
6.7. Close sock removal hook (13).



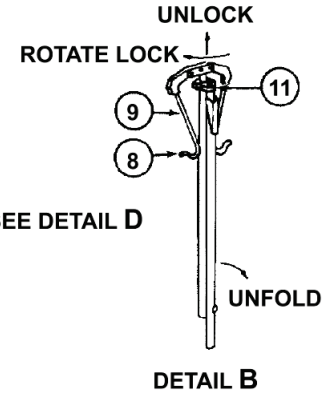
NOTE

The main rotor nonfolding blade tie-down consists of four tie-down sock assemblies (2) P/N 412-782-113-103 and one wand assembly (3) P/N 412-782-118-101.

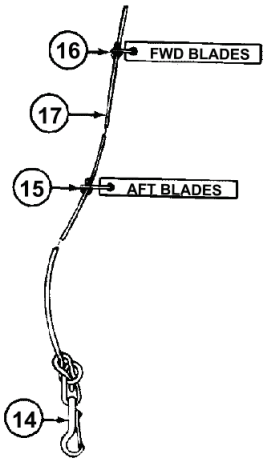
1. Blade
2. Sock assembly (not installed)
3. Wand assembly
4. Aft mooring shackle
5. Forward mooring shackle
6. Sock assembly (installed)
7. Grommet
8. Spreader arm guide
9. Spreader arm
10. Upper tube
11. Lock
12. Lower tube
13. Sock removal hook
14. Snap hook
15. Tie-down ring and AFT BLADE flag
16. Tie-down ring and FWD BLADE flag
17. Tie-down line
18. Sock removal loop



DETAIL A



DETAIL B



DETAIL C

ICN-412-A-100000-B-97499-00002-A-001-01

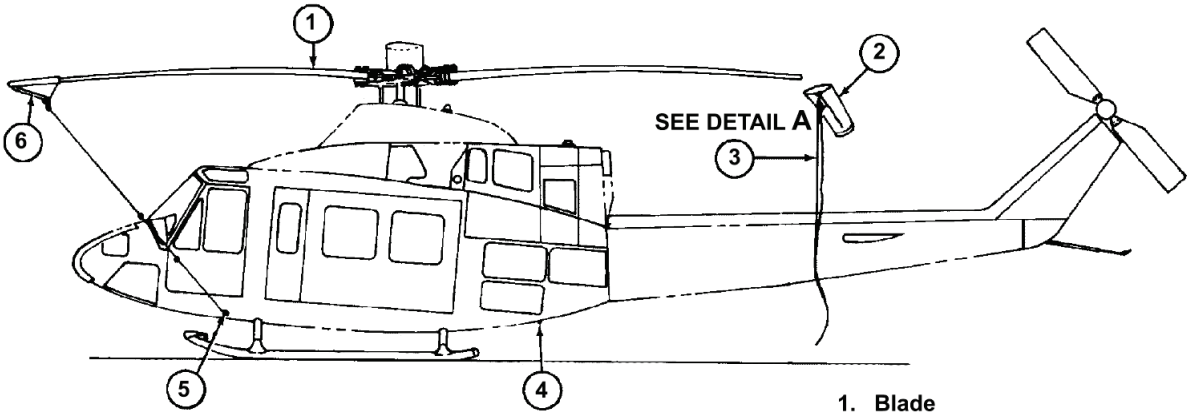
Figure 1. Main Rotor Non-folded Blade Tie-down (Sheet 1 of 1)

Main Rotor Non-folded Blade Tie-down - Inspection

Procedure

1. Inspect four sock assemblies (2, Figure 1) for deterioration of fabric, deteriorated tie-down line (17), broken or missing snap hooks (14), and deteriorated or missing tie-down rings and flags (15 and 16).

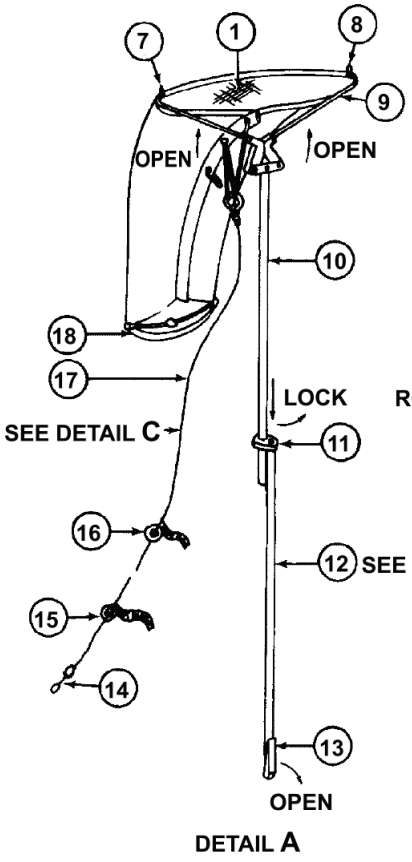
2. Inspect wand assembly (3) for distortion and for proper operation of spreader arms (9), lock (11), and sock removal hook (13).



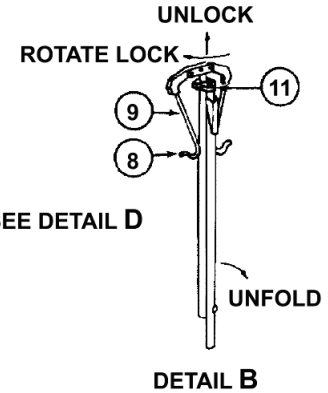
NOTE

The main rotor nonfolding blade tie-down consists of four tie-down sock assemblies (2) P/N 412-782-113-103 and one wand assembly (3) P/N 412-782-118-101.

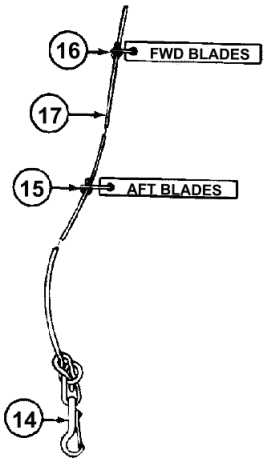
1. Blade
2. Sock assembly (not installed)
3. Wand assembly
4. Aft mooring shackle
5. Forward mooring shackle
6. Sock assembly (installed)
7. Grommet
8. Spreader arm guide
9. Spreader arm
10. Upper tube
11. Lock
12. Lower tube
13. Sock removal hook
14. Snap hook
15. Tie-down ring and AFT BLADE flag
16. Tie-down ring and FWD BLADE flag
17. Tie-down line
18. Sock removal loop



DETAIL A



DETAIL B



DETAIL C

ICN-412-A-100000-B-97499-00002-A-001-01

Figure 1. Main Rotor Non-folded Blade Tie-down (Sheet 1 of 1)

Main Rotor Non-folded Blade Tie-down - Installation

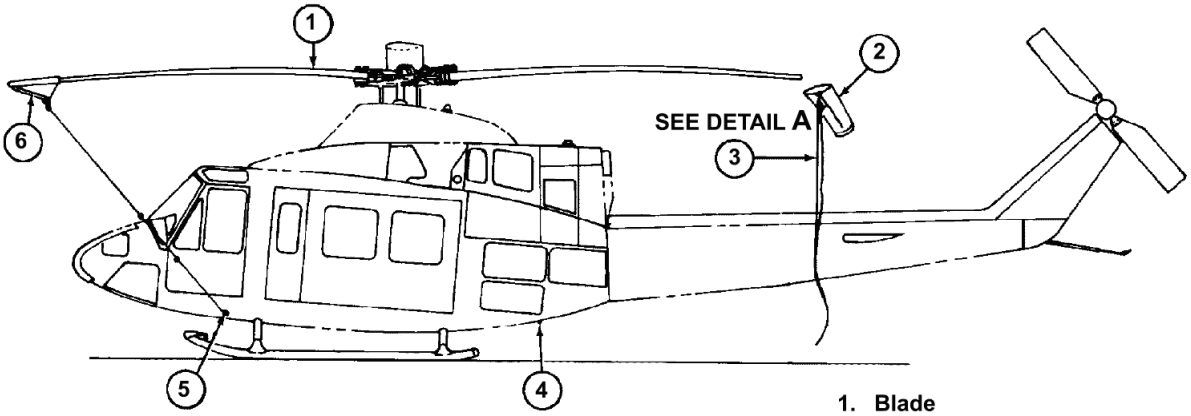
Procedure

1. Position blades (1, Figure 1) at 45° angle to fuselage so tie-down lines (17) can be secured at aft and forward mooring shackles (4 and 5). Vary position slightly if necessary to align tail rotor blade for installation of tie-down strap.
2. Apply rotor brake, if installed.
3. Unfold wand assembly (3).
4. Unlock lock (11) by pulling up and rotating lock clockwise 90°.
5. Rotate lower tube (12) counterclockwise 180°.
6. Rotate lock (11) counterclockwise 90°.
7. Slide lock (11) down and engage end of lower tube (12).
8. Open spreader arms (9).
9. Spring spreader arms (9) toward each other, and at same time insert spreader arm guides (8) into grommets (7) in sock assembly (2).
10. Use wand assembly (3) to position sock assembly (2) on rotor blade. Remove wand assembly by pulling down to disengage spreader arms (9) and spreader arm guides (8) from sock assembly.



TIE-DOWN RING (15) MUST BE IN CORRECT POSITION ON TIE-DOWN LINE (17) TO RESULT IN NOT MORE THAN 60 POUNDS (27 KG) VERTICAL TENSION IN TIE-DOWN LINE AND NOT EXCEEDING 40 INCHES (101.6 CM) DEFLECTION OF MAIN ROTOR HUB AND BLADE (MEASURED FROM STATIC DROOP AT BLADE TIP), WHEN SNAP HOOK (14) IS FASTENED TO AFT MOORING SHACKLE (4). TIE-DOWN RING (15) ON TIE-DOWN LINE CAN BE MOVED TO OBTAIN CORRECT TIE-DOWN LINE TENSION AND HUB AND BLADE DEFLECTION.

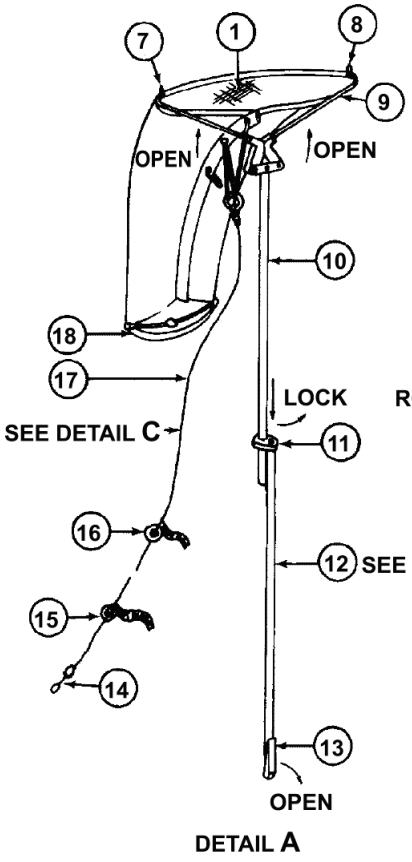
11. Pass snap hook (14) through aft mooring shackle (4) and fasten to tie-down ring (15).
12. Install remaining three sock assemblies (2) in same manner, but fasten snap hook (14) to tie-down ring (16) when securing forward blades.



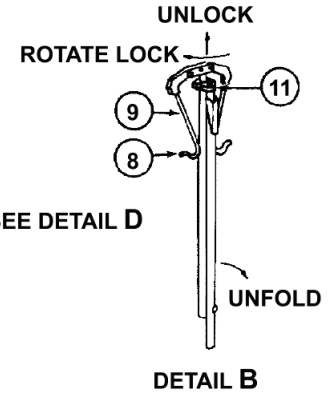
NOTE

The main rotor nonfolding blade tie-down consists of four tie-down sock assemblies (2) P/N 412-782-113-103 and one wand assembly (3) P/N 412-782-118-101.

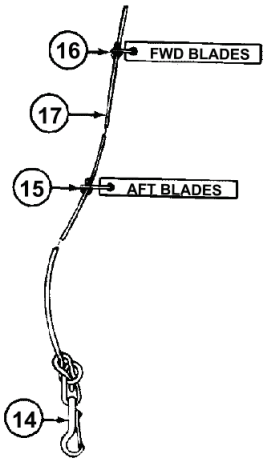
1. Blade
2. Sock assembly (not installed)
3. Wand assembly
4. Aft mooring shackle
5. Forward mooring shackle
6. Sock assembly (installed)
7. Grommet
8. Spreader arm guide
9. Spreader arm
10. Upper tube
11. Lock
12. Lower tube
13. Sock removal hook
14. Snap hook
15. Tie-down ring and AFT BLADE flag
16. Tie-down ring and FWD BLADE flag
17. Tie-down line
18. Sock removal loop



DETAIL A



DETAIL B



DETAIL C

ICN-412-A-100000-B-97499-00002-A-001-01

Figure 1. Main Rotor Non-folded Blade Tie-down (Sheet 1 of 1)

Folding/Unfolding (Using Droop Restraint Tube Assembly) - Description

The main rotor droop restraint tube assembly consists of two spindle pins, two blade lock assemblies, two locking pins, and a tube assembly with attached lifting ring and cables. The main rotor droop restraint supports two opposing blades while in folded position and prevents folded blade spindles from rotating.

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Folding/Unfolding (Using Droop Restraint Tube Assembly) - Folding

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Main Rotor Droop Restraint Assembly	T101261-101	1	

Procedure



WHEN main rotor droop restraint assembly (T101261-101) IS INSTALLED, DO NOT TURN HUB ASSEMBLY WITHOUT FIRST DISCONNECTING PITCH CHANGE LINKS TO PREVENT LOADING OF PITCH CHANGE BEARINGS.

1. Position main rotor hub and blades, as shown in Figure 1, using suitable hoist position tube assembly (8) on main rotor hub.

2. With hydraulic systems 1 and 2 off and blades in normal (not folded) position, manually position cyclic stick to position holes through spindle grips (6) and holes through yoke (9) and fittings (10), with mismatch as equal as possible (Detail A).

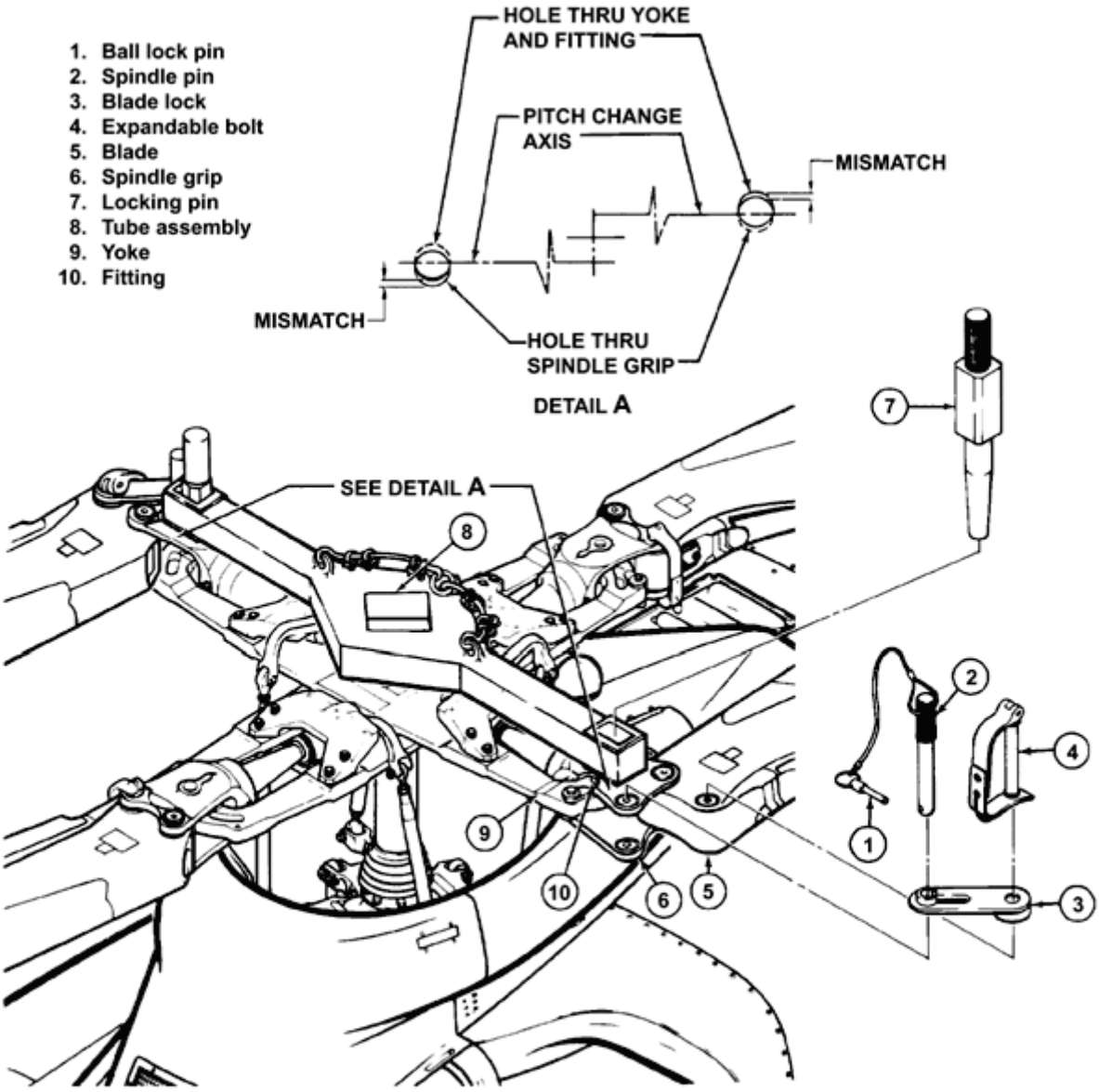
3. Install tube assembly (8) using locking pins (7). Raise or lower collective stick as necessary to align holes in spindle grip, yoke, and fitting.

4. Disconnect pitch change links associated with blades to be folded.

5. Remove safety lock hair pin (not illustrated) from lower end of expandable bolt (4). Raise bolt handle and remove bolt. Partially rotate blade. Place blade lock (3) in position and insert expandable bolt through blade lock and blade (5). Install spindle pin (2) through blade lock and spindle grip (6).

6. With spindle pin (2) raised slightly, rotate blade (5) until pin can be pushed down into locking hub of blade lock (3). Secure spindle pin by installing ball lock pin (1) through hole in lower end of spindle pin.

7. Opposite blade is folded in same manner.



NOTE

The (T101261-101) main rotor droop restrain assembly consists of two locking pins (7), two spindle pins (2), two blade locks (3) and a tube assembly (8).

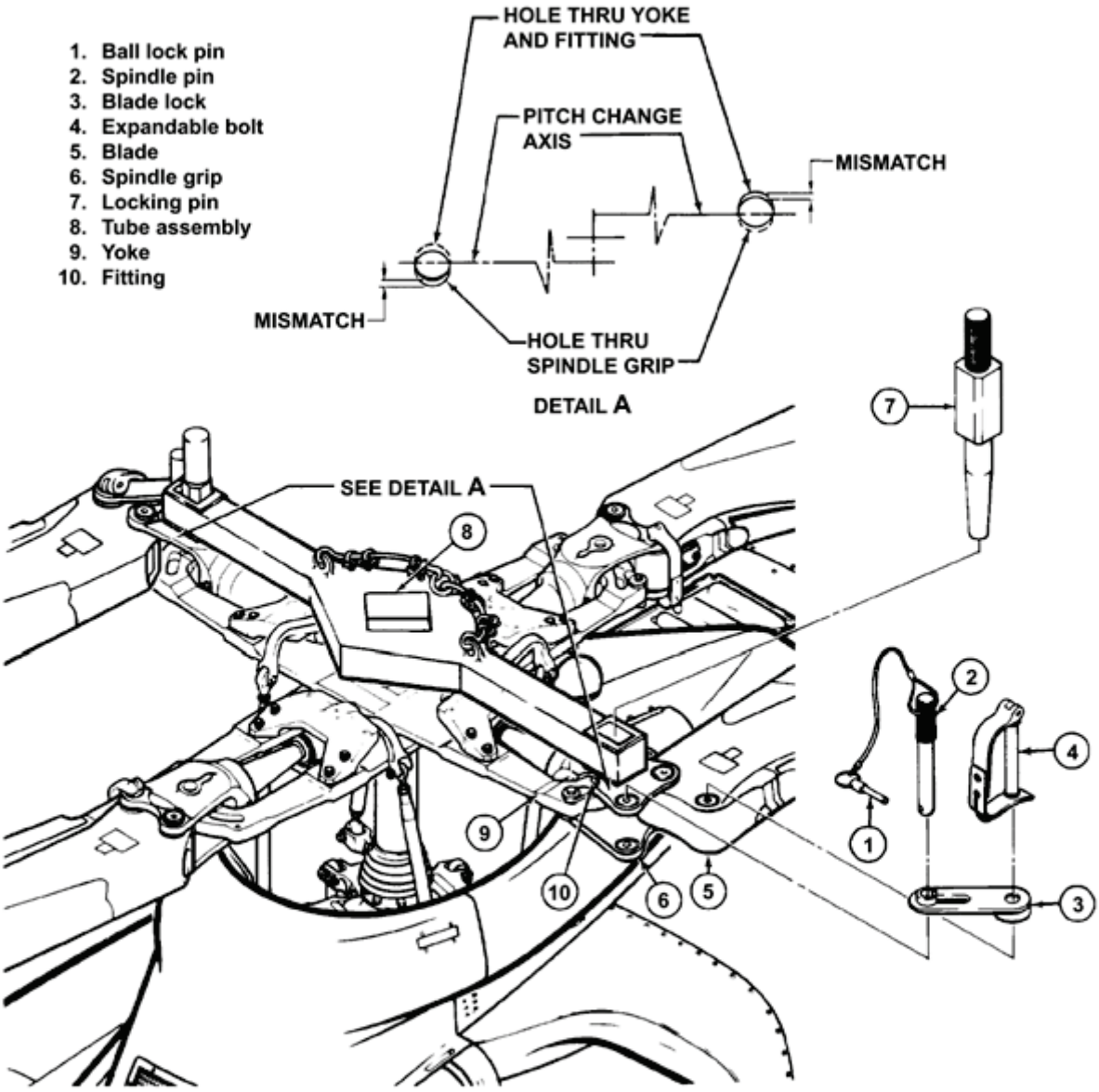
ICN-412-A-100000-B-97499-00003-A-001-01

Figure 1. Main Rotor Droop Restraint Tube Assembly T101261-101 (Sheet 1 of 1)

Folding/Unfolding (Using Droop Restraint Tube Assembly) - Unfolding

Procedure

1. Remove ball lock pin (1, Figure 1) from spindle pin (2). Remove spindle pin from blade lock (3) and spindle grip (6).
2. Remove expandable bolt (4) from blade (5) and blade lock (3) and rotate blade into position in spindle grip (6).
3. Raise handle on expandable bolt (4). Install expandable bolt in trailing edge hole of spindle grip (6) and blade (5). Push down handle of expandable bolt. Latch spring clip of handle. Install safety lock hair pin (not illustrated) to hold handle in down position.
4. Connect pitch change links associated with folded blades.
5. With hydraulic systems 1 and 2 off, manually raise or lower collective stick to align holes through spindle grip (6), yoke (9), and fitting (10) (Detail A). Remove locking pin (7) from tube assembly (8).
6. Repeat Step 1. through Step 4. for opposite blade (5).
7. Using suitable hoist, remove tube assembly (8) from main rotor hub.



NOTE

The (T101261-101) main rotor droop restrain assembly consists of two locking pins (7), two spindle pins (2), two blade locks (3) and a tube assembly (8).

ICN-412-A-100000-B-97499-00003-A-001-01

Figure 1. Main Rotor Droop Restraint Tube Assembly T101261-101 (Sheet 1 of 1)

Folding/Unfolding (Using Blade Folding Racks B67-43001-02) - Description

The main rotor blade folding kit consists of two blade folding racks with attached struts and blade cradles and cradle lock handles. The aft blade folding rack attaches to hard points on tail boom and the forward blade folding rack attaches to hard points on nose.

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Folding/Unfolding (Using Blade Folding Racks B67-43001-02) - Folding

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Blade Folding Racks and Support Equipment	B67-43001-02	1	(refer to BHT-4R-SI-59)

Procedure

1. Ensure collective is lowered and cyclic is approximately centered.

2. Install forward and aft blade folding racks (1 and 2, Figure 1) on the nose and tail.

3. Position blue blade in forward upper cradle (3) and red blade in aft upper cradle (6). Secure blades with cradle lock handles (7).

4. Raise cradle height adjustment rods (8) to support blade weight to remove pressure on expandable bolts (9).

5. Remove safety pin lock (10, Detail C) from bottom of expandable bolt (9). Raise bolt handle and remove expandable bolts from trailing edge of blue and red blades.



BLUE AND RED BLADES WILL REMAIN IN THEIR CRADLES.

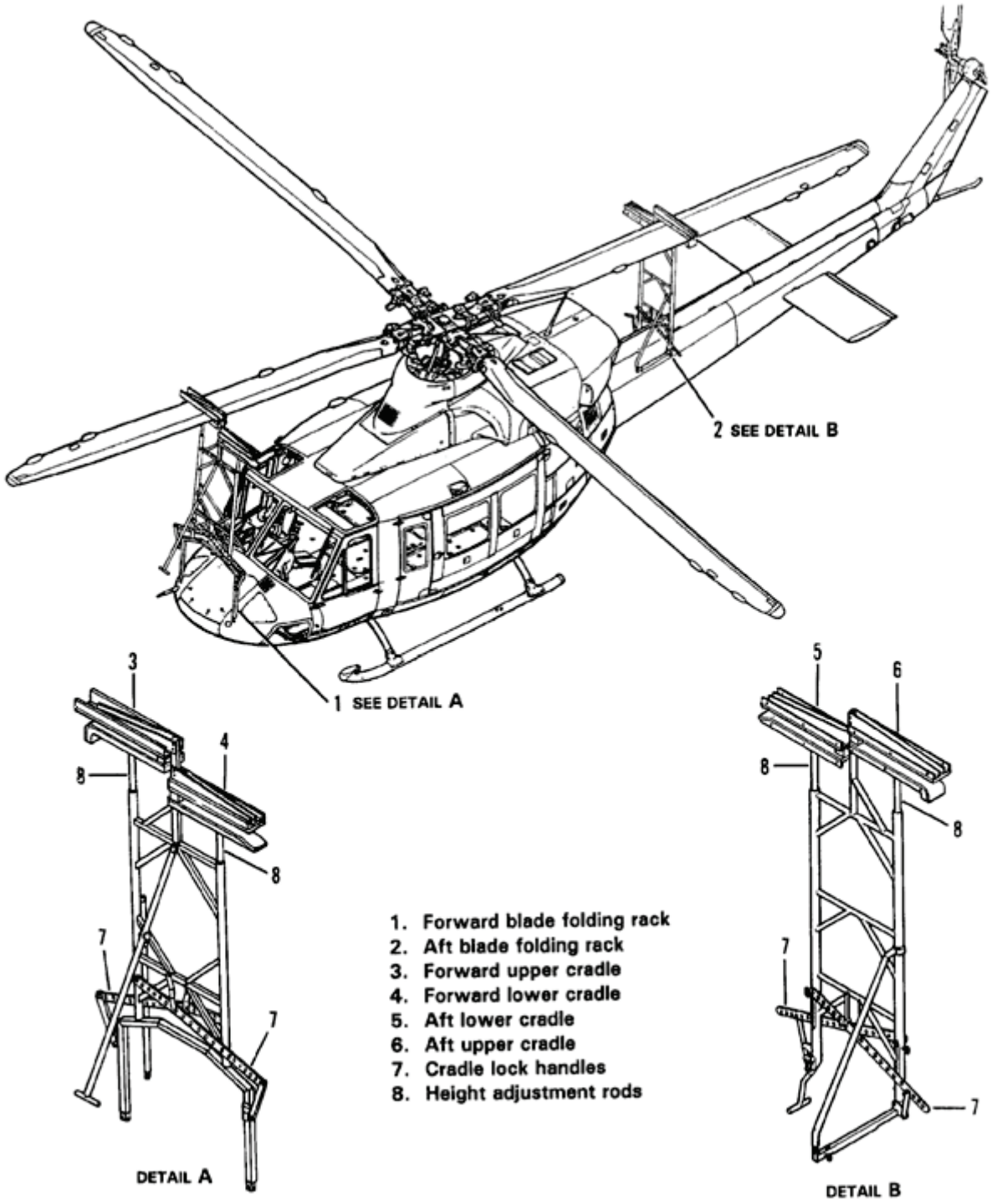
6. Turn main rotor clockwise (viewing from top) so that green blade moves to the front and orange blade moves to rear (Figure 1, Sheet 2).

7. Secure blades in forward and aft lower cradles (4 and 5) with cradle lock handles (Figure 1, Sheet 3).



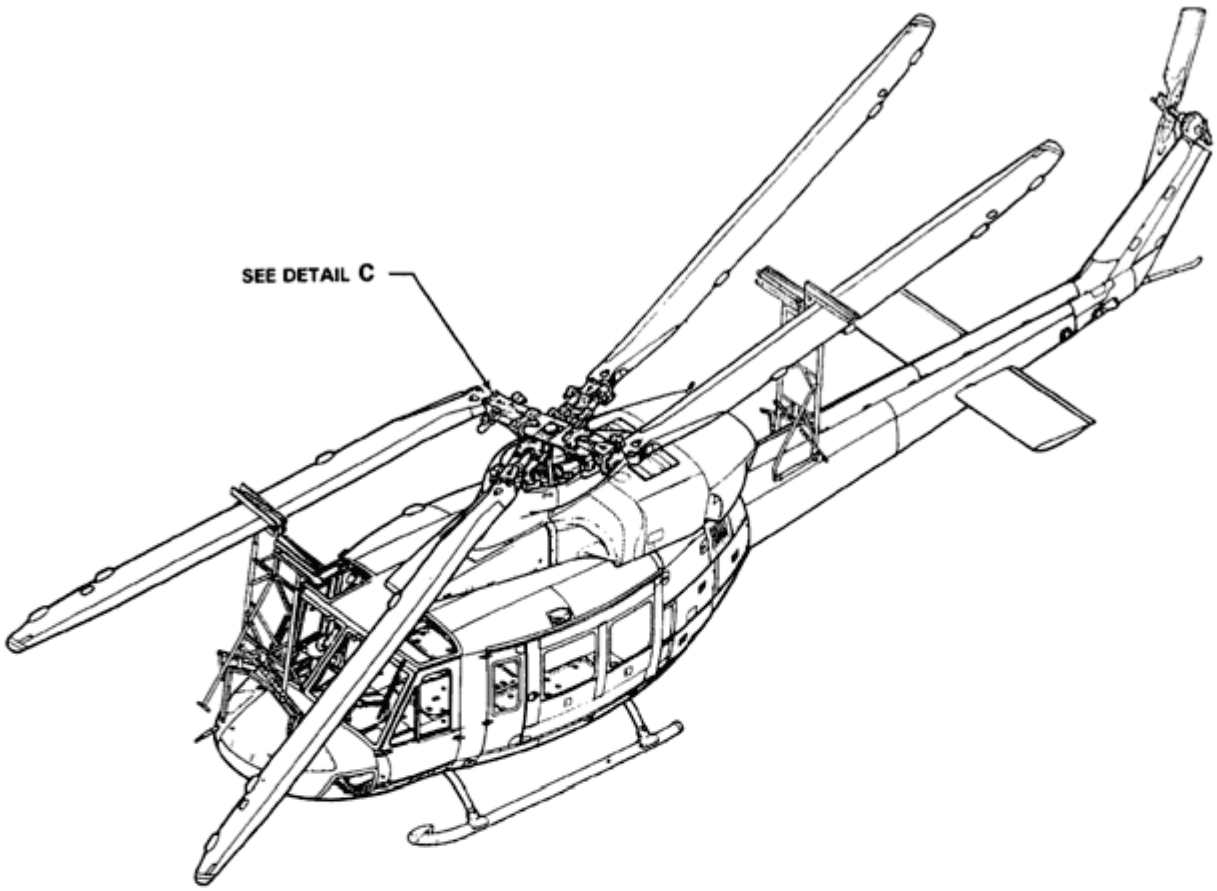
EXPANDABLE BOLT HANDLE SHALL NOT BE CLOSED WHEN THE BOLT IS NOT INSTALLED. FAILURE TO FOLLOW THIS CAUTION COULD RESULT IN DAMAGING THE BOLT RING SEGMENTS.

8. Store expandable bolts (9, Detail C) by inserting bolts through trailing holes in spindle grips (11).

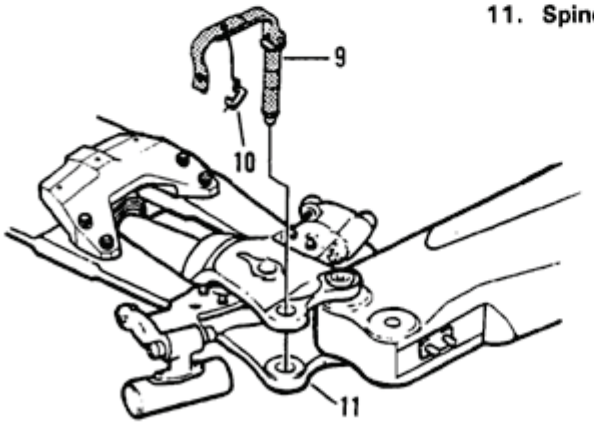


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Figure 1. Main Rotor Blade Folding Kit (Sheet 1 of 3)



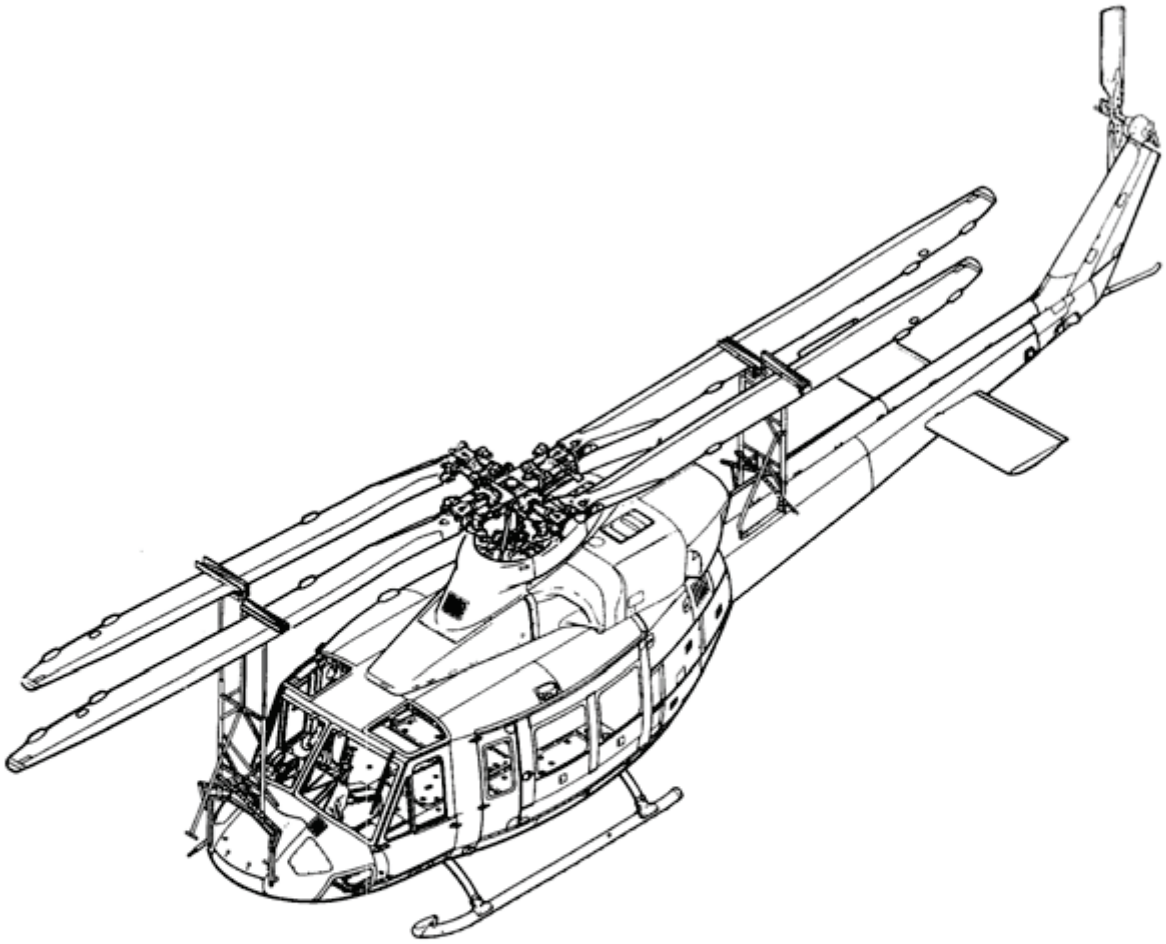
9. Expandable bolts
10. Safety pin lock
11. Spindle grip



DETAIL C

ICN-412-A-100000-B-97499-00005-A-001-01

Figure 1. Main Rotor Blade Folding Kit (Sheet 2 of 3)



ICN-412-A-100000-B-97499-00006-A-001-01

Figure 1. Main Rotor Blade Folding Kit (Sheet 3 of 3)

Folding/Unfolding (Using Blade Folding Racks B67-43001-02) - Unfolding Procedure



FAILURE TO FOLLOW THIS PROCEDURE WILL CAUSE DAMAGE TO ROTOR AND CONTROL SYSTEMS.

1. Release tail rotor tie-down if installed.

2. Release straps securing green and orange blades.



EXPANDABLE BOLT HANDLE SHALL NOT BE CLOSED WHEN THE BOLT IS NOT INSTALLED. FAILURE TO FOLLOW THIS CAUTION COULD RESULT IN DAMAGING THE BOLT RING SEGMENTS.

3. Remove stored expandable bolts (9, Figure 1) from spindle grips (11).



BLUE AND RED BLADES WILL REMAIN IN THEIR CRADLES.

4. Turn the main rotor counterclockwise (viewing from top) so that green blade moves aft and orange blade moves forward to their original positions (Figure 1, Sheet 2).

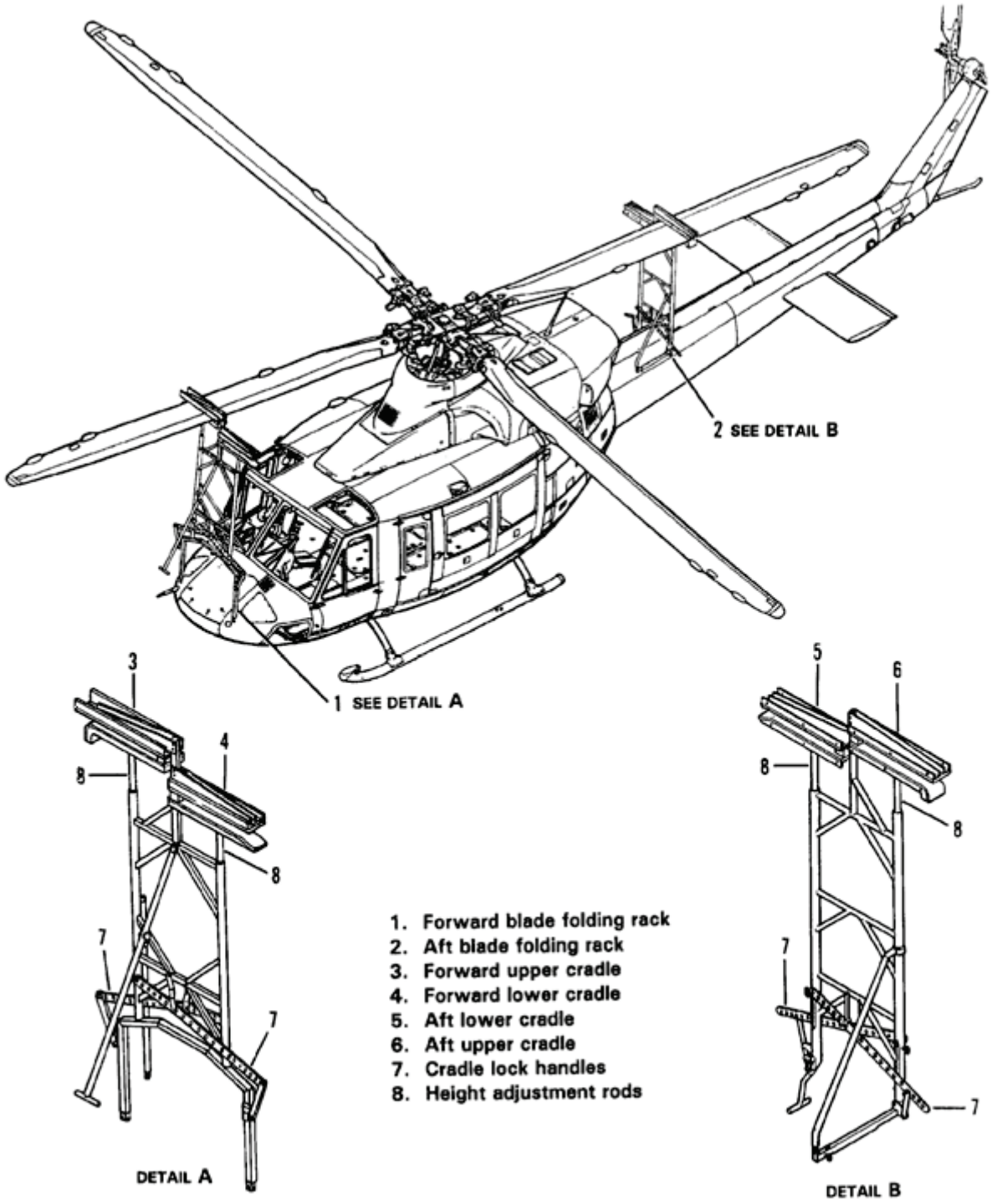
5. Install the expandable bolts (9) in blue and red blades as follows:

5.1. Insert bolts through trailing holes in spindle grips (11). Push down handle of expandable bolts (9).

5.2. Latch spring clip of handle over end of expandable bolts. Secure with safety pin lock (10).

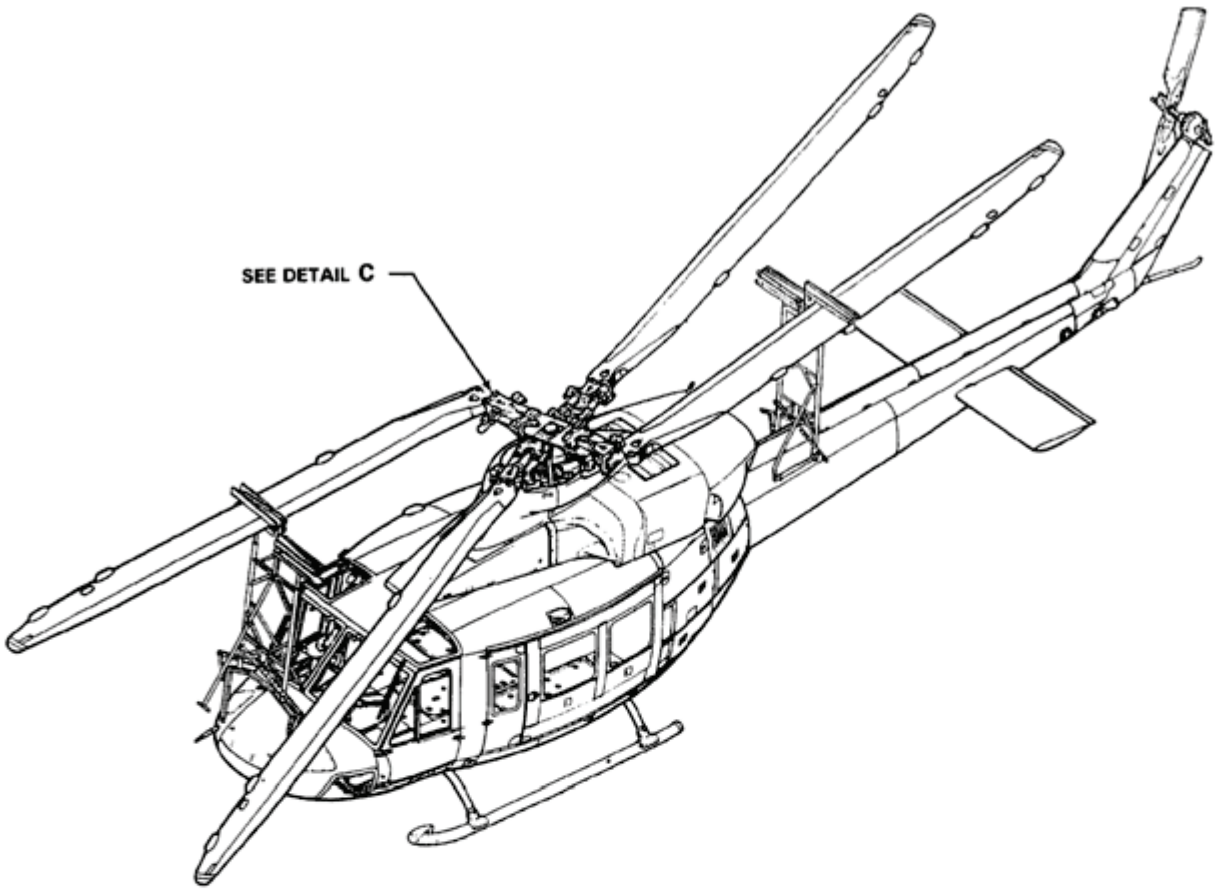
6. Lower height adjustment rods (8).

7. Remove forward and aft blade folding racks (1 and 2) from helicopter.

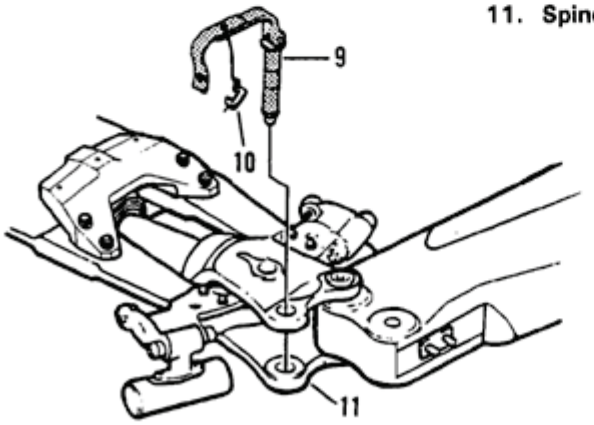


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Figure 1. Main Rotor Blade Folding Kit (Sheet 1 of 3)



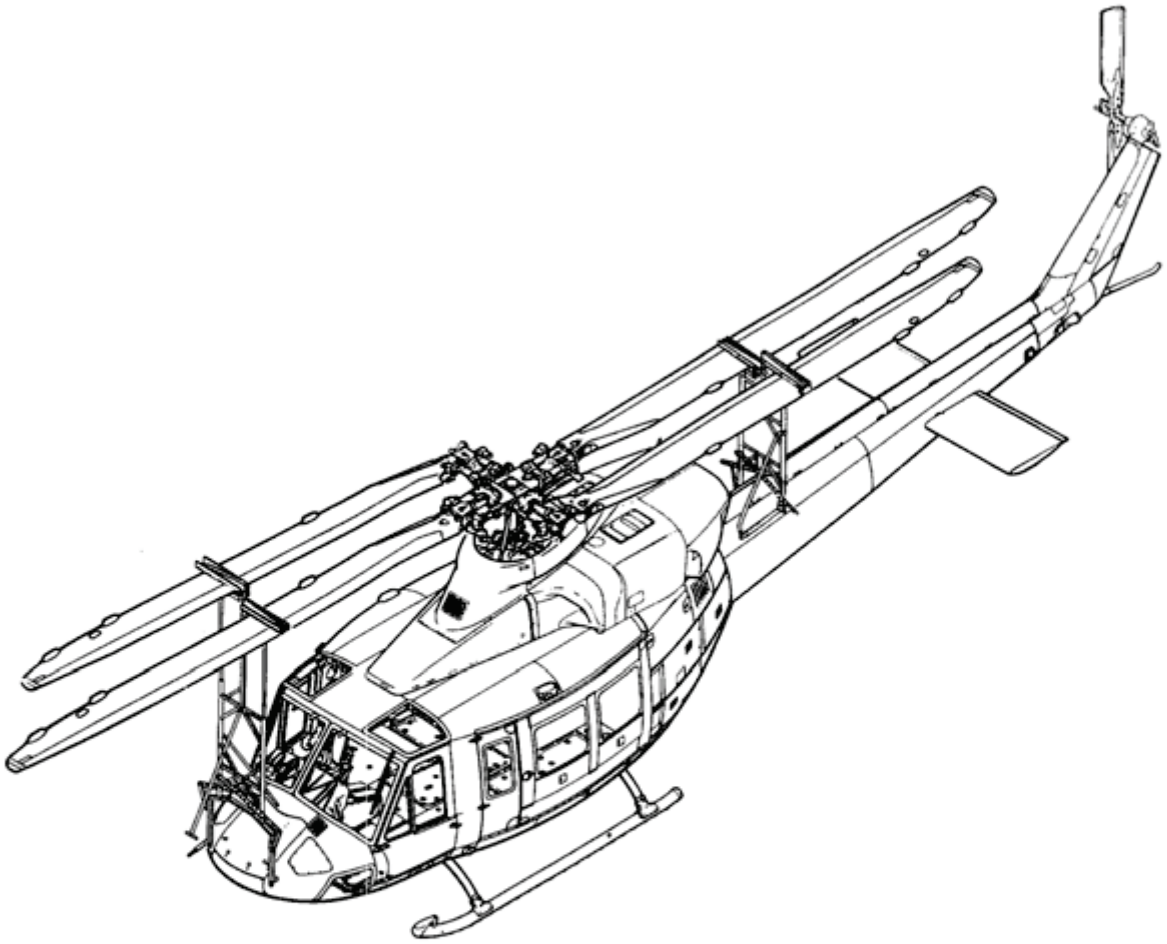
9. Expandable bolts
10. Safety pin lock
11. Spindle grip



DETAIL C

ICN-412-A-100000-B-97499-00005-A-001-01

Figure 1. Main Rotor Blade Folding Kit (Sheet 2 of 3)



ICN-412-A-100000-B-97499-00006-A-001-01

Figure 1. Main Rotor Blade Folding Kit (Sheet 3 of 3)

Helicopter - Storage

Procedure

1. Refer to the BHT-ALL-SPM, Standard Practices Manual for helicopter storage procedures.

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CHAPTER 11
PLACARDS AND MARKINGS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

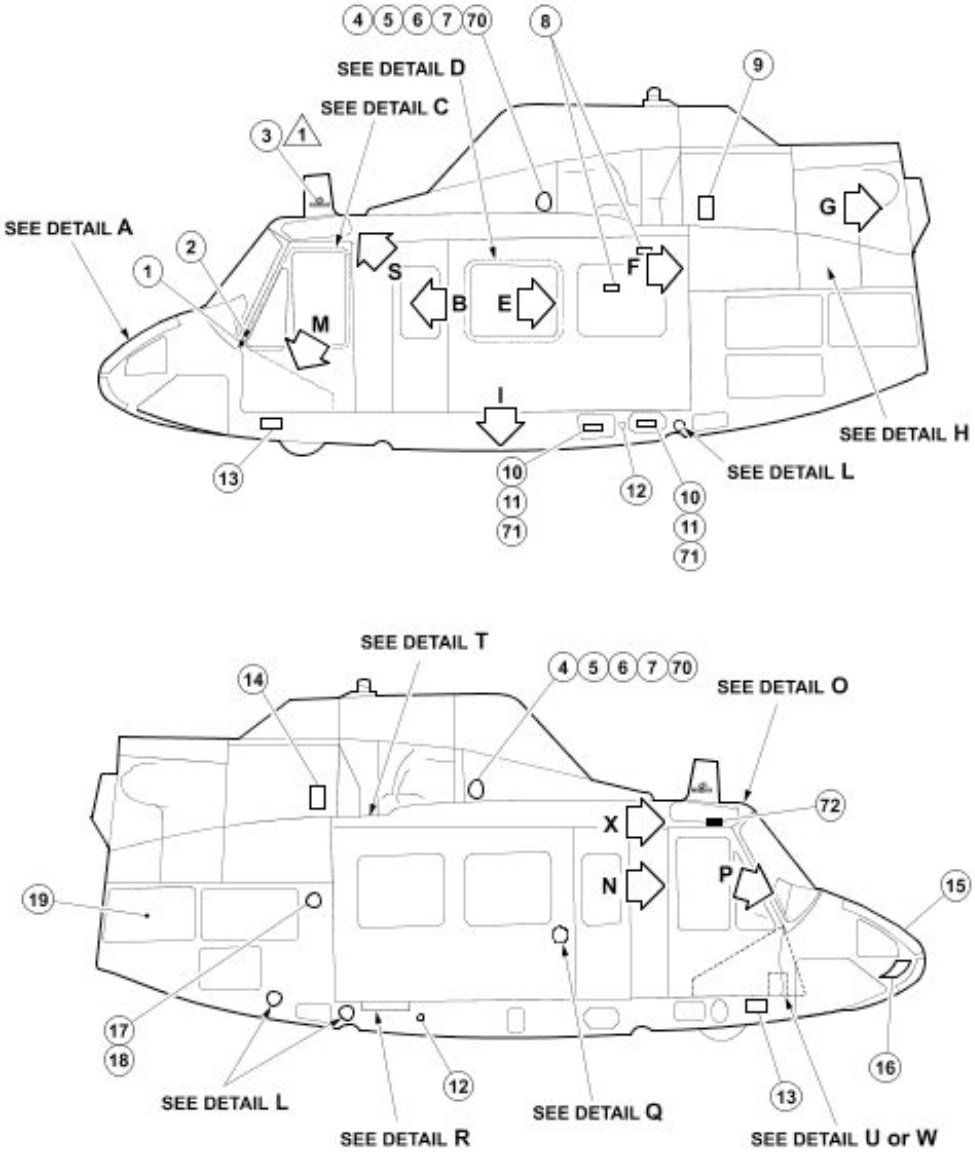
C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Placards and Markings - Description	DMC-412-A-11-00-00-00A-028A-A / 00123	2023-05-31	ALL
Placards and Markings - Decals - Application	DMC-412-A-11-00-00-00A-067A-A / 00124	2023-05-31	ALL

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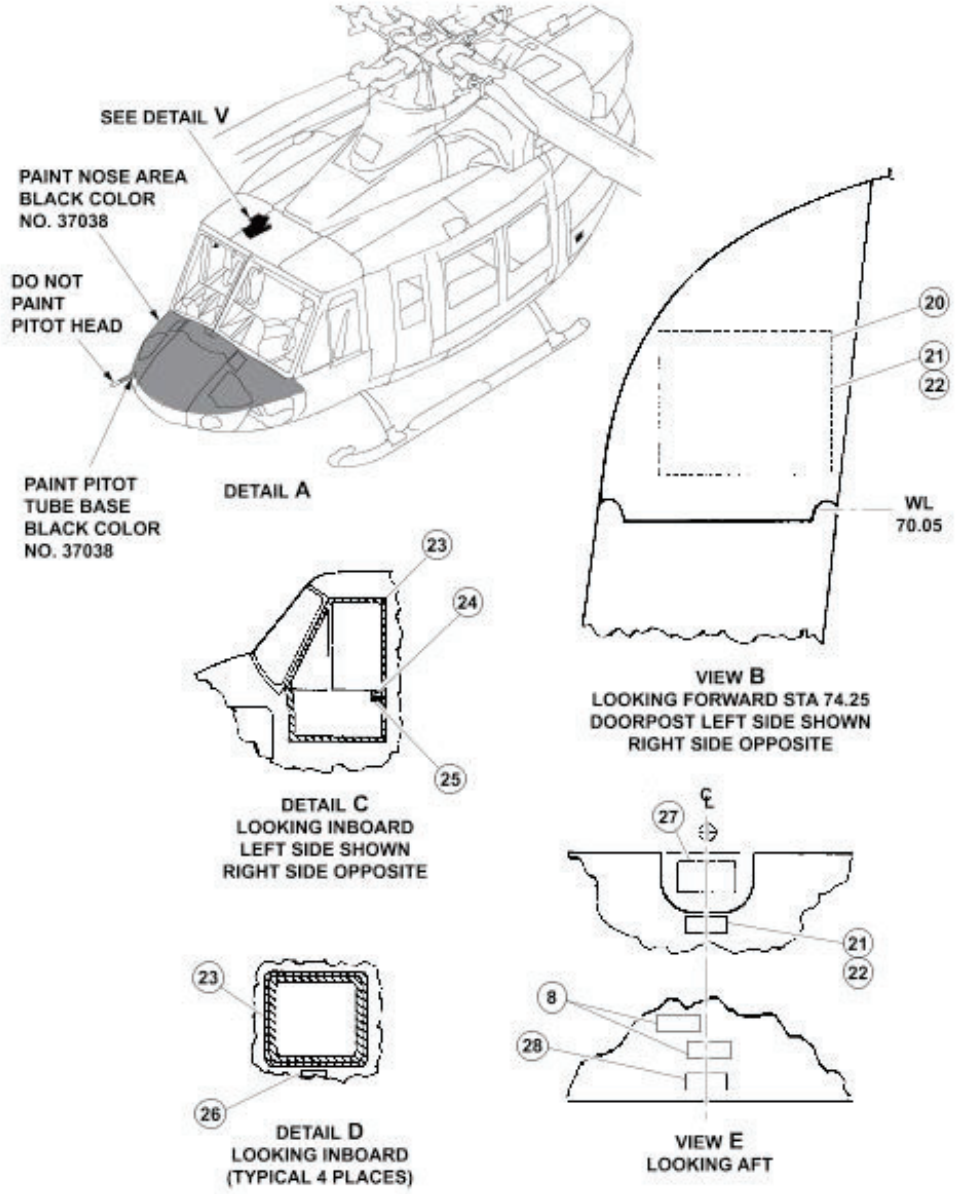
Placards and Markings - Description

Decals, stencils, and markings are shown in Figure 1 through Figure 5. The first part of the figures shows sections, views, and details on the helicopter. Subsequent parts of the figures show detail appearance, quantity, and location of each unit by item number.



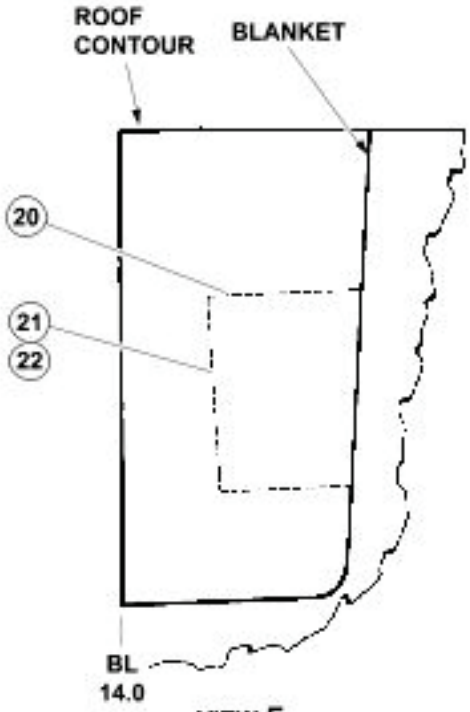
ICN-412-A-110000-B-97499-00001-A-001-01

Figure 1. Decal and Stencil Markings, Forward (Sheet 1 of 19)



ICN-412-A-110000-B-97499-00002-A-001-01

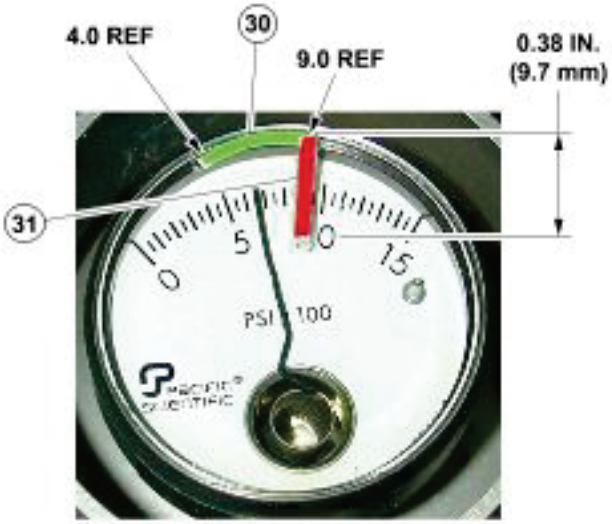
Figure 1. Decal and Stencil Markings, Forward (Sheet 2 of 19)



VIEW F
LOOKING AFT STA 155.0
PYLON ISLAND LEFT SIDE SHOWN
RIGHT SIDE OPPOSITE



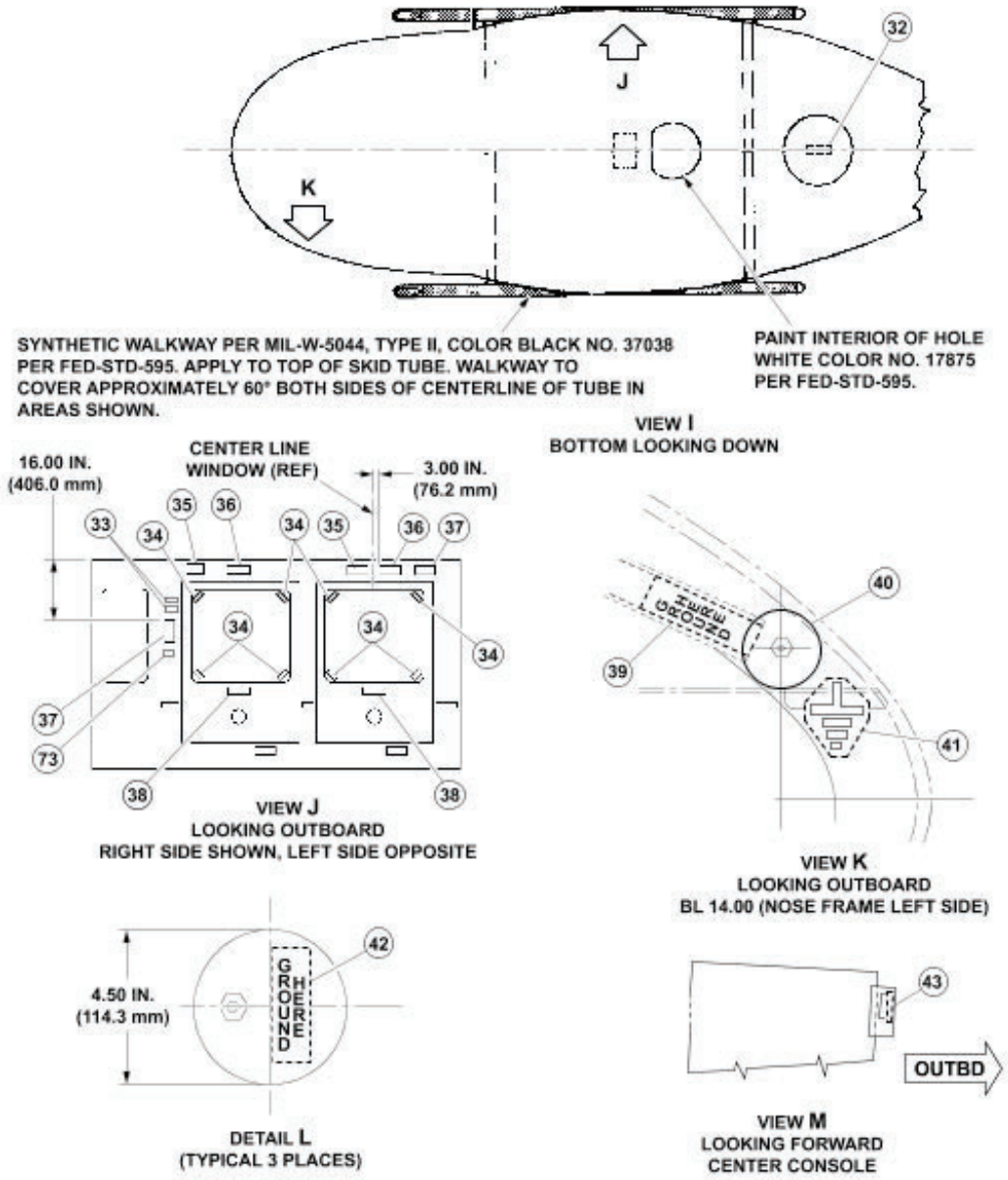
VIEW G
LOOKING AFT LEFT SIDE SHOWN
RIGHT SIDE OPPOSITE



DETAIL H
FIRE EXTINGUISHER CONTAINER
PRESSURE INDICATING GAUGE (2 PLACES)

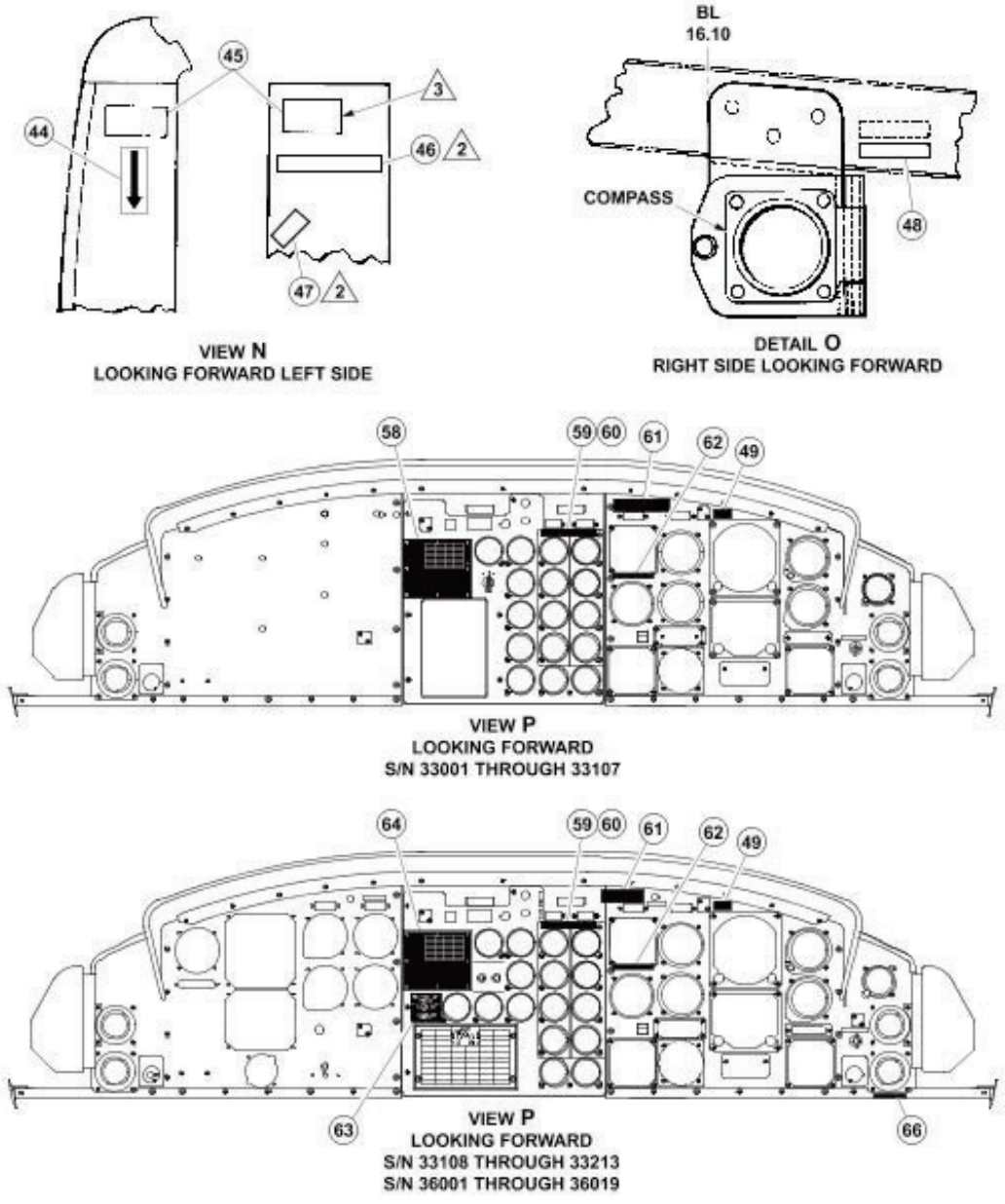
ICN-412-A-110000-B-97499-00003-A-001-01

Figure 1. Decal and Stencil Markings, Forward (Sheet 3 of 19)



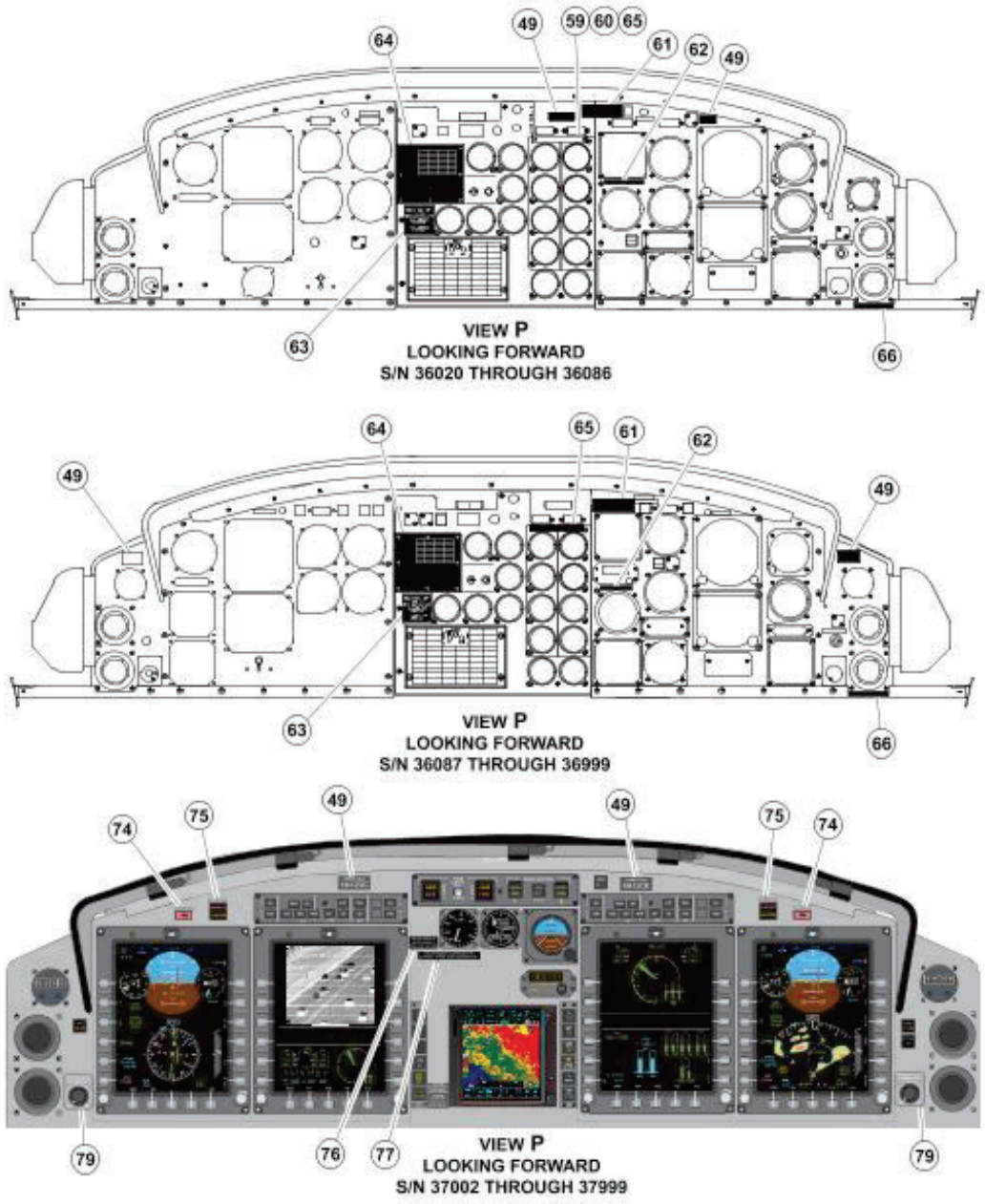
ICN-412-A-110000-B-97499-00004-A-001-01

Figure 1. Decal and Stencil Markings, Forward (Sheet 4 of 19)



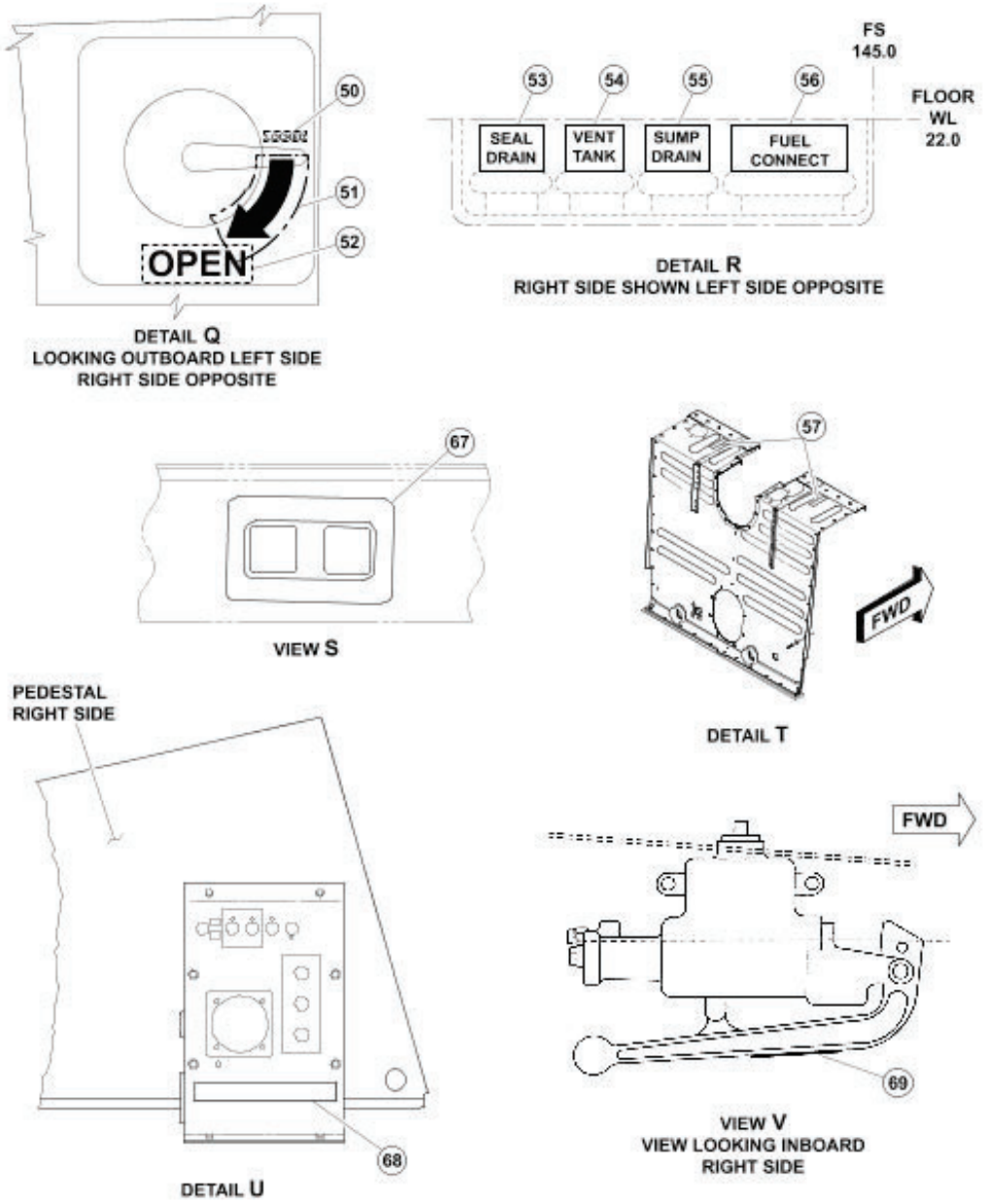
ICN-412-A-110000-B-97499-00005-A-001-01

Figure 1. Decal and Stencil Markings, Forward (Sheet 5 of 19)



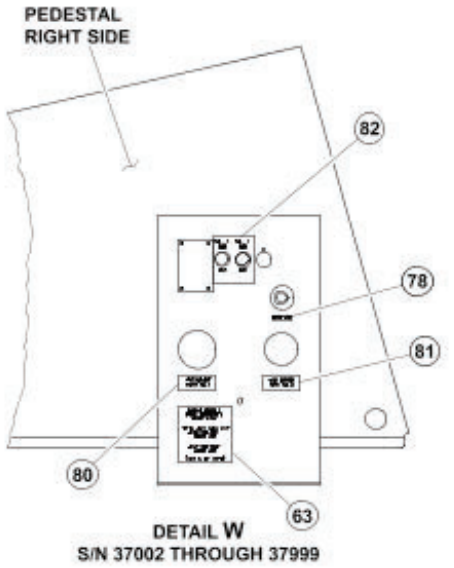
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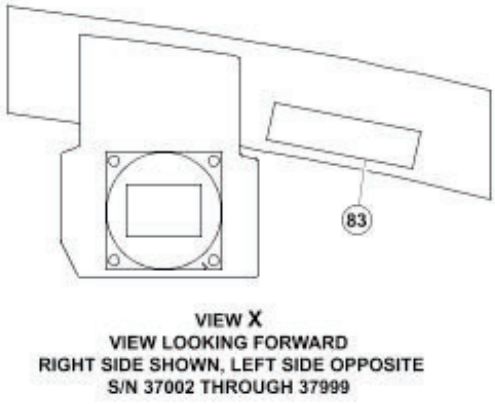
Figure 1. Decal and Stencil Markings, Forward (Sheet 6 of 19)



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Figure 1. Decal and Stencil Markings, Forward (Sheet 7 of 19)





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Figure 1. Decal and Stencil Markings, Forward (Sheet 8 of 19)



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Figure 1. Decal and Stencil Markings, Forward (Sheet 9 of 19)

205-076-058 RESERVOIR
TOTAL SYSTEM CAP. 10 U.S. PTS=4.7 LITERS
RESERVOIR CAP. 5.3 U.S. PTS=2.5 LITERS
REFILL LEVEL CAP. 2.5 U.S. PTS=1.2 LITERS
USE MIL-PRF-5606 OR MIL-PRF-87257
HYDRAULIC FLUID. SERVICE INSTRUCTIONS:
WHEN FLUID LEVEL IS VISIBLE
FILL TO OVERFLOW

7. DECAL 205-076-060-105
APPLICABLE FOR HELICOPTERS
POST TB 412-12-229 EQUIPPED WITH HYDRAULIC RESERVOIR P/N 205-076-058-XXX (PEEK PLASTIC).
LOCATE ON THE HYDRAULIC RESERVOIRS.

STRUCTURAL PANEL
REQUIRED GROUND
RUN AND FLIGHT

8. DECAL 31-044-3CHP.
12 REQUIRED. LOCATE ON BOTH SIDES OF APPROPRIATE PANELS WHICH MUST BE INSTALLED BEFORE GROUND RUN OR FLIGHT.



9. DECAL 212-070-662-003
LOCATE ON BRACKET ADJACENT TO LEFT ENGINE OIL SIGHT GAUGE.

INSTRUCTIONS FOR USE OF
HYDRAULIC GROUND TEST STAND
FLUID PER MIL-H-5606
TEST STAND SETTINGS

USE TEST STAND RESERVOIR	
RELIEF VALVE	1300 PSI
VOLUME OUTPUT	6 GPM
PRESSURE COMPENSATOR	1000 PSI

10. DECAL 205-076-224-001.
(S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36636 PRE TB 412-12-229).

INSTRUCTIONS FOR USE OF
HYDRAULIC GROUND TEST STAND
FLUID PER MIL-H-5606 OR MIL-PRF-87257
TEST STAND SETTINGS

USE TEST STAND RESERVOIR	
RELIEF VALVE	1300 PSI
VOLUME OUTPUT	6 GPM
PRESSURE COMPENSATOR	1000 PSI

11. DECAL 205-076-224-101.
(S/N 36637 THROUGH 36999 OR S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36636 POST TB 412-12-229).

PUSH FOR FUEL
SUMP DRAIN

12. DECAL 31-054-5DFHP.
2 REQUIRED. LOCATE BOTH SIDES ABOVE SWITCH.

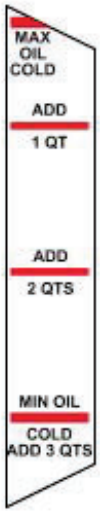
CAUTION

WHEN TOWING AT OR ABOVE 8900 LBS, REFER TO MAINTENANCE MANUAL CHAPTERS 4 & 9

13. DECAL 212-070-600-143
(S/N 36493 THROUGH 36999 AND S/N 37002 THROUGH 37999).

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Figure 1. Decal and Stencil Markings, Forward (Sheet 10 of 19)



14. DECAL 212-070-662-004.
LOCATE ON BRACKET ADJACENT TO RIGHT ENGINE OIL SIGHT GAUGE.

MAXIMUM ALLOWABLE BALLAST
LEFT SIDE 62 POUNDS
RIGHT SIDE 62 POUNDS

15. DECAL 31-044-7DHF.
LOCATE INSIDE UPPER NOSE DOOR.

28 V
D.C.

16. DECAL 31-054-6DFHP.
LOCATE ON OUTSIDE OF POWER DOOR.

FUEL
FUEL SYSTEM USEABLE CAPACITY
BASIC AIRCRAFT 211 US GAL = 798.6 LITERS
WITH 412-709-007 AUX KIT 374 US GAL = 1415.5 LITERS
WITH 412-709-009 AUX KIT 248 US GAL = 938.7 LITERS
FUEL TYPE PER PWA SPEC 522
SEE FLIGHT MANUAL FOR APPROVED FUEL

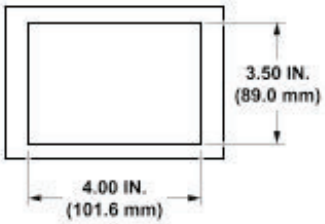
17. DECAL 412-070-609-103.
LOCATE ABOVE FUEL FILLLER CAP ON RIGHT SIDE AFT OF PASSENGER DOOR.
(S/N 33001 THROUGH 33107).

FUEL JP-4 OR JP-5

18. DECAL 31-053-9DFHP.
(S/N 33108 THROUGH 33213, S/N 36001 THROUGH 36999, AND S/N 37002 THROUGH 37999).

STRUCTURAL TUBE - MUST BE INSTALLED PRIOR TO FLIGHT

19. DECAL 205-030-159-001.
INSTALL ON BOTH SIDES OF TENSION ROD.



20. STENCIL.
5 REQUIRED. LOCATE AS SHOWN.
USE STENCIL WITH STANDARD INTERIOR ONLY.
PAINT AREA 3.50 BY 4.00 INCHES (88.9 BY 101.6 mm), WHITE COLOR NO. 37875 PER FED-STD-595.

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Figure 1. Decal and Stencil Markings, Forward (Sheet 11 of 19)



21. STENCIL 412-070-884-101.
5 REQUIRED. LOCATE CENTERED WITHIN STENCIL 20A. USE STENCILS FOR STANDARD INTERIOR TRIM ONLY. PAINT RED COLOR NO. 31136 PER FED-STD-595.



22. DECAL 31-056-10DKKP.
5 REQUIRED. LOCATE SAME PLACES AS STENCILS 20 AND 21. USE DECALS FOR DELUXE INTERIOR TRIM ONLY.

23. MARKINGS EMERGENCY EXIT (5 OR 6 REQUIRED). LOCATE OUTSIDE AROUND EMERGENCY OPENINGS. PAINT CONTRASTING COLOR BAND 2.0 INCHES (50.8 mm) WIDE PER FAA ADVISORY CIRCULAR 20.47 FAR 25.811 (h) (2) AND FAR 121.310 (j). PAINT AROUND LEFT CREW DOOR IF SINGLE PILOT CONTROLS AND INSTRUMENTS ARE INSTALLED.



24. DECAL 31-053-2DFHP.
2 REQUIRED. LOCATE ON BOTH CREW DOOR OUTSIDE HANDLES.



25. DECAL 31-042-12CFK.
2 REQUIRED. LOCATE BELOW DECAL 24 ON COPILOT AND PILOT OUTSIDE DOOR HANDLE.



26. SIGN 212-072-636-101.
4 REQUIRED. LOCATE OUTSIDE, CENTERED BELOW EACH PASSENGER WINDOW. (S/N 33001 THROUGH 33160).



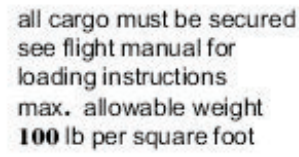
26. SIGN 212-072-636-109.
4 REQUIRED. LOCATE OUTSIDE, CENTERED BELOW EACH PASSENGER WINDOW. (S/N 33161 THROUGH 33213, S/N 36001 THROUGH 36999, AND S/N 37002 THROUGH 37999).



27. STENCIL 31-030-67U10.
LOCATE ON PYLON FORWARD BULKHEAD AT STA 127.00, 2.00 INCHES (50.8 mm) BELOW ROOFLINE. CENTER HORIZONTALLY. COLOR RED NO. 31136 (LUSTERLESS). (S/N 33190 THROUGH 33213, S/N 36001 THROUGH 36999, AND S/N 37002 THROUGH 37999).

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Figure 1. Decal and Stencil Markings, Forward (Sheet 12 of 19)



28. STENCIL 205-070-889-005.
2 REQUIRED. LOCATE ON CENTERLINE OF BULKHEAD AND BLANKET AT STA 129.00 (S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36329).

LOCATE ON CENTERLINE OF BULKHEAD AND BLANKET AT STA 129.00 (S/N 36330 THROUGH 36999 AND S/N 37002 THROUGH 37999).

LETTERS 0.50 INCH (12.7 mm) HIGH, ON BULKHEAD, COLOR BLACK NO. 17038 PER FED-STD-595 ON BLANKET INSULATION, COLOR WHITE NO. 37875 PER FED-STD-595.



29. STENCIL 212-072-601-001.
LOCATE ON LEFT FORWARD SIDE OF BULKHEAD AT STA 225.86.



29. STENCIL 212-072-601-003.
LOCATE ON RIGHT FORWARD SIDE OF BULKHEAD AT STA 225.86.



30. DECAL 31-113-1D.
2 REQUIRED. ONE ON EACH ENGINE FIRE EXTINGUISHER PRESSURE INDICATOR GAUGE. LOCATE GREEN DECAL ON ARC OF GAUGE DIAL BETWEEN NUMBER 4 AND DECAL 31.



31. DECAL 31-113-6A.
2 REQUIRED. ONE ON EACH ENGINE FIRE EXTINGUISHER PRESSURE INDICATOR GAUGE. LOCATE RED DECAL 0.38 INCH (9.6 mm) LONG ALIGNED BETWEEN THE NUMBER 9 ON THE SCALE AND THE CENTER OF THE GAUGE.



32. DECAL 31-055-1DFHP.
2 REQUIRED. LOCATE INSIDE AND OUTSIDE SURFACE OF LARGE ROUND DOOR AT APPROXIMATELY STA 200.00 UNDER AFT CABIN.



33. DECAL 31-030-61F22.



34. DECAL 212-072-635-101.
16 REQUIRED. LOCATE INSIDE AND OUTSIDE LOWER CORNERS 0.10 TO 0.40 INCH (2.5 TO 10.2 mm) FROM RETAINER OF EACH PASSENGER WINDOW.

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Figure 1. Decal and Stencil Markings, Forward (Sheet 13 of 19)



35. SIGN CAN/AC/4-0.
4 REQUIRED. LOCATE INSIDEIN PAN ABOVE EACH PASSENGER WINDOW.
(S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36688).



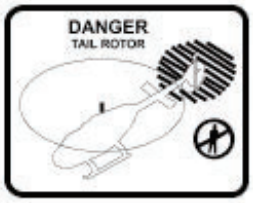
35. SIGN 9004030-1.
4 REQUIRED. LOCATE INSIDEIN PAN ABOVE EACH PASSENGER WINDOW.
(S/N 36689 THROUGH 36999).



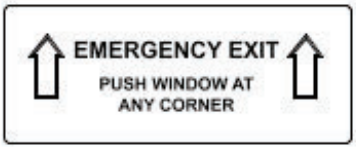
36. STENCIL 31-030-67U5.
4 REQUIRED. LOCATE INSIDE LEFT AND RIGHT PASSENGER DOORS. COLOR RED NO. 31136 (LUSTERLESS). (S/N 33119 THROUGH 33213, S/N 36001 THROUGH 36999, AND S/N 37002 THROUGH 37999).



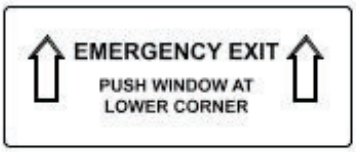
36. DECAL 31-084-1.
(ALTERNATE TO DECAL 31-030-67U5).



37. DECAL 31-072-1.
4 REQUIRED. LOCATE INSIDE LEFT AND RIGHT PASSENGER DOORS.



38. DECAL 212-072-637-101.
4 REQUIRED. LOCATE INSIDE, CENTERED BELOW EACH PASSENGER WINDOW.
(S/N 33001 THROUGH 33160).



38. DECAL 212-072-637-107.
4 REQUIRED. LOCATE INSIDE, CENTERED BELOW EACH PASSENGER WINDOW.
(S/N 33161 THROUGH 33213, S/N 36001 THROUGH 36999, AND S/N 37002 THROUGH 37999).

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Figure 1. Decal and Stencil Markings, Forward (Sheet 14 of 19)

GROUND

39. DECAL 204-070-626-011.
LOCATE ON LEFT SIDE NOSE BL 14 FRAME INSIDE UPPER NOSE DOOR.

40. MARKING - PAINT GROUND AREA 3.00 INCH (76.2 mm) DIAMETER, COLOR BLACK NO. 37038 PER FED-STD-595, ADJACENT TO DECAL 39.



41. DECAL 31-052-1HP.
LOCATE BELOW GROUND AREA ON NOSE FRAME BL 14 INSIDE NOSE DOOR.

GROUND

42. DECAL 204-070-626-009.
3 REQUIRED. LOCATE ON GROUND OUTSIDE LEFT AFT CORNER BELOW PASSENGER DOOR AND TWO GROUNDS ON RIGHT SIDE SAME AREA. PAINT 4.50 INCHES (114.3 mm) DIAMETER OF GROUND BLACK COLOR NO. 17038 PER FED-STD-595, THEN APPLY DECAL.

ON DEFROST OFF

43. DECAL 205-070-854-007.
(S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36557).
DECAL 205-070-854-101.
(S/N 36558 THROUGH 36999 AND S/N 37002 THROUGH 37999).
LOCATE VERTICALLY ON DEFROSTER CONTROL HANDLE ON RIGHT SIDE ABOVE CONSOLE.



44. DECAL 31-039-3MD.
LOCATE BELOW DECAL 45 ON AFT SIDE OF BULKHEAD STA 63.38 ADJACENT TO FIRE EXTINGUISHER.

FIRE EXTINGUISHER

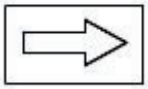
45. DECAL 31-043-14CFK.
2 REQUIRED. LOCATE ABOVE DECAL 44 ON AFT SIDE OF BULKHEAD STA 63.38. LOCATE OTHER ONE ON INBOARD SIDE OF SAME BULKHEAD.

FIRE EXTINGUISHER UNDER SEAT

46. DECAL 31-043-96CLF.
LOCATE AS SHOWN.
(INSTALLED WITH AIRLINE SEAT KIT (BHT-412-SI-43).

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Figure 1. Decal and Stencil Markings, Forward (Sheet 15 of 19)



47. DECAL 31-039-3DM.
LOCATE AS SHOWN.
INSTALLED WITH AIRLINE SEAT KIT (BHT-412-SI-43).

CAUTION
COMPASS UNRELIABLE WITH WINDSHIELD WIPERS ON

48. DECAL 31-030-63G10.
LOCATE ON RIGHT SIDE OF COMPASS ABOVE WINDSHIELD IN COCKPIT.

RADIO CALL

49. PLATE 100-029-1.
LOCATE AS SHOWN ON INSTRUMENT PANEL.

LOCKED

50. DECAL 31-042-13DHF.
2 REQUIRED. LOCATE INSIDE ABOVE END OF BOTH PASSENGER DOOR HANDLES.



51. DECAL 31-057-3EH.
LOCATE INSIDE UNDER END OF LEFT PASSENGER DOOR HANDLE.



51. DECAL 31-057-4EH.
LOCATE INSIDE UNDER END OF RIGHT PASSENGER DOOR HANDLE.

OPEN

52. DECAL 31-042-7BHF.
2 REQUIRED. LOCATE INSIDE UNDER SHAFT OF PASSENGER DOOR HANDLES (S/N 33001 THROUGH 33213, AND S/N 36001 THROUGH 36644).

SEAL
DRAIN

53. DECAL 31-043-28DMP.
2 REQUIRED. LOCATE BOTH SIDES UNDER THE FLOOR NEAR AFT CORNER PASSENGER DOOR (S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36644).

VENT
TANK

54. DECAL 31-043-11DMP.
2 REQUIRED. LOCATE BOTH SIDES UNDER FLOOR NEAR AFT CORNER PASSENGER DOOR (S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36644).

SUMP
DRAIN

55. DECAL 31-043-13DMP.
2 REQUIRED. LOCATE BOTH SIDES UNDER FLOOR NEAR AFT CORNER PASSENGER DOOR (S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36644).

FUEL
CONNECT

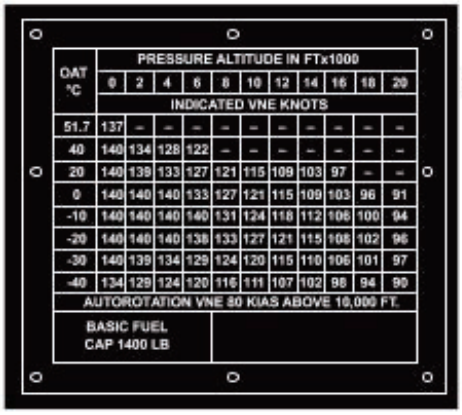
56. DECAL 31-043-12DMP.
2 REQUIRED. LOCATE BOTH SIDES UNDER FLOOR NEAR AFT CORNER PASSENGER DOOR (S/N 33001 THROUGH 33213 AND S/N 36001 THROUGH 36644).

NO STEP

57. DECAL 31-042-6CKF.
LOCATE AS SHOWN.

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Figure 1. Decal and Stencil Markings, Forward (Sheet 16 of 19)



58. DECAL 412-075-215-111.
LOCATE AS SHOWN.



59. PANEL 212-075-043-103.
LOCATE AS SHOWN
FOR GAS PRODUCER
INSTRUMENT PART NUMBER
212-075-037-101.



60. PANEL 212-075-043-105.
LOCATE AS SHOWN
FOR GAS PRODUCER
INSTRUMENT PART NUMBER
212-075-037-113.



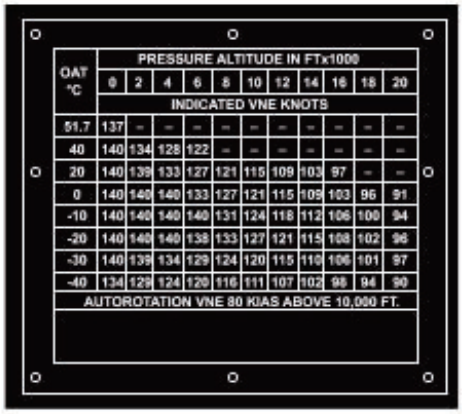
61. DECAL 205-070-159-001.
LOCATE AS SHOWN.



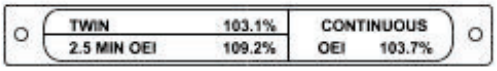
62. DECAL 412-075-151-183.
LOCATE AS SHOWN.



63. DECAL 412-075-015-107.
LOCATE AS SHOWN.



64. DECAL 412-375-235-103.
LOCATE AS SHOWN.



65. PANEL 412-375-038-101.
LOCATE AS SHOWN.

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Figure 1. Decal and Stencil Markings, Forward (Sheet 17 of 19)

IN ALTN POSITION MAINTAIN INSTRUMENT ACCURACY BY CLOSING WINDOWS AIRVENTS AND TURNING HEATER OFF

66. DECAL 412-075-200-103.
LOCATE AS SHOWN.



67. SIGN 10-0852-3.
LOCATE AS SHOWN.

COLD WEATHER START PROCEDURE NOT REQUIRED
AFCU P/N 3118098-12 OR LATER ARE
INSTALLED ON BOTH ENGINES

68. DECAL 31-044-66RFH.
LOCATE ON PEDESTAL LOWER RIGHT SIDE
(PLACARD APPLICABLE ONLY WHEN BOTH
AUTOMATIC FUEL CONTROL UNITS ARE
P/N 3118098-12 OR SUBSEQUENT).

DO NOT APPLY ROTOR BRAKE
ABOVE 40% RPM

69. DECAL 204-076-336-001.
LOCATE AS SHOWN.

205-076-135 RESERVOIR
TOTAL SYSTEM CAP 10 U.S. PTS = 4.7 LITERS
RESERVOIR CAP. 5.3 U.S. PTS = 2.5 LITERS
REFILL LEVEL CAP. 2.5 U.S. PTS = 1.2 LITERS
USE MIL-PRF-87257 HYDRAULIC FLUID
NOTE
WHEN FLUID LEVEL IS VISIBLE
FILL TO OVERFLOW

70. DECAL 412-075-818-109.
LOCATE ON THE HYDRAULIC RESERVOIRS.
(S/N 37002 THROUGH 37999).

INSTRUCTIONS FOR USE OF
HYDRAULIC GROUND TEST STAND
FLUID PER MIL-PRF-87257

TEST STAND SETTINGS	
USE TEST STAND RESERVOIR	
RELIEF VALVE	1300 PSI
VOLUME OUTPUT	6 GPM
PRESSURE COMPENSATOR	1000 PSI

71. DECAL 412-075-818-111.
LOCATE AS SHOWN. LEFT SIDE ONLY.
(S/N 37002 THROUGH 37999).



72. DECAL 412-075-818-107.
LOCATE ON OVERHEAD CIRCUIT BREAKER
PANEL PILOT SIDE.
(S/N 37002 THROUGH 37999).



73. DECAL SG8833-1.
LOCATE INSIDE LEFT AND RIGHT
PASSENGER DOORS
(S/N 37002 THROUGH 37999).

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Figure 1. Decal and Stencil Markings, Forward (Sheet 18 of 19)

PUSH TO MUTE

74. DECAL 412-075-818-105. LOCATE AS SHOWN ON INSTRUMENT PANEL.

PUSH TO RESET

75. DECAL 412-075-818-103. LOCATE AS SHOWN ON INSTRUMENT PANEL.

DO NOT OPERATE HEATER ABOVE 21°C OUT AIR TEMP

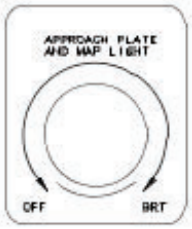
76. DECAL 412-075-818-117. LOCATE AS SHOWN ON INSTRUMENT PANEL.

THIS ROTORCRAFT IS APPROVED FOR DAY/NIGHT VFR/IFR, NON-ICING OPERATION

77. DECAL 412-075-818-119. LOCATE AS SHOWN ON INSTRUMENT PANEL.

CRTSY LTG

78. DECAL 412-075-818-101. LOCATE AS SHOWN ON GSE MAINTENANCE PANEL. RIGHT SIDE.



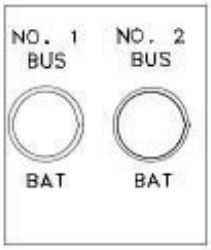
79. DECAL 412-075-818-113. LOCATE AS SHOWN ON INSTRUMENT PANEL.

GSE MAINT PORT NO.1

80. DECAL 31-006-528. LOCATE AS SHOWN ON GSE MAINTENANCE PANEL.

GSE MAINT PORT NO.2

81. DECAL 31-006-529. LOCATE AS SHOWN ON GSE MAINTENANCE PANEL.



82. DECAL 212-075-419-001. LOCATE AS SHOWN ON GSE MAINTENANCE PANEL.

CAUTION
COMPASS UNRELIABLE WITH WINDSHIELD HEAT OR WIPERS ON

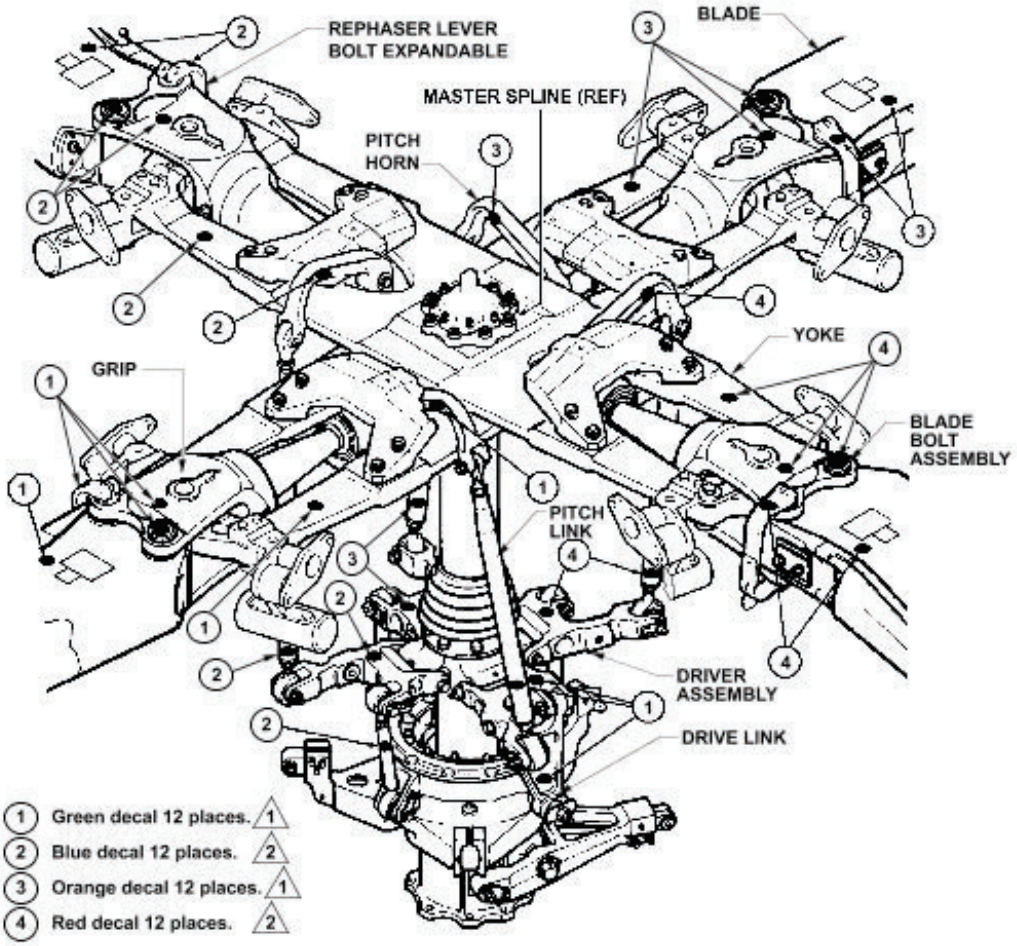
83. DECAL 31-030-277F12. LOCATE AS SHOWN.

NOTES

1. Apply to cover Collins decal.
2. Part of the airline seat kit (412-706-028).
3. Remove decal when airline seat kit (412-706-028) is installed.

ICN-412-A-110000-B-97499-00019-A-001-01

Figure 1. Decal and Stencil Markings, Forward (Sheet 19 of 19)



NOTES

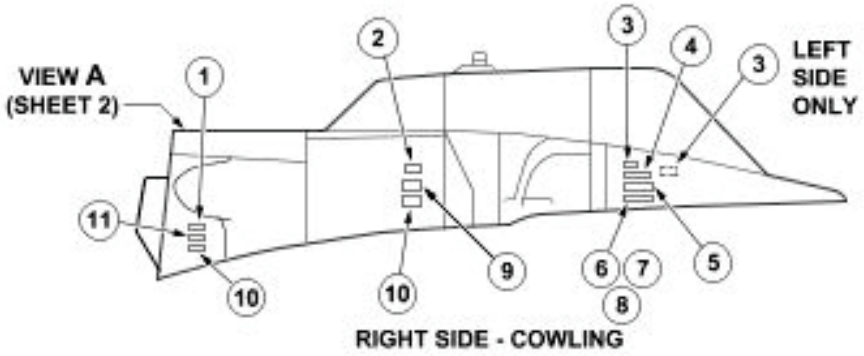
1 Locate spot decals on pitch horn, pitch links (two places), driver assembly, drive link, yoke, upper and lower grip, upper and lower surface of blade, rephaser lever bolt expandable, and blade bolt assembly to ensure that the parts are reinstalled in same position.

2 Locate spot decals on pitch horn, pitch links (two places), driver assembly, yoke, upper and lower grip, upper and lower surface of blade, rephaser lever bolt expandable, and blade bolt assembly to ensure that the parts are reinstalled in same position.

3. Ensure that the red decals are applied to main rotor blade and hub adjacent to the main rotor hub master spline.

ICN-412-A-110000-B-97499-00020-A-001-01

Figure 2. Decal Markings, Main Rotor (Sheet 1 of 1)



(1) GEAR BOX
FILLER CAP ON TOP

Decal -2 places. Locate both sides engine oil cooler cowling (upper decal).

(2) ENGINE OIL
FILLER CAP INSIDE

Decal -2 places. Locate both sides engine cowling (upper decal).

(3) HYDRAULIC RESERVOIR INSIDE

Decal -2 places. Locate both sides forward pylon fairing (upper decal).

(4) TRANSMISSION
OIL ACCESS

Decal -1 place. Locate right side forward pylon fairing (under decal 3).

(5) TRANSMISSION OIL
CAPACITY 11.0 US QTS = 10.4 LITERS

Decal -1 place. Locate right side forward pylon fairing (under decal 4) (S/N 33001 through 33213, S/N 36008 through 36999, and S/N 37002 through 37999).

(6) SERVICE WITH MIL-L-7808 OIL
OR MIL-L-23699 OIL
DO NOT MIX
SEE FLIGHT MANUAL

Decal -1 place. Locate right side of forward pylon fairing (under decal 5) (S/N 33001 to 33213 and 36001 to 36019).

(7) SERVICE WITH DOD-L-85734 (AS)
AEROSHELL TURBINE OIL 555,
EXXON TURBO OIL 25, OR
ROYCO OIL 555.
DO NOT MIX
SEE FLIGHT MANUAL

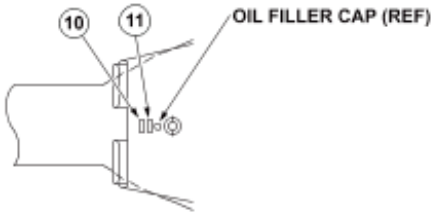
Decal -1 place. Locate right side forward pylon fairing (under decal 5) (S/N 36020 to 36247).

(8) SERVICE XMS WITH DOD-L-85734
(AS) AEROSHELL TURBINE OIL 555
EXXON TURBO OIL 25, OR
ROYCO OIL 555.
DO NOT MIX
SEE FLIGHT MANUAL

Decal -1 place. Locate right side forward pylon fairing (under decal 5) (S/N 36248 through 36999 and S/N 37002 through 37999).

ICN-412-A-110000-B-97499-00021-A-001-01

Figure 3. Decal Markings, Cowling (Sheet 1 of 2)



VIEW A
TOP LOOKING DOWN

9

ENGINE OIL CAPACITY 6.4 US QTS
= 6.0 LITERS

Decal -2 places. Locate both sides engine cowling (center decal) (S/N 33001 through 33213, S/N 36008 through 36999, and S/N 37002 through 37999).

10

SEE FLIGHT MANUAL FOR
APPROVED LUBRICANTS

Decal -5 places. Locate both sides of engine oil cooler cowling and engine cowling (lower decal) and on top of reduction gearbox cowling (Detail A aft decal) (S/N 36302 through 36999 and S/N 37002 through 37999).

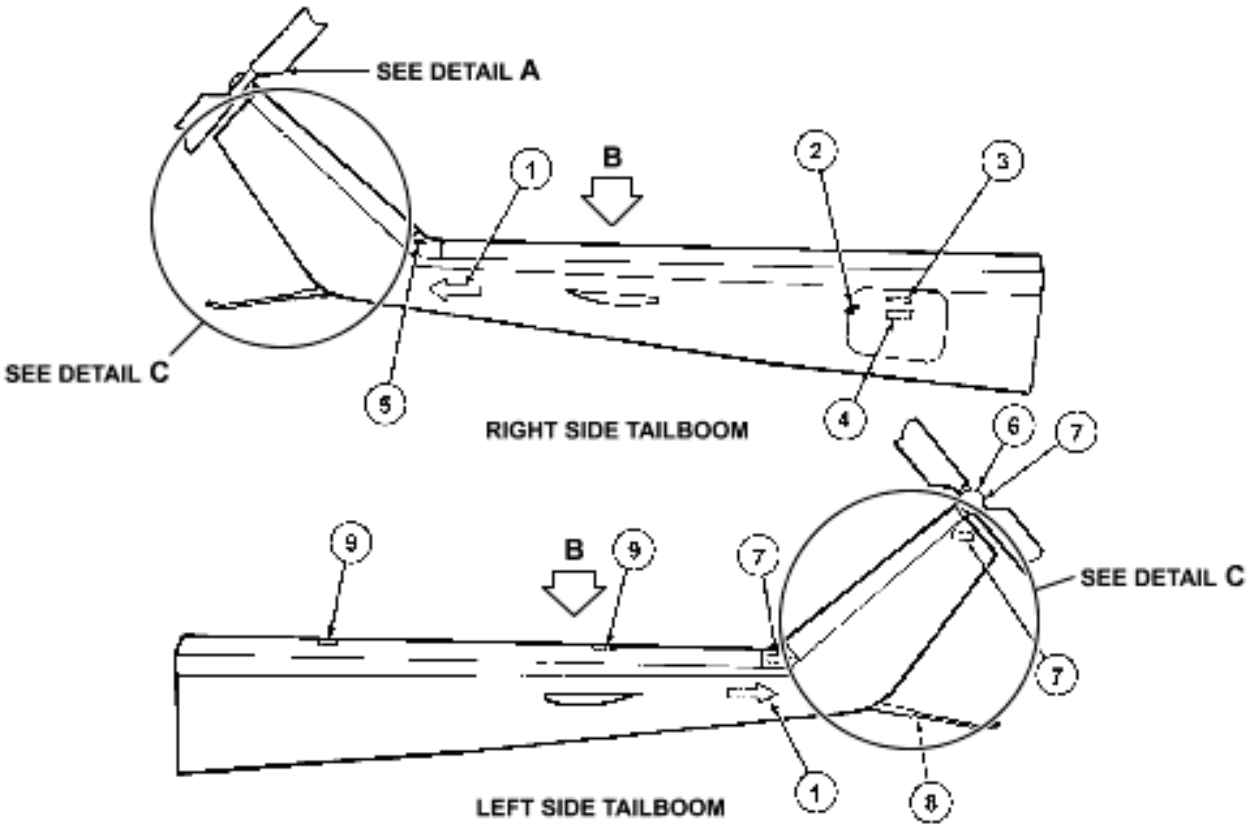
11

COMBINING GEARBOX
OIL CAPACITY 5.0 US QTS = 4.7 LITERS

Decal -3 places. Locate both sides engine oil cooler cowling (center decal) and on top of reduction gearbox cowling (Detail A forward decal) (S/N 33001 through 33213, S/N 36008 through 36999, and S/N 37002 through 37999).

ICN-412-A-110000-B-97499-00022-A-001-01

Figure 3. Decal Markings, Cowling (Sheet 2 of 2)



(1) DANGER

(Left side shown, right side opposite)

Decal -1 place (left), 1 place (right). Locate on center line of tailboom. 21.5 to 22.5 inches (546.1 to 571.5 mm) from aft edge of elevator to forward end of decal.

(2)

CLOSE OPEN

Decal -1 place. Locate outside baggage door above handle.

(3)

ALL CARGO MUST BE LOADED IN ACCORDANCE WITH FLIGHT MANUAL INSTRUCTIONS

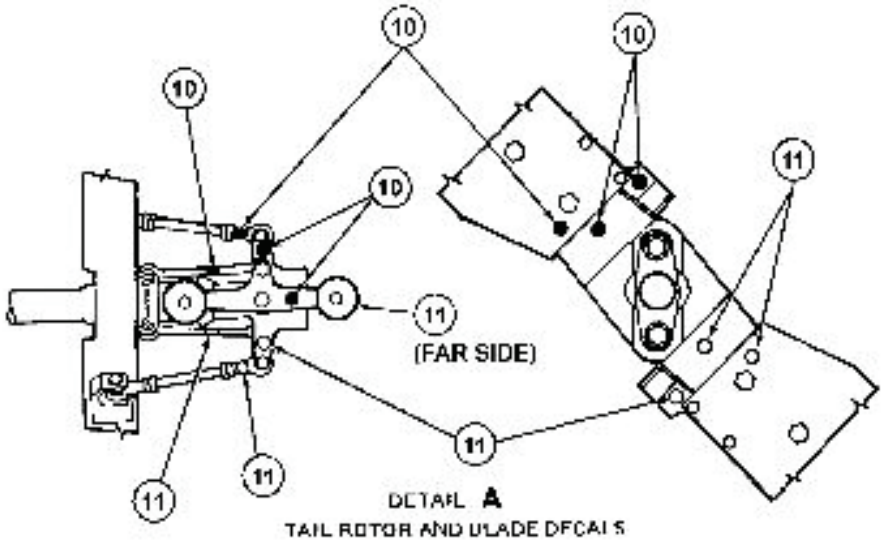
Stencil -1 place. Locate inside baggage door. Letters 0.5 inch (12.7 mm) high. Black color No. 17038 on white background No. 17875 per Fed Std. 595.

*(4) maxim um all owab le
wei tht **400** l bs.
100 l bs. per sq ft

Stencil -1 place. Locate inside baggage door. Letters 0.5 inch (12.7 mm) high. Black color No. 17038 on white background No. 17875 per Fed Std. 595.

ICN-412-A-110000-B-97499-00023-A-001-01

Figure 4. Decal and Stencil Markings, Aft (Sheet 1 of 3)



DETAIL A
TAIL ROTOR AND BLADE DECALS

*(5)

42° GEAR BOX OIL
CAPACITY 0.2 US QTS= 0.2 LITERS

Decal -1 place. Locate right side cover.

*(6) 90° GEAR BOX OIL
CAPACITY 0.4 QTS = 0.4 LITERS

Decal -1 place. Locate on top of gearbox.

(7)

BEFORE WITH DOOL-
DETAIN, ATTACH TI.
[illegible] - OIL SEE [illegible]
[illegible] OIL 2h, OR [illegible]
OIL SEE DO NOT [illegible] SEE
FLIGHT MANUAL

Decal -3 places. Locate on top of 90 degree gearbox, left side fin and left side cover.

(8)

BALLAST INSIDE

Decal -1 place. To be applied when ballast installed inside tailskid (See Chapter 8). Locate on top of tailskid 1.0 inch (25.4 mm) aft of tailskid step. In lieu of decal, stencil may be used. Letters 0.25 inch (6.35 mm) high, gloss black on transparent background.

(9) INSTALL SHAFT CLAMPS INDEXED 90° TO EACH OTHER TORQUE TO 30-35 IN.-LB

Decal -2 places. Locate inside top center of driveshaft cover adjacent to clamp.

(10)

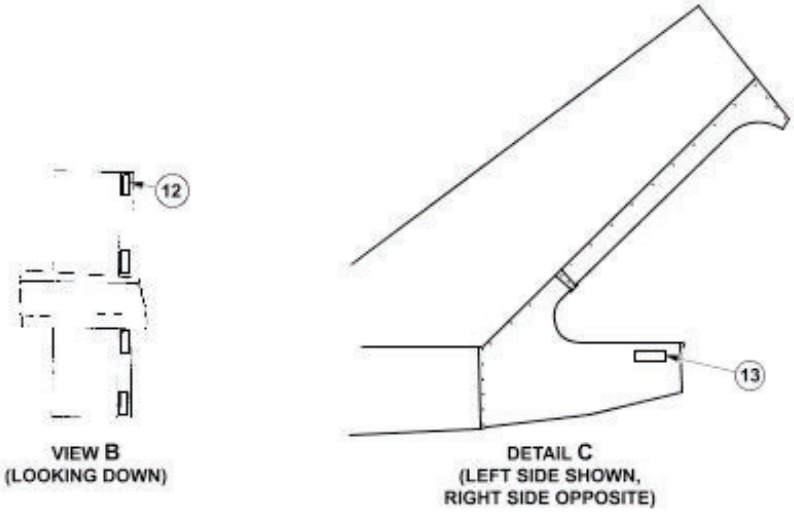
Red ● decal -7 places. See detail A.

(11)

White ○ decal -7 places. See detail A.

ICN-412-A-110000-B-97499-00024-A-001-01

Figure 4. Decal and Stencil Markings, Aft (Sheet 2 of 3)



(12) NO PUSH

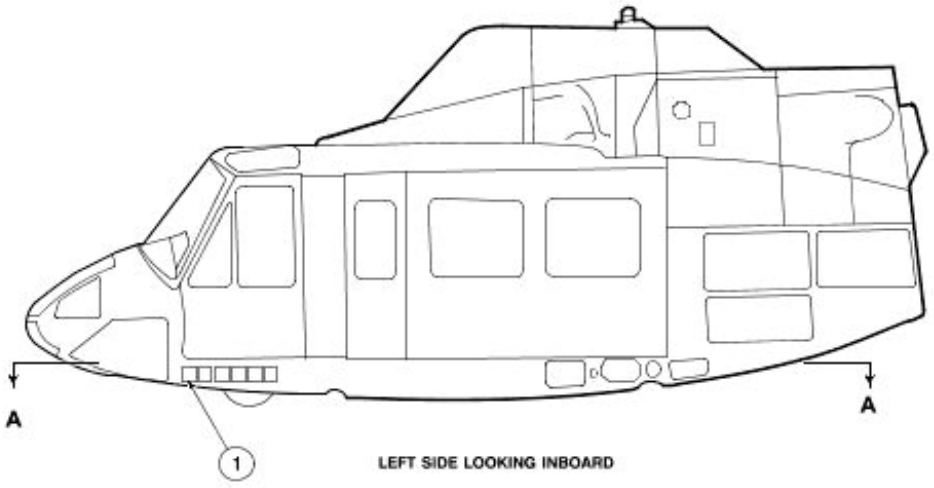
Decal - 4 places. Locate approximately as shown on upper surface of elevator trailing edge. (S/N 33121 through 33213, S/N 36001 through 36999, and S/N 37002 through 37999).

(13) NO PUSH

Decal - 2 places. Locate approximately as shown on each side of the lower tail fairing. (For helicopters equipped with the fast fin system kit 412-705-040-101).

ICN-412-A-110000-B-97499-00025-A-001-01

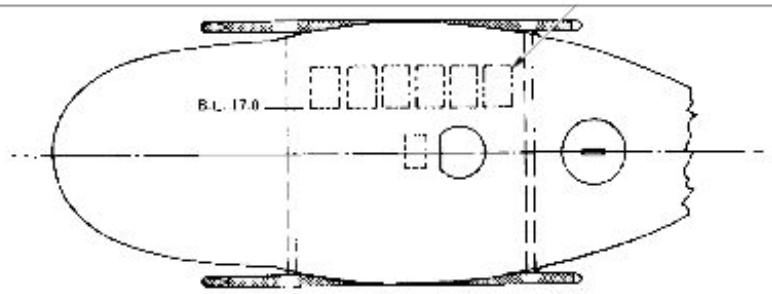
Figure 4. Decal and Stencil Markings, Aft (Sheet 3 of 3)



(1) Identification No. stencil located both sides below crew door. 4.0 inches (101.6 mm) high 2.6 inches (66.04 mm) wide, 0.6 inch (15.24 mm) thick, and 0.6 inch (15.24 mm) spacing.

SIX CHARACTER IDENTIFICATION MARKING STENCILS TO BE 16.0 INCHES (406.4 mm) HIGH, 10.7 INCHES (271.78 mm) WIDE, EXCEPT THE NUMBER "1" IS TO BE 2.7 INCHES (68.58 mm) WIDE AND THE LETTERS "M" AND "N" ARE TO BE 16.0 INCHES (406.4 mm) WIDE, 2.6 INCHES (66.04 mm) SPACING, CHARACTER THICKNESS TO BE 2.7 INCHES (68.58 mm).

FIVE OR LESS CHARACTER STENCILS TO BE 20.0 INCHES (508.00 mm) HIGH, 13.3 INCHES (337.82 mm) WIDE EXCEPT THE NUMBER "1" IS TO BE 3.3 INCHES (83.82 mm) WIDE AND THE LETTERS "M" AND "N" ARE TO BE 20.0 INCHES (508.00 mm) WIDE, 3.3 INCHES (83.82 mm) SPACING, CHARACTER THICKNESS TO BE 3.3 INCHES (83.82 mm). LOCATE WITH TOP OF CHARACTERS NEAR LEFT SIDE CENTER BETWEEN LANDING GEAR CROSS TUBES.



VIEW A-A

NOTE

Location and dimensions shown are for reference only. Refer to local regulations for actual location and dimensions of identification markings.

ICN-412-A-110000-B-97499-00026-A-001-01

Figure 5. Identification Markings (Sheet 1 of 1)

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Placards and Markings - Decals - Application

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Polyurethane enamel	C-233	AR	
Aliphatic naphtha	C-305	AR	
Toluene	C-306	AR	
Edge sealer	C-349	AR	
Isopropyl alcohol	C-385	AR	
Masking tape	C-426	AR	
Clean cloth	C-516	AR	

Procedure

NOTE

Bonding surfaces must be nonporous.

1. Remove dirt, grease, wax, or other contaminants from the surfaces to be bonded, as follows:

1.1. For non-metallic and painted surfaces, clean with a clean cloth moistened with aliphatic naphtha (C-305). Dry the surfaces with a clean cloth (C-516) before the aliphatic naphtha evaporates.



DO NOT ALLOW toluene (C-306) TO MAKE CONTACT WITH PAINTED OR NON-METALLIC SURFACES. DAMAGE TO THE SURFACE MAY OCCUR.

1.2. For bare metal surfaces, clean with a clean cloth moistened with toluene (C-306). Dry the surfaces with a clean cloth (C-516) before the toluene evaporates.

2. Apply the pressure-sensitive (adhesive-backed) decal to the bonding surface, as follows:

2.1. Apply the decal at temperatures above 60°F (16°C) for best results. Otherwise, first apply solvent (refer to the manufacturer's instructions) or isopropyl alcohol (C-385) to the decal.

2.2. Remove the adhesive protection from one edge of the decal with a quick smooth movement.

2.3. Carefully align the decal in the correct position.

2.4. Put the peeled edge of the decal on the edge of the bonding surface and apply firm pressure with your finger along the same edge.

NOTE

For the application of large decals, use a plastic squeegee with a firm pressure.

2.5. While you continue to remove the adhesive protection, apply finger pressure to the remaining part of the decal until the entire decal is applied. For the best result, hold the opposite edge away from the surface until the entire decal is applied.

2.6. Make sure that there are no air bubbles trapped under the decal. Otherwise, make a hole in the bubble with a pin and press with your finger or a squeegee to remove the air.

3. Edge seal or fully coat the decal with the applicable coating, as follows:

3.1. Apply masking tape (C-426) 1/8 inch (3 mm) away from the edge and all around the decal.

NOTE

A clear polyurethane enamel (C-233) may be used to seal decals applied to all types of exterior finishes.

3.2. For the sealing of exterior decals, apply the same material used for the exterior painting. For example, if the exterior paint is polyurethane enamel (C-233), then use a clear polyurethane enamel (C-233).

3.3. For the sealing of interior decals, apply a clear decal edge sealer (C-349).

3.4. Remove the masking tape (C-426) from around the decal when the coating is dry.

CHAPTER 12
SERVICING

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number		Issue date	Applicable to
Servicing - General	DMC-412-A-12-00-00-00A-040A-A / 00126	C	2025-11-18	ALL
Servicing Tolerance - Description	DMC-412-A-12-00-00-00A-042A-A / 00127		2023-05-31	ALL
Main Fuel System - Description	DMC-412-A-12-10-01-00A-042A-A / 00128		2023-05-31	ALL
Main Fuel System - Fueling/Defueling Preparation	DMC-412-A-12-10-01-00A-913A-A / 00129		2023-05-31	ALL
Main Fuel System - Fueling	DMC-412-A-12-10-01-00A-210A-A / 00130		2023-05-31	ALL
Main Fuel System - Defueling	DMC-412-A-12-10-01-00A-221A-A / 00131		2023-05-31	ALL
Main Fuel System - Purging Fuel Cells	DMC-412-A-12-10-01-00A-255A-A / 00132		2023-05-31	ALL
Main Fuel System - Long Range Fuel Cells (160 Gallon (606 L)), Kit No. 412-706-007-101/103 (If Installed)	DMC-412-A-12-10-01-00A-042B-A / 00133		2023-05-31	ALL
Engine Oil System - Description	DMC-412-A-12-10-05-00A-040A-A / 00134		2023-05-31	ALL
Transmission Oil System - Description	DMC-412-A-12-10-03-00A-040A-A / 00135		2024-12-19	ALL
Tail Rotor and Intermediate Gearbox Oil Systems - Description	DMC-412-A-12-10-04-00A-040A-A / 00136		2023-05-31	ALL
Tail Rotor and Intermediate Gearbox Oil Systems - Changing Oil Brands or Type	DMC-412-A-12-10-04-00A-292A-A / 00137		2023-05-31	ALL
Hydraulic Systems and Flight Controls - Description	DMC-412-A-12-10-02-00A-040A-A / 00138		2023-05-31	ALL
Rotor Brake - Servicing	DMC-412-A-12-10-06-00A-210A-A / 00139		2023-05-31	ALL
Hydraulic Fluid Conversion - Change of Specification	DMC-412-A-12-10-02-00A-292A-A / 00140		2023-05-31	ALL
Lubrication - Description	DMC-412-A-12-01-00-00A-040A-A / 00141	C	2025-11-18	ALL
Lubrication - Lubricant Restrictions (204-040-755-005)	DMC-412-A-12-01-00-00A-240A-A / 00142		2023-05-31	ALL
Lubrication - Flex Coupling Lubrication Log	DMC-412-A-12-01-00-00A-240B-A / 00143	C	2025-11-18	ALL

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Servicing - General

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	
Hydraulic fluid	C-072	AR	
Lubricating oil	C-030	AR	
Lubricating oil	C-011	AR	
Lubricating oil	C-010	AR	

Procedure



STRAINED OR DISCOLORED SIGHT GAUGE GLASSES MAY GIVE FALSE INDICATION OF OIL/FLUID QUANTITY. IF FALSE INDICATION IS SUSPECTED, SHAKE HELICOPTER BY TAIL SKID AND OBSERVE SIGHT GAUGE OIL/FLUID MOVEMENT. REPLACE ANY SIGHT GAUGE GLASS THAT DOES NOT PROVIDE ADEQUATE INDICATION OF FLUID LEVEL.

1. This chapter contains instructions to replenish fuel, lubricating oil, and hydraulic fluid, and to lubricate the helicopter. Defueling and fuel cell purging instructions are also included.

2. Figure 1 identifies servicing points, defueling, and drain valve locations. Table 1 specifies usable materials for fuel, lubricating oil, hydraulic fluid, battery, and fire extinguishers. Table 2 presents the drain map legend.

Table 1. Servicing Chart

SYSTEM	MATERIAL	REMARKS
Fuel	Refer to Flight Manual.	Comply with Note 1.
Engine Oil, Left and Right Power Sections and Combining Gearbox	Refer to Flight Manual.	Comply with Notes 2, 3, and 4.
Transmission Oil	Refer to Flight Manual.	Comply with Notes 4, 5, 6, 7, and 8.
Intermediate Gearbox	Refer to Flight Manual.	Comply with Notes 4, 5, 6, 7, and 9.
Tail Rotor Gearbox	Refer to Flight Manual.	Comply with Notes 4, 5, 6, 7, and 9.
Hydraulic Systems 1 and 2	hydraulic fluid (C-002) or hydraulic fluid (C-072)	Do not substitute other hydraulic fluids. Comply with Note 10.
Rotor Brake	hydraulic fluid (C-002) or hydraulic fluid (C-072)	Do not substitute other hydraulic fluids. Comply with Note 10.
Battery		Refer to BHT-ALL-SPM, Standard Practices Manual for instructions to service battery.
Fire Extinguishers, Engine and Portable		Refer to 412-A-26-20-00-00A-028A-A / 00308 for servicing of fire extinguisher.

NOTES:

1. Use fuels listed in authorized fuels list in appropriate BHT-412-FM and/or applicable Pratt and Whitney Canada Engine Maintenance Manual.

2. Use lubricating oils listed in Flight Manual and/or other oils listed in Pratt and Whitney Aircraft Engine Service Bulletin No. 5001 in the engine power sections and combining gearbox.

3. Change oil in engine power sections and combining gearbox at intervals stated in applicable Pratt and Whitney Canada Engine Maintenance Manual.

4. Make entry in helicopter and engine log books to indicate type of oil used to service engine, transmission, etc. to prevent subsequent servicing with different type of oil.

Table 1. Servicing Chart (continued)

SYSTEM	MATERIAL	REMARKS
SN: 33001-33213 or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed): *5. Use* lubricating oil (C-030) *or* lubricating oil (C-011) *(NATO-O-148). When not available, use* lubricating oil (C-010) *Type E and subsequent suffixes.*		
SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: *6. Use only* lubricating oil (C-030).		
7. Do not mix lubricating oil (C-010), lubricating oil (C-011), lubricating oil (C-030) type oils, or other lubricating oils. If oil does become mixed (by adding oil of a different type) or if oil type must be changed due to availability of oil currently being used, or oil change is required due to climatic conditions, comply with the following: a. Drain lubricating oil from system. Inspect and clean system oil filters and strainers. b. Fill oil system with lubricating oil to be used. c. Operate system for a period of 30 minutes to 1 hour to heat oil to operating temperature. Shut down engine. d. Inspect and clean system oil filters and strainers. If heavy contamination of filters and strainer is noted, repeat step a through step d. If little or no contamination of filters and strainers is noted, release helicopter for service and proceed with step e and step f. e. Operate helicopter for 5 hours and inspect and clean system oil filters and strainers. If no unusual contamination is noted, continue with step f. f. Operate helicopter for 10 hours (for a total of 15 hours after completion of step d) and inspect and clean system oil filters and strainers. If no unusual contamination is noted, return to normal inspection interval.		
8. Change transmission oil and replace external oil filter at 1/2 transmission TBO or 12 months, whichever comes first.		
9. Change oil in intermediate and tail rotor gearboxes at 300 hours or 180 days, whichever comes first.		
NOTE When adding hydraulic fluid, the same specification of hydraulic fluid already used in the hydraulic systems shall be used. However, in circumstances where emergency top-off or inadvertent mixing may occur, it is acceptable to use hydraulic fluid of the other approved specification. No further maintenance action will be required.		
10. Helicopters S/N 33001 through 33213, 34001 through 34036, and 36001 through 36636 were delivered with hydraulic fluid (C-002). For initial hydraulic fluid conversion from hydraulic fluid (C-002) to hydraulic fluid (C-072) and to retrofit helicopters with appropriate decals, refer to TB 412-12-229. Helicopters S/N 36637 and Subsequent are delivered with hydraulic fluid (C-072). To change the specification of hydraulic fluid, refer to 412-A-12-10-02-00A-292A-A / 00140. Make an entry in the helicopter log book to indicate the specification of hydraulic fluid used to service hydraulic systems and the rotor brake to prevent subsequent servicing with a different specification of hydraulic fluid.		

Table 2. Drain Map Legend

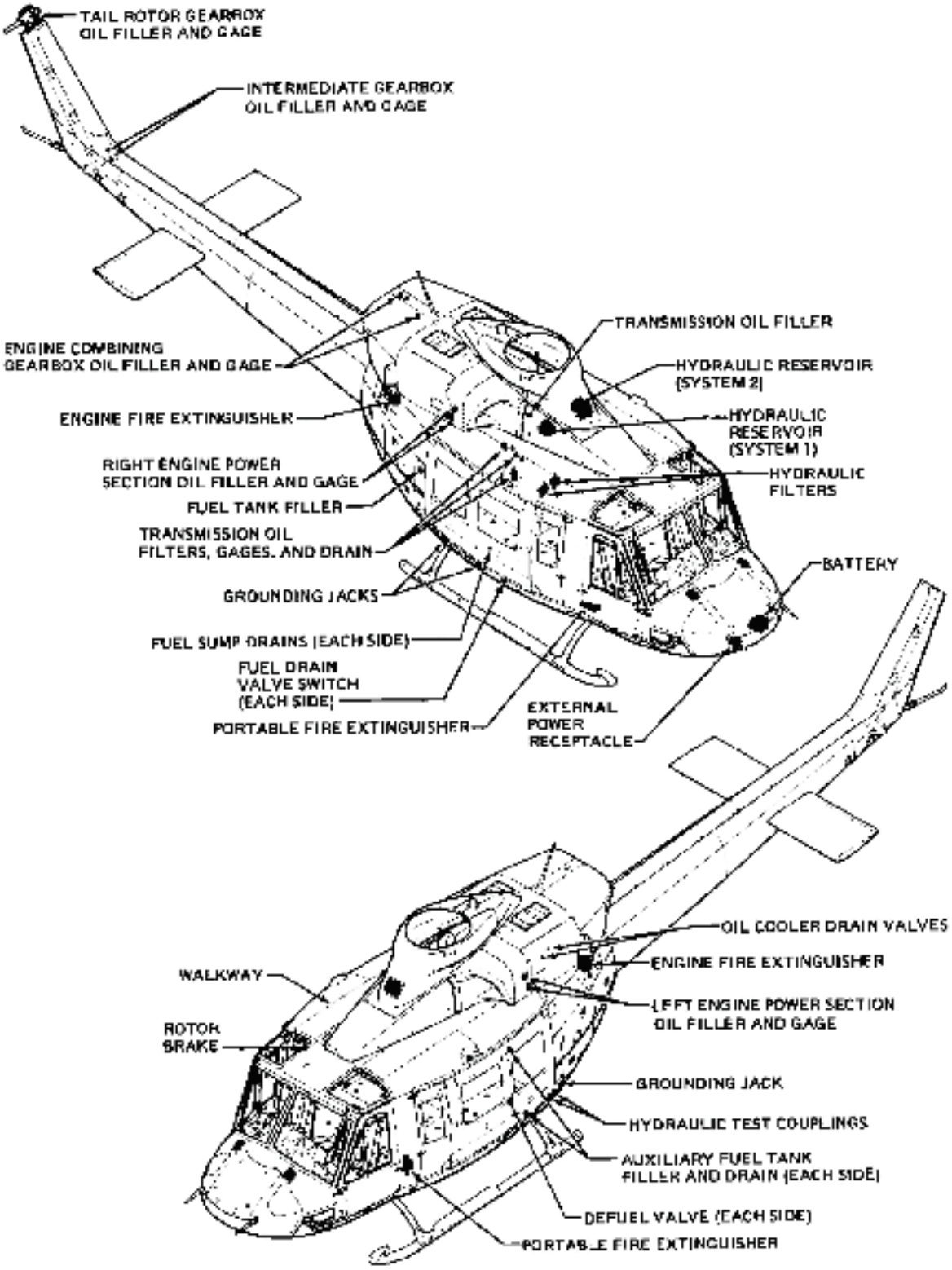
DRAIN NUMBER	DESCRIPTION	MEET IN/AT	LIQUIDS
1,2	Combining gearbox deck drain valve		Water
3	#1 Engine oil cooler drain valve #2 Engine oil cooler drain valve Combining gearbox oil cooler inlet line drain valve Combining gearbox oil cooler outlet line drain valve #1 Engine blower seal drain #2 Engine blower seal drain Combining gearbox oil filler scupper drain	Combining gearbox area and tail rotor servo compartment	Oil, Water
4	#1 Center engine deck drain		Water
5	#2 Center engine deck drain		Water

Table 2. Drain Map Legend (continued)

DRAIN NUMBER	DESCRIPTION	MEET IN/AT	LIQUIDS
6	#1 Engine burner drain valve #1 Engine exhaust pipe drain #2 Engine burner drain valve #2 Engine exhaust pipe drain Combining gearbox output seal drain	ADF panel and left side of combining gearbox deck	Fuel, Oil, Water
7	#1 Engine oil scupper drain #1 Engine accessory gearbox drain valve #1 Engine FCU seal drain #1 Engine started seal drain #2 Engine oil scupper drain #2 Engine accessory gearbox drain valve #2 Engine FCU seal drain #2 Engine starter seal drain	ADF panel and front of engine accessory gearbox area (corner panels in cabin)	Oil
8	Right main fuel cell vent system	Top of fuel cell cavity	Fuel
9	Left main fuel cell system	Top of fuel cell cavity	Fuel
10	#1 Engine fuel filter drain valve #1 Engine MFCU internal seal drains #2 Engine fuel filter drain valve #2 Engine MFCU internal seal drains	Engine accessory gearbox areas and ADF panel	Fuel
11	Combining gearbox deck drain		Water
12	#1 Center engine deck drain		Water
13	#2 Center engine deck drain		Water
14	Combining gearbox deck drain		Water
15, 16	Left main fuel cell cavity drains		Fuel, Water
17, 18	#1 Engine accessory gearbox deck drains		Water
19	Aft main fuel cell cavity drain		Fuel, Water
20, 21	#2 Engine Accessory gearbox deck drains		Water
22, 23	Main fuel cell cavity drains		Fuel, Water
24, 25	Right main fuel cell cavity drains		Fuel, Water
26, 27	Transmission oil cooler line drain valves		Oil
28	Aft main fuel cell cavity drain		Fuel
29	Transmission sump drain valve Transmission filler scupper drain Debris monitor scupper drain Transmission input seal drain	Under and around main transmission	Oil, Water
30	#1 Hydraulic oil reservoir scupper drain #1 Hydraulic pump seal drain	Forward of main transmission case	Oil, Hydraulic
31	#2 Hydraulic oil reservoir scupper drain		Oil, Hydraulic
32	#1 Center engine deck drain		Water
33	#1 Engine accessory gearbox deck drain		Water
34	#2 Center engine deck drain		Water
35	#2 Engine accessory gearbox deck drain		Water

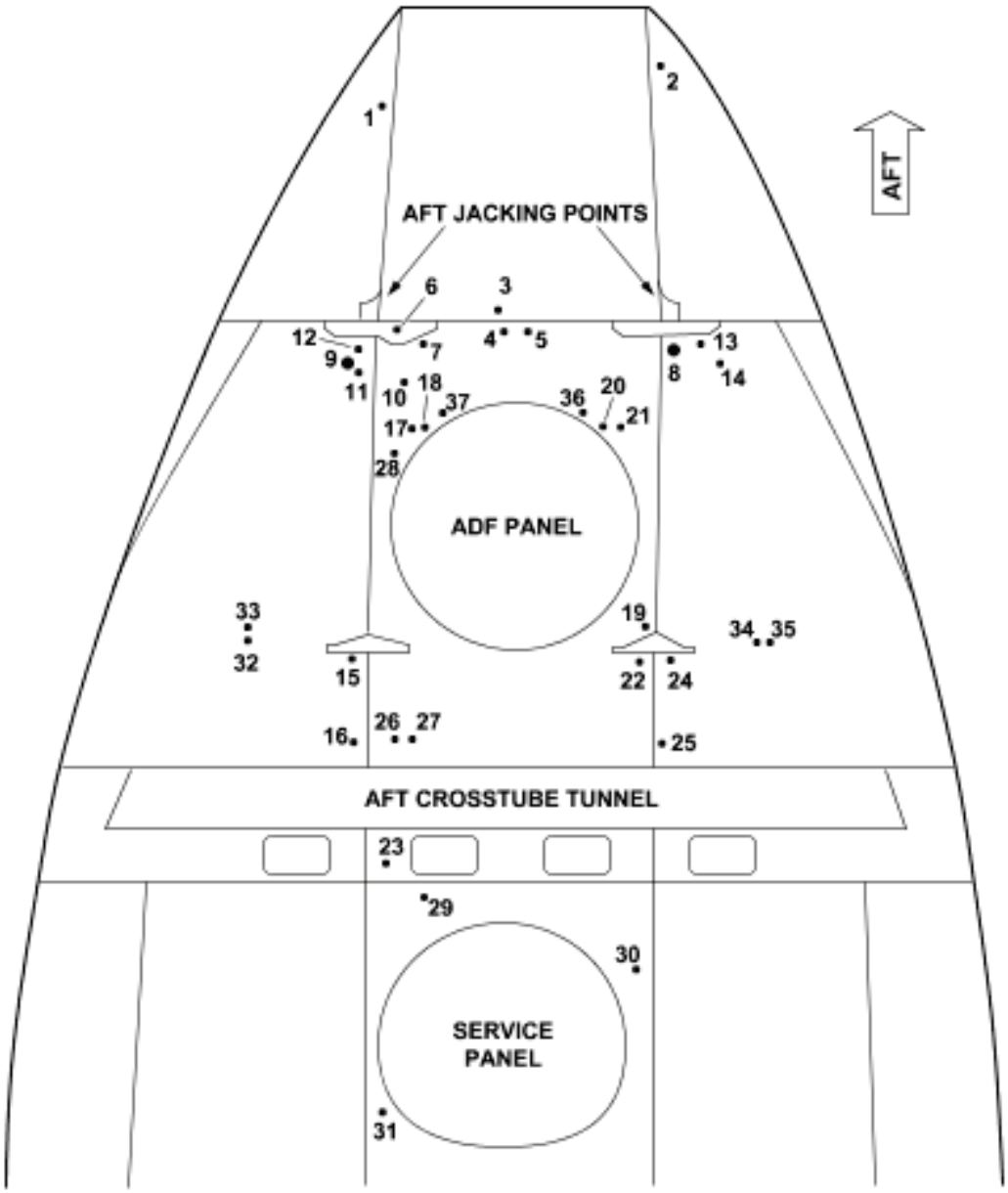
Table 2. Drain Map Legend (continued)

DRAIN NUMBER	DESCRIPTION	MEET IN/AT	LIQUIDS
36	#2 Center engine deck drain		Water
37	#1 Center engine deck drain		Water



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Figure 1. Servicing Locations and Belly Drain Map (Sheet 1 of 2)



DRAIN MAP
(BELLY, LOOKING UP)

NOTE

Drain locations are typical for 412EP S/N 36087 and subsequent.

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Figure 1. Servicing Locations and Belly Drain Map (Sheet 2 of 2)

Servicing Tolerance - Description



DO NOT APPLY THESE TOLERANCES IF THE HELICOPTER IS OPERATED IN EXTREME ENVIRONMENTAL CONDITIONS REQUIRING AN INCREASED FREQUENCY OF SERVICING. OTHERWISE, DAMAGE TO THE PARTS MAY OCCUR.

The Bell Helicopter Textron approved tolerance for 412-A-12-00-00-00A-040A-A / 00126 servicing intervals, unless otherwise stated, is 10% of the specified hourly or calendar time interval, whichever is less. The tolerances are established for maintenance scheduling convenience only.

Servicing intervals required beyond the stated tolerances must be approved by Product Support Engineering.

NOTE

The following is only applicable for those operators whose governing aviation authority requires to specifically approve the servicing tolerance.

If approval of the servicing tolerance is required by the applicable governing aviation authority, this is the responsibility of the owner/operator.

Refer to the Pratt & Whitney PT6T-3 Series Maintenance Manual for the engine servicing tolerances.

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Main Fuel System - Description

The main fuel system includes interconnected cells which are serviced through a single filler located on right side of helicopter. Two grounding jacks are located near the filler port.

An electrically operated sump drain valve, a manual sump drain valve, and a defuel valve are mounted on each sump plate. The sump plate is accessible through access panels on each side of the bottom of helicopter. Button-type fuel drain valve switches which actuate sump drain valves are mounted under passenger/cargo doors on each side of the helicopter.

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Main Fuel System - Fueling/Defueling Preparation

Procedure

WARNING

USE EXTREME CAUTION DURING FUELING/DEFUELING PROCEDURES. FUEL IS EXTREMELY FLAMMABLE AND MAY BE IGNITED BY STATIC OR FRICTION SPARKS, HOT EXHAUST PIPES, LIGHTED CIGARETTES, ELECTRICAL DEVICES, OR SIMILAR IGNITION SOURCES.

PERSONNEL SHOULD NOT BE ABOARD HELICOPTER DURING FUELING/DEFUELING PROCEDURES.

WHEN HELICOPTER FUEL COMES IN CONTACT WITH SKIN, A SOLVENT ACTION OCCURS WHICH REMOVES NATURAL FATS AND OIL THAT MAY EXPOSE THE SKIN TO INFECTIOUS DERMATOSIS. EXTENSIVE VAPOR INHALATION MAY CAUSE SERIOUS ILLNESS. ACCIDENTAL SWALLOWING OF FUELS WILL RESULT IN INTERNAL INJURY AND POSSIBLE DEATH. PERSONNEL WHO ARE SUBJECTED TO SPLASHED OR SPRAYED FUEL SHOULD REMOVE CONTAMINATED CLOTHING AS SOON AS POSSIBLE AND WASH DOWN/SHOWER WITH LARGE AMOUNTS OF WATER. DO NOT REMOVE CONTAMINATED CLOTHING NEAR POTENTIAL IGNITION SOURCES.

1. Allow only qualified personnel engaged in fueling/defueling operations in the area. Allow no one to carry matches, cigarette lighters, or other sparking or flame producing devices in the area.

2. Do not perform fueling/defueling operations when high winds are considered hazardous or when electrical storms are within 3 miles (5 km) of the area.

3. Do not allow personnel fueling/defueling the helicopter to wear shoes with protruding nails or taps, or static producing clothing such as nylon, rayon or wool.

4. Do not perform fueling/defueling near a drainage ditch or similar low place where combustible vapors may accumulate.

5. Do not perform fueling/defueling in a hangar. Position helicopter at least 50 feet (15 m) from any building or smoking area, and at least 500 feet (152 m) from any radar system.

6. Do not allow another aircraft to operate within 100 feet (30 m) of the area.

7. Position servicing unit as far from helicopter as hose will permit, and such that it may be driven or towed away in case of an emergency. Set parking brake.

8. Position a fully charged 50-pound (22 kg) CO_2 fire extinguisher with an extension assembly in an accessible place. Make sure fire extinguishers on service unit are readily available.

9. Maintain clear paths for immediate evacuation of personnel and vehicles in case of emergency.

10. Do not allow other aircraft within 20 feet (6 m) of helicopter.

11. If fuel cell filler cap is to be removed for any reason other than fueling, grasp helicopter grounding jack adjacent to fuel cell filler cap with bare hand to dissipate static electricity prior to removing cap.

WARNING

ACCOMPLISH THE BONDING PROCEDURES IN THE FOLLOWING STEPS IN ORDER LISTED OR STATIC ELECTRICITY MAY CAUSE FUEL TO IGNITE AND CAUSE INJURY OR DAMAGE TO EQUIPMENT.

12. Bond fueling/defueling equipment and helicopter as follows prior to removing fuel hose filler dust cap and/or helicopter fuel cell filler cap or prior to using defueling valve.

12.1. Bond fueling equipment to the helicopter by use of a cable.

12.2. Bond fuel hose nozzle with a nozzle bonding cable to one of the helicopter grounding jacks.

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Main Fuel System - Fueling

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	
Hydraulic fluid	C-072	AR	
Lubricating oil	C-030	AR	
Lubricating oil	C-011	AR	
Lubricating oil	C-010	AR	

Procedure

1. Accomplish fueling preparation steps outlined in 412-A-12-10-01-00A-913A-A / 00129.

2. Make sure helicopter electrical power is off. If electrical power is required, turn BATTERY BUS 1 and 2 switches ON and position remaining switches as required prior to start of fueling operation. Do not accomplish further switching until fueling is complete, except as required by an emergency condition.

3. Confirm fuel in fueling vehicle is correct type prior to refueling. Refer to 412-A-12-00-00-00A-040A-A / 00126 for list of approved fuels.

NOTE

Comply with bonding instructions in step 12 of 412-A-12-10-01-00A-913A-A / 00129 prior to accomplishing following steps.

4. Remove fuel hose nozzle dust cap.

5. Remove fuel cell filter cap.



DO NOT LEAVE NOZZLE UNATTENDED AT ANY TIME DURING REFUELING OPERATION.

6. Fill fuel cells and then remove hose nozzle from fuel cell.

7. Install filler cap and nozzle dust cap.

8. Place BATTERY BUS switches and required system switches to OFF, if required.

9. Remove fuel hose nozzle bonding cable from helicopter grounding jack.

10. Remove bonding cable from helicopter and fueling equipment.

11. In the event of a fire emergency, accomplish the following as quickly as possible:

11.1. Stop fuel flow.

11.2. Separate helicopter and service unit.

11.3. Sound alarm.

11.4. Attempt rescue and contain fire.

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Main Fuel System - Defueling

Procedure

1. Accomplish defueling preparation steps outlined in 412-A-12-10-01-00A-913A-A / 00129.

NOTE

If helicopter is being defueled with defueling type vehicle, accomplish Step 2. through Step 15..

If helicopter is being defueled with boost pumps, accomplish Step 16..

2. Notify fire department of defueling operation, location, anticipated start time, and estimated time of completion.

3. Make sure that all electrical power is off.

NOTE

Comply with bonding instructions in step 12 of 412-A-12-10-01-00A-913A-A / 00129 prior to accomplishing following steps.

4. If helicopter is being defueled because of an accident or incident, or if quality of fuel is questionable, sample and test drained fuel. Dispose of fuel or return fuel to storage as test results indicate.

NOTE

Comply with bonding instructions in step 12 of 412-A-12-10-01-00A-913A-A / 00129 prior to accomplishing following steps.

5. Make sure that all helicopter electrical power is off. If electrical power is required, turn BATTERY BUS switches ON and position other switches as required prior to starting defueling operation. Do not accomplish additional switching until defueling is complete, except as required by an emergency condition.

6. Remove filler cap and insert defueling tube/hose into helicopter fuel cell or attach defueling hose to defueling valve on sump with MS33656E12 fitting.

7. Pump fuel from helicopter. Stop pump as soon as fuel flow stops.

8. Remove defueling tube/hose and replace filler cap. Disconnect defuel tube/hose ground wire from helicopter.

9. Remove access plates under fuel cell sumps.

10. Position suitable container under helicopter.

11. Remove plugs from defuel valves.

12. If aft cells were defueled through filler cap in Step 5., attach defueling hose to defueling valve on sump with MS33656E12 fitting and remove remaining fuel.

13. Disconnect cable from helicopter and defueler equipment. Move defueler away from helicopter.

14. Dispose of fuel as noted in Step 4..

15. Purge fuel cells as outlined in 412-A-12-10-01-00A-255A-A / 00132.

16. Defuel with helicopter boost pumps as follows:

16.1. Accomplish Step 2. through Step 5..

16.2. Disconnect engine fuel pressure hose above engine deck. Connect a suitable hose to engine fuel pressure connection that will transfer fuel to a suitable container or servicing unit.

16.3. Open fuel cell interconnect valves and actuate appropriate boost pump to pump fuel out of helicopter.

16.4. Shut off boost pump as soon as fuel flow stops.

16.5. Attach defueling hose to defueling valve on sump with MS33656E12 fitting and remove remaining fuel.

16.6. Install hose disconnected in Step 16.2..

16.7. Disconnect grounding cables.

16.8. Dispose of fuel as noted in Step 4..

16.9. Purge fuel cells (412-A-12-10-01-00A-255A-A / 00132).

Main Fuel System - Purging Fuel Cells

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lubricating oil	C-009	AR	

Procedure



ON A STILL DAY (NO WIND CONDITION), FUEL VAPOR CAN ACCUMULATE IN AREA OF HELICOPTER (EVEN IN AN OPEN AREA), POSSIBLY PRODUCING AN EXPLOSIVE MIXTURE. UNDER THIS CONDITION, PERSONNEL SHOULD BE CLEARED FROM AREA EVEN AFTER PURGE LINES ARE INSTALLED AND INERT GAS IS FLOWING UNTIL 8 HOURS OF PURGING IS COMPLETED.

1. Defuel helicopter (412-A-12-10-01-00A-221A-A / 00131).

NOTE

Use dry, filtered compressed air or inert gas (nitrogen or carbon dioxide) as outlined in Step 2. and Step 3..

2. Purge fuel cells using dry, filtered, compressed air as follows:

2.1. Open all sump drains and drain lines.

2.2. Insert compressed air hose in cell filler inlet under cap and seal with masking tape.

2.3. Blow air through cell at 60 to 100 PSI (414 to 689 kPa) for a period of 8 hours.

2.4. Remove air hose from cell inlet. Wait 1 hour.

2.5. Check interior of fuel cell with explosimeter (combustible gas indicator calibrated for fuel vapors) for level of vapors.

2.6. Explosimeter reading shall be less than 20%. If reading is more than 20%, repeat Step 2.1. through Step 2.3. until reading is less than 20%.

3. Purge fuel cells using inert gas (nitrogen or carbon dioxide) as follows:



REMOVE FIBER HORN WHEN USING A FIRE EXTINGUISHER BOTTLE AS A SOURCE OF CARBON DIOXIDE (CO_2) FOR PURGING FUEL CELLS. GROUND NOZZLE TO HELICOPTER. DISCHARGE CO_2 INTO FUEL CELL SLOWLY (1 POUND PER MINUTE). RAPID PASSAGE OF GAS THROUGH A HOSE CAN GENERATE STATIC ELECTRICAL CHARGES. RAPID DISCHARGE OF GAS ALLOWS RAPID EXPANSION OF GAS (LOWERING THE TEMPERATURE WITH POSSIBLE DAMAGE TO CELL AS A RESULT). NITROGEN OR OTHER INERT GAS MAY BE USED WITH THE SAME PRECAUTIONARY MEASURES OUTLINED ABOVE.

3.1. Open all sump drains and drain lines.

3.2. Insert gas hose in cell filler inlet and seal with masking tape.

3.3. Blow inert gas through cells for a period of 8 hours.

3.4. Shut off inert gas flow and remove gas hose from cell inlet. Wait 1 hour.

3.5. Check interior of fuel cell with explosimeter (combustible gas indicator calibrated for fuel vapors) for level of fuel vapors.

3.6. Explosimeter reading shall be less than 20%. If reading is more than 20%, repeat Step 3.1. through Step 3.3. until reading is less than 20%.

4. Fog interior of fuel cell with lubricating oil (C-009). Close drains and replace filler cap. Do not close vents.

Main Fuel System - Long Range Fuel Cells (160 Gallon (606 L)), Kit No. 412-706-007-101/103 (If Installed)

Refer to BHT-412-SI-4, Auxiliary Fuel Kit (163.4 Gallon) for installation/maintenance instructions and 412-A-00-77-00-00A-942A-A for illustrated parts list.

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Engine Oil System - Description

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	
Hydraulic fluid	C-072	AR	
Lubricating oil	C-030	AR	
Lubricating oil	C-011	AR	
Lubricating oil	C-010	AR	

Procedure

1. The engine assembly has three independent oil systems, each with its own filler and oil level sight gauge. Fillers and gauges for left and right engine power sections are outboard of accessory gearboxes and are accessible by opening engine cowling. The third gauge is on aft side of reduction gearbox, with access through aft right engine cowl below exhaust ejectors. The third system oil filler is located in aft top fairing between exhaust ejectors. The third oil system cooler and strainer may be drained by opening drain valve located on engine deck. Use extreme care to avoid excessive oil spillage if chip detector plug is removed to completely drain system. Refer to 412-A-12-00-00-00A-040A-A / 00126 for list of approved lubricating oils and for location of fillers. Refer to 412-A-79-00-00-00A-028A-A / 01766 for additional engine oil system information.

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Transmission Oil System - Description

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	
Hydraulic fluid	C-072	AR	
Lubricating oil	C-030	AR	
Lubricating oil	C-011	AR	
Lubricating oil	C-010	AR	

Procedure

1. The transmission sump case serves as the reservoir for the transmission. The filler is located on upper right side of transmission and is accessible when forward pylon fairing is opened. Oil level sight gauges may be viewed through right side of pylon support structure in cabin. A sump drain valve is located directly beneath the sump. An external filter, in return line from oil cooler, is located inside pylon structure at right side and has a red pop-out indicator which is visible when filter is in impending bypass condition. Drain valves in oil cooler lines are accessible through bottom fuselage skin inboard of station 169.0. Refer to 412-A-12-00-00-00A-040A-A / 00126 for list of approved lubricating oils and for location of filler. Refer to 412-A-63-22-02-00A-028A-A / 01117 for additional transmission oil system information.

1.1. Transmission Oil System - Seepage Rates

Main transmission oil seepage shall not exceed 10 drops per minute from any single source or shall not exceed a total of 15 drops per minute from all sources on transmission.

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Tail Rotor and Intermediate Gearbox Oil Systems - Description

Procedure

1. Oil level in intermediate and tail rotor gearboxes may be checked on sight gauges. Oil is replenished through integral filler ports. Refer to 412-A-12-00-00-00A-040A-A / 00126 for list of approved lubricating oils and for location of fillers.

1.1. Tail Rotor and Intermediate Gearbox Oil Systems - Seepage Rates

Gearbox oil seepage shall not exceed two drops per minute from any single source or shall not exceed a total of six drops per minute from all sources on a gearbox.

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Tail Rotor and Intermediate Gearbox Oil Systems - Changing Oil Brands or Type

Procedure

1. If changing oil types, refer to Servicing Chart (412-A-12-00-00-00A-040A-A / 00126) for procedure and requirements.

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Hydraulic Systems and Flight Controls - Description

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	
Hydraulic fluid	C-072	AR	
Lubricating oil	C-030	AR	
Lubricating oil	C-011	AR	
Lubricating oil	C-010	AR	

Procedure

1. Access to hydraulic reservoirs is gained by opening forward pylon fairing. System 1 reservoir is on right side and System 2 reservoir is on left side. Pressure and return filters with red pop-out indicators which will be visible when filter requires change, are located in integrated valve and filter assemblies in pylon. System 1 integrated valve and filter assembly is located at lower front center of lift beam. System 2 integrated valve and filter assembly is on aft side of left beam.

NOTE

When adding hydraulic fluid, the same specification of hydraulic fluid already used in the system shall be used. However, in circumstances where emergency top-off or inadvertent mixing may occur, it is acceptable to use hydraulic fluid of the other approved specification. No further maintenance action will be required. To change the specification of hydraulic fluid used in the system, refer to 412-A-12-10-02-00A-292A-A / 00140.

2. For servicing fluids, refer to 412-A-12-00-00-00A-040A-A / 00126.

3. The filter may be checked from inside cabin area through an access door on forward side of pylon structure. A supplemental indicator is located to right of helicopter centerline at W.L. 22.0, and is visible through the lower right nose window. With engine operating and hydraulic circuit breakers closed, a green or black indication verifies systems are operating without clogging. Under the same circumstances, a red or alternating black and white indication warns of an impending bypass condition.

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Rotor Brake - Servicing

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	
Hydraulic fluid	C-072	AR	

Procedure

1. The rotor brake hydraulic system is a separate, self-contained system. Refer to 412-A-12-00-00-00A-040A-A / 00126 for approved hydraulic fluid and for location of fillers.

1.1. Fully apply rotor brake.

1.2. Remove reservoir filler cap (1, Figure 1) of rotor brake master cylinder (3), located on top, right forward area of cabin roof.

1.3. Check fluid level of reservoir is 0.50 inch (12.7 mm) below bottom of filler neck.

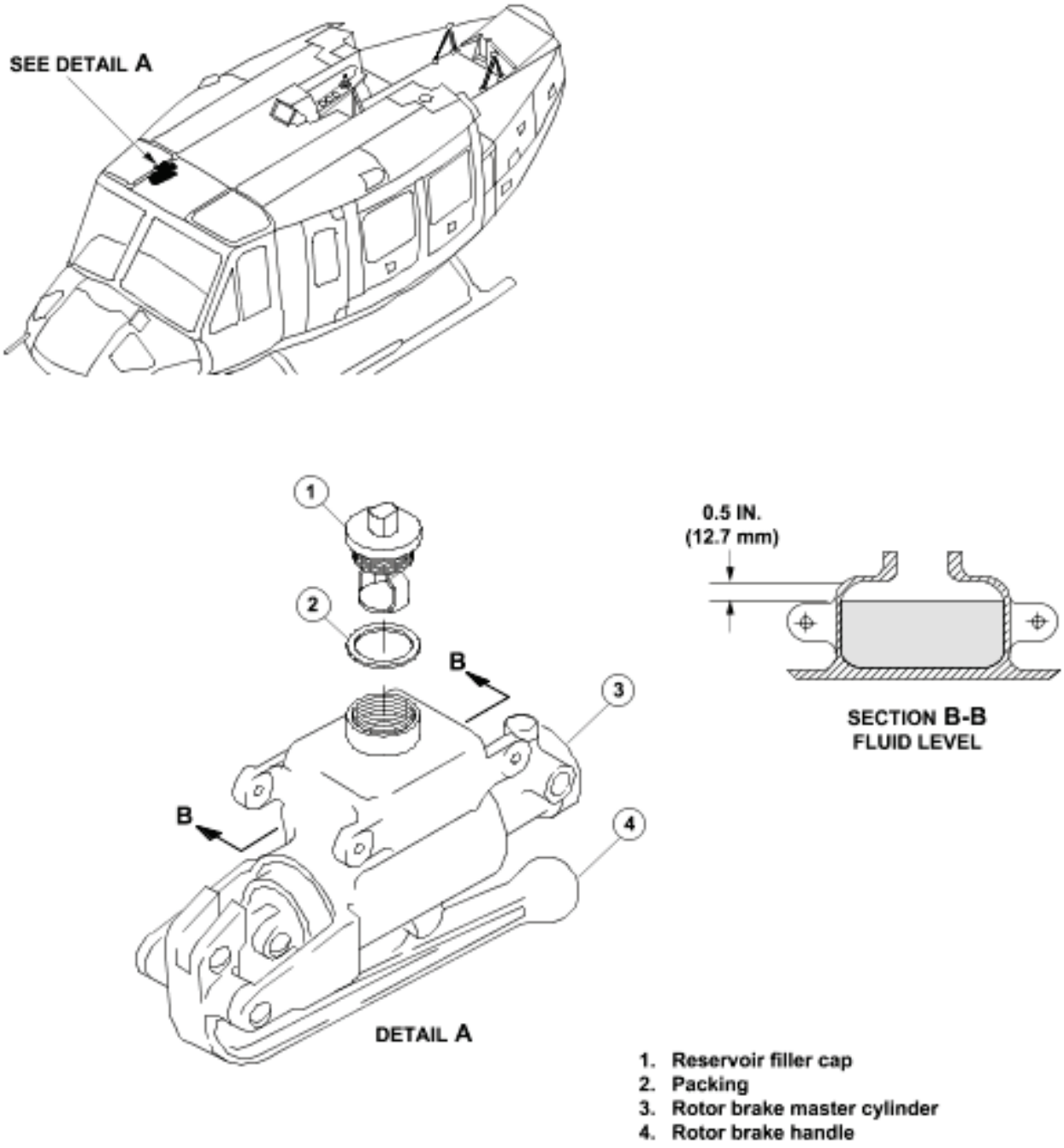
NOTE

When adding hydraulic fluid, the same specification of hydraulic fluid already used in the system shall be used. However, in circumstances where emergency top-off or inadvertent mixing may occur, it is acceptable to use hydraulic fluid of the other approved specification. No further maintenance action will be required. To change the specification of hydraulic fluid used in the system, refer to 412-A-12-10-02-00A-292A-A / 00140.

1.4. If fluid level is low, replenish with hydraulic fluid (C-002) or hydraulic fluid (C-072).

1.5. Install reservoir filler cap (1).

1.6. Release rotor brake by returning rotor brake handle (4) of rotor brake master cylinder (3) to stow position.



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Figure 1. Rotor Brake Servicing (Sheet 1 of 1)

Hydraulic Fluid Conversion - Change of Specification

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	
Hydraulic fluid	C-072	AR	

Procedure

NOTE

hydraulic fluid (C-002) and hydraulic fluid (C-072) are compatible and can be mixed.

NOTE

SN: 36637-Subsequent:

hydraulic fluid (C-072) was introduced in production, or as a customer option on previous S/Ns per TB 412-12-229. If a conversion from hydraulic fluid (C-002) to hydraulic fluid (C-072) is being accomplished for the first time, and as decal changes are required, accomplish the conversion in accordance with TB 412-12-229.

NOTE

In the event operators subsequently wish to convert their hydraulic systems to use hydraulic fluid (C-002) or hydraulic fluid (C-072), the following procedure is provided. Although the procedure is written with respect to converting from hydraulic fluid (C-002) to hydraulic fluid (C-072), the hydraulic fluid types specified in each step may be reversed to convert from hydraulic fluid (C-072) to hydraulic fluid (C-002).

1. Complete each of the three hydraulic fluid conversion phases as follows:

1.1. Phase 1:

NOTE

The hydraulic cart must conform to the requirements of 412-A-29-00-00-00A-042A-A / 00612.

1.1.1. Prepare the hydraulic cart as follows:

NOTE

If the hydraulic cart does not have dual system hoses, do the following procedure before doing Step 1.1.2. for the hydraulic system No. 1 and before doing Step 1.1.2. for the hydraulic system No. 2.

1.1.1.1. Drain the hydraulic cart until 50% of the hydraulic fluid (C-002) stays in the cart. Add hydraulic fluid (C-072) in the hydraulic cart until it is full.

NOTE

If the hydraulic cart does not have dual system hoses, do the following on the hydraulic system No. 1 and then repeat the procedure on the hydraulic system No. 2.

1.1.2. Do the following steps on the hydraulic system No. 1 and No. 2 and the flight controls:

1.1.2.1. Connect the hydraulic cart to the hydraulic test couplings of the hydraulic system No. 1 and No. 2 under the left sliding door of the helicopter.

1.1.2.2. Turn electrical power on or apply external power to the helicopter.

1.1.2.3. Turn the hydraulic cart on and apply 1000 PSI (6895 kPa) of pressure to both hydraulic systems.

1.1.2.4. Let the hydraulic cart run for 30 minutes.

1.1.2.5. Move the collective stick, cyclic stick, and pedals to operate all the flight control actuators through full travel for a minimum of five cycles.

1.1.2.6. Operate the HYDR SYS No. 1 and HYDR SYS No. 2 switches between the ON and OFF positions five times in approximately 2-second intervals.

1.1.2.7. Turn the hydraulic cart off.

1.1.2.8. Turn the electrical power off or disconnect external power from the helicopter.

1.1.2.9. Disconnect the hydraulic cart from the hydraulic test couplings of the hydraulic system No. 1 and No. 2.

1.1.3. For the rotor brake do as follows:

1.1.3.1. Remove the applicable doors and panels to get access to the bleed valves of the rotor brake assembly.

NOTE

There is more than one type of rotor brake assembly. Use the correct instructions.

1.1.3.2. Bleed each bleed valve of the brake assembly in accordance with the instructions provided in (412-A-63-23-00-01A-231A-A / 01207, or 412-A-63-23-00-01A-231B-A / 01208) then service the master cylinder with hydraulic fluid (C-072) (412-A-12-10-06-00A-210A-A / 00139).

1.1.3.3. Repeat Step 1.1.3.2..

1.1.3.4. Install the previously removed doors and panels.

1.2. Phase 2: After 3 Flight Hours or 2 Weeks Following Phase 1.

1.2.1. Prepare the hydraulic cart as follows:

NOTE

If the hydraulic cart does not have dual system hoses, do the following procedure before doing Step 1.2.2. for the hydraulic system No. 1 and before doing Step 1.2.2. for the hydraulic system No. 2.

1.2.1.1. Fully drain the hydraulic cart.

1.2.1.2. Fill the hydraulic cart to 75% of the total capacity with hydraulic fluid (C-072) and the remaining 25% with hydraulic fluid (C-002).

NOTE

If the hydraulic cart does not have dual system hoses, do the following on the hydraulic system No. 1 and then repeat the procedure on the hydraulic system No. 2.

1.2.2. Do the following steps on the hydraulic system No. 1 and No. 2 and the flight controls:

1.2.2.1. Connect the hydraulic cart to the hydraulic test couplings of the hydraulic system No. 1 and No. 2 under the left sliding door of the helicopter.

1.2.2.2. Turn electrical power on or apply external power to the helicopter.

1.2.2.3. Turn the hydraulic cart on and apply 1000 PSI (6895 kPa) of pressure to both hydraulic systems.

1.2.2.4. Let the hydraulic cart run for 30 minutes.

1.2.2.5. Move the collective stick, cyclic stick, and pedals to operate all the flight control actuators through full travel for a minimum of five cycles.

1.2.2.6. Operate the HYDR SYS No. 1 and HYDR SYS No. 2 switches between the ON and OFF positions five times in approximately 2-second intervals.

1.2.2.7. Turn the hydraulic cart off.

1.2.2.8. Turn the electrical power off or disconnect external power from the helicopter.

1.2.2.9. Disconnect the hydraulic cart from the hydraulic test couplings of the hydraulic system No. 1 and No. 2.

1.2.3. For the rotor brake do as follows:

1.2.3.1. Remove the applicable doors and panels to get access to the bleed valves of the rotor brake assembly.

NOTE

There is more than one type of rotor brake assembly. Use the correct instructions.

1.2.3.2. Bleed each bleed valve of the brake assembly in accordance with the instructions provided in (412-A-63-23-00-01A-231A-A / 01207, or 412-A-63-23-00-01A-231B-A / 01208), then service the master cylinder with hydraulic fluid (C-072) (412-A-12-10-06-00A-210A-A / 00139).

1.2.3.3. Repeat Step 1.2.3.2..

1.2.3.4. Install the previously removed doors and panels.

1.3. Phase 3: After 3 Flight Hours or 2 Weeks Following Phase 2.

1.3.1. Prepare the hydraulic cart as follows:

1.3.1.1. Fully drain the hydraulic cart.

1.3.1.2. Fill the hydraulic cart with hydraulic fluid (C-072) until it is full.

NOTE

If the hydraulic cart does not have dual system hoses, do the following on the hydraulic system No. 1 and then repeat the procedure on the hydraulic system No. 2.

1.3.2. Do the following steps on the hydraulic system No. 1 and No. 2 and the flight controls:

1.3.2.1. Remove the forward pylon fairing to get access to the hydraulic reservoirs.

1.3.2.2. Disconnect the hose from the suction outlet fitting of the hydraulic system No. 1 and No. 2 reservoirs and drain the hydraulic fluid in a suitable container.

1.3.2.3. Connect the hose to the suction outlet fitting of the hydraulic system No. 1 and No. 2 reservoirs. Tighten the hose.

1.3.2.4. Fill the hydraulic system No. 1 and No. 2 reservoirs with hydraulic fluid (C-072) up to the correct level.

NOTE

If the hydraulic cart does not have dual system hoses, do the following on the hydraulic system No. 1 and then repeat the procedure on the hydraulic system No. 2.

1.3.2.5. Connect the hydraulic cart to the hydraulic test couplings of the hydraulic system No. 1 and No. 2 under the left sliding door of the helicopter.

1.3.2.6. Disconnect the return hoses from the hydraulic cart and route them into an appropriate container to collect the discarded hydraulic fluid.

1.3.2.7. Turn electrical power on or apply external power to the helicopter.

NOTE

To avoid wasting fluid, perform all steps below in tight sequence without leaving the system pressurized unnecessarily. If stops are required to prepare the next step, turn the hydraulic cart off.

1.3.2.8. Turn the hydraulic cart on and apply 1000 PSI (6895 kPa) of pressure with a flow of 3 gal/min (11 L/min) only to the hydraulic system No. 1. The hydraulic system No. 2 must stay unpressurized.

1.3.2.9. Move the collective stick, cyclic stick, and pedals to operate all the flight control actuators through full travel for a minimum of five cycles.

1.3.2.10. Operate the HYDR SYS No. 1 and HYDR SYS No. 2 switches between the ON and OFF positions five times in approximately 2-second intervals.

1.3.2.11. Let the hydraulic cart run for 1 minute.

1.3.2.12. Turn the hydraulic cart off.

1.3.2.13. Turn the hydraulic cart on and apply 1000 PSI (6895 kPa) of pressure with a flow of 3 gal/min (11 L/min) only to the hydraulic system No. 2. The hydraulic system No. 1 must stay unpressurized.

1.3.2.14. Move the collective stick, cyclic stick, and pedals to operate all the flight control actuators through full travel for a minimum of five cycles.

1.3.2.15. Operate the HYDR SYS No. 1 and HYDR SYS No. 2 switches between the ON and OFF positions five times in approximately 2-second intervals.

1.3.2.16. Let the hydraulic cart run for 1 minute.

1.3.2.17. Turn the hydraulic cart off.

1.3.2.18. Connect the return hoses to the hydraulic cart.

1.3.2.19. Turn the hydraulic cart on and apply 1000 PSI (6895 kPa) of pressure with a flow of 5 gal/min (19 L/min) to both hydraulic systems.

1.3.2.20. Let the hydraulic cart run for 5 minutes.

1.3.2.21. Turn the hydraulic cart off.

1.3.2.22. Turn the electrical power off or disconnect external power from the helicopter.

1.3.2.23. If required, add hydraulic fluid (C-072) in the reservoirs of the hydraulic system No. 1 and No. 2 up to the correct level.

1.3.2.24. Replace the four hydraulic filter assemblies (412-A-29-10-00-05A-042A-A / 00638).

1.3.2.25. Disconnect the hydraulic cart from the hydraulic test couplings of the hydraulic system No. 1 and No. 2.

1.3.2.26. Install the previously removed doors and panels.

1.3.3. For the rotor brake do as follows:

1.3.3.1. Remove the applicable doors and panels to get access to the bleed valves of the rotor brake assembly.

NOTE

There is more than one type of rotor brake assembly. Use the correct instructions.

1.3.3.2. Bleed each bleed valve of the brake assembly in accordance with the instructions provided in (412-A-63-23-00-01A-231A-A / 01207, or 412-A-63-23-00-01A-231B-A / 01208), then service the master cylinder with hydraulic fluid (C-072) (412-A-12-10-06-00A-210A-A / 00139).

1.3.3.3. Repeat Step 1.3.3.2..

1.3.3.4. Install the previously removed doors and panels.

Lubrication - Description

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-561	AR	
Grease	C-172	AR	
Corrosion preventive compound	C-101	AR	
Grease	C-001	AR	

Procedure

1. Figure 1 illustrates each area that requires lubrication with grease-type lubricant. The legend of this illustration lists items to be lubricated. Table 1 lists required lubrication, interval, and lubricant type.

2. The intervals specified are the maximum permitted under normal operation. Do not exceed these intervals. It may be necessary to decrease the interval of servicing when the operating conditions are more severe.

2.1. After each day of operation in rain, snow, or after washing helicopter, all exposed control bearings should be purge lubricated to remove trapped moisture and ensure a lubricant film is applied to susceptible surfaces.

2.2. Parking helicopters outside in a heavy dew environment requires all exposed control bearings be purge lubricated every 7 days to ensure no voids exist, which could trap moisture.

2.3. If helicopter is stored for periods in excess of 45 days without operation or service, purge-lubricate all bearings.



PRIOR TO LUBRICATION, VERIFY THE CONTENT OF GREASE DISPENSING EQUIPMENT. IF THERE IS PARTIAL SEPARATION OF THE LUBRICATION OIL FROM THE THICKENING AGENT, OIL MAY BE REMIXED THOROUGHLY WITH THE THICKENING AGENT TO RESTORE THE ORIGINAL LOAD CARRYING CAPABILITIES. IF THERE IS A LOSS OF LUBRICATING OIL FROM THE THICKENING AGENT, DISCARD THE ENTIRE CONTENTS OF THE GREASE DISPENSING EQUIPMENT AND REPLACE WITH A SUITABLE SUPPLY OF APPROVED LUBRICANT (IL GEN-03-93).



DO NOT USE HIGH CAPACITY/HIGH VELOCITY GREASE GUNS ON THE COMPONENTS. THE USE OF HIGH CAPACITY/HIGH VELOCITY GREASE GUNS CAN RESULT IN DAMAGE.

3. If it becomes necessary to change the brand of grease, remove the old grease by purging until only new grease is present, except as otherwise specified.

4.

NOTE

Prior to and following grease lubrication, visually examine grease fittings for presence of the spring loaded steel ball. If the steel ball is not visible, does not spring back to the closed position, the grease fitting indicates excessive leakage, or it is difficult to inject grease during lubrication, replace the grease fitting (BHT-ALL-SPM, Chapter 8, Standard Practices Manual). If the grease fitting is removed due to the spring loaded steel ball not being visible, insert a small steel probe into the back of the grease fitting to confirm the presence of the steel ball. If it is identified that the steel ball is not present in the grease fitting, further investigation is required to ensure that the steel ball has not migrated into the component and caused damage. Discard grease fittings that have been removed.

Table 1. Lubrication Chart

INTERVAL NOMENCLATURE LUBRICANT REMARKS
25 hours, more frequent if required by environmental conditions

Table 1. Lubrication Chart (continued)

Tail rotor trunnion bearings (two)
Grease MIL-G-81322
Do not intermix grease types. Purge until same grease appears.
50 hours and on assembly
Tail rotor crosshead bearing
Grease MIL-G-81322
Two shots every 50 hours. Do not intermix grease types.
25 hours, more frequent if required by environmental conditions
Drive link (two places per link)
Grease MIL-G-81322
Do not intermix grease types. Purge until same grease appears.
25 hours, more frequent if required by environmental conditions
Rephasing lever bearings (four places)
Grease MIL-G-81322
Do not intermix grease types. Purge until same grease appears.
25 hours, more frequent if required by environmental conditions
Swashplate inner ring trunnion rod ends (two) and collective lever trunnion rod end (one)
Grease MIL-G-81322
Do not intermix grease types. Purge until same grease appears. Comply with Figure 1, note 9.
50 hours, more frequent if required by environmental conditions
Anti-torque flight control tube bearings
Grease MIL-G-81322
Fittings not present on all helicopters. Do not intermix grease types. Purge until same grease appears.
25 hours, more frequent if required by environmental conditions
Collective hub and sleeve (two fittings)
Grease MIL-G-81322
Do not intermix grease types. Purge until same grease appears.
25 hours, more frequent if required by environmental conditions
Swashplate bearings (two)
Grease MIL-G-81322
Do not intermix grease types. Purge until same grease appears.
100 hours
Cargo hook link assembly, clevis assembly, load bolt assembly

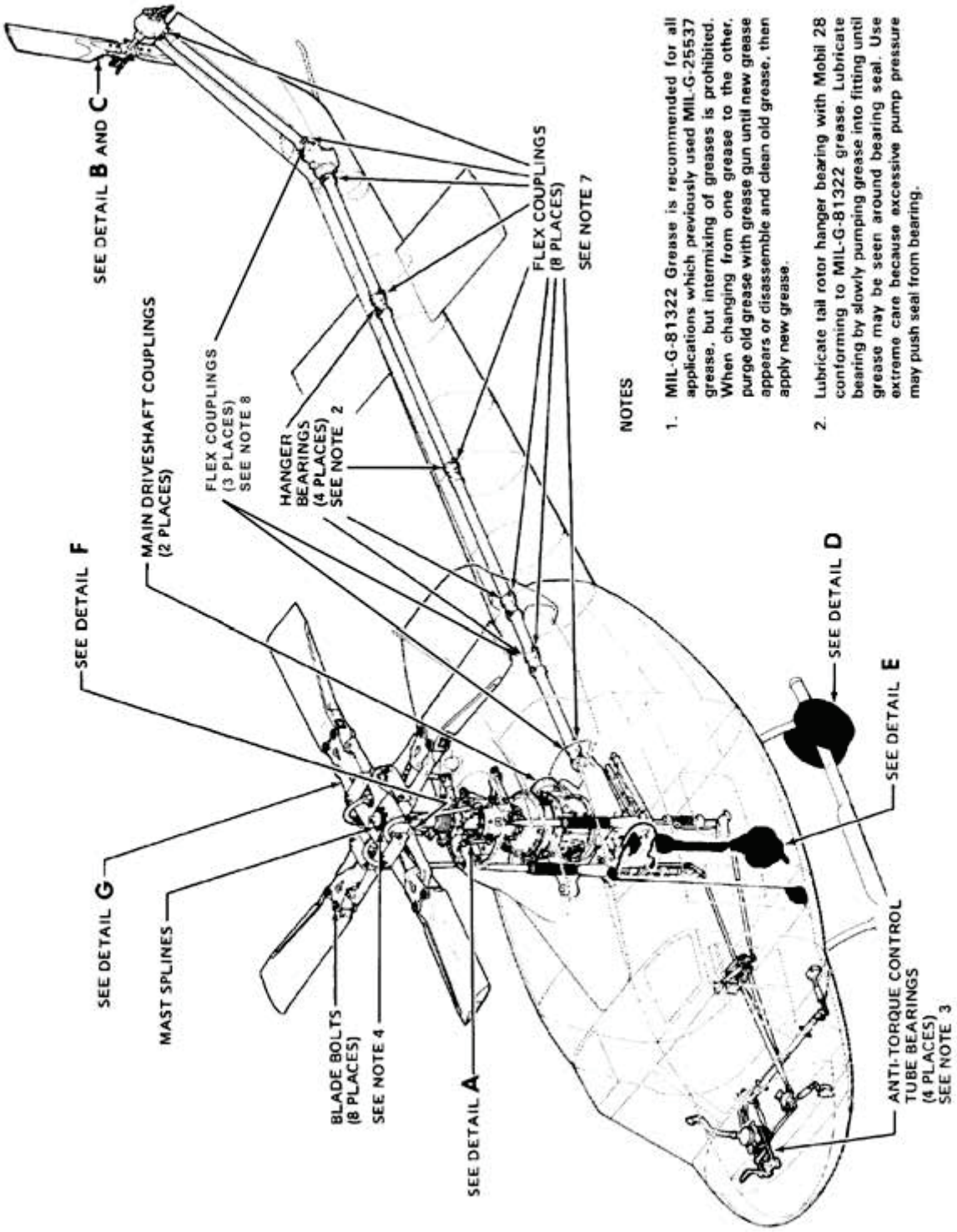
Refer to Service Instruction BHT-212-SI-5 and BHT-412-SI-17 (P/N 212-706-103-XXX).
300 hours
Tail rotor hanger bearings (four)
Grease - MOBIL 28 no substitution
Slowly pump grease into fitting until it may be seen around bearing seal.
600 hours/12 months (refer to 412-A-05-47-00-11A-283A-A / 00044 Special Inspections)
Tail rotor driveshaft flex couplings
Grease 204-040-755-005
Comply with Figure 1, notes 6, 7, and 8 as applicable.
600 hours/12 months (refer to 412-A-05-47-00-11A-283A-A / 00044 Special Inspections)
Intermediate gearbox flex couplings
Grease 204-040-755-005
Comply with Figure 1, notes 6, 7, and 8 as applicable.
600 hours/12 months (1) (refer to 412-A-05-47-00-11A-283A-A / 00044 Special Inspections)
Tail rotor gearbox flex coupling (one)
Grease 204-040-755-005
Comply with Figure 1, notes 6 and 7.
600 hours/12 months (refer to 412-A-05-47-00-11A-283A-A / 00044 Special Inspections)
Transmission tail rotor drive quill flex coupling (one)
Grease 204-040-755-005
Comply with Figure 1, notes 6, 7, and 8 as applicable.
600 hours/12 months (refer to 412-A-05-47-00-12A-283A-A / 00046 Special Inspections)
Main driveshaft couplings (two)
Grease 204-040-755-005
Comply with Figure 1, note 6.
On assembly
Main rotor blade bolts
grease (C-561) or grease (C-172)
After final torque, overcoat non-expanding blade bolt head, nut, washer, and exposed threads with corrosion preventive compound (C-101).
On assembly

Table 1. Lubrication Chart (continued)

Main rotor hub to mast
grease (C-561) or grease (C-172)
Apply thin coat of grease (C-561) or grease (C-172) to mating surfaces of mast splines, upper and lower cones, and cap assembly taking care that grease is not allowed on Teflon of cone or mating surfaces with Teflon.
On assembly
Mast and splined plate splines
grease (C-561) or grease (C-172)
Apply grease (C-561) or grease (C-172) to mast and splined plate splines.
On assembly
Tail rotor control tube
Lubricating oil MIL-PRF-7808 MIL-PRF-23699
Coat tube in area of bearing.
On assembly
Tail rotor control race
grease (C-001)
Hand pack. Do not intermix grease types.
On assembly
Tail rotor gearbox output shaft splines to tail rotor hub trunnion splines
grease (C-561) or grease (C-172)
Apply a thin coat of grease (C-561) or grease (C-172) to mating surface of tail rotor gearbox output shaft splines and tail rotor hub trunnion splines.
On assembly
Tail rotor crosshead splines
grease (C-001)
Apply a thin film of grease at assembly. Do not intermix grease types.
50 hours, more frequent if required by environmental conditions
Tail rotor control level (three places)
grease (C-001)
Do not intermix grease types. Purge until same grease appears.
On assembly
Ground handling wheel bearings
grease (C-001)
Do not intermix grease types. Purge until same grease appears.
On assembly and each 6 months
Ground handling wheel axle pivot bearing, trunnions, and pins
grease (C-001)
Do not intermix grease types. Purge until same grease appears.
On assembly and after each 25 hours of operation
Droop restraint arm, inboard clevis race two fittings (four places)
grease (C-001)
Do not intermix grease types. Purge until same grease appears.
Every 25 hours of operation
Bearing assembly, Pendulum absorbers main rotor hub (eight places)
grease (C-001)
Do not intermix grease types. Purge until same grease appears.

NOTE:

1 *Applicable to helicopters S/N 33001 through 33213, and S/N 36001 through 36019 not modified by 412-570-001-103 or Pre BHT-412-SI-74 (412SP to 412HP Upgrade) or Pre BHT-412-SI-97 (412 Classic to 412HP).*



ICN-412-A-120100-B-97499-00001-A-001-01

Figure 1. Lubricating Points (Sheet 1 of 4)

3. All models will not have lubrication fittings on anti-torque control tube bearings. Purge lubrication only required when grease fitting is installed.

4. Coat all bolted joints and faying surfaces on hub and swashplate with corrosion preventive compound MIL-C-16173 grade 1, external.

5. Some environmental conditions that require additional lubrication are listed below:

   If helicopter is operated in rain showers or snow, or if helicopter is washed, purge-lubricate all exposed control bearings to remove trapped moisture and ensure that a lube film is applied to all working surfaces.

   If helicopter is parked in a heavy dew environment, purge-lubricate all exposed control bearings every seven days to ensure that no voids exist that could trap moisture.

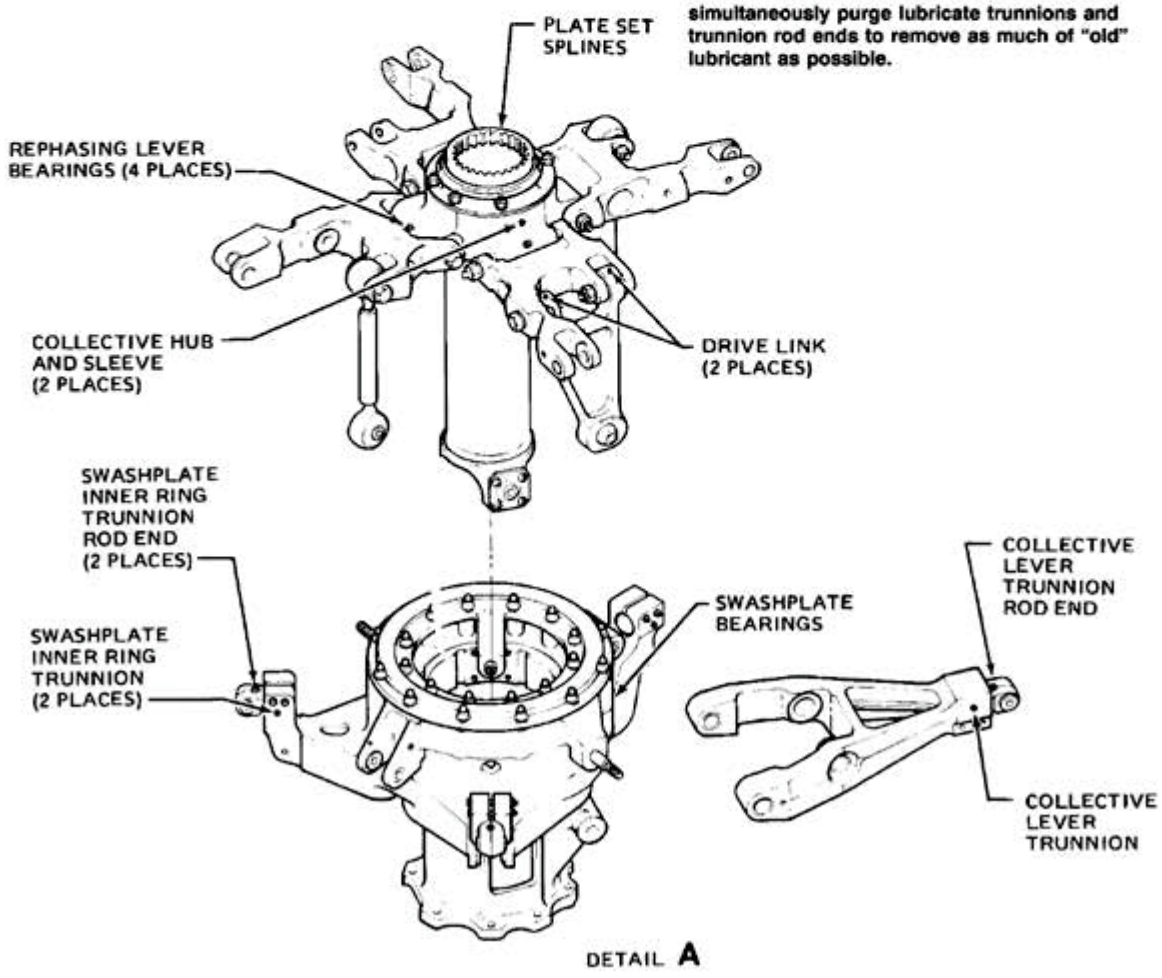
   If helicopter is stored for periods in excess of 45 days without operation or service, purge-lubricate all grease bearings.

6. Grease, 204-040-755-5, has a shelf storage life of 4 years whether stored in original container or in a component. If a component is not put in service prior to expiration of the 4 year shelf life of the grease, component shall be relubricated prior to installation on a helicopter. After initial operation of component on the helicopter, refer to lubrication chart for lubrication intervals. (The application of torque constitutes placement into service).

7. 600 hour/12 month interval applicable to helicopters S/N 33001 through 33213, and S/N 36001 through 36019 not modified by 412-570-001-103 or per BHT-412-SI-74 (412SP to 412HP upgrade) or per BHT-412-SI-97 (412Classic to 412HP upgrade). Refer to chapter 65 for instructions to lubricate flexible couplings.

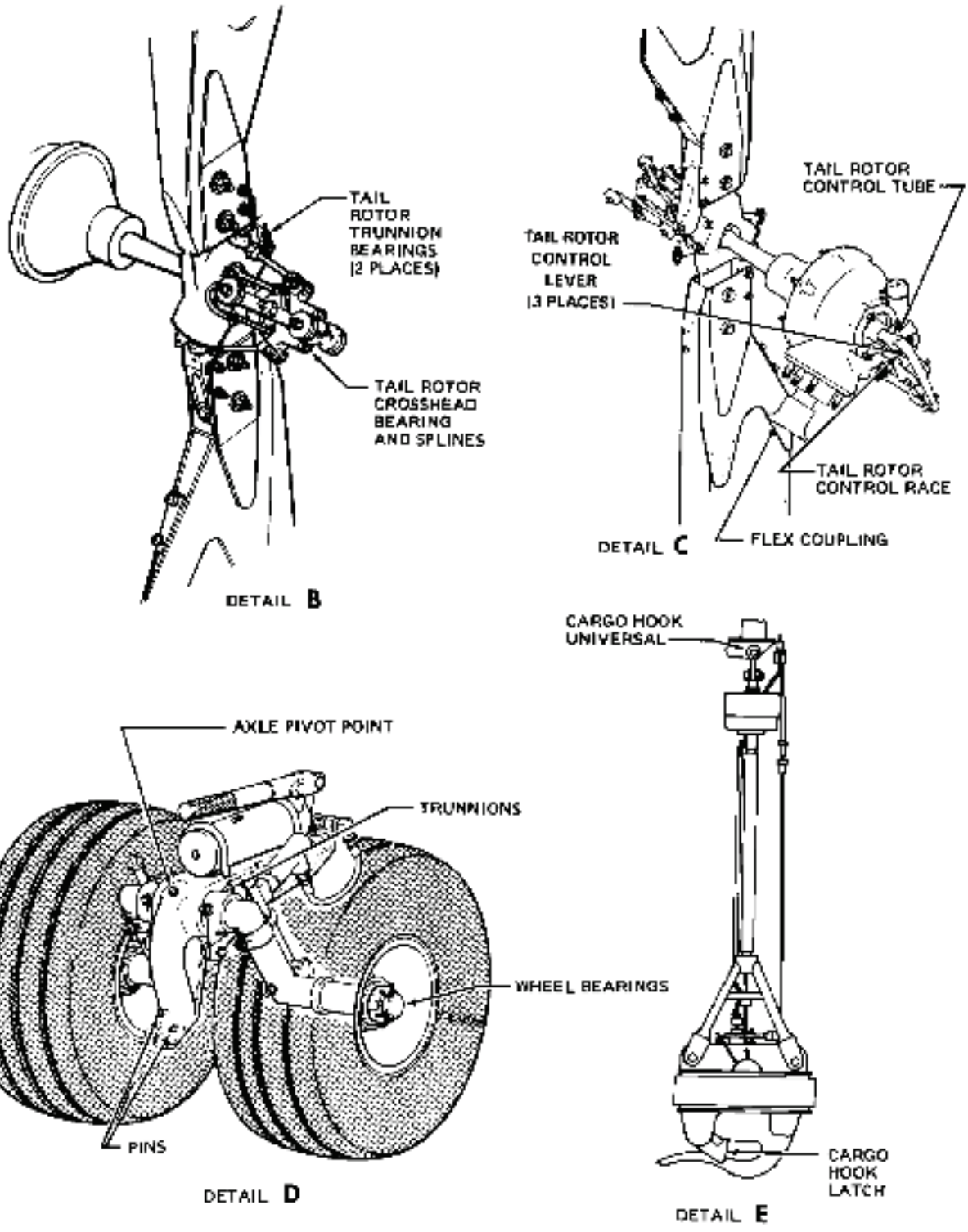
8. 600 hour/12 month interval applicable to helicopters S/N 36020 and subsequent and helicopters modified by 412-570-001-103 or post BHT-412-SI-74 (412SP to 412HP upgrade) or post BHT-412-SI-97 (412Classic to 412HP upgrade). Refer to chapter 65 for instructions to lubricate flexible couplings.

9. At 300 hour inspection, purge lubricate swashplate and collective lever trunnions and trunnion rod ends as follows: Rotate and simultaneously purge lubricate trunnions and trunnion rod ends to remove as much of "old" lubricant as possible.



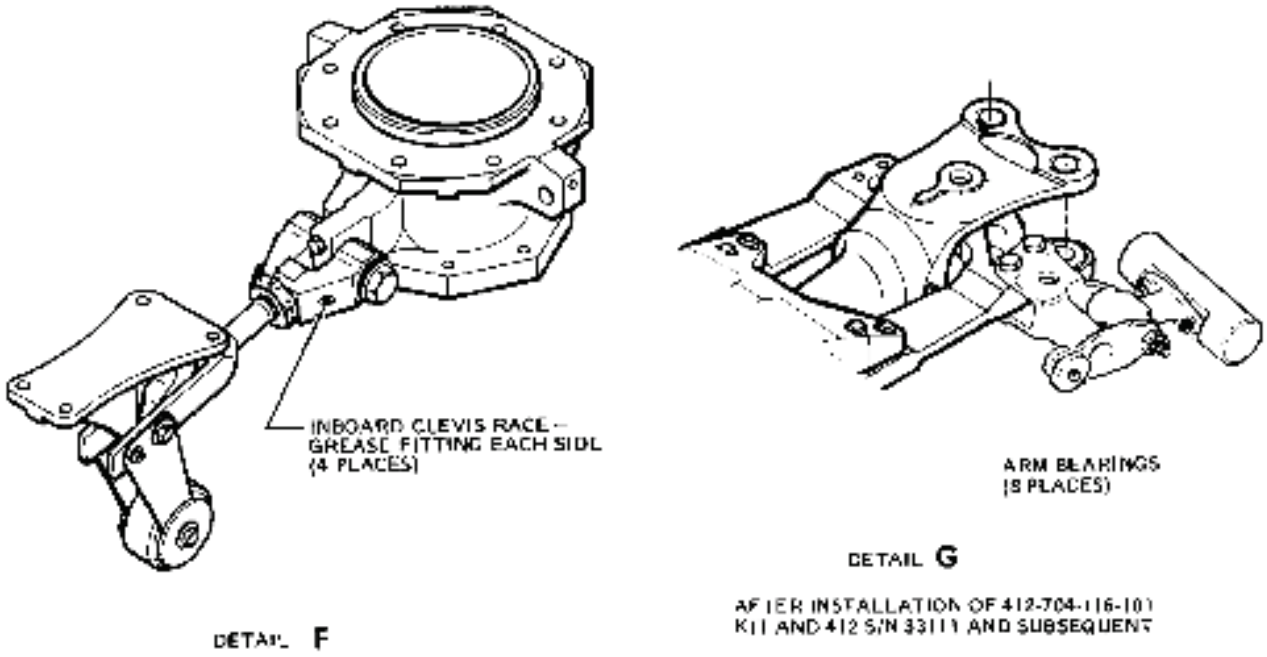
ICN-412-A-120100-B-97499-00002-A-002-01

Figure 1. Lubricating Points (Sheet 2 of 4)



ICN-412-A-120100-B-97499-00003-A-001-01

Figure 1. Lubricating Points (Sheet 3 of 4)



ICN-412-A-120100-B-97499-00004-A-001-01

Figure 1. Lubricating Points (Sheet 4 of 4)

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Lubrication - Lubricant Restrictions (204-040-755-005)

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lubricant	C-015	AR	

Procedure



LUBRICANT MAY CAUSE EYE AND SKIN IRRITATION. AVOID CONTACT WITH EYES, SKIN, AND CLOTHING. WASH THOROUGHLY AFTER HANDLING.



LUBRICANT IN ORIGINAL TUBES AND COMPONENTS CONTAINING lubricant (C-015) MUST BE STORED AT MODERATE AMBIENT TEMPERATURES, PREFERABLY LESS THAN 80°F (26.7°C). ELEVATED STORAGE TEMPERATURES PROMOTE OIL SEPARATION FROM THE LUBRICANT.

NOTE

Do not exceed lubrication intervals specified on lubrication chart.

1. lubricant (C-015) shelf life is 4 years from packing date on container.

NOTE

Make sure grease is thoroughly mixed prior to use.

2. In-stock components that utilize lubricant (C-015) must be pulled from stock and relubricated upon expiration of the original 4-year shelf life of the lubricant.

3. Once a component lubricated with lubricant (C-015) goes into service, the relubrication interval is dictated by the calendar and service-time schedule, whether the component is installed on a helicopter or is subsequently removed and returned to stock as a spare on the shelf.

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Lubrication - Flex Coupling Lubrication Log

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-561	AR	
Grease	C-172	AR	
Corrosion preventive compound	C-101	AR	
Grease	C-001	AR	

Procedure

1. The flex coupling lubrication log, as shown in Table 1, shall be maintained any time a component using lubricant 204-040-755-005 is installed on helicopter. This log shall be maintained listing date lubricant 204-040-755-005 was applied in component. This lubricant carries an operational hour and a calendar life as specified in this chapter. Lubricant shall be replaced at requirement occurring first (hours/months).

Table 1. ***Flex Couplings Lubrication Log (Example Only)***

PART NUMBER	*NOMENCLATURE*	*MODEL AFFECTED*	*DATE LUBRICATED*	*AIRFRAME HOURS*	*DATE LUBRICATED*	*AIRFRAME HOURS*
	Main Driveshaft	*All*				
	Transmission Tail Rotor Drive Output Coupling	*All*				
	Tail Rotor Driveshaft Hanger	*(1) HP/EP* *(4) SP*				
	Intermediate Gearbox Input Quill	*SP*				
	Intermediate Gearbox Output Quill	*All*				
	Tail Rotor Gearbox Input Quill	*SP*				
	Tail Rotor Driveshaft Coupling Assemblies	*HP/EP*				

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CHAPTER 18
VIBRATION AND NOISE ANALYSIS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Vibration and Noise Analysis - General Description	DMC-412-A-18-00-00-00A-028A-A / 00145	2023-05-31	ALL
Sources of Vibrations - Description	DMC-412-A-18-10-00-00A-042A-A / 00146	2023-05-31	ALL
Track and Balance Equipment - Description	DMC-412-A-18-10-00-04A-042A-A / 00147	2025-05-13	ALL
Control of Main Rotor Vibration - Main Rotor Vibration	DMC-412-A-18-10-00-05A-042A-A / 00148	2023-05-31	ALL
Rotor Balancing - Rotor Vibration Measurement Instrumentation	DMC-412-A-18-11-00-00A-372A-A / 00149	2023-05-31	ALL
Rotor Balancing - Main Rotor System Color Coding	DMC-412-A-18-11-00-00A-310A-A / 00150	2023-05-31	ALL
Rotor Balancing - RADS AT Installation	DMC-412-A-18-11-00-00A-720A-A / 00151	2023-05-31	ALL
Rotor Balancing - Flight Tests to Smooth Rotor	DMC-412-A-18-11-00-00A-372B-A / 00152	2023-05-31	ALL
Rotor Balancing - Advanced Editing Features	DMC-412-A-18-11-00-01A-042A-A / 00153	2023-05-31	ALL
Advanced Editing Features - Pitch Links	DMC-412-A-18-11-00-01A-271A-A / 00154	2023-05-31	ALL
Advanced Editing Features - Trim Tabs	DMC-412-A-18-11-00-02A-271A-A / 00155	2023-05-31	ALL
Advanced Editing Features - Hub Balance Weights	DMC-412-A-18-11-00-03A-271A-A / 00156	2023-05-31	ALL
Rotor Balancing - Product Balance	DMC-412-A-18-11-00-04A-271A-A / 00157	2023-05-31	ALL
Rotor Balancing - Setting Autorotation RPM	DMC-412-A-18-11-00-05A-271A-A / 00158	2023-05-31	ALL
Setting Autorotation RPM - Control of Main Rotor 2/Rev	DMC-412-A-18-11-00-01A-042B-A / 00159	2023-05-31	ALL
Setting Autorotation RPM - Control of Main Rotor 4/rev	DMC-412-A-18-11-00-01A-042C-A / 00160	2023-05-31	ALL
Rotor Balancing - Ride Quality Guidelines	DMC-412-A-18-11-00-01A-042D-A / 00161	2023-05-31	ALL
Rotor Balancing - Troubleshooting Problems	DMC-412-A-18-11-00-00A-422A-A / 00162	2023-05-31	ALL
Balancing the Tail Rotor - Control of Tail Rotor Vibration	DMC-412-A-18-12-00-00A-042A-A / 00163	2023-05-31	ALL
Balancing the Tail Rotor - Installation of the RADS	DMC-412-A-18-12-00-01A-720A-A / 00164	2023-05-31	ALL
Balancing the Tail Rotor - Tests to Balance the Tail Rotor	DMC-412-A-18-12-00-02A-271A-A / 00165	2023-05-31	ALL
Balancing the Tail Rotor - Making Tail Rotor Balance Adjustments	DMC-412-A-18-12-00-03A-271A-A / 00166	2023-05-31	ALL
Balancing the Tail Rotor - Tracking Tail Rotor	DMC-412-A-18-12-00-00A-042B-A / 00167	2023-05-31	TB 412-01-173: Pre

Document title	Data module code / Seq number		Issue date	Applicable to
Balancing the Tail Rotor - Troubleshooting Tail Rotor Balance	DMC-412-A-18-12-00-00A-422A-A / 00168		2023-05-31	ALL
Balancing Main Driveshaft - Control of Main Driveshaft Vibration	DMC-412-A-18-11-00-00A-042B-A / 00169	C	2025-11-18	ALL
Balancing Main Driveshaft - Installation of RADS	DMC-412-A-18-11-00-06A-720A-A / 00170		2023-05-31	ALL
Balancing Main Driveshaft - Tests to Balance Driveshaft	DMC-412-A-18-11-00-07A-271A-A / 00171		2023-05-31	ALL
Balancing Main Driveshaft - Using Diagnostics	DMC-412-A-18-11-00-08A-271A-A / 00172		2023-05-31	ALL
Balancing Main Driveshaft - Making Adjustments	DMC-412-A-18-11-00-09A-271A-A / 00173	C	2025-11-18	ALL
Balancing Main Driveshaft - Troubleshooting	DMC-412-A-18-11-00-01A-422A-A / 00174		2023-05-31	ALL
Main Rotor Balancing (Chadwick-Helmuth Vibrex Equipment, Model 177M or Equivalent) - Description	DMC-412-A-18-11-01-00A-028A-A / 00175		2023-05-31	ALL
Rotor Vibration Measurement Instrumentation - Installation	DMC-412-A-18-11-01-00A-720A-A / 00176		2023-05-31	ALL
Main Rotor Blades - Strobe Tracking	DMC-412-A-18-11-01-00A-373A-A / 00177		2023-05-31	ALL
Main Rotor - Track and Balance	DMC-412-A-18-11-01-00A-373B-A / 00178		2023-05-31	ALL
Track and Balance - Preparation	DMC-412-A-18-11-01-00A-913A-A / 00179		2023-05-31	ALL
Track and Balance - Working Main Rotor	DMC-412-A-18-11-01-00A-724A-A / 00180		2023-05-31	ALL
Track and Balance - Main Rotor Autorotation RPM Adjustment	DMC-412-A-18-11-01-00A-271A-A / 00181		2023-05-31	ALL
Track and Balance - Working Main Rotor 2/Rev Vibration	DMC-412-A-18-11-01-00A-271B-A / 00182		2023-05-31	ALL
Tail Rotor Hub and Blade - Operational Check	DMC-412-A-18-12-01-00A-320A-A / 00183		2023-05-31	ALL
Tail Rotor - Troubleshooting	DMC-412-A-18-12-01-00A-422A-A / 00184		2023-05-31	ALL
Tail Rotor Hub and Blade - Flag Tracking	DMC-412-A-18-12-01-00A-373A-A / 00185		2023-05-31	ALL
Tail Rotor - Tracking	DMC-412-A-18-12-01-01A-373A-A / 00186		2023-05-31	ALL
Tail Rotor - Balancing	DMC-412-A-18-12-01-00A-271A-A / 00187		2023-05-31	ALL
Main Driveshaft - Vibration Control	DMC-412-A-18-11-00-14A-271A-A / 00188	C	2025-11-18	ALL

Vibration and Noise Analysis - General Description

This chapter provides information and procedures required to maintain helicopter with respect to quality of the ride and control of rotating component generated vibration. The instrument recommended to support the Model 412 is the Smiths Aerospace RADS AT.

For convenience, balancing of the tail rotor and main driveshaft are also included in this chapter. Maintenance procedures for these components will be found in the appropriate chapters.

Refer to 412-A-62-00-00-00A-028A-A / 00933 for the complete field maintenance instructions for the main rotor and rotating control components.

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Sources of Vibrations - Description

1. Sources of Vibrations

Many types of vibrations are present in the helicopter at low to moderate levels. The main sources of these vibrations are the main rotor, tail rotor, and engine-to-transmission (main) driveshaft. Other dynamic components (hydraulic pump, oil cooler fan, etc.) can generate vibrations at various frequencies. These vibrations are generally at such a low level that they are seldom perceived (Table 1). The primary source of vibrations is the main rotor. Main rotor vibrations occur at 1/rev (once per revolution), 2/rev (twice per revolution), 4/rev (four times per revolution), and 8/rev (eight times per revolution) frequencies. The magnitude of 1/rev and 2/rev vibrations in the cabin area is influenced by the condition of main rotor blades, main rotor hub, and control system components.

2. Main Rotor Vibrations

Vibrations can be reduced in magnitude or eliminated by repairing or replacing worn or unserviceable components and by tracking and balancing the main rotor. The installation, condition, and compatibility of lead-lag dampers have the most impact upon the ability to obtain, and maintain, a low level of 1/rev and 2/rev vibration. Additionally, degradation in any of the hub or control system elastomeric bearings will affect the level of 1/rev vibration. As opposed to the earlier version of the main rotor blade, P/N 412-015-200-111, the later version blades, P/N 412-015-300-109, will seldom cause an uncorrectable 1/rev vibration unless damaged or improperly repaired. The 4/rev vibrations are controlled by the pendulum absorbers in the rotor and by the tightness of the instrument panel and nose structure. An increase in 4/rev in cabin area can signify degradation of pendulum support bearings or even a loosening of helicopter nose structure. This increase can also be caused by improperly accomplished nose structure modifications and loose components in nose such as the battery or avionic equipment. Main rotor 8/rev vibrations are always present. However, vibrations at this frequency generally do not affect ride quality.

3. Tail Rotor Vibrations

Vibrations from the tail rotor can occur at the 1, 2, and 4/rev levels. In general, only the 1/rev vibration can be corrected or reduced by tracking and balancing the tail rotor.

4. Other Vibrations

A high frequency vibration and pulsation noise in cabin can be caused by the engine-to-transmission driveshaft. This vibration can be eliminated by proper balancing of the engine-to-transmission driveshaft (412-A-18-11-00-00A-042B-A / 00169, 412-A-18-11-00-06A-720A-A / 00170, 412-A-18-11-00-07A-271A-A / 00171, 412-A-18-11-00-08A-271A-A / 00172, 412-A-18-11-00-09A-271A-A / 00173 and 412-A-18-11-00-01A-422A-A / 00174 or 412-A-18-11-00-14A-271A-A / 00188).

Table 1. Excitation Frequencies and Sources

VIBRATION SOURCE	APPROX RPM	APPROX HZ
1/Rev Main Rotor	324	5.40
2/Rev Main Rotor	648	10.80
4/Rev Main Rotor	1,296	21.60
1/Rev Tail Rotor	1,662	27.70
6/Rev Main Rotor	1,944	32.40
8/Rev Main Rotor	2,592	43.20
2/Rev Tail Rotor	3,324	55.30
1/Rev Lower Hydraulic Pump	4,300	71.70
1/Rev Tail Rotor Driveshaft	4,302	71.70
1/Rev Transmission Input Shaft	6,600	110.00
1/Rev Upper Hydraulic Pump	6,600	110.00
4/Rev Tail Rotor	6,648	110.60
Oil Cooler	8,200	136.00
Rotor Brake	8,300	138.00
Lower Hydraulic Pump Piston Passage Frequency	30,120	502.00
1/Rev Power Turbine	33,020	550.33

Table 1. Excitation Frequencies and Sources (continued)

VIBRATION SOURCE	APPROX RPM	APPROX HZ
Upper Planetary Gear Mesh	38,556	642.60
Upper Hydraulic Pump Piston Passage	46,200	770.00
Lower Hydraulic Pump 2 x Piston	60,240	1,004.00
Oil Cooler Fan Blade Passage	87,960	1,466.00
90° Gearbox Gear Mesh	94,620	1,577.00

Track and Balance Equipment - Description

NOTE

It is acceptable to use a different system, if the measurement is taken at the locations specified in this chapter. However, the level of technical support provided for vibration issues where a RADS-AT system is used will have better due to experience and software availability. The Zing Test Elite (ZTE) is a Bell Helicopter Textron (BHT) accepted alternative to the Rotor Analysis Diagnostics System - Advanced Technology (RADS-AT). The equipments list and installation instructions for the ZTE are found on the units PC-GBS interface. Go to Bellcustomer.com for the latest software download.

Bell Helicopter has tested and supports the following equipment for main rotor tracking and balancing and main driveshaft balancing. The primary diagnostic equipment is the Rotor Analysis and Diagnostic System Model AT (RADS AT) with procedures included in the BHT-412-MM and the Bell Helicopter Vibration Monitor (BHVM) system installed by Supplemental Type Certificate (STC) , which is not included in the BHT-412-MM, but covered by separate documentation. Also usable is the analog Model 177M Vibrex equipment.

1.1. Main Rotor Track and Balance

For main rotor track and balance adjustments using RADS AT equipment, refer to 412-A-18-11-00-00A-372A-A / 00149.

1.2. Tail Rotor Track and Balance

For tail rotor track and balance procedures using RADS AT equipment, refer to 412-A-18-12-00-00A-042A-A / 00163.

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Control of Main Rotor Vibration - Main Rotor Vibration

The most common vibration felt by crew and passengers is main rotor generated 1/rev (one beat per revolution), 2/rev (two beats per revolution), and 4/rev (four beats per revolution). 412-A-18-10-00-00A-042A-A / 00146 shows the frequencies of these vibrations. In general only the 1/rev and 2/rev can be controlled by adjusting the main rotor system. The 4/rev levels are dependent on the condition of the pendulum absorbers and the condition of the fuselage (primarily the nose and instrument panel structure) and are not significantly effected by standard track and balance adjustments or rotor system component changes.

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Rotor Balancing - Rotor Vibration Measurement Instrumentation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Basic RADS AT kit with basic software Basic RADS operating system 3.10AO or greater	058R3 P/N 29333300	1	GE Aviation Systems - Clearwater, 14200 Roosevelt Blvd., Dock A, Clearwater, FL 33762. 412_41 Version 5.22 or greater for use on helicopters with 204-031-927-105 pylon mounts. 412_50 Version 5.24 or greater for use on helicopters with 204-031-927-107 pylon mounts.
212/412 Optical tracker bracket	P/N 29203300D	1	
212/412 Optical tail rotor adapter	P/N 29338500A	1	
Accelerometer cable (additional)	P/N 29105600 or 29105605	1	
Accelerometer	P/N 991D	3	
Accelerometer brackets (stiff)	P/N 29329700	1	
Basic kit	P/N 29333306	1	
Bell 212/412 Adapter kit	P/N 29335401	1	
Bolts	NAS6603-7	5	
Bolts	NAS1305-32	2	
Nuts	MS21042L5	2	
Washers	AN970-5	AR	
Shim, Laminated, Aluminum Alloy	P/N 120-035E44-12	AR	

Procedure

1. The primary instrumentation for measuring rotor vibration is the Rotor Analysis and Diagnostic System, Model AT (RADS AT) . The RADS AT automates the acquisition of blade track and vibration data as well as corrective actions required to accomplish tracking and balancing. Also, the RADS AT automatically obtains blade sweep (lead-lag) information which helps to locate problem components in the hub and rotating control system. RADS AT consists of the following:

1.1. A harmonic analysis system to measure main rotor, tail rotor, and engine-to-transmission driveshaft induced vibrations at multiple frequencies.

1.2. A passive and/or an active optical tracker to measure individual blade track and lead-lag variations. The passive optical tracker is used for day-only operations. The active optical tracker can be used for either day or night operations.

1.3. A diagnostic program, which defines rotor adjustments required to track and balance main rotor and to balance tail rotor and engine-to-transmission (main) driveshaft.

1.4. A spectrum analysis program (FFT), which allows the operator to determine magnitude and frequency of any vibration present in helicopter.

NOTE

To change from Vibrex equipment to RADS AT, most sensors and cables can be used. Contact Smiths Aerospace, Clearwater, FL, for help in making the conversion.

1.5. Three main versions of Model 412 rotor analysis and diagnostics software are available as follows:

1.5.1. Basic RADS operating system should be 3.10AO or greater.

1.5.2. 412-50 Version 5.24 or greater for use on helicopters with 204-031-927-107 pylon mounts.

1.6. Determine version of software loaded into Control and Data Unit (CADU) of RADS AT as follows:

1.6.1. Select MANAGER option from MAIN MENU screen and then select STATUS.

1.6.2. Use down arrow key to view pages of CADU status.

1.6.3. Find Page 3 which lists current helicopter software that is loaded.

1.7. Various accelerometers may be used with RADS AT. Match accelerometers and software as follows:

1.7.1. Using CADU select MANAGER option from MAIN MENU screen and then select STATUS.

1.7.2. Use down arrow key to view pages of CADU status.

1.7.3. Find page listing accelerometers versus software. Select accelerometers which are compatible with software or change types of accelerometers in software by selecting SETUP in MANAGER section. Scroll to ACCELEROMETERS and, by pressing right or left cursor keys, various accelerometers may be chosen.

NOTE

Failure to select and verify accelerometers match software will result in inaccurate vibration data or no vibration data measured at all.

Rotor Balancing - Main Rotor System Color Coding

Procedure

CAUTION

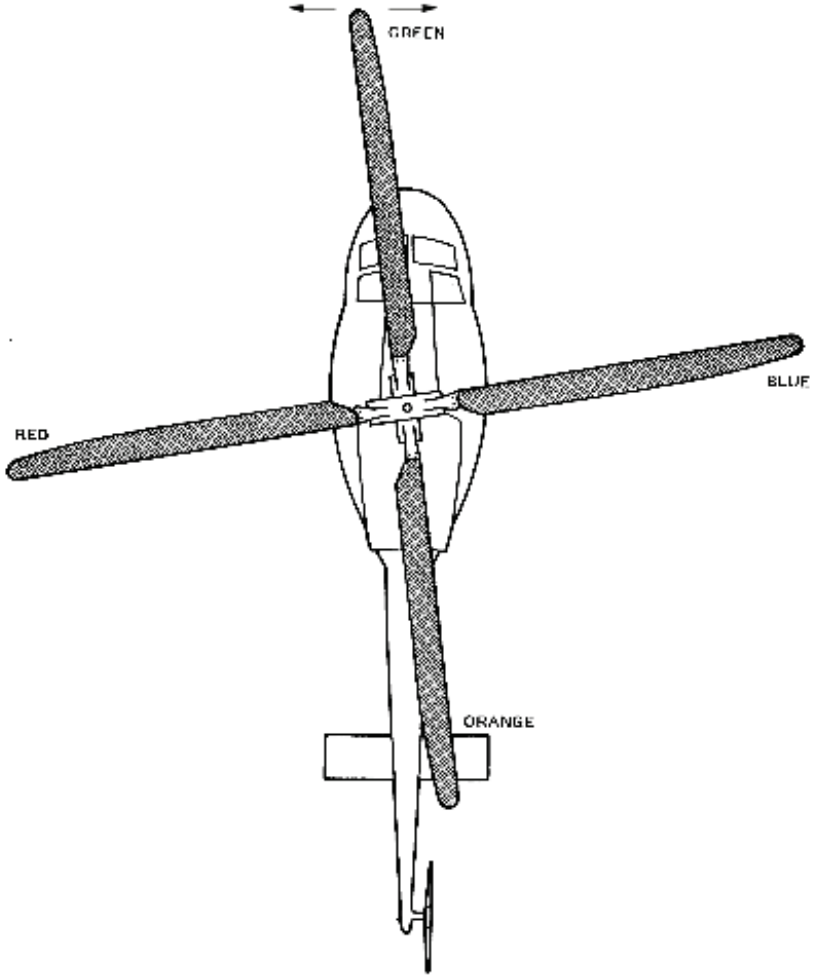
ENSURE COLOR CODE INTEGRITY IS MAINTAINED ON MAIN ROTOR BLADES, HUB, AND ROTATING CONTROL COMPONENTS.

1. When installing main rotor hub and blade, verify color code as follows:

1.1. Red color code on hub and blade is adjacent to master spline on mast and hub splined plate.

1.2. Remaining hub and blades are installed (Figure 1) using direction of rotation as a guide. When properly color coded, upper blades will be red and blue, lower blades will be green and orange.

1.3. Rotating control components are coded accordingly.



NOTE: The interrupter should be over the magnetic pickup when the rotor is in the above position.

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Figure 1. Main Rotor Color Coding (Sheet 1 of 1)

Rotor Balancing - RADS AT Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	
Tape	C-482	AR	

Procedure

1. Locate DAU (29328200 or 29481400) in cabin aft of crew compartment in a convenient area.

2. Using a 29313000 or 29329700 accelerometer bracket (or equivalent) install one 991D accelerometer on bottom edge of instrument panel in front of pilot cyclic with connector toward right side of helicopter. Connect end of accelerometer cable (291056000 and 29105605) to lateral accelerometer. Route to DAU so it does not foul controls. Connect to ACC #1 port (Figure 1).

3. Using a 29329700 accelerometer bracket (or equivalent), install one 991D accelerometer on front face of instrument panel inboard of pilot ash tray oriented vertically with connector down. Connect end of an accelerometer cable to vertical accelerometer, route to DAU so it cannot foul controls. Connect to ACC #2 port (Figure 1).

NOTE

This installation is for helicopters with the 204-031-927-107 pylon mounts only.

4. Using a 29313000 or 29329700 accelerometer bracket (or equivalent), install one 991D fore-and-aft accelerometer on bottom edge of instrument panel in front of pilot cyclic with connector pointed forward. Connect accelerometer cable to fore-and-aft accelerometer, route to DAU so it cannot foul controls. Connect to ACC #3 port (Figure 1).

5. Remove outer jamnut from magnetic RPM sensor (27288400) and insert into magnetic sensor support bracket at rear of swashplate inner ring (5, Figure 2). Reinstall outer jamnut and loosely tighten so only one thread is showing beyond jamnut. Rotate rotor so magnetic interrupter is next to sensor (Figure 3). Adjust both jamnuts until gap between sensor and interrupter is 0.025 inch (0.63 mm). Tighten both jamnuts while maintaining this spacing and secure with lockwire (C-405). Connect helicopter magnetic sensor cable to sensor and tighten to secure.

6. Connect magnetic sensor cable (29105403) to magnetic sensor port on copilot side of center pedestal (Figure 4). Connect opposite end of cable to TACHO #1 on DAU.

7. Connect 28 VDC power cable (29104700) to 28 VDC port on copilot side of pedestal (Figure 4). Connect opposite end of cable to 28 VDC connector on the DAU.

8. Install passive optical tracker (4, Figure 5) as follows:

8.1. Remove screws (1) left side of upper nose panel.

8.2. Position tracker bracket (8) on left nose panel and secure using bolts (7).

8.3. Torque bolts (7) 15 to 20 inch-pounds (1.69 to 2.26 Nm).

8.4. Install the passive optical tracker (4) on bracket (8) using bolts (9), washers (3), and nut (5). Tighten upper bolt enough to allow tracker to be moved by hand.

NOTE

Use an inclinometer to verify the angle of tracker when first installed on a new mount. Scribe mark angle on mount so subsequent installations will not require an inclinometer for alignment.

8.5. Set angle of tracker (4) as shown in Figure 6 depending on mode of operation to be used (day or night). Once tracker (4, Figure 5) is angled properly, torque nuts (5) 80 to 120 inch-pounds (9.04 to 13.56Nm). Reverify and correct angle if necessary.

9. Connect tracker cable (6) to tracker (4) and route through lower front edge of copilot door. Secure cable to nose using tape (C-482). Route cable in cabin to avoid flight controls and connect to Tracker #1 on DAU.

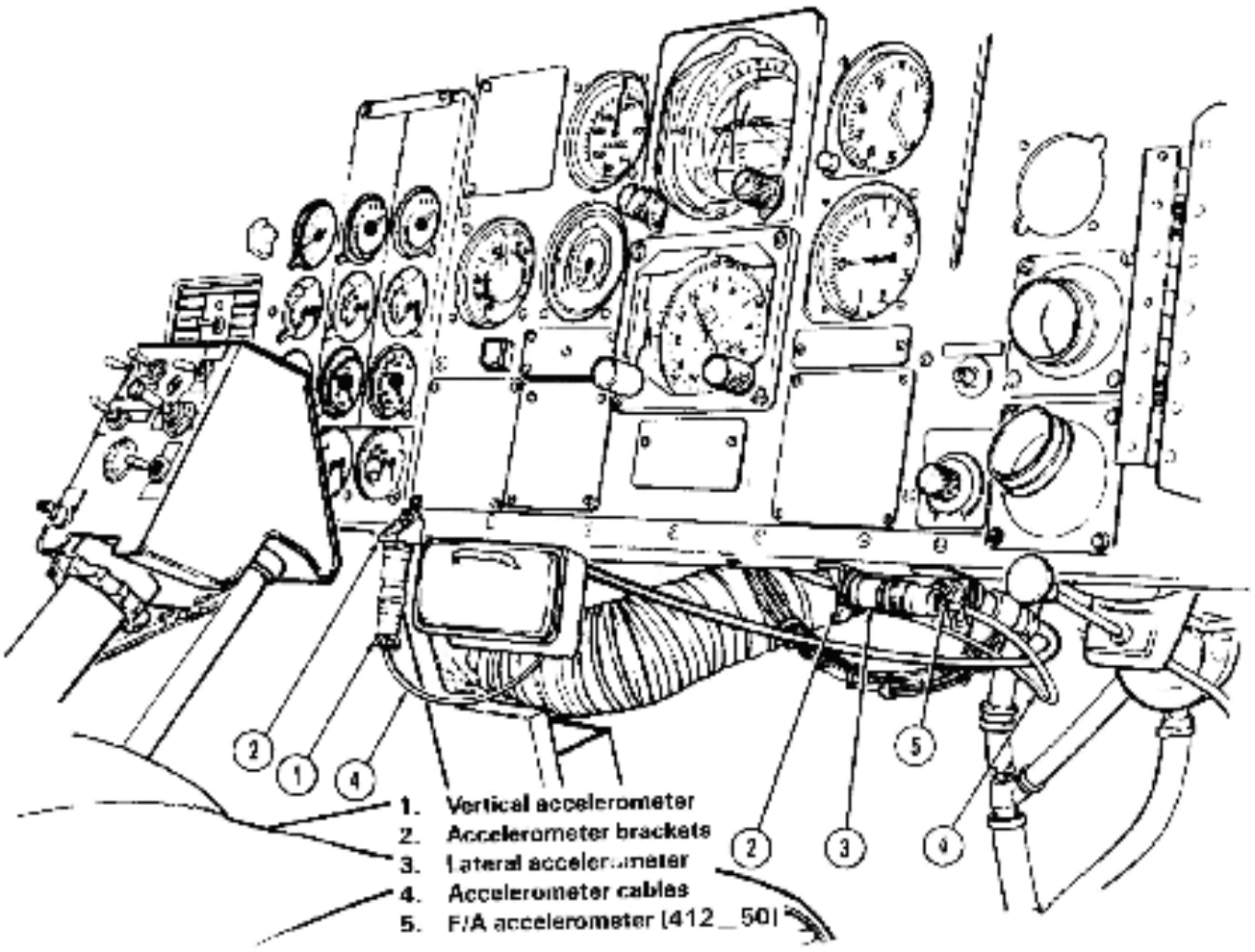


DO NOT BLOCK STATIC PORTS ON SIDE OF NOSE WHEN ROUTING CABLE.

10. Connect CADU (29314101 or 29481301) to DAU using communications cable (29325601).

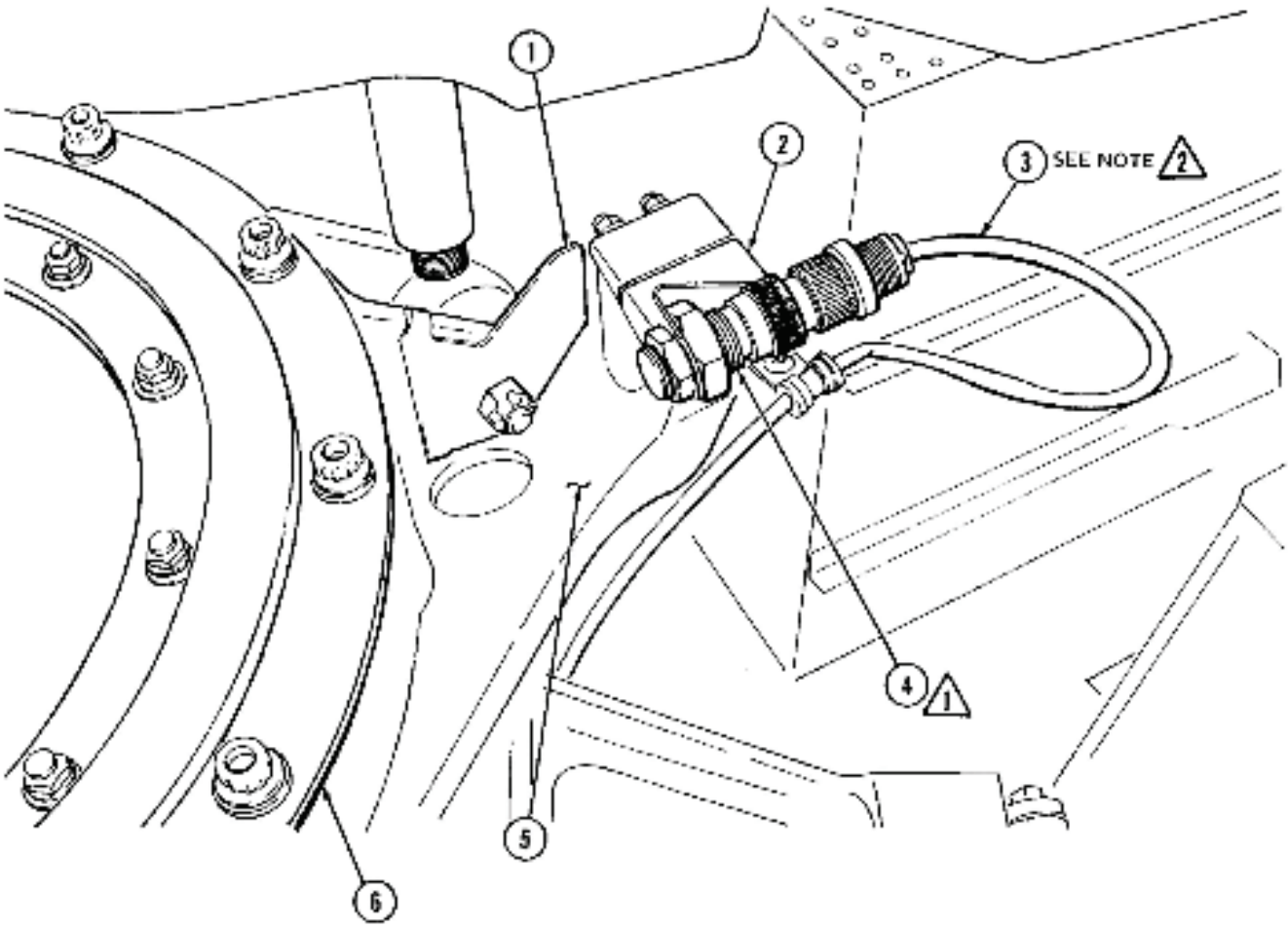
11. Install reflective tape (10605000) on trailing edge of each blade for night tracking (Figure 7). Paint leading edge of each blade black for daylight tracking.

12. The RADS is now installed and ready to smooth the rotor system. When correctly installed, RADS should be configured as shown in Figure 8.



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Figure 1. Accelerometer Locations (Sheet 1 of 1)



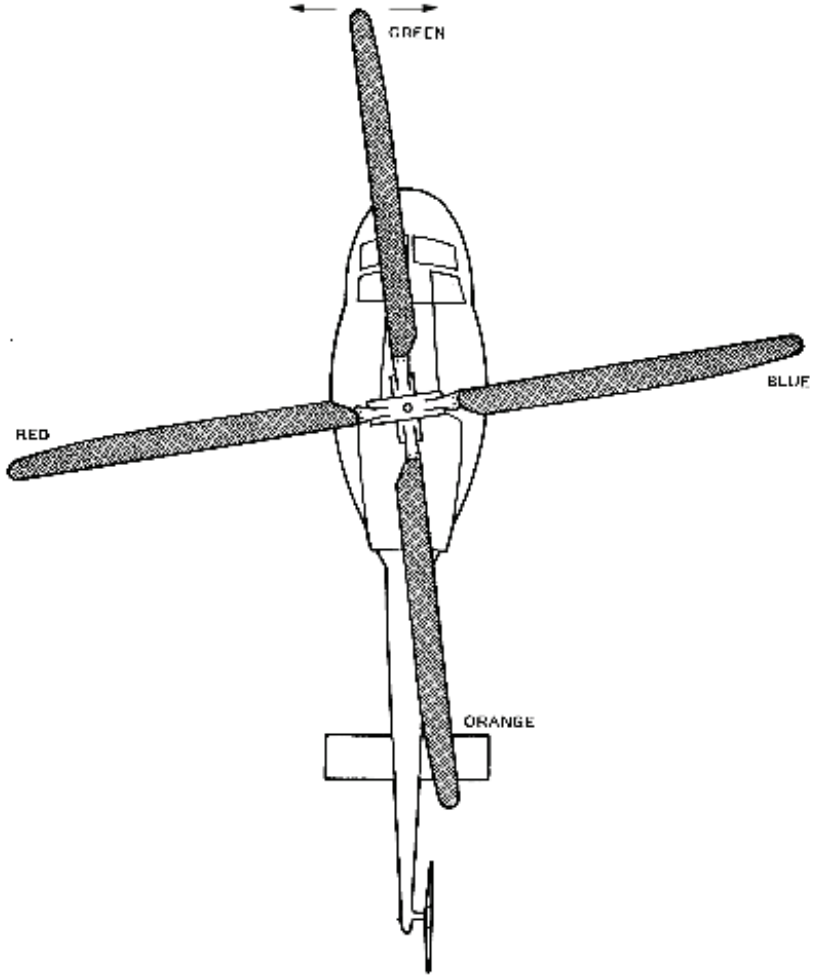
1. Magnetic field interrupter (red swashplate horn)
2. Magnetic pickup suport bracket
3. Pickup cable 2
4. Magnetic pickup 1
5. Swashplate inner ring
6. Swashplate outer ring

NOTES:

1. Part of RADS AT kit or Chadwick-Helmuth basic Vibrex system C.
2. Other end connects to the ROTO TRACK SIGNAL/BLADE BALANCER RECPT. on right side of pedestal.

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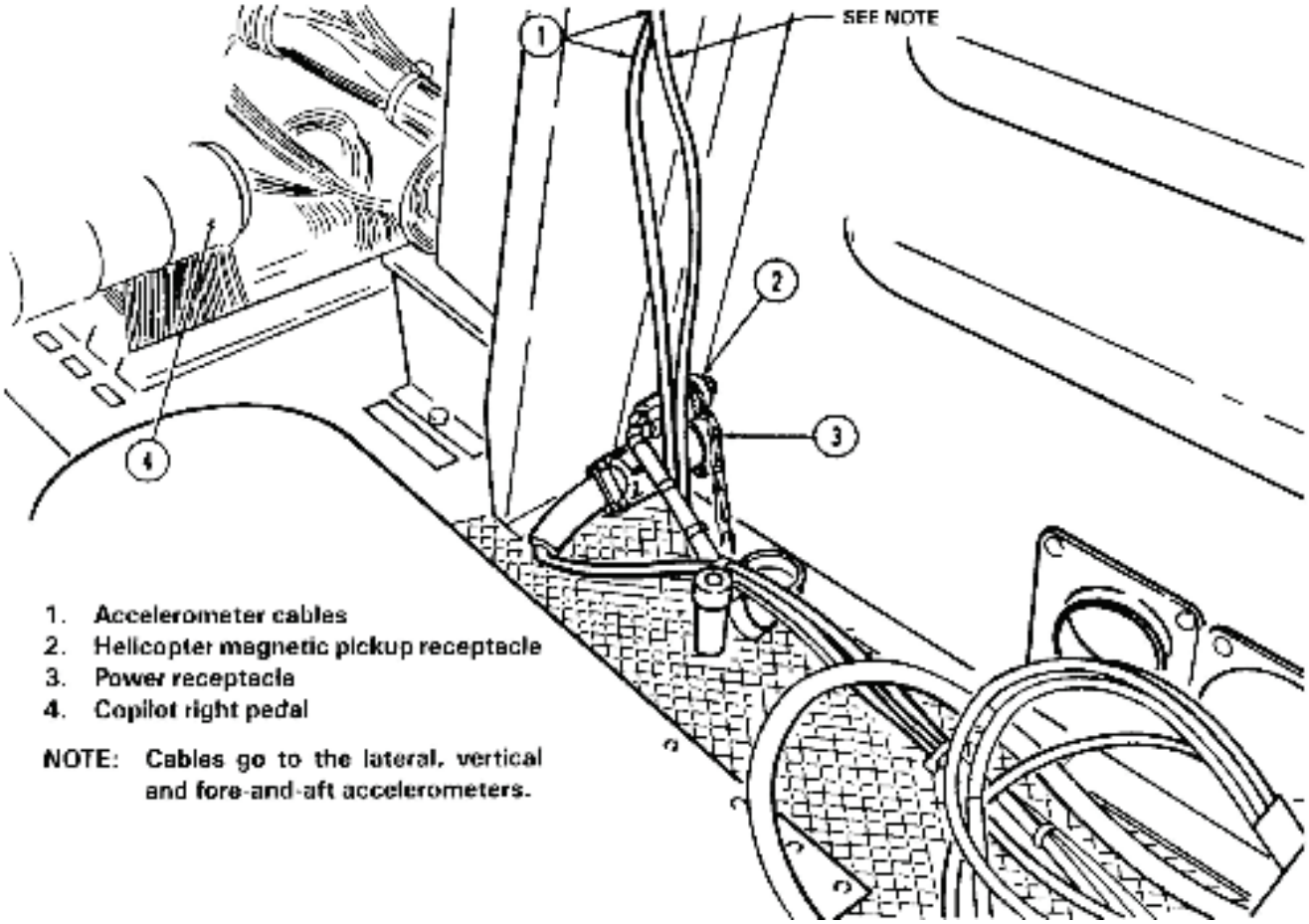
Figure 2. Magnetic Pickup and Interrupter (Sheet 1 of 1)



NOTE: The interrupter should be over the magnetic pickup when the rotor is in the above position.

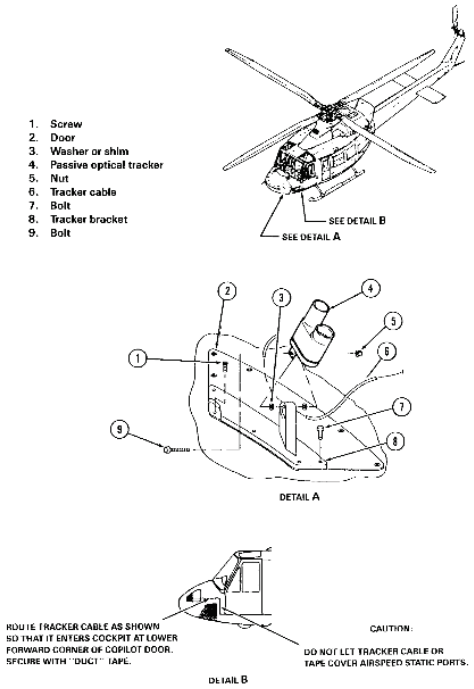
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Figure 3. Main Rotor Color Coding (Sheet 1 of 1)



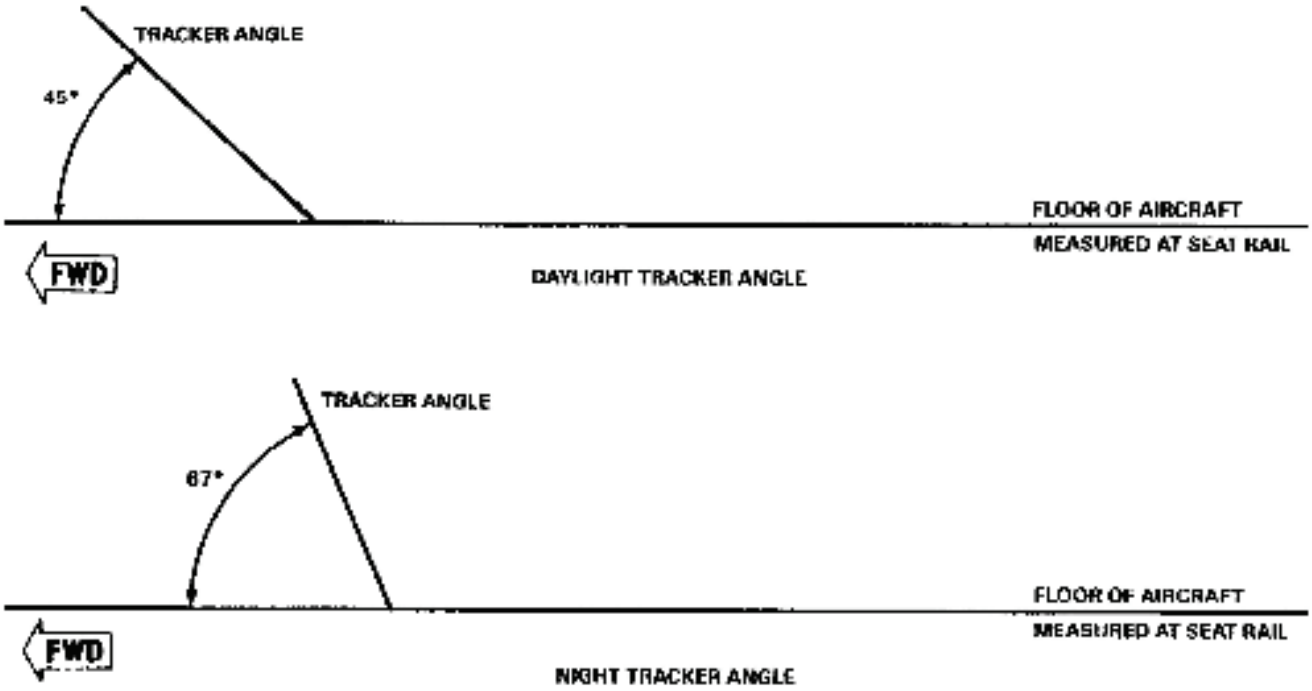
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Figure 4. Magnetic Port/Azimuth (Sheet 1 of 1)



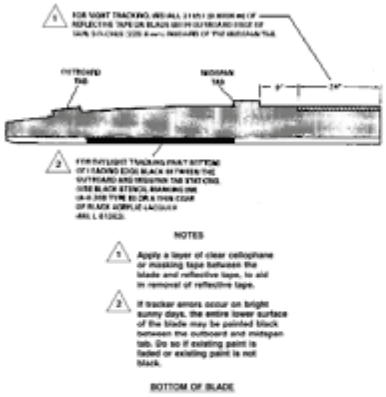
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Figure 5. Passive Optical Tracker Location and Cable Routing (Sheet 1 of 1)



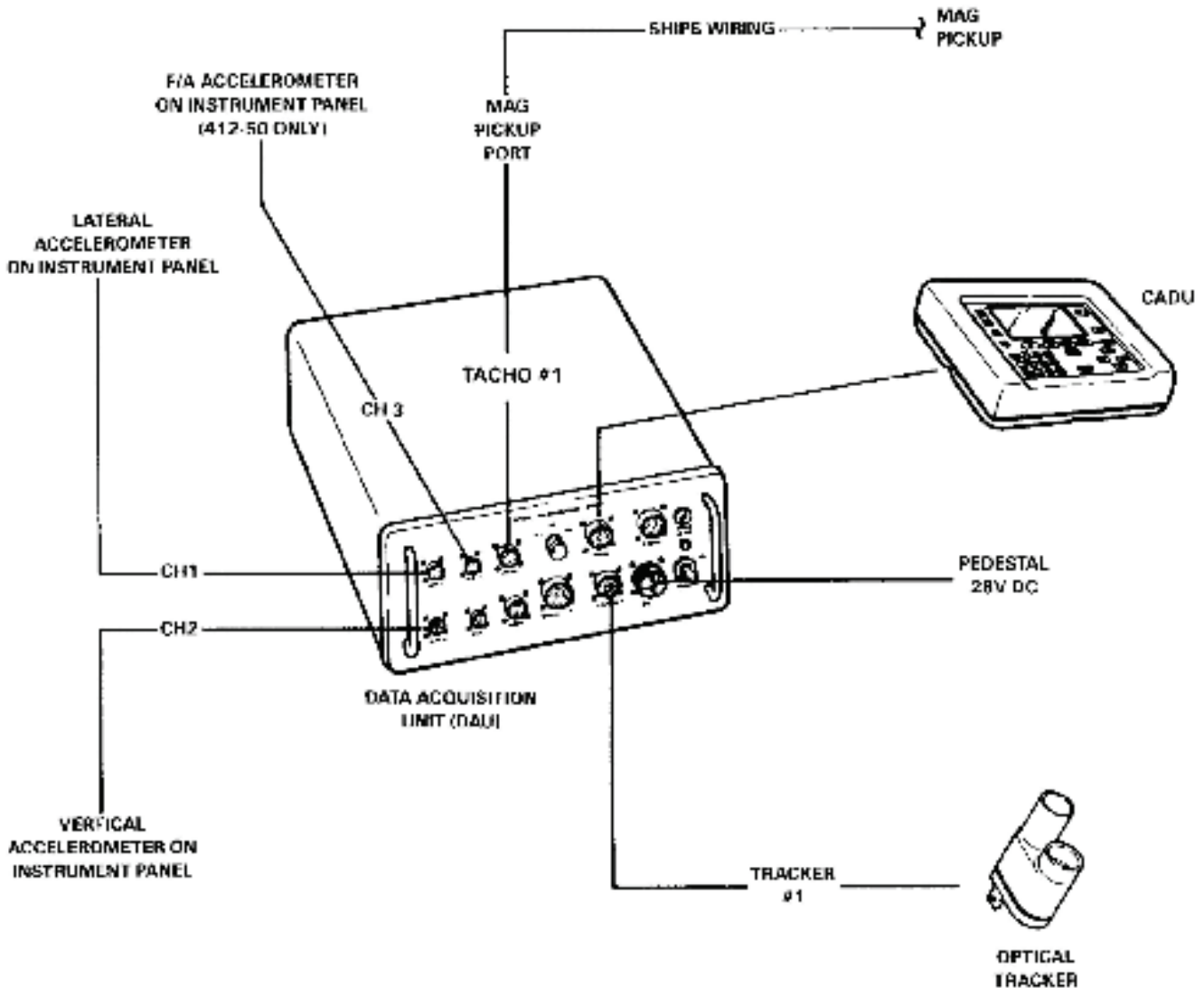
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Figure 6. Optical Tracker Angle (Sheet 1 of 1)



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Figure 7. Location of Paint and Tape for Tracking (Sheet 1 of 1)



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Figure 8. Main Rotor Proper RADS Configuration (Sheet 1 of 1)

Rotor Balancing - Flight Tests to Smooth Rotor

Procedure

1. Table 1 shows test conditions used by various flight plans to smooth the rotor. All test conditions should be performed at 100% RPM, unless otherwise noted. It is recommended all flight tests be conducted at a light gross weight as this is the configuration for which RADS diagnostics have been optimized. The recommended configuration is two/three crew and 500 to 1500 pounds of fuel. Two flight plans are used to smooth the rotor system. These are:

INITIAL: The purpose of this mode is to track rotor on ground and in hover and to balance rotor while on the ground. This mode is used to set up rotors with one or more major component changes prior to flight (i.e. blade, hub, pitch link, etc.). This mode does not reduce vibration in hover and is designed to provide level of 1/rev on first flight, which allows complete flight envelope to be checked. This mode has been successfully completed when balance on ground is below 0.2 IPS and track of blades at idle and FPG100 are within 2.0 mm of ± 20 mm position and hover track is within 4 mm of ± 20 mm.

FLIGHT: The purpose of this mode is to smooth the rotor system. This mode works best when the INITIAL has been accomplished first. This mode is used alone when a rotor is being resmoothed as part of routine maintenance (no major component changes). This mode also optimizes track spread versus airspeed which normally will minimize 2/rev levels. This mode has been completed when flight crew is satisfied with the ride or diagnostics indicate that no further improvement can be made. Refer to 412-A-18-11-00-01A-042D-A / 00161 on ride quality guidelines.

1.1. In the 412-41 program for helicopters with 204-031-927-105 mounts, these two flight plans are called INITIAL and FLIGHT. In 412-50 program for helicopters with 204-031-927-107 mounts, these programs are:

1.1.1. INT-LOW: INITIAL mode with low landing gear.

1.1.2. INT-HIGH: INITIAL mode with high/float landing gear.

1.1.3. FLI-LOW: FLIGHT mode with low landing gear.

1.1.4. FLI-HIGH: FLIGHT mode with high/float landing gear.

1.2. Smooth rotor with RADS as follows:

1.2.1. Ensure system is installed (412-A-18-11-00-00A-720A-A / 00151).

1.2.2. Turn on CADU to view main screen. Use arrow keys to select AIRCRAFT TYPE and press DO. Use arrows to highlight either 412-41 or 412-50 (depending on pylon mount configuration) and press DO to select.

1.2.3. With aircraft type selected, press DO to select or define TAIL NUMBER of helicopter (BHTI prefers operator to use S/N and not registration number). When TAIL NUMBER has been highlighted or defined, press DO to select.

1.2.4. With aircraft type and tail number defined, press DO to select FLIGHT PLAN. In this case, a determination to run an Initial or Flight to smooth the rotor is required. INITIAL sets up new or overhauled rotors for flight and FLIGHT smooths rotor in flight. When using 412-50 program, define type of skid gear.

1.2.5. With type, number, and plan selected, CADU screen should look similar to examples in Figure 1. With CADU properly configured, press F1 to initiate MEASUREMENT mode of RADS.

1.2.6. Once measurement mode has been activated, the list of test conditions to be conducted will appear. Refer to Table 1 for test definition if required.

1.2.7. With list of test conditions displayed, operate helicopter and obtain data at each designated test condition, which can be achieved by highlighting test condition and pressing DO once to arm system and a second time to initiate data collection sequence when helicopter has stabilized on the test condition. Repeat individual test at least three times if a warning appears before terminating the test flight and troubleshooting the system.

1.2.8. Once all data has been obtained, shut down helicopter and select DIAGNOSTICS (if all test conditions have been obtained) or press QUIT, then select SAVE and EXIT. Select DIAGNOSTICS to determine what, if any, corrective action is required (Step 2.).

1.2.9. Repeat Step 1.2.5. through Step 1.2.8. until an acceptable ride is obtained. If an acceptable ride cannot be obtained or other problems are encountered, refer to 412-A-18-11-00-00A-422A-A / 00162 for recommended corrective action.

1.2.10. All data has been obtained for a test condition if the word DONE appears after the test. Partial data has been obtained if PARTIAL appears and no data has been obtained if FAIL appears.

1.2.11. Note what error messages say. Do not continue if failure is an obvious system fault as this will only waste time. Take corrective actions displayed on screen and refer to 412-A-18-11-00-00A-422A-A / 00162 for additional troubleshooting steps.

1.2.12. Tracker faults are most likely when pointed directly at the sun. Adjust orientation of the helicopter if a tracker fault occurs.

2. Using Diagnostics

2.1. Once data has been obtained and Diagnostics entered:

2.1.1. A screen comparing measured levels to predefined limits will appear on screen. These limits are not true limits or vibration target levels, but are programmed into RADS to allow the use of RESOLVE TO LIMIT feature in RADS diagnostics to optimize number of adjustments specified to smooth rotor. This feature eliminates unnecessary adjustments, which can be called out to work rotor when all adjustments are specified without control. If RADS indicates all measurements are within limits, and/or ride is acceptable to the crew, press QUIT to return to main menu. If any measurements are outside limits and ride is considered unacceptable to crew, press DO to begin diagnostic analysis.

2.1.2. The default setting for diagnostics is RESOLVE TO LIMIT. As diagnostics are entered, RADS will begin to count down number of moves required to reach target levels below predefined limits. This counting down eliminates less meaningful adjustments and takes approximately 3 minutes with newer type CADU's.

2.1.3. Upon completion of analysis, a list of adjustments will be displayed. Record these adjustments on a form similar to that shown in Figure 2. Make adjustments specified as described in 412-A-18-11-00-01A-271A-A / 00154, 412-A-18-11-00-02A-271A-A / 00155 and 412-A-18-11-00-03A-271A-A / 00156. If adjustment cannot be made as defined, approximately the same result can be obtained by making opposite move on opposite blade. If the operator does not want to make a specific move (such as a product balance move), adjustment can be turned off using EDIT ADJUSTABLES mode in DIAGNOSTICS menu. To deselect this adjustment, select Edit Adjustables option and press DO. Select adjustment to be turned-off (only Y adjustments are on), and press F1 to change from on to off (Y to N). Press DO to recalculate adjustments to be made.

2.1.4. Once adjustments to be made have been defined and recorded, return to main menu by selecting MAIN MENU from DIAGNOSTICS MENU.

2.1.5. When in INITIAL mode if all test conditions are not measured, the following editing adjustments are required to calculate correct set of adjustments:

2.1.5.1. Idle Only: Turn off all adjustments except pitch links.

2.1.5.2. 100% RPM only: Turn off all adjustments except hub weights.

2.1.5.3. Idle and 100% only: Turn off product balance weights.

2.1.6. When in FLIGHT mode, never turn off hub weights. Ignore weight moves below 0.3 weights.

2.2. No track data: Turn off the adjustments for the red and green blades.

2.2.1. On initial runs with new blades or pitch links, it is best to make at least one idle run to correct large track errors prior to conducting full initial test plan.

2.2.2. As experience is gained with the RADS, the operator may want to use other diagnostic options available. These options are described in paragraph 412-A-18-11-00-01A-042A-A / 00153.

2.2.3. Never turn on red or green hub weights.

Table 1. Test Conditions

TEST I.D.	DESCRIPTION	412_41/412_50 FLIGHT PLAN				
		INITIAL	FLIGHT	FRAHM	D/S	TAIL
Idle	Flat Pitch at 62% RPM	X	X			
FPG100	Flat Pitch at 100% RPM	X	X			

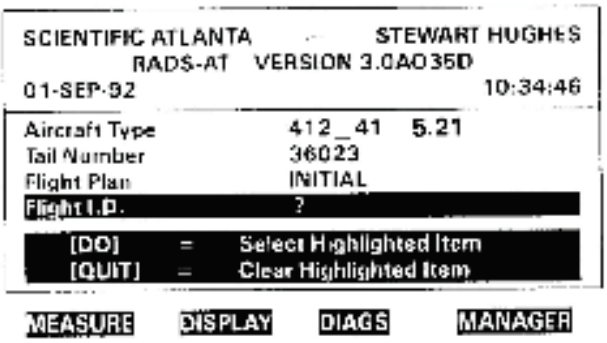
Table 1. Test Conditions (continued)

TEST I.D.	DESCRIPTION	412_41/412_50 FLIGHT PLAN				
		INITIAL	FLIGHT	FRAHM	D/S	TAIL
100TQ	100% RPM at 27% Mast Torque	O	O			
Hover	Hover into wind; IGE at 100% RPM	X	X			
Climb	1000 FPM Climb at 60 Kts at 100% RPM		X			
120K	Level Flight at 120 Kts at 100% RPM		X			
120/97	Level Flight at 120 Kts at 97% RPM		O			
VNE	Dive to 140 Knots at 100% RPM		X			
L/Down	1000 FPM Descent at 60 Knots at 100% RPM		X			
97%	Level Flight at 120 Kts at 97% RPM			X		
98%	Level Flight at 120 Kts at 98% RPM			X		
99%	Level Flight at 120 Kts at 99% RPM			X		
100%	Level Flight at 120 Kts at 100% RPM			X		
101%	Level Flight at 120 Kts at 101% RPM			X		
102%	Level Flight at 120 Kts at 102% RPM			X		
FPGDS	Flat Pitch 100% RPM - Twin Engine				X	
FPGTL	Flat Pitch 97% - 100% RPM - Single or Twin Engine					X

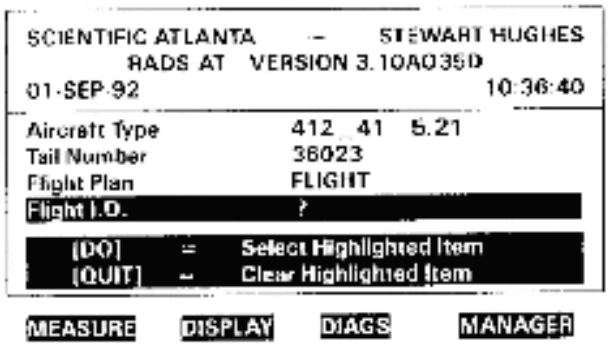
X = Test conditions for both 412_41 and 412_50

O = Additional Test for 412_50

412-41

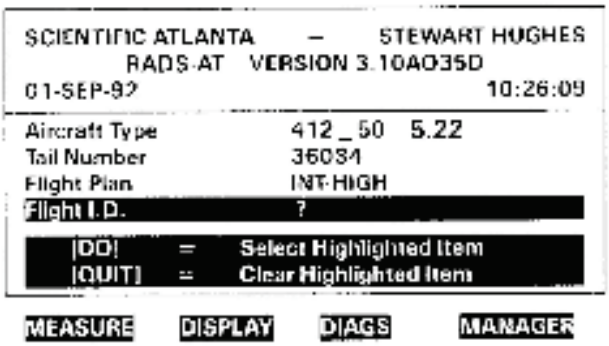


a) Initial Mode.

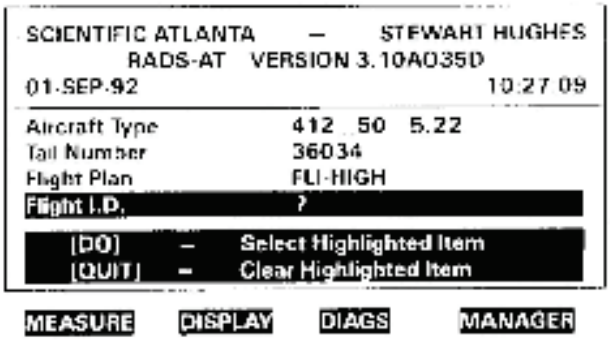


b) Flight Mode.

412-50



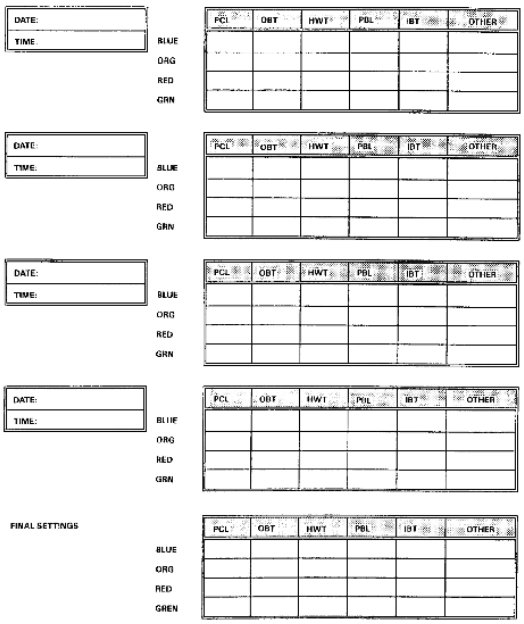
a) Initial Mode.



b) Flight Mode.

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Figure 1. Main Rotor Smoothing CADU Screen (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00010-A-001-01

Figure 2. Main Rotor Move Log (Sheet 1 of 1)

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Rotor Balancing - Advanced Editing Features

The RADS incorporates a number of editing features, which the operator can use to optimize diagnostics. The 412-41 and 412-50 program use the option called RESOLVE TO LIMIT as default analysis. Listed below are the basic features of options available from the DIAGNOSITCS MENU:

EDIT ADJUSTABLES: This feature allows operator to specify which adjustments are valid to be used in analysis. A ‘Y’ allows adjustments to be used and ‘N’ disallows its use. F1 allows changing of an adjustment status for use in an additional calculation. F2 and F3 disallow either a blade or an adjustment. F4 returns setting to original default. When editing has been completed and DO is pressed, the RADS will recalculate adjustments using only the new adjustments allowed.

EDIT DEFAULTS: This feature allows operator to specify maximum number of moves, based upon predefined LIMITS or by a set number specified by the RADS. The features available under this option are:

MAXIMUM NUMBER OF ADJUSTMENTS: This specifies maximum number of adjustments to be made. A ‘00’ allows RADS to use all adjustments. Placing a number will limit RADS to specify that number of moves only (the best combination). For example, if ‘4’ is specified, the four best moves will be displayed.

RESOLVE TO LIMIT: The standard mode for INITIAL is OFF, which allows all adjustments to be used to obtain optimum track and ground balance. The standard mode for FLIGHT is ON, which allows the system to calculate the least number of adjustments to reduce 1/rev levels to below predefined limits. This is a significant time saver as it allows only meaningful adjustments to be used. It is possible to use arrow keys to select OFF in FLIGHT mode. This allows all adjustments to be used to target an optimum 1/rev level. This should only be done in conjunction with selecting 6 or less adjustments as MAXIMUM NUMBER to minimize needless adjustments. In combination (i.e. OFF and 6), the system will calculate the best set of moves (limited to number selected) to reduce 1/rev levels.

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Advanced Editing Features - Pitch Links

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Lockwire	C-405	AR	

Procedure

1. Pitch links are used to track rotor at idle and to minimize lateral 1/rev experienced during flight. RADS AT equipment or Vibrex charts are used to determine pitch link adjustments. When adjusting a pitch link, the smallest meaningful move is one flat (0.6 or above one flat). All moves greater than one flat are rounded to the nearest 0.5 flat.

If a very large adjustment is required, use upper rod end bearing. The ratio is 1/2 turn of upper rod end bearing equals a 12 flat turn of pitch link.



THE MAXIMUM ALLOWABLE EXPOSED THREADS SHALL NOT EXCEED 11 ON PITCH LINK UPPER ROD END BEARING AND 8 ON THE LOWER ROD END BEARING.

NOTE

When working newly installed rotors, set pitch links to nominal setting (refer to 412-A-62-30-00-02A-720A-A / 00965).

1.1. Review RADS AT correction data to determine which pitch link(s) is (are) to be adjusted. Ensure amount and direction for each pitch link adjustment is recorded.

NOTE

Using a grease pencil or a felt tip marker, mark position of pitch link to rod end bearing before making any adjustment.

1.2. Remove lockwire and loosen nuts of selected pitch link.

1.3. Refer to decal on pitch link (412-A-62-30-00-02A-720A-A / 00965) and turn tube in required direction while counting off number of flats needed to make adjustment.



ENSURE ROD END BEARINGS REMAIN CENTERED IN CLEVISES WHILE TIGHTENING NUTS.

1.4. Center upper rod end bearing in clevis of applicable pitch horn. Center lower rod end bearing in clevis of rephasing lever. Hold tube and torque nuts (412-A-62-30-00-02A-720A-A / 00965).

1.5. To adjust upper rod end bearing only, loosen nut on rod end bearing, remove cotter pin, nut, and washer. Use work aid (412-A-62-30-00-02A-720A-A / 00965) to remove bolt and washer from pitch horn and pitch link.

NOTE

Install all bolts in rotating controls with heads in direction of rotation.

1.6. After adjustment of upper rod end bearing, center bearing in pitch horn. Install bolt, washers, and nut. Torque nut. Torque nut on upper rod end bearing 412-A-62-30-00-02A-720A-A / 00965. Install cotter pin.



ENSURE ROD END BEARINGS REMAIN CENTERED IN CLEVIS WHILE TIGHTENING NUTS.

1.7. If upper rod end bearing did not require adjustment, center bearing in clevis of pitch horn and torque nut (412-A-62-30-00-02A-720A-A / 00965).

1.8. Lockwire lower nut to insert using lockwire (C-405).

1.9. After completion of rotor smoothing, apply corrosion preventive compound (C-101) to threads of pitch link.

Advanced Editing Features - Trim Tabs

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Trim Tab Bender	T101654-101	1	
Trim Tab Gauge	T101656-145	1	

Procedure

1. Trim tab locations are shown in Figure 1. Outboard trim tabs (9) are used for initial ground tracking and are the primary adjustment for reducing vertical 1/rev in flight. Midspan trim tabs (6) are used as a secondary adjustment. The main purpose is to maintain proper track separation in flight. The next purpose is to supplement outboard trim tab if a very large adjustment of outboard trim tab is required during both initial tracking and vibration phases. Maintaining proper track separation with airspeed ensures low 2/rev levels. The inboard trim tabs are used if vertical 1/rev at VNE and/or during a 60 knot/1000 RPM letdown cannot be made acceptable while maintaining an acceptable vertical 1/rev at 120 knots IAS. The trim tab adjustments, as indicated by RADS AT equipment, are made equal and opposite in both blades of the pair.

2. Trim tabs are adjusted using the trim tab gauge (T101656-145) (Figure 2) and trim tab bender (T101654-101).

2.1. Review RADS AT correction data to identify trim tabs to be adjusted and amount of adjustment.

2.2. Install trim tab bender (T101654-101) on trim tab to be adjusted as follows:

NOTE

Install trim tab bender so angle indicator is on bottom side of trim tab.

2.2.1. Slightly loosen wingnuts on trim tab bender until bender can be slipped onto trim tab.

2.2.2. Position trim tab bender so there is a 0.125 inch (3.2 mm) gap between leading edge of trim tab bender and trailing edge of blade. Ensure trim tab bender is square with trim tab.

2.2.3. Tighten wingnuts until trim tab bender is tight on trim tab.

2.3. Place trim tab gauge outboard of outboard trim tab, outboard of midspan trim tab, or inboard of inboard trim tab as applicable. The spring clip of trim tab gauge shall be on bottom. Adjust position of trim tab gauge on blade until trailing edge of blade and angle of angle indicator of trim tab bender is flush with trim tab gauge surface.

2.4. Determine angle of trim tabs using trim tab gauge. Use trim tab gauge marked STA258 (-200BLD) to read outboard trim tab angle. Use trim tab gauge marked STA 200.1 (-300 BLD) to read angle of midspan trim tab (Figure 2). Use inboard tab side of trim tab gauge to read angle of inboard trim tab.

NOTE

When making small adjustments (less than 1.5°), slightly overcorrect, then bend trim tab back to desired setting. This ensures trim tab setting will hold without creeping back to original position.

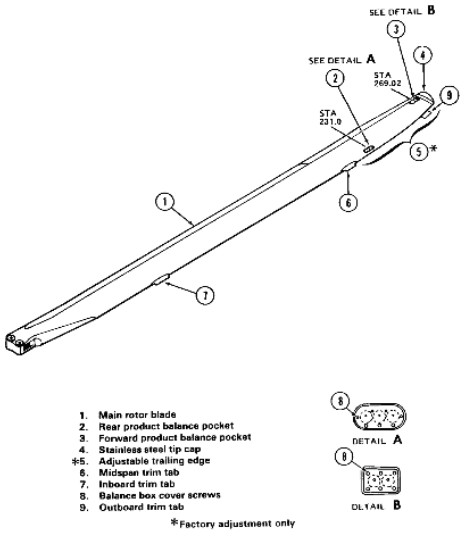
2.5. Use trim tab bender to adjust trim tab to required angle.

NOTE

Maximum allowable adjustment for midspan and outboard tab is ± 7°. Maximum allowable adjustment for inboard tab is ± 15°.

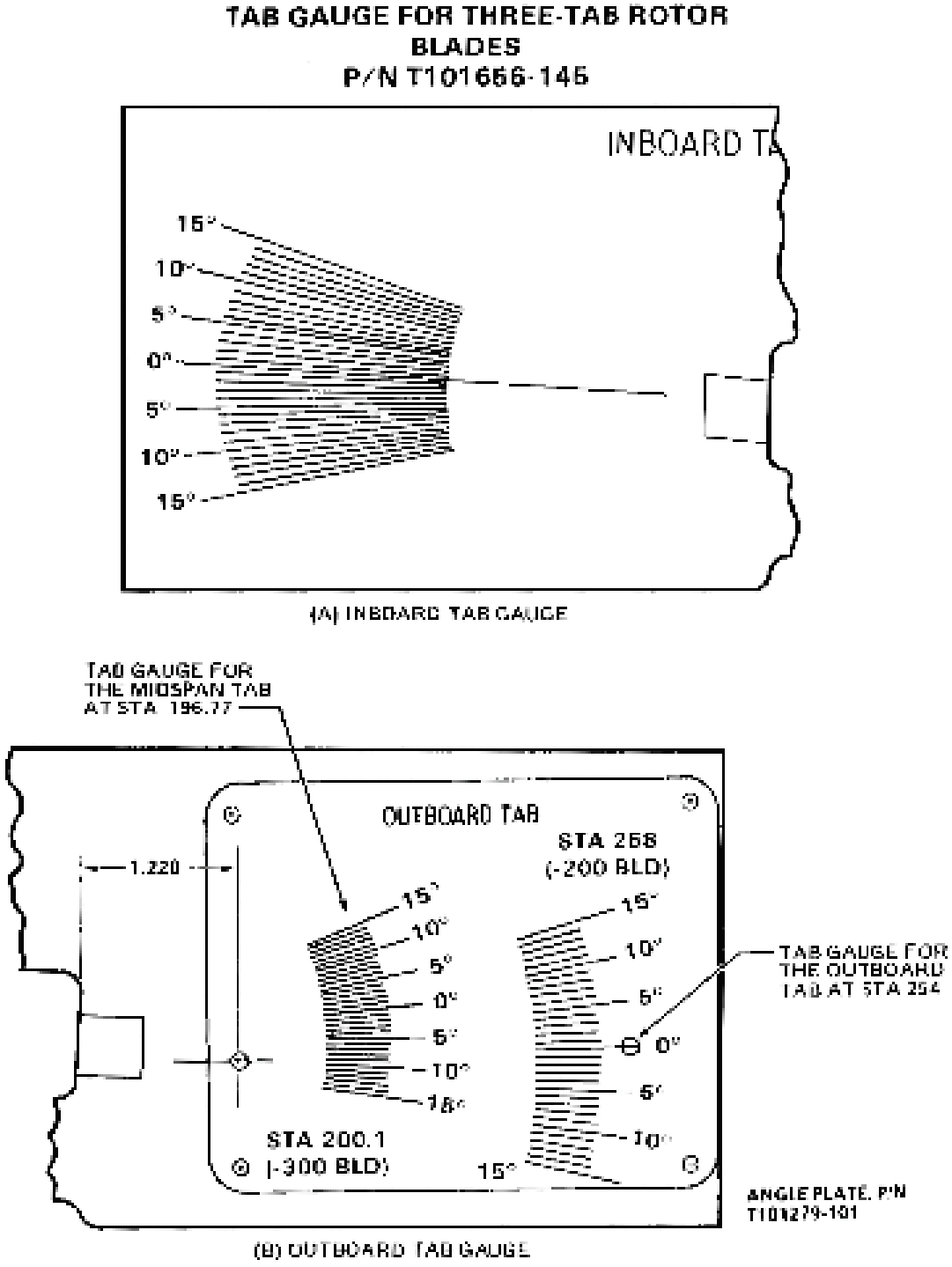
2.6. Repeat Step 2.1. through Step 2.5. until all outboard, midspan, and inboard trim tabs are adjusted.

2.7. After all trim tab adjustments have been made and verified, loosen wingnuts on trim tab bender and remove bender and trim tab gauge from blade.



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Figure 1. Adjustments on 412-015-300 Main Rotor Blade (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00012-A-001-01

Figure 2. Tab Gauge for Three-Tab Rotor Blades (Sheet 1 of 1)

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Advanced Editing Features - Hub Balance Weights

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Balance Weights	212-010-710-003	1	
Balance Weights	212-010-710-105	1	

Procedure

1. Hub balance weights (Figure 1) are called for by the number of balance weights (212-010-710-003) (0.25 lb. or 113 g) required. On RADS AT equipment, these are identified as WEIGHTS. A 0.5 weight adjustment is achieved by using one balance weights (212-010-710-105) (0.14 lb. or 63 g). In general, weight moves are made only to trailing edge location. The leading edge locations are used only when eight large weights have been added to trailing edge. After eight weights have been added to trailing edge, add remaining weights to leading edge location.

1.1. Review RADS AT correction data to determine which hub balance weight adjustments are required. Round adjustments to nearest 0.5 weight.

1.2. Inspect hub to determine current balance weight configuration.

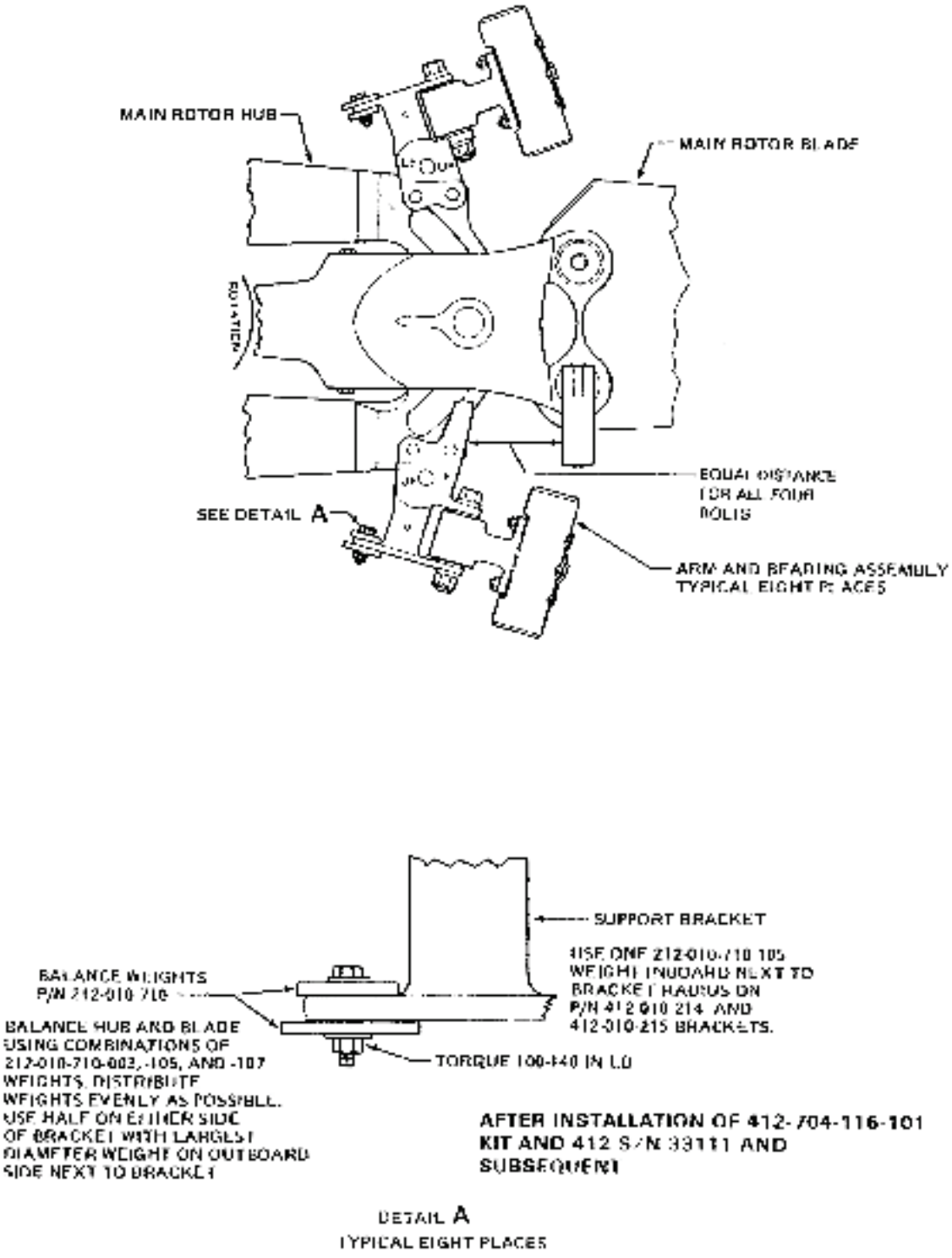
1.3. Determine required addition or subtraction of weights to make indicated adjustment.

1.4. Install required number of weights in locations shown in Figure 1. Equalize number of weights on each side of balance location arm. Do not exceed eight weights at any balance location.

NOTE

Use small steel washers to compensate for bolts slightly longer than needed for hub balance weights installed.

1.5. Use required length of NAS6205 bolts to secure weights to balance arm. Ensure a minimum of two threads extend beyond MS21042L5 nut. Torque bolt 100 to 140 inch-pounds (11.30 to 15.82 N-m).



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Figure 1. Main Rotor Hub Balance Weights (Sheet 1 of 1)

Rotor Balancing - Product Balance

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Product Balance Weights	412-015-052-101	1	
Product Balance Weights	412-015-053-101	1	
Product Balance Weights	412-015-056-101	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-307	AR	

Procedure

1. Once a main rotor has been product-balanced, it should not require additional product balancing until a blade is replaced, or an existing blade is refinished, repaired, or otherwise altered. Therefore, when trimming a previously worked main rotor, use pitch link adjustments and/or hub balance weight adjustments to correct balance. Product balance adjustments are defined as the number of product balance weights (412-015-053-101) added or removed from rear product balance pocket (2, Figure 1). The forward product balance pocket (3) and rear product balance pocket are at different blade stations. A different amount of weight must be adjusted in each pocket to achieve the same balancing result.

To simplify product balance adjustment, Table 1 shows combinations of weights needed to adjust product balance while maintaining mass balance of blade. The smallest meaningful adjustment is a 0.5 weight adjustment. The following is an example of how to use Table 1.

EXAMPLE: The RADS AT data indicate to product balance the blue coded blade -0.4 weight. Table 1 shows this adjustment is rounded to a -0.5 weight. Remove balance box covers screws (8, Figure 1). Remove balance box covers. The rear product balance pocket (2) is found to have one product balance weights (412-015-053-101) installed. The forward product balance pocket (3) is found to have one product balance weights (412-015-056-101) installed.

Two possible combinations of adjustments will yield a -0.5 weight move. The table shows the first option is to remove product balance weights (412-015-053-101) from rear product balance pocket, to add three product balance weights (412-015-052-101) to this pocket, and to add two product balance weights (412-015-052-101), one AN970-3 washer, and one AN970-4 washer to forward product balance pocket (3). The table shows the second option is to remove three product balance weights (412-015-052-101) from rear product balance pocket and add two product balance weights (412-015-052-101), one AN970-3 washer, and one AN970-4 washer to forward product balance pocket. However, the rear product balance pocket does not have three product balance weights (412-015-052-101) installed. Therefore, the second option can not be accomplished. Use first option to make required product balance adjustment. Refer to actual procedure for removal and installation of weight hold down nuts, balance box covers, and balance box cover screws.

NOTE

Use the following conversions to convert from one weight (or set of weights) to another.

- Six 412-015-052-101 weights = one 412-015-053-101 weight.
- Five 412-015-052-101 weights = one 412-015-056-101 weight.
- Three AN970-4 washers = one 412-015-052-101 weight.
- Two AN970-3 washers = one AN970-4 washers.
- Four 412-015-052-101 weights + two AN970-4 washers + two AN970-3 washers = one 412-015-056-101 weight.

1.1. Review RADS AT correction data to determine which blades require product balancing. Make balance adjustments in increments no smaller than 0.5 weight.

1.2. For each blade requiring product balancing, remove cover screws (8, Figure 1) from covers of forward product balance pocket (3) and rear product balance pocket (2). Remove covers.

1.3. Identify and note weights installed in both pockets and refer to Table 1 for required adjustment of weights.

NOTE

If required adjustment of weights can be made, proceed to Step 1.4.. If required adjustment of weights cannot be made, perform Step 1.7.. Recalculate corrections (412-A-18-11-00-00A-372A-A / 00149) using RADS AT equipment, and return to Step 1.1. above.

1.4. Remove weight hold down nuts from forward product balance pocket and/or rear product balance pocket, as required, to permit removal and addition of weights.

1.5. Add or remove weights to make adjustment described in Table 1.

1.6. Install hold down nuts removed in Step 1.4.. Torque nuts 15 to 20 inch-pounds (1.69 to 2.26 N-m).

1.7. Install cover on forward product balance pocket and on rear product balance pocket. Use balance box cover screws removed during Step 1.2..

1.8. Perform test flight (BHT-412-FM). If ride is satisfactory, proceed to Step 1.9.. If ride is not satisfactory, repeat Step 1.1. through Step 1.8..

1.9. Upon completion of product balancing, perform the following steps as soon as practical.

1.9.1. Remove and discard all balance box cover screws (8, Figure 1). Remove cover from forward product balance pocket (3) and rear product balance pocket (2).

1.9.2. Replace all hold down nuts (MS21042L3) which were removed and installed during product balancing. Torque nuts 15 to 20 inch-pounds (1.69 to 2.26 N-m). Apply adhesive (C-307) to new hold down nuts and screw heads.

1.9.3. Apply adhesive (C-307) to mating surfaces of cover and blade pocket. Install cover on forward product balance pocket and on rear product balance pocket.

1.9.4. Install new balance box cover screws. Apply adhesive (C-307) to screw heads before installation. Smooth adhesive squeeze-out flush with contour of blade. Allow adhesive to cure before flight.

1.10.

NOTE

Curing sealant at lower temperatures requires more time.

Table 1. Product Balance Adjustment Combinations

			FORWARD PRODUCT BALANCE POCKET		REAR PRODUCT BALANCE POCKET	
RADS AT INCREMENTS	ROUNDED VALUE	OPTION NO.	ADD	REMOVE	ADD	REMOVE
0 to 0.3	0	1	---	---	---	---
0 to -0.3	0	1	---	---	---	---
0.3 to 0.7	0.5	1	Two product balance weights (412-015-052-101) One AN970-3 One AN970-4	One product balance weights (412-015-056-101)	Three product balance weights (412-015-052-101)	---
0.3 to 0.7	0.5	2	---	Two product balance weights (412-015-052-101) One AN970-3 One AN970-4	Three product balance weights (412-015-052-101)	---

Table 1. Product Balance Adjustment Combinations (continued)

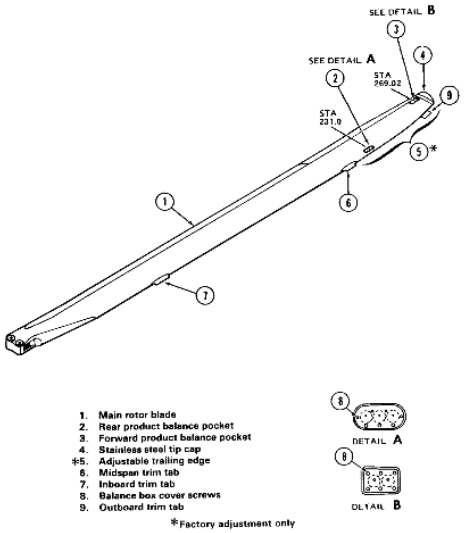
			FORWARD PRODUCT BALANCE POCKET		REAR PRODUCT BALANCE POCKET	
RADS AT INCREMENTS	**ROUNDED VALUE**	**OPTION NO.**	**ADD**	**REMOVE**	**ADD**	**REMOVE**
-0.3 to -0.7	-0.5	1	Two product balance weights (412-015-052-101) One AN970-3 One AN970-4	---	Three product balance weights (412-015-052-101)	One product balance weights (412-015-053-101)
-0.3 to -0.7	-0.5	2	Two product balance weights (412-015-052-101) One AN970-3 One AN970-4	---	---	Three product balance weights (412-015-052-101)
0.8 to 1.3	1	1	---	One product balance weights (412-015-056-101)	One product balance weights (412-015-053-101)	---
-0.8 to -1.3	-1	1	One product balance weights (412-015-056-101)	---	---	One product balance weights (412-015-053-101)

NOTES:

1. The standard load of product balance for the 412-015-300-109 blade is a total of 0.3 lb. (135 g), one product balance weights (412-015-056-101) and one product balance weights (412-015-052-101), in the forward product balance pocket and 0.35 lb. (158 g), one product balance weights (412-015-053-101) and one product balance weights (412-015-052-101), in the rear product balance pocket. If serious problems occur with balancing the rotor, return the blades to the standard load of product balance and rework the rotor.

2. One product balance weights (412-015-052-101) weighs 0.05 lb. (23 g). One product balance weights (412-015-053-101) weighs 0.30 lb. (135 g). One product balance weights (412-015-056-101) weighs 0.25 lb. (117 g).

3. Decide on the best move(s) that can be made from the options presented. Make the entire group of moves shown for the option chosen. It may be necessary to add and subtract weights from the same product balance pocket to accomplish the adjustment.



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Figure 1. Adjustments on 412-015-300 Main Rotor Blade (Sheet 1 of 1)

Rotor Balancing - Setting Autorotation RPM

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Lockwire	C-405	AR	

Procedure

NOTE

Autorotation RPM should be set during first flight.

1. Check autorotation (rotor) RPM (Figure 1).

2. If required, correct autorotation (rotor) RPM as follows:



MAXIMUM EXPOSED THREADS SHALL NOT EXCEED 11 ON PITCH LINK UPPER ROD END BEARING AND 8 ON LOWER ROD END BEARING.

NOTE

An airspeed of 65 KNOTS is used to set autorotation RPM.

2.1. If autorotation RPM is high, increase length of all pitch link assemblies an equal amount.

2.2. If autorotation RPM is low, decrease length of all pitch link assemblies an equal amount.

2.3. Remove lockwire and loosen jamnuts on pitch link. Adjust pitch link as required.

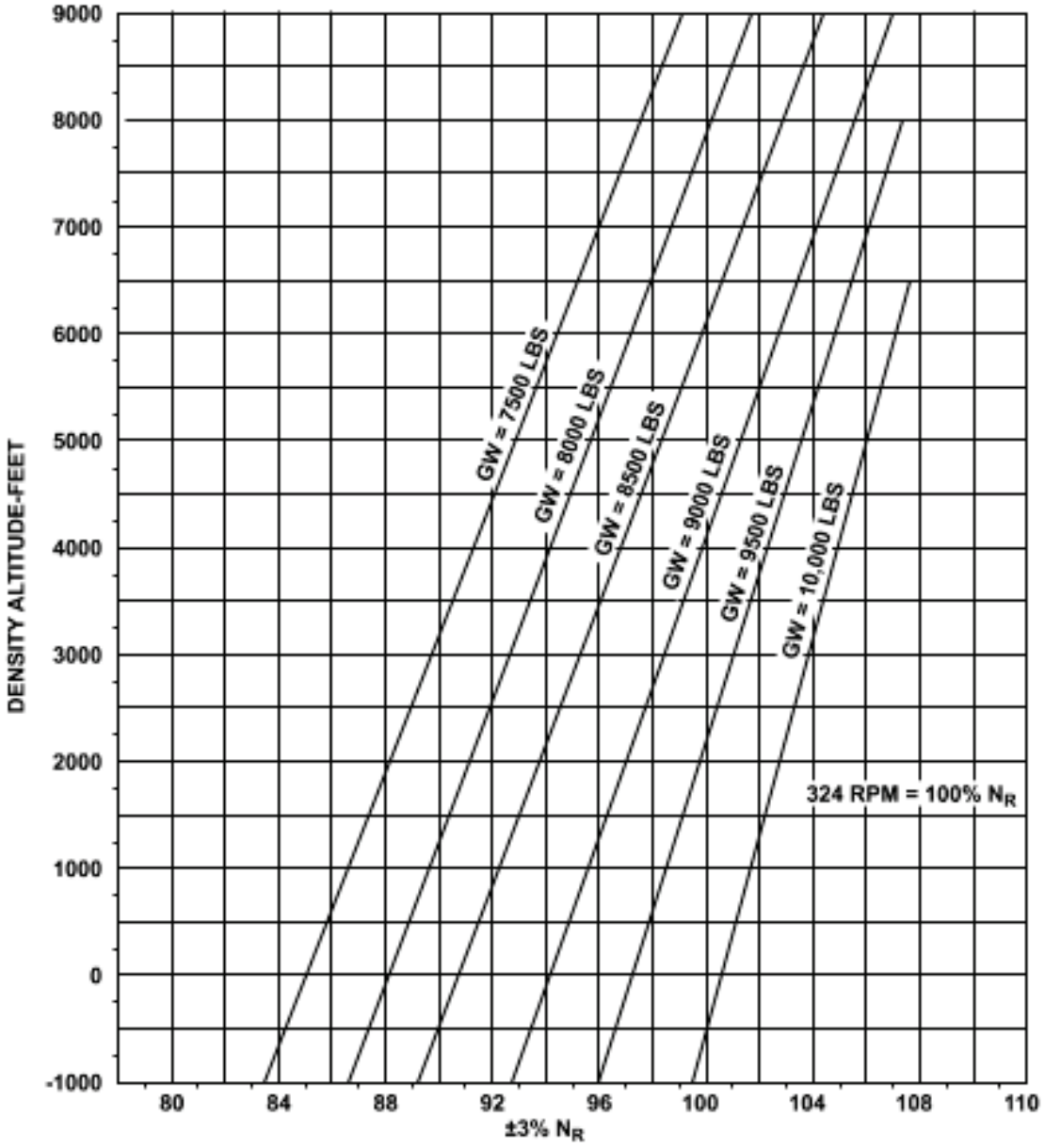
NOTE

One full turn of all four pitch link tubes will change autorotation RPM approximately 1% N_R. One full turn of all four pitch link upper rod ends will change autorotation RPM approximately 3%.

2.4. Center upper rod end bearing in pitch horn clevis. Center lower rod end bearing in rephasing lever clevis. Hold pitch link tube and torque jamnuts 120 to 150 inch-pounds (13.56 to 16.95 N-m). Secure lower jamnut to pitch link tube insert using lockwire (C-405).

2.5. Test fly helicopter to confirm autorotation RPM is set correctly (BHT-412-FM).

2.6. After final adjustment apply corrosion preventive compound (C-101) to threads of pitch link.



NOTE

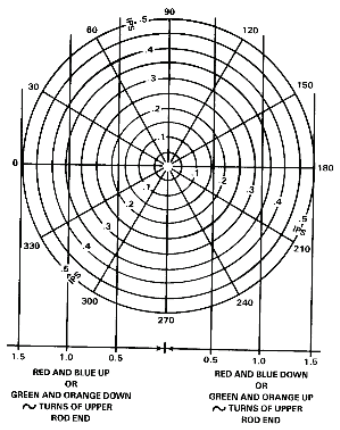
Refer to model 412 Flight Manual for density altitude chart.

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Figure 1. Autorotation Descent Rotor RPM Chart (Sheet 1 of 1)

Setting Autorotation RPM - Control of Main Rotor 2/Rev

The standard accelerometer fit for RADS is adequate to monitor 2/rev levels. Main rotor 2/rev vibration results when one pair of blades is not generating the same lift as the other pair of blades. This is normally reduced to acceptable levels by flying the rotor as close to the ±20 mm track spread as possible. This track spread is automatically targeted by the desired spread. When normal tracking does not reduce vibration to an acceptable level, refer to Figure 1 for further corrective action.



1. Adjust tabs to obtain stable track readings from hover to 120 kts.
2. Plot 120 kt lateral 2/rev on chart if above .15 IPS.
3. Make indicated pitch link adjustment on one pair of blades to minimize 2/rev.

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Figure 1. 2/Rev Chart (Sheet 1 of 1)

Setting Autorotation RPM - Control of Main Rotor 4/rev

The next most predominant vibration (besides 1/rev) is main rotor 4/rev. This vibration is perceived by the crew and passengers as a “buzz” in the cabin. The primary device for maintaining low 4/rev levels is the pendulum absorber assembly on the main rotor hub. Secondary devices are instrument panel fore-and-aft struts (being delivered on new helicopters) and instrument panel Frahm absorber. These devices are designed to reduce instrument panel vibration which can blur instruments and cause crew to perceive higher vibration levels.

Many factors can affect the 4/rev levels. These include:

Degradation of the pendulum absorbers.

Loose battery or avionics in nose.

Loose instrument housings in the instrument panel.

Low break-out friction in the elevator.

Cracks or looseness in the instrument panel support structure (cracks in glare shield, loose side brace attachments to the airframe, and the sympathy of special kits to the 4/rev vibration).

If 4/rev is perceived or measured to be higher than normal, refer to Table 1 for recommended corrective action.

Table 1. Troubleshooting Main Rotor Vibration

PROBLEM	SOURCE	CORRECTIVE ACTION
Unworkable or sensitive lateral 1/rev	Soft pylon mounts	If 204-031-927-105 are installed replace with 204-031-927-107 mount.
	Lead-lag dampers	Replace lead lag damper pair which show erratic trends as described in Figure 1.
	Degraded -107 mounts	Inspect -107 pylon mounts for degradation.
	Degraded skid gear supports	Inspect cross tube support rubber for degradation. Replace separated supports.
Changing 1/rev	Creeping tabs	Check tab angles. If tab is washing out, correct using RADS to specify corrective adjustment. Overbend tab and set to proper position to minimize tab creep.
	Hub shifting on the cones	Check upper and lower cone torques. Incorporate T.B. 412-92-104 improvements. Refer to Table 2 for signature of hub shift.
Unworkable vertical 1/rev in flight	Mismatched rotor blades	Turn on inboard tabs (blue and orange only), eliminate VNE condition when working rotor if not required for normal missions. Replace blades which show erratic track with airspeed.
Unworkable vertical and lateral 1/rev	Failed inboard spindle bearing	Inspect inboard spindle bearings; replace those beyond limits.
Main rotor 2/rev	Unequal lift between blade pairs	Retrack rotor to minimize change in track between pairs with airspeed using midspan tabs (1° = 11 mm).
		Use pitch links to minimize 2/rev using the chart in Figure 2.
Main rotor 4/rev	Degraded pendulums	Grease pendulums.
		Inspect pendulums for worn bearings.
	Loose airframe structure	Tighten battery. Tighten instruments. Repair instrument panel support, glareshield cracks. Install F/A panel braces.

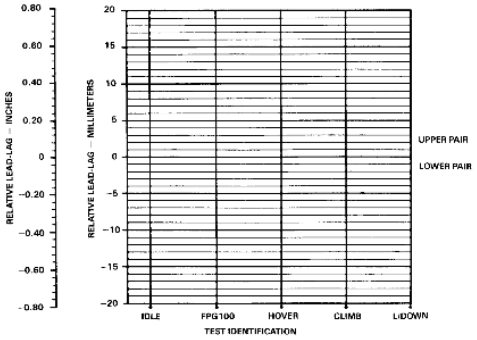
Table 2. Example of Track Change with a Shifting Hub

A) TRACK FOR ORIGINAL FLIGHT				
TEST	**BLADE COLOR CODE**			
ID	BLU	ORG	RED	GRN
IDL	16	-17	11	-10
FPG100	12	-17	15	-10
HOV	16	-18	14	-11
CLI	13	-15	15	-12
100	14	-8	11	-17
120	10	-6	16	-20
VNE	10	-7	17	-20
LET	11	-17	18	-20

B) TRACK AFTER RIDE CHANGED				
TEST	**BLADE COLOR CODE**			
ID	BLU	ORG	RED	GRN
IDL	17	-17	9	-10
FPG100	9	-22	20	-7
HOV	10	-10	19	-9
CLI	6	-17	22	-11
100	7	-13	21	-14
120	3	-10	24	-17
VNE	4	-12	26	-18
LET	6	-19	23	-10

C) CHANGE IN TRACK				
TEST	BLADE COLOR CODE			
ID	BLU	ORG	RED	GRN
IDL	1	0	-2	0
FPG100	-3	-5	5	3
HOV	-6	-8	5	2
CLI	-7	-2	7	1
100	-7	-5	10	3
120	-7	-4	8	3
VNE	-6	-5	9	2
LET	-5	-2	5	10

Change in track is obtained by subtracting the track numbers obtained in the original flight from those obtained after the ride changed. To determine if the hub is shifting on its cones, look at the change in track at idle and at 120 knots. If the track does not change at idle, but has changed significantly at 120 knots with the blades of a pair changing equally and oppositely, the hub is probably shifting. This is an actual example of a hub shifting on the cones. Notice the track change of the blue and red blades at 120 knots.

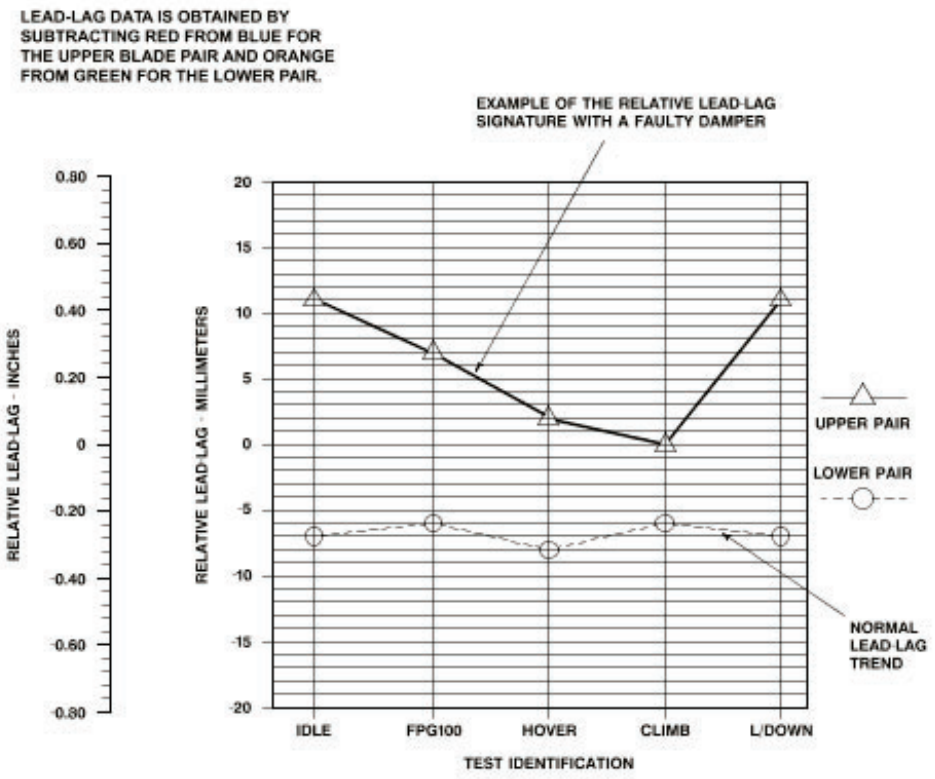


NOTES

1. Subtract the ORG lead-lag from the GRN and plot on a copy of the chart.
2. Subtract the BLUE lead-lag from the RED and plot on a copy of the chart.
3. Properly operating dampers will have a straight tread.
4. Possible faulty dampers will have a tread of changing lead-lag with the idle and letdown near the same value.

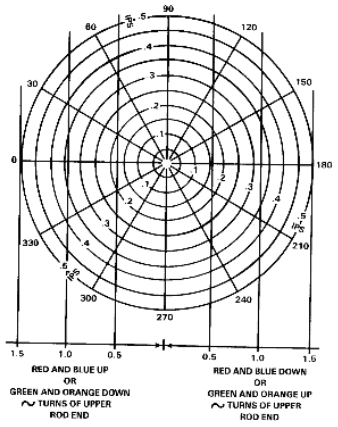
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Figure 1. Lead - Lag Trending (Sheet 1 of 2)



ICN-412-A-181100-B-97499-00017-A-001-01

Figure 1. Lead - Lag Trending (Sheet 2 of 2)



1. Adjust tabs to obtain stable track readings from hover to 120 kts.
2. Plot 120 kt lateral 2/rev on chart if above .15 IPS.
3. Make indicated pitch link adjustment on one pair of blades to minimize 2/rev.

ICN-412-A-181100-B-97499-00015-A-001-01

Figure 2. 2/Rev Chart (Sheet 1 of 1)

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Rotor Balancing - Ride Quality Guidelines

The following guidelines are provided as recommended action to be followed depending upon existing ride quality.

Main rotor 1/rev in flight:

0 - 0.2 ips: No action required.

0.2 - 0.5 ips: 1/rev levels are serviceable with no corrective action required.

0.5 - 1.0 ips: 1/rev levels higher than recommended. Corrective action should be taken as soon as possible.

Over 1.0 ips: 1/rev levels unacceptable for continued service. Helicopter should be removed from service as soon as possible and corrective action taken.

Main rotor 2/rev: The RADS is programmed to alert the operator if 2/rev levels exceed 0.15 ips. This is only a guide to ride quality. If 2/rev is above 0.15 ips, refer to 412-A-18-11-00-01A-042B-A / 00159 for corrective action. Normally, the RADS will correct 2/rev automatically by controlling spacing between rotor blade pairs.

Main rotor 4/rev: Main rotor 4/rev is dependent on kits installed and condition of nose structure. As measured on the instrument panel, 4/rev levels at 120 knots below 1.0 ips should be considered normal. 4/rev levels above 1.0 ips at 120 knots are above normal, refer to 412-A-18-11-00-00A-422A-A / 00162 for corrective action.

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Rotor Balancing - Troubleshooting Problems

Procedure

1. Table 1 lists the corrective actions if problems are encountered reducing vibration to acceptable levels. Problems with RADS are addressed in flow chart (Figure 1) and Table 2 through Table 5.

Table 1. Troubleshooting Main Rotor Vibration

PROBLEM	SOURCE	CORRECTIVE ACTION
Unworkable or sensitive lateral 1/rev	Soft pylon mounts	If 204-031-927-105 are installed replace with 204-031-927-107 mount.
	Lead-lag dampers	Replace lead lag damper pair which show erratic trends as described in Figure 2.
	Degraded -107 mounts	Inspect -107 pylon mounts for degradation.
	Degraded skid gear supports	Inspect cross tube support rubber for degradation. Replace separated supports.
Changing 1/rev	Creeping tabs	Check tab angles. If tab is washing out, correct using RADS to specify corrective adjustment. Overbend tab and set to proper position to minimize tab creep.
	Hub shifting on the cones	Check upper and lower cone torques. Incorporate T.B. 412-92-104 improvements. Refer to Table 6 for signature of hub shift.
Unworkable vertical 1/rev in flight	Mismatched rotor blades	Turn on inboard tabs (blue and orange only), eliminate VNE condition when working rotor if not required for normal missions. Replace blades which show erratic track with airspeed.
Unworkable vertical and lateral 1/rev	Failed inboard spindle bearing	Inspect inboard spindle bearings; replace those beyond limits.
Main rotor 2/rev	Unequal lift between blade pairs	Retrack rotor to minimize change in track between pairs with airspeed using midspan tabs (1° = 11 mm).
		Use pitch links to minimize 2/rev using the chart in Figure 3.
Main rotor 4/rev	Degraded pendulums	Grease pendulums.
		Inspect pendulums for worn bearings.
	Loose airframe structure	Tighten battery. Tighten instruments. Repair instrument panel support, glareshield cracks. Install F/A panel braces.

Table 2. Troubleshooting RADS Main Rotor

PROBLEM	SOURCE	CORRECTIVE ACTION
Adjustments do not reduce 1/rev	Accelerometer cables crossed	Verify proper installation
CADU will not communicate with DAU	DAU not getting power	Check switch, cable, and power port
	Tracker cable internally shorted	Disconnect cable, if CADU now communicates with DAU. Repair or replace tracker cable.
TACHO failure	Incorrect hook-up	Connect azimuth to TACHO #1
	Magnetic pickup to interrupter gap incorrect	Correct gap

Table 2. Troubleshooting RADS Main Rotor (continued)

PROBLEM	SOURCE	CORRECTIVE ACTION
	Airframe wiring failed	Bypass ship's wiring and route TACHO cable directly to sensor
Bad daylight track data	Bare blade leading edge	Paint leading edges
	Data corrupted by sunlight	Add sunshield, take data flying away from sun.
	Tracker backwards	Position tracker with lens forward.
Track swaps (upper to lower)	Corrupted track data	Repeat. If persistent, replace tracker.
No track data at night	Incorrect tracker angle	Correct tracker angle
	Old reflective tape	Replace tape
	DAU switch in wrong position	Set to "Night"
	Tracker bulb burned out	Replace tracker

Table 3. Troubleshooting Chart When no Changes are Identified

PROBLEM	PROBABLE CAUSE	CORRECTIVE ACTION
High lateral 1/rev at idle, and at 100% RPM. Vibration will not respond normally to balance moves. Vibration could be intermittent from run to run.	Hung droop stops	1. Inspect droop stops for binding and ensure correct rigging. Correct any discrepancies found. (Note: RADS can be used to guide inspections. Check phase of lateral 1/rev at idle: ORG=250°, BLUE=340°, GRN=160°, RED=70°). Remove all hub balance weights and rebalance with RADS to verify correction of problem. 2. If problem persists, remove droop stop arms and operate helicopter without them for maintenance tests only. Balance rotor using RADS. Once rotor has been balanced, reinstall droop stops and rig per maintenance manual.
High ground bounce at approx. 80% rotor RPM. Idle 1/rev not excessive.	Swashplate support	1. Inspect swashplate support bearings for proper installation and serviceability.
Excessive ground bounce only at 100% RPM.	Rotor track and balance	1. Reinstall RADS and balance rotor at 100% RPM. Have flight crew position collective and cyclic to minimize vibration during normal operation.
Vertical 1/rev at 120 knots will not respond to tab moves.	Cracked trim tab	1. Inspect all tabs and replace any found cracked.
Vertical 1/rev in hover and climb, but not at 120 knots.	Elastomeric failure	1. Inspect spindles for signs of elastomer delamination. Replace any rubber bearing delaminated beyond maintenance manual limits. (Note: Use following hover vertical phase as a guide: ORG=190°, BLUE=280°, GRN=100°, RED=10°.)
	Control system looseness	2. Inspect control system rod end bearings for excessive wear. Replace any worn bearing. (Note: Use following hover vertical phase as a guide: ORG=0°, BLUE=90°, GRN=180°, RED=270°.)
Rotor will not stay tracked.	Hub shift	1. Incorporate improved hub to mast joint (T.B. 412-92-104/-110). 2. If above has been incorporated, retorque rotor head cones and thru bolts.
Lateral 1/rev in flight which will not respond properly to adjustments.	Soft pylon mounts	1. Install -107 pylon mounts (T.B. 412-92-111).

Table 3. Troubleshooting Chart When no Changes are Identified (continued)

PROBLEM	PROBABLE CAUSE	CORRECTIVE ACTION
	Lead-lag damper looseness failure	1. Inspect lead-lag damper to spindle joint for black “smoke” which would indicate looseness. To correct, bond joint on all dampers per T.B. 412-90-92.
Lateral 1/rev in flight which will not respond properly to adjustments.	Lead-lag damper looseness/failure	2. Review lead-lag trends as defined in maintenance manual. Replace pair which shows “problem” trend.
Lateral 1/rev in letdown which has appeared to degrade over time.	Control system looseness	1. Inspect control system rod end bearings for excessive wear. Replace any worn bearing.
Vertical 1/rev in flight which appeared to degrade over time.	Soft trim tabs	1. Inspect trim tabs and reset to previous levels. 2. Replace trim tabs that are too soft and washout or tabs that are too stiff and creep.

Table 4. Troubleshooting Rotor Blade Problems

PROBLEM	PROBABLE CAUSE	CORRECTIVE ACTION
Excessive lateral 1/rev at idle and higher rpms which cannot be balanced with available hub weight provisions.	Main rotor blade unbalance	Note: Verify droop stop rigging prior to conducting following steps. 1. Check main rotor product balance pockets to verify that blades have proper loading. Reset blade product balance as necessary to return blade to proper loading. 2. If resetting product balance does not fix balance problem, check lateral 1/rev phase at idle. 245° is BLUE heavy, 135° is ORANGE heavy, 45° is RED heavy, 315° is GREEN heavy. Statistically, the heavy blade is the problem as it is hard to make a blade lighter than master balance. To correct problem: A) Remove suspect blade(s) and balance on scale balancer. B) Replace suspect blade with another. Note: In an emergency when additional blades are not available it is possible to adjust the product balance weights to balance the rotor. This, however, will limit the ability to product balance the rotor in flight. To balance the blade using product balance either add a -056 weight to the light blade’s leading edge pocket, or remove a -056 weight from the heavy blade’s trailing edge pocket. One -056 weight is worth eleven (-11) hub weights.
Excessive track change with RPM.	Blade outboard trailing edge shape problem	Note: Verify RADS tracker, install as described in flow chart. 1. Use midspan tab to help outboard tab correct track change. 2. Observe other blades. If another blade has similar track trend with RPM, repair the rotor blades so that blades with similar RPM trends are opposite of each other.

Table 4. Troubleshooting Rotor Blade Problems (continued)

PROBLEM	PROBABLE CAUSE	CORRECTIVE ACTION
		3. Contact Bell service rep. about bending the outboard trailing edge of the blade to correct RPM trend.
		4. Replace blade.
Excessive vertical on first flight after ground tracking which cannot be worked to an acceptable ride with RADS and standard tab adjustments.	Inboard contour of blade	1. Inspect trim tabs to ensure that they are not cracked or damaged.
		2. Use midspan tab to help outboard tab correct/reduce vibration levels.
		3. Observe track change from FPG100 to 120 knots and from 120 knots to VNE. Repair rotor blades to match observed track trends. Rework rotors. If this cannot be done or does not work, then replace blades with abnormal track trend.
Excessive lateral in flight.	Soft pylon mounts	1. Install -107 pylon mounts as per T.B. 412-92-111.
Lateral and/or F/A 1/rev in flight.	Mismatched blades	1. Observe track characteristics of blades. Change blade(s) which do not behave as the others.
Excessive track change from ground to hover and vibration in hover.	Product balance	1. Verify RADS tracker installation.
		2. Reset blade product balance to nominal.
		3. Replace blade(s) which do not have sufficient product balance capability to obtain an acceptable ride.

Table 5. Troubleshooting Main Rotor Hub/Control System Faults

PROBLEM	PROBABLE CAUSE	CORRECTIVE ACTION
High lateral 1/rev at idle, during runup, and at 100% RPM. Vibration will not respond normally to balance moves. Vibration could be intermittent from run to run.	Hung droop stops	1. Inspect droop stops for binding and ensure correct rigging. Correct any discrepancies found. (Note: RADS can be used to guide inspections. Check phase of lateral 1/rev at idle: ORG=250°, BLUE=340°, GRN=160°, RED=70°.)
		Remove all hub balance weights and rebalance with RADS to verify correction of problem.
		2. If problem persists, remove droop stop arms and operate helicopter without them for maintenance tests only. Balance rotor using RADS. Once rotor has been balanced, reinstall droop stops and rig per maintenance manual.
Excessive ground bounce only at 100% RPM.	Rotor track and balance	1. Reinstall all RADS and balance rotor at 100% RPM. Have flight crew position collective and cyclic to minimize vibration during normal operation.
Vertical 1/rev in hover and climb, but not at 120 knots.	Elastomeric failure	1. Inspect spindles for signs of elastomer delamination. Replace any rubber bearing delaminated beyond limits. (Note: use following hover vertical phase as a guide: ORG=190°, BLUE=280°, GRN=100°, RED=10°.)
Rotor will not stay tracked.	Hub shift	1. Incorporate improved hub to mast joint (T.B. 412-92-104).
		2. If above has been incorporated, retorque rotor head cones and thru bolts.

Table 5. Troubleshooting Main Rotor Hub/Control System Faults (continued)

PROBLEM	PROBABLE CAUSE	CORRECTIVE ACTION
Lateral 1/rev in flight which will not respond properly to adjustments.	Lead-lag damper looseness/failure	1. Inspect lead-lag damper to spindle joint for black “smoke” which would indicate looseness. To correct, bond joint on all dampers as per T.B. 412-90-92. 2. Review lead-lag trends as defined in maintenance manual. Replace damper pair which shows “problem” trend.
High ground bounce at approx. 80% rotor RPM. Idle 1/rev not excessive.	Swashplate support	1. Inspect swashplate support bearings for proper installation and serviceability.
Vertical 1/rev in hover and/or climb, but not at 120 knots.	Control system looseness	1. Inspect control system rod end bearings for excessive wear. Replace any worn bearing. Look at track to note any blade that climbs from hover to climb then dives from climb to 120 knots. Inspect this blade’s control system. (Note: Use following hover vertical phase as a guide: ORG=0°, BLUE=90°, GRN=180°, RED=270°.)

Table 6. Example of Track Change with a Shifting Hub

A) TRACK FOR ORIGINAL FLIGHT

TEST	BLADE COLOR CODE			
ID	BLU	ORG	RED	GRN
IDL	16	-17	11	-10
FPG100	12	-17	15	-10
HOV	16	-18	14	-11
CLI	13	-15	15	-12
100	14	-8	11	-17
120	10	-6	16	-20
VNE	10	-7	17	-20
LET	11	-17	18	-20

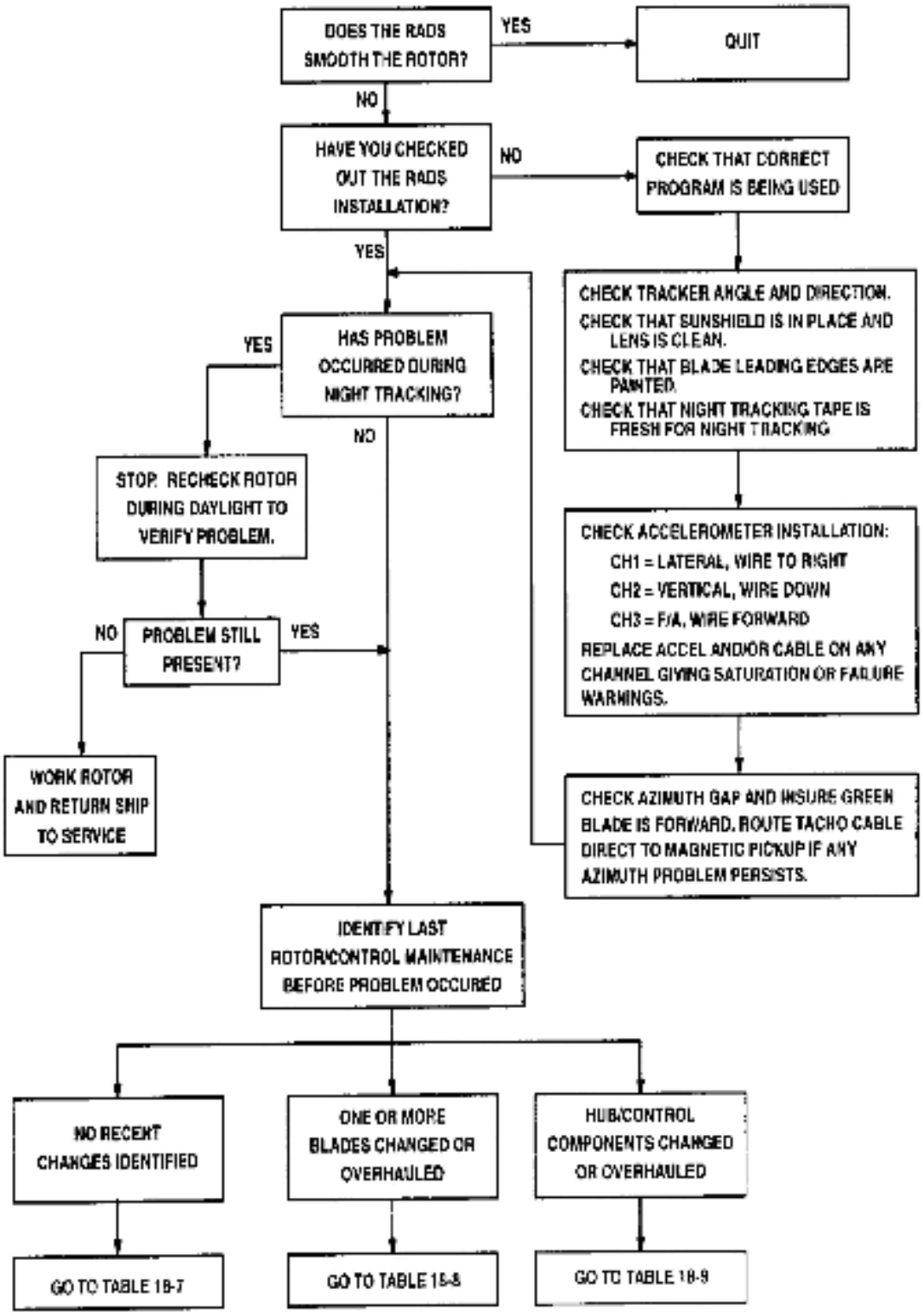
B) TRACK AFTER RIDE CHANGED

TEST	BLADE COLOR CODE			
ID	BLU	ORG	RED	GRN
IDL	17	-17	9	-10
FPG100	9	-22	20	-7
HOV	10	-10	19	-9
CLI	6	-17	22	-11
100	7	-13	21	-14
120	3	-10	24	-17
VNE	4	-12	26	-18
LET	6	-19	23	-10

C) CHANGE IN TRACK

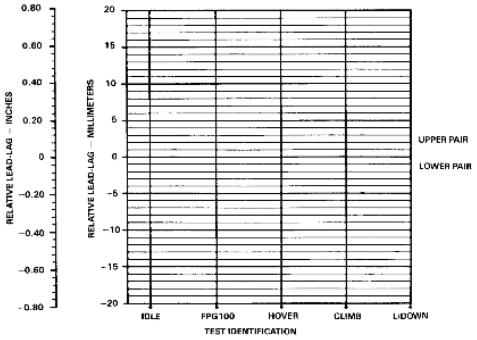
TEST	BLADE COLOR CODE			
ID	BLU	ORG	RED	GRN
IDL	1	0	-2	0
FPG100	-3	-5	5	3
HOV	-6	-8	5	2
CLI	-7	-2	7	1
100	-7	-5	10	3
120	-7	-4	8	3
VNE	-6	-5	9	2
LET	-5	-2	5	10

Change in track is obtained by subtracting the track numbers obtained in the original flight from those obtained after the ride changed. To determine if the hub is shifting on its cones, look at the change in track at idle and at 120 knots. If the track does not change at idle, but has changed significantly at 120 knots with the blades of a pair changing equally and oppositely, the hub is probably shifting. This is an actual example of a hub shifting on the cones. Notice the track change of the blue and red blades at 120 knots.



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Figure 1. Flow Chart for Troubleshooting Main Rotor (Sheet 1 of 1)

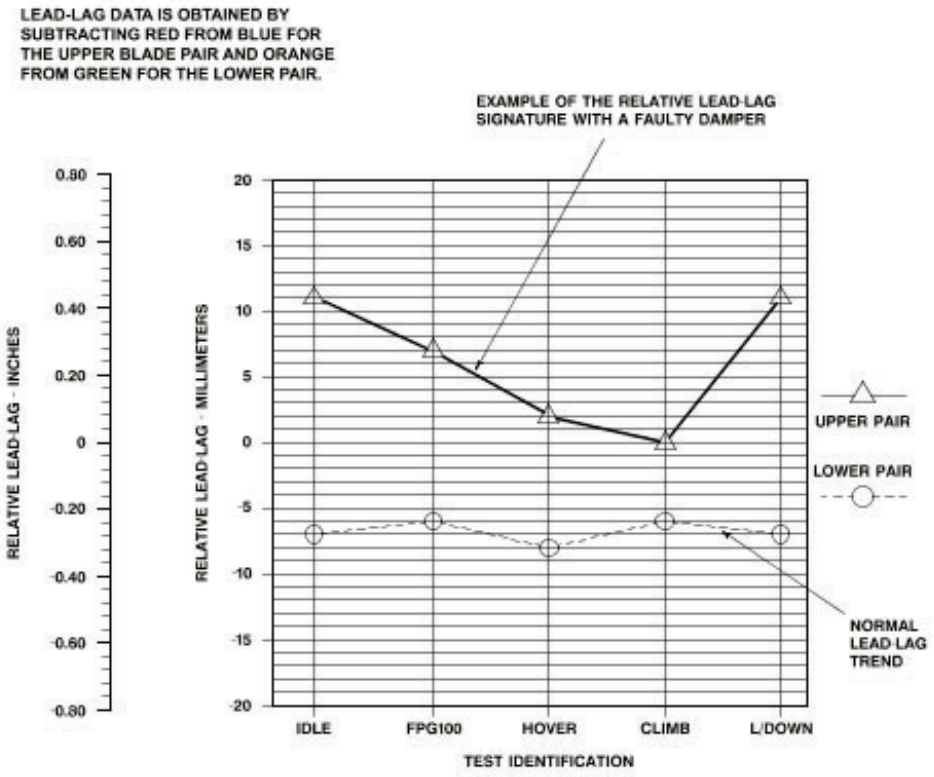


NOTES

1. Subtract the ORG lead-lag from the GRN and plot on a copy of the chart.
2. Subtract the BLUE lead-lag from the RED and plot on a copy of the chart.
3. Properly operating dampers will have a straight tread.
4. Possible faulty dampers will have a tread of changing lead-lag with the idle and letdown near the same value.

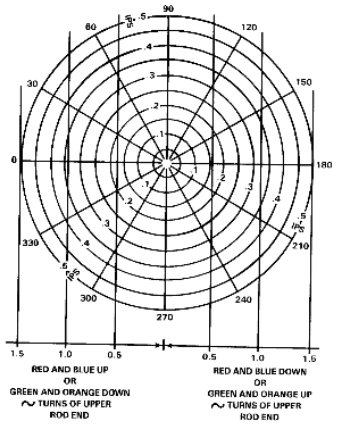
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Figure 2. Lead - Lag Trending (Sheet 1 of 2)



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Figure 2. Lead - Lag Trending (Sheet 2 of 2)



1. Adjust tabs to obtain stable track readings from hover to 120 kts.
2. Plot 120 kt lateral 2/rev on chart if above .15 IPS.
3. Make indicated pitch link adjustment on one pair of blades to minimize 2/rev.

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Figure 3. 2/Rev Chart (Sheet 1 of 1)

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Balancing the Tail Rotor - Control of Tail Rotor Vibration

Vibration from tail rotor can be perceived as a medium to high frequency vibration in the airframe. The following procedures should be conducted when:

Flight crew feels abnormal medium to high frequency vibration.

After any component changes in tail rotor/90° gearbox assembly.

After 300 hour and annual inspections.

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Balancing the Tail Rotor - Installation of the RADS

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Accelerometer Bracket (Stiff)	SA# 29329700	1	
Basic RADS AT Kit	SA# 29333300/29333306	1	
212/412 Optical Tail Rotor Bracket	SA# 29338500A/29338501	1	Basic RADS Operating System, 3.10AO or greater 412_41 Version 5.22 or greater for use on helicopters with 204-031-927-105 pylon mounts 412_50 Version 5.24 or greater for use on helicopters with 204-031-927-107 pylon mounts

Procedure

1. Install RADS to balance tail rotor as follows:

1.1. Locate DAU (29328200 or 29481400) in cabin aft of crew compartment in a convenient area.

1.2. Remove nut from top stud (12:00 o'clock position) on 90° gearbox. Using accelerometer bracket (stiff) (SA# 29329700) (or equivalent), install one 991D accelerometer on stud with connector up and secure with nut. Connect accelerometer end of 50 foot long accelerometer cable (29105600) to vertical accelerometer. Route to DAU down left side of fin and tailboom. Secure so it cannot foul tail rotor or controls. Connect to ACC #4 (Figure 1).

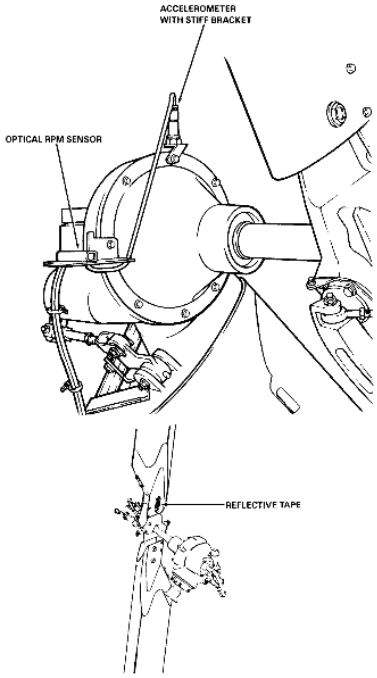
1.3. Attach optical RPM sensor (29314700) to 212/412 optical tail rotor bracket (SA# 29338500A/29338501) so wire is on flange edge of bracket. Secure sensor to bracket with #440 self-locking nuts. Remove nut from aft stud on 90° gearbox and install sensor/bracket. Secure with gearbox nut so it is level to horizon and pointed directly aft (Figure 1). Route optical sensor cable to DAU, down left side of fin and tailboom. Secure cable so it cannot foul tail rotor or controls. Connect cable to TACHO #2 on DAU.

1.4. Check the blades to determine which one does not have large balance washers on gearbox side of blade bolts. This will be target blade. Install a full width piece of reflective tape (10605000) between two blade bolts on gearbox side of blade (Figure 1).

1.5. Connect 28 Vdc power cable (29104700) to 28 Vdc port on copilot side of pedestal (Figure 2). Connect other end of cable to 28 Vdc connector on DAU.

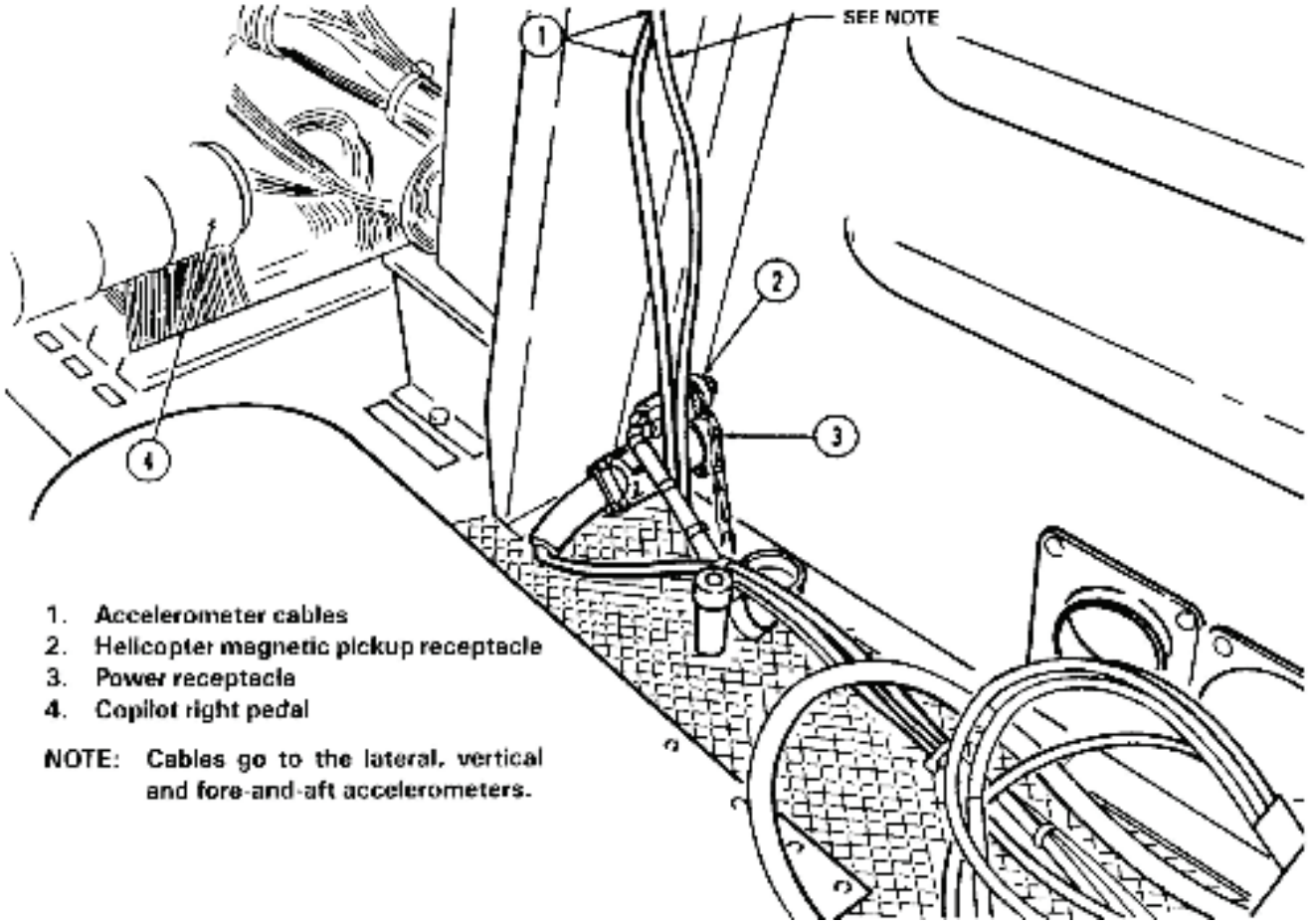
1.6. Connect CADU to DAU using communications cable (29325601).

1.7. When correctly installed, RADS should be configured as shown in Figure 3.



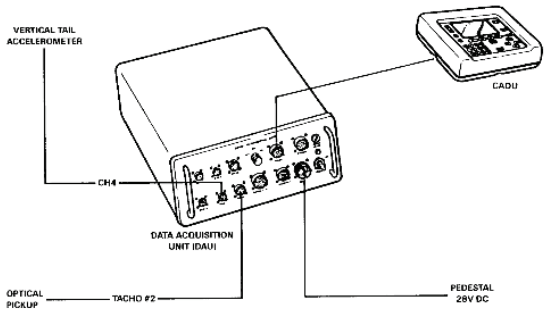
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Figure 1. Equipment Installation, Tail Rotor Balancing (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00004-A-001-01

Figure 2. Magnetic Port/Azimuth (Sheet 1 of 1)



ICN-412-A-181200-B-97499-00002-A-001-01

Figure 3. RADS Configuration, Tail Rotor Balancing (Sheet 1 of 1)

Balancing the Tail Rotor - Tests to Balance the Tail Rotor

Procedure

1. Turn on CADU. From the main screen select the proper 412 program, select or define the tail number of the helicopter and select TAIL as FLIGHT PLAN. This configures RADS to balance tail rotor. An example of a properly setup CADU is shown in Figure 1.

2. Press F1 to activate MEASUREMENT mode.

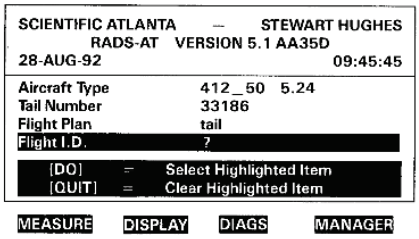
3. Operate helicopter with either or both power sections operating (operate as near 100% RPM as possible if operating both power sections or 99% with single power section operating).

4. Once RPM and temperatures have stabilized, press DO to obtain balance data. In the event any error occurs, repeat the acquisition of data up to four times before shutting down the helicopter and troubleshooting the system as described in 412-A-18-12-00-00A-422A-A / 00168.

5. Once the data has been obtained, shut down the helicopter.

6. With the data taken, select DIAGNOSTICS to define what adjustments are required to balance the tail rotor. Record the adjustments to be made and implement as described in 412-A-18-12-00-03A-271A-A / 00166.

7. Repeat the tests until the tail rotor has been balanced. To troubleshoot balancing problems, refer to 412-A-18-12-00-00A-422A-A / 00168.



ICN-412-A-181200-B-97499-00003-A-001-01

Figure 1. CADU Screen for Tail Rotor Balance (Sheet 1 of 1)

Balancing the Tail Rotor - Making Tail Rotor Balance Adjustments

Procedure

1. RADS specifies balance adjustments required to balance the tail rotor as adjustments made only to target the blade. There are three dynamic adjustments available: the chordwise balance bracket, inboard blade bolt, and outboard blade bolt. The default in the RADS software is to use only chordwise balance bracket and inboard blade bolt. Should the operator need to use the outboard bolt for balance, then from within DIAGNOSTICS, select EDIT ADJUSTABLES from the DIAGNOSTICS menu and use F1 to turn off the inboard blade bolt (by changing Y to N) and then use F1 to turn on the outboard blade bolt on the target blade only (N to Y). Pressing DO will automatically recalculate corrections using the outboard blade bolt.

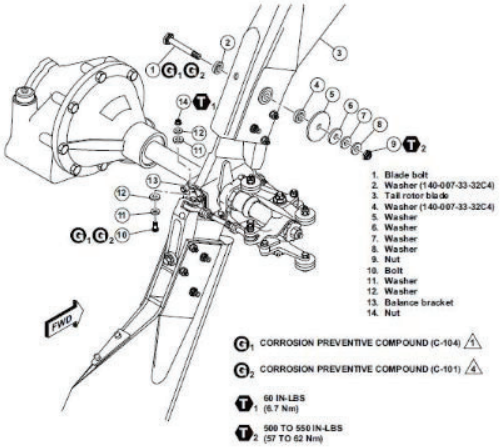
Once the adjustments to the target blade have been defined, determine what moves can be made. If RADS specifies to add weight to a location where weight is on the opposite blade, the proper course is to remove the required amount from the opposite location. If RADS specifies to remove weight from a location that does not have weight, add weight to the opposite side. Refer to Figure 1 or Figure 2 for information on making balance adjustments. Use Figure 1 for tail rotor hub and balde assemblies with tail rotor blades 212-010-750 installed. Use Figure 2 for tail rotor hub and blade assemblies with tail rotor baldes 412-016-100 installed.

1.1. The target blade is the one with the reflector.

1.2. The target chordwise balance bracket is next to the target pitch link.

1.3. It is acceptable to add chordwise weight on both the blank and target balance brackets to make moves smaller than can be made on one side only due to the weight of the bolt.

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1. Blade bolt
2. Washer (140-007-33-32C4)
3. Tail rotor blade
4. Washer (140-007-33-32C4)
5. Washer
6. Washer
7. Washer
8. Washer
9. Nut
10. Bolt
11. Washer
12. Washer
13. Balance bracket
14. Nut

G_1 CORROSION PREVENTIVE COMPOUND (C-104) △1

G_2 CORROSION PREVENTIVE COMPOUND (C-101) △4

T_1 60 IN-LBS (6.7 Nm)

T_2 500 TO 550 IN-LBS (57 TO 62 Nm)

NOTES

△1 Apply a coating of corrosion preventive compound (C-104) to the shanks of the bolts (1 and 10) prior to installation. Do not apply corrosion preventive compound to bolt threads.

△2 To accomplish spanwise dynamic balance, use blade bolts (1) as required to accommodate balance washers (5, 6, 7 or 8). Bolts (1) may be installed with head, inboard or outboard, but all four blade bolts must be installed the same. When change of balance washers and/or bolts is required, make sure that one washer (2) and one washer (4) are installed next to the blade as illustrated. Then install the heaviest balance washer next to the washer (4). Install the required combination of washers (5, 6, 7, or 8) on blade (3) or on opposite blade as required to obtain balance. After installation, make sure proper thread engagement of two complete threads is shown on bolt.

BLADE BOLT SPANWISE BALANCE HARDWARE △2

ITEM NUMBER	NOMENCLATURE	PART NUMBER	WEIGHT (GRAMS)
1	BLADE BOLT	NAS1306-34	87.5
1	BLADE BOLT	20-957-6-34	84.8
1	BLADE BOLT	NAS1306-36	99.7
1	BLADE BOLT	20-957-6-36	89.0
5	WASHER	AN970-6	37.7
6	WASHER	212-010-744001	11.3
7	WASHER	AN960-616	3.2
7	WASHER	NAS1149F0663P	5.1
8	WASHER	AN960-616L	1.6
8	WASHER	NAS1149F0632P	2.4
9	NUT	NAS1022A6	8.9
9	NUT	MS21245-L6	9.9

△3 To accomplish chordwise dynamic balance on balance brackets (13), use bolts (10) as required to accommodate the required combination of balance washers (11 and 12). Use a maximum of two washers (12) and a maximum of ten washers (11) on one bracket. Install washers (11 and 12) symmetrically. Install the required hardware on balance bracket (13) or opposite bracket (not illustrated), as required to obtain balance. It is also acceptable to install balance hardware on both balance brackets to make moves smaller than can be made on one bracket only due to the weight of the bolt.

CHORDWISE BALANCE BRACKET HARDWARE △3

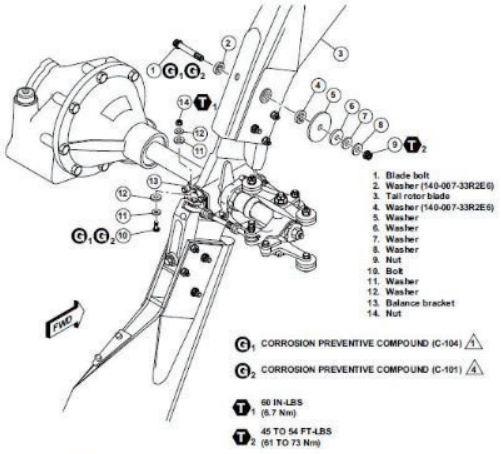
ITEM NUMBER	NOMENCLATURE	PART NUMBER	WEIGHT (GRAMS)
10	BOLT	NAS1304-1	5.4
10	BOLT	NAS6604-1	5.2
10	BOLT	NAS1304-2	5.8
10	BOLT	NAS6604-2	5.6
10	BOLT	NAS1304-3	6.2
10	BOLT	NAS6604-3	5.9
10	BOLT	NAS1304-4	6.5
10	BOLT	NAS6604-4	6.3
10	BOLT	NAS1304-5	7.3
10	BOLT	NAS6604-5	7.1
10	BOLT	NAS1304-6	8.0
10	BOLT	NAS6604-6	7.9
10	BOLT	NAS1304-10	8.7
10	BOLT	NAS6604-10	8.9
10	BOLT	NAS1304-12	9.4
10	BOLT	NAS6604-12	9.4
10	BOLT	NAS1304-14	10.2
10	BOLT	NAS6604-14	10.2
11	WASHER	AN960-416	1.2
11	WASHER	NAS1149F0463P	1.2
12	WASHER	AN970-4	7.5
14	NUT	MS21042L4	1.6

△4 Apply a coating of corrosion preventive compound (C-101) to all bolt heads, washers, nuts and exposed threads after installation.

ICN-412-A-181200-B-97499-00004-A-001-01

Figure 1. Tail Rotor Dynamic Balance Weight Adjustment With Tail Rotor Blades 212-010-750 Installed (Sheet 1 of 1)

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1. Blade bolt
2. Washer (140-007-33R2E6)
3. Tail rotor blade
4. Washer (140-007-33R2E6)
5. Washer
6. Washer
7. Washer
8. Washer
9. Nut
10. Bolt
11. Washer
12. Washer
13. Balance bracket
14. Nut

G_1 CORROSION PREVENTIVE COMPOUND (C-104) 1

G_2 CORROSION PREVENTIVE COMPOUND (C-101) 4

T_1 60 IN-LBS (6.7 Nm)

T_2 45 TO 54 FT-LBS (61 TO 73 Nm)

NOTES

1 Apply a coating of corrosion preventive compound (C-104) to the shanks of the bolts (1 and 10) prior to installation. Do not apply corrosion preventive compound to bolt threads.

2 To accomplish spanwise dynamic balance, use blade bolts (1) as required to accommodate balance washers (5, 6, 7 or 8). Bolts (1) may be installed with head, inboard or outboard, but all four blade bolts must be installed the same. When change of balance washers and/or bolts is required, make sure that one washer (2) and one washer (4) are installed next to the blade as illustrated. Then install the heaviest balance washer next to the washer (4). Install the required combination of washers (5, 6, 7, or 8) on blade (3) or on opposite blade as required to obtain balance. After installation, make sure proper thread engagement of two complete threads is shown on bolt.

BLADE BOLT SPANWISE BALANCE HARDWARE 2			
ITEM NUMBER	NOMENCLATURE	PART NUMBER	WEIGHT (GRAMS)
1	BLADE BOLT	20-065-08036C	96.7
1	BLADE BOLT	20-065-08038C	99.8
5	WASHER	AN970-8	37.7
6	WASHER	212-010-764-001	11.3
7	WASHER	NAS1149F0863P	5.1
8	WASHER	NAS1149F0832P	2.4
9	NUT	MS21245-L8	9.9

3 To accomplish chordwise dynamic balance on balance brackets (13), use bolts (10) as required to accommodate the required combination of balance washers (11 and 12). Use a maximum of two washers (12) and a maximum of ten washers (11) on one bracket. Install washers (11 and 12) symmetrically. Install the required hardware on balance bracket (13) or opposite bracket (not illustrated), as required to obtain balance. It is also acceptable to install balance hardware on both balance brackets to make moves smaller than can be made on one bracket only due to the weight of the bolt.

CHORDWISE BALANCE BRACKET HARDWARE 3			
ITEM NUMBER	NOMENCLATURE	PART NUMBER	WEIGHT (GRAMS)
10	BOLT	NAS6904-1	5.2
10	BOLT	NAS6904-2	5.6
10	BOLT	NAS6904-3	5.9
10	BOLT	NAS6904-4	6.3
10	BOLT	NAS6904-6	7.1
10	BOLT	NAS6904-8	7.9
10	BOLT	NAS6904-10	8.9
10	BOLT	NAS6904-12	9.4
10	BOLT	NAS6904-14	10.2
11	WASHER	NAS1149F0463P	1.2
12	WASHER	AN970-4	7.5
14	NUT	MS21042L4	1.6

4 Apply a coating of corrosion preventive compound (C-101) to all bolt heads, washers, nuts and exposed threads after installation.

ICN-412-A-181200-B-97499-00005-A-001-01

Figure 2. Tail Rotor Dynamic Balance Weight Adjustment With Tail Rotor Blades 412-016-100 Installed (Sheet 1 of 1)

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Balancing the Tail Rotor - Tracking Tail Rotor

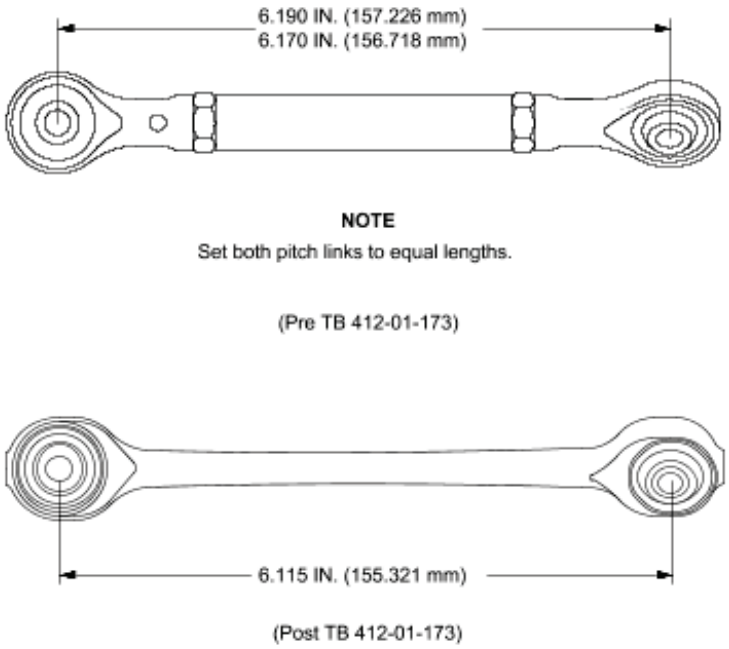
Procedure

NOTE

SN: 33214-Subsequent or 34037-Subsequent or 36225-Subsequent:

Helicopters that have TB 412-01-173 incorporated do not require tracking of the tail rotor.

1. The optimum tail rotor vibration in flight (within the current 1/2 turn pitch link adjustment capability) is achieved with the pitch links set to as nearly equal lengths as possible. Therefore, tracking tail rotors is no longer required if the operator sets both pitch links as shown in Figure 1.



ICN-412-A-181200-B-97499-00006-A-001-01

Figure 1. Tail Rotor Pitch Link Rigging (Sheet 1 of 1)

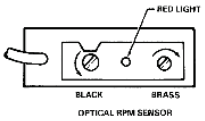
Balancing the Tail Rotor - Troubleshooting Tail Rotor Balance

Procedure

1. Table 1 lists probable causes of helicopter and RADS problems encountered during balancing of the tail rotor.

Table 1. Troubleshooting Tail Rotor Balance

PROBLEM	SOURCE	CORRECTIVE ACTION
No TACHO or TACHO out-of-bounds warnings	No/dirty reflective tape	Install or replace reflective tape.
	Optical sensor fault	Refer to Figure 1 to fine tune sensor.
Acceleration saturation warning	Accelerometer saturation	Verify accelerometer bracket is stiff. Replace accelerometer.



PROBLEM: Repeated tacho fail/high tacho/tacho out of bounds warnings.

1. Verify that reflector is in proper position. With DAU power on red light will appear when reflector is seen. Flap tail rotor through full range to ensure light stays on throughout range. Adjust reflective tape location and size as required to correct.

2. If tape fails to fix problem, remove clear cover from back of sensor. Turn black screw 1/3 turn in direction shown and brass screw in direction shown until slight resistance is felt. Repeat measurement.

ICN-412-A-181200-B-97499-00007-A-001-01

Figure 1. Troubleshooting Optical Pickup (Sheet 1 of 1)

Balancing Main Driveshaft - Control of Main Driveshaft Vibration

Most Model 412 helicopters will make a pulsating low frequency noise at flat pitch due to normal amount of freeplay in shaft coupling and gearbox bearings. If this pulsating noise is present in hover, shaft out-of-balance is indicated. High levels of shaft out-of-balance will be perceived as a buzz in the tail rotor pedals in addition to a pulsating noise.

Applicable: SN: 33001-33213 or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

Balancing the main driveshaft can minimize an out-of-balance condition. During balancing procedure, 0.2 IPS or less should be the vibration level attained to maintain the vibration levels within acceptable limits during the time in-service of the shaft. This procedure is optional and not required for any component life or serviceability requirement.

For helicopters S/N 36020 and Subsequent and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade), refer to Table 1 *for balance criteria.*

Table 1. ***Helicopters S/N 36020 and Subsequent and Helicopters Modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade) Main Drive Balance Criteria***

VIBRATION FREQUENCY	*OPERATION CONDITION*	*VIBRATION LEVEL*	*RECOMMENDED ACTION*
Main driveshaft 1/rev	*100% N_R on ground*	*0.2 IPS or less*	*Vibe. Equipment target value.*
		0.5 IPS or less	*No immediate action required.*
		Greater than 0.5 IPS	*Balance driveshaft.*

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Balancing Main Driveshaft - Installation of RADS

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Accelerometer Brackets (Stiff)	SA# 29329700	2	
Basic RADS AT Kit	SA# 29333300/29333306	1	
212/412 Optical Sensor Tail Rotor Adapter	SA# 29338500A/29338501	1	Basic RADS Operating System, 3.10AO or greater 412-41 Version 5.22 or greater for use on helicopters with 204-031-927-105 pylon mounts. 412-50 Version 5.24 or greater for use on helicopters with 204-031-927-107 pylon mounts

Procedure

1. Locate DAU (29328200 or 29481400) in cabin aft of crew compartment in a convenient area.

2. Install one 991D accelerometer using accelerometer brackets (stiff) (SA# 29329700) (or equivalent) on right side of combining gearbox (Figure 1). Orient accelerometer and bracket in a horizontal plane with connector pointing outboard. Connect accelerometer end of an accelerometer cable (291056000 and 29105605) to combining gearbox accelerometer. Route to DAU so it cannot foul any rotating components, and connect to ACC #3.

3. Using a accelerometer brackets (stiff) (SA# 29329700) (or equivalent), install one 991D accelerometer on stud of main driveshaft input quill (refer to Figure 1) by removing Allen head screw and using a bolt to secure accelerometer. Orient accelerometer and bracket in a horizontal plane with connector pointing outboard. Connect accelerometer end of an accelerometer cable (291056000 and 29105605) to transmission input quill accelerometer, route to DAU so it cannot foul any rotating components and connect to ACC #4.

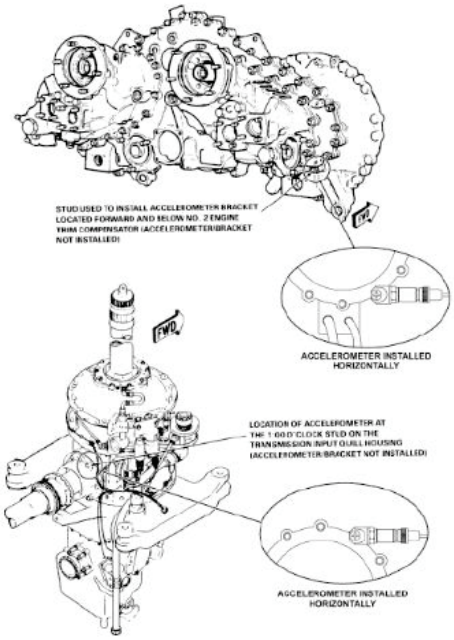
4. Attach optical RPM sensor (29314700) to 212/412 optical sensor tail rotor adapter (SA# 29338500A/29338501) so wire is opposite flange edge of bracket. Secure sensor to bracket with self-locking nuts. Remove right hand inlet cowl and install optical pickup to cowling alignment bracket using a bolt (5, Figure 2), washers (1 and 6), and nut (2). If properly installed, optics will be looking at shaft and wire will be pointed up. Route optical sensor cable up and to right and secure to a convenient location with a tie-wrap. Route cable into cabin ensuring it cannot foul any rotating components. Connect cable to TACHO #2 on DAU.

5. With optical sensor installed, turn shaft so one shaft-to-adapter bolt is directly on top. With shaft in this position, center a 2.0 inch by 1.0 inch piece of reflective tape on shaft with long axis of tape aligned with long axis of shaft and centered relative to shaft-adapter bolt. The bolt aligned with reflector is #1 bolt (Figure 3).

6. Connect 28 VDC power cable (29104700) to 28 VDC port on copilot side of pedestal (Figure 4). Connect other end of cable to 28 VDC connector on DAU.

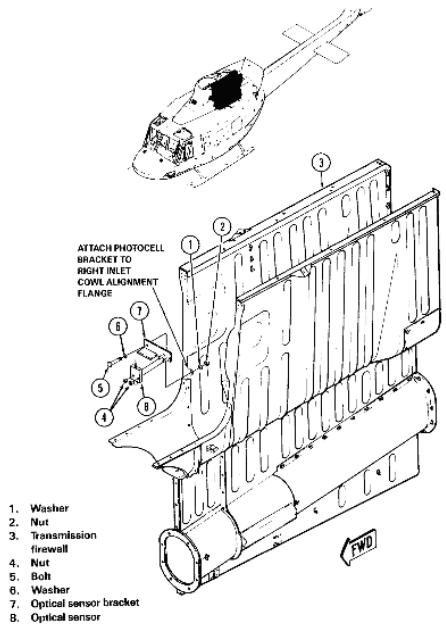
7. Connect CADU (29314101 or 29481301) to DAU using communications cable (29325601).

8. The RADS is now installed and ready to balance driveshaft. When correctly installed, RADS should be configured as shown in Figure 5.



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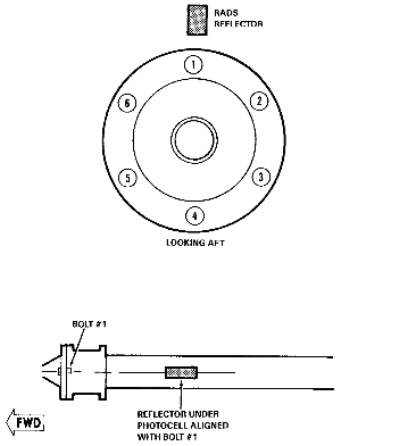
Figure 1. Equipment Installation, Main Driveshaft Balancing (Sheet 1 of 1)



1. Washer
2. Nut
3. Transmission firewall
4. Nut
5. Bolt
6. Washer
7. Optical sensor bracket
8. Optical sensor

ICN-412-A-181100-B-97499-00019-A-001-01

Figure 2. Optical Sensor and Bracket (Sheet 1 of 1)

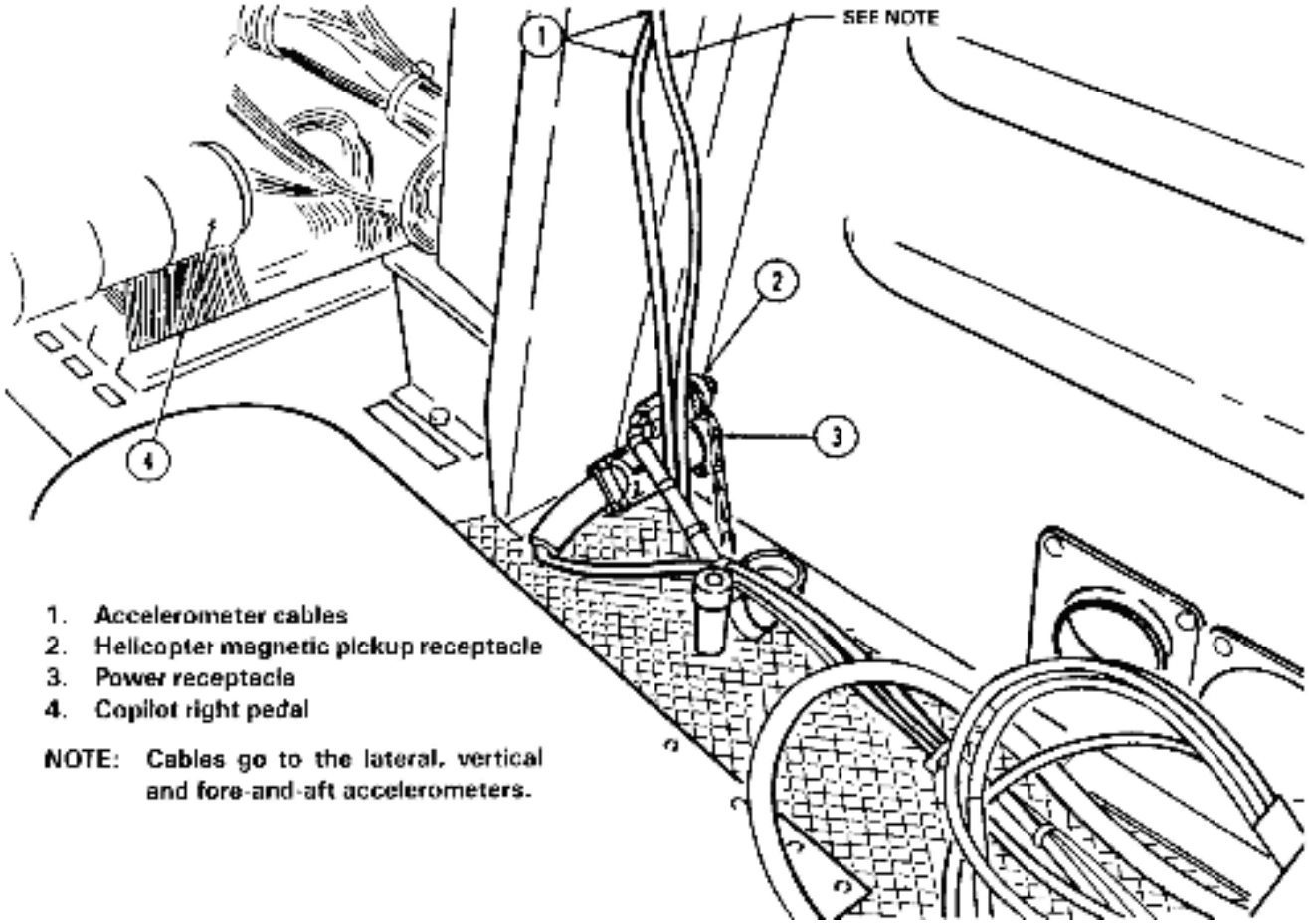


NOTES

1. Bolt orientation shown is used to determine placement of balance weights from chart information.
2. Bolt #1 is bolt aligned with the reflector on the tube.
3. Bolt 1 and 3 are active in rads diagnostics.
4. Adding weight to bolt #1 is same as removing weight from bolt #4.
5. Adding weight to bolt #3 is same as removing weight from bolt #6.

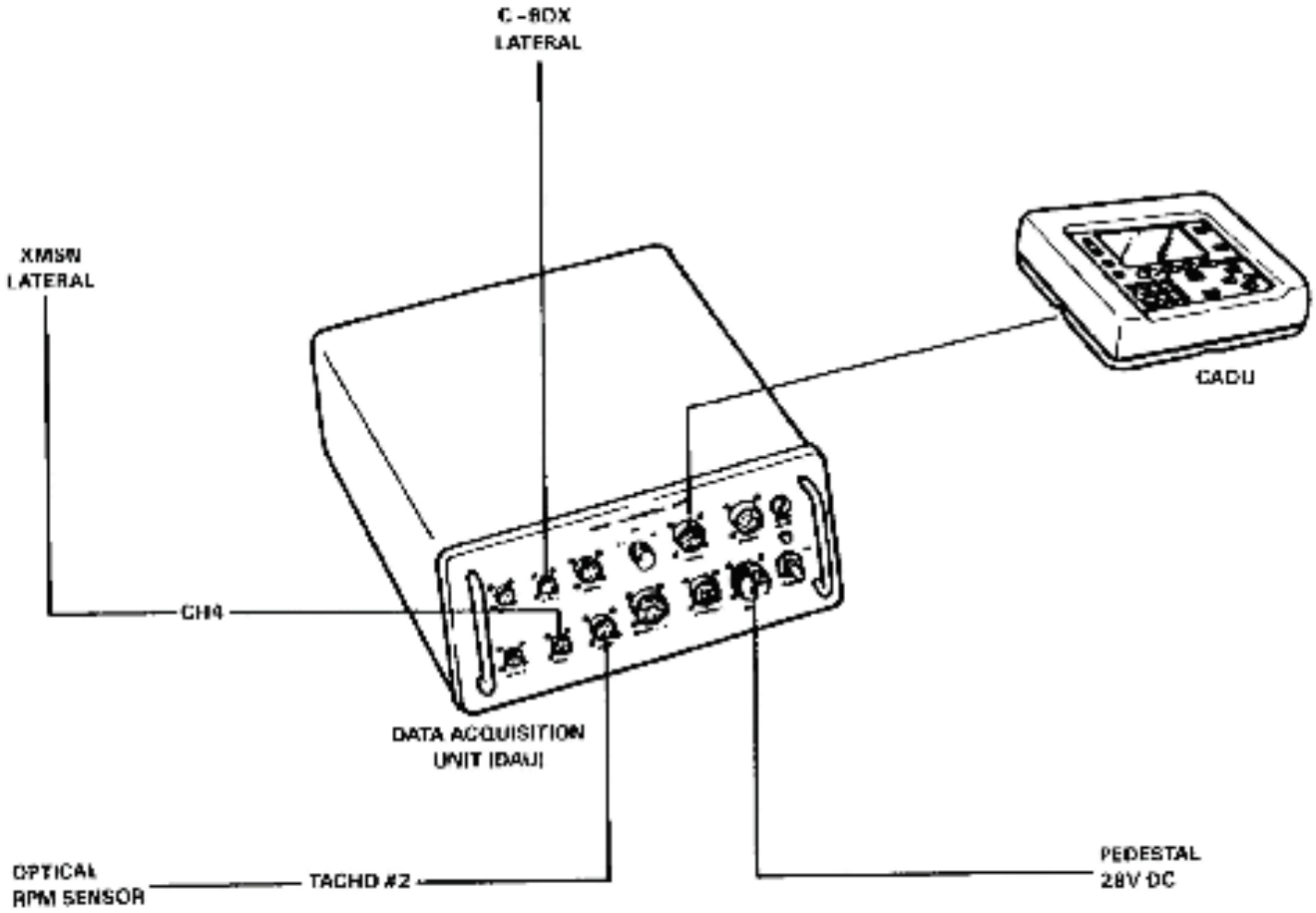
ICN-412-A-181100-B-97499-00020-A-001-01

Figure 3. Reflector Tape Installation, Main Driveshaft (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00004-A-001-01

Figure 4. Magnetic Port/Azimuth (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00021-A-001-01

Figure 5. RADS Configuration for Balancing Main Driveshaft (Sheet 1 of 1)

Balancing Main Driveshaft - Tests to Balance Driveshaft

Procedure

1. Install RADS (412-A-18-11-00-06A-720A-A / 00170).

2. Turn on CADU and, from the main screen, select proper 412 program. Select or define TAIL number of helicopter and select D/S (for standard 412 and 412SP) or D/S-HP (412HP) as FLIGHT PLAN (Figure 1). This will configure the RADS to balance main driveshaft.

3. Press F1 to activate measurement mode.

4. Operate helicopter (BHT-412-FM) with both power sections at 100% main rotor RPM.

5. Once RPM and temperatures have stabilized, press DO to obtain balance data (on initial run wait 5 minutes to warm up grease).

NOTE

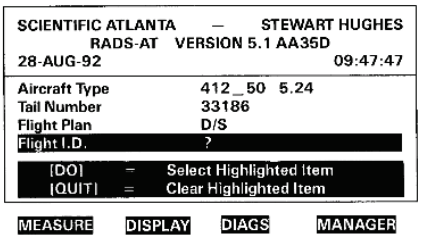
In case of any error, repeat acquisition of data up to four times before shutting down helicopter and troubleshooting system (412-A-18-11-00-01A-422A-A / 00174).

6. Once data has been obtained, shut down helicopter.

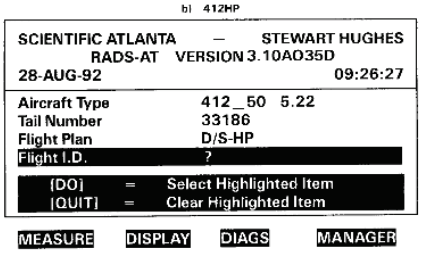
7. With data taken, select DIAGNOSTICS to define what, if any, adjustments are required to balance driveshaft (412-A-18-11-00-09A-271A-A / 00173).

8. Repeat tests until driveshaft has been balanced. To troubleshoot balancing problems, refer to 412-A-18-11-00-01A-422A-A / 00174. Upon completion, remove RADS equipment from helicopter.

a) 412 and 412SP



b) 412HP



ICN-412-A-181100-B-97499-00022-A-001-01

Figure 1. CADU Screen, Main Driveshaft Balancing (Sheet 1 of 1)

Balancing Main Driveshaft - Using Diagnostics

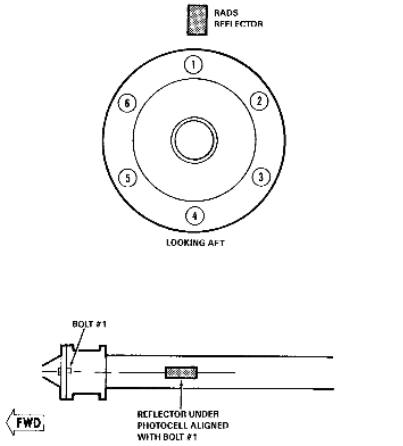
Procedure

1. Once all data has been obtained and diagnostics have been entered:

1.1. The LIMITS screen will appear and display any vibration measurement above the 0.2 limit. As previously stated, this limit is used only as a target level for RADS diagnostics and is in no way an indicator of a level of vibration that must be met to be serviceable. If operator chooses to improve the balance of shaft, press DO to calculate recommended.

1.2. When calculated, RADS will display adjustments made to aft and forward adapters of shaft. As programmed, RADS specifies adjustments only to #1, #2, and #3 bolts. This allows adjustments also to be made to #4, #5, and #6 bolts without editing adjustments. The definition of blade bolt locations and adjustment options is shown in Figure 1.

1.3. With adjustments defined, record adjustments for future reference and refer to 412-A-18-11-00-09A-271A-A / 00173 for proper way to make adjustments.



NOTES

1. Bolt orientation shown is used to determine placement of balance weights from chart information.
2. Bolt #1 is bolt aligned with the reflector on the tube.
3. Bolt 1 and 3 are active in rads diagnostics.
4. Adding weight to bolt #1 is same as removing weight from bolt #4.
5. Adding weight to bolt #3 is same as removing weight from bolt #6.

ICN-412-A-181100-B-97499-00020-A-001-01

Figure 1. Reflector Tape Installation, Main Driveshaft (Sheet 1 of 1)

Balancing Main Driveshaft - Making Adjustments

Procedure

1. Once adjustments have been defined by diagnostics, proceed as follows:

1.1. The shaft is balanced by installing or removing balance weights from shaft-to-adapter bolts (#2 and #5, Figure 1).

1.2. Once bolt to be adjusted has been located, remove nut and bolt and using bolt/weight combination shown in Table 1, configure a weight package closest to weight specified by RADS.

NOTE

Bolt weights are referenced to minimum bolt size used. Bolt AN513-12A (or NAS6604-6) shall be used when one weight is installed and bolt AN513-13A (or NAS6604-8) shall be used when two weights are stacked at one location. The maximum weight allowed at one bolt location is 1.0 ounce (28.35 g). Maximum stackup of weights is two per bolt.

Applicable: SN: 33001-33213 or (SN: 36001-36019 412-570-001-103: Not Installed BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

1.3. *Install bolt/weight package with weights under the head of the bolt. For helicopters (412Classic/SP to 412HP Upgrade) (5/16 inch bolts) torque nuts 100 to 140 inch-pounds (11.30 to 15.82 Nm).*

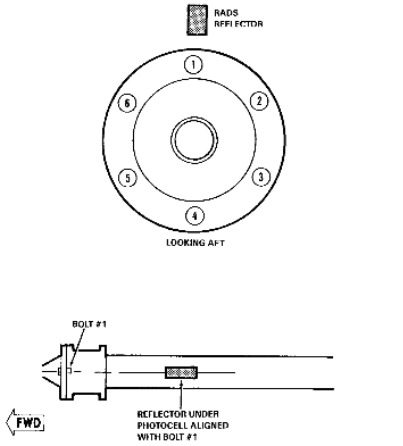
1.4. *For helicopters S/N 36020 and Subsequent and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade) (1/4 inch bolts) torque nuts 70 to 90 inch-pounds (7.91 to 10.27 Nm).*

Table 1. Hardware and Weight

PART NUMBER	NOMENCLATURE	WEIGHT	
		GRAMS	OUNCES
AN5-12A	Bolt	2.0	0.07
AN5-13A	Bolt	4.00	0.14
100-106-1	Balance Weight Plate	14.18	0.50
100-106-2	Balance Weight Plate	7.09	0.25
100-106-3	Balance Weight Plate	2.84	0.10
100-106-4 (1)	Balance Weight Plate	14.18	0.50
100-106-5 (1)	Balance Weight Plate	7.09	0.25
100-106-6 (1)	Balance Weight Plate	2.84	0.10
NAS6604-6 (1)	Bolt	2.0	0.07
NAS6604-8 (1)	Bolt	4.00	0.14

NOTE:

1 *Used on helicopters (412Classic/SP to 412HP Upgrade).*



NOTES

1. Bolt orientation shown is used to determine placement of balance weights from chart information.
2. Bolt #1 is bolt aligned with the reflector on the tube.
3. Bolt 1 and 3 are active in rads diagnostics.
4. Adding weight to bolt #1 is same as removing weight from bolt #4.
5. Adding weight to bolt #3 is same as removing weight from bolt #6.

ICN-412-A-181100-B-97499-00020-A-001-01

Figure 1. Reflector Tape Installation, Main Driveshaft (Sheet 1 of 1)

Balancing Main Driveshaft - Troubleshooting

Procedure

1. Table 1 lists probable causes of helicopter and RADS problems encountered during balancing of driveshaft.

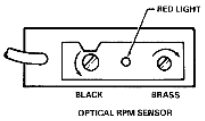
2.

NOTE

If adjustments are edited, do not turn on more than two locations at either end.

Table 1. Troubleshooting Driveshaft Balance

PROBLEM	SOURCE	CORRECTIVE ACTION
No TACHO or TACHO out-of-bounds warnings	No/dirty reflective tape	Install or replace reflective tape.
	Optical Sensor fault	Refer to Figure 1 to fine tune sensor.
Shaft balance cannot be corrected within weight limits.	Excessively out of balance	Index shaft two bolt holes and repeat.
		Regrease.
		Check adapter runout and torque.
		Check shaft installation.
		Remove shaft, inspect for correct assembly.
Erratic balance results		Make single adjustment. Refer to balance charts Figure 2 through Figure 5.



PROBLEM: Repeated tacho fail/high tacho/tacho out of bounds warnings.

1. Verify that reflector is in proper position. With DAU power on red light will appear when reflector is seen. Flap tail rotor through full range to ensure light stays on throughout range. Adjust reflective tape location and size as required to correct.

2. If tape fails to fix problem, remove clear cover from back of sensor. Turn black screw 1/3 turn in direction shown and brass screw in direction shown until slight resistance is felt. Repeat measurement.

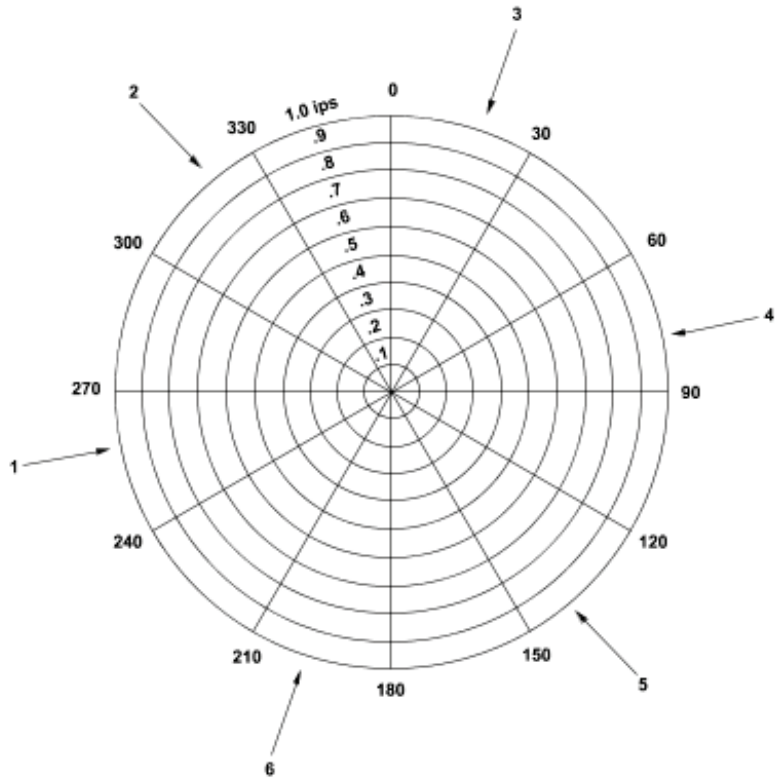
ICN-412-A-181200-B-97499-00007-A-001-01

Figure 1. Troubleshooting Optical Pickup (Sheet 1 of 1)

HELICOPTER S/N 33001 THROUGH 33213 AND 36001 THROUGH 36019
USING RADS-AT SCRIPT FILE 412-41

DATE:	FLIGHT I.D.:	SHIP S/N:

ENGINE COMBINING GEARBOX	SENSITIVITY: 0.05 ips/G	AFT - CHANNEL #3



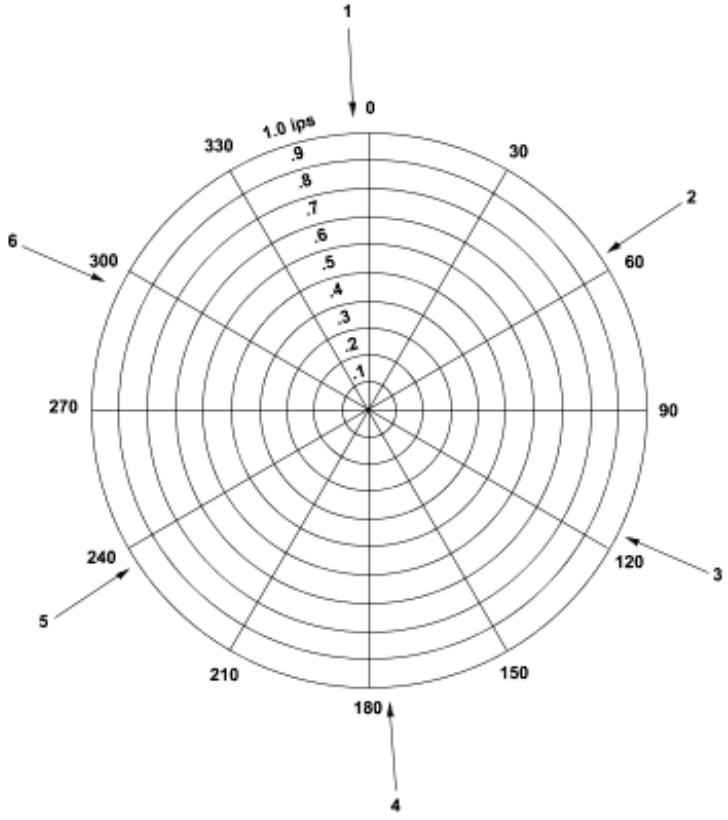
ICN-412-A-181100-B-97499-00023-A-001-01

Figure 2. 1/Rev Driveshaft Balance Chart with Script File 412-41 - Engine Combining Gearbox (Sheet 1 of 1)

HELICOPTER S/N 36001 THROUGH 36019
USING RADS-AT SCRIPT FILE 412-41

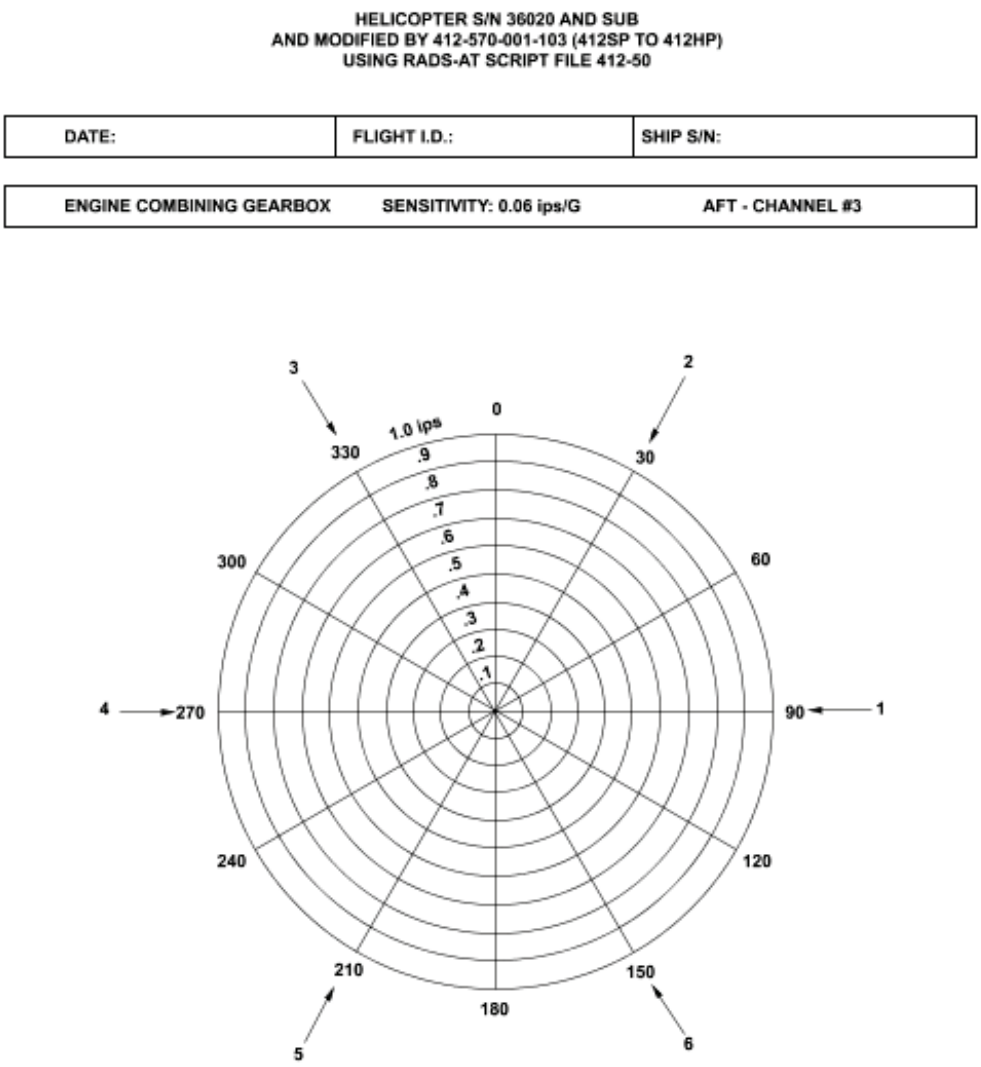
DATE:	FLIGHT I.D.:	SHIP S/N:

MAIN TRANSMISSION	SENSITIVITY: 0.07 ips/G	FWD - CHANNEL #4



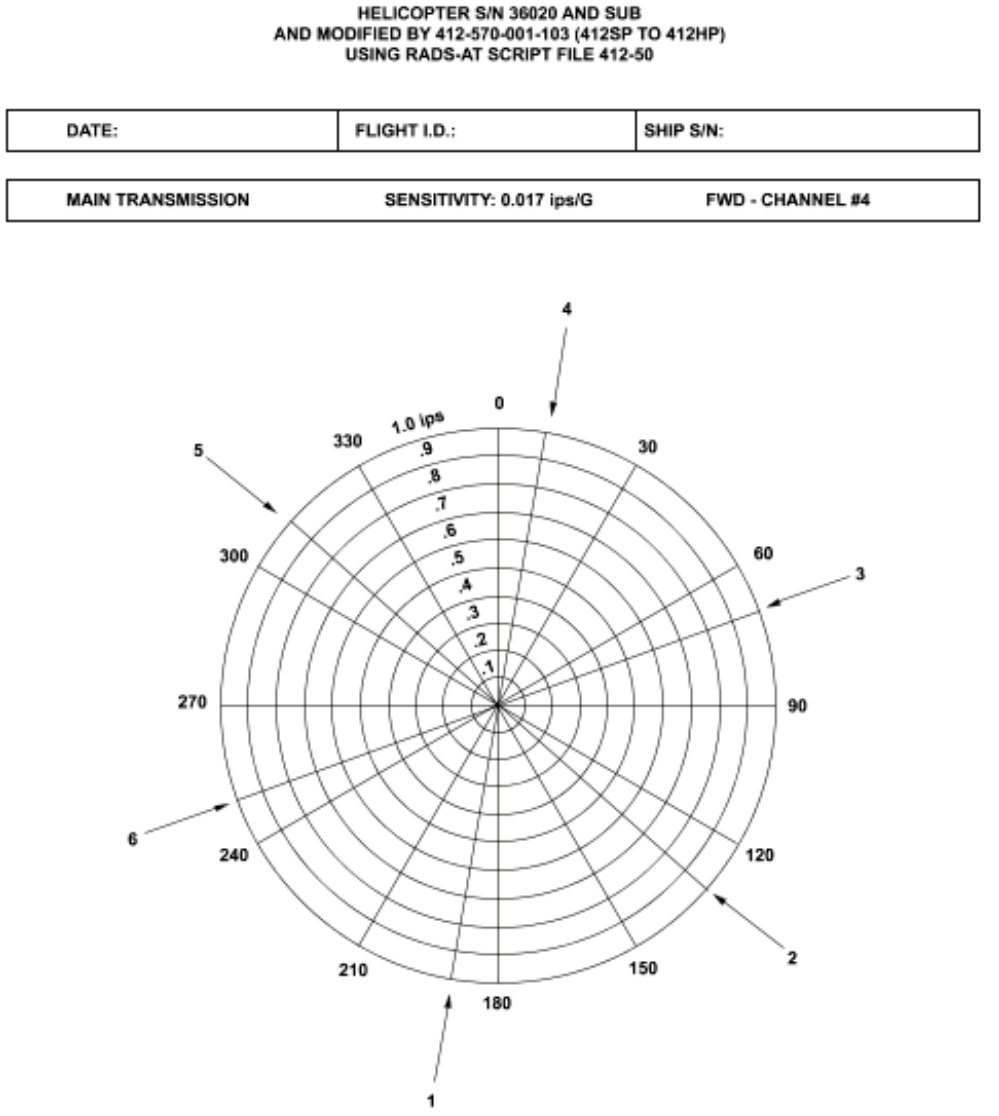
ICN-412-A-181100-B-97499-00024-A-001-01

Figure 3. 1/Rev Driveshaft Balance Chart with Script File 412-41 - Main Transmission (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00025-A-001-01

Figure 4. 1/Rev Driveshaft Balance Chart with Script File 412-50 - Engine Combining Gearbox (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00026-A-001-01

Figure 5. 1/Rev Driveshaft Balance Chart with Script File 412-50 - Main Transmission (Sheet 1 of 1)

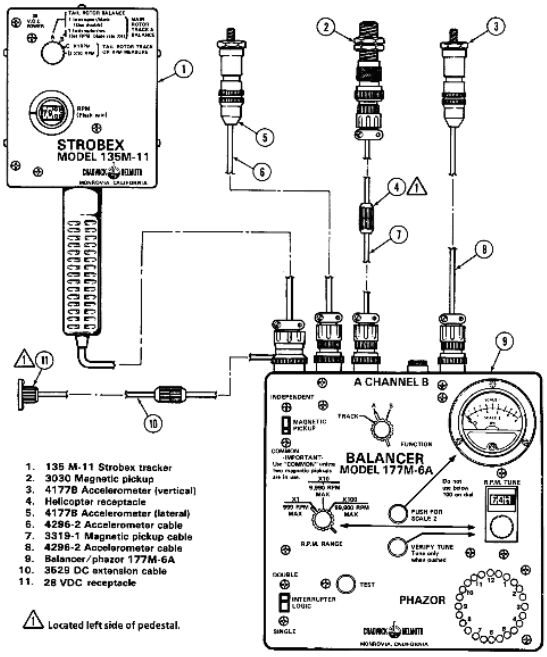
Main Rotor Balancing (Chadwick-Helmuth Vibrex Equipment, Model 177M or Equivalent) - Description

1. Main Rotor Balancing (Chadwick-Helmuth Vibrex Equipment, Model 177m Or Equivalent)

Rotor balancing using the RADS-AT equipment is the recommended method.

2. Rotor Vibration Measurement Instrumentation

The Chadwick-Helmuth Vibrex equipment is an analog system that requires all vibration and tracking data to be obtained manually. The equipment kit includes a Strobex tracker, a balancer/phazor, a magnetic pickup, two accelerometers, and various cables (Figure 1). Model 177M Vibrex equipment is recommended, but equivalent equipment is acceptable.



ICN-412-A-181101-B-97499-00001-A-001-01

Figure 1. Chadwick-Helmuth Equipment Hook-Up (Sheet 1 of 1)

Rotor Vibration Measurement Instrumentation - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Chadwick-Helmuth Vibrex Kit	Model 177M	1	or Equivalent

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Tape	C-482	AR	

Procedure

NOTE

Model 412 helicopters are delivered with the following production installed balance equipment: A magnetic pickup bracket, magnetic pickup cable and receptacle and a 28 VDC power receptacle. Remainder of equipment is contained in chadwick-helmuth vibrex kit (Model 177M).

1. Install magnetic pickup (412-A-18-11-00-00A-720A-A / 00151).

2. Connect magnetic pickup cable (7, Figure 1) to helicopter magnetic pickup receptacle and to balancer (9).

3. Connect balancer (9) to power receptacle (11) with cable (10).

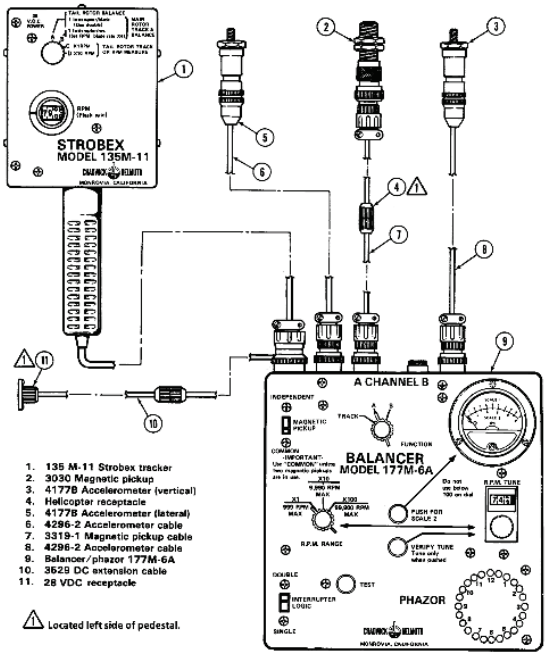
4. Install vertical accelerometer (412-A-18-11-00-00A-720A-A / 00151). Connect accelerometer cable (8, Figure 1) to channel “B” of balancer.

5. Install lateral accelerometer (412-A-18-11-00-00A-720A-A / 00151). Connect accelerometer cable (6, Figure 1) to channel “A” of balancer.

NOTE

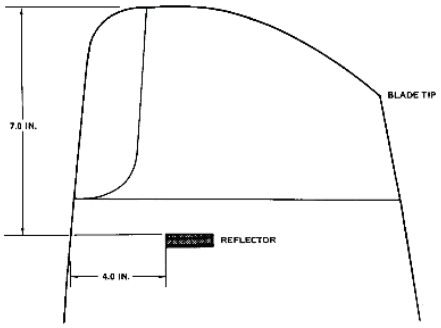
If it is necessary to measure fore-and-aft 1/rev vibration in flight, the lateral accelerometer can be rotated so it is oriented fore-and-aft (cable away from pilot).

6. Install reflective tape (C-482) on main rotor blades as shown on Figure 2. The targets shall be located exactly the same on each blade to ensure proper tracking.



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Figure 1. Chadwick-Helmuth Equipment Hook-Up (Sheet 1 of 1)



ICN-412-A-181101-B-97499-00002-A-001-01

Figure 2. Main Rotor Blade Reflective Tape Installation (Sheet 1 of 1)

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Main Rotor Blades - Strobe Tracking

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Chadwick-Helmuth Vibrex Kit	Model 177M	1	or Equivalent

Procedure



USE chadwick-helmuth vibrex kit (Model 177M) IN ACCORDANCE WITH CHADWICK-HELMUTH OPERATION AND SERVICE INSTRUCTION HANDBOOK. IF THERE ARE ANY QUESTIONS CONCERNING USE OF THIS EQUIPMENT, CONTACT CHADWICK-HELMUTH, MONROVIA, CALIFORNIA.

NOTE

Strobing blades to determine track variations during flight is essential when working main rotor. Use Chadwick-Helmuth Strobex (Model 135M-11) only.

1. It is recommended that Chadwick-Helmuth system be used to ground track rotor. A flag may be used for ground tracking (especially during ground track at 60%) to ensure perfect track and blade separation between blade pairs (upper and lower 1.75 inch separation).

1.1. Connect Strobex (1, Figure 1) to balancer (9).

1.2. To obtain a proper track picture with the Strobex, accomplish the following:

1.2.1. Set Balance/Phazor MAGNETIC PICKUP switch to COMMON.

1.2.2. Set Balancer/Phazor INTERRUPTER LOGIC switch to SINGLE.

1.2.3. Set Strobex (1) function switch to “B”.

1.2.4. Set Strobex RPM setting to less than N_R x 4 x 40%.

EXAMPLE: If N_R = 194 (60% rotor speed)

194 x 4 x 0.40 = 310.4

Set Strobex to 300

If N_R = 324 (100% rotor speed)

324 x 4 x 0.40 = 518.4

Set Strobex to 500

1.2.5. Aim Strobex at 11:30 clock position of tip path plane. Move strobe until a single target is seen; this will be green blade (Figure 2, View A).

1.2.6. Slowly increase Strobex RPM setting. As RPM setting is raised, targets on other three blades will appear to approach green blade’s target from the left. The Strobex is properly tuned when all four blades can be seen. View B shows how track should appear when Strobex is properly tuned.

NOTE

At 100% N_R, all four targets can be seen at approximately 512 RPM on the Strobex RPM selector (Figure 2, View B). Tuning Strobex too high will result in the two blade pairs merging into one target (View C).

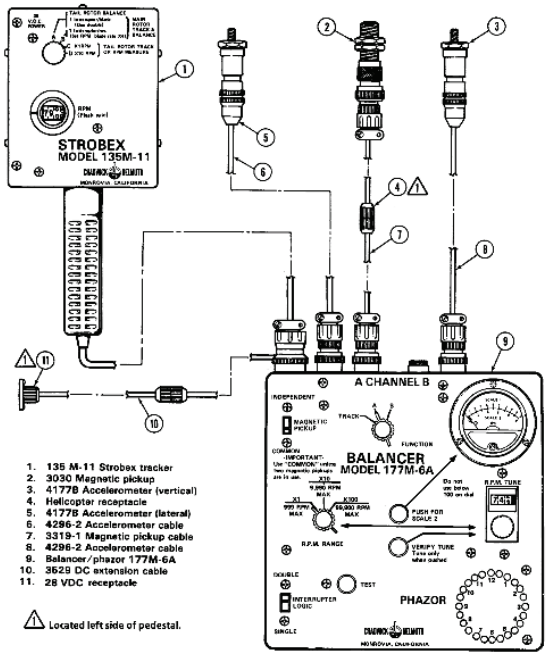
NOTE

If Strobex does not pick up blade target at 60% N_R, slightly decrease spacing between magnetic pickup and interrupter.

A sharper track picture may be obtained on Strobex by setting RPM selector on Strobex to twice recommended setting. (Example: at 60% N_R, set RPM selector on Strobex to 600 RPM instead of normal 300 RPM.) Increase setting slowly from 600 RPM until picture is obtained.

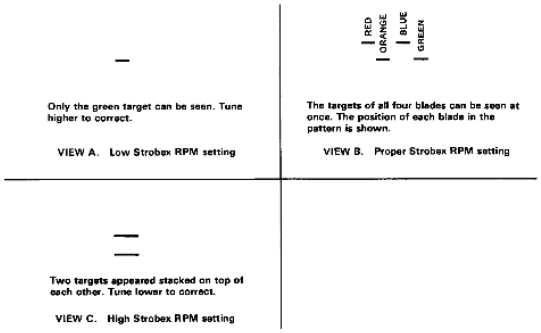
Track information from Strobex is used in conjunction with move data obtained from track and balance charts. The charts tell how much roll (pitch link adjustment) or tab is needed in each blade pair while the Strobex tells which blade of each pair should be rolled or tabbed.

2. EXAMPLE: At 120 knots IAS, the vertical 1/rev is recorded to be 0.25 ips at 12:30 clock position. As will be determined in 412-A-18-11-01-00A-724A-A / 00180, the green tab should be bent up 2° or orange tab bent down 2°. The strobe information, however, shows orange climbing as a function of airspeed with respect to other three blades. Therefore, orange blade should be tabbed down 2°.



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Figure 1. Chadwick-Helmuth Equipment Hook-Up (Sheet 1 of 1)



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Figure 2. Strobex Reflector Patterns (Sheet 1 of 1)

Main Rotor - Track and Balance

Preliminary requirements

Support equipment

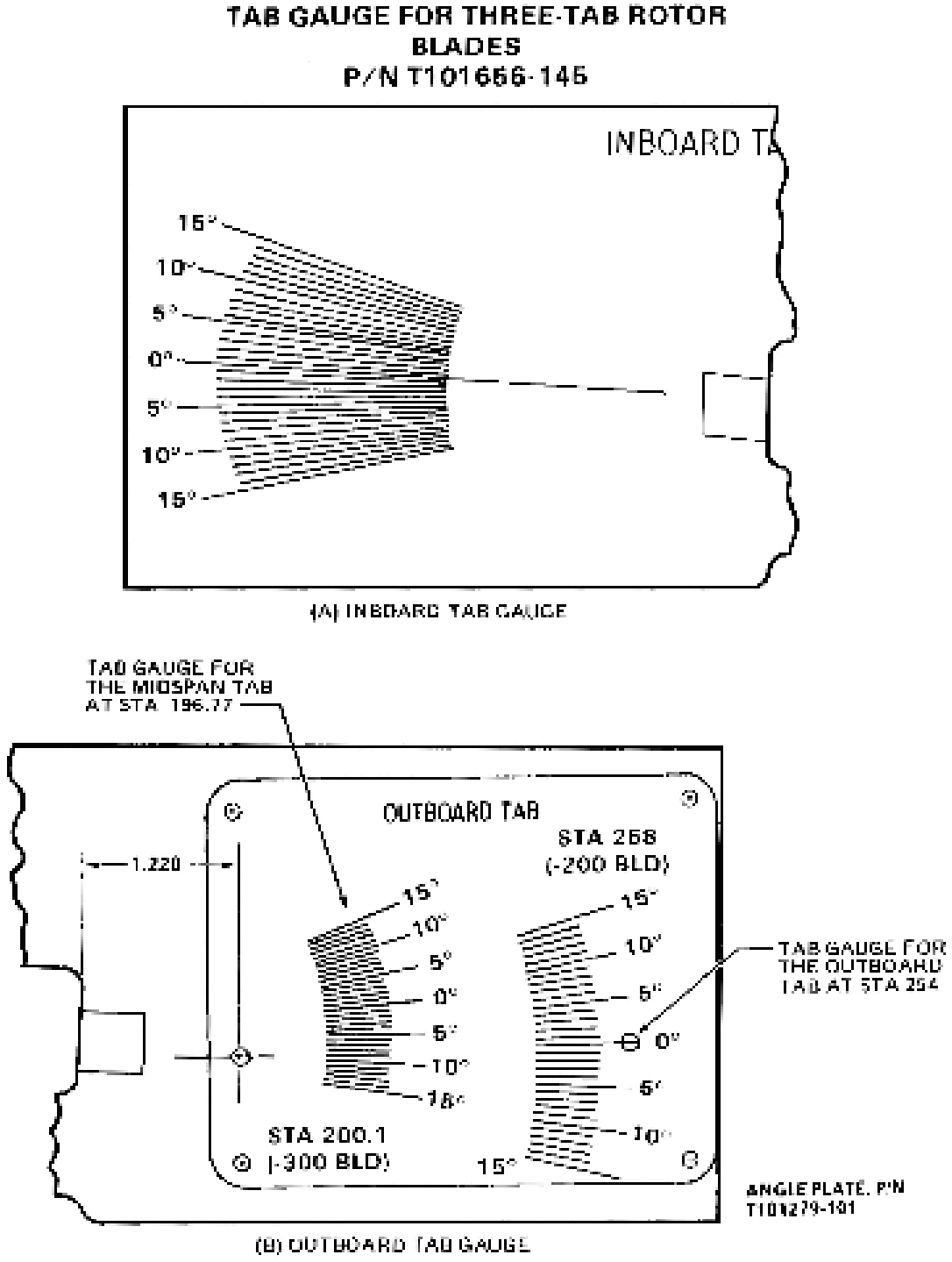
Nomenclature	Identification No.	Qty	Remarks
Chadwick-Helmuth Vibrex Kit	Model 6404	1	or Equivalent
Tab Bender	T101654-101	1	
Tab Gauge	T101656-145	1	(The -127 may be modified (Figure 1))

Procedure

1. There are four basic adjustments available to minimize main rotor 1/rev vibration. These are:

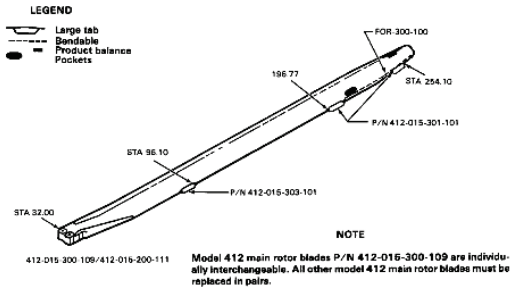
- Hub balance weights
- Pitch link length adjustment (roll)
- Blade trim tabs, and
- Blade product or dynamic balance

2. The functions of each of these adjustments are described in 412-A-18-11-00-01A-271A-A / 00154, 412-A-18-11-00-02A-271A-A / 00155, 412-A-18-11-00-03A-271A-A / 00156 and 412-A-18-11-00-04A-271A-A / 00157. The current blade designs are shown in Figure 2 with adjustments labeled in terminology used herein.



ICN-412-A-181100-B-97499-00012-A-001-01

Figure 1. Tab Gauge for Three-Tab Rotor Blades (Sheet 1 of 1)



ICN-412-A-181101-B-97499-00004-A-001-01

Figure 2. Blade Configuration (Sheet 1 of 1)

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Track and Balance - Preparation

Procedure

1. Perform the following prior to beginning work on rotor:

1.1. Ensure all trailing edge blade attachment bolts (expandable bolts, if installed) are at proper locking force and are oriented parallel to centers of blade bolts. It is extremely important that expanding bolts are all oriented the same and orientation can be repeated when blades are unfolded.

1.2. If any rotating control system links have been disconnected, ensure link is reinstalled in such a way that any elastomeric bearing has zero torsional preload at flat-pitch when swashplate is level with respect to mast. This may be accomplished as follows:

1.2.1. Center cyclic to maintain swashplate level in respect to mast.

1.2.2. Disconnect upper end of pitch links from pitch horns.

1.2.3. Lower collective lever until down stop is reached.

1.2.4. Install link in such a way that it is within one serration on elastomeric bearing of being perpendicular (90°) to the rephasing link.

1.2.5. Repeat Step 1.2.4. until all links are installed.

1.2.6. Attach upper end of pitch links to pitch horns.

1.3. If new blades are being installed, ensure tabs are set to zero or are set to angles called out on data sheet supplied with blades. When replacing one or a pair of blades in a previously worked rotor, do not “zero” tabs of original blades; leave them as a set prior to blade change. The one exception to this is when one blade is replaced, the inboard tab of mating blade should also be zeroed, since both inboard tabs of a blade pair are always bent opposite each other.

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Track and Balance - Working Main Rotor

Procedure

1. The proper sequence for working the rotor on helicopters having blades which incorporate three trim tabs and product balance is shown in Figure 1. The most important phase in working the rotor is to obtain a good ground track and balance prior to going to flight. Once in flight, track is less important, but should be recorded to provided information on which blade of a pair should be worked to maintain the best separation between the pairs. In flight, the magnitude and clock angle of the 1/rev vibration takes precedence over the track (i.e. a good ride has higher priority than a good track). All rotor track and balance data should be recorded on a copy of the form in Figure 2.

1.1. Rolling into ground track at 60%: The first phase in working the rotor is to obtain a good ground track at 60% rotor speed (N_R). To accomplish this proceed as follows:

1.1.1. Operate at approximately 60% N_R (BHT-412-FM).

1.1.2. Observe track of rotor using Strobex or flag (if Strobex is used, set initial oscillator setting at 600) and record on rotor tracking sheet.

1.1.3. Set Balance/Phazor FUNCTION selector to TRACK. Aim Strobex at the 11:30 location. Slowly increase Strobex setting until all four blades are seen.

1.1.4. Estimate amount of out-of-track and adjust length of pitch links as required (412-A-18-11-00-01A-271A-A / 00154). The amount of blade-track change per adjustment of pitch link is as follows:

- Turning barrel one flat = 0.1 inch
- One full turn of barrel (6 flats) = 0.6 inch
- One half turn of upper rod end = 1.0 inch
- One turn of upper rod end = 2.0 inches
- One turn of lower rod end = 1.4 inches

1.1.5. Repeat Step 1.1.1. through Step 1.1.4. until rotor is in proper track. The blades are in proper track when blades of upper pair (red and blue blades) are in same plane and blades of lower pair (orange and green blades) are in same plane with separation between pairs being 1.75 inches (Reflectors are 1.25 inches in length and can be used to estimate gap between blade pairs.)

1.1.6. To see track of main rotor at idle, set Balance/Phazor selector to B.

1.2. Obtaining 100% track and balance. Once 60% track has been obtained, rotor is ready to be tracked and balanced at 100% N_R. To accomplish this, proceed as follows:

1.2.1. Operate helicopter at, or near, 100% N_R (single power section operation is acceptable).

1.2.1.1. Air Strobex to view track in front of copilot seat (11:30 position).

1.2.1.2. Pull trigger and move Strobex until one target is seen.

1.2.1.3. With this target in view, increase tuning of Strobex until all four targets are seen.

1.2.1.4. Record position of targets and gauge track by length of reflective tape.

1.2.2. Measure and record magnitude and phase of lateral and vertical 1/rev vibration and record blade track (if Strobex is used, set oscillator to an initial setting of 500).

1.2.3. Plot lateral 1/rev vibration on Hub Balance Chart shown in Figure 3. Note to which blade(s) chart calls for adding weight. If track is close, make weight adjustment indicated. If rotor is significantly out of track, proceed to Step 1.2.5. prior to balancing.

1.2.4. Repeat Step 1.2.1. through Step 1.2.3. until lateral 1/rev vibration is reduced to below 0.1 ips (preferably to 0.05 ips, or below).

1.2.5. Use outboard trim tab to obtain track at 100% RPM. Bending outboard tab up causes blade to climb; bending tab down causes blade to dive (412-A-18-11-00-02A-271A-A / 00155). The sensitivity of track to outboard tab adjustment is as follows:

- Large tab: 0.25 inch per degree

1.2.6. After each tab move, repeat Step 1.2.1. through Step 1.2.5. until lateral 1/rev is below 0.1 ips and rotor is in track.

1.3. Working Lateral 1/rev Vibration in a Hover.

1.3.1. Product-balancing the Rotor (-200, -111, and all -300's). Once the rotor has been tracked and balanced on the ground, the product-balance of the rotor is checked and adjusted, if necessary. To accomplish this proceed as follows:

1.3.1.1. Observe track of rotor at 100% flat pitch on ground and record vertical and lateral 1/rev readings (if not already accomplished). The lateral 1/rev readings should be below 0.1 ips at this time.

1.3.1.2. Lift helicopter into a hover (BHT-412-FM) and orient nose into wind. Observe and record track as well as lateral 1/rev vibration levels (magnitude and phase).

1.3.1.3. If lateral 1/rev vibration is acceptable in a hover, then helicopter is ready to go into forward flight.

1.3.1.3.1. If lateral 1/rev vibration is higher than desired and at the same clock angle as lateral on the ground, use balance chart (Figure 3) to reduce lateral vibration in a hover as low as possible without causing an unacceptable 1/rev ground balance. If hover is still unacceptable, go to Step 1.3.2..

1.3.1.3.2. If the lateral 1/rev vibration in a hover is radically different from that measured on ground in either magnitude or clock angle, plot 1/rev data on Inflight Lateral 1/rev Chart shown in Figure 4. From chart, note which pair of blades should be product-balanced to minimize lateral 1/rev vibration in hover. By observing track, one can distinguish which blade of the pair should be product-balanced to maintain a proper separation between the pairs.

1.3.1.3.3. Once blade on which the product-balance will be adjusted has been identified, remove cover of product-balance pocket from which weight is to be removed (412-A-18-11-00-04A-271A-A / 00157). If exact amount, or significant fraction, of weight to be removed is present, remove cover of the other product-balance pocket and make all or part of move indicated by the chart. If insufficient weight is present in product-balance pockets of chosen blade, the opposite blade will have to be product-balanced the remainder of the move indicated by chart. Once product-balance move has been made, secure pocket covers on all pockets.

NOTE

If sufficient product-balance capability is not available in the product-balance pockets, refer to Step 1.4. below.

1.3.2. Recheck balance of rotor at 100% N_R on ground and correct if necessary.

NOTE

The mistake many make when product-balancing rotor is not rebalancing the rotor before determining full effect of product-balance move. Remember that any mistake made in the ratio of weights adjusted at blade tip is magnified by approximately a factor of eleven; then the error must be countered by weight at hub balance weight attachment locations.

1.3.3. Repeat Step 1.3.1. and Step 1.3.2. until lateral 1/rev vibration on ground is less than 0.1 ips and in hover is less than 0.2 ips. Once this has been accomplished, the helicopter is ready to go into forward flight.

NOTE

As rotor is balanced, it may be necessary to make slight adjustments of outboard trim tab to maintain an acceptable track at 100% RPM.

1.4. Rolling the Rotor for Lateral 1/rev Vibration in a Hover. On rotors with blades that do not have sufficient product balance capability, the only means available to adjust lateral 1/rev vibration in a hover semi-independently of lateral 1/rev vibration on ground is roll. Roll has more effect on lateral 1/rev vibration in a hover than it does on the lateral 1/rev vibration on ground, while hub balance weights have an equal effect on lateral 1/rev vibration on ground and in a hover. With these characteristics known, the proper method for working lateral 1/rev vibration in a hover is to roll rotor blades until lateral 1/rev vibration in a hover is approximately the same as it is on ground. Once this is done, hub balance weights are used to correct both lateral 1/rev vibration on the ground and in a hover. The proper method for working the rotor with roll is as follows:

1.4.1. Reduce lateral 1/rev vibration on the ground using Lateral 1/Rev Balance Chart shown in Figure 3.

1.4.2. Measure and record lateral 1/rev vibration in a hover. Plot this information on Lateral 1/Rev in a Hover Chart shown in Figure 4 and make roll move indicated.

1.4.3. Recheck lateral 1/rev vibration on the ground and correct using hub balance weights (412-A-18-11-00-03A-271A-A / 00156).

1.4.4. Repeat Step 1.4.1. through Step 1.4.3. continuing to use roll to correct lateral 1/rev vibration in a hover and hub balance weights to correct lateral 1/rev vibration on ground until both are 0.2 ips or lower.

1.5. Working Vertical 1/Rev Vibration.

NOTE

Check autorotation RPM (Figure 5) on the first flight and, if necessary, adjust as required (412-A-18-11-00-05A-271A-A / 00158). Once lateral 1/rev vibration has been reduced to an acceptable level, both on ground and in a hover, the rotor is ready for evaluation in forward flight. To work rotor with respect to vertical 1/rev vibration, the following steps are taken.

1.6. Once lateral vibration in hover has been reduced, conduct an airspeed sweep where track, vertical, and lateral information is obtained at conditions as follows:

1.6.1. 100% flat pitch on ground

1.6.2. Hover

1.6.3. 1000 RPM climb at 60 knots

1.6.4. 120 knots level

1.6.5. VNE

1.6.6. 1000 RPM descent at 60 knots

1.7. Review data obtained in performing Step 1.6. to determine the vibration in the helicopter. Work helicopter as follows:

1.7.1. Reduce vertical 1/rev at speeds up to and including 20 knots.

1.7.1.1. Adjust outboard trim tabs (412-A-18-11-00-02A-271A-A / 00155). Plot data on vertical 1/rev chart (Figure 6). Record outboard trim tab adjustments indicated as needed to reduce 1/rev at 120 knots. Chart provides options to bend one trim tab up or its mate down. Select option that best maintains proper track spacing for rotor.

1.7.1.2. Adjust inboard trim tabs only when vertical 1/rev at speeds above 120 knots cannot be made acceptable by adjusting outboard trim tabs (Figure 7 and 412-A-18-11-00-02A-271A-A / 00155).

1.7.1.3. Repeat Step 1.7.1.1. and Step 1.7.1.2. until vertical 1/rev vibration level is acceptable.

NOTE

If primary flight condition for helicopter is 120 knots cruise, it is not necessary to check vibration level at VNE.

1.7.2. After vertical 1/rev vibration is worked to an acceptable level, work lateral 1/rev vibration.

1.7.2.1. Using hub balance weights and M412/Pen Inflight Lateral 1/Rev Chart (Figure 4), retrim rotor at flat pitch on the ground.

1.7.2.2. Using hub balance weights (412-A-18-11-00-03A-271A-A / 00156), product balance (412-A-18-11-00-04A-271A-A / 00157), and/or pitch link adjustments (412-A-18-11-00-01A-271A-A / 00154), reduce lateral 1/rev in flight as follows:

1.7.2.2.1. If lateral 1/rev is present in a hover and at high speed (and rotor was properly tracked at idle), make product balance adjustment indicated on M412/Pen Inflight Lateral 1/Rev Chart (Figure 4). Repeat this substep until additional product balance adjustments cannot be made or lateral 1/rev at hover reaches an acceptable level.

1.7.2.2.2. Make pitch link adjustments if lateral 1/rev is not acceptable and additional product balance adjustments cannot be made. Refer to pitch adjustment (roll) portion of M412/Pen Inflight Lateral 1/Rev Chart (Figure 4). Also, refer to pitch link adjustment procedure (412-A-18-11-00-01A-271A-A / 00154).

1.7.2.2.3. After lateral 1/rev in hover is acceptable, make pitch link adjustments to obtain an acceptable lateral 1/rev in forward flight. Refer to pitch adjustment (roll) portion of M412/Pen Inflight Lateral 1/Rev Chart (Figure 4). Also, refer to pitch link adjustment procedure (412-A-18-11-00-01A-271A-A / 00154). Make one-half indicated pitch link adjustment for 120 knots.

1.7.2.2.4. Once lateral 1/rev in forward flight is acceptable, make small hub balance weight adjustments to obtain an acceptable lateral 1/rev vibration level in letdown. Since lateral 1/rev in letdown is more sensitive to hub balance weight changes than lateral 1/rev on ground and in forward flight, it is possible to make a limited improvement in the lateral 1/rev without making a significant change in lateral 1/rev on ground or in forward flight. Refer to M412/Pen Hub Balance Letdown Chart (Figure 8) and hub balance weight procedure (412-A-18-11-00-03A-271A-A / 00156) and make adjustments to reduce lateral 1/rev in letdown (Figure 9) to an acceptable level.

NOTE

As adjustments are made, it may be necessary to retrim main rotor vertical and lateral vibration by using the procedures in Step 1.6. and Step 1.7..

Do not change balance for letdown enough to make ground bounce unacceptable.

Since 1/rev vibration is primarily a ride quality factor, the acceptable level is up to the operator, and must be weighed against increased maintenance time and component replacement. Optimize ride for primary operating condition of helicopter.

1.8. Lateral 1/Rev Vibration that Changes with Gross Weight. It is important to work the rotor without making radical gross-weight changes. If lateral 1/rev vibration changes significantly when going from light to heavy gross weight, measure lateral 1/rev throughout flight regime at the increased gross weight. Use Lateral 1/Rev in Hover Chart (Figure 4) to determine proper product balance roll move to correct this condition. To date, adjustments in roll have been sufficient to correct for minor changes in 1/rev with gross weight changes. Significant changes in 1/rev vibration with gross weight changes are generally caused by degradation of elastomeric bearings in the control system; therefore, these bearings should be thoroughly inspected prior to further rotor working efforts.

1.9. Recording Rotor Configuration. To assist in maintaining rotor at an acceptable level of 1/rev, the chart shown in Table 1 should be filled out immediately upon completion of working the rotor. Save chart for future reference.

1.10. Remove all track and balance equipment from helicopter.

1.11. Trimming the Rotor. Once the rotor is worked, very little maintenance should be required to maintain acceptable vibration levels. If, after maintenance or during normal operations, the ride has deteriorated, the following steps should be taken:

1.11.1. If rotor has been folded, ensure all expandable blade bolts are torqued and aligned properly. Make any necessary corrections.

1.11.2. Verify that idle track and tab settings are the same as documented in Rotor Configuration Log (Table 1). Correct if necessary.

1.11.3. Check hub to mast installation.

1.11.4. Remove any water from product balance pockets. Verify covers for product balance pockets (2 and 3, Figure 7) are installed and sealed (412-A-18-11-00-04A-271A-A / 00157).

If the rotor still exhibits higher than desired main rotor 1/rev levels, work the rotor from its present point; do not zero everything and start over. If the rotor works successfully, log final tab settings, idle track, and weight configuration on a copy of the Rotor Configuration Log (Table 1). Assure all logs are maintained in such a way that any trend which could give indications of degradation in the control system elastomerics can be monitored. If degradation is found later, a previous log entry may allow rotor to be reset to original configuration; thus minimizing rotor working time.

NOTE

When reworking rotors, initial data should be taken with previous hub balance weights and trim tab settings. Outboard trim tabs should only be zeroed on new blades. Midspan and inboard trim tabs should be zeroed on both new blade and blade opposite new blade.

Table 1. Rotor Configuration Log

HELICOPTER TYPE: _____ DATE: ____/____

HELICOPTER S/N: _____

		TAB SETTING		
BLADE S/N	COLOR	OUTBOARD	MIDSPAN	INBOARD
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

TRACK PICTURE AT 60%

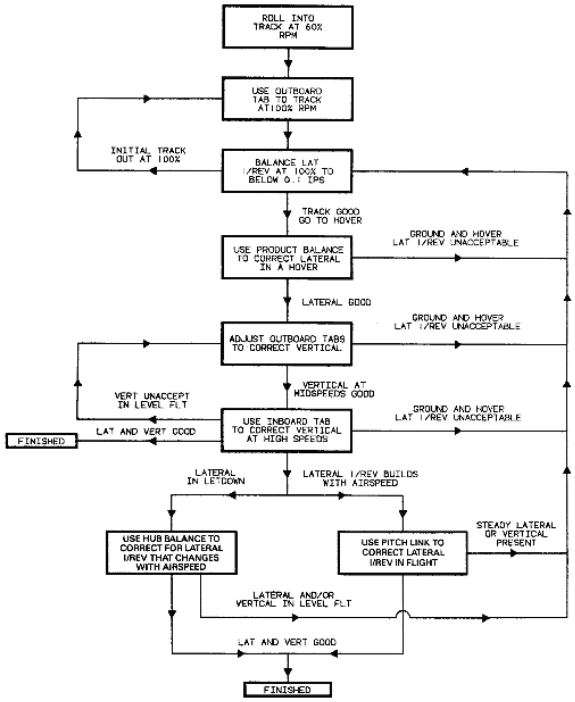
Note Relative Dimensions.

HUB WEIGHTS

Table 1. Rotor Configuration Log (continued)

BLADE S/N	COLOR	LEADING EDGE	TRAILING EDGE
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

CHANGES TO BLADE MASTER:



ICN-412-A-181101-B-97499-00005-A-001-01

Figure 1. Flow Chart Working Model 412 Rotor with Three-Tab Blades (Sheet 1 of 1)

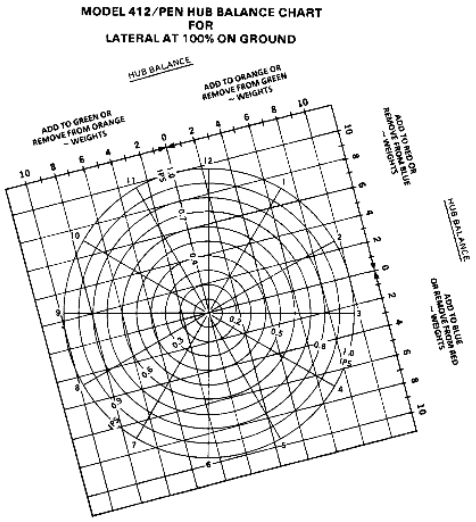
HELICOPTER S/N: _____ HELICOPTER CONFIG: _____

A: _____ B: _____ DATE: _____ PAGE: _____

	60%	100%	HOVER	CLIMB	100 KT	120 KT	VH	DESCENT	CHANGE
TRACK									
A									
B									
TRACK									
A									
B									
TRACK									
A									
B									
TRACK									
A									
B									
TRACK									
A									
B									

ICN-412-A-181101-B-97499-00006-A-001-01

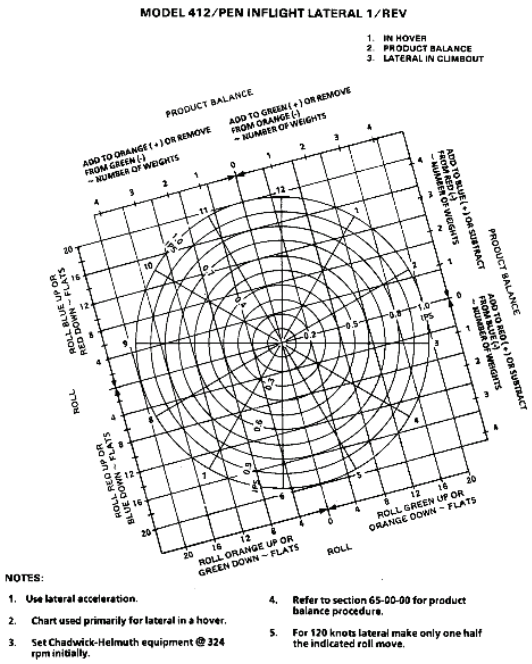
Figure 2. Sample Form Used to Record 1/Rev Vibration Levels and Rotor Moves (Sheet 1 of 1)



1. Use lateral accelerometer
2. Balance weight sizes:
   - 1.0 weight, P/N 212-010-710-003, 0.25 pound (113 grams)
   - 0.5 weight, P/N 212-010-710-105, 0.14 pound (63 grams)
3. Set Chadwick – Helmuth equipment at 324 rpm initially.

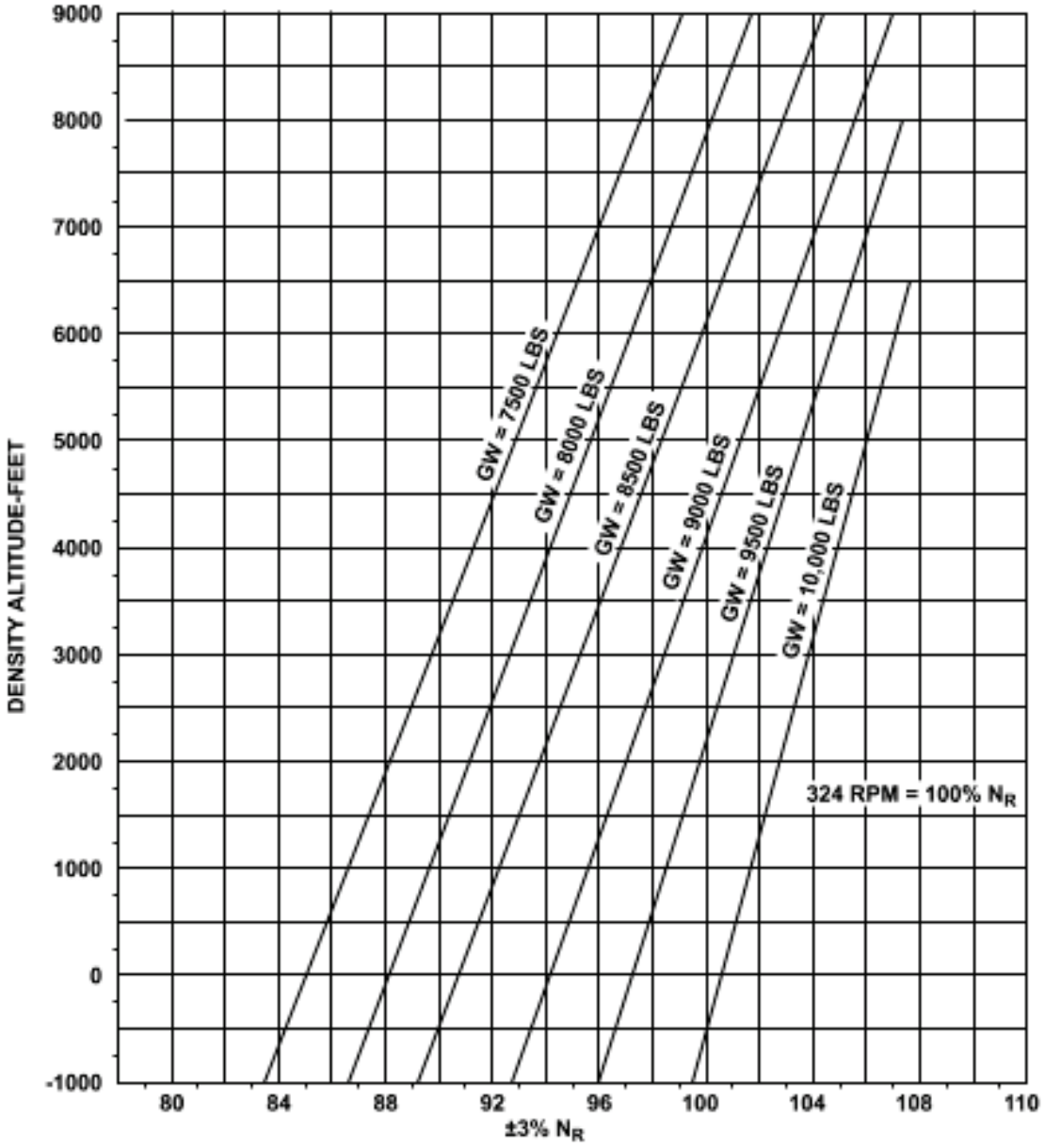
ICN-412-A-181101-B-97499-00007-A-001-01

Figure 3. M412/Pen Hub Balance Chart (Sheet 1 of 1)



ICN-412-A-181101-B-97499-00008-A-001-01

Figure 4. M412/Pen Inflight Lateral 1/Rev Chart (Sheet 1 of 1)

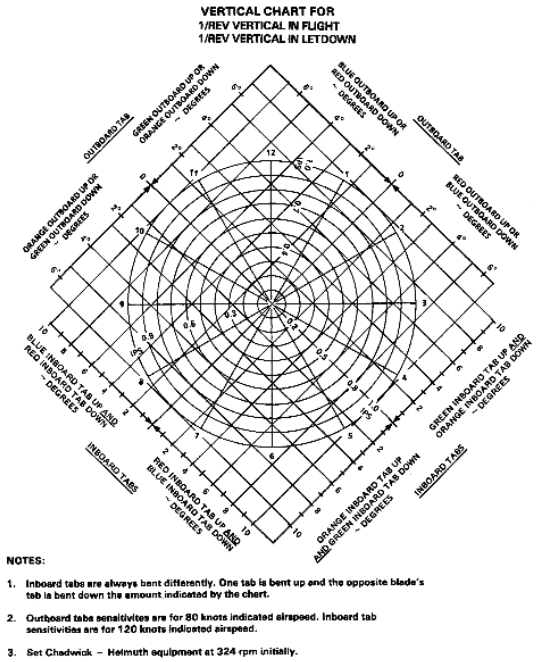


NOTE

Refer to model 412 Flight Manual for density altitude chart.

ICN-412-A-181100-B-97499-00014-A-001-01

Figure 5. Autorotation Descent Rotor RPM Chart (Sheet 1 of 1)

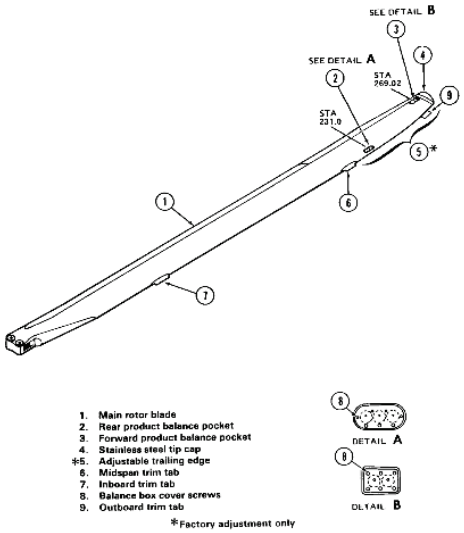


NOTES:

1. Inboard tabs are always bent differently. One tab is bent up and the opposite blade's tab is bent down the amount indicated by the chart.
2. Outboard tabs sensitivites are for 80 knots indicated airspeed. Inboard tab sensitivities are for 120 knots indicated airspeed.
3. Set Chadwick – Helmuth equipment at 324 rpm initially.

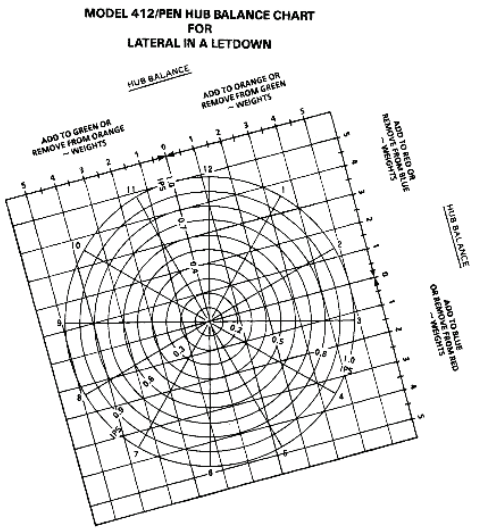
ICN-412-A-181101-B-97499-00009-A-001-01

Figure 6. Vertical 1/Rev Chart (Sheet 1 of 1)



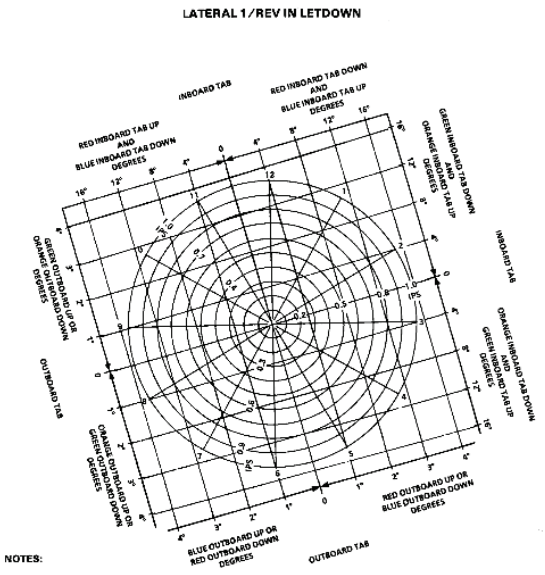
ICN-412-A-181100-B-97499-00011-A-001-01

Figure 7. Adjustments on 412-015-300 Main Rotor Blade (Sheet 1 of 1)



ICN-412-A-181101-B-97499-00010-A-001-01

Figure 8. M412/Pen Hub Balance for Lateral in Letdown Chart (Sheet 1 of 1)



NOTES:

1. Use lateral accel. to obtain data
2. To work lateral 1/rev in a letdown make both the indicated outboard and inboard tab moves
3. Sensitivities are for 80 kts @ 1000'/min descent
4. Set Chadwick-Helmuth equipment at 324 rpm initially.

ICN-412-A-181101-B-97499-00011-A-001-01

Figure 9. Lateral 1/Rev Vibration Letdown (Sheet 1 of 1)

Track and Balance - Main Rotor Autorotation RPM Adjustment

Procedure

NOTE

Autorotation RPM should be adjusted with consideration given to seasonal OAT and altitudes in the operating environment. Low gross weight, low temperature or low density altitude result in lower autorotation RPM when compared to high gross weight, high temperature or high density altitude.

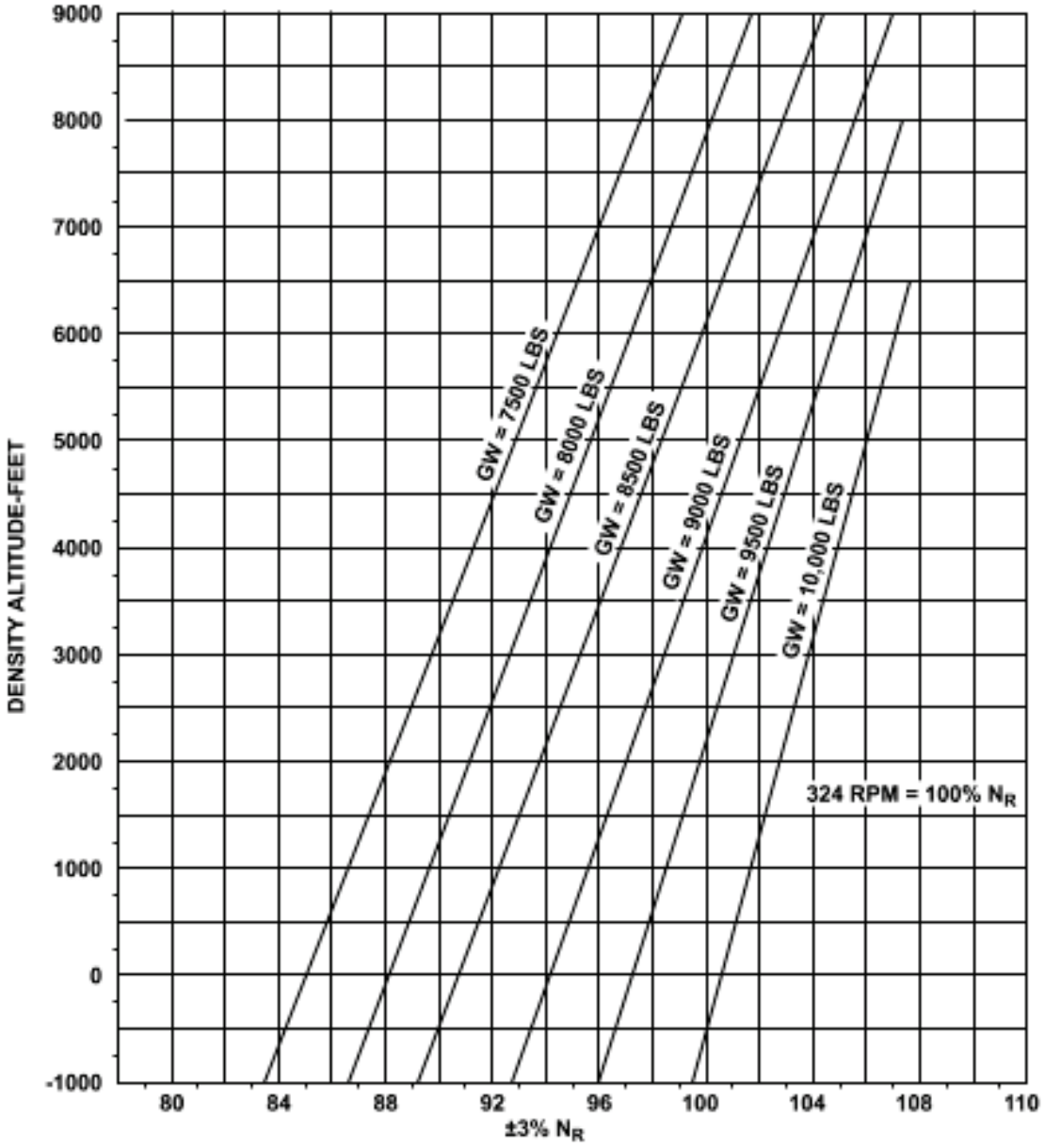
1. Fly the helicopter and establish the existing autorotation RPM (N_R) by initiating an autorotative descent at 65 KIAS (the collective full down, both the throttles at the IDLE position).

2. Once autorotation is stabilized, record pressure altitude (H_P), outside air temperature (OAT), main rotor RPM and fuel quantity on the autorotation RPM Adjustment Log (Table 1).

3. Using H_P and OAT, determine the density altitude (H_D) at which the autorotative RPM check was conducted from the density altitude chart (BHT-412-FM-1, Figure 4-2).

Table 1. AUTOROTATION RPM ADJUSTMENT LOG

	AUTOROTATION RPM ADJUSTMENT LOG
	Date: Model: 412 Helicopter S/N: Total Flight Time:
1	Conduct Autorotation RPM check (collective full down, both throttles at idle, stabilize at 65 KTS): H_P(set 29.92 (1013 mb) on altimeter) _____ OAT_____ Main Rotor RPM _____ Fuel Quantity _____
2	Calculate H_D: Apply H_P_____ and OAT _____ against (BHT-412-FM-1, Figure 4-2) to obtain H_D
3	Apply H_D and helicopter test gross weight to Autorotational Adjustment RPM Chart (Figure 1): H_D _____ Gross Weight _____ Target Main Rotor RPM _____
4	Adjust main rotor RPM if not within ±3% of target. Refer to 412-A-18-11-00-05A-271A-A / 00158.
5	Final adjusted main rotor Autorotation RPM _____



NOTE

Refer to model 412 Flight Manual for density altitude chart.

ICN-412-A-181100-B-97499-00014-A-001-01

Figure 1. Autorotation Descent Rotor RPM Chart (Sheet 1 of 1)

Track and Balance - Working Main Rotor 2/Rev Vibration

Procedure

1. Main rotor 2/rev vibration in helicopters with pendulums is similar to main rotor 1/rev vibration. Main rotor 2/rev is perceived by the flight crew as a roughness in ride which does not register on the Balancer when tuned to 324 RPM (1/rev). While main rotor 1/rev vibration is caused by the variation in lift between blades within a pair, 2/rev is caused by variations in lift between pairs. Generally maintaining proper separation between pairs will result in low 2/rev vibration levels throughout the flight regime. If, after the main rotor 1/rev has been reduced to an acceptable level, a roughness is still felt in the helicopter main rotor, 2/rev vibration should be checked. To check main rotor 2/rev vibration, the following procedure should be accomplished:

1.1. Mount an accelerometer on the floor of the cabin directly behind the aft row of passenger seats, on the right side of cabin near cargo door. Connect right-angle accelerometer brackets (Figure 1) (or make a bracket, which orients the accelerometer perpendicular to the mounting plane). Once the two brackets have been connected, attach with a convenient screw at indicated location. To assure that the brackets are sufficiently stiff, back up the bracket attached to the deck with a large washer. Once brackets are installed, attach accelerometer to bracket in such a way that it is oriented vertically with connector pointed up.

1.2. Connect accelerometer cable to accelerometer and Balancer/Phazor. Tune balancer to 647 RPM.

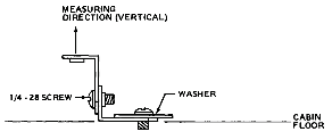
1.3. Measure 2/rev levels at flight conditions specified for checking main rotor 1/rev vibration.

1.4. Plot 2/rev levels on Vertical 2/Rev Chart (Figure 2).

1.4.1. If clock angle of 2/rev vibration is generally the same but levels build with airspeed, then indicated roll move should be sufficient to reduce 2/rev vibration.

1.4.2. If clock angle of 2/rev changes 3 hours between low speed and high speed, then roll for low speed 2/rev and use midspan tabs for high speed 2/rev.

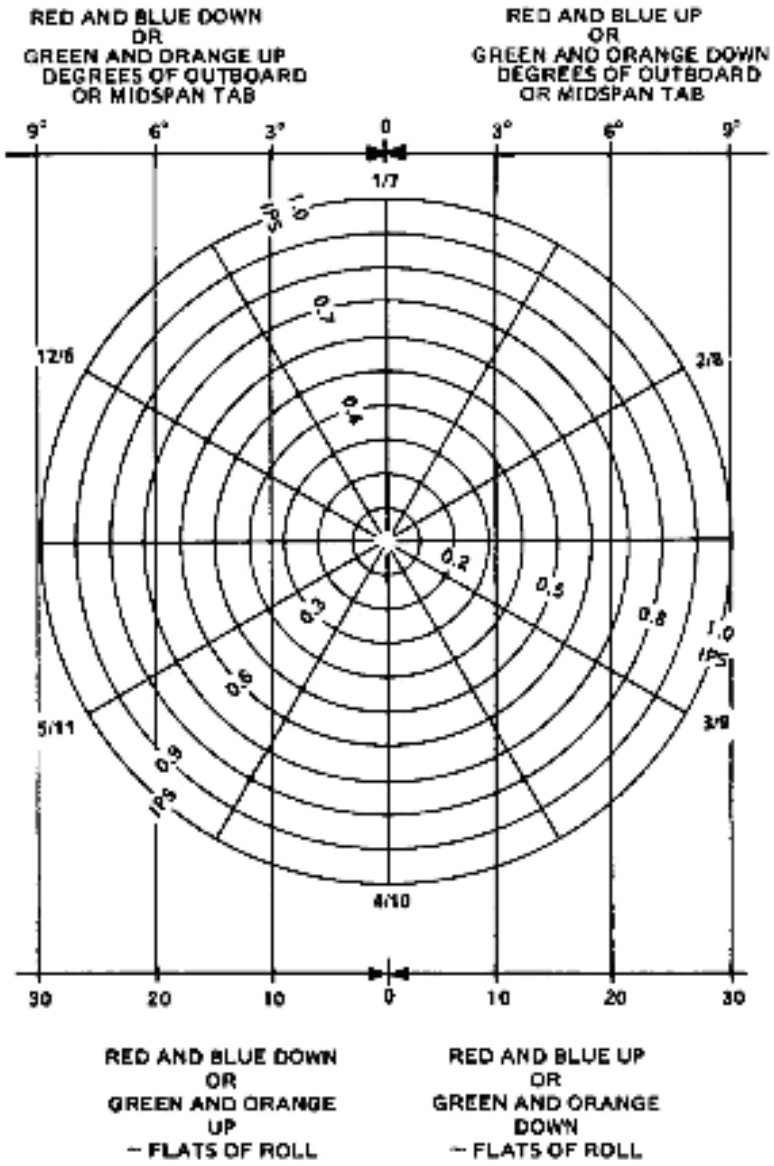
1.4.3. If 2/rev is present only in a letdown, then bend both outboard tabs in direction indicated by charts and counter it with an opposite bend in midspan tabs. Once 2/rev is adjusted so that all of the data is at the same clock angle, make a final adjustment using roll to correct.



ICN-412-A-181101-B-97499-00012-A-001-01

Figure 1. Accelerometer Support for Measuring Cabin 2/Rev Levels (Sheet 1 of 1)

MAIN ROTOR 2/REV REDUCTION CHART



NOTES:

1. **Measure vertically at right aft pass seat**
2. **1/2 turn of the upper rod end = 10 flats**
3. **Set Chadwick-Helmuth equipment at 648 rpm initially.**

ICN-412-A-181101-B-97499-00013-A-001-01

Figure 2. Main Rotor 2/Rev Reduction Chart (Sheet 1 of 1)

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Tail Rotor Hub and Blade - Operational Check

Procedure

1. Tail rotor track check by manual flag method (412-A-18-12-01-00A-373A-A / 00185) or by VIBREX Track and Balance System (dynamic method) (412-A-18-12-01-01A-373A-A / 00186 and 412-A-18-12-01-00A-271A-A / 00187).

2. Tail rotor vibration check by VIBREX Track and Balance System (dynamic method) (412-A-18-12-01-01A-373A-A / 00186 and 412-A-18-12-01-00A-271A-A / 00187).

3. Tail rotor rigging check for adequate right and left anti-torque control by flight test (412-A-64-00-00-00A-720A-A / 01232 and BHT-412-FM).

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Tail Rotor - Troubleshooting

Procedure

1. Potential troubles, which may occur in the tail rotor, are listed in 412-A-18-12-00-00A-422A-A / 00168.

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Tail Rotor Hub and Blade - Flag Tracking

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Tinting medium	C-438	AR	

Procedure

1. Attach a small piece of sponge rubber 0.125 to 0.25 inch (3 to 6 mm) thick on the tip of a 0.5 x 0.5 inch (12 x 12 mm) pine stick or any other flexible device. Coat sponge rubber with tinting medium (C-438) or similar type of coloring thinned with oil.

2. Start engine and run at 100% N_R, with pedals in neutral position (BHT-412-FM).

3. Brace marking device on forward side of tail fin assembly. Slowly move marking device toward tail rotor blades, approximately 1 inch (25 mm) from the blade tip. Move marking device just far enough to touch one blade, which will be the near blade.

4. Shut down engine and allow rotor blades to stop after marking near blade. Shorten pitch control link of unmarked blade one-half turn.

5. Repeat Step 2. and Step 3..



WHEN ADJUSTING PITCH CHANGE LINK, USE WITNESS HOLE IN BARREL TO ENSURE PROPER AMOUNT OF ROD END THREAD ENGAGEMENT.

6. Lengthen pitch control link of marked blade if further adjustment is needed. Continue adjustments alternately on opposite links by one-half turn increments until blades are in track.

7. Perform maintenance test flight (BHT-412-FM).

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Tail Rotor - Tracking

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Chadwick-Helmuth Vibrex Kit	Model 177M	1	or Equivalent

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Tape	C-483	AR	

Procedure



COMPLY WITH THE CHADWICK-HELMUTH OPERATIONS AND SERVICE INSTRUCTION HANDBOOK, AND THE FOLLOWING INSTRUCTIONS WHEN TRACKING TAIL ROTOR WITH chadwick-helmuth vibrex kit (Model 177M).

NOTE

If there are questions concerning use of Vibrex equipment, contact Chadwick-Helmuth, Monrovia, California.

1. Inspect tail rotor hub and blades for damage and proper installation.

2. Install the Balancer/Phazor for use outside the helicopter, near the left side of the tail rotor. Connect Strobex to Balancer.

NOTE

Do not connect the magnetic pickup cable. It is not required when tracking the tail rotor.

3. Mount tail rotor balance accelerometer on the tail rotor gearbox as follows:

3.1. Loosen nut on tail rotor gearbox case at the 12 o'clock position.

3.2. Install accelerometer and bracket (#3382) between gearbox case and nut, then tighten nut.

4. Connect accelerometer cable to accelerometer and route the cable forward to, and around, the elevator and then to the right side of the tail rotor, approximately 15 feet (4.57 m) from the rotor disk. Connect only the tail rotor accelerometer cable to the Balancer/Phazor at "A". Secure cable to tailboom with masking tape.

5. Select one rotor blade as "target" blade. Place a 3/4 inch (19.05 mm) strip of tape (C-483) on blade grip between blade bolts as shown on Figure 1.

6. Place a 3/4 inch (19.05 mm) strip of tape (C-483) on tail rotor blade tips as shown on Figure 2.



BEFORE PROCEEDING, ENSURE THAT ALL CABLE SLACK HAS BEEN REMOVED BY TAPING CABLE TO THE TAILBOOM AND THAT ADEQUATE CLEARANCE EXISTS BETWEEN CABLE AND ROTATING COMPONENTS.

7. Operate helicopter (BHT-412-FM) at 100% N_R at flat pitch on the ground, standing approximately 15 feet (4.57 m) from the right side of the tail rotor. Adjust Strobex oscillator to approximately 6648 RPM (four times tail rotor RPM).

8. Face the tail rotor disk, shine the Strobex at the rotor and locate the reflection target (Figure 1). Refer to operational manual to fine tune the Balancer/Phazor and thus "stop" the reflective image from rotating. The target will appear simultaneously at the 12, 3, 6, and 9 o'clock positions when the oscillator is properly adjusted.

9. Stand on the right side of the helicopter next to the pilot compartment. Aim the Strobex at the front of the tail rotor while it is at normal operating RPM. Observe the track of the target. If targets form a “+” sign as shown on Figure 3, the blades are in track.



WHEN ADJUSTING PITCH CHANGE LINK, USE WITNESS HOLE IN BARREL TO ENSURE PROPER AMOUNT OF ROD END THREAD ENGAGEMENT.

10. If blades are out of track, shut down engines, and adjust the length of the pitch change link on the target blade by one-half increments to bring target blade close to other blade. Record amount of change.

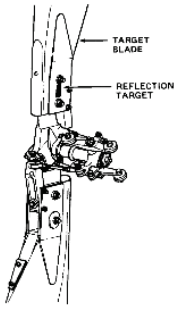
11. Repeat Step 8. if blades are still out of track, and adjust pitch link on opposite blade to bring it closer to target blade. Record amount of change.

12. Repeat Step 7. through Step 11. if required to bring blades into track.

13. Disconnect accelerometer from Balancer/Phazor. Disconnect dc power cable from balancer.

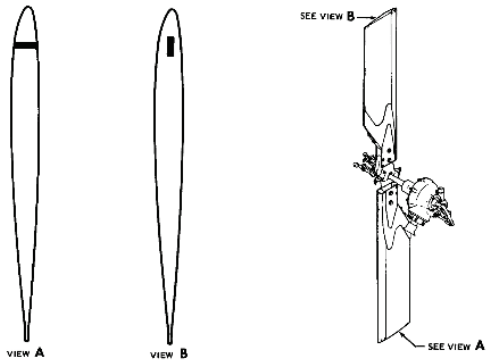
14. Remove dc power cable.

15. Remove accelerometer and bracket from tail rotor gearbox.



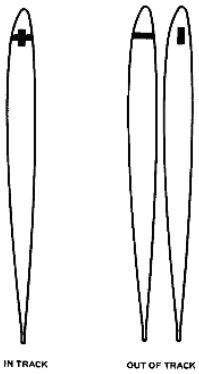
ICN-412-A-181201-B-97499-00001-A-001-01

Figure 1. Tail Rotor Reflection Target Installation at Tail Rotor Hub (Sheet 1 of 1)



ICN-412-A-181201-B-97499-00002-A-001-01

Figure 2. Tail Rotor Reflection Target Installation at Tail Rotor Tips (Sheet 1 of 1)



ICN-412-A-181201-B-97499-00003-A-001-01

Figure 3. Tail Rotor Reflection Target Images for “In Track” and “Out of Track” Condition (Sheet 1 of 1)

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Applicable to: ALL
ECCN EAR99

Tail Rotor - Balancing

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Chadwick-Helmuth Vibrex Kit	Model 177M	1	or Equivalent
Accessory Kit	6584	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Tape	C-483	AR	

Procedure



COMPLY WITH THE CHADWICK-HELMUTH OPERATIONS AND SERVICE HANDBOOK WHEN PERFORMING DYNAMIC BALANCING PROCEDURES.

1. Inspect tail rotor hub and blades for damage and proper installation.

2. Confirm that tail rotor hub and blade assembly has been statically balanced prior to starting dynamic balance procedure.

3. Install the Balancer/Phazor for use outside the helicopter, near the left side of the tail rotor. Connect Strobex to Balancer.

NOTE

Do not connect the magnetic pickup cable. It is not required when balancing the tail rotor.

4. Mount tail rotor balance accelerometer on the tail rotor gearbox as follows:

4.1. Loosen nut on tail rotor gearbox case at the 12 o'clock position.

4.2. Install accelerometer and bracket (#3382) between gearbox case and nut, then tighten nut.

5. Connect accelerometer cable to accelerometer and route the cable forward to, and around, the elevator and then to the right side of the tail rotor (approximately 15 feet (4.57 m) from rotor disk). Connect only the tail rotor accelerometer cable to Balancer/Phazor at "A". Secure cable to tailboom with masking tape.

6. Place a 3/4 inch (19.05 mm) strip of tape (C-483) on one blade grip between blade bolts (Figure 1).



BEFORE PROCEEDING, ENSURE ALL CABLE SLACK HAS BEEN REMOVED BY TAPING CABLE TO TAILBOOM AND ENSURING ADEQUATE CLEARANCE EXISTS BETWEEN CABLE AND ROTATING COMPONENTS.

7. Operate helicopter (BHT-412-FM) at 100% N_R at flat pitch on the ground. Stand approximately 15 feet (4.57 m) from right side of tail rotor. Set Balancer/Phazor function switch to 'A', RPM range switch to 'x 10', and tune filter to 166.

8. Facing tail rotor disk, shine Strobex at rotor and locate reflective target (Figure 2). Refer to operational manual to fine tune Balancer/Phazor and thus "stop" reflective image from rotating.

9. Approximate clock position of image and note ips level.

NOTE

Do not read ips level when Strobex is flashing because it changes the ips level.

10. Plot and record ips level and position on tail rotor balance chart (4685). Shut down helicopter.

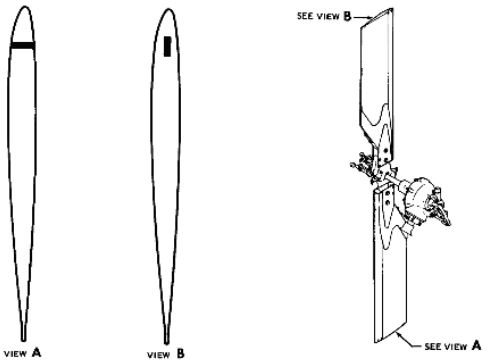
11. Determine weight, if any, and location required to balance rotor from information recorded in Step 10.. Determine correct hardware required to achieve a balanced rotor from Figure 3.

12. Install weights as illustrated on Figure 3. Comply with applicable sections of notes 1, 2, and 3 on illustration.

13. Repeat Step 8. through Step 12. until an ips level of 0.2 ips or less is achieved.

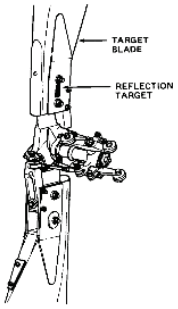
14. Remove accelerometer and bracket from tail rotor gearbox. Torque nut on gearbox 50 to 70 inch-pounds (5.65 to 7.91 N-m).

15. Remove Balancer and cables.



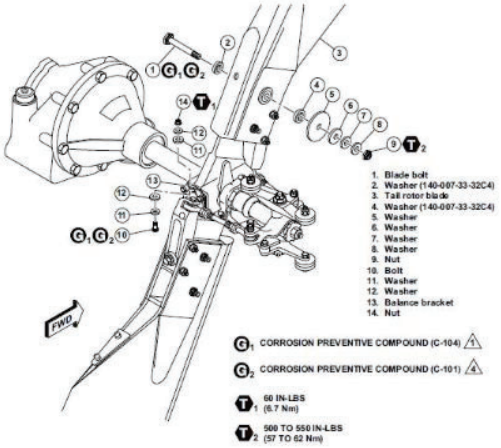
ICN-412-A-181201-B-97499-00002-A-001-01

Figure 1. Tail Rotor Reflection Target Installation at Tail Rotor Tips (Sheet 1 of 1)



ICN-412-A-181201-B-97499-00001-A-001-01

Figure 2. Tail Rotor Reflection Target Installation at Tail Rotor Hub (Sheet 1 of 1)



1. Blade bolt
2. Washer (140-007-33-32C4)
3. Tail rotor blade
4. Washer (140-007-33-32C4)
5. Washer
6. Washer
7. Washer
8. Washer
9. Nut
10. Bolt
11. Washer
12. Washer
13. Balance bracket
14. Nut

G_1 CORROSION PREVENTIVE COMPOUND (C-104) ⚠1

G_2 CORROSION PREVENTIVE COMPOUND (C-101) ⚠4

T_1 60 IN-LBS (6.7 Nm)

T_2 500 TO 550 IN-LBS (57 TO 62 Nm)

NOTES

⚠1 Apply a coating of corrosion preventive compound (C-104) to the shanks of the bolts (1 and 10) prior to installation. Do not apply corrosion preventive compound to bolt threads.

⚠2 To accomplish spanwise dynamic balance, use blade bolts (1) as required to accommodate balance washers (5, 6, 7 or 8). Bolts (1) may be installed with head, inboard or outboard, but all four blade bolts must be installed the same. When change of balance washers and/or bolts is required, make sure that one washer (2) and one washer (4) are installed next to the blade as illustrated. Then install the heaviest balance washer next to the washer (4). Install the required combination of washers (5, 6, 7, or 8) on blade (3) or on opposite blade as required to obtain balance. After installation, make sure proper thread engagement of two complete threads is shown on bolt.

BLADE BOLT SPANWISE BALANCE HARDWARE ⚠2

ITEM NUMBER	NOMENCLATURE	PART NUMBER	WEIGHT (GRAMS)
1	BLADE BOLT	NAS1306-34	87.5
1	BLADE BOLT	20-957-4-34	84.8
1	BLADE BOLT	NAS1306-36	99.7
1	BLADE BOLT	20-957-4-36	89.0
5	WASHER	AN970-8	37.7
6	WASHER	212-010-744001	11.3
7	WASHER	AN960-616	3.2
7	WASHER	NAS1149F0663P	5.1
8	WASHER	AN960-616L	1.6
8	WASHER	NAS1149F0632P	2.4
9	NUT	NAS1022A8	8.9
9	NUT	MS21245-L8	9.9

⚠3 To accomplish chordwise dynamic balance on balance brackets (13), use bolts (10) as required to accommodate the required combination of balance washers (11 and 12). Use a maximum of two washers (12) and a maximum of ten washers (11) on one bracket. Install washers (11 and 12) symmetrically. Install the required hardware on balance bracket (13) or opposite bracket (not illustrated), as required to obtain balance. It is also acceptable to install balance hardware on both balance brackets to make moves smaller than can be made on one bracket only due to the weight of the bolt.

CHORDWISE BALANCE BRACKET HARDWARE ⚠3

ITEM NUMBER	NOMENCLATURE	PART NUMBER	WEIGHT (GRAMS)
10	BOLT	NAS1304-1	5.4
10	BOLT	NAS6604-1	5.2
10	BOLT	NAS1304-2	5.8
10	BOLT	NAS6604-2	5.6
10	BOLT	NAS1304-3	6.2
10	BOLT	NAS6604-3	5.9
10	BOLT	NAS1304-4	6.5
10	BOLT	NAS6604-4	6.3
10	BOLT	NAS1304-5	7.3
10	BOLT	NAS6604-5	7.1
10	BOLT	NAS1304-6	8.0
10	BOLT	NAS6604-6	7.9
10	BOLT	NAS1304-10	8.7
10	BOLT	NAS6604-10	8.9
10	BOLT	NAS1304-12	9.4
10	BOLT	NAS6604-12	9.4
10	BOLT	NAS1304-14	10.2
10	BOLT	NAS6604-14	10.2
11	WASHER	AN960-416	1.2
11	WASHER	NAS1149F0463P	1.2
12	WASHER	AN970-4	7.5
14	NUT	MS21042L4	1.6

⚠4 Apply a coating of corrosion preventive compound (C-101) to all bolt heads, washers, nuts and exposed threads after installation.

ICN-412-A-181200-B-97499-00004-A-001-01

Figure 3. Tail Rotor Dynamic Balance Weight Adjustment With Tail Rotor Blades 212-010-750 Installed (Sheet 1 of 1)

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Main Driveshaft - Vibration Control

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Accelerometers	Chadwick-Helmuth	2	
Extension Cable (50 feet (15.24 m))	Chadwick-Helmuth	1	
Strobex	Model 135M-10 or -11	1	
Balancer	Model 6A or 7A	1	
Weight, 0.50 ounce	100-106-1	AR	
Weight, 0.25 ounce	100-106-2	AR	
Weight, 0.10 ounce	100-106-3	AR	
Weight, 0.50 ounce	100-106-4	AR	*Indicates used on helicopters S/N 36020 and Subsequent and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade).*
Weight, 0.25 ounce	100-106-5	AR	*Indicates used on helicopters S/N 36020 and Subsequent and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade).*
Weight, 0.10 ounce	100-106-6	AR	*Indicates used on helicopters S/N 36020 and Subsequent and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade).*
Retro-Reflective Targets	3300	1	
Accelerometer Bracket	6752m	2	

Procedure

1. Most Model 412 helicopters will make a pulsating low frequency noise at flat pitch due to normal amount of freeplay in shaft coupling and gearbox bearings. If this pulsating noise is present in hover, shaft out-of-balance is indicated. High levels of shaft out-of-balance will be perceived as a buzz in the tail rotor pedals in addition to a pulsating noise.

2. *Balancing the main driveshaft can minimize an out-of-balance condition. During balancing procedure, 0.2 IPS or less should be the vibration level attained to maintain the vibration levels within acceptable limits during the time in-service of the shaft. This procedure is optional and not required for any component life or serviceability requirement. For helicopters For helicopters 33001 through 33213, 36001 through 36019 and S/N 36020 and Subsequent and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade), refer to* Table 1 *for balance criteria.*

2.1. Install one accelerometer bracket (6752m) on right side of combining gearbox on stud below and forward of right power section trim compensator control (Figure 1). Position accelerometer laterally with connector pointing right.

2.2. Install second accelerometer bracket (6752m) on transmission input quill at one o'clock (looking forward) stud by removing existing Allen head screw and replacing it with a 1/4 X 28 UNF bolt (Figure 2). Position accelerometer laterally with connector pointing right.

2.3. Connect combining gearbox accelerometer to Channel A and input quill accelerometer to Channel B.

2.4. Secure accelerometer cables to prevent fouling of rotating components.

2.5. *Operate helicopter (both power sections) at 100% N_R, flat pitch. Refer to applicable Flight Manual.*

2.6. Tune balancer to 6600 RPM.

2.7. Record IPS levels of Channels A and B. A vibration level of 0.2 IPS, or below, is considered acceptable.

NOTE

If, when measuring Channel A (combining gearbox accelerometer), the IPS needle is seen to oscillate significantly, the accelerometer is being saturated. To compensate, allow both power sections to warm up for 5 minutes. If, after warm up, the oscillation still exists, shut down right power section. This should cause the oscillation to cease. Retune balancer, as necessary, if single power section operation will not operate rotor at 100% N_R.

2.8. Shut down helicopter. Return helicopter with acceptable vibration levels to flight status.

2.9. If main driveshaft out-of-balance is in excess of 0.2 IPS on either end of driveshaft, remove right engine air inlet cowling.

2.10. Install retro-reflective targets (3300) on outer rim of transmission input quill adapter (Figure 3). Since only one third of the adapter can be strobed from one location above roof line, three targets are placed at 120° intervals around adapter circumference.

2.11. Using a soft lead pencil, mark 4 o'clock target with one broad stripe and 8 o'clock target with two narrow stripes (Figure 3).

2.12. The targets on forward coupling are used to obtain phase information for balancing both ends of shaft. Should alignment of bolts between engine and transmission adapters not be the same, some reclocking of engine adapter balance chart may be required.

2.13. Position work stand at side of cabin so forward coupling is sufficiently visible to Chadwick operator to strobe at least one target.

2.14. Accomplish Step 2.4. through Step 2.6.. Record IPS levels on form (Figure 4).

2.15. Using Strobex, locate target visible from operator location and record clock angle on form.

2.16. Using strobe information from form, determine clock angle of master target. Record on form.

2.17. Plot IPS level and master target clock angle on appropriate chart (Figure 5 and Figure 6).

2.18. Using data supplied on charts, locate proper bolt on adapter and add required balance weights (and bolts) as required (refer to Table 2 for weights and usage).

NOTE

SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

The combining gearbox (engine) coupling should be balanced to 0.2 IPS or below prior to balancing the transmission end coupling.

A maximum of two weights of any combination up to 1.0 ounce (28.35 g) may be installed on a single bolt.

A maximum of 1.5 ounces (42.35 g) may be added at each coupling.

Use bolt AN5-12A when adding one weight and use bolt AN5-13A when adding two weights. For helicopters (412SP to 412HP Upgrade), use NAS6604-6 bolt when adding one weight and NAS6604-8 bolt when adding two weights.

2.19. Continue balancing until both couplings are balanced to 0.2 IPS or below. Upon completion of balancing, replace any nut removed with a new nut. Torque nuts in required sequence (412-A-63-10-01-01A-720A-A / 00999).

2.20. When balancing either coupling, if clock angle moves from side to side and IPS level change does not agree with balance move made, the following causes are suspect:

2.20.1. Excessive freeplay in combining gearbox output shaft or support bearings.

2.20.2. Excessive freeplay in transmission input quill shaft or support bearings.

2.21. When balancing either coupling, if maximum allowable weight is insufficient to reduce vibration to acceptable levels, accomplish the following:

2.21.1. Remove driveshaft and measure runout of engine output adapter or transmission input quill adapter. If runout is in excess of 0.004 inch (0.10 mm), reindex adapter on gearbox output shaft.

2.21.2. If adapter runout is acceptable, reindex female driveshaft coupling with respect to male coupling by 180°.

Table 1. ***Helicopters S/N 36020 and Subsequent and Helicopters Modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade) Main Drive Balance Criteria***

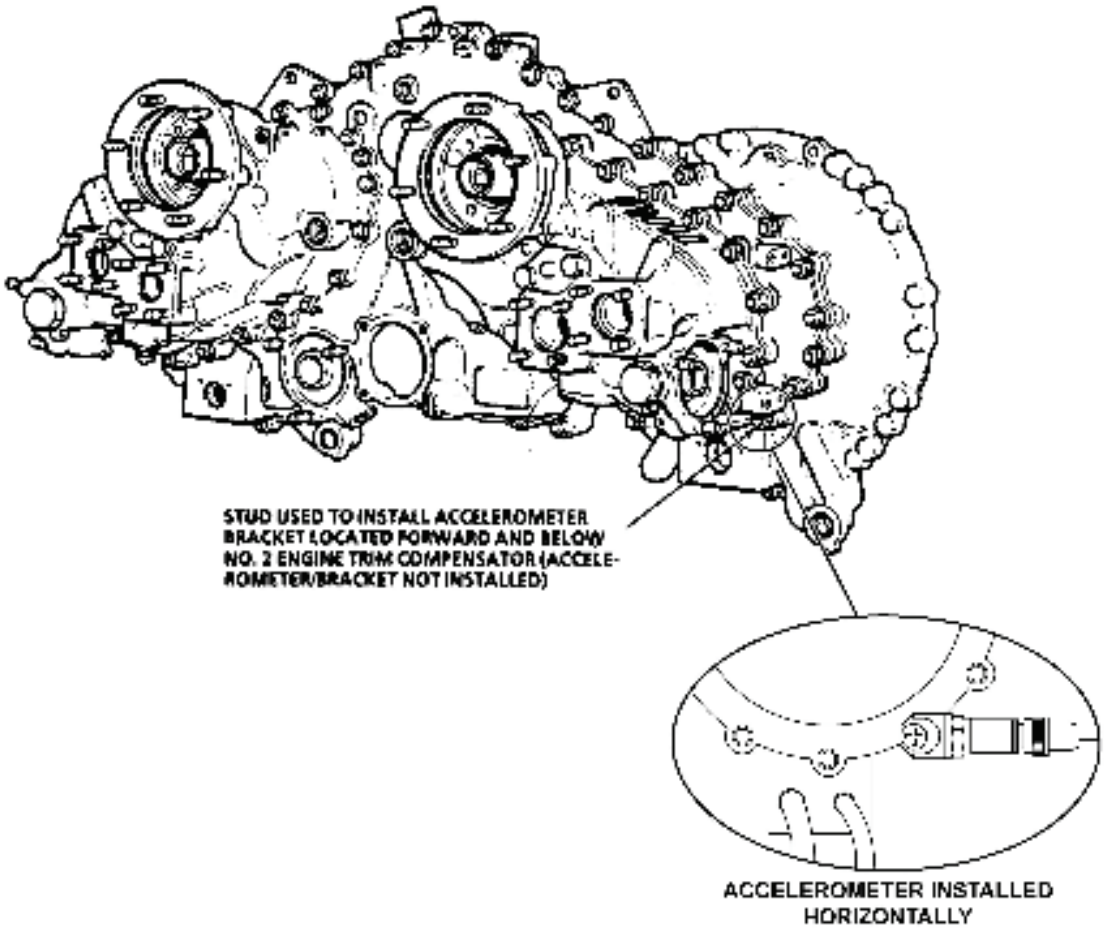
VIBRATION FREQUENCY	*OPERATION CONDITION*	*VIBRATION LEVEL*	*RECOMMENDED ACTION*
Main driveshaft 1/rev	*100% N_R on ground*	*0.2 IPS or less*	*Vibe. Equipment target value.*
		0.5 IPS or less	*No immediate action required.*
		Greater than 0.5 IPS	*Balance driveshaft.*

Table 2. Hardware and Weight

PART NUMBER	NOMENCLATURE	WEIGHT	
		GRAMS	OUNCES
AN5-12A	Bolt	2.0	0.07
AN5-13A	Bolt	4.00	0.14
weight, 0.50 ounce (100-106-1)	Balance Weight Plate	14.18	0.50
weight, 0.25 ounce (100-106-2)	Balance Weight Plate	7.09	0.25
weight, 0.10 ounce (100-106-3)	Balance Weight Plate	2.84	0.10
weight, 0.50 ounce (100-106-4) (1)	Balance Weight Plate	14.18	0.50
weight, 0.25 ounce (100-106-5) (1)	Balance Weight Plate	7.09	0.25
weight, 0.10 ounce (100-106-6) (1)	Balance Weight Plate	2.84	0.10
NAS6604-6 (1)	Bolt	2.0	0.07
NAS6604-8 (1)	Bolt	4.00	0.14

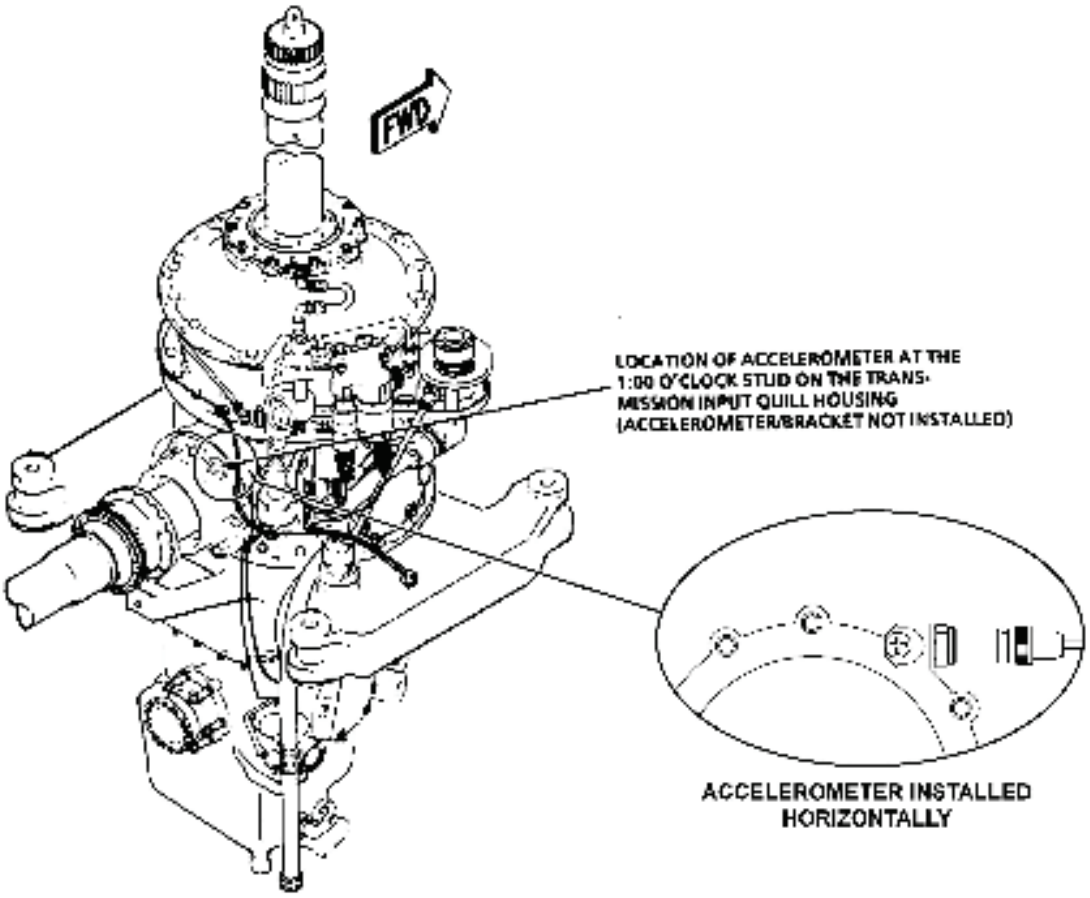
NOTE:

1 *Used on helicopters (412Classic/SP to 412HP Upgrade).*



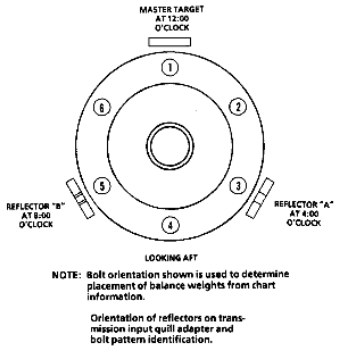
ICN-412-A-181100-B-97499-00027-A-001-01

Figure 1. Combining Gearbox Accelerometer (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00028-A-001-01

Figure 2. Transmission Input Quill Accelerometer (Sheet 1 of 1)



ICN-412-A-181100-B-97499-00029-A-001-01

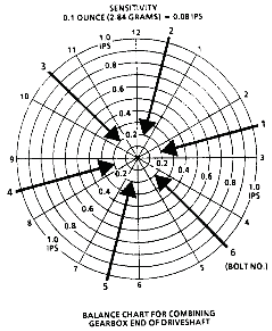
Figure 3. Reflector Positioning and Bolt Pattern Identification (Sheet 1 of 1)

LOCATION	AMPLITUDE	TARGET VISIBLE	CLOCK ANGLE OF TARGET VISIBLE	CLOCK ANGLE OF MASTER TARGET	CHANGE
TRANSMISSION					
ENGINE					
TRANSMISSION					
ENGINE					
TRANSMISSION					
ENGINE					
TRANSMISSION					
ENGINE					
TRANSMISSION					
ENGINE					

FORM FOR RECORDING BALANCE MOVES

ICN-412-A-181100-B-97499-00030-A-001-01

Figure 4. Balance Moves Recording Form (Sheet 1 of 1)



BALANCE CHART FOR COMBINING GEARBOX END OF DRIVESHAFT

EXAMPLE:

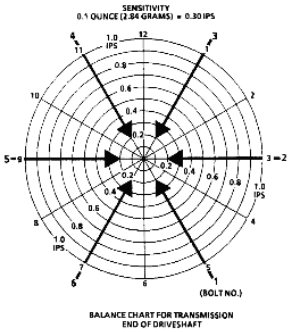
IF VIBRATION IS MEASURED TO BE 0.5 IPS WITH THE MASTER TARGET AT 8:30, ADD 0.6 OUNCES TO BOLT NO. 4

NOTES:

1. TO BALANCE AND ADD WEIGHT(S) TO INDICATED ADAPTER/COUPLING BOLTS.
2. BOLTS ORIENTED AND NUMBERED AS SHOWN IN POSITION OF REFLECTORS AND BOLT IDENTIFICATION ILLUSTRATION.
3. CLOCK ANGLE REFERS TO ORIENTATION OF MASTER TARGETS.

ICN-412-A-181100-B-97499-00031-A-001-01

Figure 5. Balance Chart - Combining Gearbox End of Engine to Transmission (Main Driveshaft) (Sheet 1 of 1)



BALANCE CHART FOR TRANSMISSION END OF DRIVESHAFT

EXAMPLE:

IF VIBRATION IS MEASURED TO BE 0.5 IPS WITH THE MASTER TARGET AT 2:00 O'CLOCK,ADD 0.3 OUNCES TO BOLT NO. 3.

NOTES:

1. TO BALANCE AND ADD WEIGHT(S) TO INDICATED ADAPTER/COUPLING BOLTS.
2. BOLTS ORIENTED AND NUMBERED AS SHOWN IN POSITION OF REFLECTORS AND BOLT IDENTIFICATION ILLUSTRATION.
3. CLOCK ANGLE REFERS TO ORIENTATION OF MASTER TARGETS.

ICN-412-A-181100-B-97499-00032-A-001-01

Figure 6. Balance Chart - Transmission End of Engine (Main Driveshaft) (Sheet 1 of 1)

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CHAPTER 21
AIR DISTRIBUTION (VENTILATION)

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Air Distribution (Ventilation) - General Description	DMC-412-A-21-20-00-00A-028A-A / 00190	2023-05-31	ALL
Bleed Air Shutoff Valve - Description	DMC-412-A-21-20-00-01A-042A-A / 00191	2023-05-31	ALL
Bleed Air Shutoff Valve - Removal	DMC-412-A-21-20-00-01A-520A-A / 00192	2023-05-31	ALL
Bleed Air Shutoff Valve - Installation	DMC-412-A-21-20-00-01A-720A-A / 00193	2023-05-31	ALL
Variable Control Mixing Valve - Description	DMC-412-A-21-20-00-02A-042A-A / 00194	2023-05-31	ALL
Variable Control Mixing Valve - Removal	DMC-412-A-21-20-00-02A-520A-A / 00195	2023-05-31	ALL
Variable Control Mixing Valve - Installation	DMC-412-A-21-20-00-02A-720A-A / 00196	2023-05-31	ALL
Remote Sensor and Positioning Motor - Description	DMC-412-A-21-20-00-03A-042A-A / 00197	2023-05-31	ALL
Remote Sensor and Positioning Motor - Removal	DMC-412-A-21-20-00-03A-520A-A / 00198	2023-05-31	ALL
Remote Sensor and Positioning Motor - Installation and Rigging	DMC-412-A-21-20-00-03A-720A-A / 00199	2023-05-31	ALL
Noise Suppressor - Description	DMC-412-A-21-20-00-04A-042A-A / 00200	2023-05-31	ALL
Noise Suppressor - Removal	DMC-412-A-21-20-00-04A-520A-A / 00201	2023-05-31	ALL
Noise Suppressor - Installation	DMC-412-A-21-20-00-04A-720A-A / 00202	2023-05-31	ALL
Housed Selector - Description	DMC-412-A-21-20-00-05A-042A-A / 00203	2023-05-31	ALL
Housed Selector - Removal	DMC-412-A-21-20-00-05A-520A-A / 00204	2023-05-31	SN: 33001-33107
Housed Selector Valve - Removal	DMC-412-A-21-20-00-05A-520B-A / 00205	2023-05-31	SN: 33108-33213 or 36001-36999
Housed Selector - Installation	DMC-412-A-21-20-00-05A-720A-A / 00206	2023-05-31	SN: 33001-33107
Housed Selector Valve - Installation	DMC-412-A-21-20-00-05A-720B-A / 00207	2023-05-31	SN: 33108-33213 or 36001-36999
Heat/Defrost Control Lever - Description	DMC-412-A-21-20-00-06A-042A-A / 00208	2023-05-31	ALL
Heat/Defrost Control Lever - Removal	DMC-412-A-21-20-00-06A-520A-A / 00209	2023-05-31	ALL
Heat/Defrost Control Lever - Installation	DMC-412-A-21-20-00-06A-720A-A / 00210	2023-05-31	ALL
Heat/Vent Air Ducts - Description	DMC-412-A-21-20-00-08A-042A-A / 00214	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Heat/Vent Air Ducts - Removal	DMC-412-A-21-20-00-08A-520A-A / 00215	2023-05-31	ALL
Heat/Vent Air Ducts - Repair	DMC-412-A-21-20-00-08A-685A-A / 00216	2023-05-31	ALL
Heat/Vent Air Ducts - Installation	DMC-412-A-21-20-00-08A-720A-A / 00217	2023-05-31	ALL
Overhead Ventilation System - Description	DMC-412-A-21-20-00-07A-042A-A / 00211	2023-05-31	ALL
Overhead Ventilation System - Removal	DMC-412-A-21-20-00-07A-520A-A / 00212	2023-05-31	ALL
Overhead Ventilation System - Installation	DMC-412-A-21-20-00-07A-720A-A / 00213	2023-05-31	ALL
Ventilation/Defog System - Description	DMC-412-A-21-20-01-00A-028A-A / 00218	2023-05-31	ALL
Ventilation Blower - Description	DMC-412-A-21-20-01-01A-042A-A / 00219	2023-05-31	ALL
Ventilation Blower - Removal	DMC-412-A-21-20-01-01A-520A-A / 00220	2023-05-31	ALL
Ventilation Blower - Installation	DMC-412-A-21-20-01-01A-720A-A / 00221	2023-05-31	ALL
Winterization Heater Kit (212-706-008) - Description	DMC-412-A-21-40-00-00A-028A-A / 00222	2023-05-31	ALL
Mixing Valve - Description	DMC-412-A-21-40-00-01A-042A-A / 00223	2023-05-31	ALL
Mixing Valve - Removal	DMC-412-A-21-40-00-01A-520A-A / 00224	2023-05-31	ALL
Mixing Valve - Cleaning	DMC-412-A-21-40-00-01A-251A-A / 00225	2023-05-31	ALL
Mixing Valve - Inspection	DMC-412-A-21-40-00-01A-280A-A / 00226	2023-05-31	ALL
Mixing Valve - Installation	DMC-412-A-21-40-00-01A-720A-A / 00227	2023-05-31	ALL
Noise Suppressor - Description	DMC-412-A-21-40-00-02A-042A-A / 00228	2023-05-31	ALL
Noise Suppressor - Removal	DMC-412-A-21-40-00-02A-520A-A / 00229	2023-05-31	ALL
Noise Suppressor - Cleaning	DMC-412-A-21-40-00-02A-251A-A / 00230	2023-05-31	ALL
Noise Suppressor - Inspection	DMC-412-A-21-40-00-02A-280A-A / 00231	2023-05-31	ALL
Noise Suppressor - Installation	DMC-412-A-21-40-00-02A-720A-A / 00232	2023-05-31	ALL
Valve assembly - Description	DMC-412-A-21-40-00-03A-042A-A / 00233	2023-05-31	ALL
Valve Assembly - Removal	DMC-412-A-21-40-00-03A-520A-A / 00234	2023-05-31	ALL
Valve Assembly - Cleaning	DMC-412-A-21-40-00-03A-251A-A / 00235	2023-05-31	ALL
Valve Assembly - Inspection	DMC-412-A-21-40-00-03A-280A-A / 00236	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Valve Assembly - Installation	DMC-412-A-21-40-00-03A-720A-A / 00237	2023-05-31	ALL
Heat Air Ducts - Description	DMC-412-A-21-40-00-04A-042A-A / 00238	2023-05-31	ALL
Heat Air Ducts - Removal	DMC-412-A-21-40-00-04A-520A-A / 00239	2023-05-31	ALL
Heat Air Ducts - Cleaning	DMC-412-A-21-40-00-04A-251A-A / 00240	2023-05-31	ALL
Heat Air Ducts - Inspection	DMC-412-A-21-40-00-04A-280A-A / 00241	2023-05-31	ALL
Heat Air Ducts - Repair	DMC-412-A-21-40-00-04A-685A-A / 00242	2023-05-31	ALL
Heat Air Ducts - Installation	DMC-412-A-21-40-00-04A-720A-A / 00243	2023-05-31	ALL

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Air Distribution (Ventilation) - General Description

The cabin air distribution system (Figure 1) includes systems that provide the use of heated air.

Heated air for the aft cabin heating is supplied by the bleed air from the engine compressors. The system pumps air through ducting to outlets at both door posts, either side of the center pedestal, each side of the instrument panel, lower forward window nozzles, and left and right windshield defrosting nozzles. Heating is controlled by switches on the overhead console, a temperature selecting dial on the right side door post, and heater defrost control lever on the right side of the center pedestal.

The winterization heater kit (212-706-008) has an additional heating duct installed in the aft area of the helicopter cabin.

The ventilating-defogging system uses ram air as a source from the intake grills on the cabin nose, with the optional use of blowers for the windshield nozzle and outlets ahead of the crew stations. Air intake is controlled by actuating a cable knob. Blowers are controlled by an overhead panel switch (Figure 2).

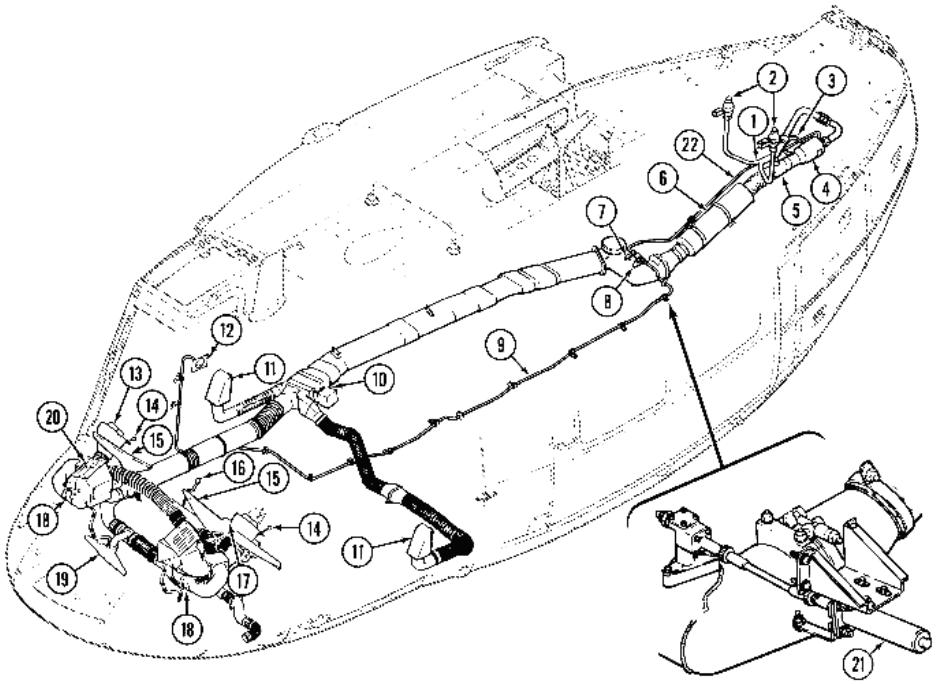
The temperature selecting dial, through an electrical positioning motor, controls a remote sensor that is mounted on the plenum chamber in the compartment on the right side below the engine deck. This remote sensor, uses a bi-metallic element, and senses the temperature in the heater duct and controls the mixing valve through a small tube that operates bleed port in the valve.

When the heater-defrost control lever on the front right pedestal is moved forward to ON, the valve in the base of the pedestal is closed by the linkage to shut off the heated air flow to four air valves in the pedestal base. Simultaneously, the lever linkage activates a micro-switch in the base and a micro-switch closes two valves on the single shaft in the air distribution outlet control valve and shuts off the heated air flow to the passenger compartment valves. When the lever is moved aft to the OFF position, the reverse actions occur to open the valves.

The engine compressor air flow noise is muffled by a noise suppressor in the rear of the fuselage forward of the mixing valve. Heated air exits from the noise suppressor into the plenum chamber aft of the cabin. A temperature limit switch in the aft cabin prevents system overheating. If the temperature in the plenum exceeds 220°F (104.4°C), the limit switch activates the MASTER CAUTION light and the HEATER AIR LINE caution segment. Cycling continues until the heater is turned off or the problem is corrected.

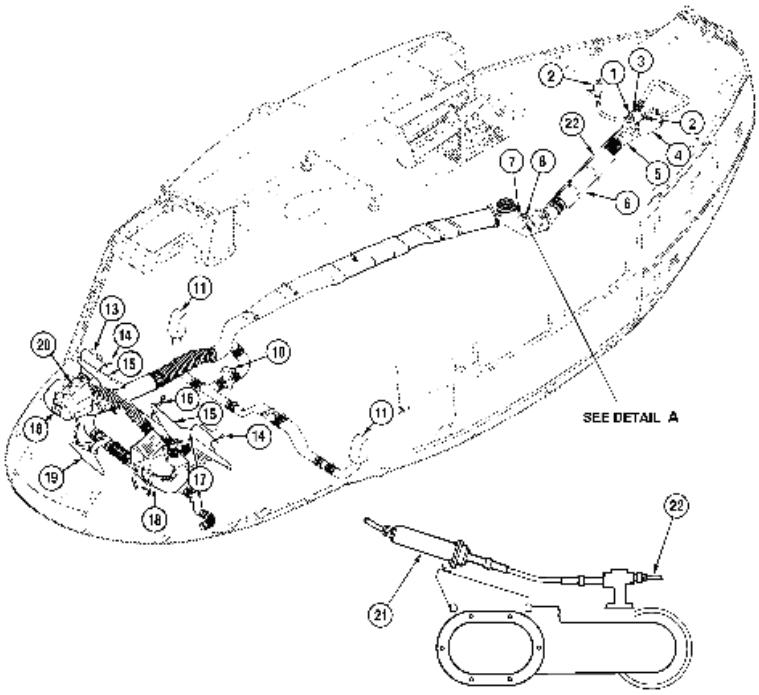
Table 1. Troubleshooting - Air Distribution (Ventilation) System

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
No bleed flow to mixing valve.	Ruptured bleed air line.	Find and repair line.
	No voltage to bleed air valve.	Refer to 412-A-96-00-00-00A-009A-A / 01870.
	Bleed air valve stuck.	Replace valve.
No bleed air flow through mixing valve.	No voltage to mixing valve.	Refer to 412-A-96-00-00-00A-009A-A / 01870.
	Inadequate air pressure.	Repair leakage in line from mixing valve to sensor.
	Inoperative mixing valve.	If bleed air and voltage exists - replace mixing valve.
Temp of heated ventilation air will not modulate.	Faulty selector dial cable.	Replace cable.
Ventilation air all hot.	Plugged line from mixing valve to sensor.	Clear line.
Ventilation air all cool.	Faulty temp selector dial cable.	Replace cable.
No air out of door post outlets.	No voltage to door post outlet valve.	Replace wiring. Refer to 412-A-96-00-00-00A-009A-A / 01870.
	Inoperative limit switch on console.	Same as above.
Unable to cut off air to door post outlets.	No voltage to door post outlet valve.	Same as above.
HEATER AIR LINES caution segment illuminated - HEATER switch ON or OFF.	Defective overheat switch or defective wiring.	Replace switch and/or repair wiring. Refer to 412-A-96-00-00-00A-009A-A / 01870.
HEATER AIR LINES caution segment illuminated when HEATER switch is ON.	Defective overheat switch.	Replace switch. Refer to 412-A-96-00-00-00A-009A-A / 01870.



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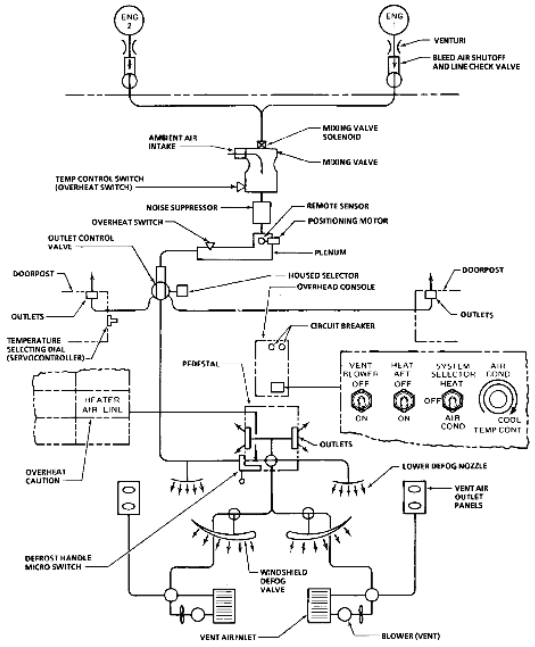
Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 2)



1. "Y" tube
2. Bleed air shut-off valves
3. Air intake
4. Variable control mixing valve
5. Overheat temperature switch
6. Noise suppressor
7. Overheat temperature switch 212-072-705-1
8. Remote sensor
9. Selecting dial to positioning motor electrical harness
10. Air distribution outlet control valve and housed selector
11. Doorpost outlet valves
12. Temperature selecting dial (servocontroller)
13. Ventilating air outlets
14. Ventilating air intake control levers
15. Windshield defroster nozzles
16. Heat-defrost control lever
17. Ram air cooling valve
18. Defogging blowers
19. Defroster nozzles
20. Ventilating air intakes
21. Positioning motor
22. Sensing tube

ICN-412-A-212000-B-97499-00002-A-001-01

Figure 1. Heating and Ventilation Air Distribution System (Sheet 2 of 2)



ICN-412-A-212000-B-97499-00003-A-001-01

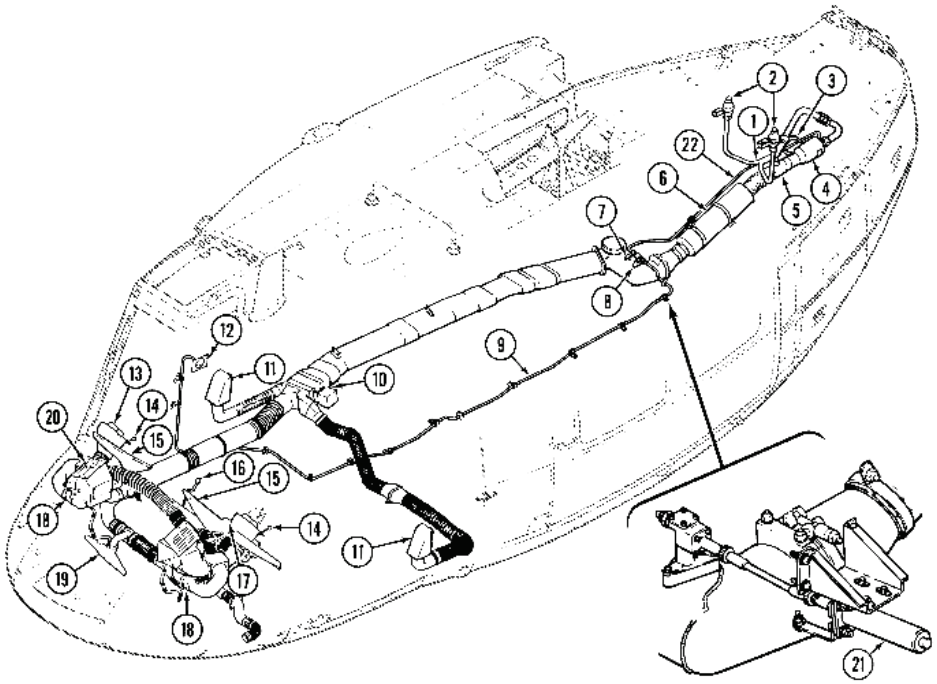
Figure 2. Heating and Ventilation Air Distribution System Schematic (Sheet 1 of 1)

Bleed Air Shutoff Valve - Description

Two bleed air shut off valves (2, Figure 1), installed on deck of engine compartment, control air flow to cabin heating system. Valves are opened by 28 VDC to actuate solenoid and are closed by air pressure.

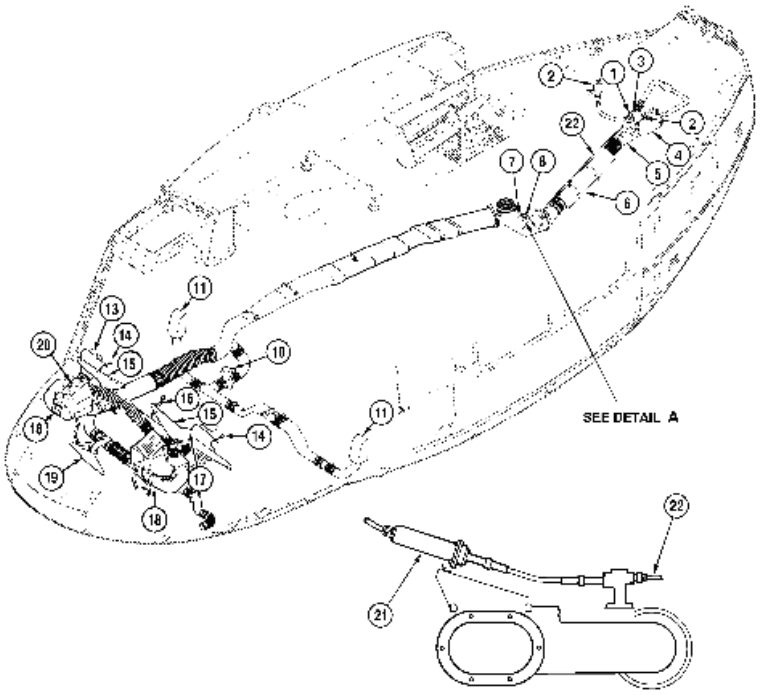
NOTE

Maintenance procedures for both bleed air shutoff valves are the same. The maintenance procedures in 412-A-21-20-00-01A-520A-A / 00192 and 412-A-21-20-00-01A-720A-A / 00193 are for bleed air shutoff valve (3, Figure 2).



ICN-412-A-212000-B-97499-00001-A-001-01

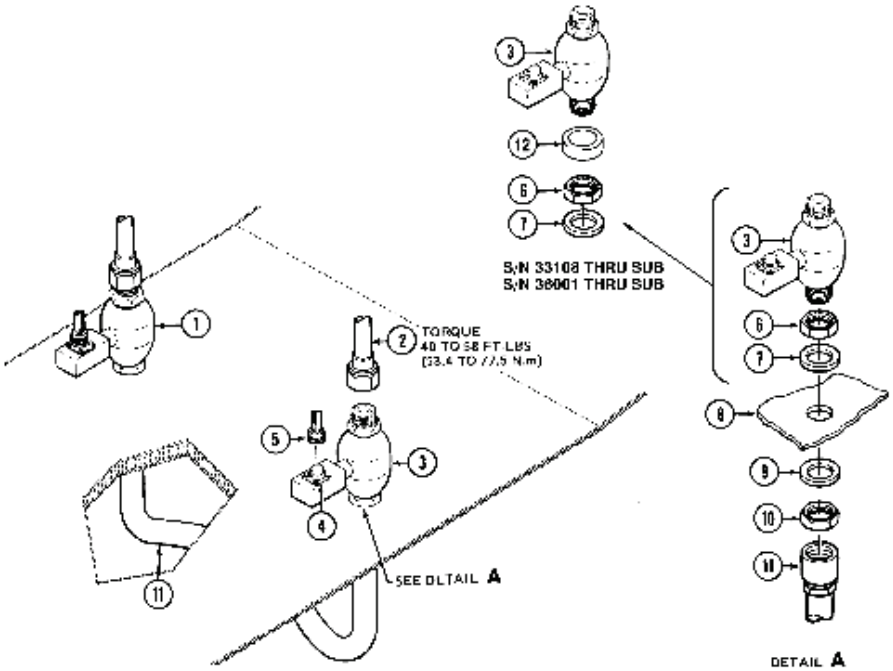
Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 2)



1. "Y" tube
2. Bleed air shut-off valves
3. Air intake
4. Variable control mixing valve
5. Overheat temperature switch
6. Noise suppressor
7. Overheat temperature switch 212-072-705-1
8. Remote sensor
9. Selecting dial to positioning motor electrical harness
10. Air distribution outlet control valve and housed selector
11. Doorpost outlet valves
12. Temperature selecting dial (servocontroller)
13. Ventilating air outlets
14. Ventilating air intake control levers
15. Windshield defroster nozzles
16. Heat-defrost control lever
17. Ram air cooling valve
18. Defogging blowers
19. Defroster nozzles
20. Ventilating air intakes
21. Positioning motor
22. Sensing tube

ICN-412-A-212000-B-97499-00002-A-001-01

Figure 1. Heating and Ventilation Air Distribution System (Sheet 2 of 2)



1. Bleed air shut-off valve (right engine power section)
2. Bleed air tube
3. Bleed air shut off valve (left engine power section)
4. Electrical receptacle
5. Electrical connector
6. Lock nut
7. Washer (AN960C2116)
8. Engine service deck
9. Washer (AN960C2116)
10. Locknut
11. "Y" tube
12. Union

ICN-412-A-212000-B-97499-00004-A-001-01

Figure 2. Bleed Air Shutoff Valve (Sheet 1 of 1)

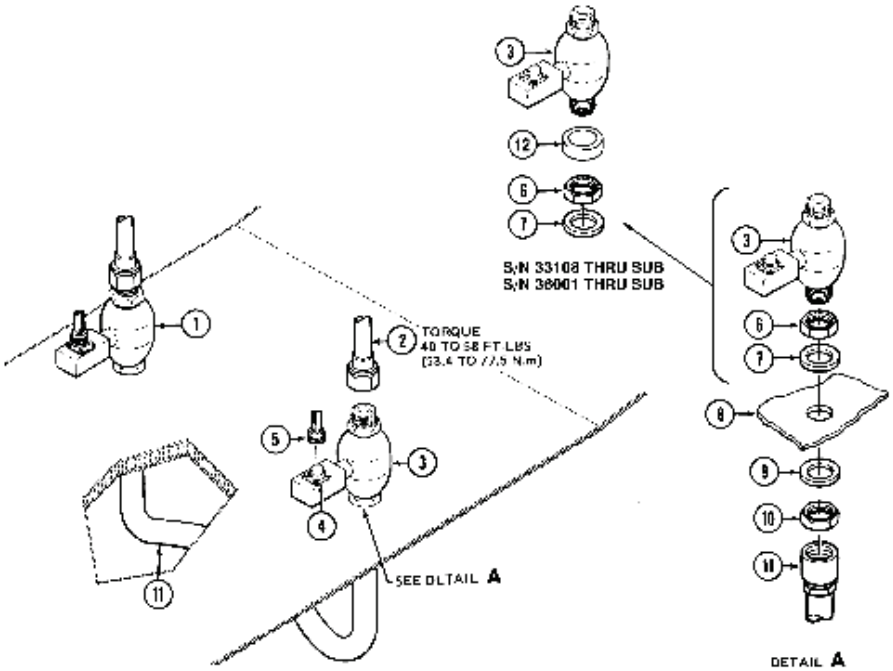
Bleed Air Shutoff Valve - Removal

Procedure

NOTE

Maintenance procedures for both bleed air shutoff valves are the same. The following removal procedures are for bleed air shutoff valve (3, Figure 1).

1. Open engine lower cowling.
2. Disconnect electrical connector (5, Figure 1) and bleed air tube (2).
3. Remove "Y" circular access door from bottom of fuselage near F.S. 211 (412-A-52-40-00-02A-520A-A / 00837).
4. Disconnect "Y" tube (11) under engine deck. Remove locknut (10) and washer (9).
5. Remove valve (3) from top of engine deck.



1. Bleed air shut-off valve (right engine power section)
2. Bleed air tube
3. Bleed air shut off valve (left engine power section)
4. Electrical receptacle
5. Electrical connector
6. Lock nut
7. Washer (AN960C2116)
8. Engine service deck
9. Washer (AN960C2116)
10. Locknut
11. "Y" tube
12. Union

ICN-412-A-212000-B-97499-00004-A-001-01

Figure 1. Bleed Air Shutoff Valve (Sheet 1 of 1)

Bleed Air Shutoff Valve - Installation

Preliminary requirements

Consumables, materials and expendables

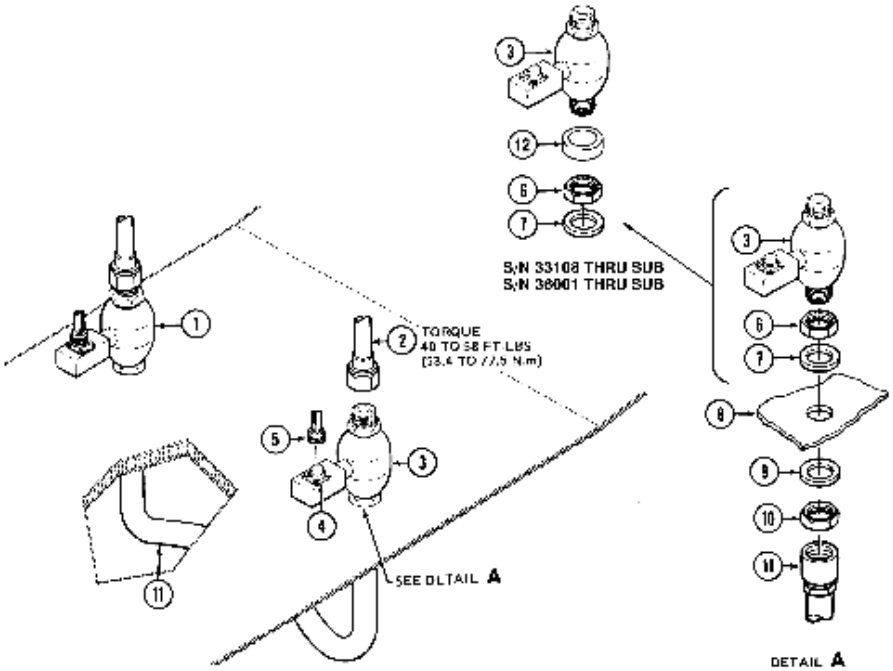
Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Maintenance procedures for both bleed air shutoff valves are the same. The following installation procedures are for bleed air shutoff valve (3, Figure 1).

1. Install locknut (6, Figure 1) on valve (3). Install washer (7) on valve (3).
2. Secure nut (6) to valve (3) with lockwire (C-405).
3. Position valve (3) on top of engine deck (8) with electrical receptacle (4) forward.
4. Insert lower end of valve (3) through engine deck (8). Install washer (9) and locknut (10).
5. Connect "Y" tube (11) to valve (3).
6. Apply lockwire (C-405) between ends of "Y" tube (11).
7. Connect tube (2) from bleed air outlet at top of engine power section to top of valve (3). Torque nut on tube (2) 40 to 58 foot-pounds (53.4 to 77.5 Nm).
8. Install electrical connector (5) to receptacle (4).



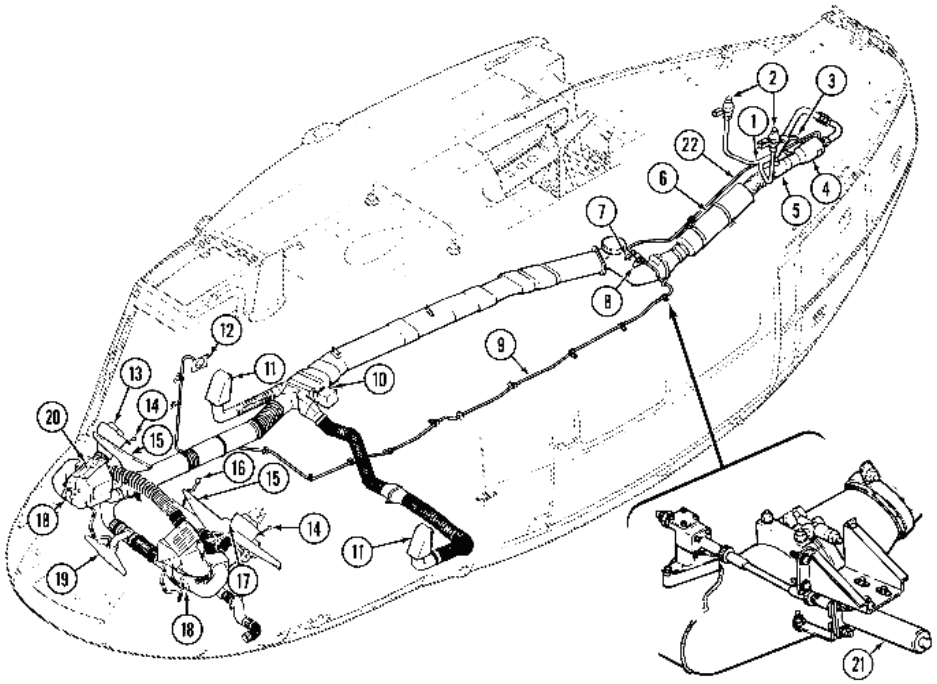
1. Bleed air shut-off valve (right engine power section)
2. Bleed air tube
3. Bleed air shut off valve (left engine power section)
4. Electrical receptacle
5. Electrical connector
6. Lock nut
7. Washer (AN960C2116)
8. Engine service deck
9. Washer (AN960C2116)
10. Locknut
11. "Y" tube
12. Union

ICN-412-A-212000-B-97499-00004-A-001-01

Figure 1. Bleed Air Shutoff Valve (Sheet 1 of 1)

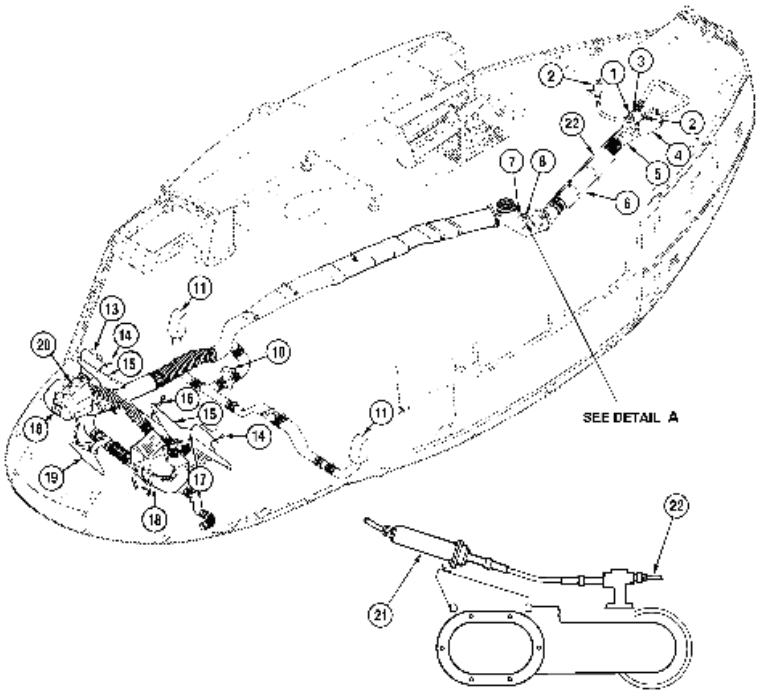
Variable Control Mixing Valve - Description

The variable control mixing valve (4, Figure 1) is installed in the compartment on right side below engine deck. This valve controls bleed air and ambient air flow as directed by the remote sensor. A bleed air nozzle inside the mixing valve acts as a jet pump through a venturi to draw in ambient air and control air flow mixture. When heated air is not at selected temperature, the remote sensor causes the variable control mixing valve to increase or decrease engine bleed air flow and regulate ambient air flow to correct the temperature variation. A temperature limit overheat switch (5) on mixing valve (4) prevents structural damage in the event of an overheat. The CABIN HTR circuit breaker will open when air in mixing valve reaches 220°F (104.4°C).



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Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 2)



1. "Y" tube
2. Bleed air shut-off valves
3. Air intake
4. Variable control mixing valve
5. Overheat temperature switch
6. Noise suppressor
7. Overheat temperature switch 212-072-705-1
8. Remote sensor
9. Selecting dial to positioning motor electrical harness
10. Air distribution outlet control valve and housed selector
11. Doorpost outlet valves
12. Temperature selecting dial (servocontroller)
13. Ventilating air outlets
14. Ventilating air intake control levers
15. Windshield defroster nozzles
16. Heat-defrost control lever
17. Ram air cooling valve
18. Defogging blowers
19. Defroster nozzles
20. Ventilating air intakes
21. Positioning motor
22. Sensing tube

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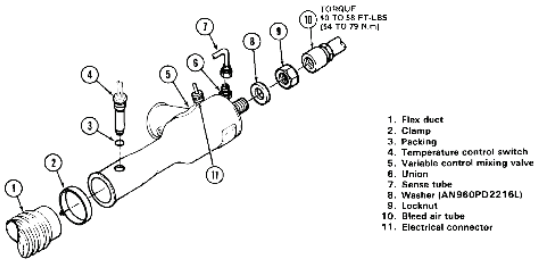
Figure 1. Heating and Ventilation Air Distribution System (Sheet 2 of 2)

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Variable Control Mixing Valve - Removal

Procedure

1. Open aft access door at right side of fuselage below engine deck.
2. Remove switch (4, Figure 1) and packing (3). Discard packing.
3. Disconnect tube (7) from valve and remove union (6).
4. Remove clamp (2) and duct (1).
5. Disconnect tube (10).
6. Disconnect electrical connector (11).
7. Remove locknut (9) and washers (8).
8. Remove mixing valve (5).



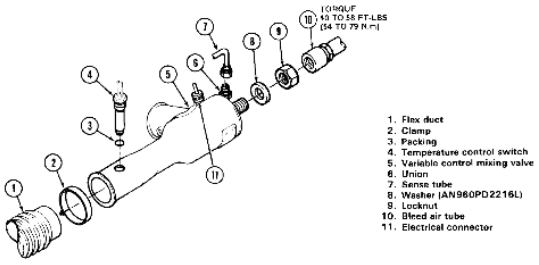
ICN-412-A-212000-B-97499-00005-A-001-01

Figure 1. Variable Control Mixing Valve (Sheet 1 of 1)

Variable Control Mixing Valve - Installation

Procedure

1. Position mixing valve (5, Figure 1) in compartment below engine deck.
2. Install washer (8) and locknut (9).
3. Install tube (10) and torque "B" nut on tube 40 to 58 foot-pounds (54 to 79 Nm).
4. Install duct (1) and clamp (2).
5. Install sense tube (7) on union (6) and install union.
6. Install packing (3) and switch (4).
7. Connect electrical connector (11).

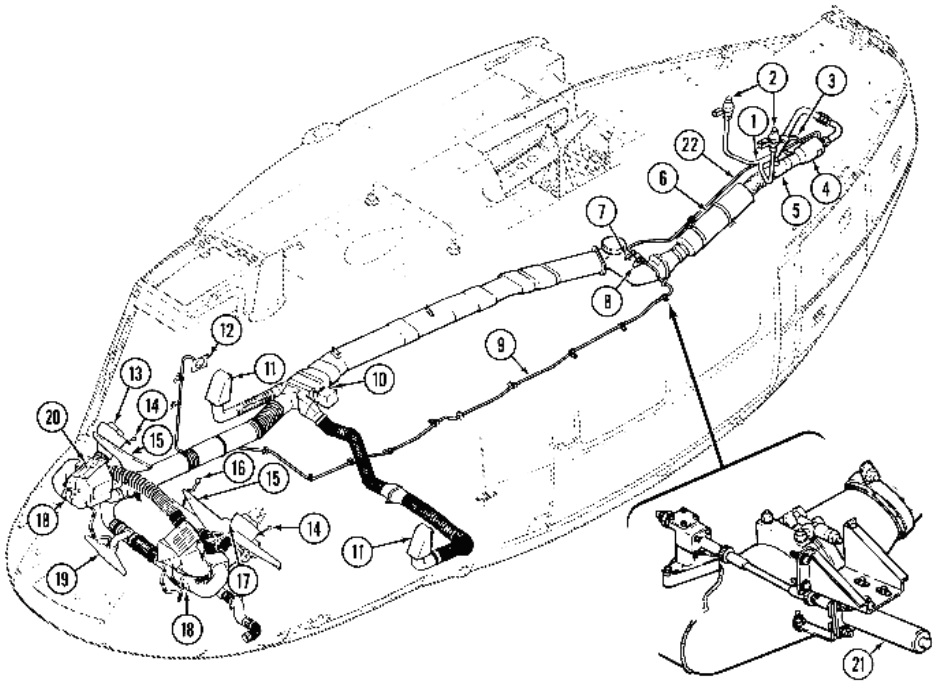


ICN-412-A-212000-B-97499-00005-A-001-01

Figure 1. Variable Control Mixing Valve (Sheet 1 of 1)

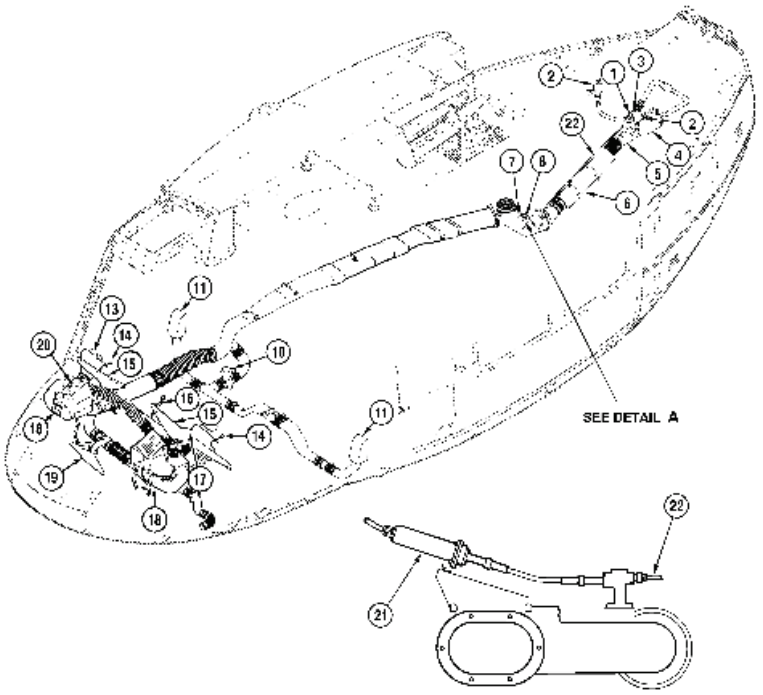
Remote Sensor and Positioning Motor - Description

The remote sensor (8, Figure 1) is installed on top of the plenum chamber. The sensor controls the mixing valve through a bleed air port. A bi-metallic element in the sensor activates the mixing valve through small diameter sensing tube (7, Figure 2) between mixing valve and sensor. The positioning motor is connected electrically to the temperature selecting dial (servo-controller).



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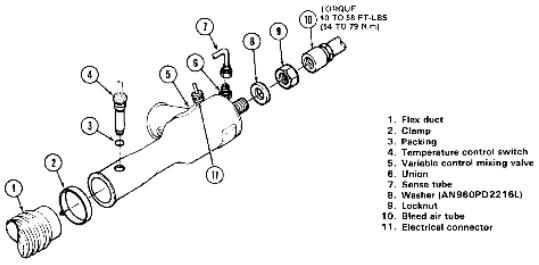
Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 2)



1. "Y" tube
2. Bleed air shut-off valves
3. Air intake
4. Variable control mixing valve
5. Overheat temperature switch
6. Noise suppressor
7. Overheat temperature switch 212-072-705-1
8. Remote sensor
9. Selecting dial to positioning motor electrical harness
10. Air distribution outlet control valve and housed selector
11. Doorpost outlet valves
12. Temperature selecting dial (servocontroller)
13. Ventilating air outlets
14. Ventilating air intake control levers
15. Windshield defroster nozzles
16. Heat-defrost control lever
17. Ram air cooling valve
18. Defogging blowers
19. Defroster nozzles
20. Ventilating air intakes
21. Positioning motor
22. Sensing tube

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Figure 1. Heating and Ventilation Air Distribution System (Sheet 2 of 2)



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Figure 2. Variable Control Mixing Valve (Sheet 1 of 1)

Remote Sensor and Positioning Motor - Removal

Procedure

1. Gain access to remote sensor (3, Figure 1) by opening forward access door on right side of fuselage below engine deck.

2. Remove remote sensor as follows:

2.1. Disconnect sensing tube (2) and flex shaft (5) from sensor (3).

2.2. Remove screws (4) through spacer (15) on plenum (1) and remove sensor.

3. Gain access to positioning motor (7) by opening forward access door on right side of fuselage below engine deck.

4. Remove positioning motor as follows:

4.1. Disconnect electrical plug (8) from positioning motor (7).

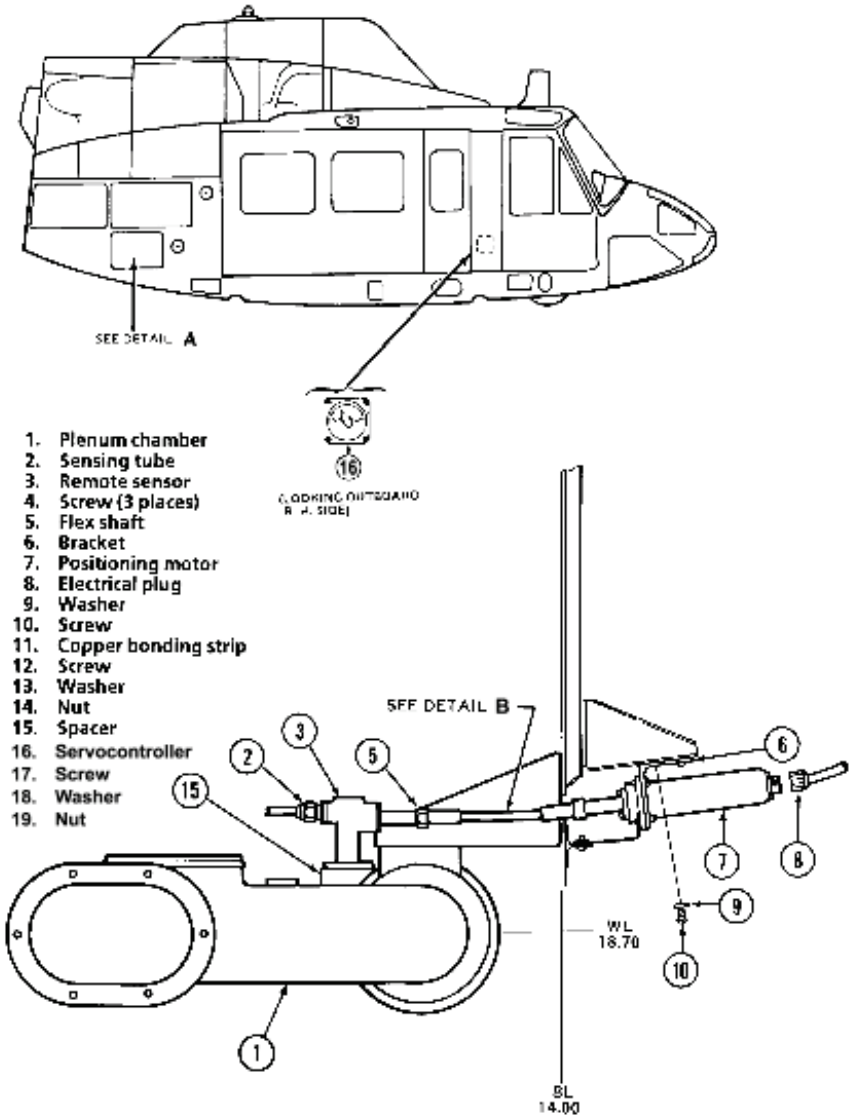
Applicable: SN: 33001-33107:

4.2. Disconnect copper bonding strip (11) by removing nut (14), washer (13), and screw (12).

4.3. Remove screws (10) and washers (9) from bracket (6) and remove positioning motor (7).

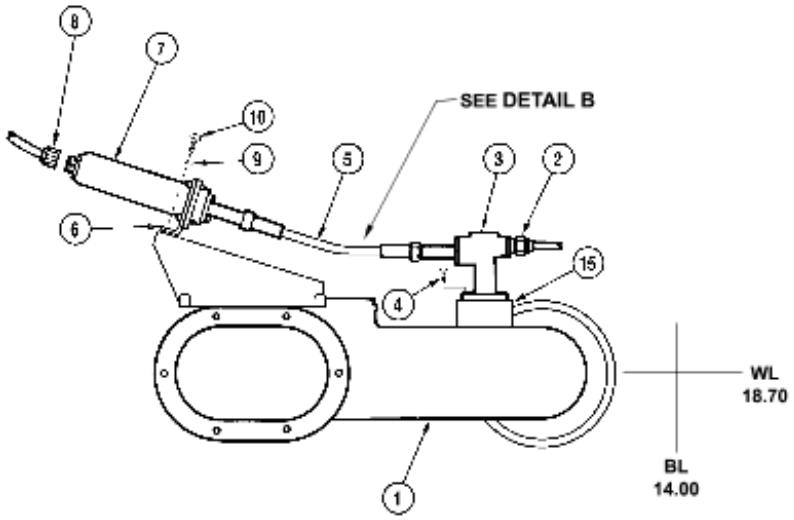
Applicable: SN: 33001-33107:

4.4. Remove copper bonding strip (11) and bracket (6) as a unit.

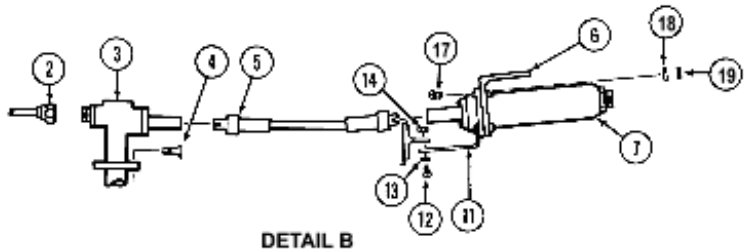


ICN-412-A-212000-B-97499-00006-A-001-01

Figure 1. Remote Sensor and Positioning Motor Installation (Sheet 1 of 2)



VIEW LOOKING AFT STA 182
S/N 33108 THROUGH SUB, 36001 THROUGH SUB
DETAIL A



DETAIL B
TYPICAL

ICN-412-A-212000-B-97499-00007-A-001-01

Figure 1. Remote Sensor and Positioning Motor Installation (Sheet 2 of 2)

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Remote Sensor and Positioning Motor - Installation and Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Seat remote sensor (3, Figure 1) onto spacer (15) on plenum chamber (1). Secure with three screws (4).

2. Connect sensing tube (2) to remote sensor (3).

3. Rotate remote sensor (3) tang counterclockwise to stop position.

4. Mount positioning motor (7) in bracket (6) with screws (17), washers (18) and nuts (19). Do not tighten screws at this time.

5. Locate bracket (6) in place and secure with washers (9) and screws (10).

Applicable: SN: 33001-33107:

6. Connect copper bonding strip (11) with screw (12), washer (13), and nut (14).

NOTE

In the following steps, the remote sensor (3) is phased with the positioning motor (7) with electrical power on (see 412-A-96-06-00-01A-320C-A / 02144).

7. Connect electrical harness to servocontroller (16). Move servocontroller knob to full COLD position.

8. Connect electrical plug (8) to positioning motor (7).

9. Close CABIN HTR circuit breaker.

10. Set HEATER switch to ON.

11. Attach flex shaft (5) between remote sensor (3) and positioning motor (7).

12. Secure positioning motor (7) in bracket (6) with screws (17), washers (18) and nuts (19).

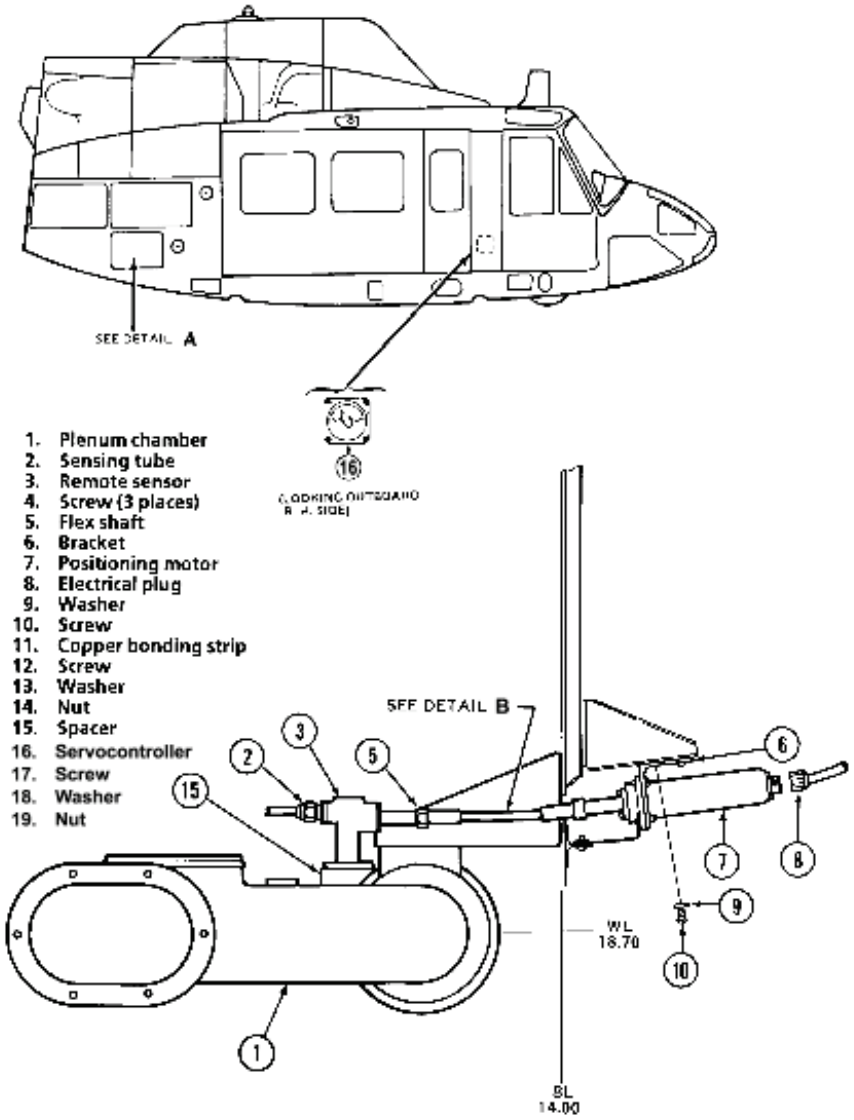
13. Secure flex shaft ends with lockwire (C-405).

14. Close access door on fuselage.

15. Set HEATER switch to OFF.

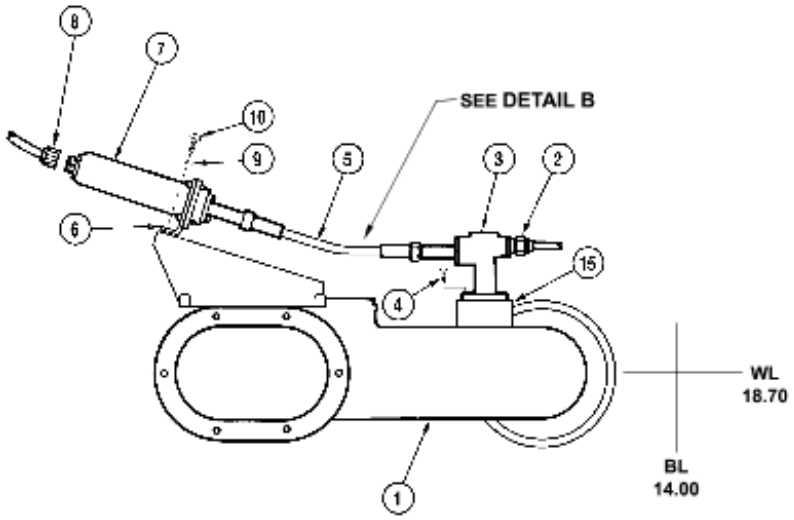
16. Turn off electrical power.

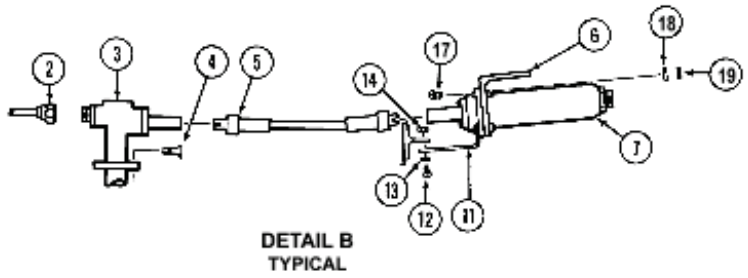
17. Refer to 412-A-96-06-00-01A-320C-A / 02144 for operational check procedure.



ICN-412-A-212000-B-97499-00006-A-001-01

Figure 1. Remote Sensor and Positioning Motor Installation (Sheet 1 of 2)





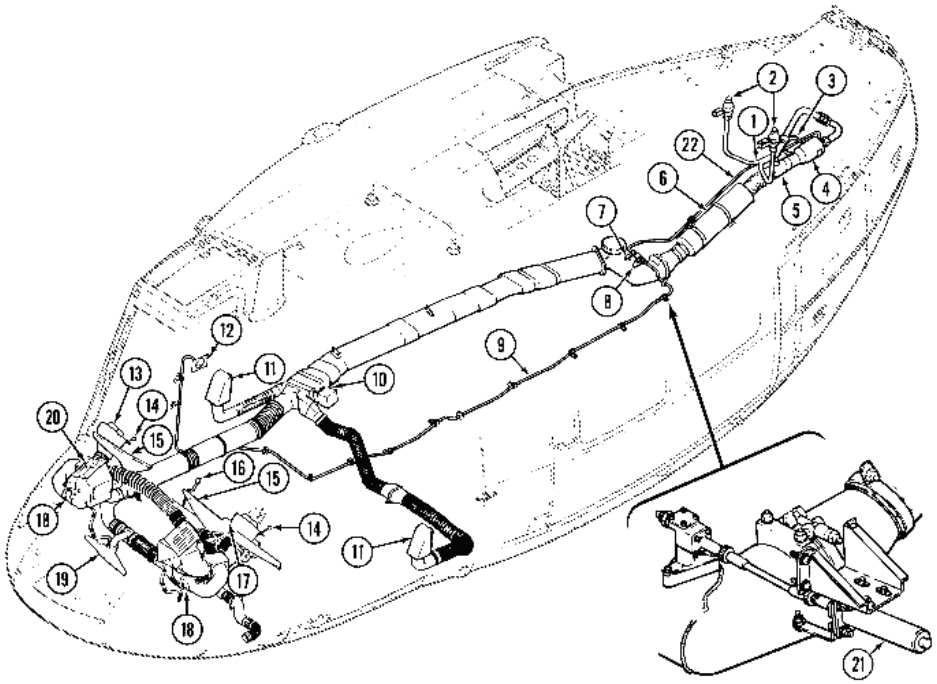
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Figure 1. Remote Sensor and Positioning Motor Installation (Sheet 2 of 2)

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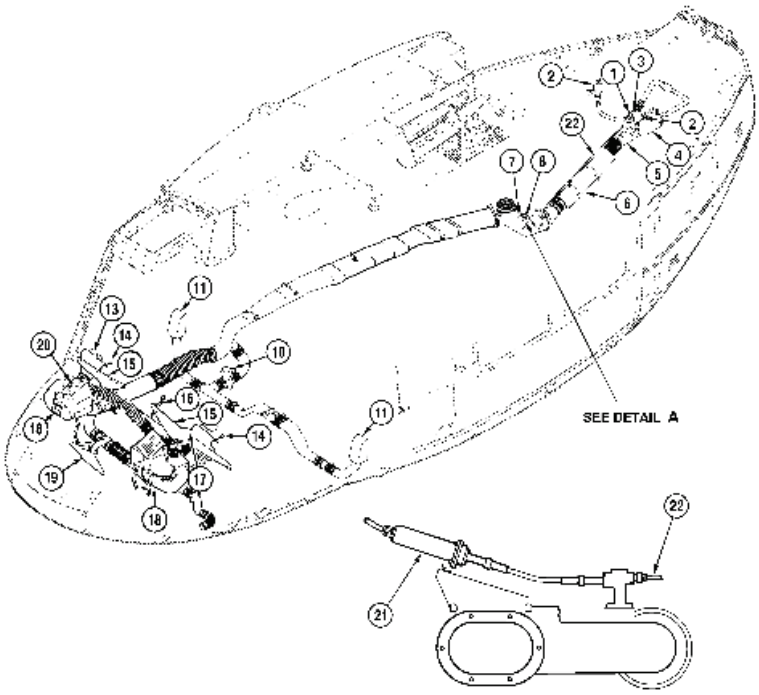
Noise Suppressor - Description

Air flow noise from the engine compressor is muffled by the noise suppressor (6, Figure 1). Heated air exits from the noise suppressor into a plenum chamber in aft cabin area. The noise suppressor is installed in the rear fuselage forward of the mixing valve.



ICN-412-A-212000-B-97499-00001-A-001-01

Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 2)



1. "Y" tube
2. Bleed air shut-off valves
3. Air intake
4. Variable control mixing valve
5. Overheat temperature switch
6. Noise suppressor
7. Overheat temperature switch 212-072-705-1
8. Remote sensor
9. Selecting dial to positioning motor electrical harness
10. Air distribution outlet control valve and housed selector
11. Doorpost outlet valves
12. Temperature selecting dial (servocontroller)
13. Ventilating air outlets
14. Ventilating air intake control levers
15. Windshield defroster nozzles
16. Heat-defrost control lever
17. Ram air cooling valve
18. Defogging blowers
19. Defroster nozzles
20. Ventilating air intakes
21. Positioning motor
22. Sensing tube

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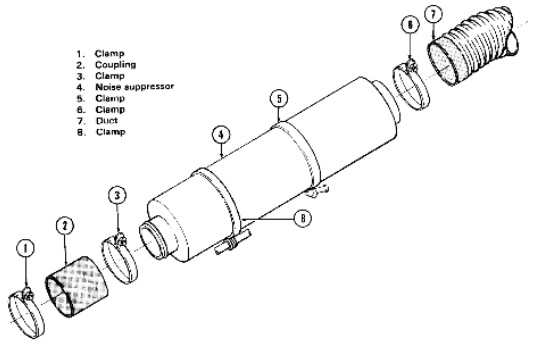
Figure 1. Heating and Ventilation Air Distribution System (Sheet 2 of 2)

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Noise Suppressor - Removal

Procedure

1. Open access door on right side of fuselage below engine deck.
2. Remove clamps (1 and 3, Figure 1). Remove coupling (2).
3. Remove clamp (6). Remove duct (7) from aft end of suppressor (4).
4. Remove clamps (5 and 8) and suppressor (4).



ICN-412-A-212000-B-97499-00008-A-001-01

Figure 1. Noise Suppressor (Sheet 1 of 1)

Noise Suppressor - Installation

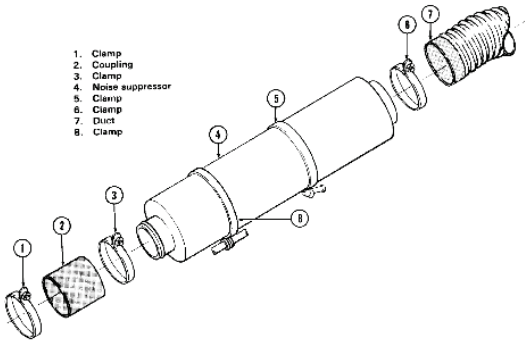
Procedure

1. Position noise suppressor (4, Figure 1) in fuselage compartment near station 211, with coupling (2) on suppressor (4) and duct (not illustrated). Install clamps (1 and 3) with screws, washers, and nuts.

2. Place clamp (5) around middle of suppressor (4) and install on bracket on forward side of station 211 bulkhead with screw, washer, and nut.

3. Place clamp (8) around middle of suppressor (4) and install clamp and sense tube with screw, washers, and nut.

4. Position duct (7) on suppressor (4). Install clamp (6) with screw, washer, and nut.



ICN-412-A-212000-B-97499-00008-A-001-01

Figure 1. Noise Suppressor (Sheet 1 of 1)

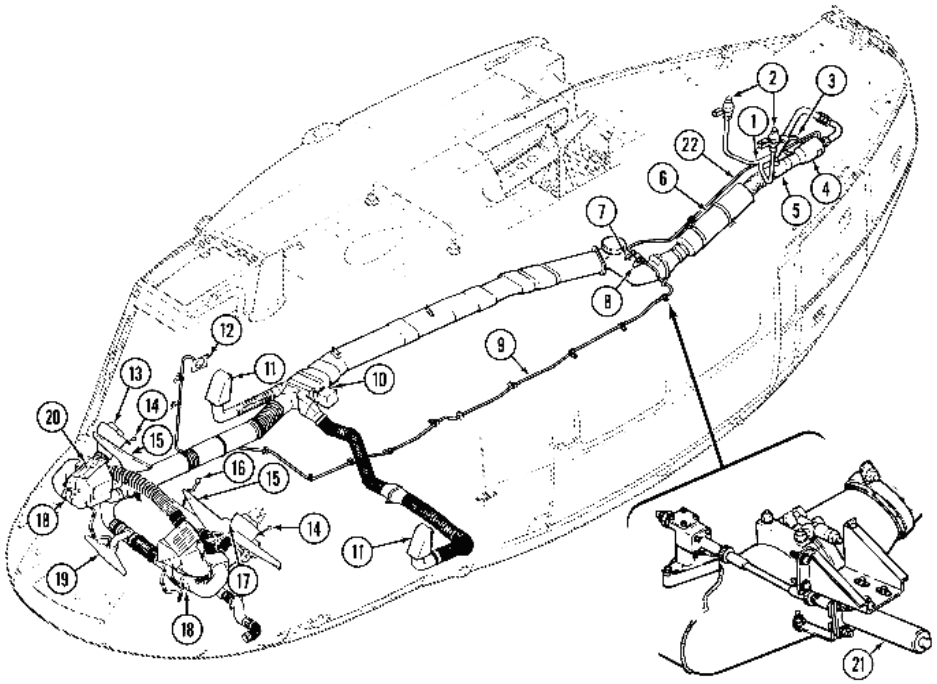
Housed Selector - Description

Applicable: SN: 33001-33107:

The housed selector (10, Figure 1) is electrically controlled to open and close two flappers in the air distribution outlet control valve. The selector and valve are interconnected by a shaft adapter.

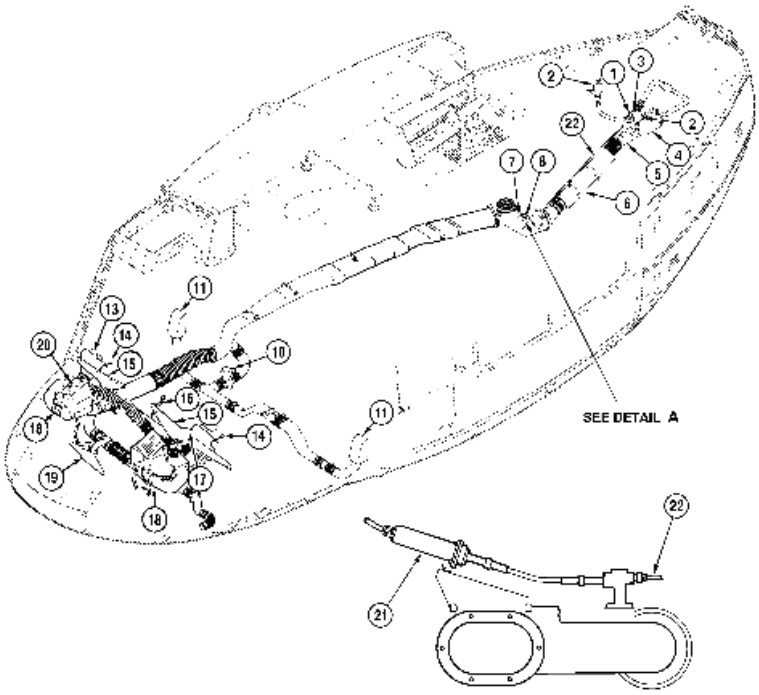
Applicable: SN: 33108-33213 or 36001-36999:

The housed selector valve (10) is electrically controlled to open and close a flapper in the air distribution housed selector valve. The housed selector, valve and flapper are built as one unit.



ICN-412-A-212000-B-97499-00001-A-001-01

Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 2)



1. "Y" tube
2. Bleed air shut-off valves
3. Air intake
4. Variable control mixing valve
5. Overheat temperature switch
6. Noise suppressor
7. Overheat temperature switch 212-072-705-1
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18. Defogging blowers
19. Defroster nozzles
20. Ventilating air intakes
21. Positioning motor
22. Sensing tube

ICN-412-A-212000-B-97499-00002-A-001-01

Figure 1. Heating and Ventilation Air Distribution System (Sheet 2 of 2)

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Housed Selector - Removal

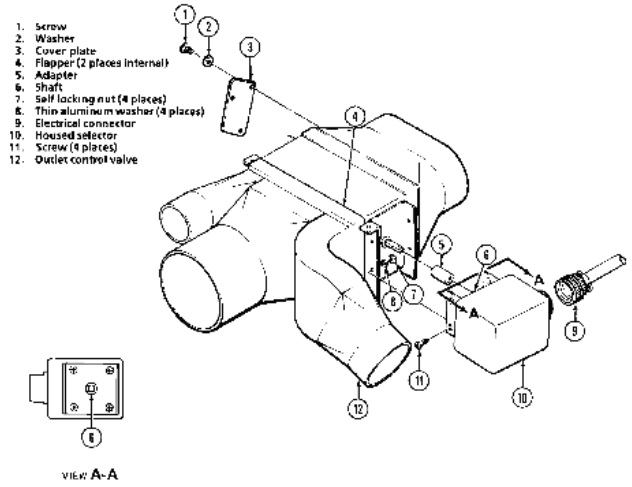
Procedure

1. Remove floor panel and access plate from lower fuselage near right B.L. 30.0, forward of F.S. 102.0.

NOTE

Ensure HEATER AFT OUTLET switch is OFF.

2. Disconnect electrical connector (9, Figure 1) from selector (10).
3. Remove four screws (11) and remove selector (10) and adapter (5).
4. Remove adapter (5) from shaft (6) of selector (10).



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Figure 1. Housed Selector (S/N 33001 through 33107) (Sheet 1 of 1)

Housed Selector Valve - Removal

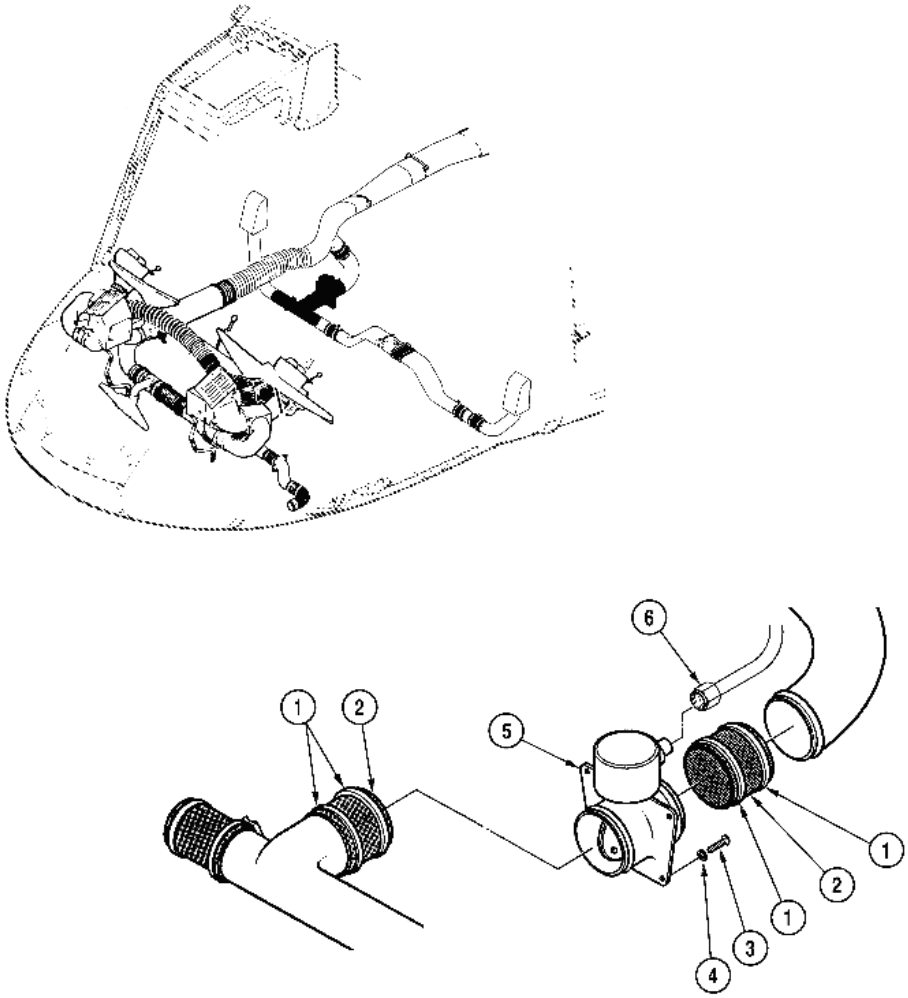
Procedure

1. Remove floor panel and access plate from lower fuselage near right B.L. 30.0, forward of F.S. 102.0.

NOTE

Ensure HEATER AFT OUTLET switch is OFF.

2. Disconnect electrical connector (6, Figure 1) from the housed selector valve (5).
3. Loosen the clamps (1) and move the shrouds (2) out of the housed selector valve (5).
4. Remove four screws (3) and washers (4) and remove the housed selector valve (5).



S/N 33108 THRU SUB, 36001 THRU SUB

1. Clamp
2. Shroud
3. Screw (4 places)
4. Washer (4 places)
5. Housed selector valve
6. Electrical connector

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Figure 1. Housed Selector Valve (S/N 33108 through 33213, 36001 through 36999) (Sheet 1 of 1)

Housed Selector - Installation

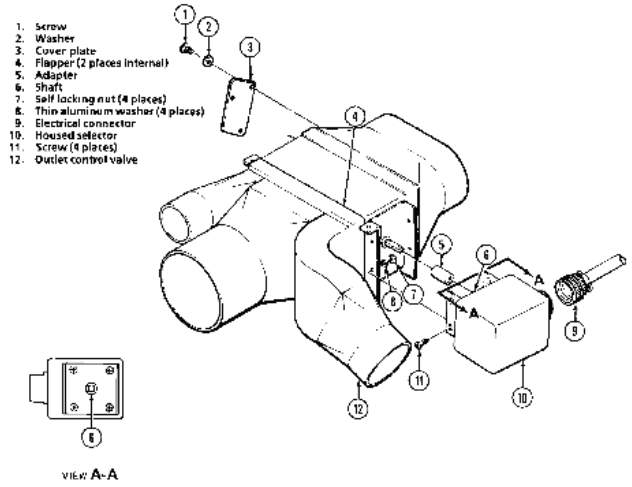
Procedure

1. Remove cover plate (3, Figure 1) so position of flapper (4) can be visually checked through opening.

NOTE

An inspection mirror will aid viewing.

2. Manually rotate flapper (4) to horizontal (open) position.
3. Position shaft (6) as shown in view A-A and install adapter (5) onto shaft (6).
4. Position housed selector (10) onto valve and secure with four screws (11), washers (8), and self-locking nuts (7).
5. Connect electrical connector (9) to selector (10).
6. Apply electrical power. Place HEATER AFT OUTLET switch on overhead panel to OFF and ensure flapper (4) rotates to vertical (closed) position. Remove electrical power.
7. Replace cover plate (3) and secure with screws (1) and washers (2).
8. Replace access plate and floor panel.



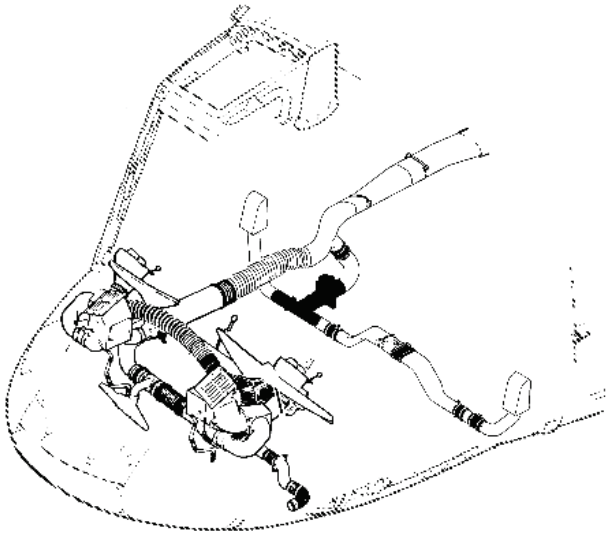
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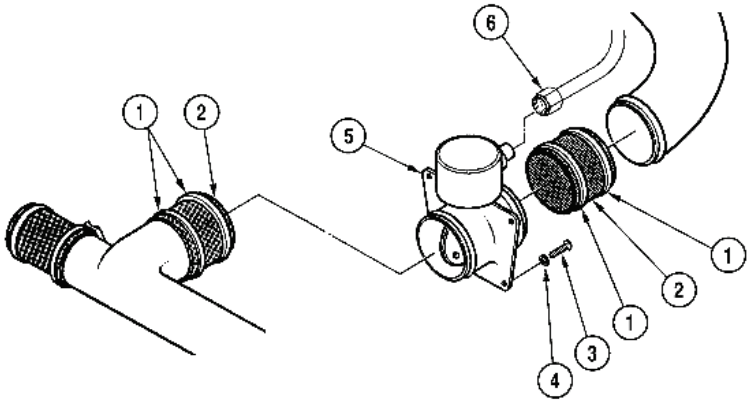
Figure 1. Housed Selector (S/N 33001 through 33107) (Sheet 1 of 1)

Housed Selector Valve - Installation

Procedure

1. Ensure position of the housed selector valve (5, Figure 1) flapper is in the horizontal (open) position.
2. Install the housed selector valve (5) with screws (3) and washer (4).
3. Connect electrical connector (6) to housed selector valve (5).
4. Apply electrical power. Place AFT OUTLET switch on overhead panel to OFF and ensure the housed selector valve (5) flapper rotates to vertical (closed) position. Remove electrical power.
5. Install shrouds (2) to housed selector valve (5) and secure with clamps (1).
6. Replace access plate and floor panel.





S/N 33108 THRU SUB, 36001 THRU SUB

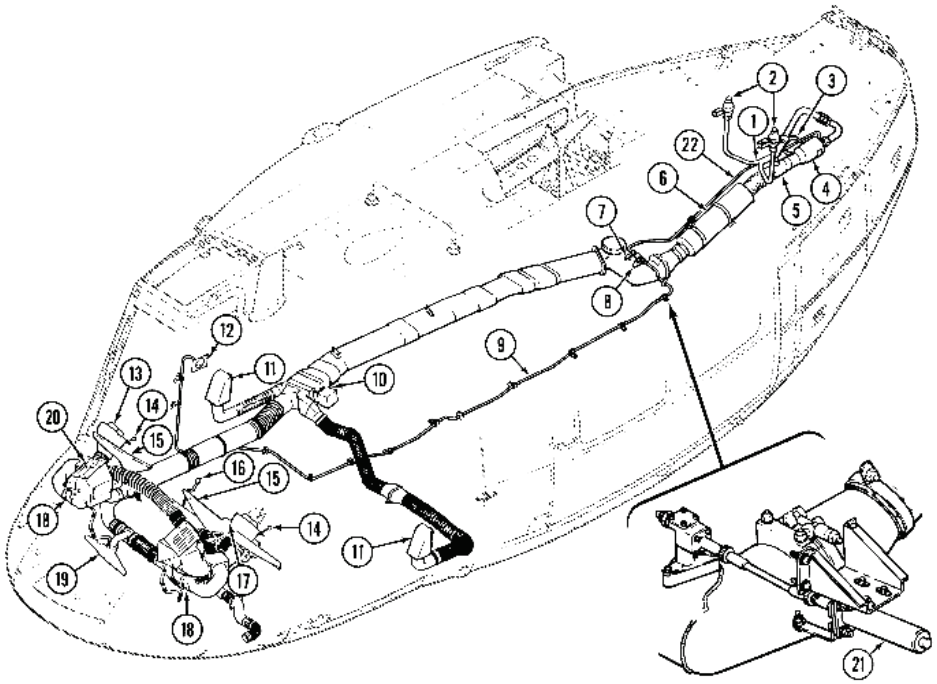
1. Clamp
2. Shroud
3. Screw (4 places)
4. Washer (4 places)
5. Housed selector valve
6. Electrical connector

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Figure 1. Housed Selector Valve (S/N 33108 through 33213, 36001 through 36999) (Sheet 1 of 1)

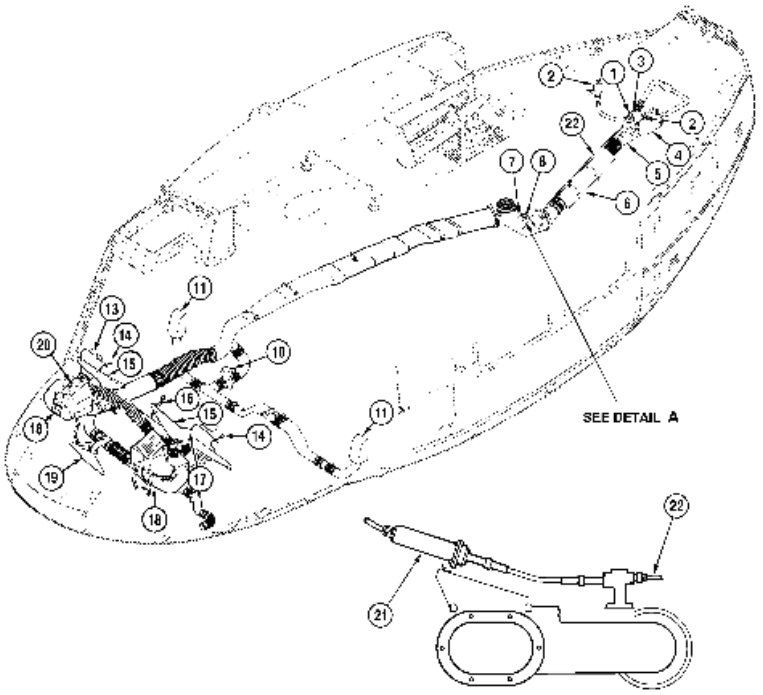
Heat/Defrost Control Lever - Description

The heat/defrost control lever (16, Figure 1) is installed on the right side of pedestal in cockpit. With lever pushed forward to ON detent, the flapper valve in the heat-defrost duct opens and directs heated air to windshield. With lever pulled aft to OFF detent, flapper valve closes and heated air is directed to crew and passengers.



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Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 2)



S/N 33108 THRU 5UB, 36001 THRU 5UB

DETAIL A

1. "Y" tube
2. Bleed air shut-off valves
3. Air intake
4. Variable control mixing valve
5. Overheat temperature switch
6. Noise suppressor
7. Overheat temperature switch 212-072-705-1
8. Remote sensor
9. Selecting dial to positioning motor electrical harness
10. Air distribution outlet control valve and housed selector
11. Doorpost outlet valves
12. Temperature selecting dial (servocontroller)
13. Ventilating air outlets
14. Ventilating air intake control levers
15. Windshield defroster nozzles
16. Heat-defrost control lever
17. Ram air cooling valve
18. Defogging blowers
19. Defroster nozzles
20. Ventilating air intakes
21. Positioning motor
22. Sensing tube

ICN-412-A-212000-B-97499-00002-A-001-01

Figure 1. Heating and Ventilation Air Distribution System (Sheet 2 of 2)

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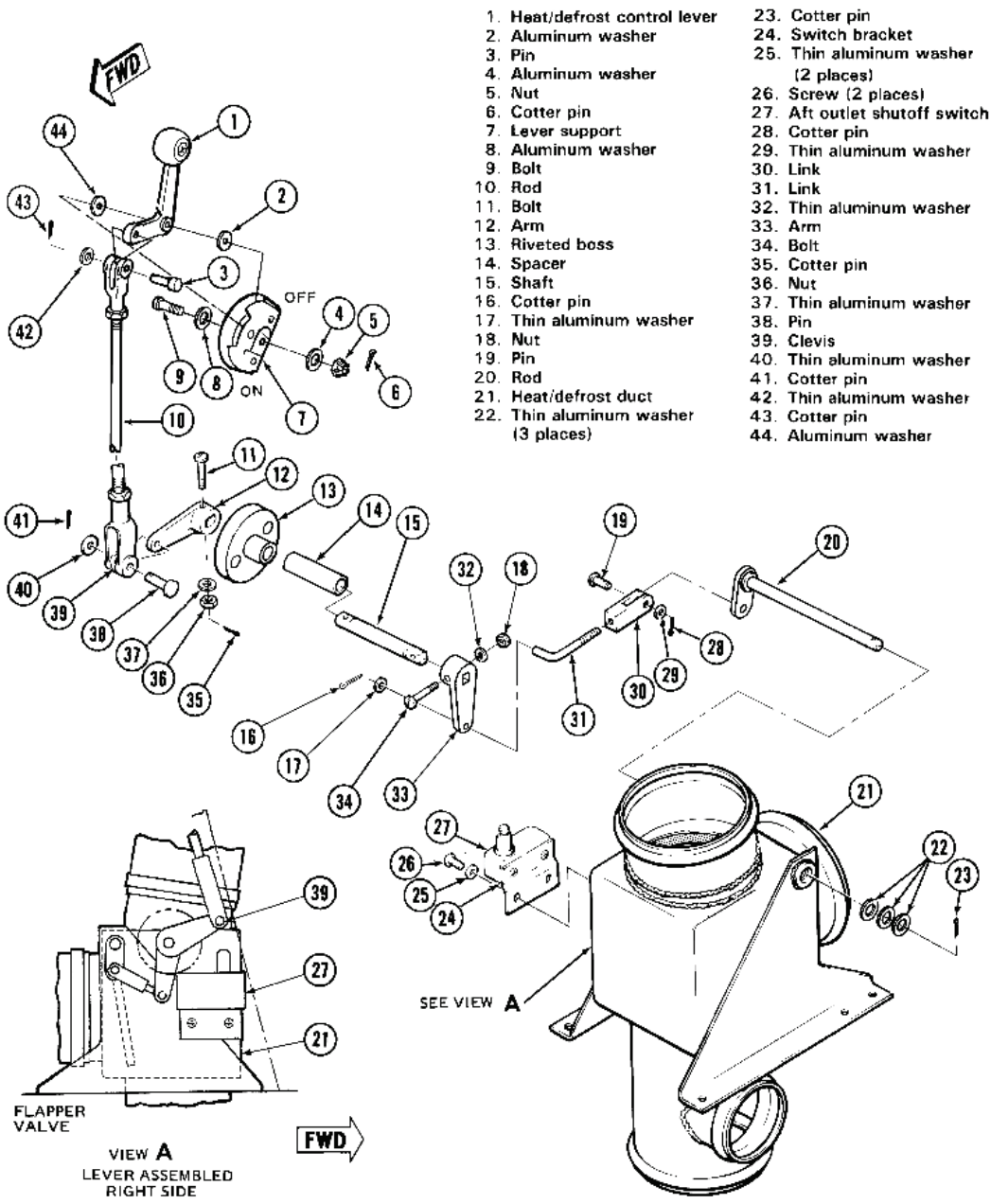
Heat/Defrost Control Lever - Removal

Procedure

1. Remove two control panels from forward right corner of center pedestal and hourmeter panel on right side of pedestal.
2. Remove bolt (9, Figure 1).
3. Remove pin (3) and lever (1). Pull lever upward through slot in support (7).
4. Remove nut (36) and bolt (11). Slide arm (12) off shaft (15) and remove with rod (10) attached.
5. Remove pin (38) and separate rod (10) from arm (12).
6. Remove two screws (26) and remove switch (27). Disconnect wiring from terminals on bottom of switch.
7. Remove pin (19). Separate link (30) from rod (20).
8. Remove bolt (34). Remove arm (33) and links (30 and 31) from shaft (15).
9. Hold spacer (14) and pull shaft (15) from outboard side of pedestal.

NOTE

Do not remove rod (20) unless flapper valve shown in View A is to be removed.



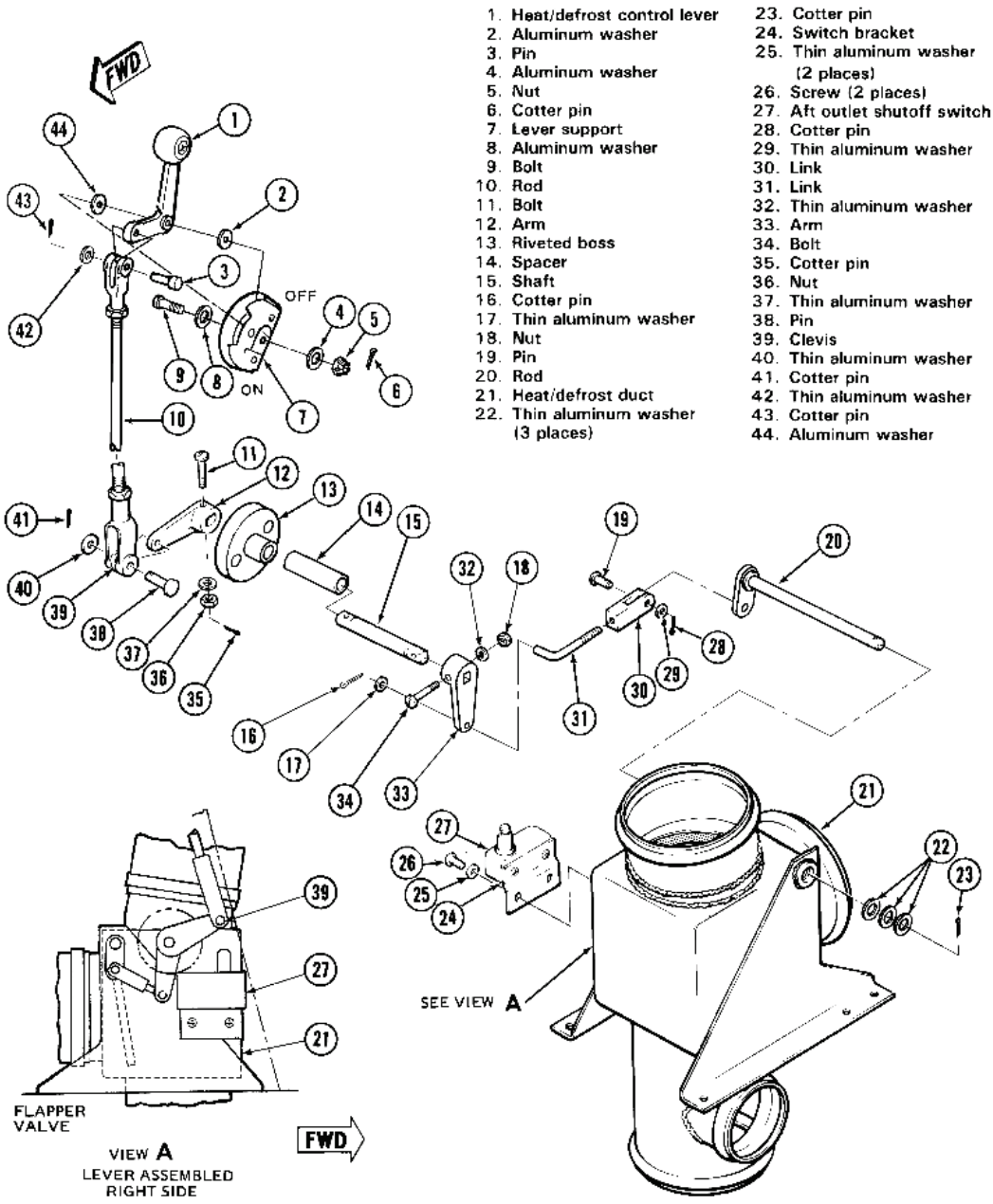
ICN-412-A-212000-B-97499-00011-A-001-01

Figure 1. Heat/Defrost Control Lever (Sheet 1 of 1)

Heat/Defrost Control Lever - Installation

Procedure

1. Insert shaft (15, Figure 1) through right side of pedestal into riveted boss (13) and install spacer (14).
2. Position arm (33) on shaft (15) and install bolt (34), washer (32), and nut (18).
3. If not previously accomplished, assemble links (30 and 31). Install link (31) on arm (33) and install washer (17) and cotter pin (16). Position assembled parts on rod (20) and install pin (19), washer (29), and cotter pin (28).
4. Attach wiring to terminals on switch (27). Position switch on duct (21) and install screws (26) and washers (25).
5. Rotate shaft (15) forward to open flapper valve to aft outlets and install arm (12), with rod (10) attached, so arm is forward.
6. Install bolt (11), washer (37), nut (36), and cotter pin (35). Ensure pin (38), washer (40), and cotter pin (41) are properly installed.
7. Position lever (1) through slot in support (7) with elbow of lever aft.
8. Install bolt (9), washers (8, 44, 2, and 4), nut (5), and cotter pin (6).
9. Push down on rod (10) until it presses switch (27) and hold in position.
10. Pivot lever (1) forward and loosen jam nuts on top and bottom ends of rod (10).
11. Adjust length of rod (10) to align with hole in lever (1). Keep exposed threads at clevis ends on rod (10).
12. Position rod (10) on lever (1) and install pin (3), washer (42), and cotter pin (43).



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Figure 1. Heat/Defrost Control Lever (Sheet 1 of 1)

Heat/Vent Air Ducts - Description

The flexible air ducts throughout the heat-vent air distribution system are made from silicone impregnated fiberglass cloth. The non-flexible air ducts are made from polycarbonate material.

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Heat/Vent Air Ducts - Removal

Procedure

NOTE

Remove individual duct sections as follows:

1. Remove tape at duct joints.

2. Remove attaching screws. Remove ducts. Cover openings to prevent entry of foreign material.

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Heat/Vent Air Ducts - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Toluene	C-306	AR	
Adhesive	C-314	AR	
Aliphatic naphtha	C-305	AR	
Glass fabric	C-404	AR	

Procedure

1. Repair flexible air ducts made from silicone-impregnated fiberglass cloth as follows:

1.1. Repair scuffs and abrasions that do not penetrate through cloth and are not more than 3.0 inches (76.2 mm) long by 1.5 inches (38.1 mm) wide. A maximum of two repairs per foot, covering up to 10% of surface area are acceptable.

1.1.1. Clean damaged area using toluene (C-306) or aliphatic naphtha (C-305). Allow area to air dry for at least 30 minutes.

1.1.2. Apply brush coat of adhesive (C-314) over damaged area, with minimum of 0.50 inch (12.7 mm) overlap beyond damage.

1.1.3. Smooth adhesive and cure by air drying for minimum of two hours at room temperature or until dry to touch.

1.2. Repair damage which penetrates fiberglass cloth as follows:

NOTE

Repairs covering up to 10% of surface area are acceptable.

1.2.1. Clean circumference of duct damage with toluene (C-306) or aliphatic naphtha (C-305). Allow area to air dry for at least 30 minutes.

1.2.2. Cut one piece of glass fabric (C-404) to cover duct circumference with at least 1.0 inch (25.4 mm) overlap of damaged area.

1.2.3. Apply one thin smooth brush coat of adhesive (C-314) around duct circumference in area to be covered by cloth.

1.2.4. Wrap cloth around duct and smooth out.

2. Allow area to air dry at least two hours or until dry to touch, before using or handling duct.

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Heat/Vent Air Ducts - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-300	AR	
Adhesive tape	C-403	AR	
Abrasive cloth or paper	C-423	AR	
Tape	C-439	AR	

Procedure

1. Position duct in its proper location as shown on Figure 1.

2. Install attaching screws.

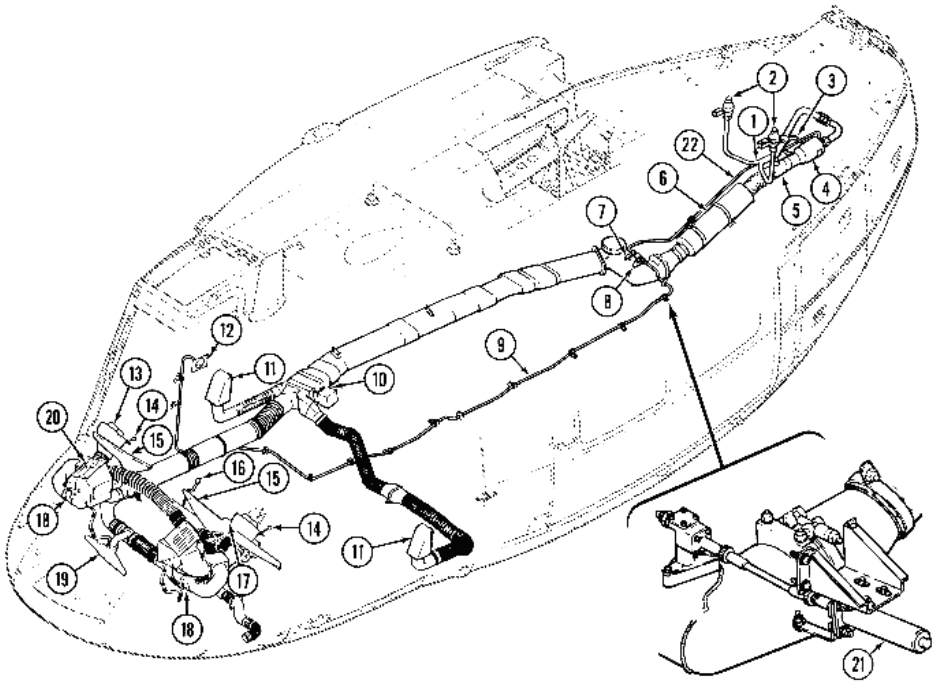
3. After joining ducts in heat/vent system to other ducts or to plenums or valves, wrap joints between units as follows:

3.1. Prepare surface of duct about 1.0 inch (25.4 mm) from outside edge by roughing up surface with 400 grit abrasive cloth or paper (C-423). Clean up using dry, pressurized air or clean dry cloths.

3.2. Brush 0.010 to 0.015 inch (0.254 to 0.381 mm) coating of adhesive (C-300) on outside surface of both ducts about 1.0 inch (25.4 mm) from their outside edges. Allow adhesive to air dry about 30 minutes, but not more than one hour.

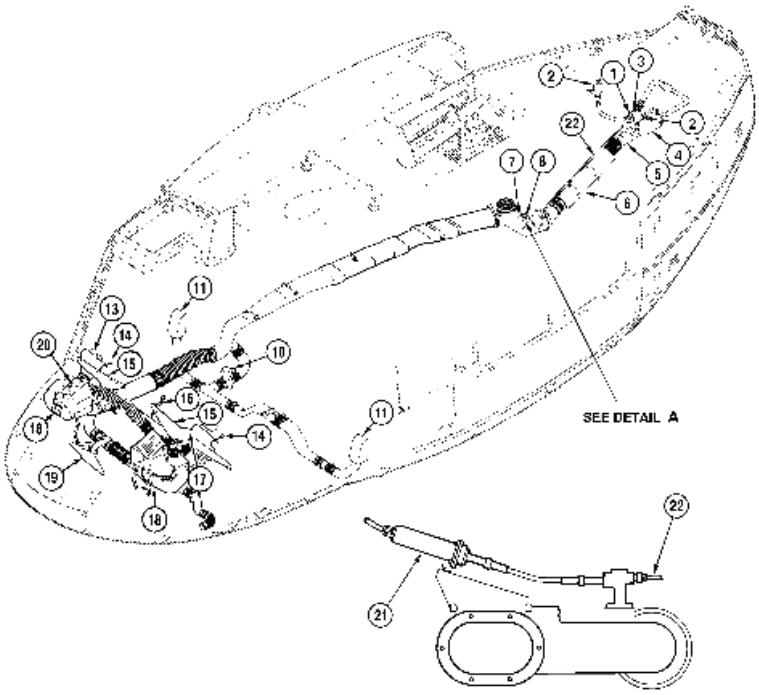
3.3. Wrap joint with three full turns of 2.0 inches (50.8 mm) wide adhesive tape (C-403). Do not stretch last two turns of tape.

3.4. Secure tape with two full turns of 2.0 inches (50.8 mm) tape (C-439). Tape edges of tape applied at preceding step together and tape ends of the tape to ducts.



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Figure 1. Heating and Ventilation Air Distribution System (Sheet 1 of 2)



S/N 33108 THRU 5UB, 36001 THRU 5UB

DETAIL A

1. "Y" tube
2. Bleed air shut-off valves
3. Air intake
4. Variable control mixing valve
5. Overheat temperature switch
6. Noise suppressor
7. Overheat temperature switch 212-072-705-1
8. Remote sensor
9. Selecting dial to positioning motor electrical harness
10. Air distribution outlet control valve and housed selector
11. Doorpost outlet valves
12. Temperature selecting dial (servocontroller)
13. Ventilating air outlets
14. Ventilating air intake control levers
15. Windshield defroster nozzles
16. Heat-defrost control lever
17. Ram air cooling valve
18. Defogging blowers
19. Defroster nozzles
20. Ventilating air intakes
21. Positioning motor
22. Sensing tube

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Figure 1. Heating and Ventilation Air Distribution System (Sheet 2 of 2)

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Overhead Ventilation System - Description

Overhead ventilation system takes ram air from screened opening in forward end of pylon fairing and directs it through air valves mounted in ceiling. Two air valves are located in crew compartment, two in forward center passenger compartment, and five along sides of passenger compartment.

NOTE

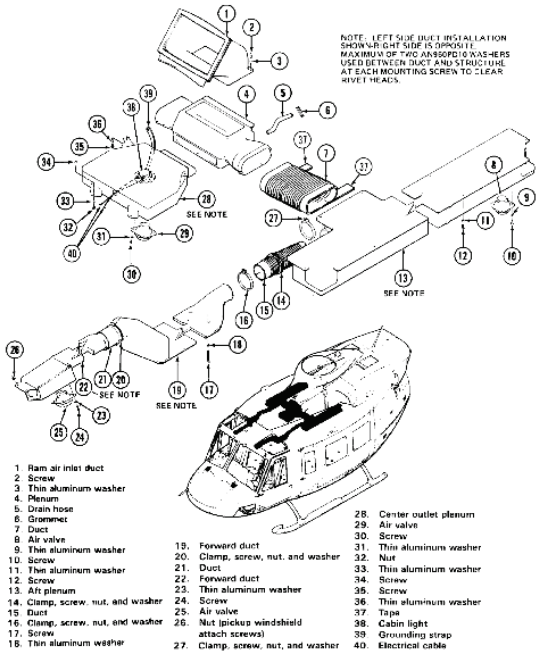
The maintenance procedures for ducts on left side is given. Maintenance procedures for ducts on right side are similar.

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Overhead Ventilation System - Removal

Procedure

1. Remove insulation blankets to gain access.
2. Remove clamps (20, 16, 14, and 27, Figure 1). Remove duct (15).
3. Remove ducts (7 and 15).
4. Remove two screws and washers from forward duct (19).
5. Remove ducts (19 and 21).
6. Remove nut (26) from windshield screws.
7. Remove forward duct (22).
8. Disconnect drain hose (5) from plenum (4).
9. Remove screws (2). Remove inlet duct (1) and plenum (4).
10. Remove grounding strap (39) and screws (35 and 34). Disconnect cabin light cable (40).
11. Remove plenum (28).
12. Remove screws (12) and remove plenum (13).



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Figure 1. Overhead Ventilation System (Sheet 1 of 1)

Overhead Ventilation System - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-314	AR	
Tape	C-401	AR	

Procedure

1. Position plenum (13, Figure 1) in helicopter and install screws (12) and washers (11). Install a maximum of two aluminum washers (not illustrated) between plenum (13) and structure with each screw (12) to clear rivet heads.

2. Position aft end of plenum (28). Route wires L170-22 and L180-22 from cabin light (38) to TB-7 and connect to pins. Install two screws (35) and washers (36). Install two screws (34), washers (33), and nut (32). Install grounding strap (39).

3. Position duct (1) and plenum (4) on structure and plenum (28). Install screws (2) and washers (3). Seal mating surfaces of duct (1) and plenum (4) with adhesive (C-314). Seal edges between plenum (4) and plenum (28) with tape (C-401).

4. Install drain hose (5). Secure hose to plenum with adhesive (C-314). Ensure grommet (6) is positioned to prevent chafing of hose seal.

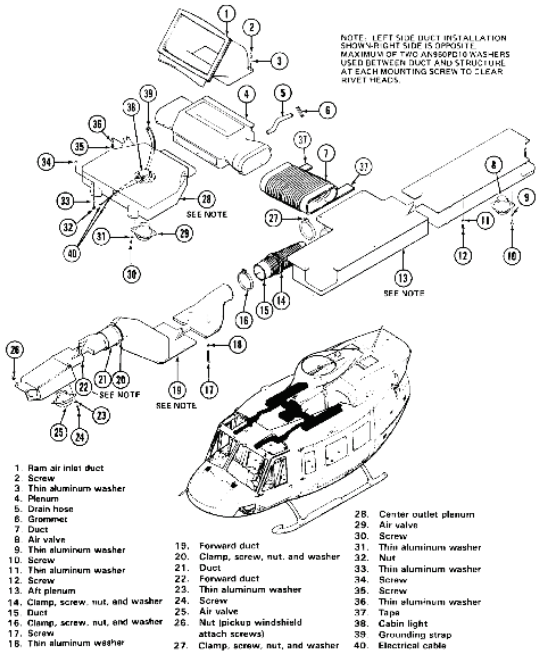
5. Position duct (19) on duct (22) and install clamp (20).

6. Position assembled ducts in helicopter. Secure forward end with two windshield attaching screws and install nuts (26) between duct and structure.

7. Secure duct (19) with washers (18) and screws (17).

8. Position duct (15) on duct (19) and plenum (13). Install clamps (14, 16, and 27).

9. Position duct (7) on plenum (13) and plenum (4). Seal edges of duct with tape (C-401).



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Figure 1. Overhead Ventilation System (Sheet 1 of 1)

Ventilation/Defog System - Description

Ventilation/defog system delivers outside air to windshield nozzles and nozzles on sides of instrument panel. Ram air intake grills are located on cabin nose below each windshield. Each grill is connected to a plenum chamber and ducts connect chambers to output nozzles. Air intake volume is controlled by knobs near instrument panel nozzles. When ram air is low, outside air may be pulled in by turning on the blower which is part of the system.

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Ventilation Blower - Description

NOTE

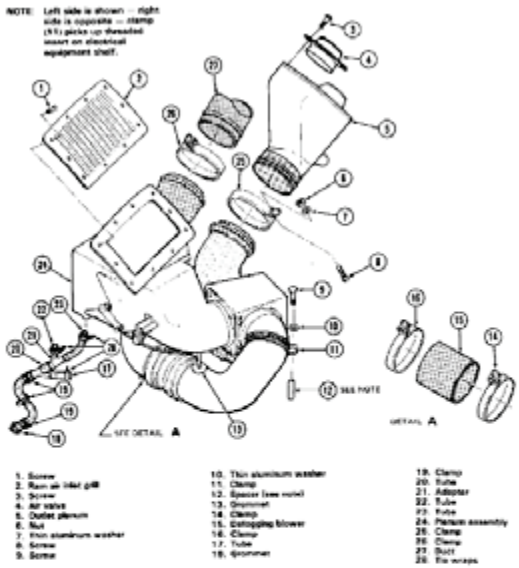
Maintenance procedures for left plenum and blower are given. Maintenance procedures for the right plenum and blower are similar.

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Ventilation Blower - Removal

Procedure

1. Remove screws (1, Figure 1) and remove grill (2) on left side of cabin nose.
2. Remove tie down wraps (28) from tubes (17, 22, and 23).
3. Remove clamps (19).
4. Remove tubes (17, 20, 22, and 23) and adapter (21) between grommet (18) and bottom of plenum assembly (24).
5. Remove clamp (25) and plenum (5).
6. Remove clamp (26) between duct (27) and plenum assembly (24). Remove clamp (11) and remove plenum assembly (24) as a unit.
7. Remove clamps (14 and 16) and remove blower (15).



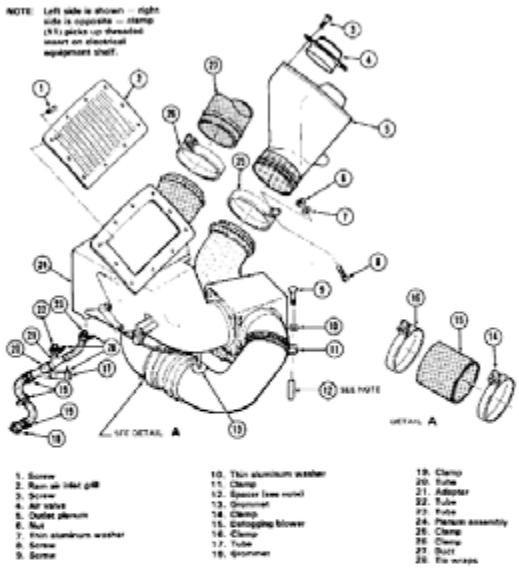
ICN-412-A-212001-B-97499-00001-A-001-01

Figure 1. Ventilation/Defog Components (Sheet 1 of 1)

Ventilation Blower - Installation

Procedure

1. Position plenum assembly (24, Figure 1) in cabin nose and install clamp (11), screw (9), washer (10), and spacer (12). Thread screw (9) into threaded insert on electrical equipment shelf.
2. Install clamp (26) between plenum assembly (24) and duct (27).
3. Position clamp (25) on plenum (5) and plenum assembly (24). Install screw (8), washer (7), and nut (6).
4. If not previously accomplished, position air valve (4) on plenum (5) and install screws (3).
5. Position tubes (23, 22, and 17) and adapter (21) on plenum (24). Install clamps (19) and install tiedown wraps (28) on tubes (23, 22, and 17).
6. If not previously accomplished, position blower (15) on plenum (24), and install clamps (14 and 16).
7. Position grill (2) on top of cabin nose left side. Install screws (1) to secure grill to plenum (24).
8. Perform functional check of blower (15).



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Figure 1. Ventilation/Defog Components (Sheet 1 of 1)

Winterization Heater Kit (212-706-008) - Description

The winterization heater kit is an optimization to the bleed air heater installation. It includes a valve assembly, noise suppressor, mixing valve, and cabin heat air ducts.

An additional cabin air distribution system is installed at the aft cabin bulkhead on the left and the right side to improve heating in the cabin. The winterization heater kit components are controlled by the existing switches on the overhead console, a temperature selecting dial on the right side door post, and heater defrost control lever on the right side of the center pedestal. These are already part of the basic installation.

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Mixing Valve - Description

The mixing valve (10L1) is installed in the compartment on the right side below the engine deck. This valve controls the bleed air and the ambient air flow as directed by the remote sensor. A bleed air nozzle inside the mixing valve acts as a jet pump through a venturi to draw in ambient air and control the air flow mixture. When heated air is not at a selected temperature, the remote sensor causes the mixing valve to increase or decrease the engine bleed air flow and regulate the ambient air flow to correct the temperature variation. A temperature limit temperature control switch on the mixing valve prevents structural damage in the event of an overheat. The CABIN HTR circuit breaker will open when the air in the mixing valve reaches 220°F (104.4°C).

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Mixing Valve - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-156	AR	
Caps or plugs	C-428	AR	

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT.

Procedure

NOTE

Remove parts only to the extent necessary to do the required maintenance.

1. Set the battery (BATT) switch to the OFF position and disconnect the external DC power from the helicopter.
2. Open the aft access door at the right side of the fuselage below the engine deck.
3. Install caps or plugs (C-428) on all the open ports as you remove the parts.
4. Remove the temperature control switch and the packing. Discard the packing.
5. Disconnect the connector (10L1P1).
6. Put caps or plugs (C-156) on the connectors.
7. Disconnect the tube assembly.
8. Remove the nut, screw, washer, and clamp and disconnect the duct from the mixing valve (10L1).
9. Cut and remove the lockwire from the tube assembly.
10. Loosen and remove the nut.
11. Remove the washer from the support.
12. Remove the mixing valve (10L1) and the washer from the support.
13. Remove the cap assembly.

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Mixing Valve - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	
Isopropyl alcohol	C-385	AR	
Clean cloth	C-516	AR	

Procedure

1. Clean the mixing valve (10L1) casing with a clean cloth (C-516) moistened with drycleaning solvent (C-304).

2. Clean the connector of the mixing valve (10L1) with isopropyl alcohol (C-385) and a soft bristle brush.

3. Clean the connector (10L1P1) with isopropyl alcohol (C-385) and a soft bristle brush.

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Mixing Valve - Inspection

Procedure

NOTE

The following inspection criteria are primarily intended when the part or component has been removed from the helicopter. To accommodate all inspection requirements, applicable criteria may also be used when the part or component is installed on the helicopter.

1. Visually examine the mixing valve (10L1) for the following:

- Casing for mechanical damage and corrosion
- Damaged threads
- Electrical leads of the connectors for evidence of chafing and damage

2. Visually examine the tube assembly for condition.

3. Visually examine the connector (10L1P1) for bent pins and evidence of chafing and damage, for wear to the contact pins, cracks, mechanical damage, and corrosion. As required, repair or replace the electrical connector.

4. Visually examine all the attaching hardware for wear, damage, and corrosion. Replace as required.

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Mixing Valve - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT.

Procedure

1. Remove the caps and/or plugs from all the ports, as you install them.

2. If previously removed, install the temperature control switch as follows:

2.1. Install new packing on the temperature control switch.

2.2. Install the temperature control switch on the mixing valve (10L1). Tighten the temperature control switch.

2.3. Safety the temperature control switch to the mixing valve (10L1) with lockwire (C-405).

3. Install the washer on the mixing valve (10L1).

4. Install the cap assembly. Tighten the cap.

5. Put the mixing valve (10L1) with the washer in position between the support and the duct in the compartment below the engine deck.

6. Install the washer and the nut on the mixing valve (10L1).

7. Align the tube assembly with the union on the mixing valve (10L1).

8. Torque the nut .

9. Connect the duct to the mixing valve (10L1) with the clamp, screw, washer, and nut. Tighten the nut.

10. Connect the tube assembly to the union on the mixing valve (10L1). Tighten the coupling nut.

11. Connect the connector (10L1P1) to the connector on the mixing valve (10L1).

12. Close the access door on the right side of the fuselage below the engine deck.

13. Do an operational check of the cabin heater system (412-A-96-06-00-01A-320B-A / 02142).

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Noise Suppressor - Description

The noise suppressor is a hollow aluminum tube with a fiberglass interior insulation. Air flow noise from the engine compressor is muffled by the noise suppressor. Heated air exits from the noise suppressor to a valve assembly (10L5) then to the plenum chamber in the aft cabin area. The noise suppressor is installed in the rear fuselage forward of the mixing valve (10L1). The operating temperature is -65 to 300°F (-18 to 149°C)

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Noise Suppressor - Removal

Procedure

1. Set the battery (BATT) switch to the OFF position and disconnect the external DC power from the helicopter.
2. Open the access door on right side of the fuselage below the engine deck.
3. Remove the nut, screw, washer, and clamps. Leave the clamps on the noise suppressor and the tube assembly, respectively.
4. Remove the nut, screw, washer, and clamp from the bracket. Leave the clamp on the noise suppressor.
5. If required, remove the nut, screw, bracket, spacer, and washer from the bulkhead at STA 211.06.
6. Remove the nut, screw, washer, and clamp and disconnect the coupling from the noise suppressor.
7. Remove the nut, screw, washer, and clamp and disconnect the duct from the noise suppressor.
8. Remove the noise suppressor from the coupling and the duct.
9. If required, remove the clamps from the noise suppressor.

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Noise Suppressor - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	
Clean cloth	C-516	AR	

Procedure

1. Clean the noise suppressor casing with a clean cloth (C-516) moistened with drycleaning solvent (C-304).

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Noise Suppressor - Inspection

Procedure

NOTE

The following inspection criteria are primarily intended when the part or component has been removed from the helicopter. To accommodate all inspection requirements, applicable criteria may also be used when the part or component is installed on the helicopter.

1. Visually examine the noise suppressor for mechanical damage and corrosion.

2. Visually examine all the attaching hardware for wear, damage, and corrosion. Replace as required.

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Noise Suppressor - Installation

Procedure

1. If the noise suppressor was replaced, install the clamps.
2. Put the noise suppressor in position between the coupling and the duct in the compartment below the engine deck.
3. Connect the coupling to the noise suppressor with the clamp, screw, washer, and nut. Tighten the nut.
4. If previously removed, install the screw, bracket, spacer, washer, and nut on the bulkhead at STA 211.06.
5. Connect the duct to the noise suppressor with the clamp, screw, washer, and nut. Tighten the nut.
6. Install the clamp with the screw, washer, and nut on the bracket. Tighten the nut
7. Install the two clamps, with the screw, washer, and nut. Tighten the nut.
8. Close the access door on the right side of the fuselage below the engine deck.

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Valve assembly - Description

The valve assembly (10L5) lets the air go to the forward section of the cabin to heat and defrost, and supplies heat to the aft cabin area. A flapper, installed inside the valve assembly, controls the quantity of air that goes to the cabin.

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Valve Assembly - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-156	AR	

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT.

Procedure

NOTE

Remove parts only to the extent necessary to do the required maintenance.

1. Set the battery (BATT) switch to the OFF position and disconnect the external DC power from the helicopter.
2. Open the aft access door at the right side of the fuselage below the engine deck.
3. Remove the nut, washer, screw, and clamp from the duct.
4. Disconnect the duct from the valve assembly (10L5).
5. Loosen the clamp and move the clamp onto the coupling.
6. Disconnect the connector (10L5P1) from the valve assembly (10L5).
7. Put caps or plugs (C-156) on the connector (10L5P1) and the valve assembly (10L5).
8. Remove the nut, washer, screw, and clamp from the coupling.
9. Disconnect the coupling from the valve assembly (10L5).
10. While holding the valve assembly (10L5), loosen and remove the clamp from the valve assembly (10L5) and the plenum heater duct.
11. Remove the valve assembly (10L5) from the plenum heater duct.

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Valve Assembly - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-285	AR	
Drycleaning solvent	C-304	AR	
Clean cloth	C-516	AR	

Procedure

1. Clean the valve assembly (10L5) casing with a clean cloth (C-516) moistened with drycleaning solvent (C-304).

2. Clean the connector on the valve assembly (10L5) with isopropyl alcohol (C-285) and a soft bristle brush.

3. Clean the connector (10L5P1) with isopropyl alcohol (C-285) and a soft bristle brush.

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Valve Assembly - Inspection

Procedure

NOTE

The following inspection criteria are primarily intended when the part or component has been removed from the helicopter. To accommodate all inspection requirements, applicable criteria may also be used when the part or component is installed on the helicopter.

1. Visually examine the valve assembly (10L5) for the following:

- Casing for mechanical damage and corrosion
- Damaged threads
- Electrical leads of the connectors for evidence of chafing and damage

2. Visually examine the connector (10L5P1) for bent pins and evidence of chafing and damage, for wear to the contact pins, cracks, mechanical damage, and corrosion. As required, repair or replace the connector.

3. Visually examine all the attaching hardware for wear, damage, and corrosion. Replace as required.

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Valve Assembly - Installation

Preliminary requirements

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT.

Procedure

1. Put the valve assembly (10L5) in position between the plenum heater duct, and couplings on the right side of the fuselage below the engine deck.
2. While holding the valve assembly (10L5), install the clamp on the valve assembly and the plenum heater duct. Torque the clamp .
3. Connect the coupling to the valve assembly (10L5) and secure with the clamp, screw, washer, and nut. Tighten the nut.
4. Install the coupling on the valve assembly (10L5) with the clamp. Tighten the clamp.
5. Connect the duct to the valve assembly (10L5) with the clamp, screw, washer, and nut. Tighten the nut.
6. Remove the caps and/or plugs from the connector (10L5P1) and the valve assembly (10L5).
7. Connect the connector (10L5P1) to the valve assembly (10L5).
8. Close the aft access door at the right side of the fuselage below the engine deck.
9. Do an operational check of the cabin heater system (412-A-96-06-00-01A-320B-A / 02142).

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Heat Air Ducts - Description

The heat air ducts include ducts, couplings, elbow assembly, duct assemblies, and air valves that direct air from the mixing valve to the cabin. The ducts are made of silicone impregnated fiberglass cloth and the other parts are made of polycarbonate.

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Heat Air Ducts - Removal

Procedure

NOTE

Remove parts only to the extent necessary to do the required maintenance.

1. Open the passenger/cargo door on the left and/or right side (412-A-52-12-00-01A-520A-A / 00812).

2. Remove the lower fuselage access door to access the components, as applicable.

3. Open the aft access door on the right side of the fuselage below the engine deck

4. On the left side, remove the air valve assembly as follows:

4.1. Remove the screws, washers, and air valve assemblies from the aft cabin bulkhead.

4.2. Remove the nut, screw, washer, and clamp from the duct.

4.3. Disconnect the duct from the duct assembly.

4.4. Remove the duct assembly from the aft cabin bulkhead.

5. Remove the duct assembly from the aft cabin bulkhead.

5.1. Remove the nut, screw, washer, and clamp from the duct.

5.2. Disconnect the duct from the elbow assembly.

5.3. Loosen and move the clamp on the duct.

5.4. Disconnect the duct from the elbow assembly.

5.5. Remove the bolts and the washers from the elbow assembly.

5.6. Remove the screws and the washers from the elbow assembly.

5.7. Remove the elbow assembly and the seal from the web. Discard the seal.

6. Remove the nut, screw, washer, and clamp from the bracket. Leave the clamp on the duct.

7. If required, remove the nut, washer, screw, and clamp from the fuel valve.

8. Remove the screw, washer, clamp, spacers from the bulkhead at STA 178.0. Leave the clamp on the duct.

9. Loosen and move the clamp on the duct.

10. Remove the duct with the clamps.

11. Remove the union from the left side beam as follows:

11.1. Loosen and move the clamp on the duct.

11.2. Disconnect the duct from the union. Leave the clamp on the duct.

11.3. Remove the nut, screw, washer, and clamp from the bracket.

11.4. If required, remove the nut, washer, screw, and bracket from the left side beam.

11.5. Remove the union from the left side beam.

12. Remove the duct as follows:

12.1. Loosen and move the clamp on the duct.

12.2. Remove the nut, washer, screw, and clamps to disconnect the duct from the oil tube assembly. Leave the clamps on the duct and tube assembly, respectively.

12.3. Remove the nut, washer, screw, spacer, and clamps to disconnect the duct from the hydraulic return tube assembly. Leave the clamps on the duct and tube assembly, respectively.

12.4. Remove the duct with the clamps.

13. Remove the elbow assembly from the right side beam as follows:

13.1. Remove the nut, washer, screw, spacer, and clamp.

13.2. Loosen and move the clamp from the coupling on the elbow assembly.

13.3. Remove the elbow assembly with the clamps.

14. Remove the elbow assembly as follows:

14.1. Loosen and move the clamp from the coupling on the elbow assembly.

14.2. Remove the coupling.

14.3. Loosen and move the clamp from the coupling on the elbow assembly.

14.4. Remove the elbow assembly with the clamps.

15. Remove the clamp and the coupling from the valve assembly (10L5).

16. Remove the duct as follows:

16.1. Remove the nut, washer, screw, and clamp to disconnect the duct from the valve assembly (10L5). Leave the clamp on the duct.

16.2. Remove the nut, washers, screw, and clamp to remove the duct from the bracket assembly. Leave the clamp on the duct.

16.3. Remove the nut, washer, screw, spacer, and clamp to remove the duct from the bracket. Leave the clamp on the duct.

16.4. If required, remove the nut, washer, screw, and bracket from the bulkhead at STA 178.0.

16.5. Remove the nut, washer, screw, and clamp to disconnect the duct from the duct assembly. Leave the clamp on the duct.

16.6. Remove the duct from the duct assembly and the valve assembly (10L5) with the clamp.

17. On the right side, remove the air valve assemblies as follows:

17.1. Remove the screws, washers, and air valve assemblies from the aft cabin bulkhead.

17.2. Remove the duct assembly from the aft cabin bulkhead.

Heat Air Ducts - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	
Clean cloth	C-516	AR	

Procedure

1. Clean the air valve assemblies, duct assemblies, elbow assemblies, and union with a clean cloth (C-516) moistened with drycleaning solvent (C-304).

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Heat Air Ducts - Inspection

Procedure

NOTE

The following inspection criteria are primarily intended when the part or component has been removed from the helicopter. To accommodate all inspection requirements, applicable criteria may also be used when the part or component is installed on the helicopter.

1. Visually examine the valve assemblies for the following:
- Mechanical damage and corrosion
- Smooth movement of the valve
- Pin is not disengaged from the valve

2. If required, replace the valve assembly(s).

3. Visually examine the ducts for surface damage. Repair or replace a damaged duct, as required (412-A-21-40-00-04A-685A-A / 00242).

4. Visually examine the couplings for condition. Replace as required.

5. Visually examine the union for general condition. Replace as required.

6. Visually examine the elbow assembly for general condition. Replace as required.

7. Visually examine all the attaching hardware for wear, damage, and corrosion. Replace as required.

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Heat Air Ducts - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Toluene	C-306	AR	
Adhesive	C-314	AR	
Aliphatic naphtha	C-305	AR	
Glass fabric	C-404	AR	

Procedure

1. Repair the ducts made from silicone-impregnated fiberglass cloth as follows:

1.1. Repair scuffs and abrasions that do not penetrate through the cloth and are not more than 3.0 x 1.5 inches (76.2 x 38.1 mm). A maximum of two repairs per foot, that cover up to 10% of the surface area are acceptable.

1.1.1. Clean the damaged area with toluene (C-306) or aliphatic naphtha (C-305). Let the area air dry for at least 30 minutes.

1.1.2. Apply a brush coat of adhesive (C-314) over the damaged area, with a minimum of 0.50 inch (12.7 mm) overlap beyond the damage.

1.1.3. Smooth the adhesive and let cure by air drying for minimum of 2 hours at room temperature or until dry to the touch.

1.2. Repair damage that is through the fiberglass cloth as follows:

NOTE

Repairs that cover up to 10% of the surface area are acceptable.

1.2.1. Clean the circumference of the duct damage with toluene (C-306) or aliphatic naphtha (C-305). Let the area air dry for at least 30 minutes.

1.2.2. Cut one piece of glass fabric (C-404) to cover the duct circumference with at least a 1.0 inch (25.4 mm) overlap of the damaged area.

1.2.3. Apply one thin smooth brush coat of adhesive (C-314) around the duct circumference in the area to be covered by the glass fabric.

1.2.4. Wind the glass fabric around the duct and smooth out.

1.2.5. Allow the area to air dry at least 2 hours or until dry to the touch, before using or handling the duct.

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Heat Air Ducts - Installation

Procedure

1. On the left side and/or right side, open the passenger/cargo door (412-A-52-12-00-01A-520A-A / 00812).

2. On the right side, install the air valve assemblies as follows:

2.1. Put the duct assembly in position on the aft cabin bulkhead.

2.2. Install the air valve assemblies on the aft cabin bulkhead with the screws and the washers. Tighten the screws.

3. Install the duct as follows:

3.1. Put the duct in position with the clamps between the duct assembly and the valve assembly (10L5).

3.2. Connect the duct to the duct assembly with the clamp, screw, washer, and nut. Tighten the nut.

3.3. If previously removed, install the bracket on the bulkhead at STA 178.0 with the screw, washer, and nut. Tighten the nut.

3.4. Install the duct on the bracket with the screw, spacer, clamp, washer, and nut. Tighten the nut.

3.5. Install the duct on bracket assembly with the screw, clamp, washers, and nut. Tighten the nut.

3.6. Connect the duct to the valve assembly (10L5) with the screw, clamp, washer, and nut. Tighten the nut.

4. Install the coupling and the clamp on the valve assembly (10L5). Tighten the clamp.

5. Install the elbow assembly as follows:

5.1. Put the clamps in position on the elbow assembly.

5.2. Install the elbow assembly on the coupling with clamp. Tighten the clamp.

5.3. Install the coupling on the elbow assembly with the clamp. Tighten the clamp.

6. Install the elbow assembly on the right side beam as follows:

6.1. Put the clamps in position on the elbow assembly.

6.2. Put the elbow assembly in position through the right side beam.

6.3. Connect the elbow assembly to the coupling with the clamp. Tighten the clamp.

6.4. Install the clamp, screw, spacer, washer, and nut on the right side beam. Tighten the nut.

7. Install the duct as follows:

7.1. Put the duct in position between the right and left side beams with the clamps.

7.2. Connect the duct to the elbow with the clamp. Tighten the clamp.

7.3. Install the duct on the hydraulic return tube assembly with the clamps, screw, spacer, washer, and nut. Tighten the nut.

7.4. Install the duct on the oil tube assembly with the clamps, screw, washer, and nut. Tighten the nut.

8. Install the union on the left side beam as follows:

8.1. Make sure the clamp is in position on the end of the duct.

8.2. Put the union through the left side beam.

8.3. Put the clamp in position on the union.

8.4. Connect the duct to the union with the clamp. Tighten the clamp.

8.5. If previously removed, install the bracket on the left side beam with the screw, washer, and nut. Tighten the nut.

8.6. Install the clamp on the bracket with the screw, washer, and nut. Tighten the nut.

9. Thread the duct through the web.

10. Put the clamps in position on the duct.

11. Install the duct on the union with the clamp. Tighten the clamp.

12. Install the screw, washer, clamp, spacers on the bulkhead at STA 178.0. Tighten the screw.

13. If previously removed, install the clamp on the fuel valve with the screw, washer, and nut. Tighten the nut.

14. Install the clamp on the bracket with the screw, washer, and nut. Tighten the nut.

15. Install the elbow assembly on the web as follows:

15.1. Make sure the clamp is in position on the duct.

15.2. Install the seal and the elbow assembly on the web with the screws and the washers. Tighten the screws.

15.3. Install the bolts and the washers on the elbow assembly. Tighten the bolts.

15.4. Connect the duct to the elbow assembly with the clamp. Tighten the clamp.

15.5. Connect the duct to the elbow assembly with the clamp, screw, washer, and nut. Tighten the nut.

16. On the left side, install the air valve assembly as follows:

16.1. Put the duct assembly in position at the back of the aft cabin bulkhead.

16.2. Install the air valve assemblies on the aft cabin bulkhead with the screws and the washers. Tighten the screws.

16.3. Install the duct on the duct assembly with clamp, screw, washer, and nut. Tighten the nut.

17. Install the lower fuselage access doors that were removed for access to the components, as applicable.

18. Close the aft access door on the right side of fuselage below the engine deck, as applicable.

19. Close the passenger/cargo door on the left and/or right side, as applicable (412-A-52-12-00-01A-720A-A / 00815).

CHAPTER 25
EQUIPMENT AND FURNISHINGS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Equipment and Furnishings - Description	DMC-412-A-25-00-00-00A-028A-A / 00245	2023-05-31	ALL
Crew and Passenger Seats - Description	DMC-412-A-25-10-00-00A-042A-A / 00246	2023-05-31	ALL
Crew Seats - Description	DMC-412-A-25-10-02-00A-042A-A / 00247	2023-05-31	SN: 33001-33213 or 36001-36419
Crew Seats - Removal	DMC-412-A-25-10-02-00A-520A-A / 00248	2023-05-31	ALL
Crew Seats - Inspection	DMC-412-A-25-10-02-00A-280A-A / 00249	2023-05-31	ALL
Crew Seats - Repairs	DMC-412-A-25-10-02-00A-685A-A / 00250	2023-05-31	ALL
Crew Seats - Installation	DMC-412-A-25-10-02-00A-720A-A / 00251	2023-05-31	SN: 33001-33213 or 36001-36419
Crew Seat Restraint Assembly - Description	DMC-412-A-25-10-02-01A-042A-A / 00252	2023-05-31	ALL
Crew Seat Restraint Assembly - Removal	DMC-412-A-25-10-02-01A-520A-A / 00253	2023-05-31	ALL
Restraint System - Check and Inspection Guidelines	DMC-412-A-25-10-00-00A-280A-A / 00254	2023-05-31	ALL
Crew Seat Restraint Assembly - Inspection	DMC-412-A-25-10-02-01A-280A-A / 00255	2023-05-31	ALL
Crew Seat Restraint Assembly - Repair	DMC-412-A-25-10-02-01A-685A-A / 00256	2023-05-31	ALL
Crew Seat Restraint Assembly - Installation	DMC-412-A-25-10-02-01A-720A-A / 00257	2023-05-31	ALL
Crew Seat Vertical Adjustment Cable - Removal and Installation	DMC-412-A-25-10-02-02A-920A-A / 00258	2023-05-31	ALL
Crew Seats - Description	DMC-412-A-25-10-02-03A-042A-A / 00259	2024-05-09	SN: 36420-Subsequent
Passenger Seats - Description	DMC-412-A-25-20-00-00A-042A-A / 00260	2023-05-31	ALL
Passenger Seats - Folding	DMC-412-A-25-20-00-00A-272A-A / 00261	2023-05-31	ALL
Passenger Seats - Inspection	DMC-412-A-25-20-00-00A-280A-A / 00262	2023-05-31	ALL
Passenger Seats - Repair	DMC-412-A-25-20-00-00A-685A-A / 00263	2023-05-31	ALL
Passenger Seat Restraint Assembly - Removal	DMC-412-A-25-20-00-01A-520A-A / 00264	2023-05-31	ALL
Passenger Seat Restraint Assembly - Inspection	DMC-412-A-25-20-00-01A-280A-A / 00265	2023-05-31	ALL
Passenger Seat Restraint Assembly - Repair	DMC-412-A-25-20-00-01A-685A-A / 00266	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Passenger Seat Restraint Assembly - Installation	DMC-412-A-25-20-00-01A-720A-A / 00267	2023-05-31	ALL
Outboard Facing Passenger Seat - Description	DMC-412-A-25-20-00-02A-042A-A / 00268	2023-05-31	ALL
Outboard Facing Passenger Seat - Removal	DMC-412-A-25-20-00-02A-520A-A / 00269	2023-05-31	ALL
Outboard Facing Passenger Seat - Installation	DMC-412-A-25-20-00-02A-720A-A / 00270	2023-05-31	ALL
Outboard Facing Passenger Seat - Bracket Preparation	DMC-412-A-25-20-00-02A-723A-A / 00271	2023-05-31	ALL
Four Place Aft Facing Passenger Seat - Removal	DMC-412-A-25-20-00-03A-520A-A / 00272	2023-05-31	ALL
Four Place Aft Facing Passenger Seat - Installation	DMC-412-A-25-20-00-03A-720A-A / 00273	2023-05-31	ALL
Four Place Aft Facing Passenger Seat - Removal from Frame	DMC-412-A-25-20-00-03A-520B-A / 00274	2023-05-31	ALL
Four Place Aft Facing Passenger Seat - Installation on Frame	DMC-412-A-25-20-00-03A-720B-A / 00275	2023-05-31	ALL
Five Place Forward Facing Passenger Seat - Removal	DMC-412-A-25-20-00-04A-520A-A / 00276	2023-05-31	ALL
Five Place Forward Facing Passenger Seat - Installation	DMC-412-A-25-20-00-04A-720A-A / 00277	2023-05-31	ALL
Five Place Forward Facing Passenger Seat - Removal from Frame	DMC-412-A-25-20-00-04A-520B-A / 00278	2023-05-31	ALL
Five Place Forward Facing Passenger Seat - Installation on Frame	DMC-412-A-25-20-00-04A-720B-A / 00279	2023-05-31	ALL
Miscellaneous Furnishings - Description	DMC-412-A-25-00-00-01A-028A-A / 00280	2023-05-31	ALL
Map and Data Case - Description	DMC-412-A-25-00-00-01A-042A-A / 00281	2023-05-31	ALL
Map and Data Case - Removal	DMC-412-A-25-00-00-01A-520A-A / 00282	2023-05-31	ALL
Map and Data Case - Inspection	DMC-412-A-25-00-00-01A-280A-A / 00283	2023-05-31	ALL
Map and Data Case - Repair	DMC-412-A-25-00-00-01A-685A-A / 00284	2023-05-31	ALL
Map and Data Case - Installation	DMC-412-A-25-00-00-01A-720A-A / 00285	2023-05-31	ALL
First Aid Kit - Description	DMC-412-A-25-60-00-01A-042A-A / 00286	2023-05-31	ALL
First Aid Kit - Removal	DMC-412-A-25-60-00-01A-520A-A / 00287	2023-05-31	ALL
First Aid Kit - Inspection	DMC-412-A-25-60-00-01A-280A-A / 00288	2023-05-31	ALL
First Aid Kit - Repair	DMC-412-A-25-60-00-01A-685A-A / 00289	2023-05-31	ALL
First Aid Kit - Installation	DMC-412-A-25-60-00-01A-720A-A / 00290	2023-05-31	ALL

Equipment and Furnishings - Description

This chapter contains maintenance information for crew and passenger seats, map and data case, and the first aid kit.

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Crew and Passenger Seats - Description

NOTE

SN: 33001-33213 or 36001-36419 or 36420-Subsequent:

The pilot and copilot (crew) seats installed are classified as energy attenuating. The crew seats installed and those modified by BHT- 412-SI-75 are classified as crash attenuating.

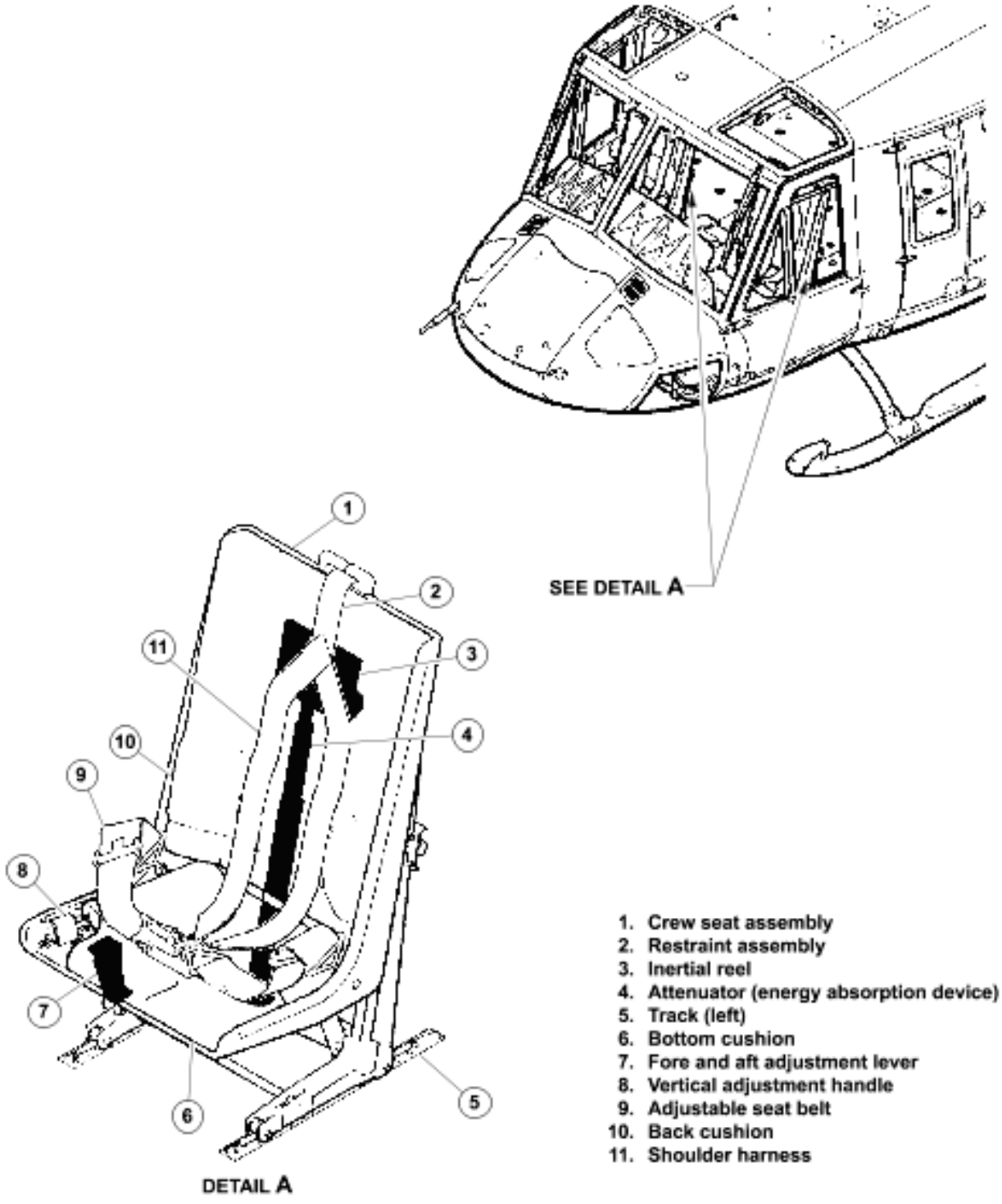
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Crew Seats - Description

The pilot and copilot (crew) seats (Figure 1) are energy-attenuating, bucket-type seats upholstered with flame-retardant materials. The seats are mounted on tracks and are adjustable in fore-and-aft and vertical directions. Each seat is equipped with a restraint assembly (2) which consists of inertia reel (3), shoulder harness (11), and seat belt (9). The inertia reel incorporates an automatic locking mechanism to prevent seat occupant from pitching forward in the event of rapid deceleration. The attenuator (4) is designed to absorb energy and thereby protect the seat occupant in event of a hard landing.

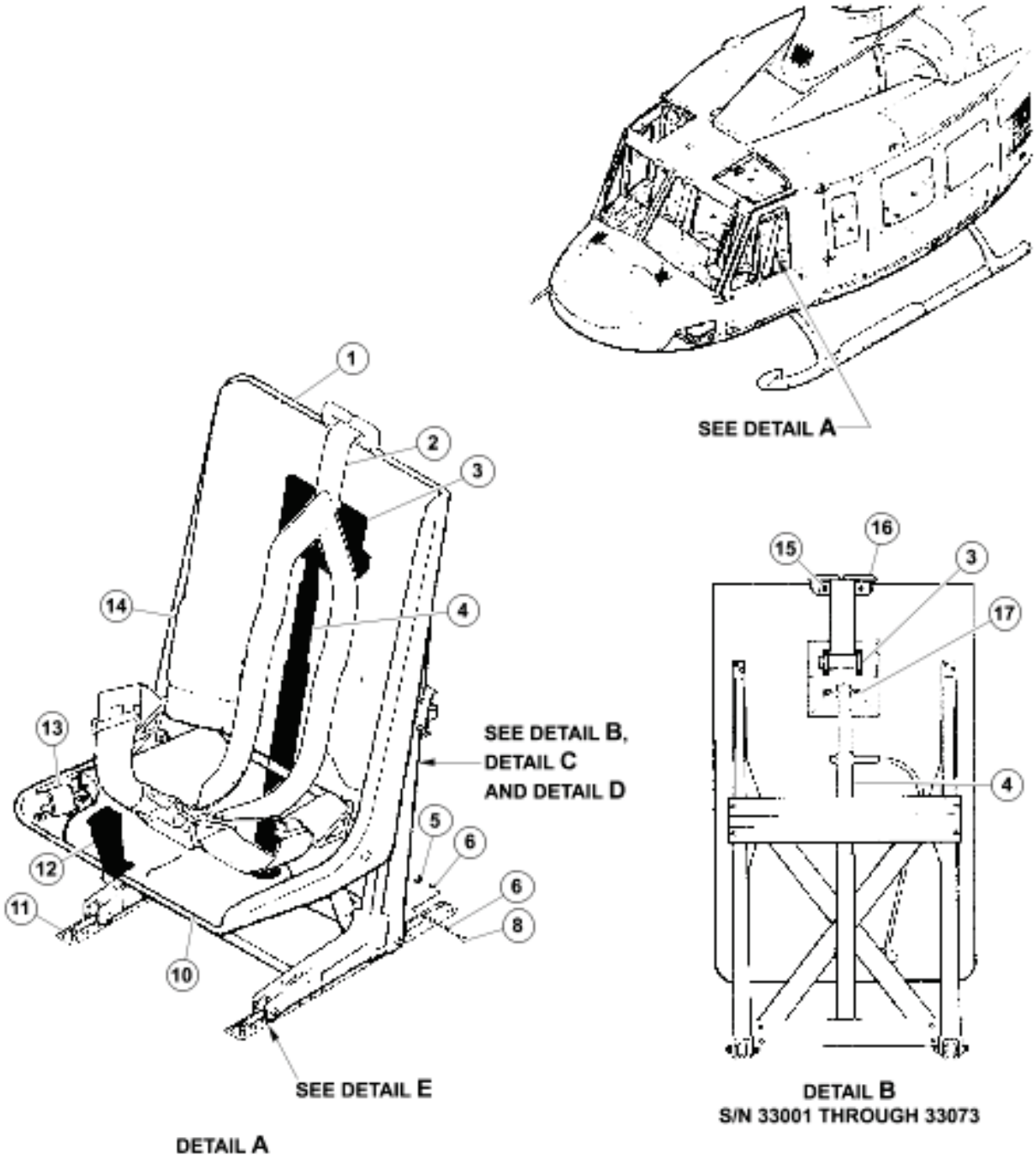
NOTE

Maintenance procedures for both pilot and copilot seats are similar, except the removal/installation procedure of the copilot seat assembly (1, Figure 2) on the 412EP S/N 36087 and subsequent helicopters has two extra forward stop screws (7) added on each track (9 and 11). This is to prevent the forward outboard corner of the copilot bottom seat cushion (10) from interfering with the copilot collective stick head when the copilot seat is in the full forward and full up position. The maintenance procedures in 412-A-25-10-02-00A-520A-A / 00248, 412-A-25-10-02-00A-280A-A / 00249, 412-A-25-10-02-00A-685A-A / 00250 and 412-A-25-10-02-00A-720A-A / 00251 are referenced to the copilot seat assembly (1, Figure 2) but can also be applied to the pilot seat.



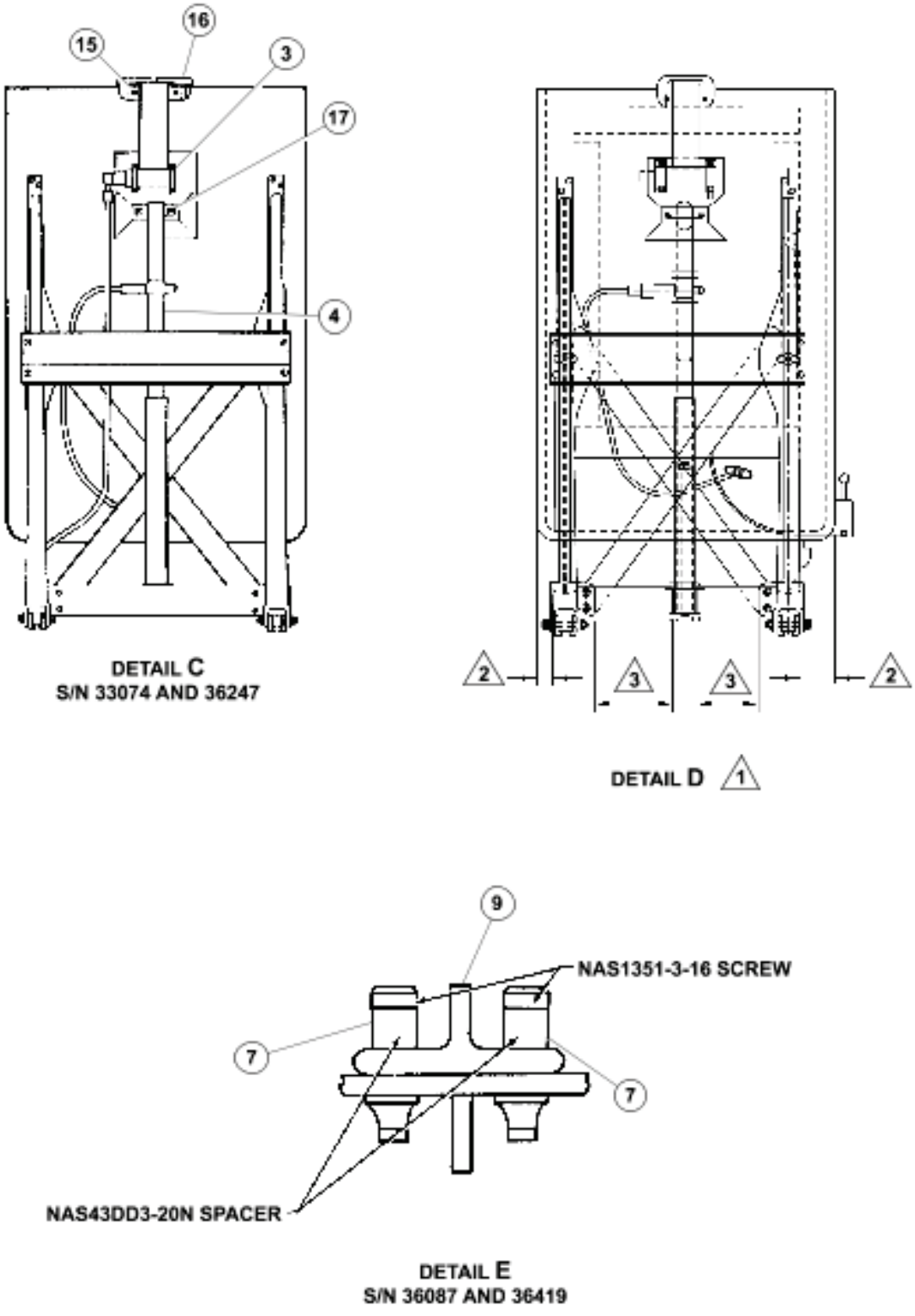
ICN-412-A-251002-B-97499-00001-A-001-01

Figure 1. Crew Seats (Sheet 1 of 1)



ICN-412-A-251002-B-97499-00002-A-001-01

Figure 2. Copilot Seat (Sheet 1 of 3)



ICN-412-A-251002-B-97499-00003-A-001-01

Figure 2. Copilot Seat (Sheet 2 of 3)

1. Copilot seat assembly
2. Restraint assembly
3. Inertia reel assembly
4. Attenuator (energy absorption device)
5. Nut
6. Washer
7. Forward stop screws
8. Screw
9. Track (left)
10. Bottom seat cushion
11. Track (right)
12. Fore and aft adjustment lever
13. Vertical adjustment handle
14. Back seat cushion
15. Screw
16. Guide (restraint assembly)
17. Screw

NOTES

1. For helicopter S/N 36248 through 36419 or with BHT-412-SI-66 incorporated.
2. Seat bucket shifted to increase clearance between copilot seat and copilot collective.
3. Attenuator assembly shifted to match position of bucket.

ICN-412-A-251002-B-97499-00004-A-001-01

Figure 2. Copilot Seat (Sheet 3 of 3)

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Crew Seats - Removal

Procedure

NOTE

The following removal procedures are referenced to the copilot seat assembly (1, Figure 1) but can also be applied to the pilot seat.

NOTE

Crew seats may be removed/installed after removing screws (8, Figure 1) on forward or aft end of tracks (9 and 11), depending on passenger seat configuration.

NOTE

SN: 36087-Subsequent: If removing copilot seat assembly (1) from forward end of tracks (9 and 11) on 412EP, the copilot seat assembly (1) may be removed/installed after removing screws (8) and forward stop screws (7) on forward end of tracks (9 and 11).

SN: 36248-36419 or BHT-412-SI-66: Installed: The pilot and copilot seats are not interchangeable. The copilot seat bucket is shifted inboard to provide additional clearance between the seat assembly and copilot collective stick assembly.

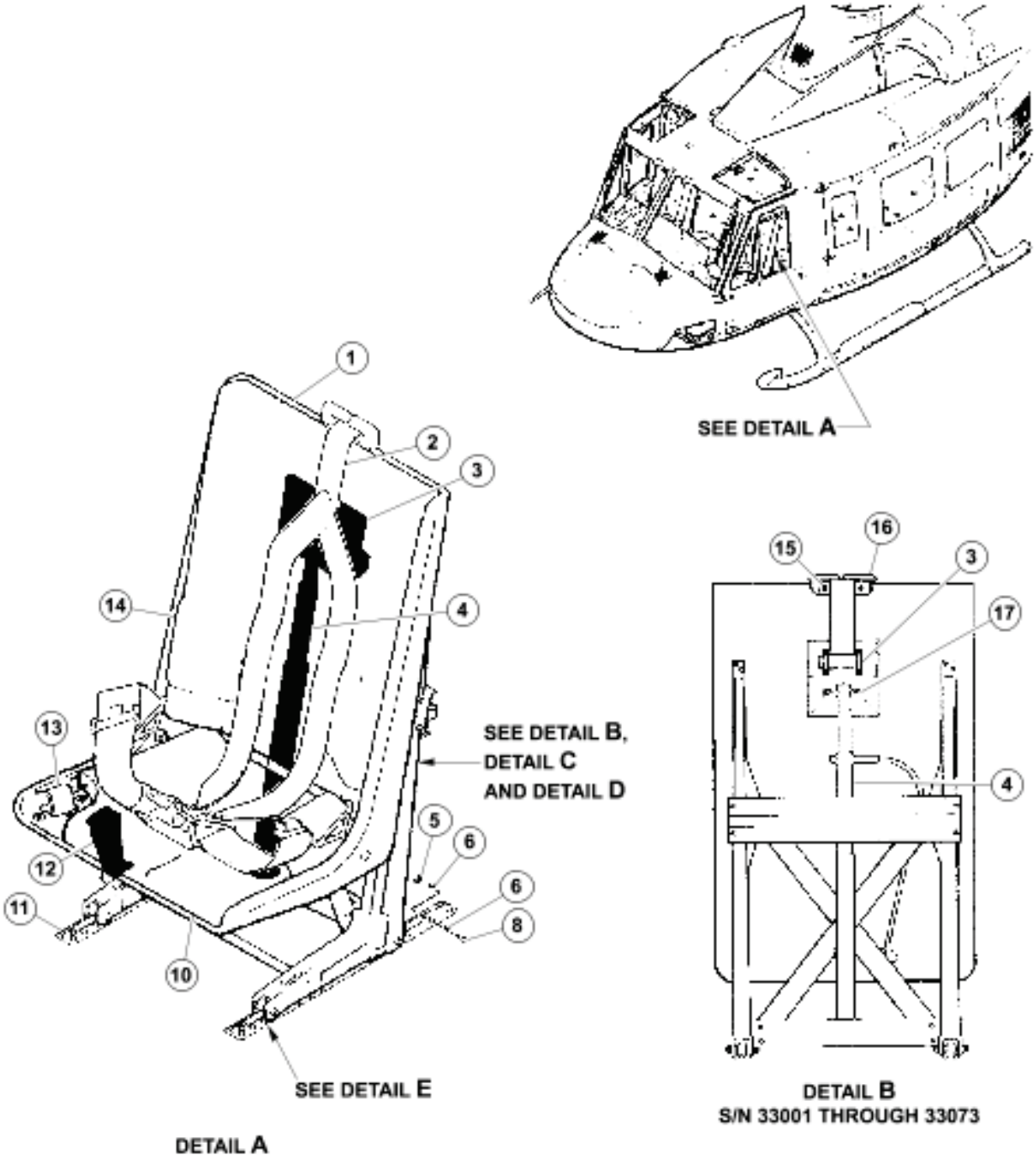
1. Remove nut (5), washers (6), screw (8) and forward stop screws (7) from track (9). Remove similar screw and forward stop screws from track (11).

WARNING

SEAT HAS HELPER SPRING TO AID VERTICAL ADJUSTMENT. SEAT WILL MOVE RAPIDLY UPWARD FROM FULL DOWN POSITION IF NO WEIGHT IS ON SEAT.

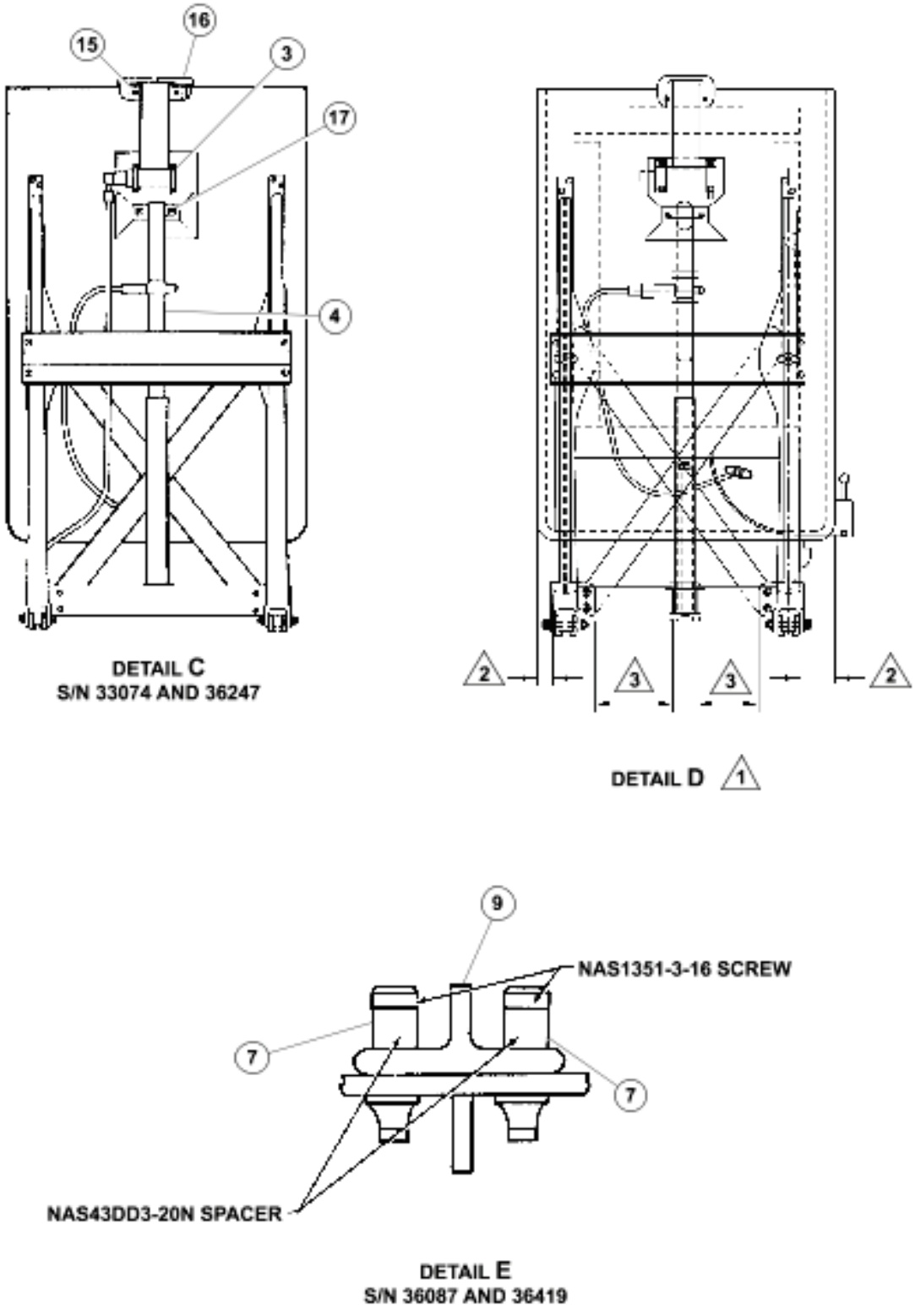
2. Lift fore and aft adjustment lever (12) on seat assembly (1) and slide seat off tracks.

3. Remove seat assembly (1) from helicopter.



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Figure 1. Copilot Seat (Sheet 1 of 3)



ICN-412-A-251002-B-97499-00003-A-001-01

Figure 1. Copilot Seat (Sheet 2 of 3)

1. Copilot seat assembly
2. Restraint assembly
3. Inertia reel assembly
4. Attenuator (energy absorption device)
5. Nut
6. Washer
7. Forward stop screws
8. Screw
9. Track (left)
10. Bottom seat cushion
11. Track (right)
12. Fore and aft adjustment lever
13. Vertical adjustment handle
14. Back seat cushion
15. Screw
16. Guide (restraint assembly)
17. Screw

NOTES

1. For helicopter S/N 36248 through 36419 or with BHT-412-SI-66 incorporated.
2. Seat bucket shifted to increase clearance between copilot seat and copilot collective.
3. Attenuator assembly shifted to match position of bucket.

ICN-412-A-251002-B-97499-00004-A-001-01

Figure 1. Copilot Seat (Sheet 3 of 3)

Crew Seats - Inspection

Procedure

NOTE

The following inspection procedures are referenced to the copilot seat assembly (1, Figure 1) but can also be applied to the pilot seat.

1. Inspect seat assemblies (1, Figure 1) for secure installation.

2. Inspect fore and aft adjustment mechanism (12) for proper operation.



SEAT HAS HELPER SPRING TO AID VERTICAL ADJUSTMENT. SEAT WILL MOVE RAPIDLY UPWARD FROM FULL DOWN POSITION IF NO WEIGHT IS ON SEAT.

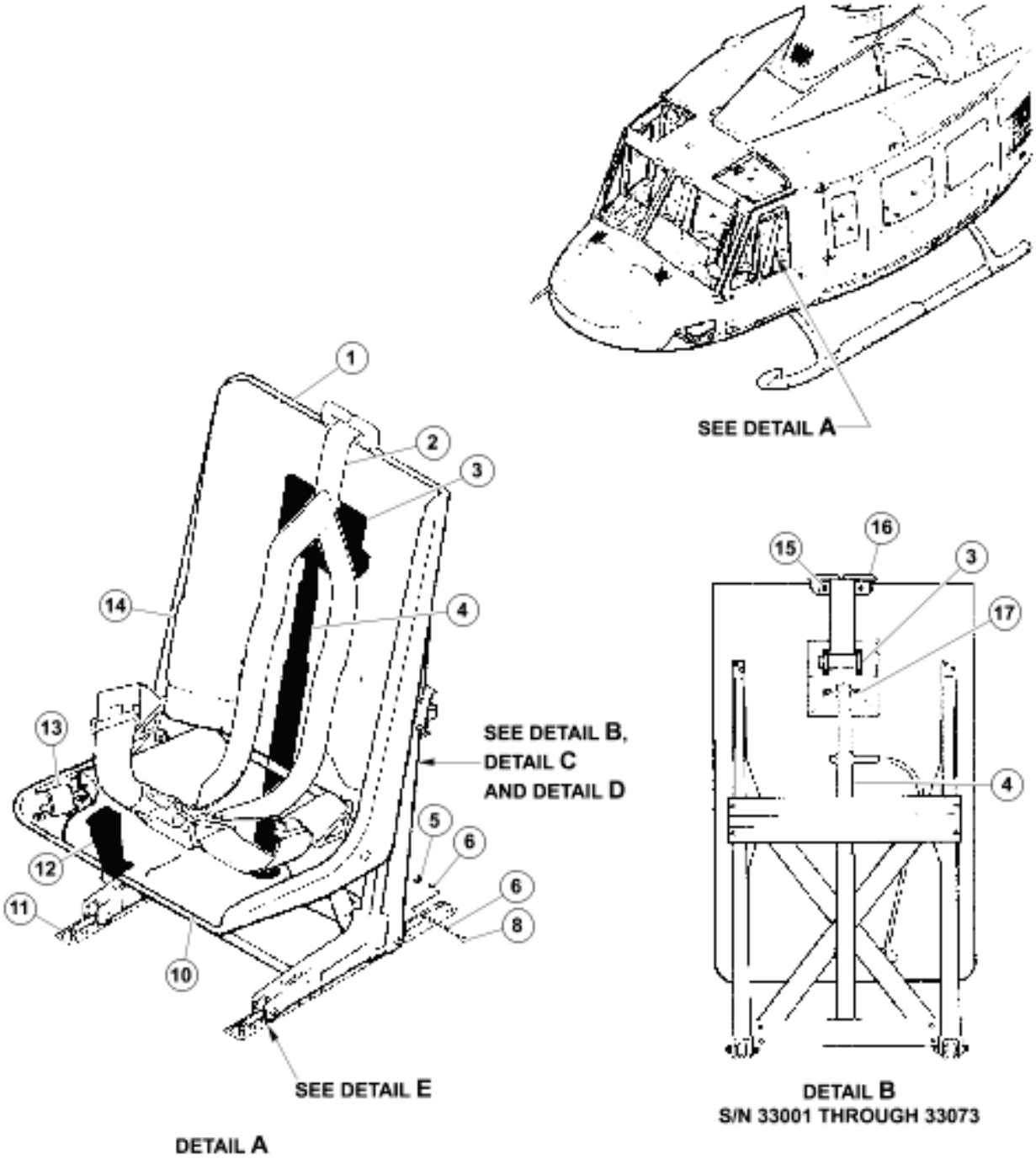
3. Inspect vertical adjustment mechanism (13) for proper operation and verify cable adjustment pin (27, Figure 2) is properly engaged into tube (5) by removing plug (4) and viewing through top of tube (detail B). If pin is not properly engaged refer to 412-A-25-10-02-02A-920A-A / 00258.

4. Inspect seat cushions (10 and 14, Figure 1) for tears, cuts, and security of upholstery at bonding areas.

5. Inspect restraint assembly (2) (412-A-25-10-02-01A-042A-A / 00252).

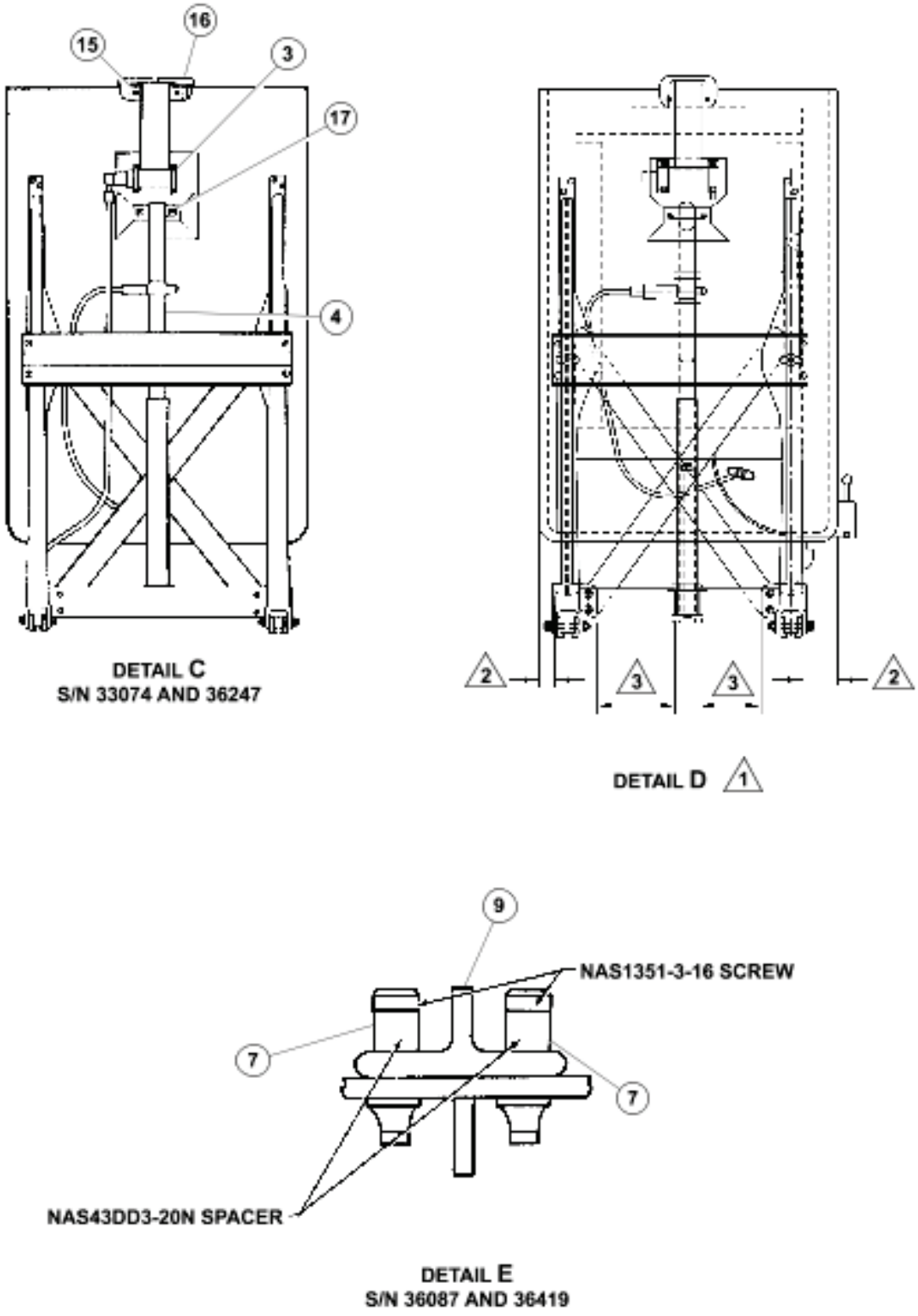
6. Inspect seat structure for obvious damage.

7. Inspect attenuator assembly (16, Figure 2) white paint stripe measurement (detail A). If stripe is not visible, refer to 412-A-25-10-02-00A-685A-A / 00250 for adjustment.



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Figure 1. Copilot Seat (Sheet 1 of 3)



ICN-412-A-251002-B-97499-00003-A-001-01

Figure 1. Copilot Seat (Sheet 2 of 3)

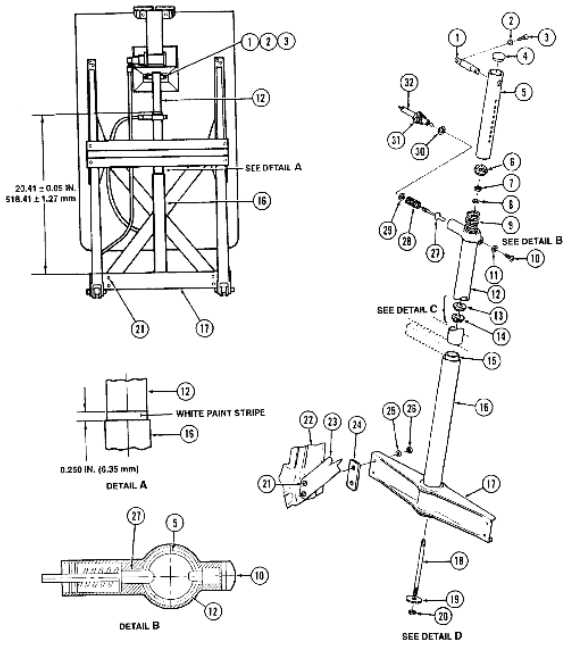
1. Copilot seat assembly
2. Restraint assembly
3. Inertia reel assembly
4. Attenuator (energy absorption device)
5. Nut
6. Washer
7. Forward stop screws
8. Screw
9. Track (left)
10. Bottom seat cushion
11. Track (right)
12. Fore and aft adjustment lever
13. Vertical adjustment handle
14. Back seat cushion
15. Screw
16. Guide (restraint assembly)
17. Screw

NOTES

1. For helicopter S/N 36248 through 36419 or with BHT-412-SI-66 incorporated.
2. Seat bucket shifted to increase clearance between copilot seat and copilot collective.
3. Attenuator assembly shifted to match position of bucket.

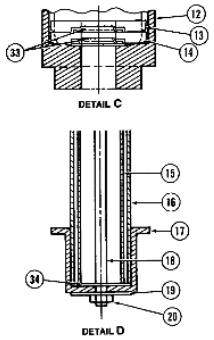
ICN-412-A-251002-B-97499-00004-A-001-01

Figure 1. Copilot Seat (Sheet 3 of 3)



ICN-412-A-251002-B-97499-00005-A-001-01

Figure 2. Crew Seat Attenuator (Sheet 1 of 2)



1. Pin
2. Washer
3. Screw
4. Plug
5. Adjustment tube
6. Spring guide
7. Nut
8. Washer
9. Spring
10. Guide pin
11. Washer
12. Guide tube assembly
13. Retainer
14. Retainer
15. Attenuator
16. Attenuator assembly
17. Support assembly
18. Rod
19. Washer
20. Nut
21. Screw
22. Seat frame
23. Strut
24. Shim
25. Washer
26. Nut
27. Pin
28. Spring
29. Washer
30. Retainer
31. Nut
32. Control Cable
33. Fasteners
34. Washer

ICN-412-A-251002-B-97499-00006-A-001-01

Figure 2. Crew Seat Attenuator (Sheet 2 of 2)

Crew Seats - Repairs

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-317	AR	
Denatured alcohol	C-326	AR	
Adhesive	C-331	AR	

Procedure

NOTE

The following repair procedures are referenced to the copilot seat assembly (1, Figure 1) but can also be applied to the pilot seat.

1. For repair procedure for restraint assembly (inertia reel, shoulder harness and seat belt) (412-A-25-10-02-01A-685A-A / 00256).

2. For replacement of vertical adjustment cable, refer to 412-A-25-10-02-02A-920A-A / 00258.

NOTE

Only limited structural repairs to bucket assembly are permitted. Contact Product Support Engineering.

3. Replace cushions as follows:

3.1. Remove two screws (6, Figure 1).

3.2. Remove bottom seat cushion (7).

3.3. Remove back seat cushion (8).

3.4. Position back seat cushion (8) in seat.

3.5. Position bottom seat cushion (7) in seat.

3.6. Install screws (6).

NOTE

Repairs to attenuator assembly are limited to replacement of defective parts and adjustments only.

4. Adjust attenuator assembly (16, Figure 2) as follows:

4.1. Remove seat assembly (412-A-25-10-02-00A-520A-A / 00248) from helicopter and place on clean work bench with bucket face down.



ENSURE THAT SEAT ADJUSTMENT IS FULLY EXTENDED TO AVOID INJURY WHEN REMOVING CONTROL CABLE FROM GUIDE TUBE ASSEMBLY.

4.2. Loosen nut (31, Figure 2) on control cable (32). Slowly extract control cable with retainer (30), washer (29), spring (28) and pin (27) from guide tube assembly (12). Save parts for re-use.

4.3. Remove attenuator assembly (16), guide tube assembly (12) and adjustment tube (5) as a unit from seat as follows:



RETAIN TAPERED SHIMS AND STRUTS IN THEIR RESPECTIVE LOCATION DURING REMOVAL OF SUPPORT FROM SEAT FRAME. MIXING OR LOSS OF SHIM ORIENTATION COULD AFFECT SEAT FRAME ALIGNMENT DURING REASSEMBLY.

4.3.1. Remove four nuts (26), washers (25) from screws (21) securing support assembly (17), shims (24), and strut (23) to left and right lower seat frame (22).

4.3.2. Remove two screws (3) and washers (2) securing pin (1) to seat back.

4.4. Remove guide pin (10) and washer (11) from guide tube assembly (12). This guide pin is securing adjustment tube (5) and once removed will allow tube to fully extend.

4.5. Remove adjustment tube (5) and spring (9) from guide tube assembly (12).



DO NOT REMOVE ROD (18) THROUGH LOWER SUPPORT ASSEMBLY (17) . RETAINERS (13 AND 14) ARE BONDED IN PLACE AND CONTAIN LOCKING FASTENERS THAT COULD BE DAMAGED.

4.6. Remove lower nut (20) and washer (19) from rod (18). Remove rod (18) from guide tube assembly end only, by pushing rod up through guide tube assembly (12).

4.7. Separate guide tube assembly (12) from support assembly (17). Remove and inspect attenuator (15) for deterioration at the bevelled end. If the bevel is worn flat or deteriorated, measure tube to confirm overall length of 9.05 to 9.20 inches (229.87 to 233.68 mm) is present. Replace tube if dimension is not met.

NOTE

These procedures are based on compliance of T.B. 412-94-126.

4.8. Inspect lower internal surface of support assembly (17) for damaged bonded washer (34, detail D). If damaged replace as follows:

4.8.1. Clean washer and lower internal surface of attenuator support assembly (17) with a cloth moistened with denatured alcohol (C-326).

4.8.2. Apply a thin bead of adhesive (C-317) to the washer (34) and carefully position washer into the support (17), taking care not to get any adhesive on the inside diameter of attenuator assembly (16).

4.9. Install attenuator (15) into support assembly (17) with bevel edge down.

4.10. If retainers (13 and 14) are loose, install replacement fasteners (33, Detail C) into retainers. Bond retainers into guide tube assembly (12) with adhesive (C-331).



TAKE CARE IN POSITIONING ROD (18) AS FASTENERS (33) INSTALLED INTO RETAINERS (13 AND 14) MAY BE DAMAGED.

4.11. Thread nut (20) onto rod (18) to allow approximately one thread to be exposed. Position washer (19) on rod and install rod through support assembly (17) lower end up through guide tube assembly (12). Ensure sufficient threads on rod (18) are present to allow installation of nut (7) and washers (8) at guide assembly (12) end. Torque nut (7) 15 to 25 in.-lbs. (1.69 to 2.82 N-m).

4.12. Install spring (9) into guide tube assembly (12) and position spring guide (6) onto spring.

4.13. Position adjustment tube (5) into the guide tube assembly (12) and align first hole on the adjustment tube (5) to enable installation of the guide pin (10). Insert guide pin (10) with washer (11) and secure in place.

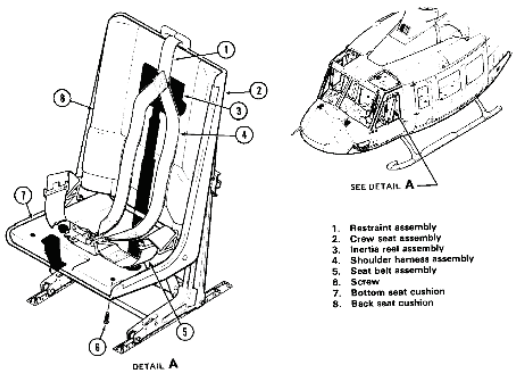
4.14. Check dimension of attenuator assembly (16) per Figure 2.

4.15. Position attenuator assembly (16) on seat back and install two screws (3) with washers (2) through pin (1). Install four screws (21) through strut (23), seat frame (22), shims (24) and support assembly (17) with washers (25) and nuts (26). Tighten screws (3 and 21).

4.16. Install guide pin (10) with washer (11) into tube assembly (12).

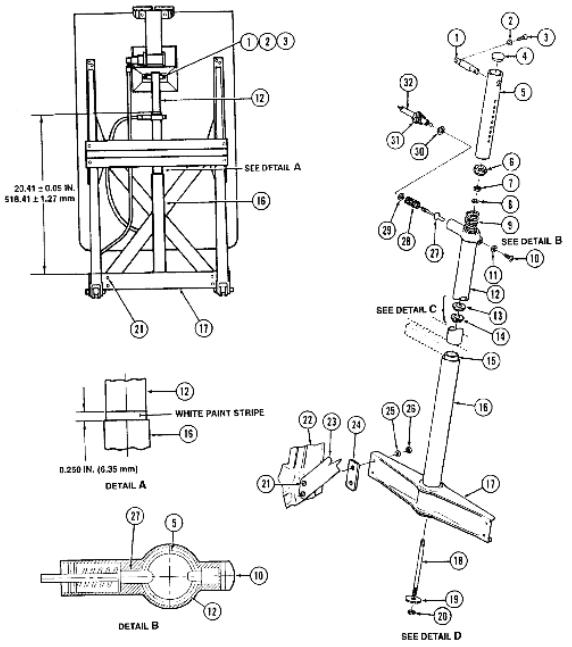
4.17. Position control cable (32) with retainer (30), washer (29), spring (28), and pin (27) assembled. Tighten nut (31).

4.18. Install seat into helicopter (412-A-25-10-02-00A-720A-A / 00251).



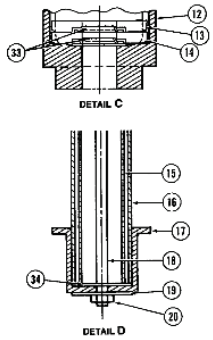
ICN-412-A-251002-B-97499-00007-A-001-01

Figure 1. Crew Seat Assembly (Sheet 1 of 1)



ICN-412-A-251002-B-97499-00005-A-001-01

Figure 2. Crew Seat Attenuator (Sheet 1 of 2)



1. Pin
2. Washer
3. Screw
4. Plug
5. Adjustment tube
6. Spring guide
7. Nut
8. Washer
9. Spring
10. Guide pin
11. Washer
12. Guide tube assembly
13. Retainer
14. Retainer
15. Attenuator
16. Attenuator assembly
17. Support assembly
18. Rod
19. Washer
20. Nut
21. Screw
22. Seat frame
23. Strut
24. Shim
25. Washer
26. Nut
27. Pin
28. Spring
29. Washer
30. Retainer
31. Nut
32. Control Cable
33. Fasteners
34. Washer

ICN-412-A-251002-B-97499-00006-A-001-01

Figure 2. Crew Seat Attenuator (Sheet 2 of 2)

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Crew Seats - Installation

Procedure

NOTE

The following installation procedures are referenced to the copilot seat assembly (1, Figure 1) but can also be applied to the pilot seat.

NOTE

SN: 36087-Subsequent:

If installing copilot seat assembly (1, Figure 1) from forward end of tracks (9 and 11) on 412EP , the copilot seat assembly (1) may be removed/installed after removing screws (8) and forward stop screws (7) on forward end of tracks (9 and 11).

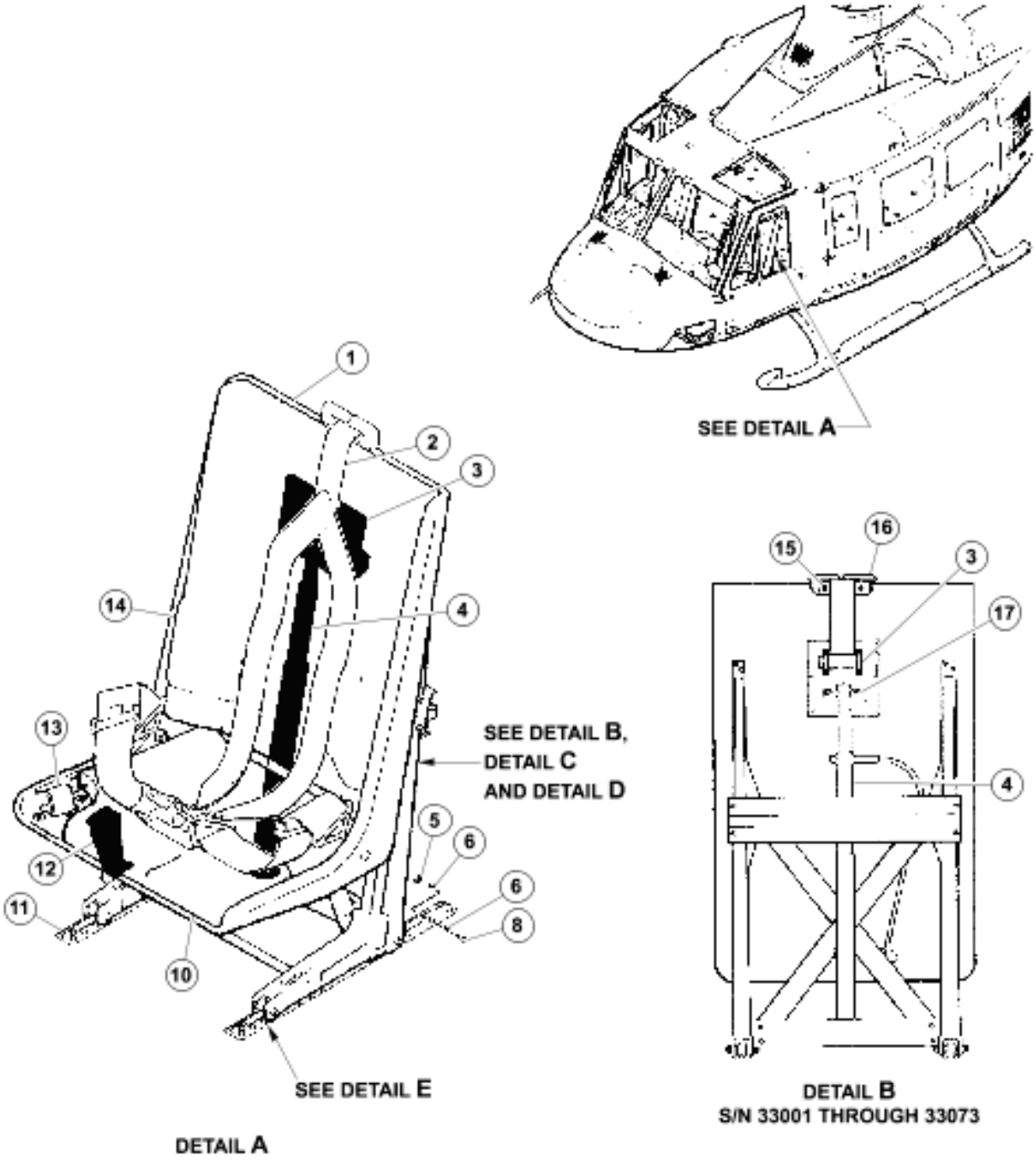
1. Position copilot seat assembly (1) on tracks (9 and 11).



SEAT HAS HELPER SPRING TO AID VERTICAL ADJUSTMENT. SEAT WILL MOVE RAPIDLY UPWARD FROM FULL DOWN POSITION IF NO WEIGHT IS ON SEAT.

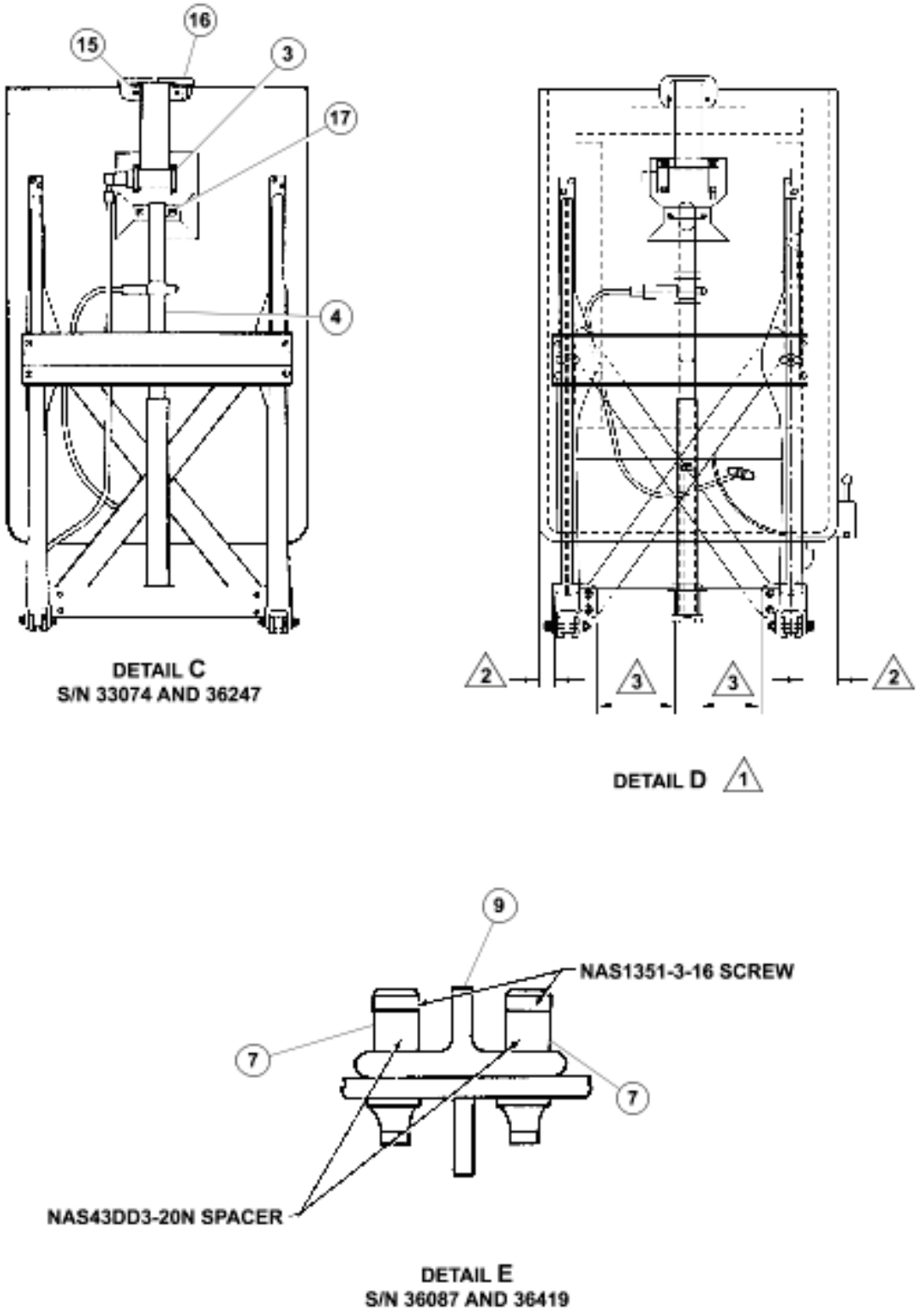
2. Move fore and aft adjustment lever (12) up and push seat assembly (1) onto tracks (9 and 11).

3. Install screw (8), washers (6), nut (5), and forward stop screws (7) on track (9). Install similar screw and forward stop screws (not illustrated) on track (11).



ICN-412-A-251002-B-97499-00002-A-001-01

Figure 1. Copilot Seat (Sheet 1 of 3)



ICN-412-A-251002-B-97499-00003-A-001-01

Figure 1. Copilot Seat (Sheet 2 of 3)

1. Copilot seat assembly
2. Restraint assembly
3. Inertia reel assembly
4. Attenuator (energy absorption device)
5. Nut
6. Washer
7. Forward stop screws
8. Screw
9. Track (left)
10. Bottom seat cushion
11. Track (right)
12. Fore and aft adjustment lever
13. Vertical adjustment handle
14. Back seat cushion
15. Screw
16. Guide (restraint assembly)
17. Screw

NOTES

1. For helicopter S/N 36248 through 36419 or with BHT-412-SI-66 incorporated.
2. Seat bucket shifted to increase clearance between copilot seat and copilot collective.
3. Attenuator assembly shifted to match position of bucket.

ICN-412-A-251002-B-97499-00004-A-001-01

Figure 1. Copilot Seat (Sheet 3 of 3)

Crew Seat Restraint Assembly - Description

NOTE

Maintenance procedures for pilot and copilot restraint assemblies are identical. The maintenance procedures in 412-A-25-10-02-01A-520A-A / 00253, 412-A-25-10-00-00A-280A-A / 00254, 412-A-25-10-02-01A-280A-A / 00255, 412-A-25-10-02-01A-685A-A / 00256 and 412-A-25-10-02-01A-720A-A / 00257 are keyed to the copilot assembly.

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Crew Seat Restraint Assembly - Removal

Procedure

NOTE

Maintenance procedures for pilot and copilot restraint assemblies are identical. The following removal procedures are keyed to the copilot assembly.

1. Unfasten buckle (7, Figure 1) and separate shoulder harness (4) om seat belt (6).

2. Disengage shoulder harness (4) from slot in guide (1).

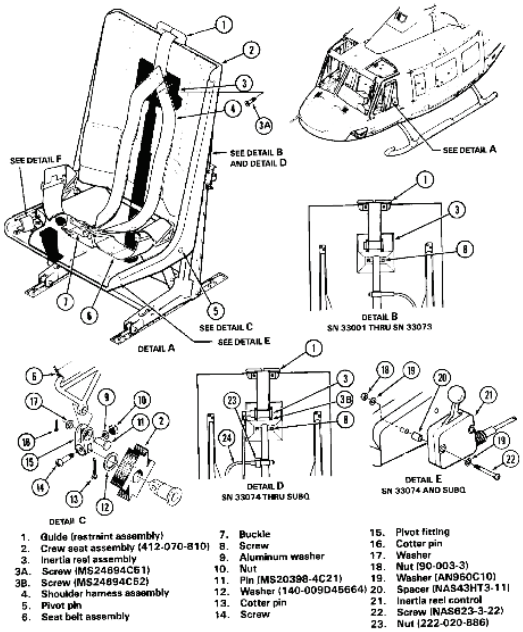
Applicable: SN: 33001-33073:

3. Remove four screws (3A) and remove shoulder harness (4) and inertia reel (3) as an assembly.

Applicable: SN: 33074-33213 or 36001-Subsequent:

4. Remove four screws (3B, detail D), two nuts (18, Detail E), four washers (19), two spacers (20), and two screws (22). Remove shoulder harness (4, Detail A) and inertia reel (3) as an assembly.

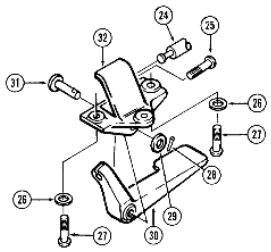
5. Remove cotter pin (16). Remove pin (11) and remove one-half of seat belt (6). Remove opposite half of seat belt in same manner.



1. Guide (restraint assembly)
2. Crew seat assembly (412-070-810)
3. Inertia reel assembly
3A. Screw (MS24694C61)
3B. Screw (MS24694C62)
4. Shoulder harness assembly
5. Pivot pin
6. Seat belt assembly
7. Buckle
8. Screw
9. Aluminum washer
10. Nut
11. Pin (MS20398-4C21)
12. Washer (140-009D45664)
13. Cotter pin
14. Screw
15. Pivot fitting
16. Cotter pin
17. Washer
18. Nut (90-003-3)
19. Washer (AN960C10)
20. Spacer (NAS43HT3-11)
21. Inertia reel control
22. Screw (NAS623-3-22)
23. Nut (222-020-886)

ICN-412-A-251002-B-97499-00008-A-001-01

Figure 1. Crew Seat Restraint (Sheet 1 of 2)



DETAIL F TYPICAL

24. Vertical adjustment cable (222-320-802)
25. Screw, cable adjustment (AN520-8R12)
26. Washer (AN960PD10L)
27. Screw (AN525-10R12)
28. Washer (AN960PD10L)
29. Cotter pin (MS24665-134)
30. Handle (222-020-884-103)
31. Pin (MS20392-2C37)
32. Support assembly (222-020-885-101)

ICN-412-A-251002-B-97499-00009-A-001-01

Figure 1. Crew Seat Restraint (Sheet 2 of 2)

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Restraint System - Check and Inspection Guidelines

Procedure

NOTE

Maintenance procedures for pilot and copilot restraint assemblies are identical. The following guidelines are keyed to the copilot assembly.

1. Make sure that the restraint system is clean and does not contain dirt, oil, or grease, or other unwanted particles or substances.

2. Make sure that all the parts are satisfactorily attached.

NOTE

Slight wear of the webbing is permitted. However, excessive web wear that has progressed to cut or worn edges must be replaced per the following step.

3. Inspect webbing for:

- Cut or worn edges
- Damaged stitching
- Broken fabric threads
- Tears
- Excessive chafe marks
- Excessive wear
- Excessive fusing

3.1. Replace webbing if any of the following are observed:

- A limited amount of frayed webbing will retain sufficient strength necessary to meet the required strength. Frayed webbing is defined as broken filaments from either the warp (longitudinal) yarns or the filler (transverse) yarns. The number of broken filaments should not be sufficient to obscure the identity of any yarn when viewed from a distance of about 8 inches.
- Any web that is cut or torn on the edge.
- The amount of fray should be limited to an amount that is less than 10% of the width of the webbing and is not to exceed 8 inches in length.
- Webbing with more than 15 broken yarns in locations other than the edge.
- Webbing that is frayed or distorted sufficiently to cause improper operation of any portion of the restraint system.
- A total of 15 stitches can be torn in a stitch pattern before the webbing needs replacement.

3.2. Make sure that you can read the markings.

NOTE

Burrs, nicks, and scratches is material raised above the normal surface. This material must be removed to make sure that the parts will fit together correctly.

Dents must not damage the finish or functional operation of any parts.

4. Examine fitting for:

- Burrs, nicks or scratches
- Cracks
- Dents
- Corrosion

5. Examine buckle and connector for:

- Burrs, nicks or scratches
- Cracks
- Dents
- Corrosion

6. Examine reel assembly for:

- Burrs, nicks or scratches
- Cracks

- Dents
- Corrosion

6.1. Make sure that movable parts operate correctly.

Crew Seat Restraint Assembly - Inspection

Procedure

NOTE

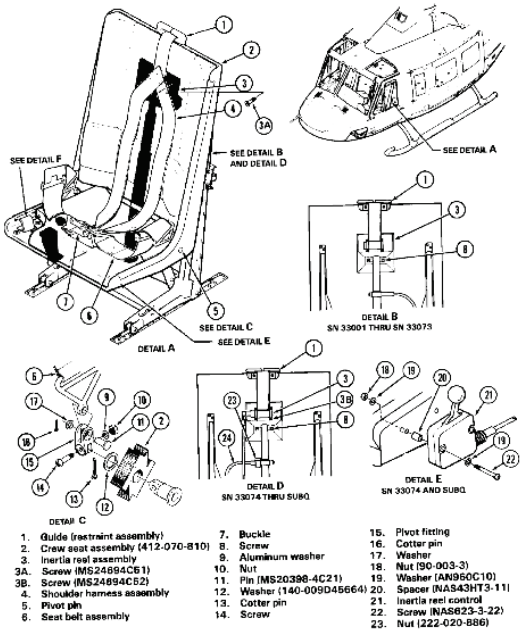
Maintenance procedures for pilot and copilot restraint assemblies are identical. The following inspection procedures are keyed to the copilot assembly.

1. Inspect seat belt buckle (7, Figure 1) for proper operation and damage. Examine buckle for burrs, nicks, scratches, cracks, dents and corrosion. Burrs, nicks, and scratches are material that is raised above the normal surface. This material must be removed to make sure the parts will fit together correctly. Dents must not damage the finish or functional operation of any part.

2. Inspect seat belt (6) shoulder harness (4) for wear, damage, and secure installation. Examine belts and harness for cut or worn edges, damaged stitching, fray, tears, chafe marks, wear. Slight wear of the webbing is permitted. However, a belt or shoulder harness that has excessive wear that has progressed to cut or worn edges must be replaced. Any belt or harness that is cut or torn deeper than 0.10 inch (2.54 mm) on the edge shall be replaced. Fray is defined as broken filaments from either the warp (longitudinal) yarns or the filler (transverse) yarns. The number of broken filaments should not be sufficient to obscure the identity of any yarn when viewed from a distance of about 8 inches (203.2 mm).

3. The amount of fray shall be limited to 10% of the width of the belt or harness, not to exceed 8 inches (203.2 mm) or a maximum of 15 broken yarns in locations other than edges. Replace any belt or harness that is frayed or distorted sufficiently to cause improper operation of any portion of the restraint assembly. A maximum of 15 torn stitches is permitted in a stitch pattern.

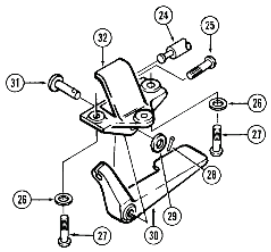
4. Inspect inertia reel assembly (3) for secure installation and smooth operation.



1. Guide (restraint assembly)
2. Crew seat assembly (412-070-810)
3. Inertia reel assembly
3A. Screw (MS24694C61)
3B. Screw (MS24694C62)
4. Shoulder harness assembly
5. Pivot pin
6. Seat belt assembly
7. Buckle
8. Screw
9. Aluminum washer
10. Nut
11. Pin (MS20398-4C21)
12. Washer (140-009D45664)
13. Cotter pin
14. Screw
15. Pivot fitting
16. Cotter pin
17. Washer
18. Nut (90-003-3)
19. Washer (AN960C10)
20. Spacer (NAS43HT3-11)
21. Inertia reel control
22. Screw (NAS623-3-22)
23. Nut (222-020-886)

ICN-412-A-251002-B-97499-00008-A-001-01

Figure 1. Crew Seat Restraint (Sheet 1 of 2)



DETAIL F TYPICAL

24. Vertical adjustment cable (222-320-802)
25. Screw, cable adjustment (AN520-8R12)
26. Washer (AN960PD10L)
27. Screw (AN525-10R12)
28. Washer (AN960PD10L)
29. Cotter pin (MS24665-134)
30. Handle (222-020-884-103)
31. Pin (MS20392-2C37)
32. Support assembly (222-020-885-101)

ICN-412-A-251002-B-97499-00009-A-001-01

Figure 1. Crew Seat Restraint (Sheet 2 of 2)

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Applicable to: ALL
ECCN EAR99

Crew Seat Restraint Assembly - Repair

Procedure

1.

NOTE

Maintenance procedures for pilot and copilot restraint assemblies are identical. The following informations are keyed to the copilot assembly.

NOTE

Repair of crew seat restraint assembly is limited to replacement of unserviceable parts (412-A-25-10-02-01A-280A-A / 00255).

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Crew Seat Restraint Assembly - Installation

Procedure

NOTE

Maintenance procedures for pilot and copilot restraint assemblies are identical. The following installation procedures are keyed to the copilot assembly.

1. Position seat belt (6, Figure 1) on seat (2). Install pin (11), washer (17), and cotter pin (16).

2. Install opposite half of seat belt (6) in the same manner outlined in preceding step.

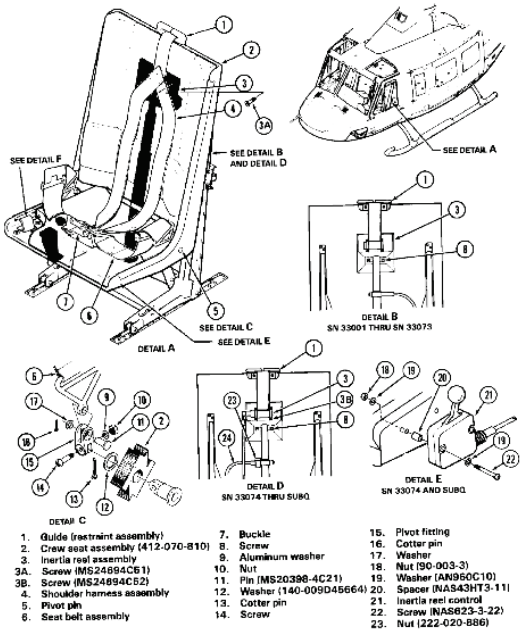
Applicable: SN: 33001-33073:

3. Position shoulder harness (4) and inertia reel (3) on seat (2). Install four screws (3A).

Applicable: SN: 33074-33213 or 36001-Subsequent:

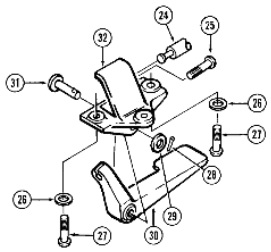
4. Position shoulder harness (4), inertia reel (3, detail D), and inertia reel control (21, detail E) on seat (2, detail A). Install two screws (22, detail E), four washers (19), two spacers (20), and two nuts (18). Install four screws (3B, detail D).

5. Install shoulder harness strap (4) in slot in guide (1).



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Figure 1. Crew Seat Restraint (Sheet 1 of 2)



DETAIL F TYPICAL

24. Vertical adjustment cable (222-320-802)
25. Screw, cable adjustment (AN520-8R12)
26. Washer (AN960PD10L)
27. Screw (AN525-10R12)
28. Washer (AN960PD10L)
29. Cotter pin (MS24665-134)
30. Handle (222-020-884-103)
31. Pin (MS20392-2C37)
32. Support assembly (222-020-885-101)

ICN-412-A-251002-B-97499-00009-A-001-01

Figure 1. Crew Seat Restraint (Sheet 2 of 2)

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Crew Seat Vertical Adjustment Cable - Removal and Installation

Procedure

1. Removal (Figure 1, detail F).

1.1. Remove nut (23) holding top of vertical adjustment cable (24) on seat back.

1.2. Remove two screws securing seat bottom cushion (10, Figure 2) in place and remove cushion (10).

1.3. Remove two retaining clips holding vertical adjustment cable (24, Figure 1) in place on support assembly (32).

1.4. Remove cotter pin (29), washer (28), and pin (31) securing seat height adjustment handle (30) located on bottom of seat (2).

1.5. Remove three screws (27) and washers (26) holding support assembly (32) located on right bottom side of seat (2).

1.6. Disconnect vertical adjustment cable (24) from seat height adjustment handle (30) located on bottom of seat (2).

1.7. Remove two retainer clips holding vertical adjustment cable (24) on support assembly (32).

NOTE

Note how cable is routed prior to removal.

1.8. Disconnect vertical adjustment cable (24) from vertical adjustment locking pin and pull cable from back of seat.

2. Installation.

2.1. Connect top of cable (24) to vertical adjustment lock pin located on seat back inside of nut (23).

2.2. Screw nut (23) in two turns to secure cable (24).

2.3. Secure support assembly (32) to bottom of seat (2) with three screws (27) and washers (26).

2.4. Insert handle (30) inside of support assembly (32) and attach vertical adjustment cable (24).

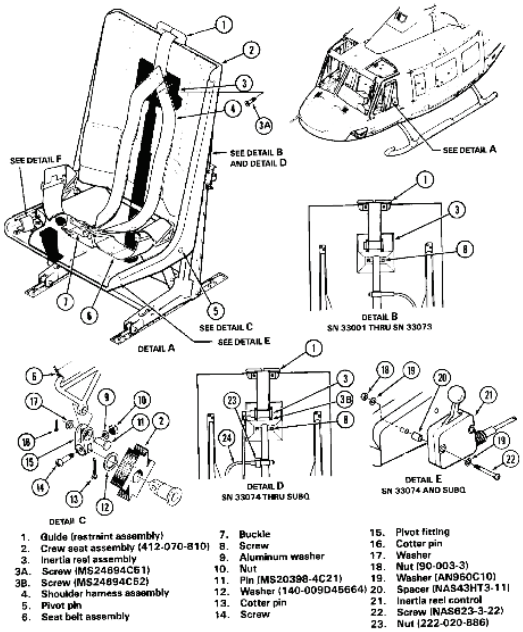
2.5. Attach handle (30) to support assembly (32) with pin (31), washer (28), and cotter pin (29).

2.6. Secure vertical adjustment cable assembly (24) to support assembly (32) with two retainer clips.

2.7. Tighten nut (23).

2.8. Adjust screw (25) to remove al l slack from vertical adjustment cable (24). After all slack has been removed from vertical adjustment cable (24), turn screw (25) 1/2 additional turn or bottom out screw, which ever occurs first.

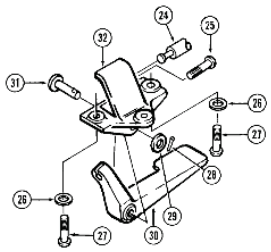
2.9. Position cushion (10, Figure 2) on seat (2, Figure 1) and secure with two screws.



1. Guide (restraint assembly)
2. Crew seat assembly (412-070-810)
3. Inertia reel assembly
3A. Screw (MS24694C61)
3B. Screw (MS24694C62)
4. Shoulder harness assembly
5. Pivot pin
6. Seat belt assembly
7. Buckle
8. Screw
9. Aluminum washer
10. Nut
11. Pin (MS20398-4C21)
12. Washer (140-009D45664)
13. Cotter pin
14. Screw
15. Pivot fitting
16. Cotter pin
17. Washer
18. Nut (90-003-3)
19. Washer (AN960C10)
20. Spacer (NAS43HT3-11)
21. Inertia reel control
22. Screw (NAS623-3-22)
23. Nut (222-020-886)

ICN-412-A-251002-B-97499-00008-A-001-01

Figure 1. Crew Seat Restraint (Sheet 1 of 2)

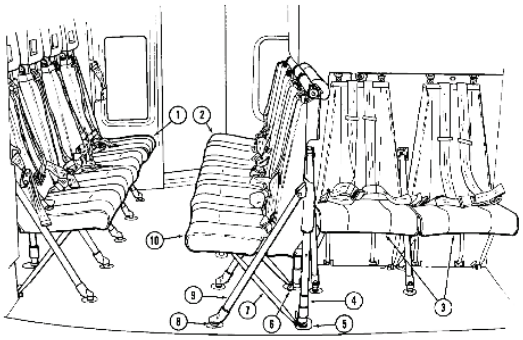


DETAIL F TYPICAL

24. Vertical adjustment cable (222-320-802)
25. Screw, cable adjustment (AN520-8R12)
26. Washer (AN960PD10L)
27. Screw (AN525-10R12)
28. Washer (AN960PD10L)
29. Cotter pin (MS24665-134)
30. Handle (222-020-884-103)
31. Pin (MS20392-2C37)
32. Support assembly (222-020-885-101)

ICN-412-A-251002-B-97499-00009-A-001-01

Figure 1. Crew Seat Restraint (Sheet 2 of 2)



1. Four place aft facing seat
2. Five place forward facing seat
3. One place outboard facing seat
4. Free-standing frame (typical)
5. Floor stud (typical)
6. Quick-release pin (typical)
7. Strut (typical)
8. Brace adapter (typical)
9. Brace (typical)
10. Cushion (typical)

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Figure 2. Passenger Seats (Sheet 1 of 1)

Crew Seats - Description

Helicopters with crash attenuating crew seats installed, refer to BHT-412-SI-75 for maintenance procedures.

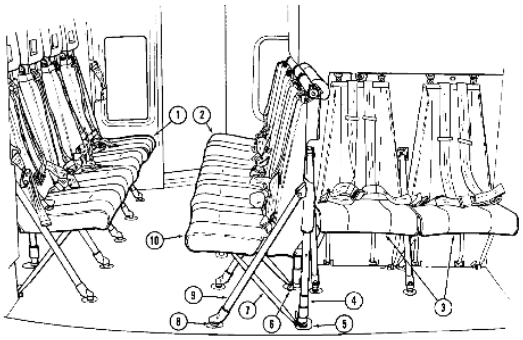
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Passenger Seats - Description

NOTE

For helicopters with Airline Seat Kit (412-706-028) installed, refer to BHT-412-SI-43 for maintenance procedures.

Seating for 13 passengers can be provided in the aft cabin (Figure 1). The forward and aft facing seats are mounted in free-standing frames secured to floor studs. The outboard facing seats are mounted on the pylon bulkhead. All seats are equipped with inertia reel restraint assembly, shoulder harness, seat belt, energy attentuators, and rebound straps. Construction of seat assembly is of low elongation polyester bottom and back panels stretched over tubular framework. Seat assemblies are upholstered with flame retardant materials. An individual seat assembly may be folded upward by removing quick-release pins at base of struts.



1. Four place aft facing seat
2. Five place forward facing seat
3. One place outboard facing seat
4. Free-standing frame (typical)
5. Floor stud (typical)
6. Quick-release pin (typical)
7. Strut (typical)
8. Brace adapter (typical)
9. Brace (typical)
10. Cushion (typical)

ICN-412-A-251002-B-97499-00010-A-001-01

Figure 1. Passenger Seats (Sheet 1 of 1)

Passenger Seats - Folding

Procedure

NOTE

Procedure is typical for all seats.

1. Folding seats.

1.1. Remove cushion (10, Figure 1).

1.2. Remove pins (6) from struts (7).



FOLDING ALL BRACES (9) OF SEATS (1 OR 2) AT ONE TIME MAY CAUSE DAMAGE TO FRAME (4). A MINIMUM OF TWO BRACES (9) PER SEAT SHALL BE INSTALLED ON STUDS (5).

1.3. Remove adapter (8) from stud (5) by pulling release snap ring and raising adapter sleeve to unlocked position.

1.4. Fold up seat bottom, brace (9), and struts (7). Secure with cord.

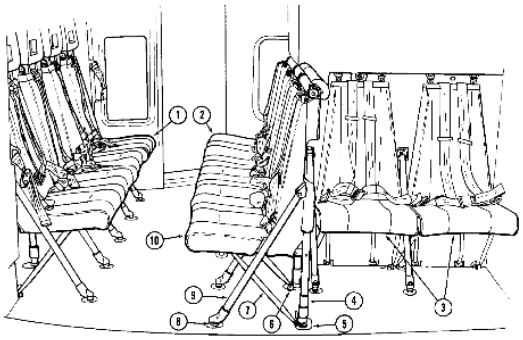
2. Unfolding of seats.

2.1. Fold down seat bottom, brace (9), and struts (7).

2.2. Install adapter (8) on stud (5) by lowering adapter sleeve to locked position.

2.3. Install struts (7) with pins (6).

2.4. Install cushion (10).



1. Four place aft facing seat
2. Five place forward facing seat
3. One place outboard facing seat
4. Free-standing frame (typical)
5. Floor stud (typical)
6. Quick-release pin (typical)
7. Strut (typical)
8. Brace adapter (typical)
9. Brace (typical)
10. Cushion (typical)

ICN-412-A-251002-B-97499-00010-A-001-01

Figure 1. Passenger Seats (Sheet 1 of 1)

Passenger Seats - Inspection

Procedure

NOTE

Procedure is typical for all seats.

1. Aft and forward facing seat free-standing frames:

1.1. Inspect cross straps (11, Figure 1) for wear, damage, and security.

1.2. Inspect frame adapters (10) for operation and security.

1.3. Inspect pins (12) for operation and security.

1.4. Inspect frame (9) for damage.

2. Braces:

2.1. Inspect brace (8) for damage and security.

2.2. Inspect brace adapter (7) for operation and security.

3. Seats:

3.1. Inspect attenuators (2) for damage and security.

3.2. Inspect rebound straps (13) for wear, damage, and security.

3.3. Inspect slides (14) for operation and security. Slides (14) should prevent straps (13) from moving downward except when keeper in slide is pressed.

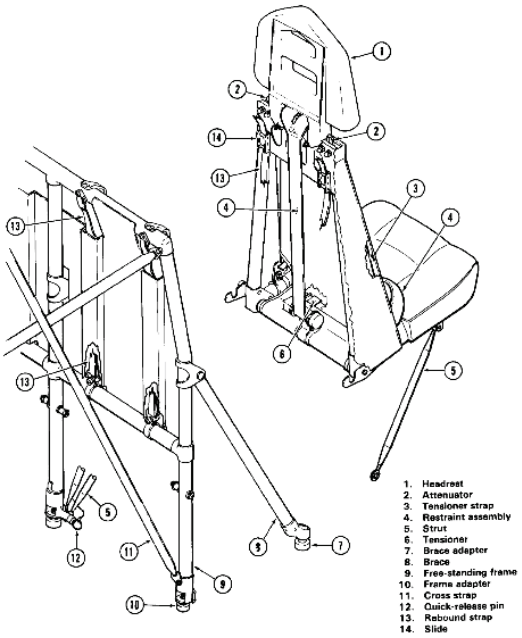
3.4. Inspect tensioner straps (3) for wear, damage, and security.

3.5. Inspect tensioner (6) for operation.

3.6. Inspect struts (5) for damage and security.

3.7. Inspect frame, upholstery, and headrest (1) for wear, damage, and security.

3.8. Inspect restraint assembly (4) (412-A-25-20-00-01A-280A-A / 00265).



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Figure 1. Passenger Seat Inspection (Sheet 1 of 1)

Passenger Seats - Repair

Procedure

1.

NOTE

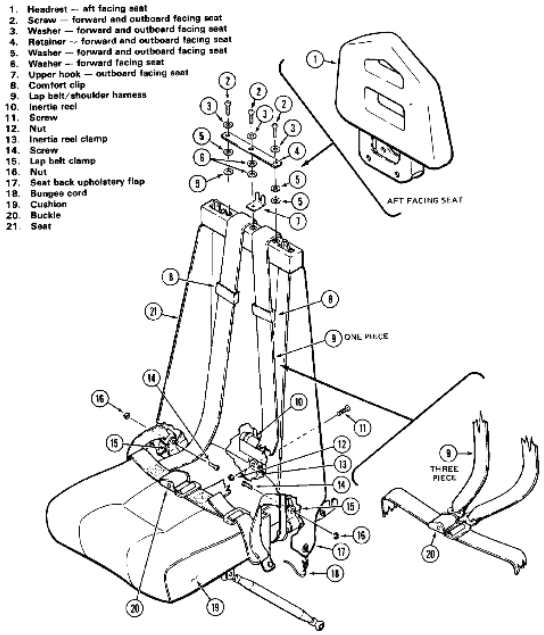
Repair is limited to replacement of unserviceable parts.

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Passenger Seat Restraint Assembly - Removal

Procedure

1. Remove cushion (19, Figure 1).
2. Remove cord (18). Position flaps (17) to expose screws (14).
3. Remove screws (14).
4. Remove screw (11).
5. Forward and outboard facing seats: Remove screws (2), retainer (4), and upper hook (7).
6. Remove lap belt/shoulder harness (9).



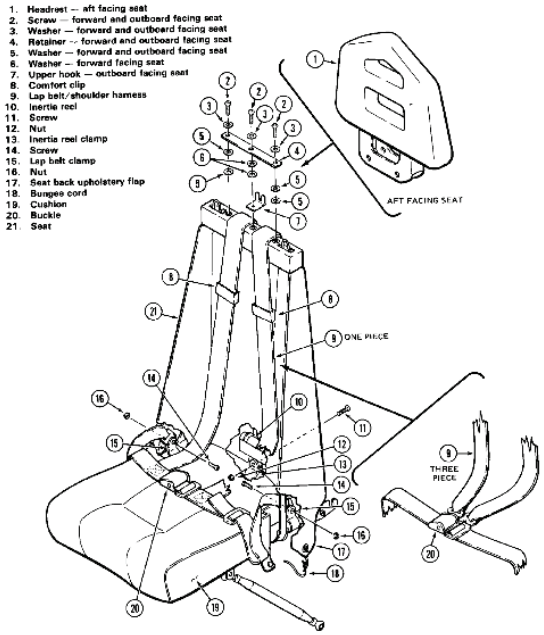
ICN-412-A-252000-B-97499-00002-A-001-01

Figure 1. Passenger Seat Restraint (Sheet 1 of 1)

Passenger Seat Restraint Assembly - Inspection

Procedure

1. Inspect buckle (20, Figure 1) for operation.
2. Inspect lap belt/shoulder harness (9) for wear, damage, and security.
3. Inspect reel (10) for operation and security. Lap belt/shoulder harness (9) should unreel smoothly when a slow continuous pull is applied and reel-up when released. A hard jerk should engage locking mechanism.
4. Inspect two clips (8) for operation and wear.



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Figure 1. Passenger Seat Restraint (Sheet 1 of 1)

Passenger Seat Restraint Assembly - Repair

Procedure

1. Repair is limited to replacement of unserviceable parts.

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Passenger Seat Restraint Assembly - Installation

Procedure

NOTE

Procedure is typical for all seats except as noted.

1. Position lap belt/shoulder harness (9, Figure 1) on seat (21).

1.1. Aft facing seats: Thread lap belt end through lower hole in headrest (1) from back to front.

1.2. Forward facing seats: Position lap belt/ shoulder harness (9) on seat. Install retainer (4) with three screws (2), three washers (3), four washers (5), and two washers (6).

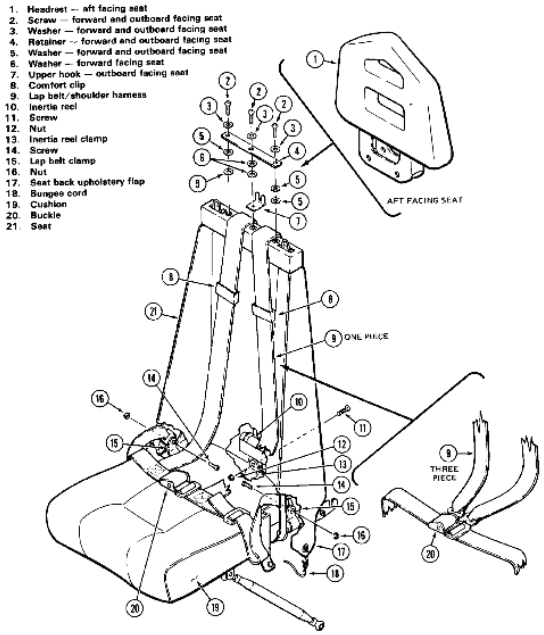
1.3. Outboard facing seats: Position lap belt/ shoulder harness (9) on seat. Install retainer (4) and hook (7) with three screws (2), three washers (3), and four washers (5).

2. Install reel (10) on clamp (13) with screws (11) and nut (12).

3. Install lap belt/shoulder harness (9) on two clamps (15) with two screws (14) and two nuts (16).

4. Position two flaps (17) over two screws (14). Install cord (18).

5. Install cushion (19).



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Figure 1. Passenger Seat Restraint (Sheet 1 of 1)

Outboard Facing Passenger Seat - Description

NOTE

Procedures for both the left and right assemblies are similar. Differences are noted in the detailed maintenance procedures that follow.

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Outboard Facing Passenger Seat - Removal

Procedure

1. Identifying removed parts by name or part number and location will aid during installation.

This procedure removes as many parts as a unit as possible.

1.1. Remove forward and aft braces (23 and 26, Figure 1).

1.1.1. Remove adapter end of braces (23 and 26) from floor studs (51 and 52).

1.1.2. Remove screws/bolts (24 and 27) from bracket end of braces (23 and 26).

1.1.3. Remove braces (23 and 26).

1.2. Disengage studs (1). Remove panel (2).

1.3. Remove forward seat (59).

1.3.1. Open tensioner (61).

1.3.2. Remove pins (38 and 42).

1.3.3. Remove screws (40).



DO NOT ALLOW SEAT TO FALL.

1.3.4. Remove bolts (17 and 19).

1.3.5. Lower seat (59) from spacer (22) and brackets (33 and 35). Remove seat.

1.4. Remove aft seat (54).

1.4.1. Open tensioner (56).

1.4.2. Remove pin (48).

1.4.3. Remove screws (46).



DO NOT ALLOW SEAT TO FALL.

1.4.4. Remove bolts (11 and 13).

1.4.5. Lower seat (54) from spacer (16) and brackets (29 and 31). Remove seat.

1.5. Remove attaching hardware as follows:

1.5.1. Remove screw (1, Figure 2), washer (2), and spacer (3).

1.5.2. Remove screw (4), washer (5), and spacer (6).

1.5.3. Remove screws (8 and 10) and bracket (7).

1.5.4. Remove screws (13) and bracket (12).

1.5.5. Remove screws (16 and 18) and bracket (15).

1.5.6. Remove screws (21 and 23) and bracket (20).

1.5.7. Remove screws (26 and 28) and bracket (25).

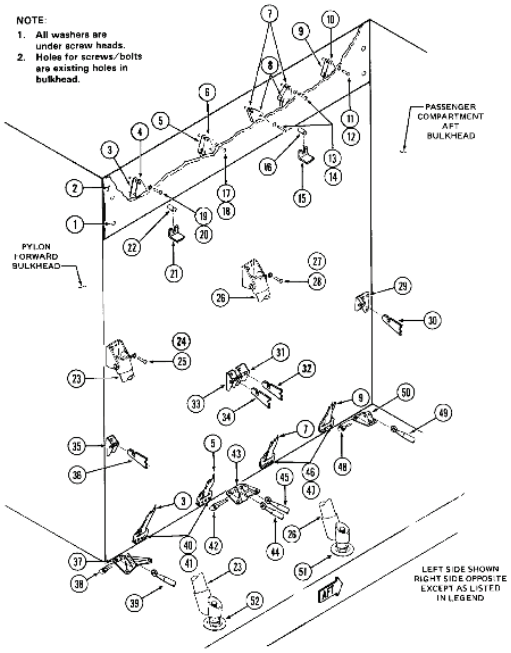
1.5.8. Remove screws (30 and 32) and bracket (29).

1.5.9. Remove screws (34 and 36) and bracket (33).

NOTE

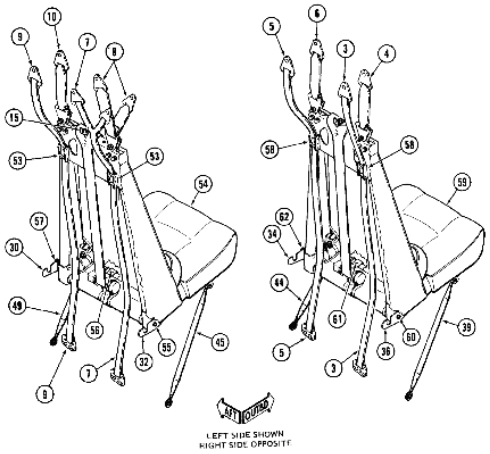
These screws vary in length. Ensure correct screw length is used in each location.

1.5.10. Install 12 screws (43) in holes which were occupied by screws (26, 28, 30, 32, 34, and 36).



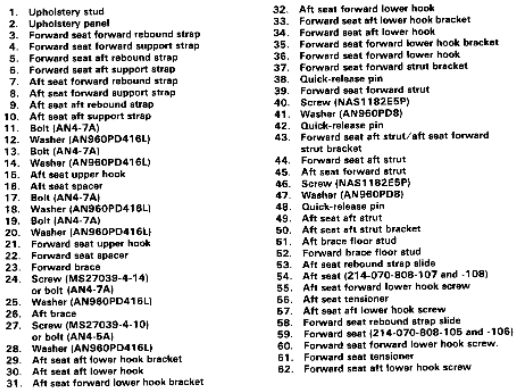
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Figure 1. One Place Outboard Facing Passenger Seat (Sheet 1 of 3)



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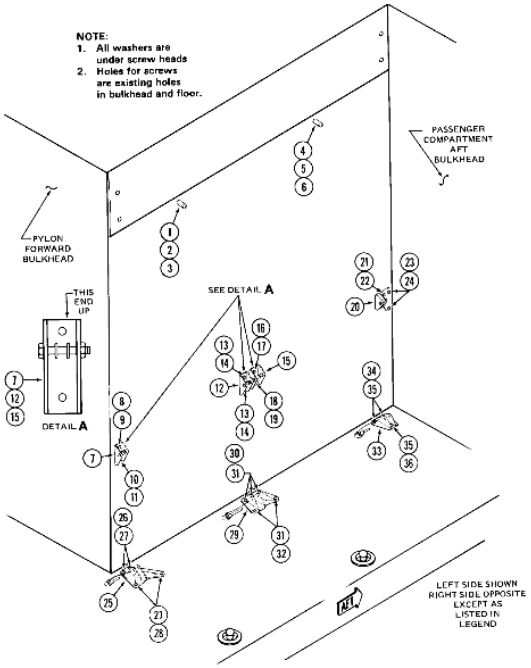
Figure 1. One Place Outboard Facing Passenger Seat (Sheet 2 of 3)



1. Upholstery stud
2. Upholstery panel
3. Forward seat forward rebound strap
4. Forward seat forward support strap
5. Forward seat aft rebound strap
6. Forward seat aft support strap
7. Aft seat forward rebound strap
8. Aft seat forward support strap
9. Aft seat aft rebound strap
10. Aft seat aft support strap
11. Bolt (AN4-7A)
12. Washer (AN960PD416L)
13. Bolt (AN4-7A)
14. Washer (AN960PD416L)
15. Aft seat upper hook
16. Aft seat spacer
17. Bolt (AN4-7A)
18. Washer (AN960PD416L)
19. Bolt (AN4-7A)
20. Washer (AN960PD416L)
21. Forward seat upper hook
22. Forward seat spacer
23. Forward brace
24. Screw (MS27039-4-14) or bolt (AN4-7A)
25. Washer (AN960PD416L)
26. Aft brace
27. Screw (MS27039-4-10) or bolt (AN4-5A)
28. Washer (AN960PD416L)
29. Aft seat aft lower hook bracket
30. Aft seat aft lower hook
31. Aft seat forward lower hook bracket
32. Aft seat forward lower hook
33. Forward seat aft lower hook bracket
34. Forward seat aft lower hook
35. Forward seat forward lower hook bracket
36. Forward seat forward lower hook
37. Forward seat forward strut bracket
38. Quick-release pin
39. Forward seat forward strut
40. Screw (NAS1182E5P)
41. Washer (AN960PD8)
42. Quick-release pin
43. Forward seat aft strut/aft seat forward strut bracket
44. Forward seat aft strut
45. Aft seat forward strut
46. Screw (NAS1182E5P)
47. Washer (AN960PD8)
48. Quick-release pin
49. Aft seat aft strut
50. Aft seat aft strut bracket
51. Aft brace floor stud
52. Forward brace floor stud
53. Aft seat rebound strap slide
54. Aft seat (214-070-808-107 and -108)
55. Aft seat forward lower hook screw
56. Aft seat tensioner
57. Aft seat aft lower hook screw
58. Forward seat rebound strap slide
59. Forward seat (214-070-808-105 and -106)
60. Forward seat forward lower hook screw.
61. Forward seat tensioner
62. Forward seat aft lower hook screw

ICN-412-A-252000-B-97499-00005-A-001-01

Figure 1. One Place Outboard Facing Passenger Seat (Sheet 3 of 3)



ICN-412-A-252000-B-97499-00006-A-001-01

Figure 2. One Place Outboard Facing Seat Attachment Hardware (Sheet 1 of 2)

1. Screw (MS27039-1-16)
2. Washer (AN960-10)
3. Spacer (NAS43HT3-32)
4. Screw (MS27039-1-18)
5. Washer (AN960-10)
6. Spacer (NAS43HT3-32)
7. Left side — Bracket (412-070-889-105)
   Right side — Bracket (412-070-889-106)
8. Screw (MS27039-1-16)
9. Washer (AN960C10L)
10. Screw (MS27039-4-18)
11. Washer (AN960C416L)
12. Bracket (412-070-889-101)
13. Screw (MS27039-1-08)
14. Washer (AN960C10L)
15. Left side — bracket (412-070-889-103)
    Right side — bracket (412-070-889-101)
16. Left side — screw (MS27039-5-08)
    Right side — screw (MS27039-1-08)
17. Left side — washer (AN960C516L)
    Right side — washer (AN960C10L)
18. Screw (MS27019-1-08)
19. Washer (AN960C10L)
20. Left side — bracket (412-070-842-101 or 412-070-890-101)
    Right side - bracket (412-070-842-102 or 412-070-890-102)
21. Screw (MS27039-5-10)
22. Washer (AN960C516L)
23. Screw (MS27039-4-10)
24. Washer (AN960C416L)
25. Left side — bracket (412-070-887-101)
    Right side — bracket (412-070-887-102)
26. Screw (MS27039-1-12)
27. Washer (AN960C10L)
28. Screw (MS27039-1-08)
29. Left side — bracket (412-070-888-101)
    Right side — bracket (412-070-888-102)
30. Screw (MS27039-1-12)
31. Washer (AN960C10L)
32. Screw (MS27039-1-08)
33. Left side — bracket (412-070-886-101)
    Right side — bracket (412-070-886-102)
34. Screw (MS27039-1-09)
35. Washer (AN960C10L)
36. Screw (MS27039-1-08)

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Figure 2. One Place Outboard Facing Seat Attachment Hardware (Sheet 2 of 2)

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Outboard Facing Passenger Seat - Installation

Procedure

NOTE

Procedure is typical for left and right seats, except as noted in legends on Figure 1 and Figure 2.

1. Preparation for seat installation, if not previously accomplished.

NOTE

The length of these screws vary according to use in brackets. Ensure correct length screw is used in each location.

1.1. Remove existing 12 screws from positions to be occupied by screws 26, 28, 30, 32, 34, and 36, (Figure 2). Identify screws by positions and retain for installation.

1.2. Prepare braces (23 and 26, Figure 1) (412-A-25-20-00-02A-723A-A / 00271).

1.3. Prepare seats (54 and 59) (412-A-25-20-00-02A-723A-A / 00271).

1.4. Prepare brackets (29, 31, 33, and 35) (412-A-25-20-00-02A-723A-A / 00271).

1.5. Disengage four studs (1). Remove panel (2).

2. Install attaching hardware as follows:

2.1. Install spacer (13, Figure 2) with washers (2) and screw (1).

2.2. Install spacer (6) with washer (5) and screw (4).



BRACKETS (7, 12, AND 15) SHALL BE INSTALLED AS SHOWN ON Figure 2 TO ENSURE ENGAGEMENT WITH HOOKS (32, 34, AND 36, Figure 1).

2.3. Install bracket (7, Figure 2), screws (8 and 10) and washers (9 and 11).

2.4. Install bracket (12) with two screws (13) and two washers (14).

2.5. Install bracket (15) with screws (16 and 18) and washers (17 and 19).

2.6. Install bracket (20) with screw (21), two screws (23), washer (22), and two washers (24).

2.7. Install bracket (25) with two screws (26), two screws (28), and four washers (27).

2.8. Install bracket (29) with screws (30), screws (32), and washers (31).

2.9. Install bracket (33) with screws (34), screw (36), and washers (35).

3. Install aft seat (54, Figure 1) as follows:

3.1. Loosen screws (55 and 57). Tap two hooks (30 and 32) downward as far as possible with nonmetallic hammer.

3.2. Release keepers of two slides (53). Pull straps (7 and 9) downward as far as possible.

3.3. Open tensioner (56).



SEAT (54) SHALL BE SUPPORTED UNTIL STRAPS (7, 8, 9, AND 10) ARE INSTALLED ON BULKHEAD.

3.4. Position seat (54) with two hooks (30 and 32) around spacer in bracket (29). Center spacer in bracket (31) and hook (15) around spacer (16).

NOTE

Do not pull straps (7 and 9) upward any more than necessary.

3.5. Pull strap (9) upward to align upper screw hole with upper screw hole of strap (10). Install straps (9 and 10) on bulkhead with bolt (11) and washer (12).

3.6. Pull strap (7) upward to align upper screw holes with upper screw holes of strap (8). Install straps (7 and 8) on bulkhead with two bolts (13) and washers (14).

3.7. Install lower end of straps (7 and 9) to bulkhead as far forward as hole patterns will permit. Use two screws (46) and washers (47).

3.8. Apply downward pressure on seat (54) and install struts and engage hooks.

3.8.1. Install aft strut (49) in bracket (50) with pin (48).

3.8.2. Install forward strut (45) in aft side of bracket (43) with pin (42).

NOTE

A 90 degree screwdriver is required to tighten screw (57).

3.8.3. Tap two hooks (30 and 32) upward as far as possible with non-metallic hammer to engage spacer in bracket (29) and center spacer in bracket (31). Tighten screws (55 and 57).

3.9. Close tensioner (56).

4. Forward seat installation:

4.1. Loosen screws (60 and 62, Figure 1). Tap hooks (34 and 36) downward as far as possible with non-metallic hammer.

4.2. Release keepers of two slides (58). Pull straps (3 and 5) downward as far as possible.

4.3. Open tensioner (61).



SEAT (59) MUST BE SUPPORTED UNTIL STRAPS (3, 4, 5, AND 6) ARE INSTALLED ON BULKHEAD.

4.4. Position seat (59) with two hooks (34 and 36) around center spacers in brackets (33 and 35) and hook (21) around spacer (22).

NOTE

Do not pull straps (3 and 5) upward any more than necessary.

4.5. Pull strap (5) upward to align upper screw hole with upper screw hole of strap (6). Install straps (5 and 6) on bulkhead with bolt (17) and washer (18).

4.6. Pull strap (3) upward to align upper screw hole with upper screw hole of strap (4). Install straps (3 and 4) on bulkhead with bolt (19) and washer (20).

4.7. Install lower end of straps (3 and 5) to bulkhead as far forward as hole patterns will permit. Use two screws (40) and washer (41).

4.8. Apply downward pressure on seat (59) and install struts and engage hooks.

4.8.1. Remove pin (42).

4.8.2. Position aft strut (44) in forward end of bracket (43). Install pin (42) in bracket (43) forward seat aft strut (44) and aft seat forward strut (45).

4.8.3. Install forward strut (39) in bracket (37) with pin (38).

NOTE

A 90 degree screwdriver is required to tighten screws (62).

4.8.4. Tap two hooks (34 and 36) upward as far as possible with non-metallic hammer to engage center spacer in brackets (33 and 35). Tighten two screws (60 and 62).

4.9. Close tensioner (61).

5. Install panel (2, Figure 1) with four studs (1).

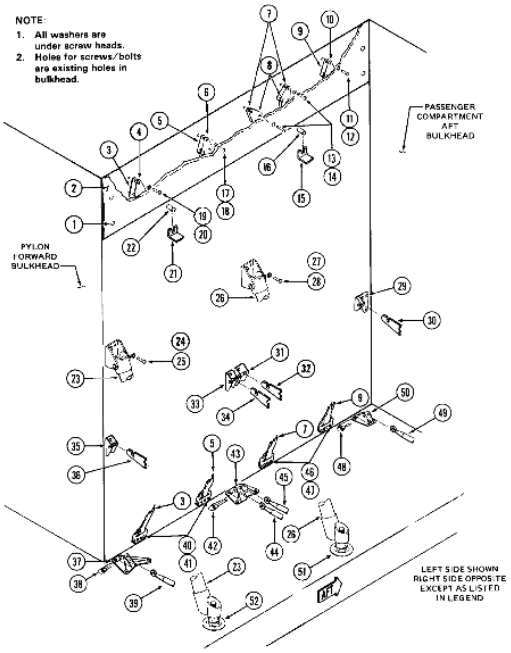
6. Install aft and forward braces:

6.1. Install bracket end of aft brace (26, Figure 1) on bulkhead using aft holes in bracket with two screws/bolts (27) and washers (28).

6.2. Install adapter end of aft brace (26) on floor stud (51).

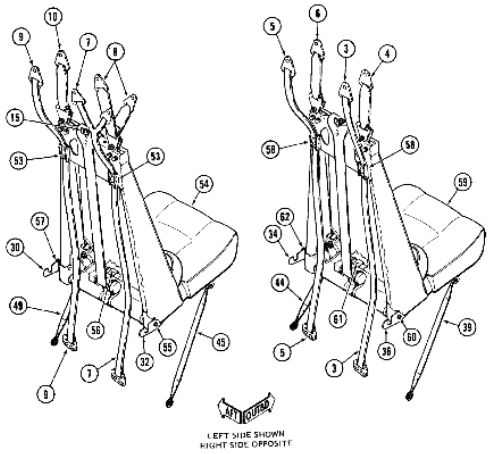
6.3. Install bracket end of forward brace (23) on bulkhead using aft holes in bracket with two screws/ bolts (24) and washers (25).

6.4. Install adapter end of forward brace (23) on floor stud (52).



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Figure 1. One Place Outboard Facing Passenger Seat (Sheet 1 of 3)



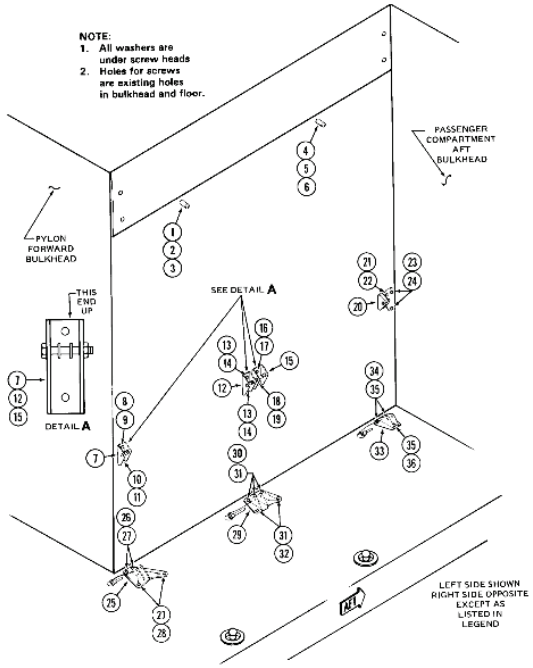
ICN-412-A-252000-B-97499-00004-A-001-01

Figure 1. One Place Outboard Facing Passenger Seat (Sheet 2 of 3)

1. Upholstery stud
2. Upholstery panel
3. Forward seat forward rebound strap
4. Forward seat forward support strap
5. Forward seat aft rebound strap
6. Forward seat aft support strap
7. Aft seat forward rebound strap
8. Aft seat forward support strap
9. Aft seat aft rebound strap
10. Aft seat aft support strap
11. Bolt (AN4-7A)
12. Washer (AN960PD416L)
13. Bolt (AN4-7A)
14. Washer (AN960PD416L)
15. Aft seat upper hook
16. Aft seat spacer
17. Bolt (AN4-7A)
18. Washer (AN960PD416L)
19. Bolt (AN4-7A)
20. Washer (AN960PD416L)
21. Forward seat upper hook
22. Forward seat spacer
23. Forward brace
24. Screw (MS27039-4-14) or bolt (AN4-7A)
25. Washer (AN960PD416L)
26. Aft brace
27. Screw (MS27039-4-10) or bolt (AN4-5A)
28. Washer (AN960PD416L)
29. Aft seat aft lower hook bracket
30. Aft seat aft lower hook
31. Aft seat forward lower hook bracket
32. Aft seat forward lower hook
33. Forward seat aft lower hook bracket
34. Forward seat aft lower hook
35. Forward seat forward lower hook bracket
36. Forward seat forward lower hook
37. Forward seat forward strut bracket
38. Quick-release pin
39. Forward seat forward strut
40. Screw (NAS1182E5P)
41. Washer (AN960PD8)
42. Quick-release pin
43. Forward seat aft strut/aft seat forward strut bracket
44. Forward seat aft strut
45. Aft seat forward strut
46. Screw (NAS1182E5P)
47. Washer (AN960PD8)
48. Quick-release pin
49. Aft seat aft strut
50. Aft seat aft strut bracket
51. Aft brace floor stud
52. Forward brace floor stud
53. Aft seat rebound strap slide
54. Aft seat (214-070-808-107 and -108)
55. Aft seat forward lower hook screw
56. Aft seat tensioner
57. Aft seat aft lower hook screw
58. Forward seat rebound strap slide
59. Forward seat (214-070-808-105 and -106)
60. Forward seat forward lower hook screw.
61. Forward seat tensioner
62. Forward seat aft lower hook screw

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Figure 1. One Place Outboard Facing Passenger Seat (Sheet 3 of 3)



ICN-412-A-252000-B-97499-00006-A-001-01

Figure 2. One Place Outboard Facing Seat Attachment Hardware (Sheet 1 of 2)

1. Screw (MS27039-1-16)
2. Washer (AN960-10)
3. Spacer (NAS43HT3-32)
4. Screw (MS27039-1-18)
5. Washer (AN960-10)
6. Spacer (NAS43HT3-32)
7. Left side — Bracket (412-070-889-105)
   Right side — Bracket (412-070-889-106)
8. Screw (MS27039-1-16)
9. Washer (AN960C10L)
10. Screw (MS27039-4-18)
11. Washer (AN960C416L)
12. Bracket (412-070-889-101)
13. Screw (MS27039-1-08)
14. Washer (AN960C10L)
15. Left side — bracket (412-070-889-103)
    Right side — bracket (412-070-889-101)
16. Left side — screw (MS27039-5-08)
    Right side — screw (MS27039-1-08)
17. Left side — washer (AN960C516L)
    Right side — washer (AN960C10L)
18. Screw (MS27019-1-08)
19. Washer (AN960C10L)
20. Left side — bracket (412-070-842-101 or 412-070-890-101)
    Right side - bracket (412-070-842-102 or 412-070-890-102)
21. Screw (MS27039-5-10)
22. Washer (AN960C516L)
23. Screw (MS27039-4-10)
24. Washer (AN960C416L)
25. Left side — bracket (412-070-887-101)
    Right side — bracket (412-070-887-102)
26. Screw (MS27039-1-12)
27. Washer (AN960C10L)
28. Screw (MS27039-1-08)
29. Left side — bracket (412-070-888-101)
    Right side — bracket (412-070-888-102)
30. Screw (MS27039-1-12)
31. Washer (AN960C10L)
32. Screw (MS27039-1-08)
33. Left side — bracket (412-070-886-101)
    Right side — bracket (412-070-886-102)
34. Screw (MS27039-1-09)
35. Washer (AN960C10L)
36. Screw (MS27039-1-08)

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Figure 2. One Place Outboard Facing Seat Attachment Hardware (Sheet 2 of 2)

Outboard Facing Passenger Seat - Bracket Preparation

Procedure

1. Build-up following brackets as shown on Figure 1.

1.1. 412-070-889-101.

1.2. 412-070-889-103.

1.3. 412-070-889-105.

1.4. 412-070-889-106.

1.5. 412-070-442-101 or 412-070-890-101.

1.6. 412-070-442-102 or 412-070-890-102.

2. Brace Preparation

Build-up brace (214-070-803-117) as shown on Figure 2.

3. Seat Preparation

NOTE

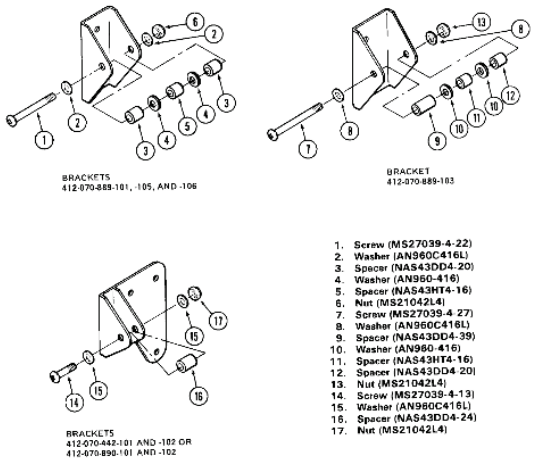
Procedure is typical for all seats, except as noted in legend on Figure 3.

3.1. Determine part number of seat (1, Figure 3) by part number on plate (11) or length or position of straps (10 and 12, detail A).

3.2. Determine part number of straps (2 and 4) by part number on shipping packages or lengths shown in detail B.

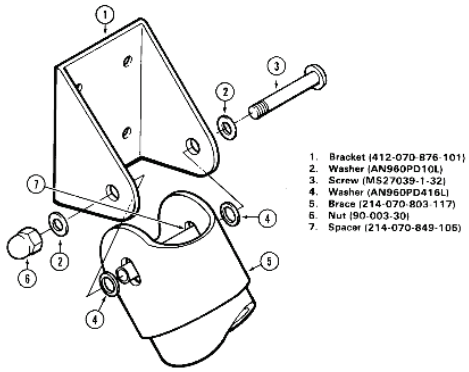
3.3. Install strap (2) on attenuator (3) with screw (9), two washers (7), bearing (8), and nut (6).

3.4. Install strap (4) on attenuator (5) with screw (9), two washers (7), bearing (8), and nut (6).



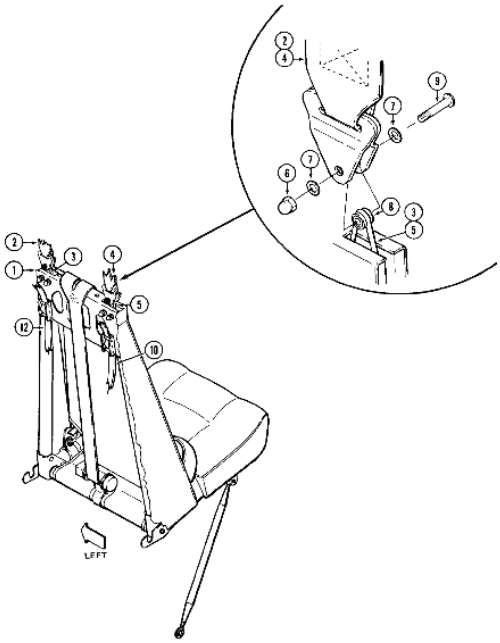
ICN-412-A-252000-B-97499-00008-A-001-01

Figure 1. One Place Outboard Facing Passenger Seat Bracket Preparation (Sheet 1 of 1)



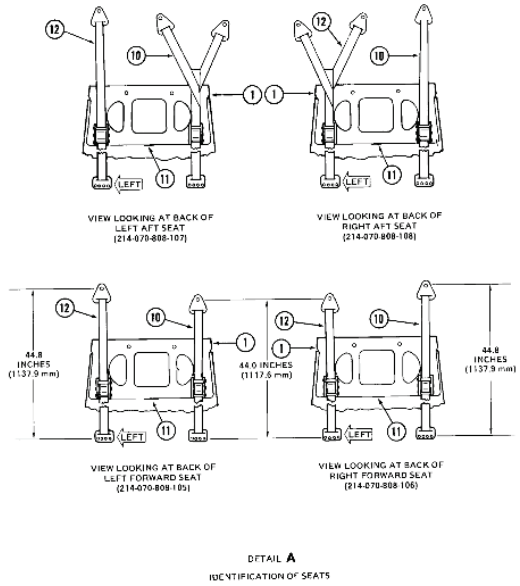
ICN-412-A-252000-B-97499-00009-A-001-01

Figure 2. One Place Outboard Facing Passenger Seat Brace Preparation (Sheet 1 of 1)



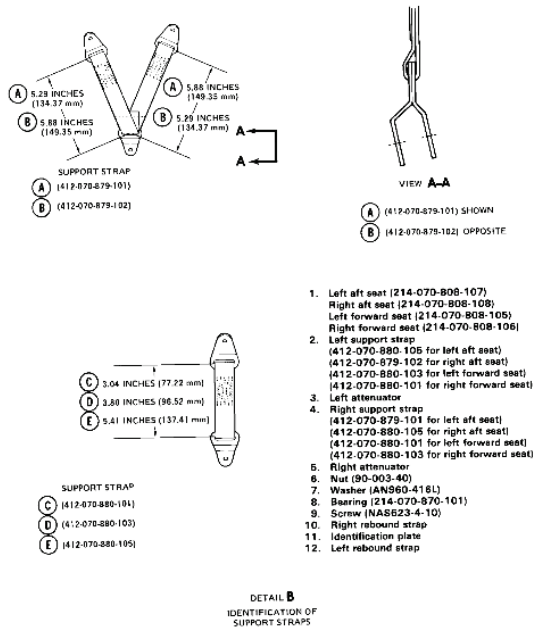
ICN-412-A-252000-B-97499-00010-A-001-01

Figure 3. One Place Outboard Facing Passenger Seat Preparation (Sheet 1 of 3)



ICN-412-A-252000-B-97499-00011-A-001-01

Figure 3. One Place Outboard Facing Passenger Seat Preparation (Sheet 2 of 3)



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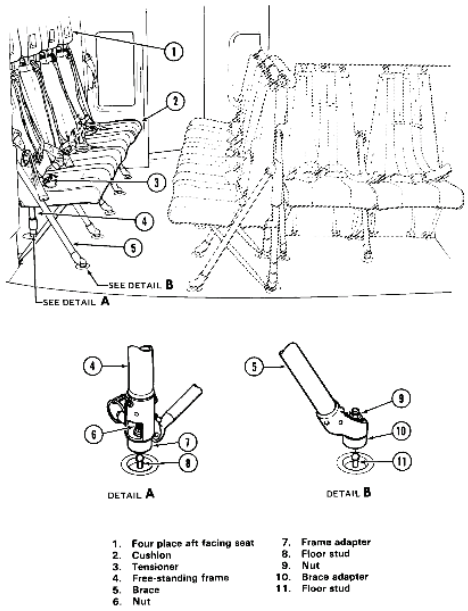
Figure 3. One Place Outboard Facing Passenger Seat Preparation (Sheet 3 of 3)

Four Place Aft Facing Passenger Seat - Removal

Procedure

1. Disengage adapters (7 and 10, Figure 1) from studs (8 and 11) by pulling snap rings and raising adapter to unlocked position.

2. Lift seat (1). Remove seat.



1. Four place aft facing seat
2. Cushion
3. Tensioner
4. Free-standing frame
5. Brace
6. Nut
7. Frame adapter
8. Floor stud
9. Nut
10. Brace adapter
11. Floor stud

ICN-412-A-252000-B-97499-00013-A-001-01

Figure 1. Four Place Aft Facing Passenger Seat (Sheet 1 of 1)

Four Place Aft Facing Passenger Seat - Installation

Procedure

1. Position seat (1, Figure 1) with adapters (7 and 10) over studs (8 and 11).

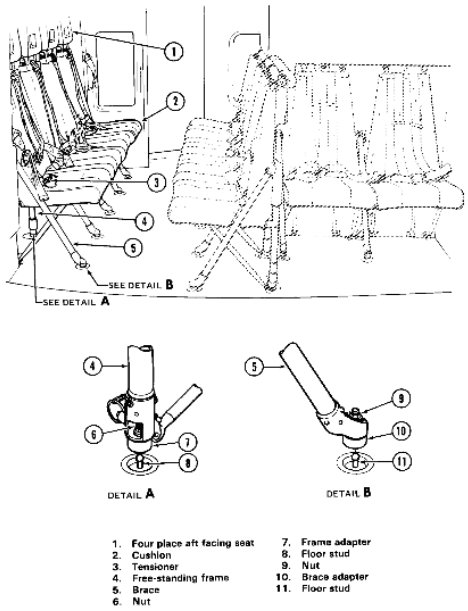
2. Apply downward pressure on frames (4) and braces (5) to engage adapters (7 and 10) on studs (8 and 11).

3. If adapters (7 and 10) and studs (8 and 11) are misaligned, perform one or more of the following to engage adapters (7 and 10).

3.1. Open tensioner (3). Apply downward pressure on frames (4) and braces (5). Close tensioner (3).

3.2. Loosen nuts (6 and 9). Apply downward pressure on cushion (2). Tighten nuts (6 and 9).

3.3. Open tensioner (3). Loosen nuts (6 and 9). Apply downward pressure on cushion (2). Tighten nuts (6 and 9). Close tension (3).



1. Four place aft facing seat
2. Cushion
3. Tensioner
4. Free-standing frame
5. Brace
6. Nut
7. Frame adapter
8. Floor stud
9. Nut
10. Brace adapter
11. Floor stud

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Figure 1. Four Place Aft Facing Passenger Seat (Sheet 1 of 1)

Four Place Aft Facing Passenger Seat - Removal from Frame

Procedure

NOTE

Procedure is typical for all seats.

1. Remove screws (15, Figure 1) and upper end of straps (3) from frame (11).

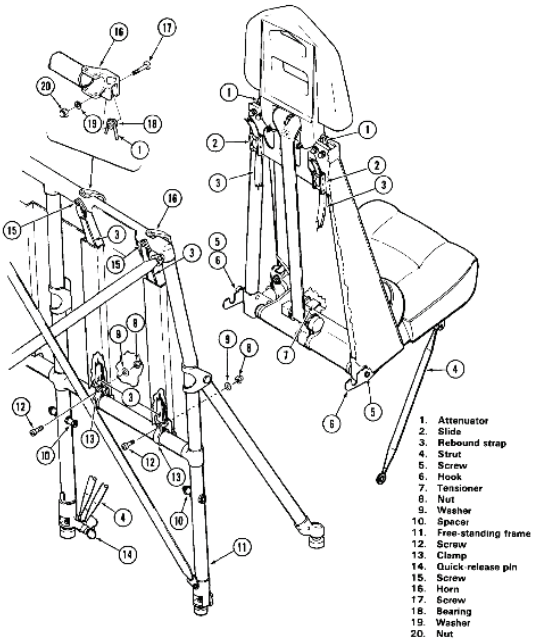
2. Remove screws (12) and lower end of straps (3) from clamps (13).

3. Remove pins (14) and struts (4) from frame (11).



PREVENT SEAT FROM FALLING.

4. Remove screws (17) from horns (16) and attenuators (1). Remove seat.



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Figure 1. Four Place Aft Facing Passenger Seat to Frame Installation (Sheet 1 of 1)

Four Place Aft Facing Passenger Seat - Installation on Frame

Procedure

NOTE

Procedure is typical for all seats.

1. Loosen two screws (5, Figure 1). Tap two hooks (6) downward as far as possible with a nonmetallic hammer.

2. Release keepers in two slides (2). Pull two straps (3) downward as far as possible.

3. Open tensioner (7).

4. Position two bearings (18) in two attenuators (1).

5. Position seat on frame (11) with attenuators (1) in horns (16) and hooks (6) around spacers (10).

6. Install two screws (17), washers (19), and nuts (20).

7. Install lower end of two straps (3) in clamps (13) with two screws (12), washers (9), and nuts (8).

NOTE

Do not pull straps (3) upward any more than necessary.

8. Pull upper end of straps (3) upward to align mounting holes. Install straps (3) on top rail of frame (11) with two screws (15).

9. Apply downward pressure on seat and install struts and engage hooks.

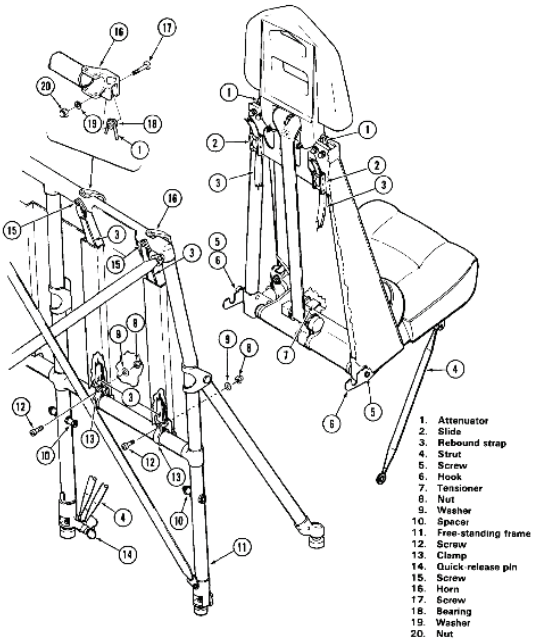
9.1. Install two struts (4) on frame (11) with two pins (14).

NOTE

A 90 degree screwdriver is required to tighten screws (5).

9.2. Tap two hooks (6) upward as far as possible with non-metallic hammer to engage spacers (10). Tighten screws (5).

10. Close tensioner (7).



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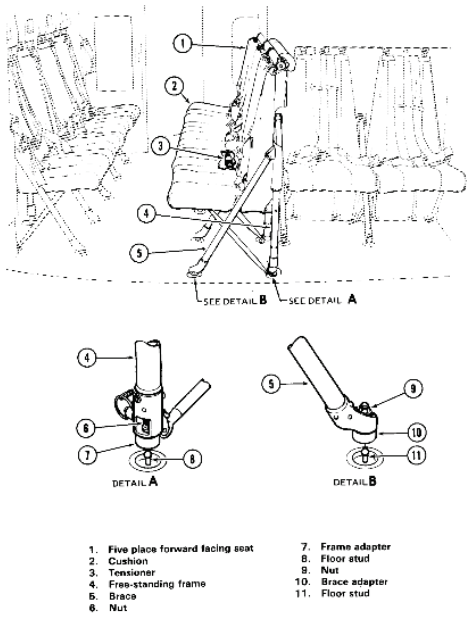
Figure 1. Four Place Aft Facing Passenger Seat to Frame Installation (Sheet 1 of 1)

Five Place Forward Facing Passenger Seat - Removal

Procedure

1. Disengage adapters (7 and 10, Figure 1) from studs (8 and 11) by pulling snap rings and raising adapter to unlocked position.

2. Lift seat (1). Remove seat.



1. Five place forward facing seat
2. Cushion
3. Tensioner
4. Free-standing frame
5. Brace
6. Nut
7. Frame adapter
8. Floor stud
9. Nut
10. Brace adapter
11. Floor stud

ICN-412-A-252000-B-97499-00015-A-001-01

Figure 1. Five Place Forward Facing Passenger Seat (Sheet 1 of 1)

Five Place Forward Facing Passenger Seat - Installation

Procedure

1. Position seat (1, Figure 1) with adapters (7 and 10) over studs (8 and 11).

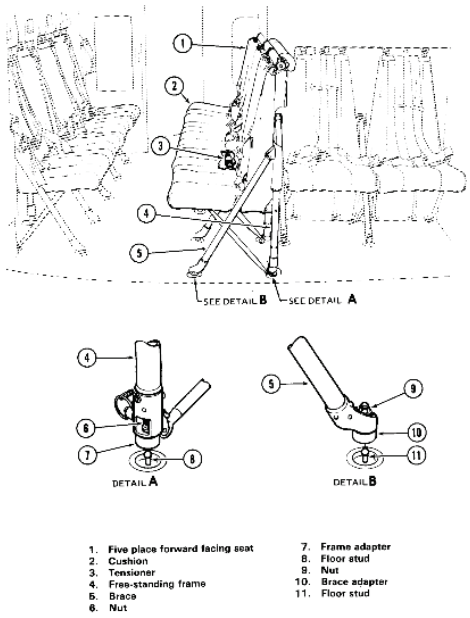
2. Apply downward pressure on frames (4) and braces (5) to engage adapters (7 and 10) on studs (8 and 11).

3. If adapters (7 and 10) and studs (8 and 11) are misaligned, perform one or more of the following to engage adapters (7 and 10).

3.1. Open tensioner (3). Apply downward pressure on frames (4) and braces (5). Close tensioner (3).

3.2. Loosen nuts (6 and 9). Apply downward pressure on cushion (2). Tighten nuts (6 and 9).

3.3. Open tensioner (3). Loosen nuts (6 and 9). Apply downward pressure on cushion (2). Tighten nuts (6 and 9). Close tensioner (3).



1. Five place forward facing seat
2. Cushion
3. Tensioner
4. Free-standing frame
5. Brace
6. Nut
7. Frame adapter
8. Floor stud
9. Nut
10. Brace adapter
11. Floor stud

ICN-412-A-252000-B-97499-00015-A-001-01

Figure 1. Five Place Forward Facing Passenger Seat (Sheet 1 of 1)

Five Place Forward Facing Passenger Seat - Removal from Frame

Procedure

NOTE

Procedure is typical for all seats.

1. Remove screws (8, Figure 1) and upper end of straps (9) from clamps (2).

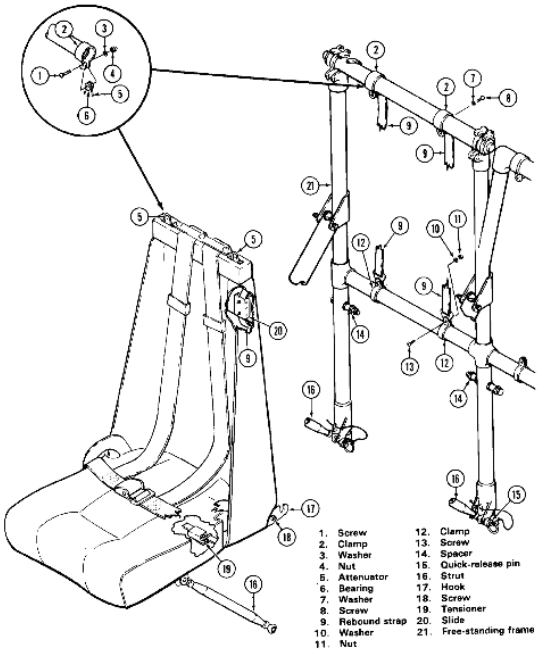
2. Remove screws (13) and lower end of straps (9) from clamps (12).

3. Remove pins (15) and struts (16) from frame (21).



PREVENT SEAT FROM FALLING.

4. Remove screws (1) from clamps (2) and attenuator (5). Remove seat.



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Figure 1. Five Place Forward Facing Passenger Seat to Frame Installation (Sheet 1 of 1)

Five Place Forward Facing Passenger Seat - Installation on Frame

Procedure

NOTE

Procedure is typical for all seats.

1. Loosen two screws (18, Figure 1). Tap two hooks (17) downward as far as possible with a nonmetallic hammer.

2. Release keepers in two slides (20). Pull two straps (9) downward as far as possible.

3. Open tensioner (19).

4. Position two bearings (6) in two attenuators (5).

5. Position seat on frame (21) with attenuators (5) in clamps (2) and hooks (17) around spacers (14).

6. Install two screws (1), washers (3), and nuts (4).

7. Install lower end of two straps (9) in clamps (12) with two screws (13), washers (10), and nuts (11).

NOTE

Do not pull straps (9) upward any more than necessary.

8. Pull upper end of straps (9) upward to align mounting holes. Install straps (9) on clamps (2) with two screws (8) and washers (7).

9. Apply downward pressure on seat and install struts and engage hooks.

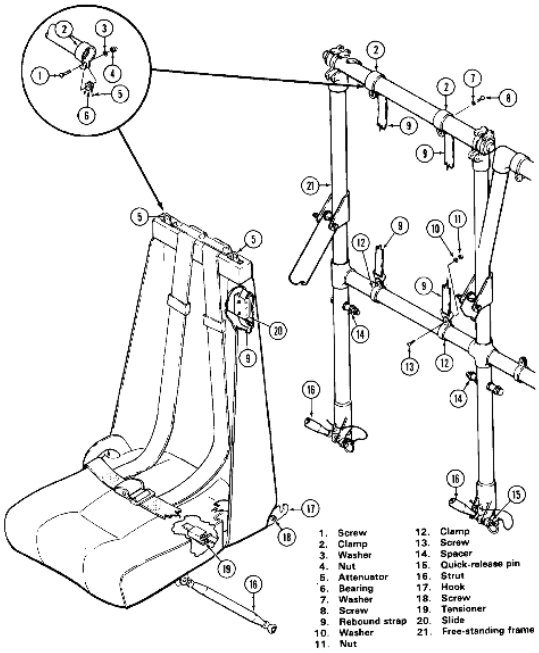
9.1. Install two struts (16) on frame (21) with two pins (15).

NOTE

A 90 degree screwdriver is required to tighten screws (18).

9.2. Tap two hooks (17) upward as far as possible with a non-metallic hammer to engage spacers (14). Tighten screws (18).

10. Close tensioner (19).



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Figure 1. Five Place Forward Facing Passenger Seat to Frame Installation (Sheet 1 of 1)

Miscellaneous Furnishings - Description

Miscellaneous furnishings consist of the map and data case and first aid kit.

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Map and Data Case - Description

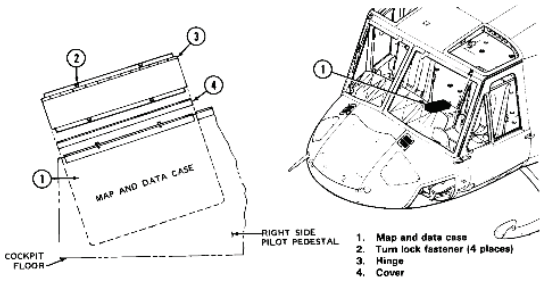
Map and data case is a plastic box with a hinged cover. The case is inserted in top of pilot pedestal on right side and is secured by four turn lock fasteners.

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Map and Data Case - Removal

Procedure

1. Unlock four fasteners (2, Figure 1) and remove case (1).



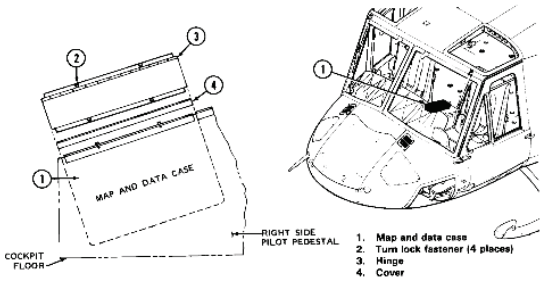
ICN-412-A-250000-B-97499-00001-A-001-01

Figure 1. Map and Data Case (Sheet 1 of 1)

Map and Data Case - Inspection

Procedure

1. Inspect case (1, Figure 1) for loose or damaged fasteners (2).
2. Inspect case (1) and hinge (3) for cracks and loose or missing rivets.



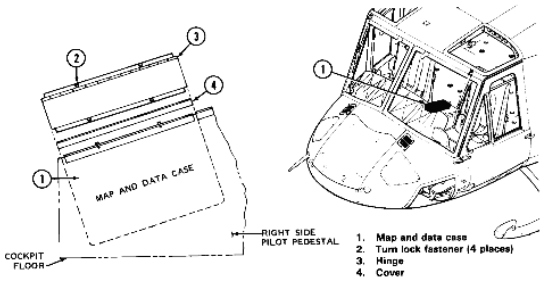
ICN-412-A-250000-B-97499-00001-A-001-01

Figure 1. Map and Data Case (Sheet 1 of 1)

Map and Data Case - Repair

Procedure

1. Repair case (1, Figure 1) in accordance with instructions in FAA Publication AC 43.13-1, Aircraft Inspection and Repair.



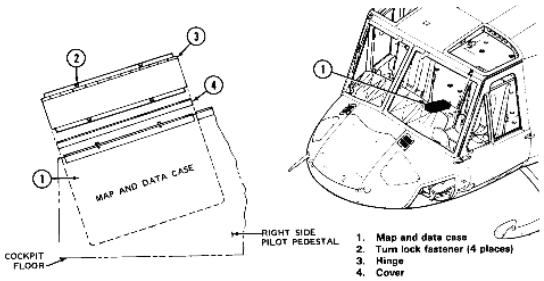
ICN-412-A-250000-B-97499-00001-A-001-01

Figure 1. Map and Data Case (Sheet 1 of 1)

Map and Data Case - Installation

Procedure

1. Insert case (1, Figure 1) into top of pedestal and secure with four fasteners (2).



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Figure 1. Map and Data Case (Sheet 1 of 1)

First Aid Kit - Description

The first aid kit is located on the pedestal between the pilot and copilot seats. The kit is held in place by three strips of pile (two at top and one at bottom center of the case).

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First Aid Kit - Removal

Procedure

1. Pull kit from pile strips.

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First Aid Kit - Inspection

Procedure

1. Inspect kit for completeness and coverage of contents.
2. Inspect contents for condition.
3. Inspect kit for security of installation.

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First Aid Kit - Repair

Procedure

1. Replace damaged or missing pile strips.
2. Replace damaged or missing components.

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First Aid Kit - Installation

Procedure

1. Push kit onto pile strips.

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CHAPTER 26
FIRE PROTECTION

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Fire Protection Systems - Description	DMC-412-A-26-00-00-00A-028A-A / 00292	2023-05-31	ALL
Engine Fire Detection - Description	DMC-412-A-26-10-00-00A-028A-A / 00293	2023-05-31	ALL
Engine Upper Fire Detection Sensing Elements - Description	DMC-412-A-26-10-00-01A-042A-A / 00294	2023-05-31	ALL
Engine Upper Fire Detection Sensing Elements - Removal	DMC-412-A-26-10-00-01A-520A-A / 00295	2023-05-31	ALL
Engine Upper Fire Detection Sensing Elements - Inspection and Repair	DMC-412-A-26-10-00-01A-280A-A / 00296	2023-05-31	ALL
Engine Upper Fire Detection Sensing Elements - Installation	DMC-412-A-26-10-00-01A-720A-A / 00297	2023-05-31	ALL
Engine Firewall Fire Detection Sensing Elements - Description	DMC-412-A-26-10-00-02A-042A-A / 00298	2023-05-31	ALL
Engine Firewall Fire Detection Sensing Elements - Removal	DMC-412-A-26-10-00-02A-520A-A / 00299	2023-05-31	ALL
Engine Firewall Fire Detection Sensing Elements - Inspection and Repair	DMC-412-A-26-10-00-02A-280A-A / 00300	2023-05-31	ALL
Engine Firewall Fire Detection Sensing Elements - Installation	DMC-412-A-26-10-00-02A-720A-A / 00301	2023-05-31	ALL
Engine Firewall Fire Detection Sensing Elements - Operational Check	DMC-412-A-26-10-00-02A-320A-A / 00302	2023-05-31	ALL
Baggage Compartment Smoke Detection System - Description	DMC-412-A-26-10-00-03A-042A-A / 00303	2023-05-31	ALL
Baggage Compartment Smoke Detector - Removal	DMC-412-A-26-10-00-03A-520A-A / 00304	2023-05-31	ALL
Baggage Compartment Smoke Detector - Inspection	DMC-412-A-26-10-00-03A-280A-A / 00305	2023-05-31	ALL
Baggage Compartment Smoke Detector - Installation	DMC-412-A-26-10-00-03A-720A-A / 00306	2023-05-31	ALL
Baggage Compartment Smoke Detector - Operational Check	DMC-412-A-26-10-00-03A-320A-A / 00307	2023-05-31	ALL
Fire Extinguishers - Description	DMC-412-A-26-20-00-00A-028A-A / 00308	2023-05-31	ALL
Engine Compartment Fire Extinguishers - Description	DMC-412-A-26-20-00-01A-042A-A / 00309	2023-05-31	ALL
Engine Compartment Fire Extinguishers - Removal	DMC-412-A-26-20-00-01A-520A-A / 00310	2023-05-31	ALL
Engine Compartment Fire Extinguishers - Inspection	DMC-412-A-26-20-00-01A-280A-A / 00311	2023-05-31	ALL
Engine Compartment Fire Extinguishers - Repair/Replacement	DMC-412-A-26-20-00-01A-685A-A / 00312	2023-05-31	ALL
Engine Compartment Fire Extinguishers - Installation	DMC-412-A-26-20-00-01A-720A-A / 00313	2023-05-31	ALL
Discharge Indicator Disc - Description	DMC-412-A-26-20-00-02A-042A-A / 00314	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Discharge Indicator Disc - Removal	DMC-412-A-26-20-00-02A-520A-A / 00315	2023-05-31	ALL
Discharge Indicator Disc - Inspection	DMC-412-A-26-20-00-02A-280A-A / 00316	2023-05-31	ALL
Discharge Indicator Disc - Installation	DMC-412-A-26-20-00-02A-720A-A / 00317	2023-05-31	ALL
Portable Fire Extinguishers and Brackets - Description	DMC-412-A-26-20-00-03A-042A-A / 00318	2024-12-19	ALL
Portable Fire Extinguishers and Brackets - Removal	DMC-412-A-26-20-00-03A-520A-A / 00319	2024-12-19	ALL
Portable Fire Extinguishers and Brackets - Inspection	DMC-412-A-26-20-00-03A-280A-A / 00320	2024-12-19	ALL
Portable Fire Extinguishers and Brackets - Repair	DMC-412-A-26-20-00-03A-685A-A / 00321	2023-05-31	ALL
Portable Fire Extinguishers and Brackets - Installation	DMC-412-A-26-20-00-03A-720A-A / 00322	2024-12-19	ALL

Fire Protection Systems - Description

Fire protection items include engine fire detection elements and extinguishers, baggage compartment smoke detector, and two cabin mounted, hand-held, portable fire extinguishers.

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Engine Fire Detection - Description

Engine compartment fire detection consists of two heat sensing elements and two flexible cable assemblies for each power section of the engine. A press-to-test switch is provided to test the system.

If a fire or overheat condition occurs, the heat sensing elements in the affected power section close a circuit, which causes the corresponding FIRE PULL handle (located on the instrument panel) to illuminate. Pulling the handle turns off the heater by de-energizing both bleed air valves and causes the engine air management bypass door and the fuel shutoff valve to close, shutting off both air and fuel to the affected power section. Pulling the FIRE 1 PULL or FIRE 2 PULL handle also arms the main and reserve fire extinguisher circuits for the corresponding power section. Placing the FIRE EXT switch to MAIN causes the main fire extinguisher bottle to discharge extinguishing agent (Halon and nitrogen) into the selected power section compartment. Placing the FIRE EXT switch to RESERVE discharges the reserve fire extinguisher bottle into the same compartment.

Pressing the FIRE EXT PRESS TO TEST switch places a low resistance across each of the fire sensing elements to simulate a fire/overheat condition. If sensing elements are operating properly, both FIRE 1 PULL and FIRE 2 PULL handles will illuminate.

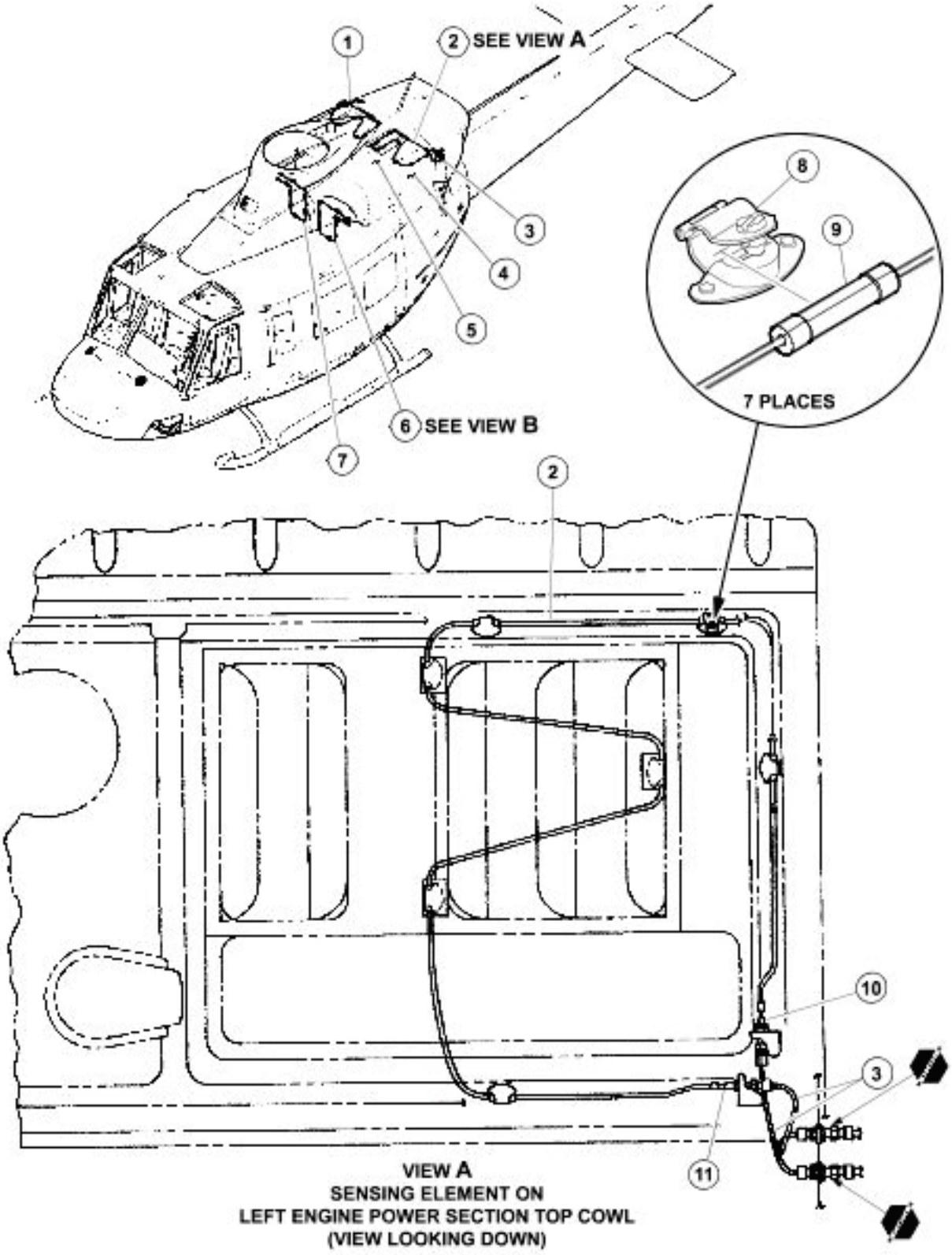
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Engine Upper Fire Detection Sensing Elements - Description

NOTE

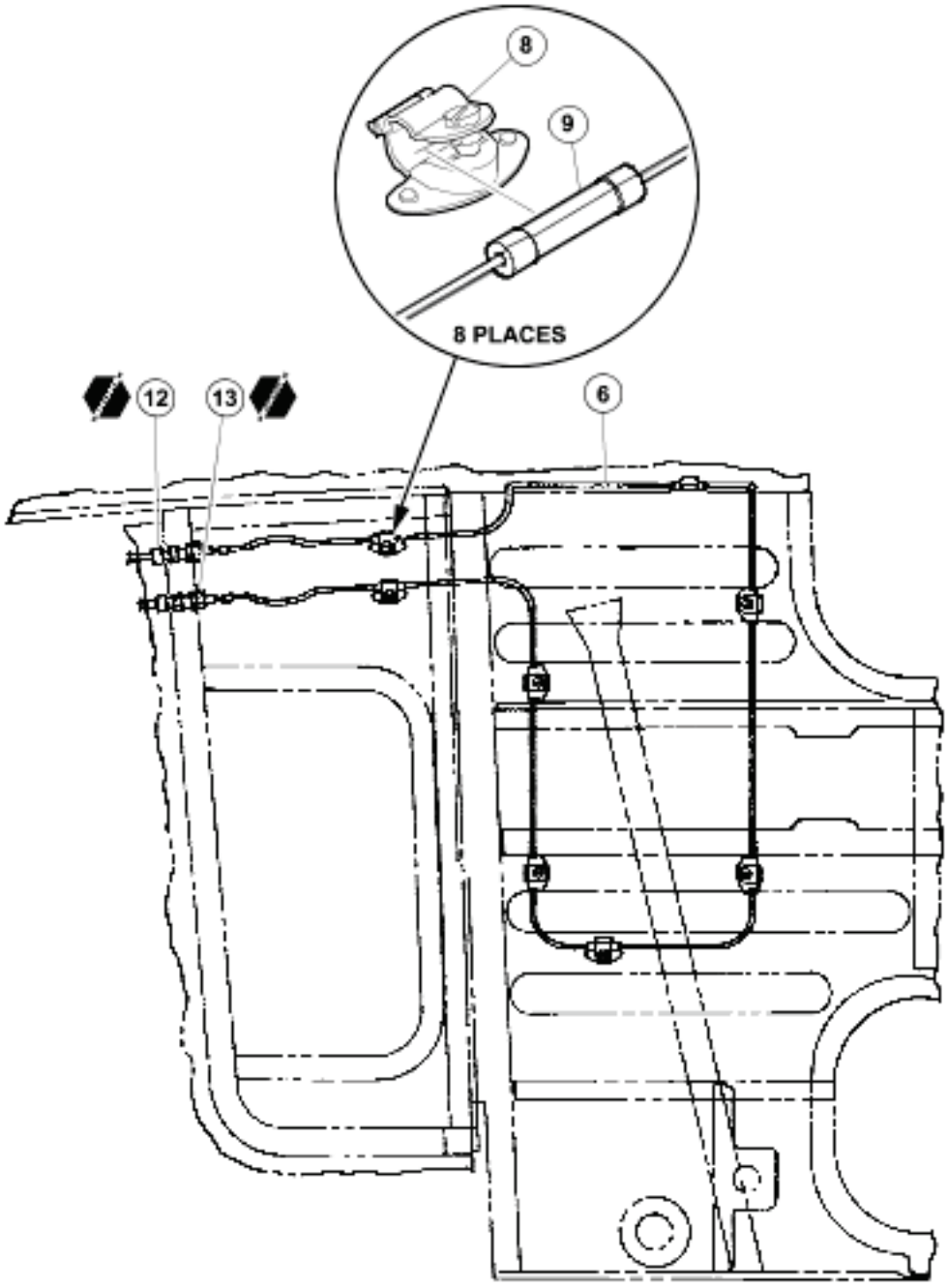
Maintenance procedures are the same for both left and right upper sensing elements.

The upper fire detection sensing elements are mounted on the underside of the upper engine cowling (Figure 1).



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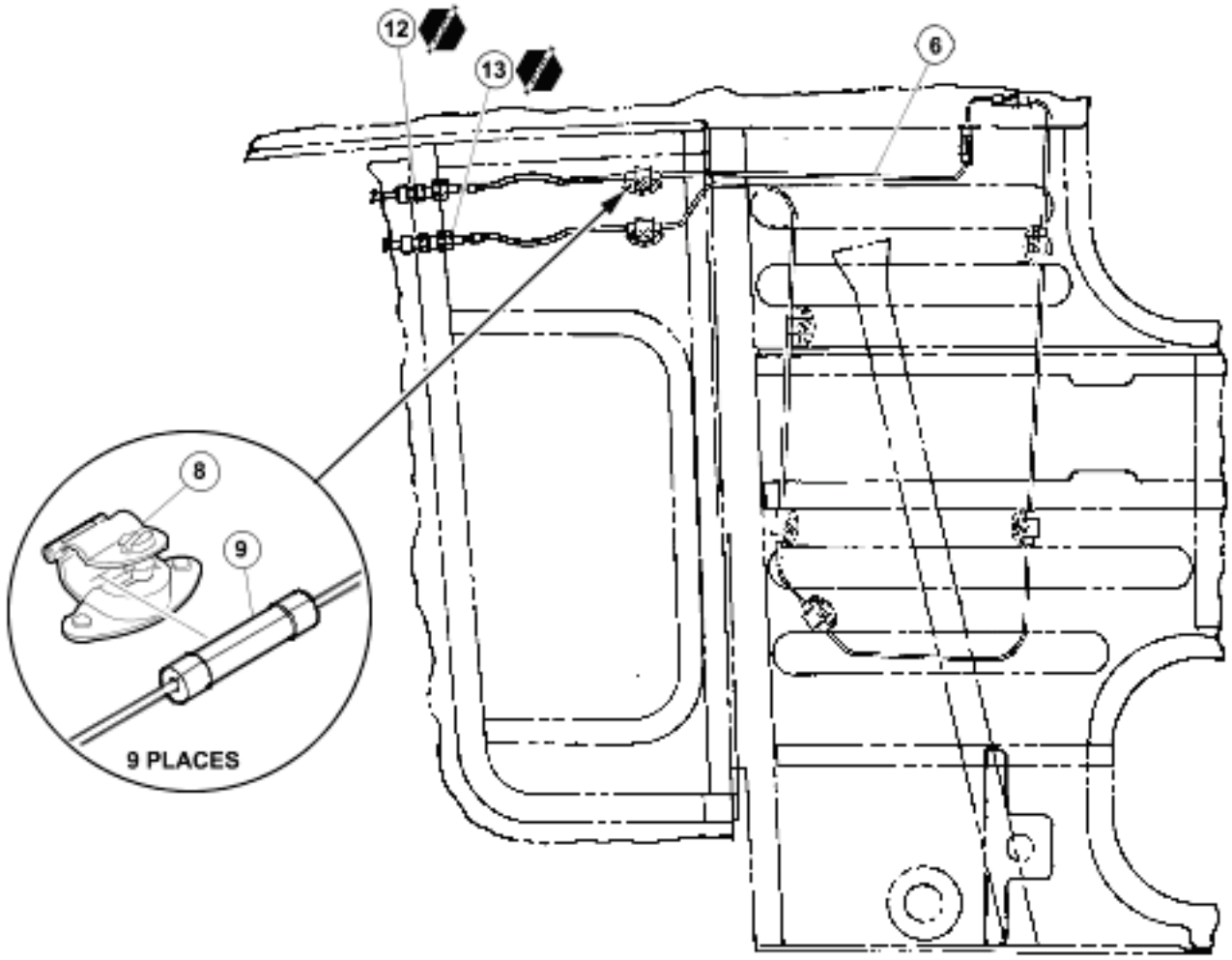
Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 1 of 3)



VIEW B
SENSING ELEMENT ON
LEFT ENGINE POWER SECTION FIREWALL
(VIEW LOOKING FORWARD)
(S/N 33001 THROUGH 36693)

ICN-412-A-261000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 2 of 3)



VIEW B
SENSING ELEMENT ON LEFT ENGINE
POWER SECTION FIREWALL
(S/N 36694 AND SUBSEQUENT)

1. Sensing element (right engine power section top cowl)
2. Sensing element (left engine power section top cowl)
3. Flex cable
4. Upper engine cowling
5. Engine cowl top panel
6. Sensing element (left power section firewall)
7. Sensing element (right power section firewall)
8. Clamp
9. Grommet
10. Electrical connector (8Z7P1)
11. Electrical connector (8Z7P2)
12. Electrical connector (8Z9P1)
13. Electrical connector (8Z9P2)

LOCKWIRE (C-405)

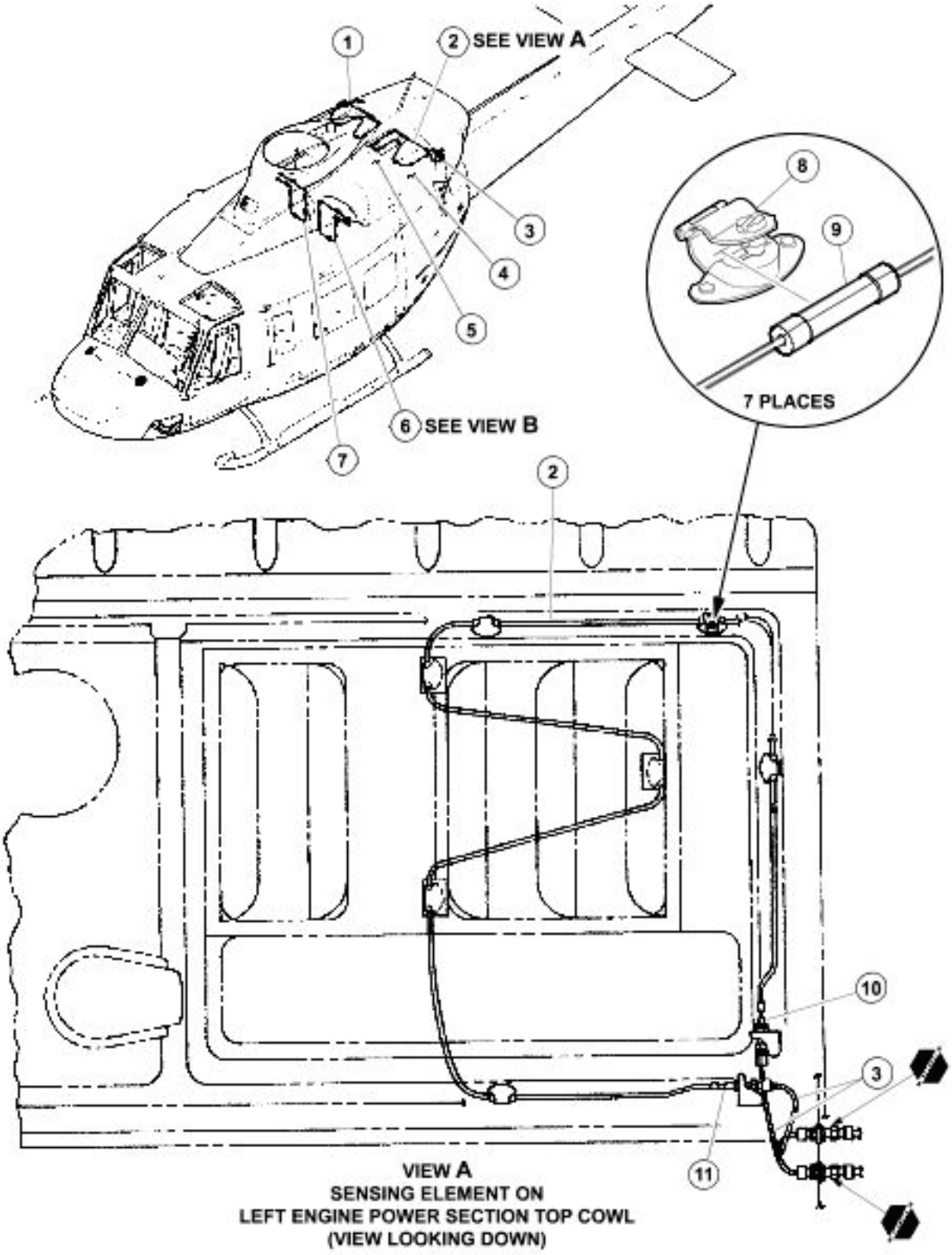
ICN-412-A-261000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 3 of 3)

Engine Upper Fire Detection Sensing Elements - Removal

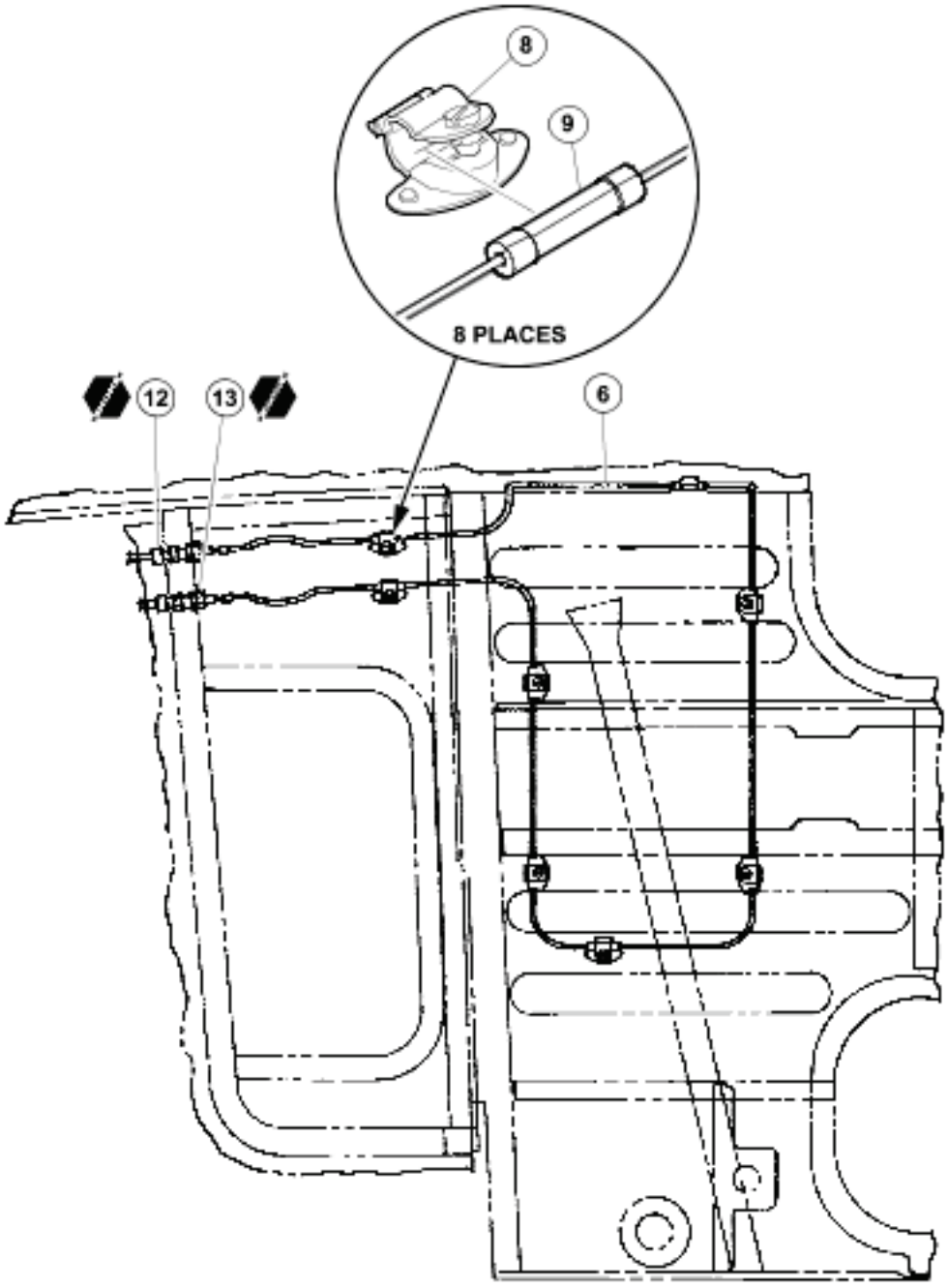
Procedure

1. Verify all electrical power is removed from the helicopter.
2. Disconnect the electrical connectors (10 and 11, Figure 1).
3. Unlock the studs of the clamps (8). Remove the screws from the six remaining clamps in the same manner.
4. Spread the clamps as little as possible and remove the sensing element (2).



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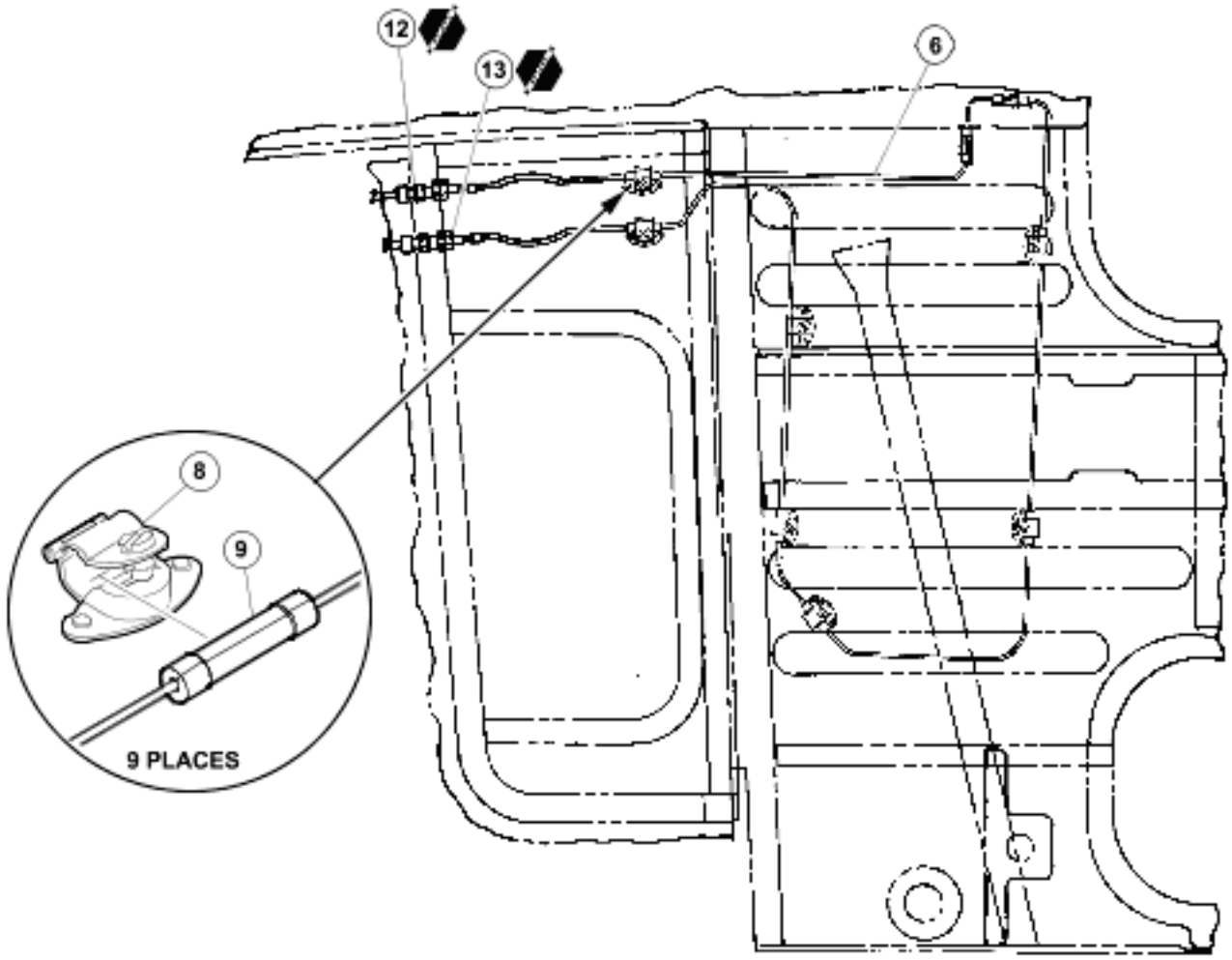
Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 1 of 3)



VIEW B
SENSING ELEMENT ON
LEFT ENGINE POWER SECTION FIREWALL
(VIEW LOOKING FORWARD)
(S/N 33001 THROUGH 36693)

ICN-412-A-261000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 2 of 3)



VIEW B
SENSING ELEMENT ON LEFT ENGINE
POWER SECTION FIREWALL
(S/N 36694 AND SUBSEQUENT)

1. Sensing element (right engine power section top cowl)
2. Sensing element (left engine power section top cowl)
3. Flex cable
4. Upper engine cowling
5. Engine cowl top panel
6. Sensing element (left power section firewall)
7. Sensing element (right power section firewall)
8. Clamp
9. Grommet
10. Electrical connector (8Z7P1)
11. Electrical connector (8Z7P2)
12. Electrical connector (8Z9P1)
13. Electrical connector (8Z9P2)

LOCKWIRE (C-405)

ICN-412-A-261000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 3 of 3)

Engine Upper Fire Detection Sensing Elements - Inspection and Repair

Procedure

1. Refer to 412-A-96-00-00-00A-009A-A / 01870 for the inspection of sensing elements. Replace sensing elements and/or retaining hardware if unserviceable.

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Engine Upper Fire Detection Sensing Elements - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Verify that all electrical power is removed from the helicopter.

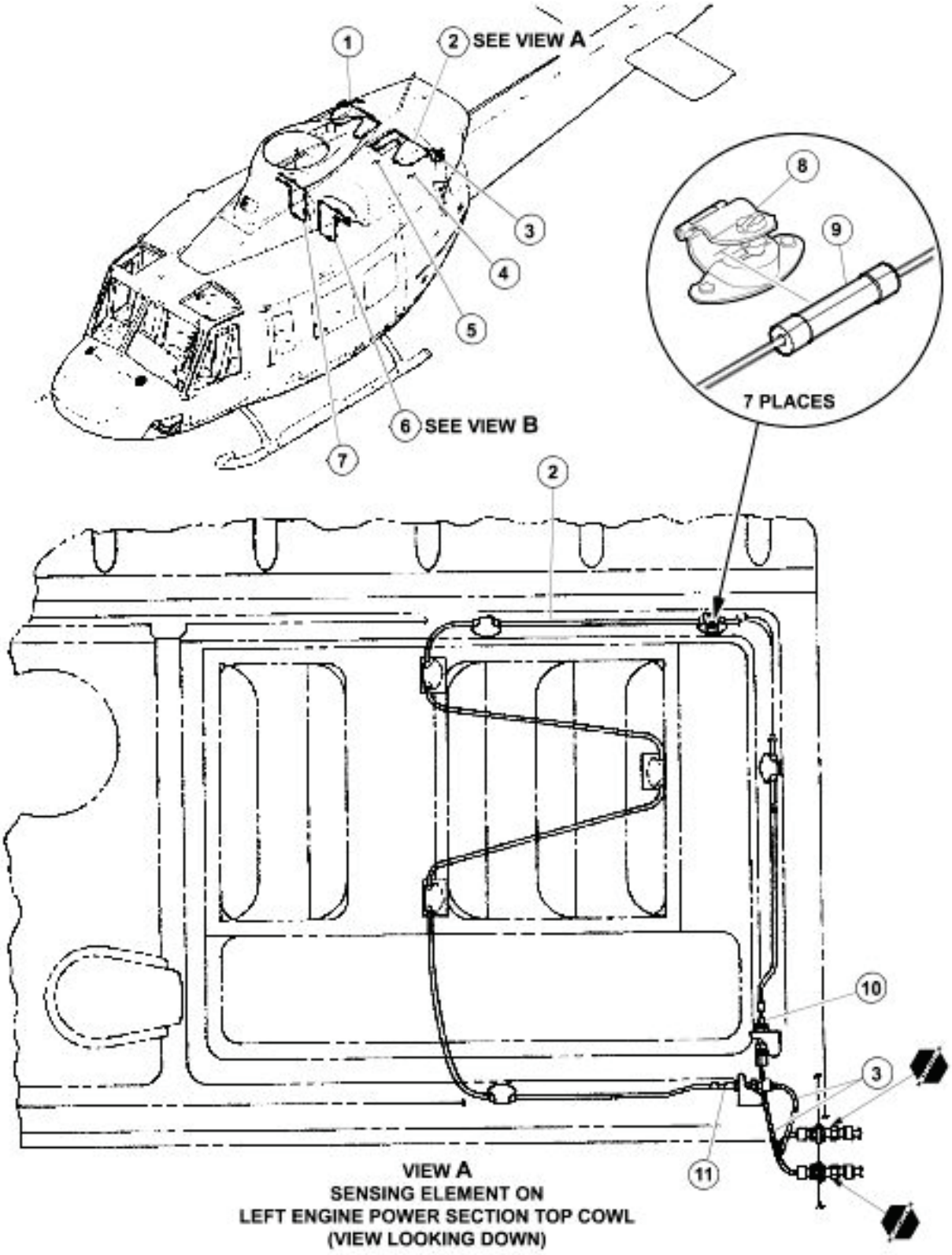
2. Position the sensing element (2, Figure 1) on the upper cowl, as shown in View A.

3. Spread the clamps (8) as little as possible and install the sensing element (2) in the clamps.

4. Lock the studs of the clamps (8).

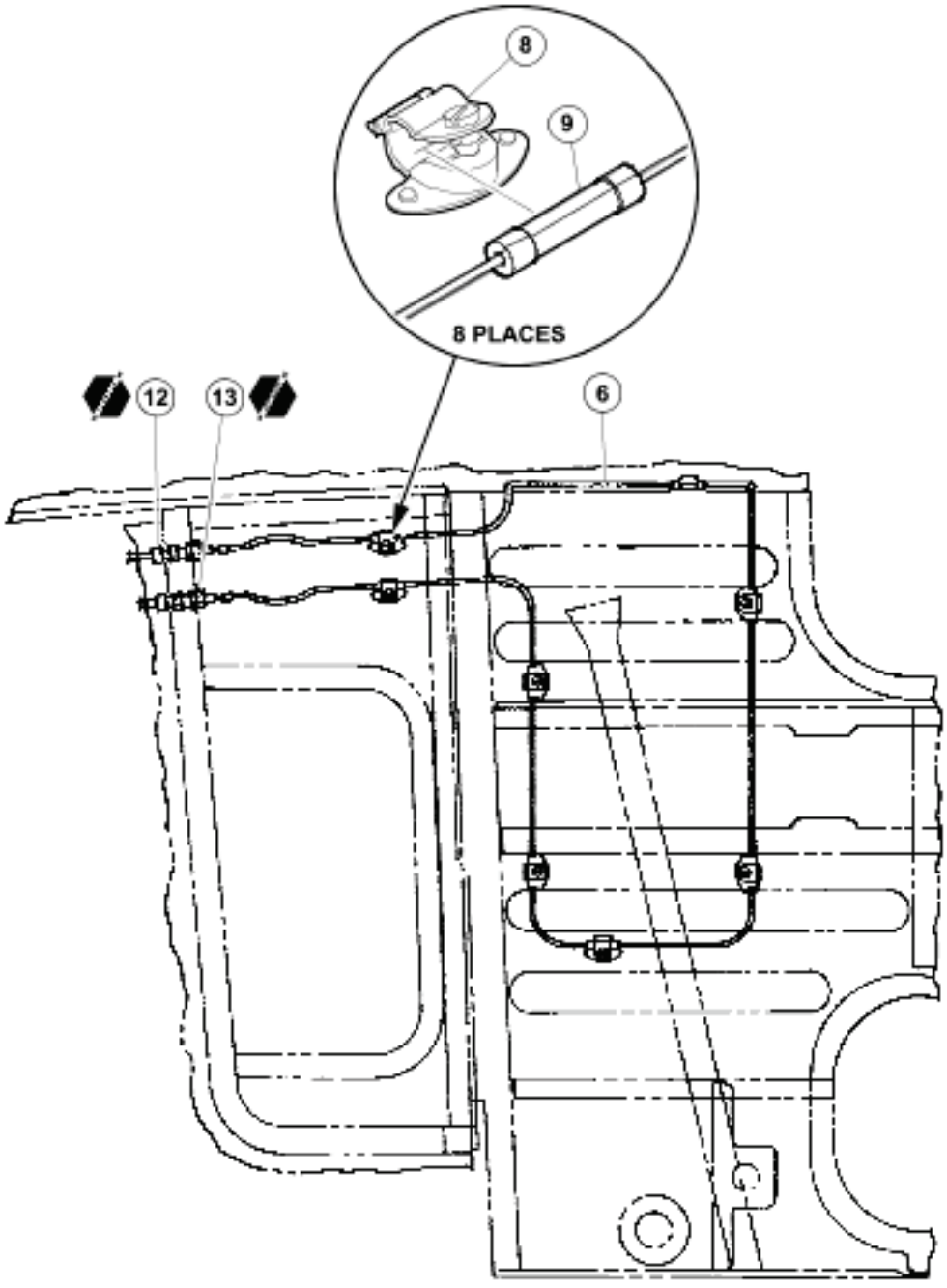
5. Connect the electrical connectors (10 and 11). Secure with lockwire (C-405).

6. Perform an operational check (412-A-26-10-00-02A-320A-A / 00302).



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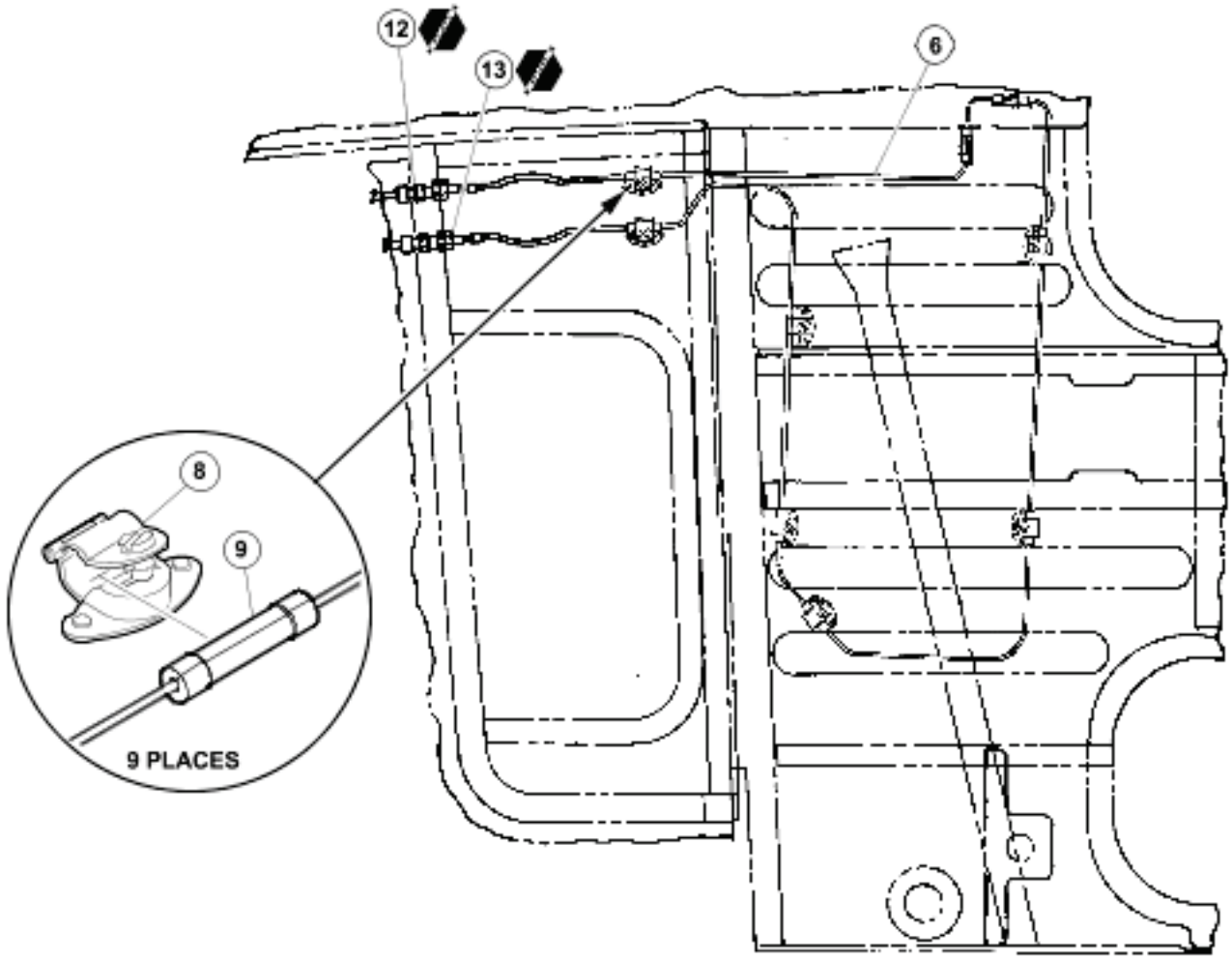
Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 1 of 3)



VIEW B
SENSING ELEMENT ON
LEFT ENGINE POWER SECTION FIREWALL
(VIEW LOOKING FORWARD)
(S/N 33001 THROUGH 36693)

ICN-412-A-261000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 2 of 3)



VIEW B
SENSING ELEMENT ON LEFT ENGINE
POWER SECTION FIREWALL
(S/N 36694 AND SUBSEQUENT)

1. Sensing element (right engine power section top cowl)
2. Sensing element (left engine power section top cowl)
3. Flex cable
4. Upper engine cowling
5. Engine cowl top panel
6. Sensing element (left power section firewall)
7. Sensing element (right power section firewall)
8. Clamp
9. Grommet
10. Electrical connector (8Z7P1)
11. Electrical connector (8Z7P2)
12. Electrical connector (8Z9P1)
13. Electrical connector (8Z9P2)

LOCKWIRE (C-405)

ICN-412-A-261000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 3 of 3)

Engine Firewall Fire Detection Sensing Elements - Description

NOTE

Maintenance procedures are the same for both left and right sensing elements.

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